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Electrical, Electronic and Magnetic Properties of Solids

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Electrical, Electronic and Magnetic Properties of Solids

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Preface

In the Preface to our last book (*Atomistic Properties of Solids*, 2011) we made a statement: “....important topics like like electrical, electronic and magnetic properties of solids which could not be included in this book may form another volume”. In presenting this book, *Electrical, Electronic and Magnetic Properties of Solids*, we are happy to redeem that hesitant promise.

The book starts with a chapter called “Preliminaries” which contains a discussion of some general topics like atomic structure, quantum mechanics etc, the knowledge of which would facilitate understanding of the following text. This chapter is optional reading for readers who have gone through a graduate course in physics but is recommended reading for readers who have not gone through such a course.

In Chap. 2 we develop the classical theory of electrons which assumes the presence of free electrons. The concepts of electron collisions, scattering, mean free path and relaxation time are introduced. Combining these with the Maxwell–Boltzmann statistics, expressions are derived for several properties like the electrical conductivity, Hall effect, Weidemann–Franz ratio etc with moderate agreement with experimental results. Sommerfeld’s theory (Chap. 3) makes use of quantum theory and the Fermi–Dirac statistics. The concept of the Fermi level is introduced. The results from the Sommerfeld theory are in better agreement with experiment.

The main framework of the band theory of solids is developed in Chap. 4. Using the one-dimensional Kronig–Penney model, Bloch’s theorem and Bloch functions are discussed. Concepts of Brillouin zones, energy bands, forbidden gap, effective mass and holes are introduced. A great success of the band theory is that it explains the existence of insulators, metals and semiconductors. In Chap. 5, we discuss the application of band theory to real crystals. Brillouin zones, Fermi surfaces and effective masses of three dimensional crystals are worked out. Also, experiments to determine the band theory parameters are described.

Our treatment of electronic properties is limited to semiconductors. Properties of intrinsic and extrinsic semiconductors like electrical conductivity, Hall effect are worked out in Chap. 6. Semiconductors have led to the discovery of many electronic devices; some typical devices are discussed in Chap. 7.

Chapter 8 begins with definitions of the basic parameters of magnetism and their experimental determination. The classical theory of diamagnetism is discussed. The classical as well as quantum theories of paramagnetism are treated. Chapter 9 deals with magnetic phenomena due to ordering of spins. Theories of ferromagnetism, antiferromagnetism and ferrimagnetism are discussed. Chapter 10 starts with a brief introduction to magnetic symmetry and magnetic structures. Their experimental determination with the help of neutron diffraction is described. Theory, instrumentation and applications of nuclear magnetic resonance and electron spin resonance form the subject of Chap. 11.

Aspects of fast ion conduction, an emerging area are discussed in Chap. 12. A comprehensive treatment of superconductivity is given in Chap. 13. Several exciting experiments are described. The treatment of theories follows a historical perspective. Some of the important applications are considered.

We quote from the preface to our earlier book: “Comprehensiveness is the main feature of this book. Each chapter is, in a way, an exhaustive essay on the topic. Another feature is that the book gives equal importance to experiment and theory. Experimental methods are described in detail and experimental data are quoted so that readers can get a feel of magnitude as well as accuracy. In discussing theories a logical approach is adopted leading the reader from the simpler to the more elaborate theories”. These remarks apply to the present book too. Yet another feature of the book is the full-length treatment of the applications of band theory to real crystals (Chap. 5), experimental determination of magnetic structures (Chap. 10) and fast ion conduction (Chap. 12).

We have benefitted from the support and encouragement received from Prof. K.G. Bansigir (Jiwaji University, Gwalior), Prof. YVGS Murti (Indian Institute of Technology, Chennai) and Prof. K.G. Prasad (Tata Institute of Fundamental Research, Bombay). We also acknowledge Prof. B.J. Rao (Tata Institute of Fundamental Research, Bombay), Dr. K. Ravi Kumar (Indian Institute of Chemical Technology, Hyderabad), Dr. Harish Dixit (Jawaharlal Nehru Centre for advanced Scientific Research, Bangalore), Dr. P. Hima Bindu (Osmania University, Hyderabad) and Dr. D. Rajesh (University of Hyderabad) for their cooperation.

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This is our fourth (and probably last) contribution to the *Springer Series in Materials Science*. Throughout this work, from the conceptual stage right up to the appearance of the finished book, Dr. C.E. Ascheron, Executive Editor, Springer, has been our friend, advisor and guide. We are grateful to him for all the help and we will always cherish our association with him and with Springer.

In the preface to his celebrated book on X-ray diffraction, B.E. Warren quotes a line of unknown origin: “if you would a teacher be, by your students you will be taught”. Each of us has spent close to four decades in the company of students. This book is appropriately dedicated to all our students.

Hyderabad, India

D.B. Sirdeshmukh
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Chapter 1

Preliminaries

Abstract The electrical, electronic and magnetic properties of solids are quite complex. On the basis of electrical behavior, solids can be classified into ionic, metallic, semiconducting, superionic and superconducting solids. The behaviour of each type is explained by a separate theory. Similarly, on the basis of the magnetic behavior, solids can be classified into diamagnetic, paramagnetic, ferromagnetic, antiferromagnetic and ferrimagnetic types. Again, the behavior of each type is accounted for by separate theories. In this chapter some general topics like atomic structure, quantum mechanics etc. are discussed. We believe that familiarity with these topics will facilitate the understanding of the theories discussed in the following text. This discussion should be particularly useful to readers who have not gone through a graduate course in physics. The Bohr model, the Sommerfeld model and the vector model of the hydrogen atom are discussed in detail. They lead to the quantum numbers associated with the orbital and spin angular momentum. Concepts of space quantization and Larmor precession lead to the magnetic quantum numbers. Quantum mechanics is discussed. The Schrödinger equation is formulated and is applied to the problems of the linear oscillator, the particle in a box and the hydrogen atom. Perturbation theory and the method of variation are also developed. Pauli's exclusion principle is discussed and applied to arrive at the electronic structure of atoms. Statistical mechanics and electromagnetic theory are treated briefly.

1.1 General

Several properties of solids are atomistic in nature. They have their origin in factors like crystal structure, interatomic forces and atomic displacements. Thermal expansion, elastic constants, dielectric constants and dislocations are examples of atomistic properties. On the other hand, there are properties which depend on other factors like electron states and electron spin; electrical properties of metals,

semiconduction, superconductivity and ferromagnetism are examples. The present book deals with electrical and magnetic properties of solids.

There is a wide variety in the electrical behaviour of solids. Insulators do not conduct electricity (there is a feeble conduction of the order of 10^{-14} – 10^{-22} /ohm-cm but that is due to transport of ions and not electrons). Metals have high conductivity of the order of 10^6 /ohm-cm. Further the conductivity of metals decreases with increasing temperature. On the other hand, semiconductors have an intermediate value of conductivity (10^{-9} – 10^2 /ohm-cm) at ordinary temperatures. In contrast to metals, the conductivity of semiconductors increases by several orders with increasing temperature. Superconductors have enormous conduction below a certain (generally) low temperature. Classical theories could account for only some features of the conducting behaviour of metals. In particular, they could not account for the difference between the electrical behaviour of metals and semiconductors. This difference was explained only by the band theory of solids.

According to the band theory, the energy levels of electrons in solids form bands (Fig. 1.1). The topmost completely filled band is called the valence band (VB) and the band above it is called the conduction band (CB). The energy difference between the bottom of the CB and the top of the VB is called the energy gap (E_g). The value of E_g is crucial in determining the electrical behaviour of solids.

In insulators, the energy gap has a large value. If thermal energy is provided to the electrons at the top of VB, it is not enough to help them to jump into the CB. Hence insulators cannot conduct. In semiconductors, E_g is small and the thermally energized electrons at the top of VB can jump into the CB band; once they are in the CB, they can conduct. The CB in metals is partially occupied or the CB and VB overlap. The electrons in this case readily respond to the electric field and conduct electricity. Besides explaining the difference between insulators, metals and semiconductors, the band theory provides explanation for several other properties of solids.

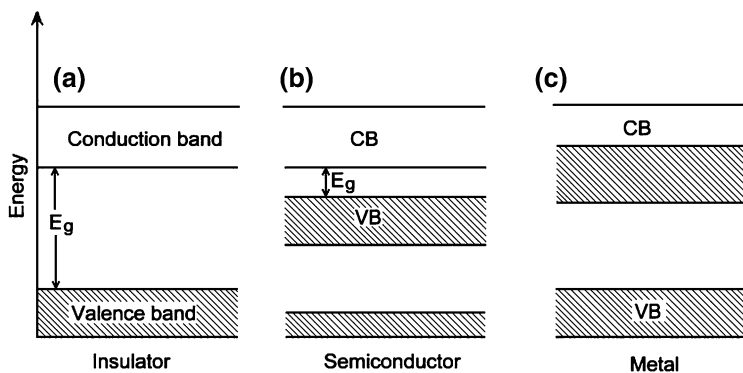


Fig. 1.1 Electron distribution in **a** an insulator, intrinsic **b** semiconductor and **c** metal at 0 K. The shaded regions are occupied by electrons. E_g is the forbidden gap

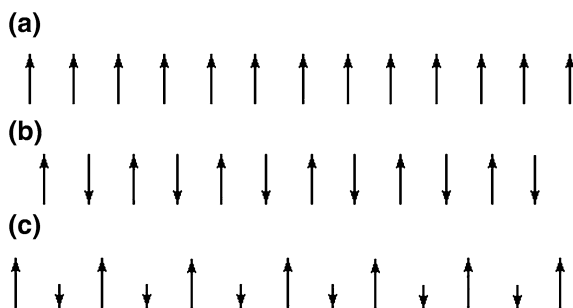


Fig. 1.2 Linear arrays of spins illustrating **a** Ferromagnetic, **b** Antiferromagnetic and **c** Ferrimagnetic orderings

On the basis of their magnetic behaviour, solids are characterized as diamagnetic, paramagnetic, ferromagnetic, antiferromagnetic and ferrimagnetic. Diamagnetic solids have a small negative susceptibility which is independent of temperature. In the other four types, the susceptibility is positive and temperature dependent. But the temperature variation of susceptibility is different in each case. The temperature dependence of the susceptibility is given by

$$\begin{aligned}\chi &= C/T && \text{for paramagnetics} \\ \chi &= C/(T - \theta) && \text{for ferromagnetics and} \\ \chi &= C/(T + \theta') && \text{for antiferromagnetics.}\end{aligned}$$

While it is possible to explain paramagnetism in terms of classical concepts, the other forms need a new concept viz. ordering of spins and their interaction which is a quantum mechanical phenomenon. The ordering of spins in different types is shown in Fig. 1.2. In ferromagnetics, spins of equal magnitude are ordered in parallel orientation. In antiferromagnetics, the spins of equal magnitudes are ordered in antiparallel orientation. In ferrimagnetics, the ordering is antiparallel but the spins are of unequal magnitude.

In order to understand and appreciate the electrical, electronic and magnetic properties of solids, it is desirable to have some familiarity with atomic structure, quantum mechanics, statistical mechanics and electromagnetic theory. These topics are discussed in the sections that follow. This discussion is **optional reading** for readers who have gone through a graduate physics course. However, for readers who have not gone through such a course, it is **recommended reading**.

1.2 Atomic Structure

J.J. Thomson's experiments indicated that an atom consists of a part with positive charge and another with negative charge. Rutherford's experiments on scattering of α -rays by matter showed that the atom is mostly empty space. The atomic mass is

concentrated in a region of radius $\sim 10^{-13}$ cm; this is called the nucleus. The nucleus carries the positive charge. The negative charge called electrons was supposed to exist in the form of a 'mist' around the nucleus. A clearer picture of the position of the electrons with respect to the nucleus emerged from spectroscopic experiments on hydrogen and their interpretation by Neils Bohr in 1913.

1.2.1 The Hydrogen Spectrum

The hydrogen spectrum is shown in Fig. 1.3. It consists of a number of lines. Each line is characterized by a wavelength λ or a frequency $\nu (=c/\lambda)$ or a wave number $\bar{\nu}(=1/\lambda)$. These lines form groups or 'series'. The first series to be observed was the Balmer series. It occurs mostly in the visible part of the spectrum. It is found that the wave numbers of the lines in the Balmer series follow a simple and elegant relation:

$$\bar{\nu} = R \left(\frac{1}{2^2} - \frac{1}{n^2} \right) \quad (1.1)$$

Here, R is a constant called the Rydberg constant and n is an integer that takes values 3, 4, 5, By fitting the observed wave numbers to the equation, it was found that $R = 109677 \text{ cm}^{-1}$. Subsequently, other series were discovered in the far ultraviolet and infrared (Fig. 1.3) which also followed equations similar to (1.1). Thus the general formula for the series in the hydrogen spectrum is:

$$\bar{\nu} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad (1.2)$$

where the integral index n_1 has a fixed value for a given series and the index n_2 takes running integral values greater than n_1 . The series and the corresponding indices are given in Table 1.1.

It may be mentioned that each line in the hydrogen spectrum has a fine structure which means that each line is actually made up of closely spaced components which can be observed only with equipment having a high resolution.

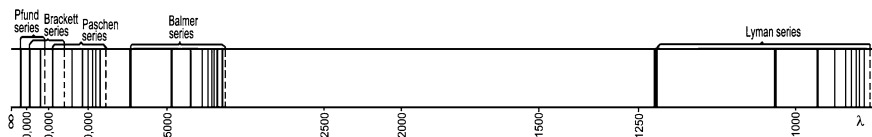


Fig. 1.3 Schematic representation of the hydrogen atom spectrum

Table 1.1 Description of the series in hydrogen spectrum

Name of the series	Year of discovery	Region of spectrum	n_1	n_2
Lyman	1916	Far ultraviolet	1	2, 3, 4...
Balmer	1884	Visible	2	3, 4, 5...
Paschen	1908	Infrared	3	4, 5, 6...
Brackett	1922	Infrared	4	5, 6, 7...
Pfund	1924	Infrared	5	6, 7, 8...

1.2.2 The Bohr Model

The Bohr model of the atom is shown in Fig. 1.4. The electron with charge $-e$ rotates with velocity v in a circular orbit of radius r about the heavy nucleus with charge $+Ze$. If $Z = 1$, the model represents the hydrogen atom. If $Z > 1$, it represents a singly charged ion of an atom with atomic number Z . The mass of the electron is m . Bohr made the following three assumptions.

- (i) The electron rotates in the Coulomb field of the nucleus. The centripetal force that keeps the electron in motion is

$$\frac{mv^2}{r} = -\frac{Ze^2}{r^2}. \quad (1.3)$$

It can be easily shown that the total energy W is

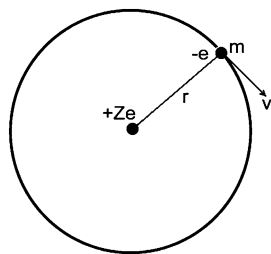
$$W = -\frac{Ze^2}{2r}. \quad (1.4)$$

- (ii) So far, we have not put any restriction on the radius of the orbit. Bohr made the important assumption that the angular momentum p_ϕ is quantized i.e.

$$\oint p_\phi d\phi = nh \quad (1.5)$$

where n is an integer and h is Planck's constant. We may write (1.5) as

Fig. 1.4 Bohr's atomic model



$$p_\phi = mvr = nh/2\pi. \quad (1.6)$$

From (1.3), (1.4) and (1.6) we get

$$W = -\frac{2\pi^2 mZ^2 e^4}{n^2 h^2}. \quad (1.7)$$

Thus, we have two new concepts: the angular momentum is quantized and the electron energy is not continuous but is discrete. The quantum number n is called the principal quantum number.

- (iii) According to classical theory of radiation, an accelerated charge emits energy in the form of radiation. Thus the electron in accelerated motion loses energy. From (1.4) it can be seen that as the energy decreases, r decreases. Finally, when the electron loses all its energy, it will collapse into the nucleus. To overcome this difficulty, Bohr made his third assumption according to which an electron in an orbit defined by (1.5) and (1.6) does not radiate energy. For this reason, such an orbit is called a stationary orbit. An electron emits radiation only when it jumps from one stationary orbit to another. This assumption was a complete deviation from accepted ideas at that time.

The electron in an orbit with larger n value has higher energy than that in an orbit with smaller n (note negative sign in (1.7)). Thus, when an electron jumps from an orbit n_2 to orbit n_1 ($n_2 > n_1$), it emits the excess energy in the form of a photon of energy $h\nu$. Thus

$$h\nu = \frac{2\pi^2 mZ^2 e^4}{h^2} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right). \quad (1.8)$$

Converting ν into $\bar{\nu}$, we get

$$\bar{\nu} = \frac{2\pi^2 mZ^2 e^4}{ch^3} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right). \quad (1.9)$$

For hydrogen with $Z = 1$, (1.9) becomes

$$\bar{\nu} = \frac{2\pi^2 me^4}{ch^3} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right). \quad (1.10)$$

Comparing (1.10) with (1.2), we get

$$R = \frac{2\pi^2 me^4}{ch^3}. \quad (1.11)$$

Substituting the values of the fundamental constants, we get a value of 109737 cm^{-1} for R , which agrees very well with the empirical value.

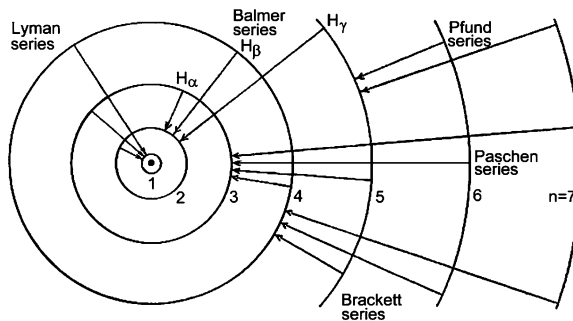


Fig. 1.5 Bohr circular orbits of hydrogen and the origin of series

We now see how the Bohr model explains the features of the hydrogen spectrum. The electron is in the ground state ($n = 1$). When it is excited, it goes to the higher orbits $n > 1$. It then relaxes i.e. jumps into the inner orbits. In that process, it emits energy. This energy is the energy associated with the various lines in the spectrum. The formation of lines in the several series is shown in Fig. 1.5.

1.2.3 The Sommerfeld Model

The planets in the solar system, moving in the gravitational inverse square field, move in elliptical orbits. Since the electron also moves in the Coulomb inverse square field, Sommerfeld proposed in 1916 that the electron moves in elliptical orbits.

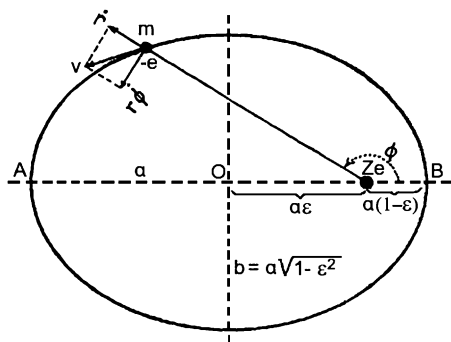
Sommerfeld's model is shown in Fig. 1.6. The nucleus is now located at one focus of the ellipse which has semi-major and semi-minor axes a and b respectively. The eccentricity of the ellipse is ϵ . The electron now has two coordinates r and ϕ . The associated momenta are

$$p_r = m\dot{r} \quad \text{and} \quad p_\phi = mr^2\dot{\phi}. \quad (1.12)$$

Sommerfeld pointed out that both the degrees of freedom have to be quantized. Thus,

$$\oint p_r dr = rh \quad \text{and} \quad \oint p_\phi d\phi = kh. \quad (1.13)$$

Here r is called the radial quantum number and k the azimuthal quantum number. The following relations exist for r and k .

Fig. 1.6 Elliptic orbits

$$\begin{aligned}
 (r + k) &= n = 1, 2, 3, \dots \\
 r &= 0, 1, 2, 3, \dots (n - 1), \\
 k &= 1, 2, 3, \dots n, \text{ and} \\
 \frac{k^2}{(k + r)^2} &= \frac{b^2}{a^2} = 1 - \varepsilon^2.
 \end{aligned}
 \tag{1.14}$$

From the last equation in (1.14), it can be seen that the values of k and $(k + r)$ define the shape of the elliptical orbit. The shape of some orbits for hydrogen are shown in Fig. 1.7.

It may be noted that Sommerfeld's model leads to the following expression for the total energy W :

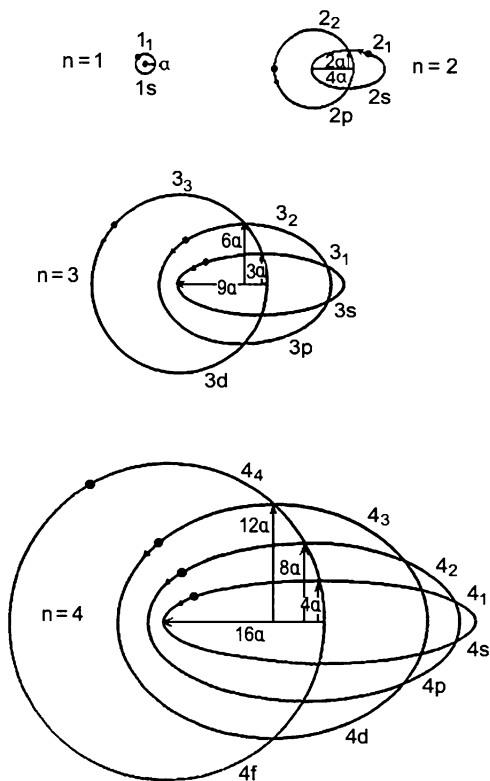
$$W = -\frac{2\pi^2 m Z^2 e^4}{h^2 (k + r)^2} = -\frac{2\pi^2 m Z^2 e^4}{h^2 n^2}
 \tag{1.15}$$

which is the same as the expression in Bohr's theory.

Thus, if the electron is in the quantum state with principal quantum number n , the quantum numbers k and r may take different values such that $(k + r) = n$. For each combination of k and r , the ellipticity of the orbit differs but the energy is the same. The state of the electron is denoted by the two quantum numbers n and k . To denote the state, the number n is mentioned first followed by the value of k as a suffix. Thus for the combination $n = 1, k = 1$, the symbol is 1_1 and for $n = 2, k = 2$, it is 2_2 . However, as a convention, the k values are represented by the letters s, p, d, f etc. Thus for $n = 1, k = 1$, the symbol is $1s$ and for $n = 2, k = 2$, it is $2p$.

Before closing the discussion on the Sommerfeld model, let us reconsider Sommerfeld's expression for the total energy W (1.15). In deriving this expression, it was assumed that the velocity v of the electron is much less than the velocity of light i.e. $v \ll c$. Sommerfeld removed this assumption and rederived the expression for the total energy using relativity. The new expression, which we shall now call W_{rel} , is

Fig. 1.7 Relative dimensions of the Bohr-Sommerfeld orbits for hydrogen. (a is the radius of the first Bohr orbit)



$$W_{\text{rel}} = -\frac{2\pi^2 m e^4 Z^2}{h^2 n^2} \left[1 + \frac{\alpha^2 Z^2}{n^2} \left(\frac{1}{k} - \frac{3}{4n} \right) \right]. \quad (1.16)$$

Here the constant α is called Sommerfeld fine structure constant; it is equal to $2\pi e^2/hc$ and has the value 7.2977×10^{-3} .

It was mentioned in Sect. 1.2.1, that the lines in the hydrogen spectrum have a fine structure, which Bohr's theory could not account for. Equation (1.16) shows that the orbital energy is now dependent on n as well as k . Thus, each state with quantum number n is now split into n sublevels. Transitions can take place from the sublevels of n_1 state to sublevels of n_2 state but their number is restricted by the selection rule $\Delta k = \pm 1$. The difference between the energy level diagrams and the transitions according to the Bohr model and the Sommerfeld model are shown in Fig. 1.8.

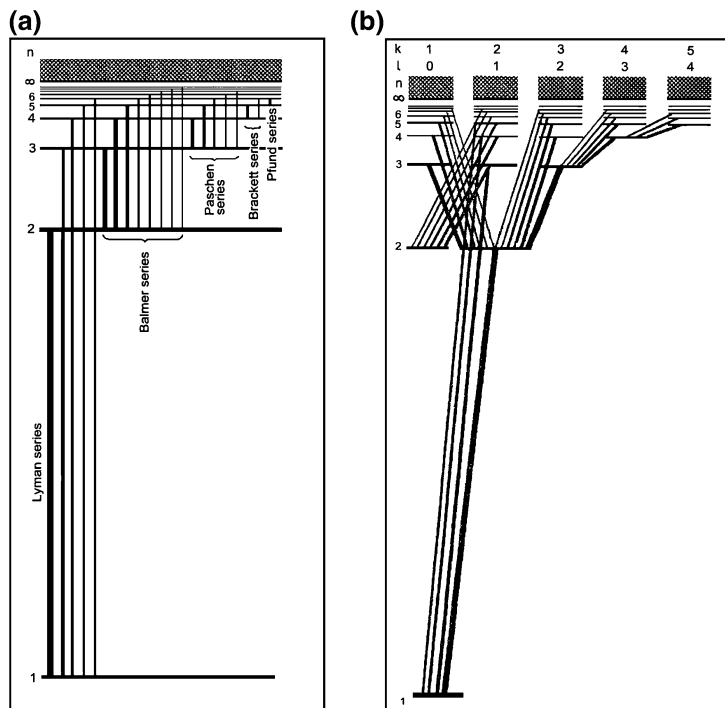


Fig. 1.8 Energy level diagram of the hydrogen atom **a** Bohr model **b** Sommerfeld model (the meaning of l explained in Sect. 1.2.6)

1.2.4 Space Quantization

In the Bohr model, we quantized one degree of freedom. In the Sommerfeld model, we quantized two degrees of freedom. Since the electron moves in three-dimensional space, we should quantize the third degree of freedom also. In effect, this process leads to one more quantum number viz. m . The physical significance is that if we represent the quantum number k by a vector in space, it assumes all orientations which have integral projections on any fixed direction; this projection is m . The fixed direction may for example, be an external magnetic field. It may be seen that m can be $\pm m$. The space-quantized projections for $k = 1, 2$ and 3 are shown in Fig. 1.9.

1.2.5 Electron Spin

In 1925, Uhlenbeck and Smith took into consideration the fact that the electron spins about its own axis and assigned a quantum number s for the angular

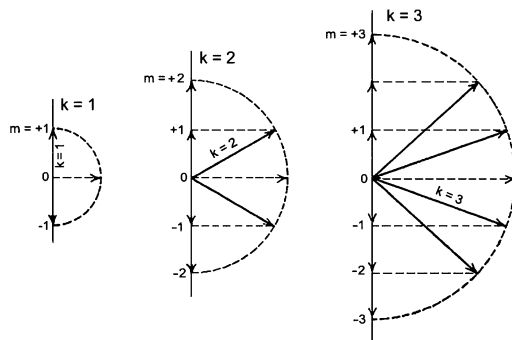


Fig. 1.9 Space-quantization diagrams for the Bohr-Sommerfeld orbits with $k = 1, 2$ and 3

momentum due to the spin of the electron. The magnitude of the spin angular momentum is $\frac{1}{2}(\frac{h}{2\pi})$. Like the quantum number k , s is also space-quantized i.e. its projection takes values $m_s = \pm\frac{1}{2}$.

1.2.6 The Vector Atom Model

In the vector atom model, the resultant of the orbital and spin angular momenta of an electron are obtained vectorially. It may be mentioned that following results from quantum mechanics, orbital angular momentum k is replaced by l where $l = k - 1$ i.e. l takes values $0, 1, 2, 3, \dots$. The vector relationship between l and s is shown in Fig. 1.10. Let us denote the vector resultant of l and s by j . This is another quantum number. In Fig. 1.11, two possible situations are shown. In (a), l and s are parallel. Here j is either $l + \frac{1}{2}$ or $l - \frac{1}{2}$ but in both cases j is parallel to l and s . In (b), l and s are not parallel. Here, again, j is $l + \frac{1}{2}$ or $l - \frac{1}{2}$ but now j is not parallel to either l or s .

Let us consider the situation when there are two electrons. Let their orbital angular momentum quantum numbers be l_1 and l_2 and their spin angular quantum numbers be s_1 and s_2 . In one scheme, l_1 and l_2 are added vectorially to yield a resultant L . Similarly, s_1 and s_2 are added to yield a resultant S . Finally, L and S are added to yield J . This is called l - s coupling. In another scheme, l_1 and s_1 are added to yield j_1 . Also, l_2 and s_2 are added to yield j_2 . Finally, j_1 and j_2 are added to yield the resultant J . This is called j - j coupling. However, following quantum mechanics the total quantum numbers are represented by L^* , S^* and J^* , they are defined as

$$L^* = \sqrt{L(L+1)}; \quad S^* = \sqrt{S(S+1)}; \quad J^* = \sqrt{J(J+1)}. \quad (1.17)$$

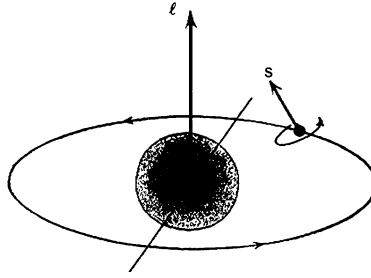
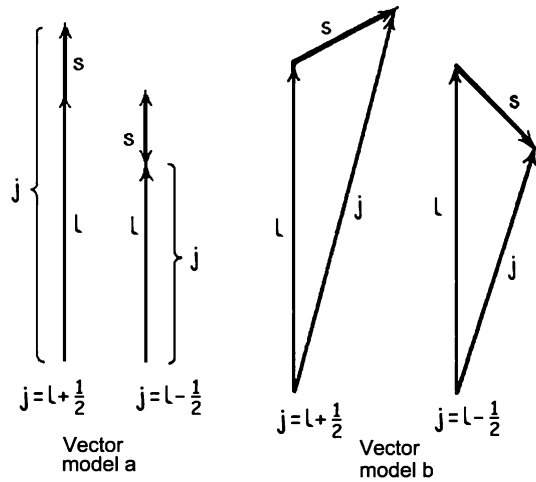


Fig. 1.10 Spin and orbital motion of the electron according to the classical theory

Fig. 1.11 Vector models *a* and *b* for the addition of the spin and orbital angular momenta



1.2.7 Larmor Precession and Magnetic Moment

Let us consider the effect of an applied magnetic field H upon an atomic electron. As shown in Fig. 1.12, let the orbital angular momentum vector make an angle with the magnetic field. Then, the effect of the field is that the l vector rotates about the magnetic field direction such that the angle between them remains constant. Such a motion is called 'precession'. Larmor showed that the frequency of precession ν_L is

$$\nu_L = \left(\frac{e}{4\pi mc} \right) H. \quad (1.18)$$

The magnetic moment μ_l due to the orbital angular momentum l is given by

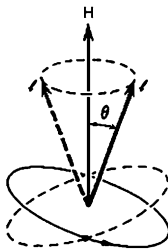


Fig. 1.12 Larmor precession of an electron orbit in a magnetic field

$$\mu_l = -\left(\frac{eh}{4\pi mc}\right)\sqrt{l(l+1)}. \quad (1.19)$$

The quantity $(eh/4\pi mc)$ is called the Bohr magneton; its value is 0.9273×10^{-20} erg/oersted. The negative sign in (1.19) implies that the magnetic moment vector is opposite relative to the orbital angular momentum. On the other hand, the magnetic moment μ_s due to spin angular momentum s is

$$\mu_s = -2\left(\frac{eh}{4\pi mc}\right)\sqrt{s(s+1)}. \quad (1.20)$$

Finally, the magnetic moment μ_j due to the resultant angular momentum due to both l and s is

$$\mu_j = -\left(\frac{eh}{4\pi mc}\right)\sqrt{j(j+1)}g. \quad (1.21)$$

The factor g called the Lande splitting factor is given by

$$g = 1 + \frac{j(j+1) + s(s+1) - l(l+1)}{2j(j+1)}. \quad (1.22)$$

It plays an important role in magneto-spectroscopic effects and also in magnetic resonance.

1.2.8 Pauli's Principle and Electronic Structure

We have seen that several quantum numbers are associated with an electron in an atom. These are $n, l, s, j, m_l, m_s, m_j$. However, some of them are related so that only four may be taken as independent. We shall consider n, l, m_l and m_s as the set of

quantum numbers that completely describe an electron. The possible values which they can assume are given in Table 1.2.

Pauli empirically proposed an important principle which states that no two electrons in an atom can have the same set of quantum numbers (n, l, m_l, m_s). This principle determines the distribution of electrons in an atom.

Let us start with hydrogen. There is only one electron. It will have $n = 1, l = 0$ and $m_l = 0$. Its m_s value may be $+\frac{1}{2}$ or $-\frac{1}{2}$. This is a 1 s electron. The next atom is He with two electrons. Both the electrons will have $n = 1, l = 0, m_l = 0$. One of the two electrons will have $m_s = +\frac{1}{2}$ and the other $m_s = -\frac{1}{2}$. The two He electrons are designated as $1s^2$. The next element is Li with 3 electrons. The first two electrons occupy states with $n = 1, l = 0$ and $m_s = \pm\frac{1}{2}$ i.e. the states occupied by the He electrons. According to Pauli's principle, it is not possible to accommodate the third electron in the state with $n = 1$. Hence, the third electron of the Li atom has necessarily to be placed in the state with $n = 2$. Here, l can take two values, $l = 1$ and $l = 0$. But the state with $l = 0$ has lower energy and, so, the third electron assumes the $l = 0$ state. Again, it assumes values $m_l = 0$ and $m_s = \pm\frac{1}{2}$. Thus, the electronic structure of Li is $1s^2 2s$. In Be, with four electrons, the outer two electrons assume values $l = 0, m_l = 0, m_s = +\frac{1}{2}$ and $l = 0, m_l = 0, m_s = -\frac{1}{2}$. Thus its electronic structure is $1s^2 2s^2$. In this way, the Pauli exclusion principle is the guiding principle in determining the electronic structure of atoms. The electronic structure of various other elements is summarized in Table 1.3.

1.2.9 The Periodic Table

In 1869, the Russian scientist Mendeleev pointed out that the physical and chemical properties of elements show a periodicity. Considering this periodic variation, he arranged the elements in the form of a table. The periodic table in the present form is given in Table 1.4. It consists of rows and columns. The rows are called periods whereas the columns are called groups. As we go from the left to the right, the atomic number of the elements increases. Similarly, from the top of a column to its bottom, the atomic number increases. The number of columns was eight to start with. In due course, subdivisions had to be created in the groups. When the original periodic table was proposed, the inert gas elements were not discovered. Later,

Table 1.2 Possible values of quantum numbers

Quantum number	Values
Principal (n)	1, 2, 3, 4,...
Orbital (l)	0, 1, 2, 3,...($n - 1$)
Orbital magnetic (m_l)	$-l, -(l-1), -(l-2), \dots, (l-1), 1$
Spin magnetic (m_s)	$-\frac{1}{2}, +\frac{1}{2}$

Table 1.3 Electronic structure of the elements

Z	Element	Structure of the ground state ^a	Z	Element	Structure of the ground state ^a	Z	Element	Structure of the ground state ^a
1	H	1s	32	Ge	– 3d ¹⁰ 4s ² 4p ²	63	Eu	– 4f ⁷ 6s ²
2	He	1s ²	33	As	– 3d ¹⁰ 4s ² 4p ³	64	Gd	– 4f ⁷ 5d 6s ²
3	Li	[He] 2s	34	Se	– 3d ¹⁰ 4s ² 4p ⁴	65	Tb	– 4f ⁸ 5d 6s ²
4	Be	– 2s ²	35	Br	– 3d ¹⁰ 4s ² 4p ⁵	66	Dy	– 4f ⁹ 5d 6s ²
5	B	– 2s ² 2p	36	Kr	– 3d ¹⁰ 4s ² 4p ⁶	67	Ho	– 4f ¹⁰ 5d 6s ²
6	C	– 2s ² 2p ²	37	Rb	[Kr] 5s	68	Er	– 4f ¹¹ 5d 6s ²
7	N	– 2s ² 2p ³	38	Sr	– 5s ²	69	Tm	– 4f ¹² 5d 6s ²
8	O	– 2s ² 2p ⁴	39	Y	– 4d5s ²	70	Yb	– 4f ¹³ 5d 6s ²
9	F	– 2s ² 2p ⁵	40	Zr	– 4d ² 5s ²	71	Lu	– 4f ¹⁴ 5d 6s ²
10	Ne	– 2s ² 2p ⁶	41	Nb	– 4d ⁴ 5s	72	Hf	– 4f ¹⁴ 5d ² 6s ²
11	Na	[Ne] 3s	42	Mo	– 4d ⁵ 5s	73	Ta	– 4f ¹⁴ 5d ³ 6s ²
12	Mg	– 3s ²	43	Tc	– 4d ⁵ 5s ²	74	W	– 4f ¹⁴ 5d ⁴ 6s ²
13	Al	– 3s ² 3p	44	Ru	– 4d ⁷ 5s	75	Re	– 4f ¹⁴ 5d ⁵ 6s ²
14	Si	– 3s ² 3p ²	45	Rh	– 4d ⁸ 5s	76	Os	– 4f ¹⁴ 5d ⁶ 6s ²
15	P	– 3s ² 3p ³	46	Pd	– 4d ¹⁰	77	Ir	– 4f ¹⁴ 5d ⁷ 6s ²
16	S	– 3s ² 3p ⁴	47	Ag	– 4d ¹⁰ 5s	78	Pt	– 4f ¹⁴ 5d ⁹ 6s
17	Cl	– 3s ² 3p ⁵	48	Cd	– 4d ¹⁰ 5s ²	79	Au	– 4f ¹⁴ 5d ¹⁰ 6
18	A	– 3s ² 3p ⁶	49	In	– 4d ¹⁰ 5s ² 5p	80	Hg	– 4f ¹⁴ 5d ¹⁰ 6s ²
19	K	[A] 4s	50	Sn	– 4d ¹⁰ 5s ² 5p ²	81	Tl	– 4f ¹⁴ 5d ¹⁰ 6s ² 6p
20	Ca	– 4s ²	51	Sb	– 4d ¹⁰ 5s ² 5p ³	82	Pb	– 4f ¹⁴ 5d ¹⁰ 6s ² 6p ²
21	Sc	– 3d4s ²	52	Te	– 4d ¹⁰ 5s ² 5p ⁴	83	Bi	– 4f ¹⁴ 5d ¹⁰ 6s ² 6p ³
22	Ti	– 3d ² 4s ²	53	I	– 4d ¹⁰ 5s ² 5p ⁵	84	Po	– 4f ¹⁴ 5d ¹⁰ 6s ² 6p ⁴
23	V	– 3d ³ 4s ²	54	Xe	– 4d ¹⁰ 5s ² 5p ⁶	85	At	– 4f ¹⁴ 5d ¹⁰ 6s ² 6p ⁵
24	Cr	– 3d ⁵ 4s	55	Cs	[Xe] 6s	86	Rn	– 4f ¹⁴ 5d ¹⁰ 6s ² 6p ⁶
25	Mn	– 3d ⁵ 4s ²	56	Ba	– 6s ²	87	Fr	[Rn] 7s
26	Fe	– 3d ⁶ 4s ²	57	La	– 5d 6s ²	88	Ra	– 7s ²
27	Co	– 3d ⁷ 4s ²	58	Ce	– 4f 5d6s ²	89	Ac	– 6d7s ²
28	Ni	– 3d ⁸ 4s ²	59	Pr	– 4f ³ 6s ²	90	Th	– 6d ² 7s ²
29	Cu	– 3d ¹⁰ 4s	60	Nd	– 4f ⁴ 6s ²	91	Pa	– 6d ³ 7s ²
30	Zn	– 3d ¹⁰ 4s ²	61	Pm	– 4f ⁵ 6s ²	92	U	– 5f ³ 6d 7s ²
31	Ga	– 3d ¹⁰ 4s ² 4p	62	Sm	– 4f ⁶ 6s ²			

^a Rare gas or closed shell structures are designated as brackets and electrons outside of these shells are indicated as –

when they were discovered, they were found to belong somewhere between IA and VIIB. But for convenience, they are shown as an extra group with number zero.

Mendeleev's classification was purely empirical. If we look at Table 1.4 in the context of Table 1.3, we see an underlying relationship. Thus, the outermost electron in the elements in the first group is a single *s* electron. Again, the outermost electrons in the elements in group II are two *s* electrons. A similar observation applies to the elements in the other groups. Thus, the electronic structure of the elements is responsible for the periodic classification of elements.

1.3 Quantum Mechanics

In this section we shall develop Schrödinger's equation and consider some of its applications.

1.3.1 The Schrödinger Equation

In 1923, de Broglie proposed the concept of waves associated with particles. These particle waves (also called matter waves or de Broglie waves) have wavelength λ given by

$$\lambda = \frac{h}{mv} \quad (1.23)$$

where h is Planck's constant, m the mass of the particle and v its velocity. The existence of these waves was demonstrated by recording the diffraction of electrons by crystals. The question then arose as to what equation represents these waves. On the one hand, such an equation must have similarity with the waves in an elastic medium. On the other hand, the equation for particle waves must also differ in some ways from the elastic wave equation.

An equation for particle waves was proposed by E. Schrödinger in 1926. The Schrödinger equation cannot be 'derived' as such but was set up by making assumptions and drawing comparison with the elastic wave equation. Let us start with the following standard relations.

$$\begin{aligned} \mathbf{p} &= \hbar \mathbf{k}, & |\mathbf{k}| &= 2\pi/\lambda, \\ E &= \hbar\omega, & \omega &= 2\pi\nu. \end{aligned} \quad (1.24)$$

Here, $\mathbf{p}$ is the momentum of the particle, $\mathbf{k}$ the propagation vector with magnitude $|\mathbf{k}|$, E the energy associated with the wave of angular frequency ω or frequency ν and $\hbar = h/2\pi$. We expect the wave function Ψ to be of the form $[\exp i(\mathbf{k} \cdot \mathbf{r} - \omega t)]$ where $\mathbf{r}$ is the position vector of the wave at time t .

Let us consider the energy of a particle. If p is the momentum and m the mass, the kinetic energy is $p^2/2m$. If V is the potential energy and W the total energy, we have

$$\frac{p^2}{2m} + V = W. \quad (1.25)$$

Schrödinger made the bold assumption that some of the quantities can be replaced by operators. Thus he denoted p and W as

$$p = -i\hbar\nabla \text{ and } W = i\hbar\frac{\partial}{\partial t} \quad (1.26)$$

where $i = \sqrt{-1}$ and ∇ is the operator $(\mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z})$ in Cartesian coordinates. Substituting in (1.25), and operating on the wave function Ψ , we have

$$-\frac{\hbar^2}{2m}\nabla^2\Psi + V\Psi = i\hbar\frac{\partial\Psi}{\partial t}. \quad (1.27)$$

This is called Schrödinger's time-dependent equation.

The wave function Ψ is a function of coordinates and time i.e. Ψ is $\Psi(r, t)$. Let us assume that it is a product of two functions $\psi(r)$ depending on r alone and $\phi(t)$ which depends on t alone i.e.

$$\Psi(r, t) = \psi(r)\phi(t). \quad (1.28)$$

Substituting in (1.27), we get

$$\phi(t) = C \exp\left(\frac{iWt}{\hbar}\right) \quad (1.29)$$

and

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + V\right]\psi(r) = W\psi(r) \quad (1.30)$$

Equation (1.30) is called Schrödinger's time-independent equation. The step-by-step derivation of (1.29) and (1.30) can be found in standard books on quantum mechanics. Equation (1.30) is sometimes written as

$$\nabla^2\psi + \frac{2m}{\hbar^2}(W - V)\psi = 0. \quad (1.31)$$

Further, $-\frac{\hbar^2}{2m}\nabla^2$ is the kinetic energy operator and V is the potential energy operator. Together they define the Hamiltonian operator $\mathcal{H}$. Thus, (1.30) can also be expressed as

$$\mathcal{H}\psi = W\psi. \quad (1.32)$$

Equation (1.30) can be solved only for certain values of the total energy. These energy values W_n are called eigen values. The solutions ψ_n corresponding to the eigen values are called eigen functions.

We shall note some important properties of the wave functions. Firstly, the product $\psi_n \psi_n^* d\tau$ represents the probability of existence of the particle in the

volume element $d\tau$; ψ_n^* is the complex conjugate of ψ_n . Since the particle must definitely exist within the total volume V , we have

$$\int_V \psi_n^* \psi_n d\tau = 1. \quad (1.33)$$

This result is called the normalization condition. Secondly, the wave functions follow the condition

$$\int_V \psi_m \psi_n d\tau = 0, \quad m \neq n. \quad (1.34)$$

The eigen functions obeying (1.34) are called orthogonal. Finally, for every function $f(r)$ there is an expectation value $\langle f(r) \rangle$ given by

$$\langle f(r) \rangle = \int_V \psi_n f(r) \psi_n^* d\tau. \quad (1.35)$$

1.3.2 Some Applications of the Schrödinger Equation

1.3.2.1 Linear Harmonic Oscillator

Let us consider a particle of mass m which can move only in the x -direction. If it is acted on by a restoring force $F = -Kx$, its potential energy V is $\frac{1}{2} Kx^2$. As such its Schrödinger equation is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + \frac{1}{2} Kx^2 \psi = W\psi \quad (1.36)$$

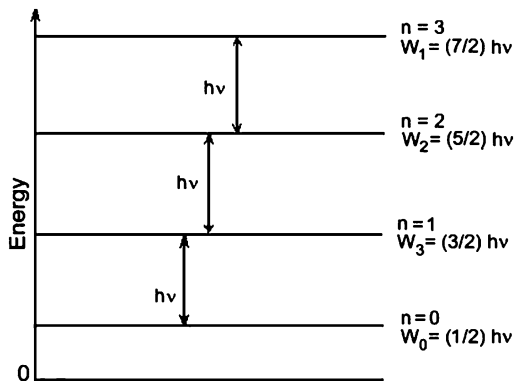
For convenience, we shall introduce a coordinate ξ such that

$$\xi = \alpha x \quad (1.37)$$

Then, (1.36) becomes

$$\frac{d^2\psi}{d\xi^2} + (\lambda - \xi^2)\psi = 0 \quad (1.38)$$

Fig. 1.13 Energy bands for a linear harmonic oscillator



Here,

$$\alpha^4 = \frac{mK}{\hbar^2}, \lambda = \frac{2W}{\hbar} \left(\frac{m}{K}\right)^{1/2} = \frac{2W}{\hbar} \frac{1}{\omega_0}, \text{ and } \omega_0 = \left(\frac{K}{m}\right)^{1/2}. \quad (1.39)$$

Equation (1.38) is solvable only if $\lambda = 2n + 1$ ($n = 0, 1, 2, 3, \dots$). The solution is worked out in books on quantum mechanics. We shall quote the results. The solution of (1.38) is

$$\psi_n(\xi) = H(\xi) \exp\left(-\frac{1}{2}\xi^2\right) \quad (1.40)$$

Here, $H(\xi)$ is a Hermite polynomial. The most important result is that the eigenvalues W_n are

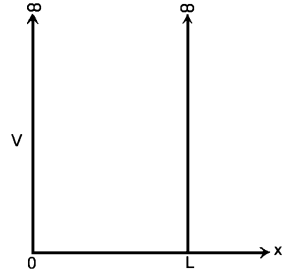
$$W_n = \left(n + \frac{1}{2}\right)\hbar\omega_0 \quad (1.41)$$

The energy level diagram for this system is shown in Fig. 1.13. It consists of equally spaced levels with a spacing of ω_0 . The lowest level for $n = 0$ is not zero but $\frac{1}{2}\hbar\omega_0$. This is a new concept—the concept of zero-point energy.

1.3.2.2 Particle in a Box

Let us consider a particle of mass m in a one-dimensional box of length L . As shown in Fig. 1.14, the potential V experienced by the particle is

Fig. 1.14 Potential experienced by a particle in a one dimensional box



$$\begin{aligned} V &= 0 & \text{at } 0 < x < L \\ &= \infty & \text{at } x < 0, x > L \end{aligned} \quad (1.42)$$

The Schrödinger equation for the particle in the region $0 < x < L$ is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} W \psi = 0 \quad (1.43)$$

Note that here W is entirely kinetic, since $V = 0$. We shall assume solutions of the type

$$\psi(x) = A e^{ikx} + B e^{-ikx} \quad \text{with } k^2 = \left(\frac{2m}{\hbar^2} W \right). \quad (1.44)$$

Let us apply the first boundary condition:

$$\psi(x) = 0 \quad \text{at } x = 0. \quad (1.45)$$

This yields

$$\begin{aligned} A + B &= 0, \\ A &= -B, \\ \psi(x) &= A(e^{ikx} - e^{-ikx}) \quad \text{or} \\ \psi(x) &= 2iA \sin kx \end{aligned} \quad (1.46)$$

Now let us apply the other boundary condition viz.,

$$\psi(x) = 0 \quad \text{at } x = L. \quad (1.47)$$

Then

$$0 = \sin kL \quad (1.48)$$

This is possible for discrete values of k given by

$$k_n = n\frac{\pi}{L}. \quad (1.49)$$

Thus, the eigen functions are

$$\psi_n = C \sin n\left(\frac{\pi}{L}\right)x \quad (1.50)$$

The value of the constant C can be found by using the normalization condition $\int_L \psi_n \psi_n^* dx = 1$. From (1.44) and (1.49), we get for the energy

$$W_n = \frac{\hbar^2 k_n^2}{2m} = \left(\frac{\hbar^2 \pi^2}{2mL^2}\right) n^2 \quad (1.51)$$

Here, $\hbar k_n$ is the momentum of the particle. The energy level diagram is shown in Fig. 1.15. It can be seen that the levels are not equally spaced; instead the energy difference increases according to n^2 .

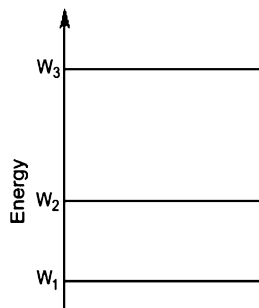
The above treatment was for a one-dimensional box. It can be extended to a 3-dimensional box of side L . The wave functions become

$$\psi_n(x, y, z) = C \sin\left[n_x\left(\frac{\pi}{L}\right)x\right] \sin\left[n_y\left(\frac{\pi}{L}\right)y\right] \sin\left[n_z\left(\frac{\pi}{L}\right)z\right] \quad (1.52)$$

where n_x, n_y, n_z are integers. The energy values are

$$W_{n_x n_y n_z} = \left(\frac{\hbar^2 \pi^2}{mL^2}\right) (n_x^2 + n_y^2 + n_z^2) \quad (1.53)$$

Fig. 1.15 Discrete energy levels of the infinite square well



There is an energy level for each combination n_x, n_y, n_z . On the other hand, the energy is the same for more than one combination. Thus the energy is the same for $n_x, n_y, n_z = 1, 2, 3; 2, 3, 1; 3, 1, 2$. This is called degeneracy.

1.3.2.3 The Hydrogen Atom

We shall start with the Schrödinger equation

$$\nabla^2 \psi + \frac{2m}{\hbar^2} (W - V) \psi = 0. \quad (1.31)$$

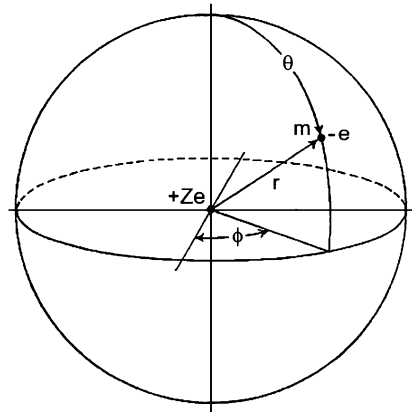
The electron with mass m and charge $-e$ is located at a distance r from the nucleus with charge $+Ze$. Hence the potential energy V is $-Ze^2/r$. To facilitate a solution of (1.31), it is necessary to designate the position of the electron by polar coordinates r, ϕ and θ (Fig. 1.16). Expressing ∇^2 in these coordinates, (1.31) takes the form

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2} + \frac{1}{r^2 \sin^2 \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{2m}{\hbar^2} (W - V) \psi = 0. \quad (1.54)$$

To solve (1.54), the method of separation of variables is employed. We assume that $\psi(r, \phi, \theta)$ is a product of three functions $R(r)$, $\Phi(\phi)$ and $\Theta(\theta)$ such that each is a function of only one coordinate. Thus,

$$\psi(r, \phi, \theta) = R(r) \Phi(\phi) \Theta(\theta). \quad (1.55)$$

Fig. 1.16 Position coordinates of the electron



This leads to three differential equations each depending on only one coordinate. The detailed solution can be found in books on quantum mechanics and on atomic spectroscopy. We quote the expression for ψ :

$$\psi(r, \theta, \phi) = \psi_{n,l,m} = Ce^{-\rho}(2\rho)^l L_{n+1}^{2l+1}(2\rho) P_l^{|m|}(\cos \theta) e^{im\phi}. \quad (1.56)$$

Here ρ is (Zr/na_1) where a_1 is the radius of the first orbit in Bohr's theory. L is a Laguerre polynomial and P is an associated Legendre polynomial. Interestingly, the energy again turns out to be the same as in Bohr's theory viz.,

$$W_n = -\frac{2\pi^2 m Z^2 e^4}{n^2 h^2}. \quad (1.7)$$

The numbers n , l and m are integers occurring in the solutions (1.56); these have to be identified as the quantum numbers empirically introduced earlier.

1.3.2.4 Tunneling

Let us consider a particle of mass m and total energy E moving from left to right (Fig. 1.17) and incident on a one-dimensional rectangular potential barrier of height V_0 and width a . As shown in the figure, the potential is:

$$V(x) = 0 \quad \text{for } x < 0, \quad (1.57)$$

$$V(x) = 0 \quad \text{for } x > a, \quad (1.58)$$

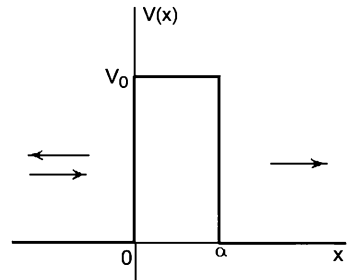
and

$$V(x) = V_0 \quad \text{for } 0 < x < a. \quad (1.59)$$

For a one-dimensional system, the Schrödinger equation is

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi(x)}{dx^2} + [V(x) - E] \psi(x) = 0 \quad (1.60)$$

Fig. 1.17 A one-dimensional rectangular potential barrier



We shall consider the solution for the three regions (i) $x < 0$, (ii) $x > a$ and (iii) $0 < x < a$.

Region (i)

For this region ($x < 0$) and (1.60) reduces to

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} - E\psi(x) = 0 \quad (1.61)$$

The solution for (1.61) is

$$\psi(x) = Ae^{ikx} + Be^{-ikx} \quad (1.62)$$

where Ae^{ikx} represents the wave function for the particle moving in the positive x -direction and Be^{-ikx} that for the particle moving in the negative x -direction. Further, the propagation constant k is given by

$$k = (2mE/\hbar^2)^{1/2} \quad (1.63)$$

Region (ii)

For this region ($x > a$) also, the Schrödinger equation is (1.61). However, it is not possible for a particle moving in the negative x -direction to exist in this region. Hence the solution is

$$\psi(x) = Ce^{ikx} \quad (1.64)$$

Region (iii)

For this region ($0 < x < a$), (inside the potential barrier), $V(x) = V_0$ and the Schrödinger equation has the full form

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + [V_0 - E]\psi(x) = 0 \quad (1.65)$$

The solution is given by

$$\psi(x) = Fe^{i\alpha x} + Ge^{-i\alpha x} \quad (1.66)$$

where α is the propagation constant given by

$$\alpha = [2m(E - V_0)/\hbar^2]^{1/2} \quad (1.67)$$

In (1.62), (1.64) and (1.66) we have five unknowns A , B , C , F and G . Fortunately, in this problem, we are interested not in A , B and C individually, but rather in the ratios B/A and C/A . This reduces the unknowns to four. The eigenfunctions are subject to the continuity conditions (or boundary conditions)

$$\psi(x)_{x=0} = \psi(x)_{x=a} \quad (1.68)$$

and

$$\psi'(x)_{x=0} = \psi'(x)_{x=a} \quad (1.69)$$

where $\psi'(x) = (d\psi/dx)$. The application of these boundary conditions results in four equations from which we get the values of B/A and C/A .

We are interested in the reflexion coefficient R and the transmission coefficient T . These are defined as

$$R = \left| \frac{B}{A} \right|^2 \quad \text{and} \quad T = \left| \frac{C}{A} \right|^2 \quad (1.70)$$

Using the above procedure we get

$$R = \left| \frac{B}{A} \right|^2 = \left[1 + \frac{4E(E - V_0)}{V_0^2 \sin^2 \alpha a} \right]^{-1} \quad (1.71)$$

and

$$T = \left| \frac{C}{A} \right|^2 = \left[1 + \frac{V_0^2 \sin^2 \alpha a}{4E(E - V_0)} \right]^{-1}. \quad (1.72)$$

It is convenient to substitute $-i\beta$ for α for $0 < E < V_0$. Then (1.72) becomes

$$T = \left[1 + \frac{V_0^2 \sinh^2 \beta a}{4E(V_0 - E)} \right]^{-1}. \quad (1.73)$$

Finally, for $\beta a \gg 1$, we have

$$T \approx \frac{16E(V_0 - E)}{V_0^2} e^{-2\beta a}. \quad (1.74)$$

Qualitatively, we see that $T \neq 0$ for $E < V_0$. Quantitatively, we get $T \approx 0.1$ for $(mV_0 a^2 / \hbar^2)$ and $(E/V_0) \approx 0.7$. Thus the transmission coefficient T has a positive real value, however small, for $E < V_0$. This result is very different from the classical result which is that T is exactly zero for $E < V_0$. It means that the particle has finite probability, however small, of penetrating across a potential barrier; this is called 'tunneling'. To start with, this result found application in explaining β -decay but in course of time, it has gained relevance in semiconduction and superconductivity.

1.3.3 Perturbation Theory

Perturbation theory helps to find small changes in the energy of a system caused by a small disturbance (called perturbation) acting on the system. Let $\mathcal{H}^{(0)}$ be the Hamiltonian of the unperturbed system and $W_n^{(0)}$ and $\psi_n^{(0)}$ the eigen values and eigen functions. Then, we have

$$\mathcal{H}^{(0)}\psi_n^{(0)} = W_n^{(0)}\psi_n^{(0)} \quad (1.75)$$

Let $\mathcal{H}$, W_n and ψ_n be the Hamiltonian, eigen value and eigen function of the perturbed system. Then

$$\mathcal{H}\psi_n = W_n\psi_n \quad (1.76)$$

Further. Let us express $\mathcal{H}^{(0)}$ as

$$\mathcal{H} = \mathcal{H}^{(0)} + \lambda\mathcal{H}^{(1)} \quad (1.77)$$

where $\mathcal{H}^{(1)}$ is the perturbation and λ a constant parameter. We shall express ψ_n and W_n as a series in powers of λ as

$$\psi_n = \psi_n^{(0)} + \lambda\psi_n^{(1)} + \lambda^2\psi_n^{(2)} + \dots \quad (1.78)$$

$$W_n = W_n^{(0)} + \lambda W_n^{(1)} + \lambda^2 W_n^{(2)} + \dots \quad (1.79)$$

Substituting (1.59–1.61) in (1.58), we have

$$\begin{aligned} & [\mathcal{H}^{(0)} + \lambda\mathcal{H}^{(1)}][\psi_n^{(0)} + \lambda\psi_n^{(1)} + \lambda^2\psi_n^{(2)} + \dots] \\ &= [W_n^{(0)} + \lambda W_n^{(1)} + \lambda^2 W_n^{(2)} + \dots][\psi_n^{(0)} + \lambda\psi_n^{(1)} + \lambda^2\psi_n^{(2)} + \dots] \end{aligned} \quad (1.80)$$

Equating coefficients of different terms in λ on either side, we get

$$\mathcal{H}^{(0)}\psi_n^{(0)} = W_n^{(0)}\psi_n^{(0)}$$

which is the same as (1.75). Further, we have

$$\mathcal{H}^{(0)}\psi_n^{(1)} + \mathcal{H}^{(1)}\psi_n^{(0)} = W_n^{(0)}\psi_n^{(1)} + W_n^{(1)}\psi_n^{(0)} \quad (1.81)$$

From this equation, we finally get the result

$$W_n^{(1)} = \int_V \psi_n^{(0)*} \mathcal{H}^{(1)} \psi_n^{(0)} d\tau. \quad (1.82)$$

Thus, the first order correction for the n th eigenvalue due to perturbation $\mathcal{H}^{(1)}$ is the expectation value of $\mathcal{H}^{(1)}$ over the n th state. This treatment is valid only for non-degenerate systems.

1.3.4 Variation Principle

Like the perturbation theory, the variation principle provides an approximate method for solving the Schrödinger equation.

Let a system have several wave functions ψ_k . Then a general solution Ψ can be written as a linear superposition of the several wave functions. Thus, we have

$$\Psi = \sum_k A_k \psi_k. \quad (1.83)$$

where

$$\mathcal{H}\psi_k = E\psi_k. \quad (1.84)$$

and the A_k 's are coefficients. The expectation value of the Hamiltonian $\mathcal{H}$ is given by

$$\begin{aligned} \langle \mathcal{H} \rangle &= \int \Psi^* \mathcal{H} \Psi d\tau \\ &= \sum_k E |A_k|^2. \end{aligned} \quad (1.85)$$

Here, it is assumed that the ψ_k 's are an orthonormal set. Let E_0 be the lowest eigenvalue. Then,

$$\begin{aligned} \langle \mathcal{H} \rangle &\geq \sum_k E_0 |A_k|^2 \\ &= E_0 \sum_k |A_k|^2 \end{aligned} \quad (1.86)$$

Since $\sum_k |A_k|^2 = 1$ for orthonormal wave functions, (1.86) leads to

$$\begin{aligned}
 E_0 &\leq \langle \mathcal{H} \rangle \\
 &\leq \int \Psi^* \mathcal{H} \Psi \, d\tau
 \end{aligned}
 \tag{1.87}$$

Thus, the procedure in the variation method is to evaluate the integral $\int \Psi^* \mathcal{H} \Psi \, d\tau$ with an assumed (trial) wave function Ψ and repeat the procedure by varying Ψ till a minimum value is obtained. This minimum value is the upper limit for the ground state energy E_0 .

1.3.5 Uncertainty Principle

Let us consider a pair of related quantities which are canonically conjugate. The simplest example is the position coordinate x and the corresponding momentum component p_x . Let the uncertainty in determining these quantities be Δx and Δp_x . Heisenberg proposed the important principle that these uncertainties are not independent but are related to each other by the relation

$$\Delta x \Delta p_x \geq \hbar \tag{1.88}$$

This is known as Heisenberg's uncertainty principle. According to (1.70), if the uncertainty Δx is low, the uncertainty Δp_x is large and vice-versa while the product of the two uncertainties is always of the order of $\hbar$.

Other quantities related by the uncertainty principle are (i) the angular position ϕ and the corresponding angular momentum component J_ϕ and (ii) the energy E and the time t at which E is measured. Thus,

$$\begin{aligned}
 \Delta \phi \Delta J_\phi &\geq \hbar \\
 \Delta E \Delta t &\geq \hbar
 \end{aligned}
 \tag{1.89}$$

Since the magnitude of $\hbar$ is small (1.0545×10^{-27} erg-sec), the Heisenberg uncertainty principle is applicable to systems of atomic size.

1.4 Statistical Mechanics

The position and velocity of a single particle at time t can be predicted from their initial values from laws of classical mechanics. Other properties of a system, like thermodynamic properties which depend on the velocities of the constituent particles can also be estimated if the constituents can be treated as independent. But if the number of particles becomes large ($N \sim 10^{23}$), the simple classical methods fail and methods of statistical mechanics have to be employed. The modus operandi in

statistical mechanics is to find the probability of an event and estimate the average value. Somewhat different procedures are employed depending on the type of particles. The main types are:

Maxwell-Boltzmann statistics (MBS)	for molecules
Bose-Einstein statistics (BES)	for photons, phonons
Fermi-Dirac statistics (FDS)	for electrons, holes

A parameter which occurs frequently in solid state physics is the distribution function $f(\varepsilon_i)$. This is the ratio of the number n_i of occupied states to the total number of available states g_i with energy ε_i . In the different types of statistical mechanics, the distribution function is

$$\begin{aligned}
 f(\varepsilon_i) = n_i/g_i &= \frac{1}{e^{(\alpha+\beta\varepsilon_i)}} && \text{for MBS} \\
 &= \frac{1}{e^{(\alpha+\beta\varepsilon_i)} - 1} && \text{for BES} \\
 &= \frac{1}{e^{(\alpha+\beta\varepsilon_i)} + 1} && \text{for FDS.}
 \end{aligned} \tag{1.90}$$

Here α and β are constants. α is given by the condition $\sum n_i = N$, N being the total number of particles and $\beta = (1/k_B T)$. When $g_i \gg n_i$, BES and FDS approximate to MBS.

1.5 Electromagnetic Theory

The electromagnetic theory is a vast subject with several books devoted to its various aspects. However, the essence of the electromagnetic theory is the system of Maxwell's equations. We shall confine ourselves to the statement of Maxwell's equations.

Denoting the electric field by E and the magnetic field by H , Maxwell's equations are given by

$$\begin{aligned}
 \nabla \times \mathbf{E} + \frac{1}{c} \frac{\partial \mathbf{H}}{\partial t} &= 0 \\
 \nabla \times \mathbf{H} - \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} &= 4\pi \mathbf{J} \\
 \nabla \cdot \mathbf{E} &= 4\pi \rho \\
 \nabla \cdot \mathbf{H} &= 0.
 \end{aligned} \tag{1.91}$$

In the above, $\mathbf{J}$ is the current density and ρ the charge density. Combining the divergence of the second equation with the time derivative of the third equation, we get

$$\nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} = 0 \quad (1.92)$$

This is known as the equation of continuity.

Defining $\mathbf{A}$ and ϕ as the electromagnetic potentials, the electric and magnetic fields can be expressed as

$$\begin{aligned} \mathbf{E} &= -\frac{1}{c} \frac{\partial \mathbf{A}}{\partial t} - \nabla \phi \\ \mathbf{H} &= \nabla \times \mathbf{A} \end{aligned} \quad (1.93)$$

Chapter 2

Theory of Free Electrons I: Classical Theory

Abstract This chapter deals with the classical theory of free electrons in metals developed by Lorentz and Drude. The theory assumes that the electrons in a metal are in haphazard motion. Their average random velocity is assumed to be given by Maxwell–Boltzmann statistics. The electrons collide with one another and get scattered. Their motion is governed by a relaxation time. The average distance travelled between two collisions is called the mean free path. When an electric field is applied the electrons develop a drift velocity which is superposed on the random motion. With these assumptions, expressions are derived for the electrical conductivity, thermal conductivity, Wiedemann–Franz constant, thermopower, Hall effect and the electronic specific heat. Agreement of theory with experimental results was not very satisfactory because of the inadequacy of the Maxwell–Boltzmann statistics. Yet the Lorentz–Drude theory is important as the first attempt to explain the various properties of metals in terms of the basic properties of electrons.

2.1 Introduction

In 1895, Lorentz laid the foundation of the electron theory. At that point of time, nothing was known about the electron. Thomson's and Millikan's experiments were to come years later. All that was known, mostly from experiments on electrolysis of salt solutions, was that atoms contain charged particles. These particles were supposed to have a negative charge $-e$ and mass m . These particles were given the name 'electrons' by Stoner. Nothing was known or said about the positive part of the atom. The name 'nucleus' was introduced years later. According to classical electromagnetic theory, an accelerated charge emits radiation. Lorentz assumed that the spectral radiation emitted by atoms is due to the 'oscillations' of electrons. Lorentz then proceeded to consider what would happen if the atoms are in a magnetic field; he showed by a simple calculation that the frequency of the oscillation of the electron is altered. Besides its original frequency, the electron now oscillates with another two frequencies, one of lesser magnitude and the other of

larger magnitude. Lorentz could estimate the difference in frequencies of these new radiations and also obtain a value for the specific charge (e/m) of the electron. Both these results, which were sheer predictions, were confirmed in 1896 when P. Zeeman discovered the ‘Zeeman effect’ which is the splitting of spectral lines when the source of radiation is placed in a magnetic field. In 1897, J.J. Thomson discovered the electron in his cathode ray experiments.

Three years later, Drude [1] extended Lorentz’s ideas to build a theory to account for the electrical and thermal properties of solids in general and metals in particular. The picture of oscillating charged particles described by Lorentz was not enough to explain electrical and thermal properties. For this purpose, Drude had to make a few additional assumptions. These assumptions are discussed in the next section. Drude’s theory is discussed by Lorentz [2] and Ashcroft and Mermin [3].

2.2 Assumptions

- (i) Each atom in a metal contains a certain number of free electrons. If we denote this number by z , the total number of free electrons (N) in a gram-atom is given by

$$N = N_A z \quad (2.1)$$

where N_A is the Avogadro number ($=6.023 \times 10^{23}/\text{mol}$). The ‘density of electrons’ (n) is defined as the number of free electrons in unit volume; this is given by

$$n = N/V = (N_A z \rho_m / A) \quad (2.2)$$

where ρ_m is the measured density and A the atomic weight. It is useful to define a sphere having volume equal to the volume per electron. The radius r_s of this sphere is

$$r_s = (3/4\pi n)^{1/3}. \quad (2.3)$$

Values of n and r_s for some metals are given in Table 2.1. Values of n vary from 1 to 25 in units of $10^{22}/\text{cm}^3$. Values of r_s lie in the range 1–3 Å. z is the number of free electrons (also called ‘conduction electrons’ in metals). There are other electrons, which are not free; they remain attached to what Lorentz and Drude called the positive part of the atoms (the present-day nucleus). The conglomeration of the nucleus and the bound electrons which we shall call the ion-core will have a net positive charge.

- (ii) The conduction electrons move freely in the metal very much like molecules in a gas; hence the name ‘electron gas’ for this assembly of electrons (Fig. 2.1). The number of electrons in unit volume in a metal is about 1,000

Table 2.1 Values of n and r_s for some metals; assumed values of z are also included. Values of ρ_m from literature have been used except for alkali metals

Metal	z	n ($10^{22}/\text{cm}^3$)	r_s (Å)
Li (78 K)	1	4.70	1.72
Na (5 K)	1	2.65	2.08
K (5 K)	1	1.40	2.57
Rb (5 K)	1	1.15	2.75
Cs (5 K)	1	0.91	2.98
Cu	1	8.47	1.41
Ag	1	5.86	1.60
Au	1	5.90	1.59
Be	2	24.7	0.99
Mg	2	8.61	1.41
Ca	2	4.61	1.73
Sr	2	3.55	1.89
Ba	2	3.15	1.96
Nb	1	5.56	1.63
Fe	2	17.0	1.12
Mn (α)	2	16.5	1.13
Zn	2	13.2	1.22
Cd	2	9.27	1.37
Al	3	18.1	1.10
In	3	11.5	1.27
Tl	3	10.5	1.31
Sn	4	14.8	1.17
Pb	4	13.2	1.22
Bi	5	14.1	1.19
Sb	5	16.5	1.13

times the number of molecules per unit volume in a gas; yet Drude treated metals like a gas obeying classical kinetic theory of gases and following Maxwell–Boltzmann statistics. Thus, he assumed that the thermal energy U_{el} of an electron is

$$U_{\text{el}} = \frac{1}{2}mv_r^2 = \frac{3}{2}k_{\text{B}}T \quad (2.4)$$

where v_r is the root-mean-square velocity of the electron.

- (iii) As the electron moves in a metal, it collides with the ion-cores and gets deflected (Fig. 2.2); its path is thus zigzag. In between the collisions, it moves in straight lines with a velocity v_r . Because of the randomness of the motion, at any given instant, the number of electrons traveling in one direction is equal to the number traveling in the opposite direction. Once an

Fig. 2.1 Drude's model of a metal

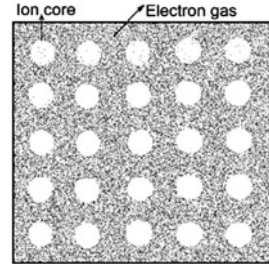
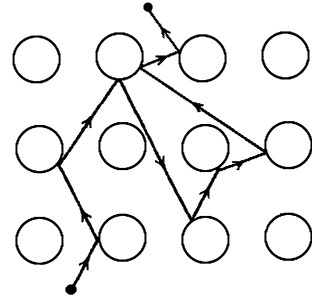


Fig. 2.2 Collisions and random motion of an electron:
filled circle Electron, *open circles* ion-cores



electron collides with an ion-core, it loses its energy, changes its direction and starts its journey as if it has lost all memory of its earlier history.

- (iv) The frequency of the electrons is governed by considerations of probability. The probability of collision of an electron per unit time with an ion-core is $1/\tau$. τ has the dimensions of time; hence it is called collision time or relaxation time. It represents the mean time between two consecutive collisions. Between two successive collisions, an electron travels an average distance Λ . Obviously,

$$\Lambda = v_r \tau \quad (2.5)$$

This distance Λ is called the mean free path.

- (v) Much after Drude's times, it was realized that electron-ion-core collision is not the only mechanism that contributes to the scattering of an electron. Other mechanisms like electron–electron interaction and interactions with thermal vibrations and structural imperfections also are responsible for electron scattering. However for the present purpose namely the explanation of thermal and electrical properties on the basis of classical theory, we shall stick to the simple picture of electron-ion-core collisions.

2.3 Applications

We shall now consider applications of the classical electron theory to some metallic properties.

2.3.1 DC Conductivity

Let us first consider the motion of a free electron in the absence of any external field. As discussed earlier, in the absence of any external field, the electrons move in a haphazard manner (Fig. 2.2) colliding, now and then, with the ion-cores. The velocity of electrons in such a random state is v_r .

If now, a DC electric field $\mathbf{E}$ is applied to the metal (Fig. 2.3), the electrons develop a velocity $\mathbf{v}_D$ parallel to the direction of the field. This motion is superposed on the random motion discussed earlier. We shall call $\mathbf{v}_D$ the drift velocity. The equation of motion for the electron is

$$m \frac{d\mathbf{v}_D}{dt} = -\mathbf{E}e. \quad (2.6)$$

The electron will be continuously accelerated. Such a situation is not stable. Remember that the electron continues to collide with the ion-cores. An electron which has developed a velocity $\mathbf{v}_D$ after a collision, loses its energy in the next collision and $\mathbf{v}_D$ reduces to zero. The time between two collisions is τ . Thus, because of collision, an effective acceleration $\mathbf{v}_D/\tau$ acts. Equation (2.6) may now be written as

$$m \frac{d\mathbf{v}_D}{dt} = -\mathbf{E}e - m \frac{\mathbf{v}_D}{\tau}. \quad (2.7)$$

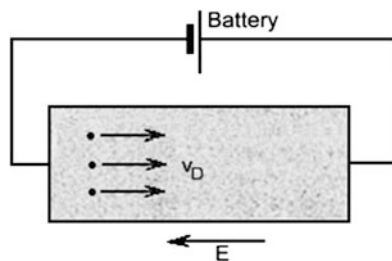


Fig. 2.3 Electrons moving with drift velocity (v_D) in an electric field E

The force $m\mathbf{v}_D/\tau$ is similar to a frictional force or damping force. The electron will move with a steady velocity when the net force on it is zero, i.e.,

$$\mathbf{E}e = -m\frac{\mathbf{v}_D}{\tau} \quad (2.8)$$

or,

$$\mathbf{v}_D = -\frac{\mathbf{E}e\tau}{m}. \quad (2.9)$$

The negative sign indicates that $\mathbf{v}_D$ is opposite in direction to $\mathbf{E}$. The current density $\mathbf{j}$ is defined as

$$\mathbf{j} = \sigma \mathbf{E} = \frac{1}{\rho} \mathbf{E} \quad (2.10)$$

where σ is the conductivity and its reciprocal ρ is the resistivity. In terms of the transported charge, $\mathbf{j}$ may be written as

$$\mathbf{j} = -ne\mathbf{v}_D. \quad (2.11)$$

From (2.9), (2.10) and (2.11), we have

$$\sigma = \frac{ne^2\tau}{m}. \quad (2.12)$$

We shall note the following:

- (i) the random velocity v_r is not affected
- (ii) the transported charge is proportional to ne
- (iii) the acceleration due to the applied field is proportional to e/m .

Values of τ calculated from room temperature data on ρ are included in Table 2.2. It can be seen that these values are typically of the order of 10^{-15} s. Let us estimate the value of the mean free path $v_r\tau$. Note that Λ has to be estimated using v_r and not v_D . To estimate v_r , Drude assumed the classical theory relation

$$\frac{1}{2}mv_r^2 = \frac{3}{2}k_B T \quad (2.4)$$

This gives a value $\sim 10^7$ cm/s for v_r . Using these values of τ and v_r , we get $\Lambda \approx 10$ Å which is about 2–3 lattice distances in metals. This is consistent with the picture of the electrons colliding with ion-cores.

Table 2.2 Resistivities (ρ) and relaxation times (τ) for some metals at room temperature

Metal	ρ ($\mu\Omega\text{-cm}$)	τ (10^{-14} s)
Li	8.55	0.88
Na	4.2	3.2
K	6.1	4.1
Rb	11.0	2.8
Cs	18.8	2.1
Cu	1.56	2.7
Ag	1.51	4.0
Au	2.04	3.0
Be	2.8	0.51
Mg	3.9	1.1
Ca	3.43	2.2
Sr	2	0.44
Ba	60	0.19
Nb	15.2	0.42
Fe	8.9	0.24
Zn	5.5	0.49
Cd	6.8	0.56
Al	2.45	0.80
In	8.0	0.38
Tl	15	0.22
Sn	10.6	0.23
Pb	19.0	0.14
Bi	107	0.023
Sb	39	0.055

2.3.2 Electronic Specific Heat of Metals

Using Maxwell–Boltzmann statistics, it can be shown that the thermal energy U_a per atom is $3k_B T$ and, so, the total energy for a mole of a metal is $N_A U_a$. Therefore the molar specific heat C_V is

$$\begin{aligned}
 C_V &= \frac{d(N_A U_a)}{dT} = 3N_A k_B = 3R \\
 &= 5.96 \text{ cal/mol/K.}
 \end{aligned}
 \tag{2.13}$$

Drude extended Maxwell–Boltzmann statistics to electrons in metals and found that the thermal energy U_{el} per electron is $\frac{3}{2}k_B T$ (2.4). Assuming that each atom contributes a single electron per atom, the total thermal energy due to electrons is $N_A U_{el}$ and the electronic specific heat $[C_V]_{el}$ is

$$[C_V]_{\text{el}} = \frac{d(N_A U_{\text{el}})}{dT} = \frac{3}{2} N_A k_B = \frac{3}{2} R. \quad (2.14)$$

Thus the total specific heat of a mole of a metal should be 50 % larger than that calculated from (2.13). Experimental values of specific heats of monovalent metals are very close to the value predicted by (2.13); exact measurements show that the electronic contribution is $\sim 1/100$ of the classical value ($3R$). This is a serious drawback of Drude's electron theory.

2.3.3 Thermal Conductivity of Metals

Let us consider a metal bar in which there is a temperature gradient ∇T . Let us define $\mathbf{j}_{\text{th}}$ as the thermal energy current. This is a vector parallel to the temperature gradient; its magnitude is the thermal energy moving from a higher temperature to a lower temperature in unit time across unit area of cross-section. The thermal energy current is proportional to the temperature gradient i.e.

$$\mathbf{j}_{\text{th}} = -K \nabla T \quad (2.15)$$

The constant of proportionality K is called the coefficient of thermal conductivity. It is taken as a positive quantity i.e. the thermal energy flows from a higher temperature to a lower temperature.

Let us consider points O, A and B lying in the x -direction in the bar (Fig. 2.4). Point O is located at x . Points A and B are located at one mean free path away from O on either side. Then the coordinates of A and B are $(x - v_x \tau)$ and $(x + v_x \tau)$ where v_x is the x -component of the velocity v_r of the electron. Since there is a temperature gradient, the thermal energy of an electron is different when it is at A or B. We shall denote these energies as $[U]_{(x-v_x\tau)}$ and $[U]_{(x+v_x\tau)}$. We have to find the energy transfer at the point O. As the electrons are in random motion, we expect half the number of electrons ($n/2$) to cross O from left to right and another half ($n/2$) to cross from right to left. The number of electrons crossing point O in either direction in one second is $(n/2)v_x$. The electrons from the left derive their energy from the collision that has occurred τ seconds earlier at A; they carry energy $(n/2) v_x [U]_{(x-v_x\tau)}$. Similarly, the thermal energy carried by electrons from right to left is $(n/2) v_x [U]_{(x+v_x\tau)}$. The energy transfer at O is the difference between these two energies. Thus we may write j_{th} as

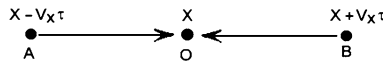


Fig. 2.4 Electrons traveling from A and B towards O

$$j_{\text{th}} = (n/2)v_x \left\{ [U]_{(x-v_x\tau)} - [U]_{(x+v_x\tau)} \right\}. \quad (2.17)$$

The quantity in the curly bracket is the change in the energy of an electron due to change in the x -coordinate by an amount $2v_x\tau$. Thus,

$$\left\{ [U]_{(x-v_x\tau)} - [U]_{(x+v_x\tau)} \right\} = (-2v_x\tau) \left(\frac{dU_{\text{el}}}{dx} \right). \quad (2.18)$$

We may write (2.17) as

$$\begin{aligned} j_{\text{th}} &= -(n/2)v_x(2v_x\tau) \left(\frac{dU_{\text{el}}}{dx} \right) \\ &= -n\tau(v_x)^2 \left(\frac{dU_{\text{el}}}{dT} \right) \left(\frac{dT}{dx} \right). \end{aligned} \quad (2.19)$$

This result can be extended to three dimensions by making the following changes in (2.19):

- (i) The random velocity of an electron is v_r . On an average, the random velocity is related to its components as $v_x^2 = v_y^2 = v_z^2 = v_r^2/3$. Combining this with (2.4) we get

$$v_x^2 = \frac{k_B T}{m}. \quad (2.20)$$

(ii)

$$\frac{d[U_{\text{el}}]}{dT} = \frac{d(\frac{3}{2}k_B T)}{dT} = \frac{3}{2}k_B. \quad (2.21)$$

(iii)

$$\frac{dT}{dx} \rightarrow \nabla T. \quad (2.22)$$

Substituting (2.20), (2.21) and (2.22) in (2.19), we get

$$\begin{aligned} \mathbf{j}_{\text{th}} &= -n\tau \left(\frac{k_B T}{m} \right) \left(\frac{3}{2}k_B \right) (\nabla T) \\ &= -\frac{3}{2} \left(\frac{n\tau}{m} \right) (k_B^2 T) (\nabla T). \end{aligned} \quad (2.23)$$

Comparing (2.23) with (2.15), we get

$$K = \frac{3}{2} \left(\frac{n\tau}{m} \right) (k_B^2 T). \quad (2.24)$$

2.3.4 Wiedemann–Franz Law

From experimental data on the thermal and electrical conductivities of metals, Wiedemann and Franz observed that the ratio K/σ is a constant for all metals at a given temperature. This is called the Wiedemann–Franz ratio. Further, the value of this constant increases with increasing temperature. On the other hand, the quantity $K/\sigma T$ is a constant independent of temperature. This is called the Lorenz number L . The values of the Lorenz number for some metals are given in Table 2.3. The values are of the same order but individual values are scattered about a mean value of $2.45 \times 10^{-8} \text{ W-}\Omega/\text{K}^2$.

Using the expressions for K and σ (2.24 and 2.12) obtained from Drude’s theory, we get

$$\begin{aligned} L = \frac{K}{\sigma T} &= \frac{3}{2} \left(\frac{n\tau}{m} k_B^2 T \right) / \left(\frac{n\tau e^2}{m} \right) T = \left(\frac{3}{2} \right) \left(\frac{k_B}{e} \right)^2 \\ &= 1.11 \times 10^{-8} \text{ W-}\Omega/\text{K}^2. \end{aligned} \quad (2.25)$$

We note that the expression for L is independent of n , m and τ ; it involves only universal constants. Thus, it predicts a constant value independent of temperature for all metals. To this extent, the agreement is fair. However, as far as the numerical value is concerned, there is a difference of a factor of 2 between the experimental value and the theoretical value.

Table 2.3 Experimental values of Lorenz number $L (= K/\sigma T)$ for some metals at 273 K

Metal	$L \text{ (W-}\Omega/\text{K}^2)$
Cu	2.20×10^{-8}
Ag	2.31
Au	2.32
Be	2.36
Mg	2.14
Fe	2.61
Zn	2.28
Cd	2.49
Bi	3.53
Sb	2.57

2.3.5 Thermopower

If a metal is subjected to a temperature gradient ∇T , it develops an electric field E parallel and proportional to the temperature gradient. Thus,

$$\mathbf{E} = Q \nabla T \quad (2.26)$$

The constant of proportionality Q is called the thermopower of the metal.

To start with, let us consider a one-dimensional case. Let the temperature gradient be along the x -direction. Figure 2.4 applies to this problem also. Let us consider the mean velocity of an electron at point O located at x . At a given instant, an electron starting from A may reach O. This electron derives its energy (and velocity) from the collision at A which is at a distance $(x - v_x \tau)$ from O. We shall call its velocity $[v_x]_{(x-v_x \tau)}$. Similarly, if an electron starting at B reaches O, its velocity will be $[v_x]_{(x+v_x \tau)}$. The mean velocity v_Q of an electron at O is therefore

$$v_Q = \frac{1}{2} \left\{ [v_x]_{(x-v_x \tau)} - [v_x]_{(x+v_x \tau)} \right\}. \quad (2.27)$$

The quantity in the floral bracket is the change in v_x due to a change in x by an amount $2v_x \tau$. Hence, we may write (2.27) as

$$\begin{aligned} v_Q &= -\frac{1}{2}(2v_x \tau) \frac{dv_x}{dx} \\ &= -\frac{\tau}{m} \left[\frac{d}{dT} \left(\frac{1}{2} m v_x^2 \right) \right] \left(\frac{dT}{dx} \right). \end{aligned} \quad (2.28)$$

We shall now extend the result to a three-dimensional model. We note that $v_x^2 = \frac{1}{3} v_r^2$, $\frac{1}{2} m v_r^2 = \frac{3}{2} k_B T$ and $\frac{dT}{dx}$ becomes ∇T . Making these changes in (2.28), we get

$$\mathbf{v}_Q = -\frac{\tau k_B}{2m} \nabla T. \quad (2.29)$$

Also, because of the electric field $\mathbf{E}$ caused by the thermal gradient, the electron develops a drift velocity $\mathbf{v}_D$ given by

$$\mathbf{v}_D = -\frac{e\tau}{m} \mathbf{E}. \quad (2.30)$$

In the steady state

$$\mathbf{v}_Q + \mathbf{v}_D = 0. \quad (2.31)$$

Thus,

$$\frac{e\tau}{m}\mathbf{E} = -\frac{\tau k_B}{2m}\nabla T, \quad (2.32)$$

or

$$\mathbf{E} = -\frac{k_B}{2e}\nabla T. \quad (2.33)$$

Comparing (2.33) with (2.26), we have

$$Q = -\frac{k_B}{2e}. \quad (2.34)$$

According to (2.34), the thermopower is independent of m , n and τ . It involves only the universal constants k_B and e . It has a single universal value -0.43×10^{-4} V/K. However, experimental values of Q are $\sim 10^{-6}$ V/K. Thus, there is a serious discrepancy of a factor of 10^{-2} between the theoretical and experimental values.

2.3.6 Hall Effect

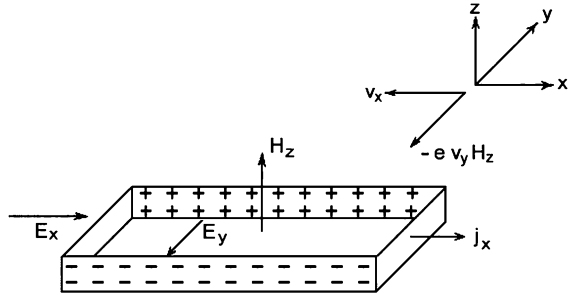
In 1879, Hall discovered that if a bar of a metal carries electric current in the x -direction and is simultaneously subjected to a magnetic field in the z -direction, a voltage develops in the y -direction. If j_x is the current density, H_z the magnetic field and E_y the voltage, the Hall coefficient is defined as

$$R_H = \frac{E_y}{j_x H_z}. \quad (2.35)$$

Its units are volt-cm/amp-oersted (volt-m³/amp-weber in SI units).

Figure 2.5 shows a Hall experiment. An electric field E_x is applied in the x -direction. Under the effect of this field, the electrons develop a drift velocity $(v_D)_x$ in the negative x -direction and a current density j_x in the positive x direction. When a magnetic field H_z is applied in the z -direction, the electrons are bent towards the y -direction and they accumulate on the y -face (nearer to the reader). Consequently, the other y -face becomes positively charged. This accumulation of charges results in an electric field E_y which opposes the motion of electrons in the y -direction. Finally, the two forces balance and the electrons move in a steady state.

Two forces act on the electron: the Lorentz force (the combined force due to the magnetic field and the electric field) and the frictional force $m\mathbf{v}_D/\tau$ due to the drift velocity (see Sect. 2.3.1). If $\mathbf{f}$ is the total force, the equation of motion is

Fig. 2.5 Hall effect experiment

$$\mathbf{f} = m \frac{d\mathbf{v}_D}{dt} = -e \left[\mathbf{E} + \frac{\mathbf{v}_D \times \mathbf{H}}{c} \right] - \frac{m \mathbf{v}_D}{\tau}. \quad (2.36)$$

In the steady state, the force components vanish. Thus,

$$0 = f_x = -eE_x - \frac{ev_y H_z}{c} - m \frac{v_x}{\tau} \quad (2.37)$$

and

$$0 = f_y = -eE_y - \frac{ev_x H_z}{c} - m \frac{v_y}{\tau}. \quad (2.38)$$

We may multiply the above equations by $n\tau e/m$. Further we have the following expressions for the conductivity $\sigma_0 = ne^2\tau/m$ and the current density $\mathbf{j} = \sigma_0 \mathbf{E} = ne\mathbf{v}_D$; here σ_0 is the conductivity in the absence of magnetic field. Introducing these substitutions appropriately, (2.37), (2.38) get modified to

$$-\sigma_0 E_x = \tau \left(\frac{eH_z}{mc} \right) j_y + j_x \quad (2.39)$$

and

$$\sigma_0 E_y = -\tau \left(\frac{eH_z}{mc} \right) j_x + j_y. \quad (2.40)$$

In (2.40), $j_y = 0$; we can then write

$$R_H = \left(\frac{E_y}{j_x H_z} \right) = -\frac{1}{nec}. \quad (2.41)$$

Thus, the Drude model predicts a negative Hall coefficient for metals. Further, R_H is independent of m and τ and depends on n .

Table 2.4 Hall coefficient for some metals at room temperature

Metal	$R_H \times 10^{11} \text{ (V-mt}^3\text{/A-Wb)}$	
	Calculated from (2.41)	Experimental
Na	-24.4	-25.0
K	-47	-42
Cs	-73	-78
Cu	-7.4	-5.5
Ag	-10.4	-8.4
Au	-10.5	-7.2
Bi	-4.1	~ -1000
Be	-2.5	+24.4
Zn	-4.6	+3.3
Cd	-6.5	+6.5

In Table 2.4, experimental values of R_H are compared with those obtained from (2.41). In alkali metals, the agreement is fair. In Ag, Cu and Au the difference in the two sets of values is $\sim 30\text{--}50\%$. The difference is too large in Bi. More serious is the case of Be, Zn and Cd where R_H is positive. The classical electron theory has no answer for this discrepancy.

2.3.7 Magnetoresistance

Magnetoresistance is the (possible) effect of a magnetic field on the resistance of a conductor. While conducting the Hall-effect experiment, Hall expected to detect the presence of magnetoresistance. But he could not detect any such effect.

As far as Drude's theory is concerned, from (2.39), we get the resistivity ρ as

$$\rho = \frac{E_x}{J_x} = \rho_0 \quad (2.42)$$

since $j_y = 0$. Thus the resistivity is independent of the magnetic field i.e., the resistivity (ρ) in the field is the same as that in the absence of the field (ρ_0). While this is in agreement with Hall's findings, it may be mentioned that subsequent experiments conducted with better accuracy have shown that in some metals $\rho \neq \rho_0$ and there is indeed a measurable magnetoresistance.

2.3.8 Transparency of Metals

Metals are generally opaque. However, it is found that some of them are transparent in certain regions of the electromagnetic spectrum. This observation was made by Wood [4]. The phenomenon was explained by Zener [5] on the basis of Drude's electron theory.

Ignoring the damping term $m\nu_D/\tau$ (Sect. 2.3.1), the equation of motion of the electron is

$$m\ddot{x} = eE. \quad (2.43)$$

The electric field of an electromagnetic wave is of the form

$$E = E_0 e^{i\omega t} \quad (2.44)$$

where E_0 is the amplitude and ω the angular frequency. In such a case, the solution of (2.43) is

$$x = -\frac{eE}{m\omega^2}. \quad (2.45)$$

The electric polarization P is

$$P = nex. \quad (2.46)$$

The complex dielectric constant ε and refractive index μ are given by

$$\varepsilon = \mu^2 = 1 + 4\pi(P/E). \quad (2.47)$$

Substituting for P and E from (2.45) and (2.46), we get

$$\mu^2 = 1 - \left[\frac{4\pi n e^2}{m\omega^2} \right]. \quad (2.48)$$

If $(4\pi n e^2/m\omega^2)$ is > 1 , μ , is imaginary, i.e. the wave is reflected back by total internal reflection. On the other hand, if $(4\pi n e^2/m\omega^2)$ is < 1 , μ is real and the electromagnetic wave can propagate in the metal. The critical value of the wavelength λ_c is given by

$$\lambda_c = 2\pi \left(\frac{mc^2}{4\pi n e^2} \right)^{1/2} \quad (2.49)$$

The critical wavelength below which the metal becomes transparent lies in the ultraviolet. The experimental values of λ_c and those obtained from (2.49) are given

Table 2.5 Critical wavelength (λ_c) values

Metal	λ_c (Å)	
	From (2.49)	From experiment
Lithium	1,550	1,550
Sodium	2,090	2,100
Potassium	2,870	3,150
Rubidium	3,220	3,400

in Table 2.5. The agreement is fair. After the development of the band theory, (2.49) was modified but the numerical values of λ_c remained very much the same.

2.4 Achievements and Shortcomings

It was an achievement on the part of Drude and Lorentz that at about the time of the discovery of the electron, they developed a theory to account for several properties of metals. Their treatment involved basic parameters of the electron like the charge, mass, velocity, number of electrons and the relaxation time. In most of the problems they could predict expressions which yielded the right order-of-magnitude results.

In certain properties, however, there were serious discrepancies in theoretical and experimental results. These properties are the specific heat, the thermopower and the Wiedemann–Franz law. These properties involve the velocities of electrons. For the velocities, Drude assumed the classical Maxwell–Boltzmann statistics. The observed discrepancies were apparently due to this reason. The application of Fermi–Dirac statistics, as will be discussed later, improved the theoretical results.

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Chapter 3

Theory of Free Electrons II: Quantum Mechanical Theory

Abstract In this chapter we discuss Sommerfeld's theory of electrons in metals. Sommerfeld's model differs from the classical Drude–Lorentz model in three respects: (i) it uses quantum mechanics, (ii) it uses Fermi–Dirac statistics and (iii) it treats the relaxation time as energy dependent. The application of quantum mechanics shows that in a metal the energy spectrum of an electron is discrete and not continuous. As we fill up the energy levels at absolute zero, we find that the levels only up to a certain energy value are filled. This critical value is the Fermi energy; it plays an important role. Corresponding to this energy value, we have the parameters, the Fermi wave vector, the Fermi temperature and the Fermi relaxation time. Expressions have been derived from the Sommerfeld model for the electrical conductivity, the Wiedemann–Franz constant, the electronic specific heat and the thermopower; they involve the Fermi energy and the Fermi relaxation time. The agreement is better than in the case of the Drude–Lorentz theory.

3.1 Introduction

We saw in the preceding chapter that the classical electron theory of Drude and Lorentz could qualitatively account for several properties of solids. Quantitatively, however, large differences were observed between theoretical and experimental results. This was particularly so in the case of the electronic specific heat, the Wiedemann–Franz ratio (or the Lorenz number) and the thermopower. These properties involve the electron velocities for which Drude and Lorentz employed the classical Maxwell–Boltzmann statistics.

After the development of quantum mechanics, Sommerfeld [1] proposed a modified theory. The Sommerfeld model retained the following features of the Drude–Lorentz model:

- (i) A metal consists of 'free electrons' which are the valence electrons.
- (ii) The electrons are in a constant state of motion.

- (iii) The electrons collide with ion-cores and get deflected; their motion is thus haphazard.
- (iv) The collisions are governed by probability. The probability of collision is equal to $1/\tau$ where τ is called the relaxation time.
- (v) In an electric field, the electrons develop a drift velocity which is superposed on the random motion.

However, Sommerfeld's model differed drastically from the Drude–Lorentz model in the following three respects:

- (i) Sommerfeld used quantum mechanics.
- (ii) He used Fermi–Dirac statistics instead of Maxwell–Boltzmann statistics
- (iii) The relaxation time was made energy-dependent.

In the next section, we shall develop the Sommerfeld model and then take up its applications. Aspects of the Sommerfeld theory are discussed by Aschroft and Mermin [2], Kittel [3] and Singleton [4].

3.2 The Sommerfeld Model

3.2.1 Energy Levels of a Free Electron in a Metal

The electrons move freely within a metal. They do not escape from the surface and are completely confined to the inside of the metal. This is possible if they are at a lower energy inside than at the surface. In effect they experience a potential V_0 (Fig. 3.1a) which acts as a potential barrier at the surface. Figure 3.1b is a one-dimensional representation of this situation. The linear dimension of the metal is L . The potential at $x = 0$ and $x = L$ is taken as $V = \infty$ for convenience. More importantly, the potential $V = 0$ in the range $0 < x < L$.

We can extend the situation to three dimensions by visualising the electron to be in a cubic box of side L with potential $V = 0$ at $0 < x < L$, $0 < y < L$, $0 < z < L$ and $V = \infty$ at $x = 0$, $y = 0$ and $z = 0$ and $x = L$, $y = L$ and $z = L$.

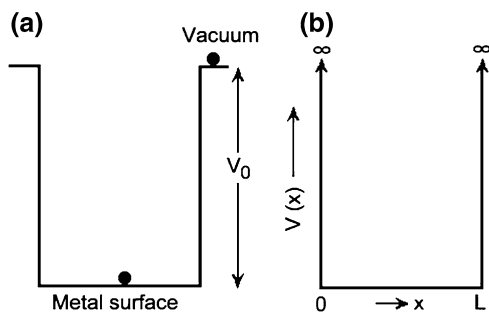


Fig. 3.1 **a** Sommerfeld's model of a metal, **b** the electron in a one-dimensional potential box

The Schrödinger equation for such an electron is

$$-\frac{\hbar^2}{2m}\nabla^2\psi = E\psi \quad (3.1)$$

We assume plane-wave solutions of the form

$$\begin{aligned} \psi_k(r) &= A \exp i(\mathbf{k} \cdot \mathbf{r}) \\ &= A \exp i(k_x x + k_y y + k_z z) \end{aligned} \quad (3.2)$$

Here, k is the propagation vector whose magnitude is $|\mathbf{k}| = \frac{2\pi}{\lambda}$, λ being the wavelength of the electron wave. The components of $|\mathbf{k}|$ are k_x, k_y, k_z . Further, denoting $|\mathbf{k}|$ by k we have

$$k^2 = \frac{2m}{\hbar^2} E_k \quad (3.3)$$

The solutions are subject to boundary conditions. We shall assume cyclic boundary conditions i.e.

$$\begin{aligned} \psi_k(x+L, y, z) &= \psi_k(x, y, z) \\ \psi_k(x, y+L, z) &= \psi_k(x, y, z) \\ \psi_k(x, y, z+L) &= \psi_k(x, y, z) \end{aligned} \quad (3.4)$$

Application of these boundary conditions restricts k_x to values

$$k_x = 0, \pm \frac{2\pi}{L}, \pm \frac{4\pi}{L}, \dots \pm \frac{2\pi}{L} n \quad (3.5)$$

with similar values for k_y and k_z ; here n is an integer.

Applying the normalising condition $\int_{-\infty}^{+\infty} \psi_k(r) \psi_k^*(r) dr = 1$ to (3.2), we get

$$A = \left(\frac{1}{L^3}\right)^{1/2} = \left(\frac{1}{V}\right)^{1/2} \quad (3.6)$$

where V is the volume of the cubic box. Finally, we have

$$\begin{aligned} \psi_k(x, y, z) &= \left(\frac{1}{L}\right)^{3/2} e^{i(k_x x + k_y y + k_z z)} \\ &= \left(\frac{1}{L}\right)^{3/2} e^{i\frac{2\pi}{L}(n_x x + n_y y + n_z z)} \end{aligned} \quad (3.7)$$

where n_x, n_y, n_z are integers. Further, from (3.3) and (3.5), we have

$$\begin{aligned}
 E_k &= \frac{\hbar^2}{2m} k^2 = \frac{\hbar^2}{2m} (k_x^2 + k_y^2 + k_z^2) \\
 &= \left(\frac{\hbar^2}{2m} \right) \left(\frac{4\pi^2}{L^2} \right) (n_x^2 + n_y^2 + n_z^2).
 \end{aligned} \tag{3.8}$$

Equation (3.7) defines the eigen functions and (3.8) the eigen values of the electron. For each combination of n_x, n_y, n_z (i.e. k_x, k_y, k_z) there is an energy value. It can also happen that several combinations have the same energy value. For example, the combinations (1, 2, 3), (2, 3, 1), (3, 1, 2) for n_x, n_y, n_z all yield the same energy value. This is called degeneracy.

From (3.8), it is seen that the energy spectrum of the electron is discrete and not continuous. However, the separation of the levels is extremely small ($\sim 10^{-15}$ eV) and for some purposes the spectrum may be treated as continuous.

3.2.2 Fermi Energy and Related Parameters

Let us consider how the energy levels defined by (3.8) are filled up. k_x, k_y, k_z are three quantum numbers of an electron. There is yet another quantum number, m_s . It arises out of the spin of the electron and can take values $+\frac{1}{2}$ or $-\frac{1}{2}$ (see Chap. 1). Although it does not enter into the expression for the energy, it has to be reckoned with while filling up of the energy levels. According to the Pauli exclusion principle, two electrons cannot have the same set of the four quantum numbers. Thus a given set of quantum numbers k_x, k_y, k_z can be associated with either $m_s = +\frac{1}{2}$ or $m_s = -\frac{1}{2}$. In other words, a maximum of two electrons can be associated with an energy level with quantum numbers k_x, k_y, k_z . The filling up of energy levels at $T = 0$ according to this scheme is shown in Fig. 3.2. Starting with the lowest level, the process of filling up is continued till all the electrons are accommodated. Let us say, the highest energy thus filled has energy E_F . This energy is called Fermi energy and the energy level is called the Fermi level. The Fermi level is the dividing level below which all levels are occupied and above which the levels are unoccupied at $T = 0$.

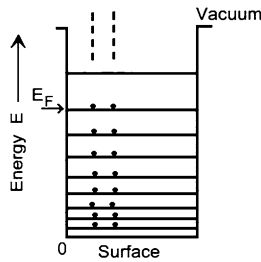
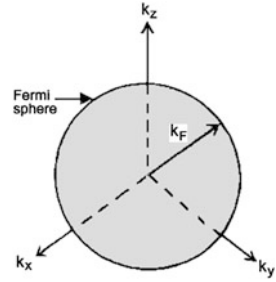


Fig. 3.2 Filling up of electrons at $T = 0$ (all levels up to E_F are filled and all levels above E_F are empty)

Fig. 3.3 The k -space and the Fermi sphere



Let us consider the matter in greater detail. We can imagine a coordinate space, the k -space, where the coordinates are k_x , k_y , k_z . Equation (3.8) is an equation to a sphere in the k -space. The separation between consecutive values of k_x (or k_y or k_z) is $2\pi/L$. Thus there is an occupied state for every cell of volume $(2\pi/L)^3$. Let us denote the vector corresponding to the highest filled level by k_F . The sphere corresponding to k_F is called the Fermi sphere and its surface is called the Fermi surface. The Fermi surface is spherical in the Sommerfeld model. However, we shall see later that it is more complicated in real crystals. From (3.8) we have

$$E_F = \frac{\hbar^2}{2m} k_F^2 \quad (3.9)$$

where k_F is the Fermi wave vector defined by $k_F^2 = (k_x)_F^2 + (k_y)_F^2 + (k_z)_F^2$. The volume of this sphere (Fig. 3.3) is $4\pi k_F^3/3$. Thus the total number of occupied states is

$$2 \frac{(4\pi/3)k_F^3}{(2\pi/L)^3} = \left(\frac{L^3}{3\pi^2}\right) k_F^3 = \left(\frac{V}{3\pi^2}\right) k_F^3 = N. \quad (3.10)$$

Here V is the volume and N the total number of electrons; the numerical 2 in the beginning of (3.10) is due to the spin of the electron. Reorganizing (3.10), we get

$$k_F = \left[3\pi^2 \left(\frac{N}{V} \right) \right]^{1/3}. \quad (3.11)$$

Substituting in (3.9), we get

$$E_F = \left(\frac{\hbar^2}{2m} \right) k_F^2 = \left(\frac{\hbar^2}{2m} \right) \left[3\pi^2 \left(\frac{N}{V} \right) \right]^{2/3}. \quad (3.12)$$

Table 3.1 Fermi surface parameters (Fermi wave vector k_F , Fermi wave velocity v_F , Fermi energy E_F and Fermi temperature T_F) for some metals, values for Li at 78 K, for Na, K, Rb, at 5 K and for the rest at room temperature

Valency	Metal	k_F (cm ⁻¹)	v_F (cm s ⁻¹)	E_F (eV)	T_F (K)
1	Li	1.11×10^8	1.29×10^8	4.72	5.48×10^4
	Na	0.92	1.07	3.23	3.75
	K	0.75	0.86	2.12	2.46
	Rb	0.70	0.81	1.85	2.15
	Cu	1.36	1.57	7.00	8.12
	Ag	1.20	1.39	5.48	6.36
	Au	1.20	1.39	5.51	6.39
2	Be	1.93	2.23	14.14	16.41
	Mg	1.37	1.58	7.13	8.27
	Ca	1.11	1.28	4.68	5.43
	Sr	1.02	1.18	3.95	4.58
	Ba	0.98	1.13	3.65	4.24
3	Al	1.75	2.02	11.63	13.49
	Ga	1.65	1.91	10.35	12.01
	In	1.50	1.74	8.60	9.98
	Pb	1.57	1.82	9.37	10.87

It may be noted that while k_F depends only on the electron concentration (N/V), E_F depends on the electron mass m also. The Fermi velocity (v_F) defined as the electron velocity at the Fermi level is given by

$$v_F = \left(\frac{\hbar}{m}\right) k_F = \left(\frac{\hbar}{m}\right) \left[3\pi^2 \left(\frac{N}{V}\right)\right]^{1/3}. \quad (3.13)$$

Finally, we define a parameter T_F as

$$T_F = \frac{E_F}{k_B} = \left(\frac{\hbar^2}{2mk_B}\right) \left[3\pi^2 \left(\frac{N}{V}\right)\right]^{2/3}. \quad (3.14)$$

T_F is called the Fermi temperature; it is not a temperature but is a quantity with dimensions of temperature. T_F is a measure of the Fermi energy.

Values of the Fermi surface parameters (k_F , v_F , E_F , T_F) calculated from (3.11–3.14) for some metals are given in Table 3.1. Values of the electron concentration n ($=N/V$) needed for the calculations are taken from Table 2.1. It can be seen that typically k_F are a few Å, $v_F \sim 0.01c$, E_F are a few eV and $T_F \sim 10^4$ – 10^5 K.

3.2.3 Density of States

We shall now introduce a quantity $D(E)$ as the density of states. It is defined as the number of allowed energy states of the electron per unit energy range. Thus the number of allowed energy states in the range between E and $E + dE$ will be $D(E) dE$. Following Singleton [4], we shall express (3.12) in a generalized form by deleting the suffix F. Then we get the number of electrons N as

$$N = \left(\frac{2m}{\hbar^2} \right)^{3/2} \left(\frac{V}{3\pi^2} \right) E^{3/2}. \quad (3.15)$$

Differentiating, we get

$$D(E) = \frac{dN}{dE} = \left(\frac{2m}{\hbar^2} \right)^{3/2} \left(\frac{V}{2\pi^2} \right) E^{1/2} = \frac{(4\pi V)(2m)^{3/2}}{h^3} E^{1/2}. \quad (3.16)$$

At $E = E_F$, (3.16) becomes

$$D(E_F) = \frac{(4\pi V)(2m)^{3/2}}{h^3} E_F^{1/2}. \quad (3.17)$$

A plot of $D(E)$ against E is shown in Fig. 3.4. The filled energy states at $T = 0$ are shown in the shaded region. At $E = E_F$, there is a cut-off.

3.2.4 Fermi–Dirac Statistics

As mentioned in Chap. 1, the distribution of electrons is governed by Fermi–Dirac statistics. The Fermi–Dirac distribution function is given by

$$F(E) = \frac{1}{[\exp(\alpha + \frac{E}{k_B T})] + 1} \quad (3.18)$$

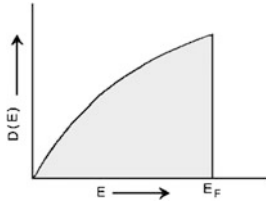


Fig. 3.4 Plot of density of states $D(E)$ against E

For electrons, $\alpha = -E_F/k_B T$. Hence (3.18) becomes

$$F(E) = \frac{1}{[\exp(\frac{E-E_F}{k_B T})] + 1}. \quad (3.19)$$

$F(E)$ is called the Fermi distribution function. It may be recalled that $F(E)$ is the probability of the energy state E . The Fermi–Dirac distribution function (3.19) has the following properties:

- (i) At $T = 0$ and for $E < E_F$, $F(E) = 1$
- (ii) At $T = 0$ and for $E > E_F$, $F(E) = 0$
- (iii) At $T > 0$ and for $E = E_F$, $F(E) = 1/2$
- (iv) At $T \rightarrow \infty$, the first term in the denominator of (3.19) $\gg 1$ and hence

$$F(E) = \exp[-(\frac{E-E_F}{k_B T})] \quad (3.20)$$

which is the Maxwell–Boltzmann distribution function.

The shapes of the Fermi–Dirac distribution function at different temperatures are shown in Fig. 3.5. It may be noted that as T rises above $T = 0$, some electrons below but close to E_F are excited to upper levels. The curve for the distribution function, which has a sharp cut-off at $E = E_F$ at $T = 0$, now gets rounded off and its tail extends into the region beyond $E = E_F$.

We shall study the role of the Fermi–Dirac distribution parameters in determining the temperature dependence of electron energy parameters. Before proceeding further let us note the following:

- (i) $N(E)$ denotes the number of electrons with energy E ,
- (ii) $D(E)$ denotes the number of electron states available in unit energy range,
- (iii) $F(E)$ denotes the probability of occupation of an energy state with energy E ,
- (iv) thus,

$$N(E) = D(E)F(E). \quad (3.21)$$

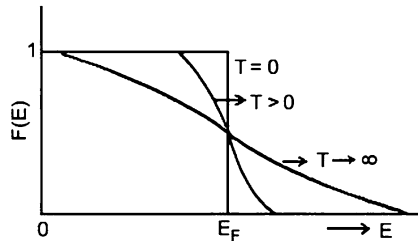


Fig. 3.5 The Fermi–Dirac distribution function $F(E)$ as a function of E at different temperatures

3.2.5 Electron Energy Parameters at $T = 0$

Let us evaluate the Fermi energy at $T = 0$. Hereafter, we shall denote the Fermi energy at $T = 0$ by E_{F_0} . Now, the total number of electrons N is

$$\begin{aligned}
 N &= \int_0^{\infty} N(E) dE \\
 &= \int_0^{\infty} D(E) F(E) dE \\
 &= \int_0^{E_{F_0}} D(E) F(E) dE + \int_{E_{F_0}}^{\infty} D(E) F(E) dE.
 \end{aligned} \tag{3.22}$$

We know that $F(E) = 1$ for $E < E_{F_0}$ and $F(E) = 0$ for $E > E_{F_0}$. Hence (3.22) becomes

$$\begin{aligned}
 N &= \int_0^{E_{F_0}} D(E) dE \\
 &= \left(\frac{V}{2\pi^2} \right) \left(\frac{2m}{\hbar^2} \right)^{3/2} \int_0^{E_{F_0}} E^{1/2} dE \\
 &= \left(\frac{V}{3\pi^2} \right) \left(\frac{2m}{\hbar^2} \right)^{3/2} E_{F_0}^{3/2}
 \end{aligned}$$

or

$$E_{F_0} = \left(\frac{\hbar^2}{2m} \right) \left(\frac{3\pi^2 N}{V} \right)^{2/3}. \tag{3.23}$$

Thus the Fermi energy at $T = 0$ depends only on the electron concentration. Let us now consider the average energy $\langle E_0 \rangle$ of an electron at $T = 0$. This is given by

$$\langle E_0 \rangle = \frac{1}{N} \int_0^{\infty} E D(E) F(E) dE. \tag{3.24}$$

Using the same procedure and arguments as in deriving (3.23), we get

$$\begin{aligned}
\langle E_0 \rangle &= \frac{1}{N} \left(\frac{V}{2\pi^2} \right) \left(\frac{2m}{\hbar^2} \right)^{3/2} \int_0^{E_{F_0}} E^{3/2} dE \\
&= \frac{1}{N} \left(\frac{V}{2\pi^2} \right) \left(\frac{2m}{\hbar^2} \right)^{3/2} \left(\frac{2}{5} \right) E_{F_0}^{5/2}
\end{aligned}$$

which, on reorganization and substitution from (3.23), gives

$$\langle E_0 \rangle = \left(\frac{3}{5} \right) E_{F_0}. \quad (3.25)$$

Both the results (3.23) and (3.25) emphasize the fact that even at $T=0$ the electrons in metals have considerable energy, a result that differs drastically from classical notions.

3.2.6 Electron Energy Parameters at $T > 0$

We shall now consider the evaluation of the Fermi energy E_F at $T > 0$. For this, we shall again start with the equation for N .

$$N = \int_0^\infty N(E) dE = \int_0^\infty D(E) F(E) dE. \quad (3.26)$$

Substituting for $D(E)$ from (3.16) and $F(E)$ from (3.19), (3.26) becomes

$$N = \left(\frac{4\pi V}{h^3} \right) (2m)^{3/2} \int_0^\infty \frac{E^{1/2} dE}{\left[\exp\left(\frac{E-E_F}{k_B T}\right) + 1 \right]}. \quad (3.27)$$

This integration is not straightforward. After a lengthy process (for details, see Wahab [5]), (3.27) becomes

$$N = \left(\frac{4\pi V}{h^3} \right) (2m)^{3/2} \times \frac{2}{3} E_F^{3/2} \left[1 + \frac{\pi^2}{8} \left(\frac{k_B T}{E_F} \right)^2 \right]. \quad (3.28)$$

For $T = 0$, (3.28) becomes

$$N = \left(\frac{4\pi V}{h^3} \right) (2m)^{3/2} \times \frac{2}{3} E_{F_0}^{3/2}. \quad (3.29)$$

From (3.28) and (3.29), we get

$$E_F = E_{F_0} \left[1 - \frac{\pi^2}{12} \left(\frac{k_B T}{E_{F_0}} \right)^2 \right]. \quad (3.30)$$

This represents the temperature variation of E_F in terms of E_{F_0} . The term $(\pi^2/12)(k_B T/E_{F_0})^2 \ll 1$. Further, this variation with T takes place over a small range of $E \sim k_B T$. Hence, unless we are interested in the temperature variation of a property that depends on E_F , the second term on the r.h.s of (3.30) is ignored and the suffix 0 is dropped from E_{F_0} .

Let us now consider the average energy $\langle E_T \rangle$ of an electron at $T > 0$. We have $\langle E_T \rangle = \frac{1}{N} \int_0^\infty E N(E) dE = \frac{1}{N} \int_0^\infty E D(E) F(E) dE$.

Substituting for $D(E)$ and $F(E)$ and carrying out the integration (for details see Singleton [4]), we get

$$\langle E_T \rangle = \langle E_0 \rangle \left[1 + \frac{5\pi^2}{12} \left(\frac{k_B T}{E_{F_0}} \right)^2 \right]. \quad (3.31)$$

3.3 Applications of the Sommerfeld Model

3.3.1 Electronic Specific Heat

It was pointed out in Chap. 2 that the Drude–Lorentz theory leads to a very large electronic specific heat of magnitude $(3/2)R$ compared to $3R$ due to thermal atomic oscillations. Further this value $(3/2)R$ is about 100 times larger than the experimentally estimated electronic specific heat.

In Sect. 3.2 we obtained the following results:

The average energy per electron at $T = 0$,

$$\langle E_0 \rangle = \frac{3}{5} E_{F_0}, \quad (3.25)$$

and average energy per electron at temperature T ,

$$\langle E_T \rangle = \langle E_0 \rangle \left[1 + \frac{5\pi^2}{12} \left(\frac{k_B T}{E_{F_0}} \right)^2 \right]. \quad (3.31)$$

The molar specific heat $[C_V]_{el}$ is

$$[C_V]_{\text{el}} = N_A \frac{d\langle E_T \rangle}{dT} = N_A k_B \langle E_0 \rangle \left[\frac{5\pi^2}{6} \left(\frac{k_B T}{E_{F_0}} \right) \right]. \quad (3.32)$$

Substituting for $\langle E_0 \rangle$ from (3.25) we get

$$[C_V]_{\text{el}} = N_A k_B \left[\frac{\pi^2}{2} \left(\frac{k_B T}{E_{F_0}} \right) \right] = \gamma T \quad (3.33)$$

where

$$\gamma = R \left[\frac{\pi^2}{2} \frac{k_B}{E_{F_0}} \right] = \frac{\pi^2 R}{2T_F}. \quad (3.34)$$

It may be mentioned that we have tacitly assumed that each atom contributes a single conduction electron which is true only for monovalent metals. According to (3.33) and (3.34), $[C_V]_{\text{el}}$ is $\sim 0.03R$ at room temperature for a typical value $T_F = 5,000$ K. This is comparable with experimental values. Thus the huge difference between experimental value of electronic specific heat and the theoretical value obtained from classical theory is now narrowed down. Experimental and theoretical values for γ for several metals are compared in Table 3.2. The two sets of values are of the same order. Detailed comparison shows that the agreement is good in the case of noble metals and alkali metals. However, the agreement is not so good in the other metals in the table, particularly in the case of Fe, Mn and Bi.

Table 3.2 Values of γ
[10^{-4} cal-mol $^{-1}$ K $^{-2}$]

Metal	γ	
	Equation (3.34)	Experimental
Li	1.8	4.2
K	4.0	4.7
Rb	4.6	5.8
Cu	1.2	1.6
Ag	1.5	1.6
Au	1.5	1.6
Mg	2.4	3.2
Ca	3.6	6.5
Sr	4.3	8.7
Fe	1.5	12
Mn	1.5	40
Bi	4.3	0.2

3.3.2 Electrical Conductivity of Metals

In Chap. 2 it was shown that according to Drude–Lorentz theory, the DC electrical conductivity (σ) of a metal is

$$\sigma = \frac{ne^2\tau}{m} \quad (3.35)$$

where n is the electron concentration (N/V) and τ the relaxation time. Sommerfeld [1] derived an expression for σ using the quantum theory model. His derivation is quite involved. It is discussed, among others, by Dekker [6].

Sommerfeld expressed the density of states not in terms of energy but in terms of the momentum p . Thus the number of electron states per unit volume in an element in momentum space $dp_x dp_y dp_z$ is $(2/h^3) dp_x dp_y dp_z$ and the number of occupied states is $(2/h^3)F_0(p) dp_x dp_y dp_z$; $F_0(p)$ is the Fermi–Dirac distribution function expressed in terms of momentum. The suffix 0 indicates absence of any external field. When an electric field X_x is applied in the x -direction, the number of occupied electron states changes to $(2/h^3)F(p) dp_x dp_y dp_z$ where $F(p)$ is the Fermi–Dirac distribution function in the presence of field X_x . We may now write the current density J_x as

$$J_x = -\frac{2e}{h^3} \iiint v_x (F - F_0) dp_x dp_y dp_z. \quad (3.36)$$

We shall note two factors that cause the change in distribution. The first factor arises out of the collisions of the electrons with the lattice. This effect is given by $(\partial F/\partial t)_{\text{coll}}$. The second factor is the applied field; its effect is given by $-eX_x(\partial F/\partial p_x)$. According to the Boltzmann transport equation [6], these two effects are equal in the steady state. Thus

$$\left(\frac{\partial F}{\partial t}\right)_{\text{coll}} = -eX_x \left(\frac{\partial F}{\partial p_x}\right). \quad (3.37)$$

We shall now introduce a relaxation time τ such that

$$\left(\frac{\partial F}{\partial t}\right)_{\text{coll}} = -\frac{(F - F_0)}{\tau}. \quad (3.38)$$

Equating (3.37) and (3.38), we get

$$\frac{(F - F_0)}{\tau} = eX_x \left(\frac{\partial F}{\partial p_x}\right). \quad (3.39)$$

For small fields, $F \approx F_0$. Introducing this change in the r.h.s of (3.39), we get

$$\frac{(F - F_0)}{\tau} = eX_x \left(\frac{\partial F_0}{\partial p_x} \right). \quad (3.40)$$

The energy E of the electron is given in terms of the momentum as

$$E = \frac{p_x^2 + p_y^2 + p_z^2}{2m}. \quad (3.41)$$

Differentiating, we get

$$\frac{\partial E}{\partial p_x} = \frac{2p_x}{2m} = \frac{p_x}{m} = v_x. \quad (3.42)$$

We can now rewrite (3.40) as

$$\begin{aligned} \frac{(F - F_0)}{\tau} &= eX_x \left(\frac{\partial F_0}{\partial p_x} \right) \\ &= eX_x \left(\frac{\partial F_0}{\partial E} \times \frac{\partial E}{\partial p_x} \right) \\ &= eX_x v_x \left(\frac{\partial F_0}{\partial E} \right). \end{aligned} \quad (3.43)$$

Substituting (3.43) in (3.36), we have

$$J_x = -\frac{2e^2}{h^3} X_x \iiint v_x^2 \tau \frac{\partial F_0}{\partial E} dp_x dp_y dp_z. \quad (3.44)$$

We shall effect the following changes in (3.44)

- (i) make τ a function of E , i.e. $\tau \rightarrow \tau(E)$
- (ii) replace v_x^2 by $v^2/3$ where v^2 is the mean square velocity
- (iii) replace $dp_x dp_y dp_z$ by $4\pi p^2 dp$ and
- (iv) replace the triple integral by a single integral.

Then (3.44) becomes

$$J_x = - \left[\frac{16\pi e^2 (2m)^{1/2}}{3h^3} \right] X_x \int_0^\infty E^{3/2} \tau(E) \frac{\partial F_0}{\partial E} dE. \quad (3.45)$$

In Fig. 3.5, we note that changes in F_0 take place mostly in the vicinity of $E = E_F$. Therefore, we effect the following changes in (3.45):

- (i) change $E^{3/2} \tau(E)$ to $E_F^{3/2} \tau_F$
- (ii) $\int_0^\infty \frac{\partial F_0}{\partial E} dE = \int_0^{E_F} \frac{\partial F_0}{\partial E} dE = 1$
- (iii) write E_F as $\frac{\hbar^2}{2m} (3\pi^2 \frac{N}{V})^{2/3}$

With these substitutions, (3.45) reduces to

$$\sigma = \frac{J_x}{X_x} = \left(\frac{N}{V} \right) \frac{e^2 \tau_F}{m} = \frac{ne^2 \tau_F}{m}. \quad (3.46)$$

We note that (3.46) obtained from the Sommerfeld theory is similar to (3.35) obtained from the Drude–Lorentz theory except that τ is replaced by τ_F .

3.3.3 Thermal Conductivity of Metals

In Chap. 2, we derived the following general equation for the coefficient of thermal conductivity (K):

$$K = n\tau v_x^2 \frac{d\langle E_T \rangle}{dT}. \quad (3.47)$$

Here n is the electron concentration (N/V), τ the relaxation time, v_x the x -component of the electron velocity and $\langle E_T \rangle$ the average electron energy at temperature T . We make the following substitutions using the results in this chapter:

- (i) we change τ to τ_F
- (ii) we replace v_x^2 by $v^2/3$ and change it to $v_F^2/3$. Further $v_F^2/3 = (2/3)(E_{F0}/m)$ and
- (iii) from (3.31), we substitute $\frac{\pi^2 k_B T}{2E_{F0}}$ for $\frac{d\langle E_T \rangle}{dT}$.

With these substitutions, (3.47) leads to

$$K = \left(\frac{\pi^2 n \tau_F}{3} \right) \left(\frac{k_B}{m} \right) T. \quad (3.48)$$

3.3.4 The Wiedemann–Franz Ratio

Another major drawback of the classical theory was that it predicted for the Lorenz number L the expression

$$L = \left(\frac{K}{\sigma T} \right) = \frac{3}{2} \left(\frac{k_B}{e} \right)^2. \quad (3.49)$$

The value of the expression on the r.h.s of (3.49) turned out to be 1.11×10^{-8} (W- Ω/K^2) which is about half the experimental values.

Using (3.46) and (3.48), we get

$$L = \left(\frac{K}{\sigma T} \right) = \frac{\pi^2}{3} \left(\frac{k_B}{e} \right)^2. \quad (3.50)$$

This is dimensionally the same as (3.49) but has a value 2.44×10^{-8} (W-Ω/K²) which is close to the experimental values.

3.3.5 Thermopower

The classical electron theory (Chap. 2) gave an expression $-(k_B/2e)$ for the thermopower Q . Its value $\sim -0.43 \times 10^{-4}$ (V-Ω/K²) was larger than typical experimental values by a factor of 10^{-2} . This is yet another drawback of the classical theory.

The general expression for the thermopower is

$$Q = - \left(\frac{1}{3e} \right) \frac{d\langle E_T \rangle}{dT}. \quad (3.51)$$

Substituting for $\langle E_T \rangle$ from (3.31), and differentiating, we get

$$Q = - \left(\frac{\pi^2}{6} \right) \left(\frac{k_B}{e} \right) \left(\frac{k_B T}{E_F} \right) = -1.42 \left(\frac{k_B T}{E_F} \right) \times 10^{-4} \text{ V-}\Omega/\text{K}^2. \quad (3.52)$$

The term $(k_B T / E_F) \sim 10^{-2}$ at room temperature. Hence Q is of the order of 10^{-6} V-Ω/K². This is of the same order as experimental values.

3.3.6 Other Properties

The Sommerfeld model yields the same results as the Drude model for properties like the Hall effect, the magnetoresistance and the transparency of metals. This is understandable since these properties do not involve electron velocities and the main difference between the two models is the use of Fermi–Dirac distribution in one and Maxwell–Boltzmann distribution in the other.

3.4 A Resume

We shall conclude this chapter with a resume of what the Sommerfeld theory has achieved and what lies beyond it.

3.4.1 *New Concepts*

The Sommerfeld model introduced the new concept of the Fermi energy and the associated parameters. The Fermi energy is the topmost occupied energy level. Electrons at energy below E_F are passive; they do not contribute to properties like specific heat and thermal conductivity. The Fermi velocity and Fermi temperature are associated parameters. Another important aspect of Fermi energy is the Fermi surface. It is the constant-Fermi-energy surface. The Fermi surface is spherical in the free electron model; it is more complicated in real structures.

The concept of relaxation time τ was introduced in the Drude–Lorentz theory. Its reciprocal represents the probability of a collision between an electron and an ion-core. In a simplistic interpretation, τ is the probable time between two consecutive collisions. The relaxation time leads to the concept of mean free path Λ through the relation $\Lambda = \tau v$, where v is the average electron velocity. The relaxation time τ enters the expressions for the electrical and thermal conductivities. Sommerfeld made an important modification by treating the relaxation time as a function of energy and pointed out the importance of τ_F , the relaxation time at the Fermi energy.

3.4.2 *Comparison of Results*

The results from the Drude–Lorentz classical model and the Sommerfeld quantum mechanical model are compared in Table 3.3. Three types of differences are noticed. In properties like electrical conductivity and thermal conductivity, the relaxation time τ in the classical theory is replaced by its Sommerfeld theory counterpart τ_F . In properties like the electron energy, specific heat and thermopower, the Fermi energy E_F finds a place. In most cases (except the electrical conductivity) the numerical coefficient is different in the two models. In the case of specific heat, the Lorenz number and the thermopower, there are large differences between the experimental values and the values calculated from the Drude–Lorentz theory. These differences have been narrowed down to a great extent in the Sommerfeld theory.

3.4.3 *Limitations of the Sommerfeld Theory*

The Sommerfeld theory gives satisfactory results for several properties, particularly those properties for which the Drude theory results were unsatisfactory. The Sommerfeld theory also accounts for the paramagnetism of metals; this topic will be discussed in a later chapter. However, the Sommerfeld model cannot account for finer, second order aspects. For example, the electrical conductivity varies with

Table 3.3 Comparison of results from Drude–Lorentz and Sommerfeld theories

Model → Aspects ↓	Classical (Drude–Lorentz)	Quantum mechanical (Sommerfeld)
Statistical mechanics	Maxwell–Boltzmann	Fermi–Dirac
Distribution function	$\exp(-E/k_B T)$	$\frac{1}{[\exp(E - E_F)/k_B T] + 1}$
Energy per electron	$\frac{3}{2} k_B T$	$\langle E_0 \rangle \left[1 + \frac{5\pi^2}{12} \left(\frac{k_B T}{E_{F0}} \right)^2 \right]$
Specific heat (Molar)	$\frac{3}{2} R$	$\left[\frac{\pi^2}{2} \left(\frac{k_B T}{E_{F0}} \right) \right] R$
Electrical conductivity	$\frac{ne^2 \tau}{m}$	$\frac{ne^2 \tau_F}{m}$
Thermal conductivity	$\frac{3}{2} \left(\frac{n\tau}{m} \right) (k_B^2 T)$	$\left(\frac{\pi^2}{3} \right) \left(\frac{n\tau_F}{m} \right) (k_B^2 T)$
Lorenz number ($K/\sigma T$)	$\frac{3}{2} \left(\frac{k_B}{e} \right)^2$	$\left(\frac{\pi^2}{3} \right) \left(\frac{k_B}{e} \right)^2$
Thermopower	$-\frac{1}{2} \left(\frac{k_B}{e} \right)$	$-\left(\frac{\pi^2}{6} \right) \left(\frac{k_B}{e} \right) \left(\frac{k_B T}{E_F} \right)$

temperature. Also, the Lorenz number varies with temperature. The Sommerfeld theory cannot account for these variations. Also, in some properties, the Sommerfeld theory results are satisfactory for some metals but not so satisfactory for some other metals.

There are two properties, Hall effect and thermoelectric effect, where the sign is not what the theory predicts. Obviously, a new approach is required to explain this observation.

Much more fundamental is the question as to why solids have to be grouped as insulators, metals and semiconductors. Why is diamond an insulator and graphite a conductor? A new theory—the band theory of solids—had to emerge to explain these phenomena.

3.5 Problems

1. Calculate the energy of an electron in the energy state above the lowest energy level in a cubic box of side 1 Å.
2. Silver has fcc structure and an atomic radius of 1.44 Å. Assuming silver to be monovalent, calculate the value of the Fermi energy, the Fermi velocity and the Fermi temperature.
3. Derive the expression

$$E_F = E_{F_0} \left[1 - \frac{\pi^2}{12} \left(\frac{k_B T}{E_{F_0}} \right)^2 \right]$$

where E_F and E_{F_0} are the Fermi energies at temperature T and $T = 0$.

4. The thermal conductivity of aluminium at 20 °C is $210 \text{ W m}^{-1} \text{ K}^{-1}$. Calculate the electrical resistivity of Al at this temperature. The Lorenz number for Al is $2.02 \times 10^{-8} \text{ W-}\Omega/\text{K}^2$.
5. Sodium has bcc structure with lattice constant $4.28 \text{ \AA}$. Calculate the Hall coefficient of sodium on the basis of the free electron model.

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Chapter 4

Band Theory of Solids I: Main Framework

Abstract The band theory of solids assumes a periodic potential unlike the Sommerfeld theory which assumes a constant potential. The solutions of the Schrödinger equation with a periodic potential are plane waves modulated by a periodic function. These are called Bloch functions. The solution of the Schrödinger equation with a one-dimensional rectangular periodic potential (the Kronig–Penney model) leads to energy values dependent on the propagation vector. The energy spectrum of the electron now consists of bands of energy levels. The gap between the top-most bands is called the forbidden gap. It plays an important role. A large gap indicates an insulator whereas a small gap indicates a semiconductor. The dynamics of electrons is examined. The concepts of Brillouin zone, effective mass and a hole are discussed. The electronic specific heat and Hall effect are discussed in the light of the band theory. Other models like the Wigner–Seitz model and the tight binding model are also considered.

4.1 Introduction

The limitations of the free electron theories were discussed at the end of Chaps. 2 and 3. It was pointed out that some of these limitations could be overcome in a new theory—the band theory of solids.

The main difference between the free electron theories and the band theory is that the latter recognizes the fact that the electron in a crystal is in a periodic potential. Application of quantum mechanics leads to the result that the energy levels of the electron are in the form of allowed and prohibited bands.

Treatments of the band theory in different degrees of detail are given by McKelvey [1], Blakemore [2], Ashcroft and Mermin [3], Dekker [4], Kittel [5], and Singleton [6].

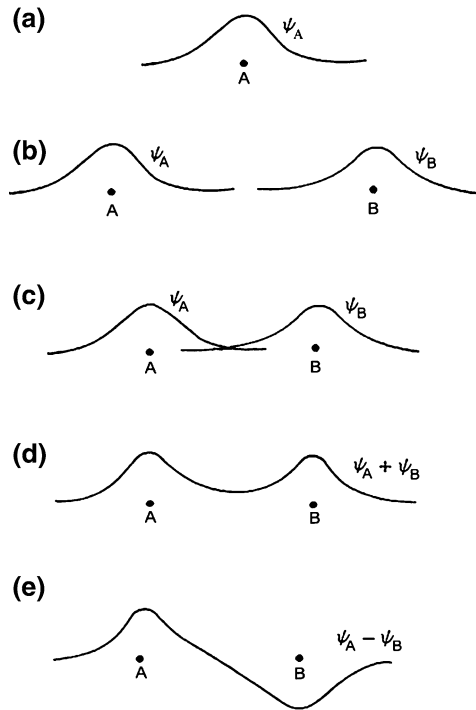
4.2 Origin of Bands

Let us consider a single hydrogen atom A (Fig. 4.1a) along with the associated wave function ψ_A . The energy levels of such an isolated atom are discrete as shown in Fig. 4.2a.

Let us now consider two well-separated hydrogen atoms A and B (Fig. 4.1b) along with their wave functions ψ_A and ψ_B . There is no overlapping of these wave functions. The energy levels of these two atoms taken together are shown in Fig. 4.2b. They are exactly the same as those for a single atom (as shown in Fig. 4.2a) with the difference that each level is now doubly degenerate. If we extend this process to, say, n well-separated H atoms, the energy level diagram will still be the same but now each level will be n -fold degenerate.

Let us bring atoms A and B nearer such that their wave functions overlap (Fig. 4.1c). Each atom perturbs the other atom. This results in the system of two atoms having two wave functions $\psi_1 = (\psi_A + \psi_B)$ and $\psi_2 = (\psi_A - \psi_B)$; this is shown in Fig. 4.1d, e. Corresponding to each degenerate level in Fig. 4.2b, there are now two separate energy levels as shown in Fig. 4.2c; the lower component corresponds to $(\psi_A + \psi_B)$. If n atoms form an array, each n -fold degenerate energy level will now split into n components. If n is large ($\sim 10^{22}$ /mole), the energy difference between the sublevels will be so small that the splitting is not resolved. Thus each

Fig. 4.1 Wave functions of hydrogen atom systems: **a** single atom A, **b** two non-interacting atoms A and B, **c** two interacting atoms A and B, and resulting wave functions **d** $\psi_A + \psi_B$ and **e** $\psi_A - \psi_B$



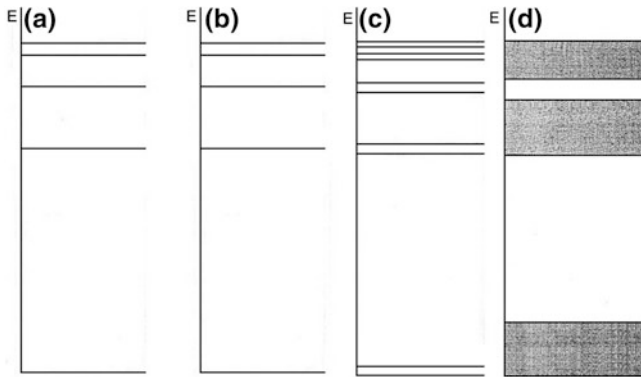


Fig. 4.2 Energy level diagram of **a** a single hydrogen atom shown in Fig. 4.1a, **b** two non-interacting atoms shown in Fig. 4.1b (this is the same as in Fig. 4.2a but the levels are now degenerate), **c** two interacting atoms (the degeneracy in Fig. 4.2b is removed and each level is now a doublet), **d** a system with several interacting atoms resulting in bands

level in Fig. 4.2a has now spread into a band (Fig. 4.2d). Each band is called an ‘allowed’ band. The energy range between two consecutive allowed bands is called a ‘prohibited’ band.

Finally, we shall consider the dependence of the splitting of levels and the width of a band upon the degree of overlapping of the atomic wave functions. We shall treat the lattice constant of the crystal as a measure of the degree of overlapping; a smaller lattice constant indicates larger overlapping and vice versa. The variation of the width of an allowed band for a six atom system with the lattice constant is shown in Fig. 4.3. At low values of the lattice constant (large overlapping), the spacing between the sublevels and the width of the band, are large. Both, the spacing between sublevels and the width of the band decrease as the lattice constant increases, (overlapping decreases) until, in the limit, for a very large lattice constant, the band narrows down into a single energy level.

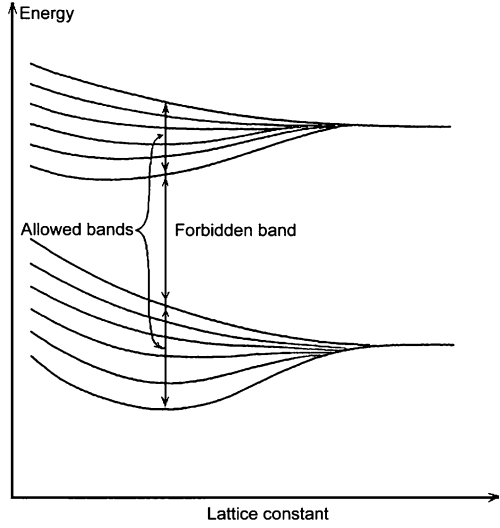
4.3 Bloch’s Theorem

4.3.1 Statement of Bloch’s Theorem

The so-called Bloch’s theorem forms the basis of the band theory of solids. Consider the one-dimensional differential equation

$$\frac{d^2\psi}{dx^2} + f(x)\psi(x) = 0 \quad (4.1)$$

Fig. 4.3 The dependence of the band-width on the lattice constant for an array of six hydrogen atoms



where $f(x)$ is a periodic function. In the literature of mathematics, (4.1) is called Floquet's equation. Floquet showed that its solution is of the form

$$\psi_k(x) = e^{ikx} u_k(x) \quad (4.2)$$

where the function $u_k(x)$ is periodic with the same periodicity as that of $f(x)$. Equation (4.2) is the statement of Floquet's theorem and the function on the r.h.s of (4.2) is called Floquet's function.

Now, let us consider an electron in a one-dimensional potential $V(x)$. The relevant Schrödinger equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} [E - V(x)]\psi(x) = 0. \quad (4.3)$$

Here E is the total energy of the electron. Let the potential $V(x)$ be periodic with periodicity ' a ' which is the periodicity of the one-dimensional lattice in which the electron is situated. It can be seen that (4.3) is similar to (4.1). Independently of Floquet, Bloch [7] proposed that (4.3) has solutions of the form

$$\psi_k(x) = e^{ikx} u_k(x) \quad (4.4)$$

where the function $u_k(x)$ is periodic with the period ' a '. If (4.3) was formulated in three dimensions, the solution would be

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k} \cdot \mathbf{r}} u_{\mathbf{k}}(\mathbf{r}). \quad (4.5)$$

Here, $\mathbf{k}$ is the propagation vector with magnitude k . We may describe the one-dimensional and three-dimensional solutions (4.4, 4.5) as plane waves modulated by a periodic function. Equations (4.4) and (4.5) are statements of Bloch's theorem and the functions on the right hand side are called Bloch functions. We shall note the following property of Bloch functions:

Let us consider the Bloch function at $(x + a)$. Then

$$\psi_k(x + a) = e^{ik(x+a)x} u_k(x + a). \quad (4.6)$$

Since u_k has period ' a ',

$$u_k(x + a) = u_k(x). \quad (4.7)$$

Substituting in (4.6), we have

$$\begin{aligned} \psi_k(x + a) &= e^{ika} [e^{ikx} u_k(x)] \\ &= e^{ika} \psi(x). \end{aligned} \quad (4.8)$$

Thus, we have

$$\psi_k(x + a) = \lambda \psi(x) \quad (4.9)$$

with

$$\lambda = e^{\pm ika}. \quad (4.10)$$

This is an important property of Bloch functions. In fact any function that satisfies (4.9, 4.10) can be taken as a Bloch function.

4.3.2 Proof of Bloch's Theorem

In the original paper, Bloch [7] gave a proof of his theorem using group theory. A simple proof is given by McKelvey [1] which we shall follow.

A second order differential equation like (4.3) will have two independent solutions; we shall call them $G(x)$ and $H(x)$. Thus, in general,

$$\psi(x) = AG(x) + BH(x) \quad (4.11)$$

where A and B are constants. Since $G(x+a)$ and $H(x+a)$ are also solutions of (4.3), they are linear combinations of the two independent solutions $G(x)$ and $H(x)$. Hence we have

$$G(x+a) = \alpha_1 G(x) + \alpha_2 H(x)$$

and

$$H(x+a) = \beta_1 G(x) + \beta_2 H(x). \quad (4.12)$$

Here the α 's and β 's are real functions of the energy E . Further like (4.11) we also have

$$\begin{aligned} \psi(x+a) &= AG(x+a) + BH(x+a) \\ &= (\alpha_1 A + \beta_1 B)G(x) + (\alpha_2 A + \beta_2 B)H(x). \end{aligned} \quad (4.13)$$

Now from (4.9) and (4.10), we have

$$\psi(x+a) = \lambda \psi(x) \quad (4.14)$$

where λ is a constant to be determined. If (4.14) is to be valid, then, using (4.11) and (4.13), we get

$$(\alpha_1 - \lambda)A + \beta_1 B = 0,$$

and

$$\alpha_2 A + (\beta_2 - \lambda)B = 0. \quad (4.15)$$

Equation (4.15) are valid if the determinant of the coefficients of A and B vanishes. Thus

$$\begin{vmatrix} \alpha_1 - \lambda & \beta_1 \\ \alpha_2 & \beta_2 - \lambda \end{vmatrix} = \lambda^2 - (\alpha_1 + \beta_2)\lambda + (\alpha_1\beta_2 - \alpha_2\beta_1) = 0. \quad (4.16)$$

This quadratic equation has two solutions which we may denote by λ_1 and λ_2 . Then, (4.14) may be written as

$$\psi(x+a) = \lambda_1 \psi(x),$$

and

$$\psi(x+a) = \lambda_2 \psi(x). \quad (4.17)$$

We shall now express the λ 's as

$$\lambda_1 = e^{ik_1 a} \text{ and } \lambda_2 = e^{ik_2 a} \quad (4.18)$$

Further, we shall define quantities $u_{k_1}(x)$ and $u_{k_2}(x)$ by

$$u_{k_1}(x) = e^{-ik_1 x} \psi(x)$$

and

$$u_{k_2}(x) = e^{-ik_2 x} \psi(x). \quad (4.19)$$

Then combining (4.17, 4.18 and 4.19), we get

$$\begin{aligned} u_{k_1}(x+a) &= e^{-ik_1(x+a)} \psi(x+a) \\ &= e^{-ik_1(x+a)} \lambda_1 \psi(x) \\ &= e^{-ik_1(x+a)} e^{ik_1 a} \psi(x) \\ &= e^{-ik_1 x} \psi(x) \\ &= u_{k_1}(x). \end{aligned} \quad (4.20)$$

Thus the function $u_{k_1}(x)$ is periodic in 'a'. Similarly, it can be shown that the function $u_{k_2}(x)$ is also periodic in 'a'. Hence in general $\psi(x)$ can be written as

$$\psi(x) = e^{ikx} u_k(x) \quad (4.21)$$

which is just (4.2). Thus Bloch's theorem is proved. Again, in three dimensions, (4.21) becomes

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k} \cdot \mathbf{r}} u_{\mathbf{k}}(\mathbf{r}) \quad (4.22)$$

We conclude that the one-electron wave function for a periodic potential can be written as a plane wave ($e^{i\mathbf{k} \cdot \mathbf{r}}$) modulated by a function $u_{\mathbf{k}}(\mathbf{r})$ which is periodic with the periodicity of the lattice.

4.4 Electron in a Periodic Potential (The Kronig–Penney Model)

4.4.1 Solution of the Schrödinger Equation

Because of the periodic nature of the crystal lattice, the potential of the ion-cores in the lattice is periodic with the same periodicity as that of the lattice. An electron in the crystal lattice then experiences a periodic potential. In order to study the behaviour of an electron in such a periodic potential, Kronig and Penney [8]

assumed a one-dimensional rectangular potential (Fig. 4.4). This is not a realistic potential. Yet, it is very useful in bringing out several important features of the band theory of solids.

In the potential assumed by Kronig and Penney, the width of the potential barrier is b and the periodicity of the lattice is $(a + b)$. The potential may be described as

$$V(x) = 0 \quad \text{for } 0 < x < a \quad (4.23)$$

and

$$V(x) = V_0 \quad \text{for } -b < x < 0. \quad (4.24)$$

The electron can be imagined to lie in the well formed by two consecutive potential barriers.

The Schrödinger equations for the electron (with mass m) in the two regions are:

$$\frac{d^2\psi}{dx^2} + \left(\frac{2m}{\hbar^2}\right)E\psi = 0 \quad \text{for } 0 < x < a \quad (4.25)$$

and

$$\frac{d^2\psi}{dx^2} + \left(\frac{2m}{\hbar^2}\right)(E - V_0)\psi = 0 \quad \text{for } -b < x < 0. \quad (4.26)$$

Since the potential is periodic, the solutions are Bloch functions, i.e.

$$\psi_k(x) = e^{ikx}u_k(x). \quad (4.27)$$

The significance has already been explained. We substitute (4.27) in (4.25) and (4.26). Further, we assume that the potential V_0 is larger than the total energy E . We introduce α^2 and β^2 as

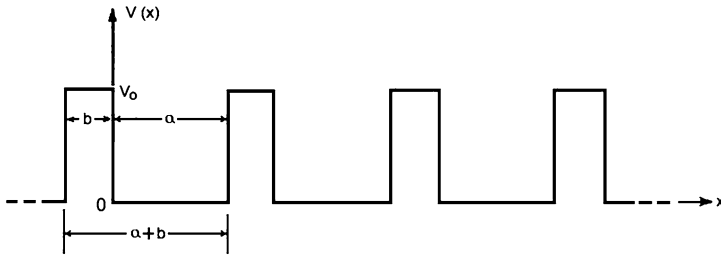


Fig. 4.4 The Kronig–Penney model of an electron in a periodic potential

$$\alpha^2 = \left(\frac{2m}{\hbar^2}\right)E \quad \text{and} \quad \beta^2 = \left(\frac{2m}{\hbar^2}\right)(V_0 - E). \quad (4.28)$$

We then get

$$\frac{d^2 u_k(x)}{dx^2} + 2ik \left[\frac{du_k(x)}{dx} \right] + (\alpha^2 - k^2)u_k(x) = 0 \quad \text{for } 0 < x < a \quad (4.29)$$

and

$$\frac{d^2 u_k(x)}{dx^2} + 2ik \left[\frac{du_k(x)}{dx} \right] - (\beta^2 + k^2)u_k(x) = 0 \quad \text{for } -b < 0 < x < 0 \quad (4.30)$$

Equations (4.29, 4.30) are standard equations. Their solutions are:

$$u_1 = Ae^{i(\alpha-k)x} + Be^{-i(\alpha+k)x} \quad \text{for } 0 < x < a \quad (4.31)$$

$$u_2 = Ce^{(\beta-ik)x} + De^{-(\beta+ik)x} \quad \text{for } -b < x < 0 \quad (4.32)$$

Although we have assumed different wave functions u_1 and u_2 for the two regions of the potential, the wave functions are subject to conditions of continuity at the boundaries of the two regions and to periodicity of $u_k(x)$. These conditions are

$$u_1(x=0) = u_2(x=0); \quad u_1(x=a) = u_2(x=-b) \quad (4.33)$$

and

$$\left(\frac{du_1}{dx}\right)_{(x=0)} = \left(\frac{du_2}{dx}\right)_{(x=0)}; \quad \left(\frac{du_1}{dx}\right)_{(x=a)} = \left(\frac{du_2}{dx}\right)_{(x=-b)} \quad (4.34)$$

Application of these conditions to (4.31, 4.32) leads to the equations

$$\begin{aligned} A + B - C - D &= 0, \\ i(\alpha - k)A - i(\alpha + k)B - (\beta - ik)C + (\beta + ik)D &= 0, \\ e^{i(\alpha-k)a}A + e^{-i(\alpha+k)a}B - e^{-(\beta-ik)b}C - e^{(\beta+ik)b}D &= 0, \\ i(\alpha - k)e^{i(\alpha-k)a}A - i(\alpha + k)e^{-i(\alpha+k)a}B - (\beta - ik)e^{-(\beta-ik)b}C + (\beta + ik)e^{(\beta+ik)b}D &= 0. \end{aligned} \quad (4.35)$$

These equations are valid only when the determinant formed from the coefficients of A, B, C, D vanishes. When the determinant is expanded we get

$$\frac{\beta^2 - \alpha^2}{2\alpha\beta} \sinh \beta b \sin \alpha a + \cosh \beta b \cos \alpha a = \cos k(a+b). \quad (4.36)$$

Equation (4.36) is not easy to handle. To give it a simpler form, Kronig and Penney introduced an approximation. They allowed $b \rightarrow 0$ and $V_0 \rightarrow \infty$ while keeping the product $V_0 b$ finite (note that β^2 involves V_0). Introducing a parameter P defined by

$$P = \frac{mV_0 b a}{\hbar^2} \quad (4.37)$$

we may write (4.36) as

$$P \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka. \quad (4.38)$$

Since b has been reduced to zero, the periodicity of the potential is reduced to ‘ a ’. Equation (4.38) is transcendental in nature; we shall call it the central equation.

4.4.2 Inferences from the Central Equation

The central equation (4.38) is rich in information. Before discussing the inferences that flow from the equation, let us note the physical significance of the various parameters which we have introduced. The parameter α^2 (4.28) is a measure of the total energy E of the electron. The parameter β^2 (4.28) is a measure of $(V_0 - E)$ but since it is assumed that $V_0 \gg E$, β^2 may be taken as a measure of the potential V_0 . The parameter P (4.37) is related to the product $V_0 b$ which may be called the effective “area” of the potential barrier [2]. It represents the influence of the potential on the electron; the influence is more if b is large or V_0 is large. Note that $V_0 b$ is finite notwithstanding the approximation made by Kronig and Penney that $b \rightarrow 0$. In fact, P represents the binding of the electron to the potential well; larger the value of P , larger is the binding and smaller the value of P , weaker is the binding. Let us now examine the information that (4.38) provides.

4.4.2.1 Allowed and Prohibited Energy Bands

One way of looking at (4.38) is that its left hand side is a function of αa (i.e. of the energy E of the electron). We shall call it $F(\alpha a)$. Let us plot $F(\alpha a)$ evaluated for an arbitrary value of P (say $3\pi/2$) against αa . The resulting plot is shown in Fig. 4.5a. It is an oscillatory curve with the amplitudes of peaks decreasing with increasing values of αa . In the figure, lines are drawn at $F(\alpha a) = \pm 1$. The curve consists of two distinct regions. In the first region (shown shaded in Fig. 4.5b), $F(\alpha a)$ has values in

excess of ± 1 . Since $F(\alpha a) = \cos ka$, this region is not acceptable. Energy values represented by αa in this region are prohibited. On the other hand, in the second region of αa (shown shaded in Fig. 4.5c), the curve is well within the limits ± 1 . These αa values (i.e. E values) are acceptable. The two regions thus represent 'prohibited' and 'allowed' bands respectively. The discontinuities occur at $\alpha a = \pm n\pi$ where n is an integer. It can be seen from the figure that the width of allowed bands increases with n whereas the width of prohibited band decreases with n . Thus (4.38) clearly leads to the idea of energy bands in solids.

4.4.2.2 The E - k Curve

We can look at (4.38) in another way. The left hand side of (4.38) is related to the energy E (through α) whereas the right hand side is related to the propagation vector

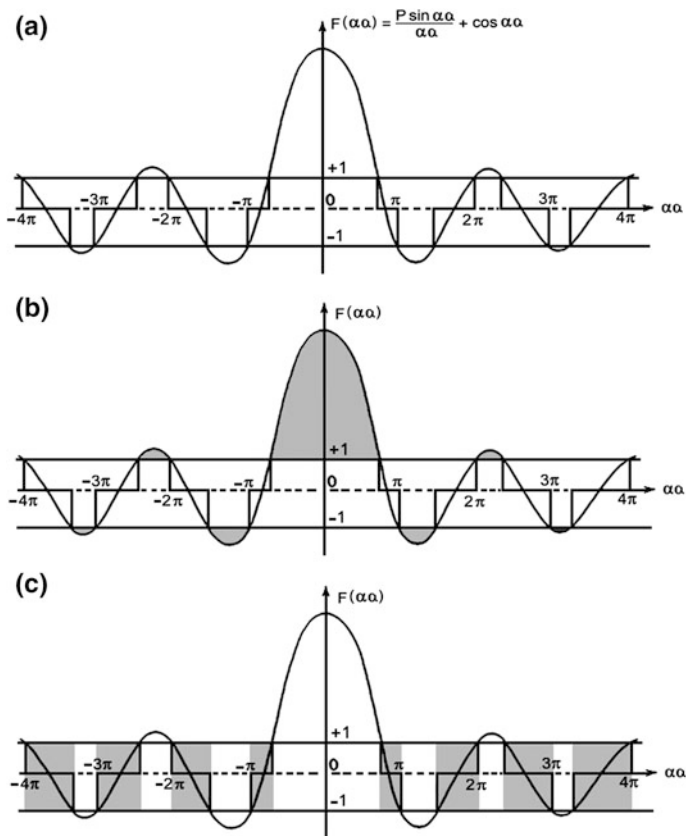
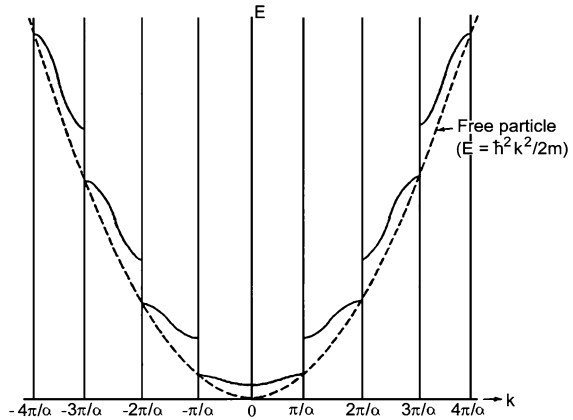


Fig. 4.5 Plot of **a** $F(\alpha a)$ versus αa , **b** showing regions of $F(\alpha a)$ where it is $> \pm 1$, **c** showing regions where it is within ± 1

Fig. 4.6 E versus k plot
(extended zone scheme)



k (through $\cos ka$). We may choose a value of E . Then calculate aa and $F(aa)$. Equating $F(aa)$ to $\cos ka$, we may find k . In this way, we may plot E versus k . The plot is shown in Fig. 4.6. We notice that the plot is discontinuous, the discontinuities occurring at $k = \pm n \pi/a$ where n is an integer. These limiting values of k define the boundaries of the ‘Brillouin zones’. Thus, the first Brillouin zone extends from $k = -\pi/a$ to $k = +\pi/a$. Likewise, the second Brillouin zone extends from $-2\pi/a$ to $-\pi/a$ and again from π/a to $2\pi/a$. A more detailed discussion of Brillouin zones will be given in the next chapter. For large values of E , we get $E = \hbar^2 k^2 / 2m$ which is the free electron value. This is shown by the dotted curve in Fig. 4.6. It may be noted from Fig. 4.6 that the curve for each allowed band has a point of inflection which means that the slope is zero at the edges of the allowed band and it has some maximum value at the middle (point of inflection) of the curve. At large energies the allowed bands become broad and the forbidden regions become narrow [1].

In the E - k curve shown in Fig. 4.6, the different bands are placed one above the other and are also laterally shifted. Such a representation is called an extended zone scheme. Another representation is shown in Fig. 4.7. In (4.38), the rhs is a cosine function which is an even function. If in the rhs k is replaced by $k + (2\pi n/a)$, $\cos ka$ remains unchanged. It follows that E is also a periodic function with period $2\pi/a$ and can be depicted over several complete periods. This representation is called the repeated zone scheme or periodic zone scheme (Fig. 4.7). Again since $k' = k + (2\pi n/a)$, we may reduce the contents of the second Brillouin zone (or any other higher zone) to the wave vector range $-\frac{\pi}{a} < k < \frac{\pi}{a}$ and represent the bands one above the other. This representation (Fig. 4.8) is called the reduced zone scheme and the wave vectors for the higher bands are called reduced vectors.

Fig. 4.7 E versus k plot
(periodic zone scheme)

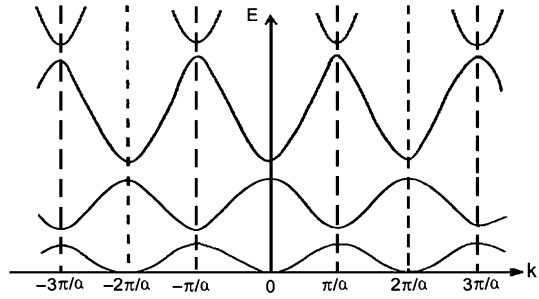
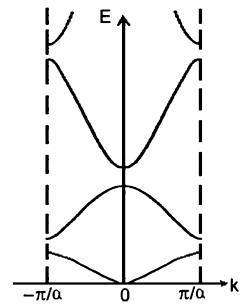


Fig. 4.8 E versus k plot
(reduced zone scheme)



4.4.2.3 Physical Origin of Discontinuities

We have seen that there is a discontinuity in energy at $k = \pm(n\pi/a)$. These k -values are the boundaries of Brillouin zones. Since by definition, $k = 2\pi/\lambda$, we get $n\lambda = 2a$. But this is the statement of Bragg's law for normal incidence. Thus, electron waves cannot propagate at the Brillouin zone boundaries. Instead, they undergo Bragg reflection at the boundary. This is the cause of the discontinuity in energy.

4.4.2.4 Total Number of Wave Functions

We shall now estimate the number of possible wave functions in a band. Let us assume a one-dimensional crystal of length L . The boundary condition for the wave function is

$$\psi(x + L) = \psi(x). \quad (4.39)$$

This is Born's cyclic condition. It was originally introduced for a circular lattice but can be applied to a linear crystal with large dimensions. Since the wave functions are Bloch functions, we have

$$e^{ik(x+L)}u_k(x+L) = e^{ikx}u_k(x). \quad (4.40)$$

Due to the periodicity of $u_k(x)$, $u_k(x+L) = u_k(x)$. Hence,

$$e^{ik(x+L)} = e^{ikx}. \quad (4.41)$$

This is possible if

$$k = \frac{2\pi n}{L} \text{ with } n = \pm 1, \pm 2, \dots \quad (4.42)$$

From (4.42), we have the number of wave functions in the range dk of the propagation vector as

$$dn = \frac{L}{2\pi} dk \quad (4.43)$$

We know that k is limited by $\pm\pi/a$. The maximum value of n is $(L/2a) = N/2$ where N is the number of unit cells. Thus, the total number of wave functions in a band is equal to the number of unit cells in the crystal. If we allow for the spin of the electron, the total number of electrons in a band is twice the number of unit cells. The significance of this is that if a band contains $2N$ number of electrons, the band is completely filled.

4.4.2.5 Effect of Parameter P on Band-width

Finally, let us consider the effect of the parameter P on the width of the allowed band. Let us recall that the parameter P is a measure of the binding of the electron to the ion-core. It depends on the potential V_0 acting over the width b of the potential barrier. Smaller the value of P , weaker is the binding and larger its value, stronger is the binding. Let us consider two extreme situations. If $P \rightarrow 0$, we have the free electron model and the electron has a quasi-continuous energy spectrum. On the other hand if $P \rightarrow \infty$, the electron behaves as if it is a particle in a box; it has discrete energy levels conforming to

$$E_n = \left(\frac{\pi^2 \hbar^2}{2ma^2} \right) n^2. \quad (4.44)$$

In between, the allowed band has a finite width (shaded area in Fig. 4.9).

Fig. 4.9 Dependence of band-width on parameter P

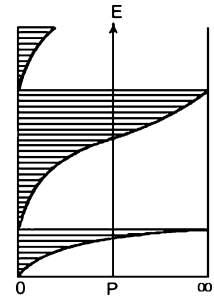
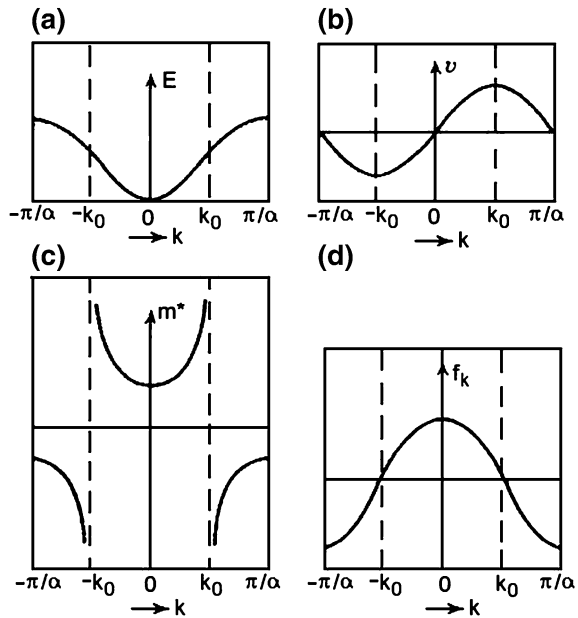


Fig. 4.10 **a** E - k curve for first allowed band; **b** v - k curve; **c** m^* - k curve; **d** f_k - k curve



4.4.3 Dynamics of Electrons in a Band

The E - k curve for the first allowed band is shown in Fig. 4.10a. It is a curve with a point of inflection at $k = k_0$. The slope (dE/dk) of the curve is zero at the bottom of the band and starts increasing till it reaches a maximum value $(dE/dk)_{\max}$ at $k = k_0$. Thereafter the slope starts decreasing, becoming zero at the top of the band. In the negative half of the Brillouin zone, the slope is, again, zero at the bottom, starts decreasing till it reaches a negative value $(-dE/dk)_{\min}$ at $k = -k_0$. Thereafter it starts increasing becoming zero at the top of the band. These features are common to all allowed bands. We shall now consider how various electron parameters vary within a band.

4.4.3.1 Velocity

The group velocity (v) of the electron is given by

$$v = d\omega/dk \quad (4.45)$$

where ω is the angular frequency of the electron wave. Since energy $E = \hbar\omega$, (4.45) may be written as

$$v = \hbar^{-1}(dE/dk) \quad (4.46)$$

Thus the velocity of the electron is directly related to the slope of the E - k curve. We have discussed how the slope varies from the bottom of the band to the top. Keeping this in view, we see that in the positive part of the Brillouin zone, the velocity increases from zero at $k = 0$ to a maximum value at $k = k_0$ and then decreases to zero at $k = \pi/a$. In the negative half of the Brillouin zone, the velocity decreases from zero at $k = 0$ to a negative minimum value at $k = -k_0$ and then increases to zero at $k = -\pi/a$. The variation of v with k is shown in Fig. 4.10b.

4.4.3.2 Acceleration

Let us consider the acceleration a^* that an electron in a band will develop if it is acted upon by an electric field F . Let the electron have velocity v at the time the field is switched on. Let its propagation vector be k . If the field F acts for time dt , the gain in the energy of the electron is

$$\begin{aligned} dE &= -eFds \\ &= -eF \frac{ds}{dt} dt \\ &= -eFvdt \\ &= -\frac{eF}{\hbar} \left(\frac{dE}{dk} \right) dt \end{aligned} \quad (4.47)$$

where we have used (4.46). But

$$dE = \frac{dE}{dk} dk. \quad (4.48)$$

Equating (4.47) and (4.48), we get

$$\frac{dk}{dt} = -\frac{eF}{\hbar}. \quad (4.49)$$

Now, the acceleration a^* of the electron is

$$\begin{aligned}
 a^* &= \frac{dv}{dt} = \left(\frac{dv}{dk} \right) \left(\frac{dk}{dt} \right) \\
 &= \frac{1}{\hbar} \left(\frac{d^2 E}{dk^2} \right) \frac{dk}{dt}.
 \end{aligned} \tag{4.50}$$

From (4.49) and (4.50), we get

$$a^* = - \left(\frac{eF}{\hbar^2} \right) \left(\frac{d^2 E}{dk^2} \right). \tag{4.51}$$

Thus the acceleration of the electron in a band is related to the second derivative of E with respect to k . It is more useful to describe a^* as related to $\frac{d}{dk} \left(\frac{dE}{dk} \right)$ i.e. the slope of the v - k curve.

4.4.3.3 Effective Mass of an Electron

Let us consider a free electron in an electric field F . Its acceleration ‘ a ’ is given by

$$a = - \frac{eF}{m}. \tag{4.52}$$

Comparing (4.51) and (4.52), we see that in a band the term $\hbar^2 / (d^2 E / dk^2)$ is equivalent to the mass of the electron. This is called the “effective mass” m^* and is given by

$$m^* = \hbar^2 / (d^2 E / dk^2). \tag{4.53}$$

The effective mass is k -dependent; it is inversely related to $\left(\frac{d}{dk} \frac{dE}{dk} \right)$ i.e. to the slope of the v - k curve. The variation of m^* over the entire first Brillouin zone is shown in Fig. 4.10c. It can be seen that m^* has some finite positive value at $k = 0$. It then increases until at $k = k_0$ it tends to ∞ . Again, it has some finite negative value at the end of the Brillouin zone at $k = \pi/a$. It decreases as k approaches k_0 where it tends to ∞ . A similar variation is seen in the region $k = 0$ to $k = -\pi/a$ with appropriate changes in sign. What does a negative m^* imply? Since the effective mass is related to the acceleration, a negative value for m^* means that the electron is decelerated by an applied field i.e. the electron behaves like a positively charged particle. From Fig. 4.10c, we conclude that the electron behaves like a negatively charged particle in the lower half of the band but it behaves like a positively charged particle in the upper half of the band. The effective mass is an important parameter of the electron and it affects several properties of solids. A more detailed discussion of the effective mass will be given later in this chapter and also in the next chapter.

4.4.3.4 Degree of ‘Freeness’ of an Electron

We shall introduce a parameter f_k defined as

$$f_k = \frac{m}{m^*} = \left(\frac{m}{\hbar^2} \right) \left(\frac{d^2 E}{dk^2} \right). \quad (4.54)$$

This parameter represents the degree of ‘freeness’ or the extent to which an electron behaves like a free electron. From the definition we see that a small f_k means that the electron is heavier than a free electron. On the other hand, if f_k is large then the electron is lighter than a free electron. It follows that when $f_k = 1$, the electron behaves like a free electron. The variation of f_k across an allowed band is shown in Fig. 4.10d. We note that f_k is positive in the lower half and negative in the upper half of the band.

4.4.3.5 Crystal Momentum

We must note a subtle conceptual difference between the momentum of the free electron and the momentum of the electron in a band. First, let us distinguish between the velocity of the free electron and that of the electron in a band. For the discussion in this section, let us call these two velocities v_F and v_B respectively. Then

$$v_F = \frac{p}{m} = \frac{mv_F}{m} = \frac{d(\frac{1}{2}mv_F^2)}{d(mv_F)} = \frac{dE}{dp}. \quad (4.55)$$

The velocity v_B of the electron in a band is defined as

$$v_B = \frac{d\omega}{dk}. \quad (4.56)$$

But

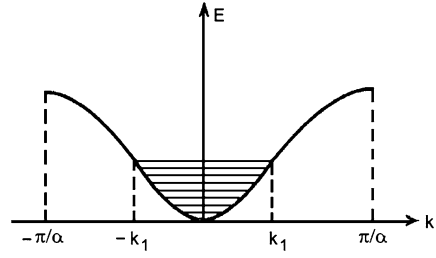
$$E = \hbar\omega. \quad (4.57)$$

Hence,

$$v_B = \frac{1}{\hbar} \frac{dE}{dk} = \frac{dE}{d(\hbar k)}. \quad (4.58)$$

Comparing (4.55) and (4.58), we see that $\hbar k$ plays the same role in the case of an electron in a band as that played by the momentum p in the case of the free electron. However, $\hbar k$ is the result of several influences acting on the electron like “the

Fig. 4.11 A partially filled energy band



reaction of the lattice, the wave function, the bands and so on” [9]. Hence, $\hbar k$ is called the crystal momentum which is the momentum of the system as a whole [6].

4.4.3.6 Effective Number of Free Electrons

Let us consider an allowed band in which levels up to $k = k_1$ are filled (Fig. 4.11). Since an electron in a band is not a free electron but has a degree of freeness f_k associated with it, the number of effective free electrons N_{eff} in the band is

$$N_{\text{eff}} = \sum f_k. \quad (4.59)$$

It has been shown in Sect. 4.4.2 that the number of states dn in an interval dk of the wave vector in a linear lattice of length L is

$$dn = \frac{L}{2\pi} dk. \quad (4.60)$$

Since each of these states is occupied by two electrons, (4.59) may be rewritten as

$$\begin{aligned} N_{\text{eff}} &= \frac{L}{\pi} \int_{-k_1}^{k_1} f_k dk \\ &= \left(\frac{2Lm}{\pi\hbar^2} \right) \int_0^{k_1} \frac{d^2 E}{dk^2} dk. \\ &= \left(\frac{2Lm}{\pi\hbar^2} \right) \left(\frac{dE}{dk} \right)_{k=k_1} \end{aligned} \quad (4.61)$$

From (4.61) the following conclusions can be drawn:

- (a) Since dE/dk vanishes at the top of an allowed band, N_{eff} for a completely filled band is zero and
- (b) since dE/dk is maximum at the point of inflexion, N_{eff} has maximum value for a band filled up to the point of inflexion.

4.4.3.7 An Electron Vacancy (Hole)

Let v_i be the velocity of any electron in a band v_j be the velocity of a chosen electron. Then, $-e$ being the charge, the current I due to electrons in a completely filled band in the absence of any external field is

$$I = -e \sum_i v_i = -e \left[\sum_{i \neq j} v_i + v_j \right] = 0. \quad (4.62)$$

If the electron with velocity v_j is missing from the band, the current I' will be

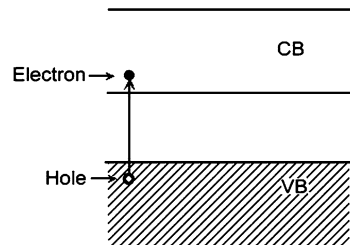
$$I' = -e \sum_{i \neq j} v_i = e v_j. \quad (4.63)$$

If an electric field F is now applied, the current I' will change according to

$$\frac{dI'}{dt} = e \left(\frac{d v_j}{dt} \right) = -\frac{e^2 F}{m_j^*}. \quad (4.64)$$

where m_j^* is the effective mass of the j th electron. Since the missing electron will generally be from the top of a band, m_j^* is negative and the r.h.s of (4.64) will be positive. Thus the missing electron behaves like a positively charged particle. Such a particle is called a 'hole'. The formation of a hole is shown in Fig. 4.12. This is a new concept. We shall see that it plays an important role, particularly in semiconductors.

Fig. 4.12 Model of 'hole' formation



4.5 Band Theory Vis-à-Vis Free Electron Theory

Some limitations of the free electron theory were pointed out at the conclusion of Chap. 3. The most prominent among them are:

- (i) Free electron theory cannot differentiate between different types of solids like, metals, insulators and semiconductors.
- (ii) Severe discrepancies persist between experimental and theoretical values of properties/parameters like electronic specific heat, electrical conductivity, Hall angle and Wiedemann–Franz ratio.
- (iii) There is a difference in sign between expected and observed value of the Hall coefficient for some metals.

We have seen that new concepts have been introduced in the band theory of solids. These are (i) existence of allowed and prohibited energy bands, (ii) effective mass of electrons and (iii) holes. We shall now examine to what extent these concepts are useful in accounting for these limitations.

4.5.1 Classification of Solids

The bands in a solid are shown in Fig. 4.13. The lower bands are completely filled. The upper-most completely filled band is called the valence band (VB) and the band above it the conduction band (CB). The energy difference between the bottom of the CB and the top of the VB is called the energy gap E_g . In the example shown the conduction band is empty and E_g is large. Since there are no electrons in the CB, conduction is not possible. Even at elevated temperatures, electrons from the top of the VB cannot jump into the CB because of the large E_g and hence conduction is not possible. Solids with such a band structure are called insulators. Diamond with $E_g = 7$ eV and NaCl with $E_g = 8.6$ eV are examples of insulators.

Let us consider the band structure shown in Fig. 4.14a. Here, again, the CB is empty at low temperatures. Hence there is no possibility of electrical conduction. But the value of E_g is less compared to that in an insulator. Therefore, as the

Fig. 4.13 Band structure of an insulator

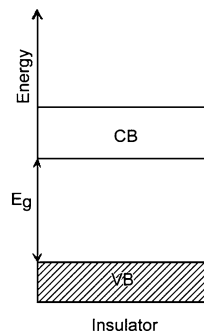
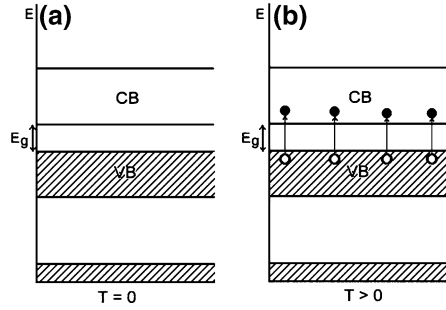


Fig. 4.14 Band structure of a semiconductor: **a** empty CB at $T = 0$; **b** partially filled CB at $T > 0$



temperature is raised, the thermal energy is enough to enable some electrons from the top of VB to jump into the CB (Fig. 4.14b). Once there are some electrons in the lower part of CB, they can accept energy from an applied electric field and pass into upper unoccupied levels in the CB; in short, they can conduct. The number of electrons jumping from VB to CB by this process increases exponentially with temperature. Solids with band structure shown in Fig. 4.14a, b are called semiconductors. Ge ($E_g = 0.7$ eV) and Si ($E_g = 1.1$ eV) are examples of semiconductors.

In Fig. 4.15a, we see that the conduction band is partially filled. If an electric field is applied to such a solid, electrons occupying lower levels in the CB get energized and move into the upper levels in the CB, thus making conduction easy. Such solids have high conductivity. Easy conduction is also possible if the top of the VB overlaps with the lower part of the CB (Fig. 4.15b). The electrons at the top of the VB will now accept energy from an applied electric field and get excited to the upper unoccupied levels in the CB. Solids with band structures shown in Fig. 4.15a, b are called metals. Al, Ag, Au, Mg, Zn, Ni, W, Fe are examples of metals.

We would like to mention one more class of solids in which the overlapping of the VB and CB is marginal; the bottom of the CB lies very slightly below the top of the valence band (Fig. 4.16). These solids also conduct like metals but because of the very slight overlap, the number of carriers is very small and, so, the conductivity is much smaller than in regular metals. Such solids are called semimetals. As, Sb and Bi are examples of semimetals.

Fig. 4.15 Band structure of a metal: **a** partially filled CB; **b** overlapping of VB and CB

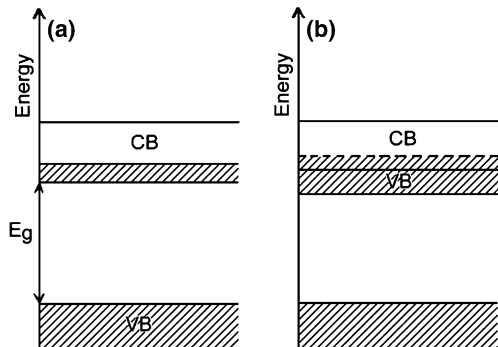
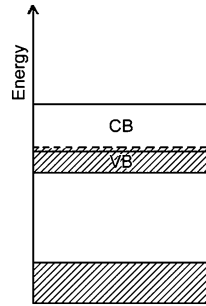


Fig. 4.16 Band structure of a semimetal (very slight overlapping of VB and CB)



Thus the band theory of solids accounts for the existence of different types of solids including semiconductors on the basis of relative dispositions of the conduction and valence bands and varying values of the energy gap.

4.5.2 Electronic Specific Heat

We have seen in Chap. 3 that the quantum free electron theory gave an expression for the electronic specific heat which was a big improvement over the one given by the Drude–Lorentz theory. Yet, there is a large difference between the experimental and theoretical values. We reproduce some of the data in Table 4.1.

The electronic specific heat is contained in the term γ which is the coefficient of the linear term in the specific heat (Chap. 3). The free-electron theory expression for γ includes the Fermi energy E_F which, in turn, includes the electron mass m . In the band theory, the electron has an effective mass m^* which is different from the free electron mass m . Thus the difference in the measured and theoretical values of γ calculated from the free electron theory is to be attributed to the use of the electron mass m instead of the effective mass m^* . The ratio of m^*/m thus calculated is given in Table 4.1.

In general, wherever the mass m occurs in the free electron theory, it should be replaced by the effective mass m^* . This leads to improved values not only for the electronic specific heat but also for other properties like the electrical conductivity and magnetic susceptibility.

4.5.3 Hall Effect

We have seen in Chap. 2 that the sign of the Hall coefficient R_H is negative assuming that the carriers are electrons. But for some metals like Be, Mg and Zn it is positive. There was no answer for this observation either in the classical electron theory or in the quantum free electron theory.

Table 4.1 Experimental values for the coefficient γ of the linear term in the molar specific heats of metals and the values given by free electron theory

Element	Free electron γ	Measured γ	Ratio (m^*/m)
	(in 10^{-4} cal-mole $^{-1}$ K $^{-2}$)		
Li	1.8	4.2	2.3
Na	2.6	3.5	1.3
K	4.0	4.7	1.2
Rb	4.6	5.8	1.3
Cs	5.3	7.7	1.5
Cu	1.2	1.6	1.3
Ag	1.5	1.6	1.1
Au	1.5	1.6	1.1
Be	1.2	0.5	0.42
Mg	2.4	3.2	1.3
Ca	3.6	6.5	1.8
Sr	4.3	8.7	2.0
Ba	4.7	6.5	1.4
Zn	1.8	1.4	0.78
Cd	2.3	1.7	0.74
Hg	2.4	5.0	2.1
Al	2.2	3.0	1.4
Ga	2.4	1.5	0.62
In	2.9	4.3	1.5
Tl	3.1	3.5	1.1
Sn	3.3	4.4	1.3
Pb	3.6	7.0	1.9

The band theory introduces a new concept of a ‘hole’ which is a missing electron in an otherwise filled valence band. The hole is equivalent to a positively charged particle and has a positive effective mass m^* . If we re-work the derivation of the Hall coefficient for a hole [5], we get the result that the Hall coefficient $R_H (=1/nec)$ is positive for holes.

Thus, the positive Hall coefficient observed for some metals is to be attributed to the predominant presence of holes in those metals.

4.6 Other Models

In Sect. 4.4, we discussed in detail the Kronig–Penney model for an electron in a periodic potential. We saw that the model introduced us to the concepts and language of the band theory of solids. At the same time, it was emphasized that the Kronig–Penney model is unrealistic since it is one-dimensional and assumes thin

rectangular potential barriers. We shall now consider some models which approximate better to real crystals.

4.6.1 The Wigner–Seitz Cellular Model

Wigner and Seitz [10] proposed a model which starts with a certain construction which we shall first describe.

Consider a two-dimensional lattice (Fig. 4.17). Point A represents an ion-core located at a lattice point. Lines are drawn from A to the nearest neighbours and next-nearest neighbours. Each of these lines is bisected and the bisectors are joined. This results in a polygon (shaded area in figure) which is called a Wigner–Seitz cell. Let us extend this construction to a three-dimensional lattice, say the bcc lattice. This lattice has an ion-core at the centre and one at each corner of the unit cell. The bisectors of the lines joining the ion-core at the centre to its nearest and next-nearest neighbours are not lines but planes. Thus the square faces (Fig. 4.18) bisect the lines from the centre to the corresponding points in the neighbouring unit cells. Similarly, the hexagonal faces bisect the lines joining the ion-core at the centre to the ion-cores at the corners of the cubes. These square and hexagonal faces form a polyhedron which is a truncated octahedron; this is the Wigner–Seitz cell for the bcc lattice. Thus the Wigner–Seitz construction divides the lattice into cells; hence it is called the cellular model.

In the bcc lattice, each unit cell contains one ion-core. The potential due to the ion-core is assumed to be spherical within each polyhedron. To start with, Wigner and Seitz consider the case for $\mathbf{k} = 0$. The wave function (Bloch function) is periodic with the period of the lattice and also symmetrical about each lattice point. This is possible if the derivative $(\partial u_0 / \partial n)$ of the periodic part of the Bloch function vanishes at the cell-boundaries; here $\partial / \partial n$ denotes differentiation normal to the surface of the polyhedron. The polyhedra are inherently nearly spherical and so, Wigner and Seitz replaced them with perfect spheres with volume equal to the

Fig. 4.17 Wigner–Seitz cell for a two-dimensional lattice

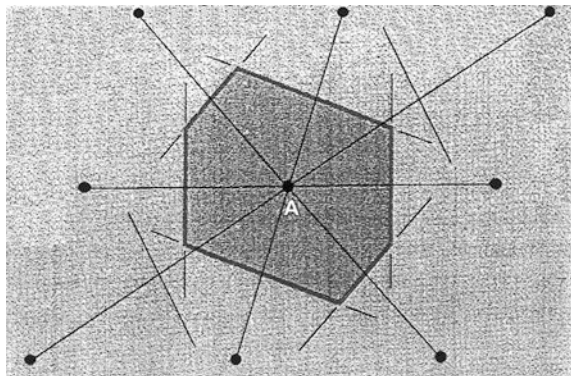
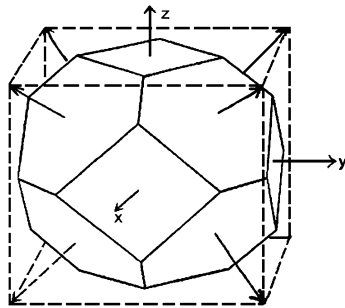


Fig. 4.18 Wigner–Seitz cell for a bcc lattice



atomic volume of the atom in the bcc unit cell. Thus, if r_0 is the radius of the Wigner–Seitz spherical cell and a the lattice constant of the bcc unit cell, we have

$$4\pi r_0^3/3 = a^3/2 \quad (4.65)$$

or, approximately,

$$r_0 \approx 0.49a. \quad (4.66)$$

The boundary condition $(\partial u_0/\partial n) = 0$ is now replaced by

$$(\partial u_0/\partial r)_{r=r_0} = 0 \quad (4.67)$$

on the surface of the sphere.

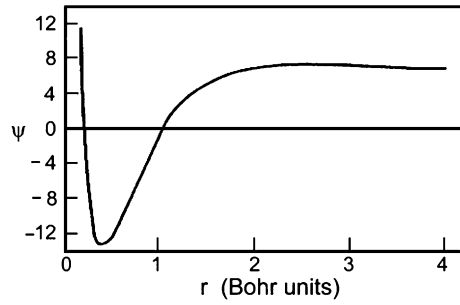
With these approximations, the Schrödinger equation for the electron in a Wigner–Seitz cell becomes

$$\left[\left(-\frac{\hbar^2}{2mr^2} \right) \frac{d}{dr} \left(r^2 \frac{d}{dr} \right) + V(r) \right] \psi = E\psi \quad (4.68)$$

For the ion-core potential $V(r)$, Wigner and Seitz assumed a known potential. The wave function calculated for different values of r for the $3s$ -state of Na is shown in Fig. 4.19. It can be seen that the wave function is constant for 90 % of the atomic volume. This means that the solutions are mostly plane waves and the oscillatory part $u_0(r)$ is observed only in the region close to the ion-core. Thus the valence electrons in Na (and other alkali metals) behave mostly like free electrons. In copper, silver and gold, however, this is not the situation.

In the next stage, Wigner and Seitz generalized the treatment taking into account the $\mathbf{k}$ -dependence of the wave function. Instead of considering only $u_0(r)$, the full Bloch function $e^{i\mathbf{k}\cdot\mathbf{r}}u_{\mathbf{k}}(\mathbf{r})$ was employed. Subjecting the Bloch function to the Hamiltonian $-\frac{\hbar^2}{2m}\nabla^2 + V(r)$, we get the differential equation

Fig. 4.19 Lowest wave function for sodium metal



$$\left[-\frac{\hbar^2}{2m} \nabla^2 - \frac{i\hbar^2}{m} \mathbf{k} \cdot \nabla + V(\mathbf{r}) \right] u_{\mathbf{k}}(\mathbf{r}) = \left(E_{\mathbf{k}} - \frac{\hbar^2 k^2}{2m} \right) u_{\mathbf{k}}(\mathbf{r}). \quad (4.69)$$

The term $\mathbf{k} \cdot \nabla$ is treated as a perturbation, the solution of this equation is beyond the scope of this book. We shall just quote the result. Using perturbation theory, the energy values are obtained as

$$E_{\mathbf{k}} = E_0 + \frac{\hbar^2 k^2}{2m} \left[1 + \frac{2}{m} \sum_{\alpha} \frac{\langle 0 | p_x | \alpha \rangle \langle \alpha | p_x | 0 \rangle}{E_0 - E_{\alpha}} \right] \quad (4.70)$$

where $\langle 0 | p_x | \alpha \rangle$ is the matrix element of the x -component of the momentum operator between the states $\mathbf{k} = 0$ in the band 0 and another band α .

Wigner and Seitz [10] used their results to calculate basic properties like the lattice constant, compressibility and binding energy of sodium metal. The Wigner and Seitz method was extended by Kimball [11] to diamond and to the transition metal elements by Slater [12]. However, because of the approximations, the cellular method has been overtaken by other methods.

4.6.2 Nearly Free Electron Model

In the nearly free electron (NFE) model, it is assumed that the total energy E of the electron is large compared to the periodic potential. We shall see that as a consequence of this assumption, the allowed bands are broad and the forbidden bands are narrow. The assumptions and results are not applicable to all metals but are reasonably applicable to alkali metals.

For convenience, we shall assume a linear lattice with period ' a '. The potential $V(x)$ is expressed as

$$V(x) = -\frac{\hbar^2}{2m}\gamma f(x) \quad (4.71)$$

where $f(x)$ is a periodic function with period 'a' and γ is a constant. The Schrödinger equation is

$$\frac{d^2\psi}{dx^2} + [k_0^2 + \gamma f(x)]\psi(x) = 0 \quad (4.72)$$

where k_0 is related to the energy E through the relation

$$E = \frac{\hbar^2}{2m}k_0^2. \quad (4.73)$$

It is known that any periodic function can be expressed as a Fourier series. Hence

$$f(x) = \sum_{n=-\infty}^{\infty} C_n e^{-2\pi i n x/a} \quad (4.74)$$

with

$$C_n = \frac{1}{a} \int_0^a f(x) e^{2\pi i n x/a} \quad (4.75)$$

Similarly,

$$V(x) = \sum_{n=-\infty}^{\infty} V_n e^{-2\pi i n x/a}. \quad (4.76)$$

From (4.71), (4.74) and (4.76), we have

$$V_n = -\frac{\hbar^2}{2m}\gamma C_n. \quad (4.77)$$

C_n and V_n are the Fourier coefficients in the Fourier series for $f(x)$ and $V(x)$ respectively.

Let us write the solution of (4.72) as a Bloch function:

$$\psi(x) = e^{ikx} u_k(x) \quad (4.78)$$

The function $u_k(x)$ is periodic and it can also be written as a Fourier series as

$$u_k(x) = \sum_{n=-\infty}^{\infty} b_n e^{-2\pi i n x/a}. \quad (4.79)$$

When $\gamma \rightarrow 0$, $u_k(x) \rightarrow b_0$ and $k \rightarrow k_0$. This reduces (4.78) to

$$\psi(x) = b_0 e^{ik_0 x} \quad (4.80)$$

The general solution of (4.72) may then be written as

$$\psi(x) = b_0 e^{ikx} + \gamma \left[e^{ikx} \sum_{n \neq 0} b_n e^{-2\pi i n x/a} \right]. \quad (4.81)$$

By comparing (4.81) with (4.78), we see that (4.81) is a Bloch function with

$$u_k(x) = b_0 + \gamma \sum_{n \neq 0} b_n e^{-2\pi i n x/a}. \quad (4.82)$$

Substituting (4.81) in (4.72), we get

$$\begin{aligned} & b_0(k_0^2 - k^2) e^{ikx} + \gamma \sum_{n \neq 0} [(k_0^2 - k_n^2) b_n + b_0 C_n] e^{ik_n x} + \gamma^2 \sum_{n \neq 0} \sum_{n' \neq 0} b_{n'} C_n e^{i(k - \frac{2\pi n}{a} - \frac{2\pi n'}{a})x} \\ & = 0 \end{aligned} \quad (4.83)$$

where

$$k_n = k - \frac{2\pi n}{a}. \quad (4.84)$$

As mentioned earlier, in the NFE model, γ is always small. Hence, at this stage we may ignore the γ^2 terms in (4.83). We shall multiply the rest of the terms by $e^{-ik_m x}$ and integrate over a period i.e. from $x = 0$ to $x = a$. Then using (4.84), we get

$$b_0(k_0^2 - k^2) \int_0^a e^{2\pi i m x/a} dx + \gamma \sum_{n \neq 0} [(k_0^2 - k_n^2) b_n + b_0 C_n] \int_0^a e^{2\pi i (m-n)x/a} dx = 0. \quad (4.85)$$

Let us consider the integrals in (4.85) for $m = 0$ and $m \neq 0$. If $m = 0$, the first integral is equal to a and the second integral becomes zero for all values of n in the summation. Then, we are left with

$$b_0(k_0^2 - k^2)a = 0 \text{ or } k = k_0. \quad (4.86)$$

On the other hand, if $m \neq 0$, the first integral vanishes and the second integral is zero except when $n = m$. In this case, we get

$$\gamma[(k_0^2 - k_n^2)b_m + b_0C_m]a = 0 \text{ or } b_m = \frac{b_0C_m}{(k_m^2 - k_0^2)} = \frac{b_0C_m}{(k_m^2 - k^2)}. \quad (4.87)$$

From (4.73) and (4.86), we get

$$E = \frac{\hbar^2}{2m}k_0^2 = \frac{\hbar^2}{2m}k^2. \quad (4.88)$$

From (4.88), we see that the first order correction to the free electron energy is zero; the E - k relationship for the electron in potential described in (4.71) is the same as for a free electron. To get the wave function, we should substitute (4.87) in (4.81). Then, we get

$$\psi(x) = b_0 e^{ikx} \left[1 + \gamma \sum_{n \neq 0} \frac{C_n}{k_n^2 - k^2} e^{-2\pi i n x / a} \right]. \quad (4.89)$$

To get the second order correction to the energy, we retain the γ^2 -term in (4.83). Then, multiplying throughout by e^{-ikx} and integrating from $x = 0$ to $x = a$, we get

$$\begin{aligned} b_0(k_0^2 - k^2)a + \gamma \sum_{n \neq 0} [(k_0^2 - k_n^2)b_n + b_0C_n] \int_0^a e^{-2\pi i n x / a} dx \\ + \gamma^2 \sum_{n \neq 0} \sum_{n' \neq 0} b_{n'} C_n \int_0^a e^{-2\pi i (n+n')x/a} dx = 0. \end{aligned} \quad (4.90)$$

The first integral in (4.90) is equal to zero for all values of n and the second integral is zero for all values of n except for $n = -n'$. Thus (4.90) becomes

$$b_0(k_0^2 - k^2)a + \gamma^2 \sum_{n' \neq 0} b_{n'} C_{-n'} a = 0. \quad (4.91)$$

In (4.74), we can substitute $-n$ for n without any change. Then

$$f(x) = \sum_{n=-\infty}^{\infty} C_{-n} e^{2\pi i n x / a} \quad (4.92)$$

We shall take the complex conjugate of both sides and noting that, since $f(x)$ is real, $f^*(x) = f(x)$, we may write

$$f(x) = \sum_{n=-\infty}^{\infty} C_n^* e^{2\pi i n x/a}. \quad (4.93)$$

Using properties of Fourier coefficients, we get from (4.74) and (4.93),

$$C_{-n} = C_n^* \quad (4.94)$$

Now, using (4.87) and (4.94), we can write (4.91) as

$$k_0^2 = k^2 + \gamma^2 \sum_{n \neq 0} \frac{C_n^* C_n}{k^2 - (k - \frac{2\pi n}{a})^2}. \quad (4.95)$$

Substituting (4.73) and (4.77) in (4.95), we finally get

$$E = \frac{\hbar^2 k^2}{2m} + \sum_{n \neq 0} \frac{|V_n|^2}{(\frac{\hbar^2 k^2}{2m}) - \frac{\hbar^2}{2m} (k - \frac{2\pi n}{a})^2}. \quad (4.96)$$

This result is satisfactory when k^2 is not close to any of the k_n^2 values. For $k^2 \approx k_n^2$, the value of E becomes too large. In this region

$$k = \pm k_n = \pm k \mp \frac{2\pi n}{a}. \quad (4.97)$$

For the upper sign, $n = 0$ which is irrelevant. For the lower sign, we get

$$k = \frac{n\pi}{a}. \quad (4.98)$$

As mentioned, at these points, E assumes very large values. To get acceptable results, we will have to start with a different wave function. One such function is

$$\psi(x) = b_0 e^{ikx} + \gamma b_n e^{ik_n x/a}. \quad (4.99)$$

Repeating the procedure followed earlier for (4.81), we will get

$$E_k = \frac{\hbar^2}{4m} \left\{ k^2 + (k - \frac{2\pi n}{a})^2 \pm \sqrt{[k^2 - (k - \frac{2\pi n}{a})^2]^2 + (\frac{4m}{\hbar^2} |V_n|)^2} \right\}. \quad (4.100)$$

At the band edge $k = k_n = n\pi/a$. At these points, (4.100) becomes

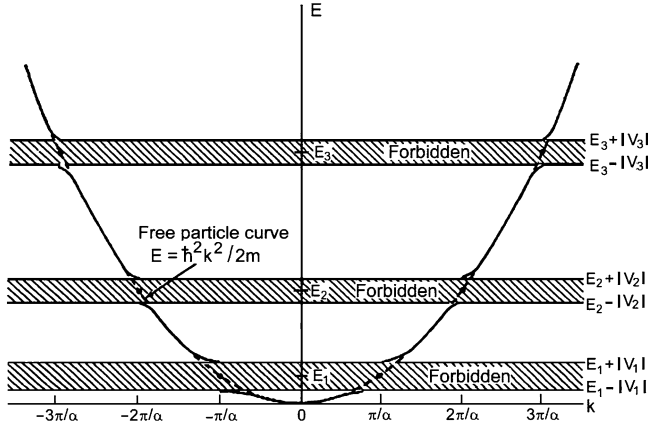


Fig. 4.20 The E - k relation for a nearly free electron

$$E = E_n \pm |V_n| \quad (4.101)$$

where

$$E_n = \left(\frac{\hbar^2}{2m} \right) \left(\frac{n\pi}{a} \right)^2. \quad (4.102)$$

The relation between E and k is shown in Fig. 4.20. There are discontinuities or energy gaps at $k = \pm n\pi/a$. The band gaps are of width $2|V_n|$ where $|V_n|$ is the n th Fourier coefficient in the Fourier expansion of the periodic potential. For $k > (n\pi/a)$, the values of E will be those given by (4.101); they are close to the free electron values given by (4.88). As mentioned earlier, the allowed bands shown in Fig. 4.20 are broad and the forbidden ranges are narrow.

We can get an expression for the effective mass m^* by differentiating (4.100). But the calculation can be simplified by introducing

$$k = \frac{n\pi}{a} + k' \quad (4.103)$$

where $k' \ll n\pi/a$. Then (4.100) approximates to

$$E_k = E_n + \frac{1}{2} \Delta E + \frac{\hbar^2}{2m} k'^2 \left(1 + \frac{4E_n}{\Delta E} \right) \quad (4.104)$$

with

$$\Delta E = 2|V_n|. \quad (4.105)$$

From (4.104), using $m^* = \hbar^2 / (d^2 E_k / dk^2) = \hbar^2 / (d^2 E_k / dk'^2)$, we get

$$m^* = m / \left(1 + \frac{4E_n}{\Delta E} \right). \quad (4.106)$$

4.6.3 The Tight Binding Model

The very first model in band theory was proposed by Bloch [7]; it is since known as the tight-binding model. In this model a major part of the total electron energy is derived from the periodic potential. It is assumed that the atoms are so separated that there is very little overlapping of the wave functions of the neighbouring atoms. Thus mutual interaction between neighbouring electrons is weak. As a result the wave functions and energy values of the crystal will be close to the wave functions and energy levels of the individual atoms. Hence, in contrast to what happens in the NFE model, the allowed bands are narrow and forbidden bands broad in the tight-binding model.

Consider an atom with an associated potential $V_0(\mathbf{r})$. The Schrödinger equation for the atom is

$$\mathcal{H}_0 \psi(\mathbf{r}) = \nabla^2 \psi(\mathbf{r}) + \frac{2m}{\hbar^2} [E - V_0(\mathbf{r})] \psi(\mathbf{r}) = 0. \quad (4.107)$$

Let $\psi_0(\mathbf{r})$ be the ground state wave function and E_0 the corresponding energy. Let us assume a periodic lattice such that the potential in a region around each atom is unaffected by the presence of other atoms. Then the wave function $\psi(\mathbf{r})$ for the crystal may be written as a linear combination of the ground state wave functions of atoms. Thus

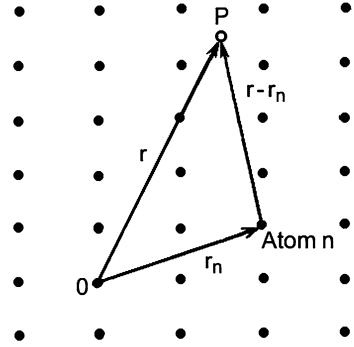
$$\psi(\mathbf{r}) = \sum_n a_n \psi_0(\mathbf{r} - \mathbf{r}_n). \quad (4.108)$$

Here $\mathbf{r}$ and $\mathbf{r}_0$ are vectors defined in Fig. 4.21. Since all atoms are equivalent, the a_n 's must have the same absolute value. We may then replace a_n by $a e^{i\phi_n}$ where ϕ_n is a phase factor. Further, $\phi_n = \mathbf{k} \cdot \mathbf{r}_n$ where $\mathbf{k}$ is a wave vector. Since ψ_0 are normalized functions, $\int \psi_0 \psi_0^* d\tau = 1$ and we should take $a = 1$. Then (4.108) takes the form

$$\psi_{\mathbf{k}}(\mathbf{r}) = \sum_n e^{i\mathbf{k} \cdot \mathbf{r}_n} \psi_0(\mathbf{r} - \mathbf{r}_n). \quad (4.109)$$

The wave functions (4.109) should satisfy the Schrödinger equation

Fig. 4.21 Vector geometry in the tight-binding model



$$\mathcal{H}\psi_{\mathbf{k}} = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r})\right]\psi_{\mathbf{k}} = E\psi_{\mathbf{k}}. \quad (4.110)$$

The potential experienced by the electron in the crystal is shown in Fig. 4.22. Two potentials are operative: (i) the potential $V_0(\mathbf{r} - \mathbf{r}_n)$ due to the ion-core at $\mathbf{r}_n$ and (ii) the potential $[V(\mathbf{r}) - V_0(\mathbf{r} - \mathbf{r}_n)]$ which is the potential due to all atoms in the lattice except the one at $\mathbf{r}_n$. The total Hamiltonian $\mathcal{H}$ is made up of two parts $\mathcal{H}_0$ and $\mathcal{H}'$ such that

$$\mathcal{H} = \mathcal{H}_0 + \mathcal{H}', \quad (4.111)$$

where

$$\mathcal{H}_0 = -\frac{\hbar^2}{2m}\nabla^2 + V_0(\mathbf{r} - \mathbf{r}_n) \quad (4.112)$$

and

$$\mathcal{H}' = V(\mathbf{r}) - V_0(\mathbf{r} - \mathbf{r}_n). \quad (4.113)$$

Here, $\mathcal{H}'$ is in the form of a perturbation.

From (4.107), we get

$$\mathcal{H}_0\psi_0 = E_0\psi_0. \quad (4.114)$$

Substituting the wave function (4.109) in (4.114), we get

$$\mathcal{H}_0\psi_{\mathbf{k}} = \sum_n e^{i\mathbf{k}\cdot\mathbf{r}_n} \mathcal{H}_0\psi_0(\mathbf{r} - \mathbf{r}_n) = E_0 \sum_n e^{i\mathbf{k}\cdot\mathbf{r}_n} \psi_0(\mathbf{r} - \mathbf{r}_n) = E_0\psi_{\mathbf{k}} \quad (4.115)$$

The energy E can be obtained as the expectation value of the total Hamiltonian $\mathcal{H}$. Thus

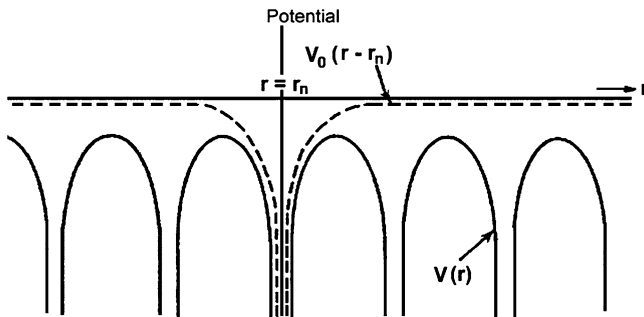


Fig. 4.22 Potentials in the tight-binding model

$$\begin{aligned}
 E &= \frac{\int_{\tau} \psi_{\mathbf{k}}^* (\mathcal{H}_0 + \mathcal{H}') \psi_{\mathbf{k}} d\tau}{\int_{\tau} \psi_{\mathbf{k}}^* \psi_{\mathbf{k}} d\tau} \\
 &= E_0 + \frac{\int_{\tau} \psi_{\mathbf{k}}^* \sum_n e^{i\mathbf{k} \cdot \mathbf{r}_n} [V(\mathbf{r}) - V_0(\mathbf{r} - \mathbf{r}_n)] \psi_0(\mathbf{r} - \mathbf{r}_n) d\tau}{\int_{\tau} \psi_{\mathbf{k}}^* \psi_{\mathbf{k}} d\tau} .
 \end{aligned} \tag{4.116}$$

Let us note that ψ_0 is a normalized wave function. Hence from (4.109), we get $\int_{\tau} \psi_{\mathbf{k}}^* \psi_{\mathbf{k}} d\tau = N$, the number of atoms in the crystal. Equation (4.116) now becomes

$$E = E_0 + \frac{1}{N} \sum_n \sum_m e^{i\mathbf{k} \cdot (\mathbf{r}_n - \mathbf{r}_m)} \int_{\tau} \psi_0^*(\mathbf{r} - \mathbf{r}_m) [V(\mathbf{r}) - V_0(\mathbf{r} - \mathbf{r}_n)] \psi_0(\mathbf{r} - \mathbf{r}_n) d\tau . \tag{4.117}$$

We shall consider the effect of only the nearest neighbours and neglect the rest of the atoms. Further, we shall assume ψ_0 to be spherically symmetric (like the s -state); this makes the contribution of all the nearest neighbours the same. For $m = 0$, the integral in (4.117) becomes

$$\int_{\tau} \psi_0^*(\mathbf{r}) [V(\mathbf{r}) - V_0(\mathbf{r})] \psi_0(\mathbf{r}) d\tau = -\alpha \tag{4.118}$$

and for the nearest neighbour atoms,

$$\int_{\tau} \psi_0^*(\mathbf{r} - \mathbf{r}_m) [V(\mathbf{r}) - V_0(\mathbf{r})] \psi_0(\mathbf{r}) d\tau = -\beta. \tag{4.119}$$

Here, $\mathbf{r}_m$ is the vector connecting the atom at the origin to a nearest neighbour atom. The actual evaluation of these integrals is complicated. We shall not go into it but simply take them as $-\alpha$ and $-\beta$.

Equation (4.117) may now be written as

$$E_k = E_0 - \alpha - \beta \sum_m e^{-i\mathbf{k} \cdot \mathbf{r}_m}. \quad (4.120)$$

The summation is over the nearest neighbours.

In order to understand the significance of (4.120), let us consider the simple cubic lattice. The components of the $\mathbf{r}_m$ vectors for this lattice are

$$\mathbf{r}_m = (\pm a, 0, 0), (0, \pm a, 0), (0, 0, \pm a) \quad (4.121)$$

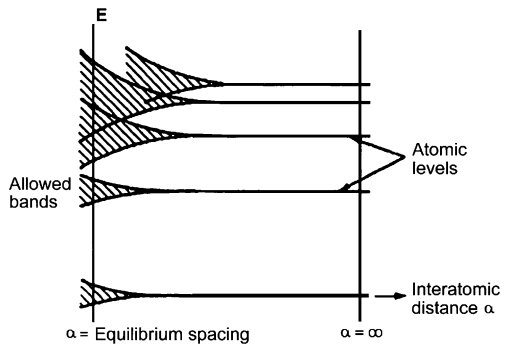
where a is the lattice constant. Substituting these values in (4.120), we get

$$E_k = E_0 - \alpha - 2\beta(\cos k_x a + \cos k_y a + \cos k_z a) \quad (4.122)$$

where k_x, k_y, k_z are components of $\mathbf{k}$. Thus, $E_{\mathbf{k}}$ takes various values for different values of k_x, k_y, k_z . This range of values determines the width of an allowed band. The minimum value of $E_{\mathbf{k}}$ occurs at $k_x = k_y = k_z = 0$. On the other hand, the maximum occurs at the corners of the cube where the k -components are $(\pm \frac{\pi}{a}, \pm \frac{\pi}{a}, \pm \frac{\pi}{a})$. At these points, the cosine terms in (4.122) take values -1 . The width of the allowed band, which is the difference between maximum and minimum values, is 12β .

Finally, let us consider how the width of a band depends on the degree of overlapping for which we shall take the lattice constant as a measure. For a large lattice constant, the nearest neighbours are far apart, the overlap between $\psi_0(\mathbf{r})$ and $\psi_0(\mathbf{r} - \mathbf{r}_n)$ is small and β becomes small; in the limit of a very large lattice constant, the band width narrows down into a single level. The dependence of band-width on the lattice constant is shown in Fig. 4.23.

Fig. 4.23 Band-width as a function of interatomic distance



4.6.4 Other Methods

In the preceding sections, we have discussed the basic models and methods of determination of band structures. However, in actual determination of band structure, various other methods are in vogue. Some of them are: the augmented plane wave method, the orthogonalised plane wave method, the pseudo-potential method, the Green's function method and the so-called K.P. method. These methods differ in (i) the potential assumed, (ii) the mathematical details and (iii) the sophistication in computation. Harrison [13] and Omar [14] have discussed these methods.

4.7 Concepts and Ideas in the Band Theory

The band theory of solids is a total departure from the earlier free electron theory. It is based on some new concepts and ideas and employs the techniques of wave mechanics. With these new concepts and new techniques, the band theory achieved success where the free electron theory failed.

The first new idea introduced in the band theory is that in a crystal the electron finds itself in a periodic potential. This was a departure from the constant potential assumed in the free electron theory. A Schrödinger equation with a periodic potential needed new functions as solutions. These solutions are the Bloch functions $e^{ik \cdot r} u_k(\mathbf{r})$; these are plane waves ($e^{ik \cdot r}$) modulated by a periodic function $u_k(\mathbf{r})$.

Solution of the Schrödinger equation for the electron in the periodic potential leads to a most unexpected result. The E - k curve is not continuous but has discontinuities at some values of the propagation vector k . Thus the energy level diagram consists of 'allowed' and 'prohibited' bands. This band formation is a result of the interaction between atoms (or the overlapping of the wave functions). It follows that greater the overlapping, greater is the band-width. Similarly, the lesser the overlapping, the less will be the band-width; in the limit of very weak interaction, the bands narrow down into single discrete atomic energy levels.

Electrons in full bands do not participate in conduction. The process of conduction depends on the electrons in the two uppermost bands. The uppermost completely filled band is called the valence band and the band above it, is called the conduction band; the latter could be empty or partially filled. The energy difference between the bottom of the conduction band and the top of the valence band is called the band gap or the forbidden gap E_g . It plays an important role in the behaviour of solids. The free electron theory totally failed to justify the existence of different types of solids like insulators and conductors. The band theory has an answer. Solids with large E_g do not conduct or they are insulators. Solids with small E_g do not conduct at $T = 0$ K, but at $T > 0$ K, electrons from the valence band jump into the conduction band and then these solids can conduct. These are new materials called semiconductors. Conduction is easy if the conduction band is partially filled

or if it overlaps with the valence band. Solids with such conduction bands conduct with great ease; they are the metals.

What is the origin of the discontinuities in the energy spectrum? The discontinuities occur close to the boundaries of the Brillouin zones. These are determined by specific values of the propagation vector. It is seen that at the Brillouin zone boundaries, the electron waves cease to propagate into the lattice; instead, they undergo Bragg diffraction and are deflected away; hence, the energy gaps.

There is a great difference between the dynamics of a free electron and that of an electron in a band. The motion of an electron in a band is k -dependent. The acceleration of a free electron in an electric field F is $(-eF/m)$. On the other hand, the acceleration of an electron in a band under the same field is $-(eF/\hbar^2)(d^2E/dk^2)$. Thus the quantity $\hbar^2/(d^2E/dk^2)$ plays the role of an effective mass m^* . This is an entirely new concept. m^* is k -dependent. It is negative in the upper half of a band and positive in the lower half. This behaviour of the electron has important consequences.

Yet another new concept is the concept of a ‘hole’. If an electron from the top of the valence band escapes, it leaves behind a vacancy. This vacancy, called a ‘hole’, behaves like a positively charged particle. Further, it has a positive effective mass. This combination of a positive charge and a positive effective mass, results in the Hall coefficient being positive. The positive Hall coefficient observed for some metals could not be explained by the free electron theory. The band theory accounts for it in terms of ‘holes’.

4.8 Problems

1. In the Kronig–Penney model, the assumption $V_0 > E$ leads to the equation

$$P \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka$$

What will be the form of this equation for $V_0 < E$?

2. Starting with the E - k relation in the Kronig–Penney model, show that the slope of the curve is zero at the band edges.
3. If $\psi(x)$ is a Bloch function for a linear lattice, show that the probability density is the same in every unit cell.
4. Construct the Wigner–Seitz cell for the fcc lattice.
5. Obtain an expression for the effective mass m^* in the tight-binding model assuming that the electron is moving in the x -direction (i.e. $k_y = k_z = 0$) and that $k_x \ll \pi/a$.

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Chapter 5

Band Theory of Solids II: Detailed Treatment of Select Topics

Abstract The main concepts of the band theory were introduced earlier on the basis of a hypothetical one-dimensional lattice. In this chapter, we elaborate these concepts by considering real three-dimensional lattices. The Brillouin zones for 2-d and 3-d lattices are worked out. The constant energy surfaces are derived for 2-d and 3-d lattices. Fermi surfaces of some real crystals are discussed. The aggregate effective mass introduced in the Kronig-Penney model is shown to become a second rank symmetric tensor in a 3-d crystal. Pioneering experiments are described which enable the determination of the band structure parameters. The techniques of soft X-ray emission, cyclotron resonance, anomalous skin effect, magnetoresistance and de Haas-van Alphen effect are described in detail. Experimentally and theoretically determined band structures of several real crystals are discussed.

5.1 Introduction

The basic framework of the band theory of solids was discussed in the preceding chapter. Important concepts were introduced and applied to simple lattices. In this chapter we present a more detailed treatment of some select topics. A fairly good discussion of these topics is to be found in McKelvey [1], Mackinnon [2], Ashcroft and Mermin [3] and Singleton [4].

5.2 Brillouin Zones

The concept of zones was introduced by Brillouin [5] and was elaborated by him in his book [6]. These zones (since called Brillouin zones) have found important applications in lattice dynamics and band theory of solids. The beauty of a Brillouin zone is that it compresses in itself information about the entire crystal.

5.2.1 Brillouin Zones of a One-Dimensional Lattice

We introduced the concept of the Brillouin zone in the preceding chapter in an empirical manner. While dealing with the behaviour of an electron in a one-dimensional periodic potential, we found that the energy of the electron, which is a function of the propagation vector k , has discontinuities at $k = \pm n\pi/a$, where n is an integer and ' a ' the lattice constant. We call these points the boundaries of the Brillouin zone. The Brillouin zones of a one-dimensional lattice are shown in Fig. 5.1. Since $k = 2\pi/\lambda$ (λ being the de Broglie wavelength of an electron), we get $n\lambda = 2a$ which is Bragg's law for normal incidence. Thus a connection is established between Bragg reflection and boundaries of Brillouin zones; in fact, this relation can be the basis for the definition of a Brillouin zone. The Brillouin zone of a crystal lattice is a polyhedron in k -space. The electron waves undergo Bragg reflection at the boundaries (faces) of a Brillouin zone and hence, they cannot propagate across the boundaries.

Obviously, the boundaries of Brillouin zones in two-dimensional and three-dimensional lattices are lines and planes respectively.

5.2.2 Brillouin Zones of a Two-Dimensional Lattice

We shall now consider Brillouin zones in a square lattice. In this discussion, we shall assume the results of reciprocal lattice theory [7, 8]. Accordingly, the reciprocal vectors can be written as $(1/a)(n_1\mathbf{i} + n_2\mathbf{j})$ where ' a ' is the lattice constant and n_1 and n_2 are integers.

We shall express Bragg's law as

$$2\mathbf{k} \cdot \mathbf{G} + G^2 = 0 \quad (5.1)$$

Here $\mathbf{k}$ is the propagation vector and $\mathbf{G}$ is 2π times a reciprocal vector. In two-dimensional space, $\mathbf{k}$ and $\mathbf{G}$ may be expressed as

$$\mathbf{k} = k_x\mathbf{i} + k_y\mathbf{j} \quad (5.2)$$

$$\mathbf{G} = G_x\mathbf{i} + G_y\mathbf{j} \quad (5.3)$$

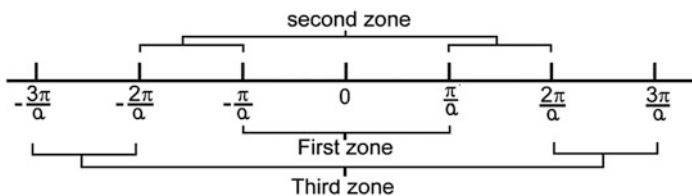


Fig. 5.1 Brillouin zones of a linear monatomic lattice with lattice constant a

Expanding (5.1) and substituting (5.2), (5.3), we get

$$2(k_x G_x + k_y G_y) + G_x^2 + G_y^2 = 0 \quad (5.4)$$

By definition $\mathbf{G}$ is of the form

$$\mathbf{G} = \frac{2\pi}{a}(n_1 \mathbf{i} + n_2 \mathbf{j}). \quad (5.5)$$

Comparing (5.3) and (5.5), we have

$$G_x = \frac{2\pi}{a}n_1, \quad G_y = \frac{2\pi}{a}n_2. \quad (5.6)$$

Substituting in (5.4), we get

$$(n_1 k_x + n_2 k_y) = -\frac{\pi}{a}(n_1^2 + n_2^2). \quad (5.7)$$

Equation (5.7) helps us in constructing Brillouin zones for the square lattice. Thus, if we make $n_1 = \pm 1$ and $n_2 = 0$, we get

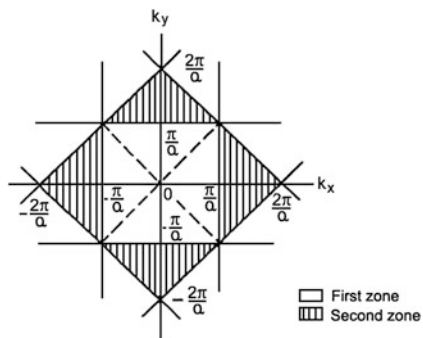
$$k_x = \pm \pi/a. \quad (5.8)$$

These are lines parallel to the k_y axis at $k_x = \pm \pi/a$. Again, if we put $n_1 = 0$ and $n_2 = \pm 1$, we have

$$k_y = \pm \pi/a. \quad (5.9)$$

These are lines parallel to the k_x axis at $k_y = \pm \pi/a$. Equations (5.8) and (5.9) define the boundaries of the first Brillouin zone; these are shown in Fig. 5.2. The first Brillouin zone of the square lattice is a square of side $2\pi/a$. Note that we have already assumed that Bragg's law holds.

Fig. 5.2 First and second Brillouin zones of a square lattice with lattice constant a



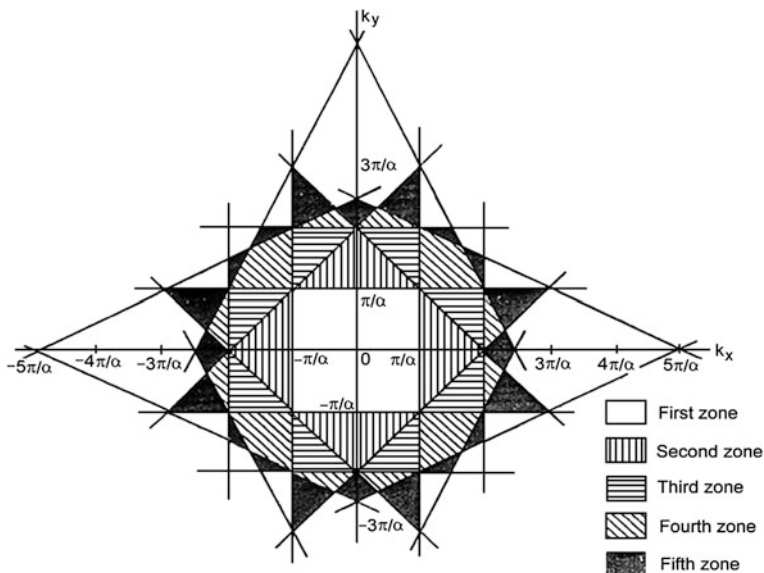


Fig. 5.3 First five zones of a square lattice

Similarly if we let $n_1 = \pm 1$, $n_2 = \pm 1$, the different combinations of n_1 and n_2 define straight lines with equations

$$\pm k_x \pm k_y = 2\pi/a. \quad (5.10)$$

These lines form the boundaries of the second Brillouin zone which is also shown in Fig. 5.2. Following this procedure, higher order Brillouin zones can be constructed. Figure 5.3 shows the first five Brillouin zones of the square lattice.

At this stage, let us note an important principle: each zone boundary is normal to the shortest G -vector at its mid-point.

5.2.3 Brillouin Zones of Three-Dimensional Lattices

Let us now consider how the Brillouin zones of three-dimensional lattices can be constructed. Before proceeding further, we note that the higher zones contain the same information as the first zone because of the periodicity of the lattice. This can be seen by subtracting from the wave vector $\mathbf{k}$ a reciprocal vector $\mathbf{G}$ to yield another wave vector $\mathbf{k}' = \mathbf{k} - \mathbf{G}$. Then, we have

$$\begin{aligned}
\psi_{\mathbf{k}}(\mathbf{r}) &= e^{i\mathbf{k}\cdot\mathbf{r}} u_{\mathbf{k}}(\mathbf{r}) \\
&= e^{i(\mathbf{k}-\mathbf{G})\cdot\mathbf{r}} u_{\mathbf{k}}(\mathbf{r}) e^{i\mathbf{G}\cdot\mathbf{r}} \\
&= e^{i\mathbf{k}'\cdot\mathbf{r}} u_{\mathbf{k}'}(\mathbf{r}) = \psi_{\mathbf{k}'}(\mathbf{r}).
\end{aligned} \tag{5.11}$$

In deriving (5.11) we have made use of the fact that $u_{\mathbf{k}}$ and $e^{i\mathbf{G}\cdot\mathbf{r}}$ both have the periodicity of the lattice and, hence, can be combined into a single function $u_{\mathbf{k}'}$ having the same period [9]. Hence, it is sufficient to know the shape of the first Brillouin zone.

5.2.3.1 The bcc Lattice

Let us consider the bcc lattice. Its primitive vectors (shown in Fig. 5.4a) are

$$\begin{aligned}
\mathbf{a} &= \frac{a}{2}(\mathbf{i} + \mathbf{j} + \mathbf{k}) \\
\mathbf{b} &= \frac{a}{2}(-\mathbf{i} + \mathbf{j} + \mathbf{k}) \\
\mathbf{c} &= \frac{a}{2}(-\mathbf{i} - \mathbf{j} + \mathbf{k})
\end{aligned} \tag{5.12}$$

The primitive vectors of the corresponding reciprocal lattice are

$$\begin{aligned}
\mathbf{a}^* &= \frac{1}{a}(\mathbf{i} + \mathbf{k}) \\
\mathbf{b}^* &= \frac{1}{a}(-\mathbf{i} + \mathbf{j}) \\
\mathbf{c}^* &= \frac{1}{a}(-\mathbf{j} + \mathbf{k})
\end{aligned} \tag{5.13}$$

If h, k, l are integers, a vector $\mathbf{G}$ in the reciprocal lattice is given by

$$\mathbf{G} = 2\pi(h\mathbf{a}^* + k\mathbf{b}^* + l\mathbf{c}^*). \tag{5.14}$$

Substituting (5.13) in (5.14), we get

$$\mathbf{G} = \frac{2\pi}{a}[(h-k)\mathbf{i} + (k-l)\mathbf{j} + (h+l)\mathbf{k}]. \tag{5.15}$$

From (5.15), we see that the shortest $\mathbf{G}$ vectors (12 in number) are

$$\mathbf{G} = \frac{2\pi}{a}(\pm\mathbf{i} \pm \mathbf{j}); \quad \frac{2\pi}{a}(\pm\mathbf{j} \pm \mathbf{k}); \quad \frac{2\pi}{a}(\pm\mathbf{i} \pm \mathbf{k}). \tag{5.16}$$

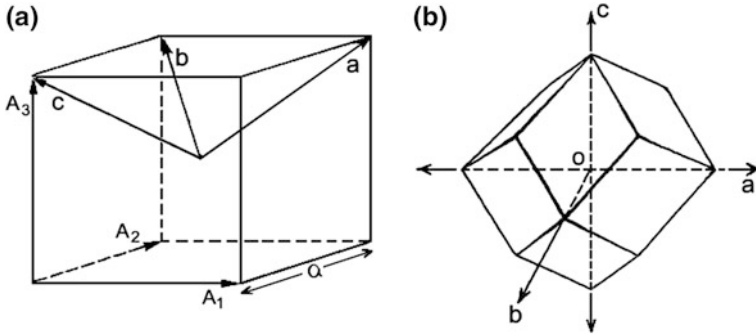


Fig. 5.4 **a** Primitive vectors of bcc lattice, **b** first Brillouin zone of bcc lattice

The first Brillouin zone is formed by drawing planes normal to these vectors at their mid-points. The resulting Brillouin zone is shown in Fig. 5.4b. It is a rhombic dodecahedron. The volume of the Brillouin zone is $2(2\pi/a)^3$.

5.2.3.2 The fcc Lattice

The primitive vectors of the fcc lattice (shown in Fig. 5.5a) are

$$\begin{aligned}\mathbf{a} &= \frac{a}{2}(\mathbf{i} + \mathbf{j}) \\ \mathbf{b} &= \frac{a}{2}(\mathbf{i} + \mathbf{k}) \\ \mathbf{c} &= \frac{a}{2}(\mathbf{j} + \mathbf{k})\end{aligned}\tag{5.17}$$

The primitive vectors of the corresponding reciprocal lattice are

$$\begin{aligned}\mathbf{a}^* &= \frac{1}{a}(-\mathbf{i} - \mathbf{j} + \mathbf{k}) \\ \mathbf{b}^* &= \frac{1}{a}(-\mathbf{i} + \mathbf{j} - \mathbf{k}) \\ \mathbf{c}^* &= \frac{1}{a}(\mathbf{i} - \mathbf{j} - \mathbf{k})\end{aligned}\tag{5.18}$$

A $\mathbf{G}$ vector in reciprocal space may be written as

$$\mathbf{G} = \frac{2\pi}{a}[(-h - k + l)\mathbf{i} + (-h + k - l)\mathbf{j} + (h - k - l)\mathbf{k}].\tag{5.19}$$

The $\mathbf{G}$ vectors required to construct the Brillouin zone are

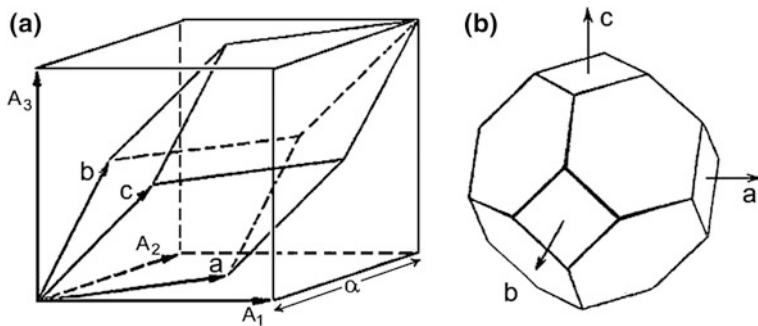


Fig. 5.5 **a** Primitive vectors of fcc lattice, **b** first Brillouin zone of fcc lattice

$$\mathbf{G} = \frac{2\pi}{a}(\pm\mathbf{i} \pm \mathbf{j} \pm \mathbf{k}); \quad \frac{2\pi}{a}(\pm 2\mathbf{i}); \quad \frac{2\pi}{a}(\pm 2\mathbf{j}); \quad \frac{2\pi}{a}(\pm 2\mathbf{k}). \quad (5.20)$$

The first Brillouin zone is constructed by drawing planes normal to these vectors at their mid-points. The resulting Brillouin zone (Fig. 5.5b) is a truncated octahedron with a combination of square and hexagonal faces. The volume of this Brillouin zone is $4(2\pi/a)^3$.

5.2.3.3 The hcp Lattice

The hcp lattice is shown in Fig. 5.6a. Its primitive vectors are

$$\begin{aligned} \mathbf{a} &= \left(\frac{a}{2}\right)\mathbf{i} + \left(\frac{3^{1/2}a}{2}\right)\mathbf{j} \\ \mathbf{b} &= \left(\frac{-a}{2}\right)\mathbf{i} + \left(\frac{3^{1/2}a}{2}\right)\mathbf{j} \\ \mathbf{c} &= c\mathbf{k} \end{aligned} \quad (5.21)$$

The corresponding reciprocal vectors are

$$\begin{aligned} \mathbf{a}^* &= \left(\frac{1}{a}\right)\mathbf{i} + \left(\frac{1}{3^{1/2}a}\right)\mathbf{j} \\ \mathbf{b}^* &= -\left(\frac{1}{a}\right)\mathbf{i} + \left(\frac{1}{3^{1/2}a}\right)\mathbf{j} \\ \mathbf{c}^* &= \left(\frac{1}{c}\right)\mathbf{k} \end{aligned} \quad (5.22)$$

The $\mathbf{G}$ vector in reciprocal lattice is

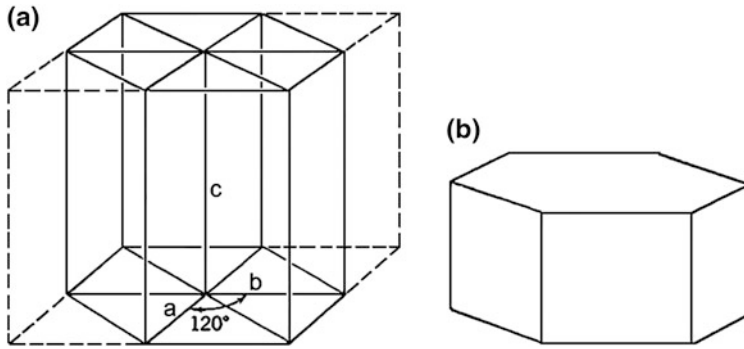


Fig. 5.6 **a** Primitive vectors of hcp lattice, **b** first Brillouin zone of hcp lattice

$$\mathbf{G} = 2\pi \left[\left(\frac{1}{a} \right) (h - k) \mathbf{i} + \left(\frac{1}{3^{1/2}a} \right) (h + k) \mathbf{j} + \left(\frac{1}{c} \right) l \mathbf{k} \right]. \quad (5.23)$$

The shortest $\mathbf{G}$ vectors are

$$\mathbf{G} = \frac{2\pi}{a} \left[\pm \mathbf{i} \pm \left(\frac{1}{3^{1/2}} \right) \mathbf{j} \right]; \quad \frac{2\pi}{a} \left[\pm \left(\frac{2}{3^{1/2}} \right) \mathbf{j} \right]; \quad \frac{2\pi}{c} (\pm \mathbf{k}). \quad (5.24)$$

The Brillouin zone is formed by drawing planes normal to these vectors at their mid-points. The resulting Brillouin zone is shown in Fig. 5.6b. It can be seen that it is a hexagonal prism much like the hcp cell.

5.3 The Fermi Surface

The term ‘Fermi surface’ is not to be found in the earliest edition of Kittel’s classic [9]. On the other hand, in recent editions [7], there is a whole chapter dedicated to the Fermi surface. In fact in the latter edition Kittel quotes Mackintosh’s suggestion that a metal could be defined as ‘a solid with a Fermi surface’. Thus the importance of the Fermi surface has been realized as solid state physics progressed.

We introduced the Fermi level in Chap. 3 as an energy level of the electron below which all energy levels are occupied and above which all levels are empty. At that stage we were dealing with free electrons for which the constant energy surface for the Fermi energy was spherical; we called it the Fermi surface.

In reality, however, electrons in solids experience a periodic potential due to which the shape of the Fermi surface is far from spherical. Yet, the significance of the Fermi surface remains the same viz., all energy levels inside the Fermi surface are filled while those outside are empty. The importance of the Fermi surface lies in the fact that only those electrons which are close to the Fermi surface can possibly

escape from it and participate in transport phenomena. Knowledge of the Fermi surface is useful in fitting parameters of the crystal potential which, in turn, enables the determination of the band structure. The Fermi surface has the same rotational symmetry as the lattice [10].

Any model of band theory will yield an expression for the energy of the electron $E_{\mathbf{k}}$ as a function of the propagation vector $\mathbf{k}$. From this, constant energy surfaces can be constructed; the constant energy surface for the Fermi energy E_F is the Fermi surface. For making sample calculations, we shall adopt the tight-binding model. From Chap. 4, the $E_{\mathbf{k}}$ relation is

$$E_{\mathbf{k}} = E_0 - \alpha - \beta \sum_m e^{-i\mathbf{k} \cdot \mathbf{r}_m}. \quad (5.25)$$

Here E_0 is the ground state energy of an atom, α and β are integrals defined in Chap. 4 and $\mathbf{r}_m$ is the vector joining the atom at the origin of the lattice to its nearest neighbours. We shall apply this equation to construct constant energy surfaces for some simple lattices.

5.3.1 Square Lattice

The square lattice with lattice constant ‘ a ’ is shown in Fig. 5.7. The components of the $\mathbf{r}_m$ vectors for the nearest neighbours are

$$\mathbf{r}_m = (\pm a, 0), (0, \pm a). \quad (5.26)$$

Substituting in (5.25), we get

$$E_{\mathbf{k}} = E_0 - \alpha - 2\beta(\cos k_x a + \cos k_y a). \quad (5.27)$$

We have seen in Sect. 5.2 that the first Brillouin zone of this lattice has boundaries at $k_x = \pm\pi/a$ and $k_y = \pm\pi/a$. We need to explore the k -space within these limits.

For very small values of k_x, k_y (5.27) can be approximated to

$$E_{\mathbf{k}} = E_0 - \alpha - 2\beta \left[2 - \frac{(k_x a)^2}{2} - \frac{(k_y a)^2}{2} \right]. \quad (5.28)$$

From (5.28), the energy at $k_x = k_y = 0$ becomes

$$E_{\mathbf{k}} = E_{\min} = E_0 - \alpha - 4\beta. \quad (5.29)$$

This is the minimum energy which we shall denote as $E_{\min}$. We may reorganize (5.28) as

Fig. 5.7 Square lattice with lattice constant a ; $\mathbf{r}_m$ vectors are shown

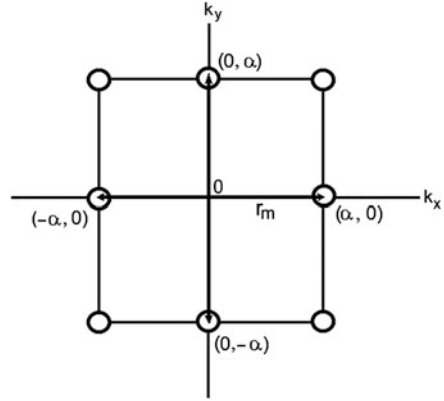
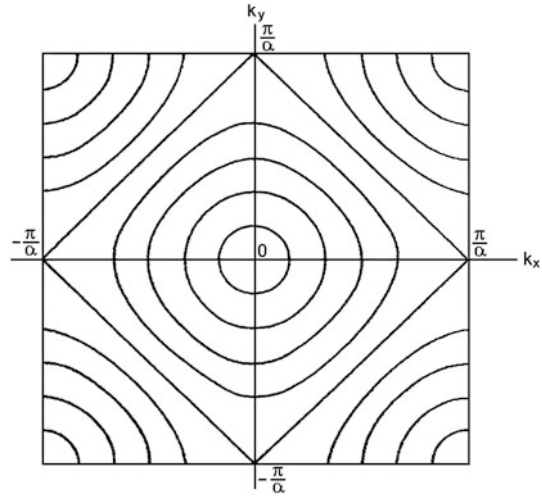


Fig. 5.8 Constant energy contours in a square lattice



$$k_x^2 + k_y^2 = \frac{E_{\mathbf{k}} - E_{\min}}{\beta a^2} = \text{constant}. \quad (5.30)$$

For a chosen value of $E_{\mathbf{k}}$, the r.h.s becomes a constant. In such a case, (5.30) is the equation to a circle. Hence, we conclude that for small values of k_x, k_y i.e. near the origin, the constant energy contours are circular as shown in Fig. 5.8.

Let us examine the constant energy contour of $E_{\mathbf{k}} = E_{\min} + 4\beta$ i.e. for $E_{\mathbf{k}} = E_0 - \alpha$. Substituting in (5.27), we have

$$\cos k_x a + \cos k_y a = 0. \quad (5.31)$$

This is possible for

$$\pm k_x \pm k_y = \frac{\pi}{a}. \quad (5.32)$$

Here all combinations of signs are permissible. Equation (5.32) represents a set of four lines joining the points of at $(\pi/a, 0)$, $(0, \pi/a)$, $(-\pi/a, 0)$, $(0, -\pi/a)$. These lines are the constant energy contours for $E_{\mathbf{k}} = E_0 - \alpha$. These are also shown in Fig. 5.8.

Let us now consider the corner points for which $k_x = \pm\pi/a$ and $k_y = \pm\pi/a$. Substituting in (5.27), we get

$$\begin{aligned} E_{\mathbf{k}} &= (E_0 - \alpha - 4\beta) + 8\beta \\ &= E_0 - \alpha + 4\beta = E_{\max}. \end{aligned} \quad (5.33)$$

This is the maximum value of $E_{\mathbf{k}}$ which we shall denote by $E_{\max}$. Thus the bandwidth is $E_{\max} - E_{\min} = 8\beta$.

Finally, let us consider the constant energy contours around the corners. For convenience, let us introduce a new vector $\mathbf{k}'$ with components k'_x, k'_y such that $k_x a = \pi - k'_x a$ and $k_y a = \pi - k'_y a$. Then we have

$$\cos k_x a = \cos(\pi - k'_x a) = -\cos k'_x a \approx -1 + \frac{(k'_x a)^2}{2} \quad \text{for small } k'_x a. \quad (5.34)$$

Similarly,

$$\cos k_y a = \cos(\pi - k'_y a) = -\cos k'_y a \approx -1 + \frac{(k'_y a)^2}{2}. \quad (5.35)$$

Substituting in (5.27), we get

$$E_{\mathbf{k}} = E_0 - \alpha - 2\beta \left[-2 + \frac{(k'_x a)^2}{2} + \frac{(k'_y a)^2}{2} \right] = E_0 - \alpha + 4\beta - \beta a^2 (k'^2_x + k'^2_y). \quad (5.36)$$

On reorganization, (5.36) becomes

$$k'^2_x + k'^2_y = \frac{E_{\max} - E_{\mathbf{k}}}{\beta a^2} = \text{constant} \quad (5.37)$$

In (5.37), the rhs is a constant for a chosen $E_{\mathbf{k}}$. Equation (5.37) is an equation for a circle with a corner as centre. Thus for small values of k'_x and k'_y the constant energy contours are circular as shown in Fig. 5.8.

To summarize, for the square lattice, the constant energy contours are circular about the central reference atom and, again, around the corners. For $E_{\mathbf{k}} > E_{\min}$ and $E_{\mathbf{k}} < E_{\max}$, the contours are curved but not exactly circular. In between, for $E_{\mathbf{k}} = E_0 - \alpha$, the constant energy contours are straight lines. These features are shown in Fig. 5.8. The minimum energy $E_{\min} = E_0 - \alpha - 4\beta$ at the origin and the maximum energy $E_{\max} = E_0 - \alpha + 4\beta$ at the corners; the band-width ($E_{\max} - E_{\min}$) is equal to 8β .

5.3.2 Simple Cubic Lattice

This is an extension of the treatment for the square lattice. The components of the $\mathbf{r}_m$ vectors of this lattice are:

$$\mathbf{r}_m = (\pm a, 0, 0); (0, \pm a, 0); (0, 0, \pm a). \quad (5.38)$$

Substitution in (5.25) yields

$$E_{\mathbf{k}} = E_0 - \alpha - 2\beta(\cos k_x a + \cos k_y a + \cos k_z a). \quad (5.39)$$

We have to explore the energy values for k -values within the limits $k_x = \pm\pi/a$, $k_y = \pm\pi/a$ and $k_z = \pm\pi/a$ which are the boundaries of the Brillouin zone of the simple cubic lattice. The k -coordinate system is shown in Fig. 5.9.

At the centre, where $k_x = k_y = k_z$, (5.39) yields

$$E_{\mathbf{k}} = E_0 - \alpha - 6\beta = E_{\min}. \quad (5.40)$$

This is the minimum value of $E_{\mathbf{k}}$; we shall denote it by $E_{\min}$.

At very small values of k_x , k_y , k_z , we have

$$(\cos k_x a + \cos k_y a + \cos k_z a) \approx \left[3 - \frac{(k_x a)^2}{2} - \frac{(k_y a)^2}{2} - \frac{(k_z a)^2}{2} \right]. \quad (5.41)$$

Substituting in (5.39) results in

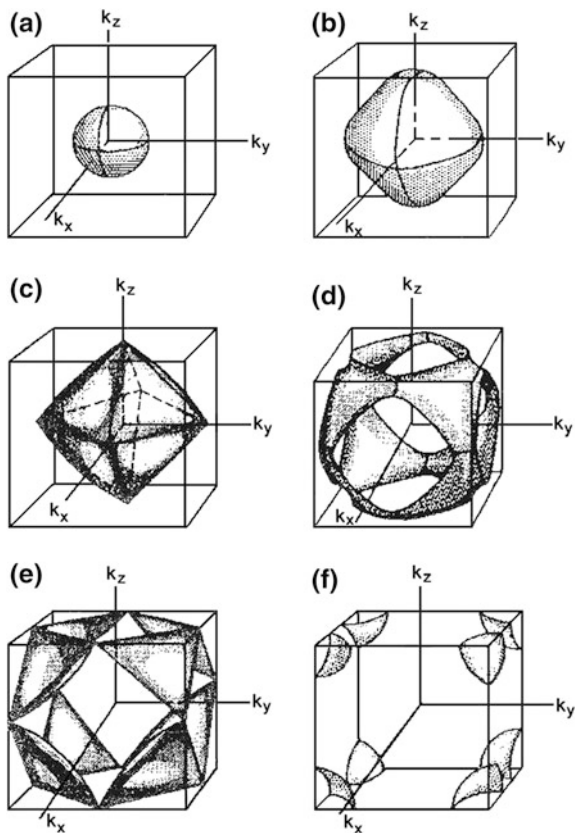
$$E_{\mathbf{k}} = E_0 - \alpha - 6\beta + \beta a^2 (k_x^2 + k_y^2 + k_z^2). \quad (5.42)$$

We may reorganize (5.42) as

$$k_x^2 + k_y^2 + k_z^2 = \frac{E_{\mathbf{k}} - E_{\min}}{\beta a^2} = \text{constant}. \quad (5.43)$$

For a chosen value of $E_{\mathbf{k}}$, the r.h.s becomes a constant. Hence, (5.43) is an equation to a sphere. Thus, we conclude that for the region close to the origin, the constant energy surfaces are spherical (Fig. 5.9a). This means that in this region the electron behaves like a free electron.

Fig. 5.9 Constant energy surfaces in a simple cubic lattice



Let us consider the energy values at the corners of the Brillouin zone. Here, $k_x = \pm\pi/a$, $k_y = \pm\pi/a$ and $k_z = \pm\pi/a$. Substitution in (5.39) results in

$$E_{\mathbf{k}} = E_0 - \alpha + 6\beta = E_{\max}. \quad (5.44)$$

This is the maximum value of $E_{\mathbf{k}}$; we shall call it $E_{\max}$. The width of the band ($E_{\max} - E_{\min}$) is 12β .

Let us consider the shape of the constant energy surfaces around the corners. As done in Sect. 5.3.1, we shall substitute $\pi - k'_x a$ for $k_x a$, $\pi - k'_y a$ for $k_y a$ and $\pi - k'_z a$ for $k_z a$. Then we have

$$\begin{aligned} \cos k_x a &= \cos(\pi - k'_x a) = -\cos k'_x a \\ &\approx -1 - \left[\frac{(k'_x a)^2}{2} \right] \quad \text{for small } k'_x a. \end{aligned} \quad (5.45)$$

Similarly

$$\cos k_y a \approx - \left[1 - \frac{(k'_y a)^2}{2} \right] \quad \text{for small } k'_y a$$

and

$$\cos k_z a \approx - \left[1 - \frac{(k'_z a)^2}{2} \right] \quad \text{for small } k'_z a. \quad (5.46)$$

Substituting (5.45), (5.46) in (5.39), we get

$$E_{\mathbf{k}} = E_0 - \alpha - 2\beta \left[-3 + \frac{(k'_x a)^2}{2} + \frac{(k'_y a)^2}{2} + \frac{(k'_z a)^2}{2} \right] \quad (5.47)$$

We can reorganize (5.47) to yield

$$k_x'^2 + k_y'^2 + k_z'^2 = \frac{(E_0 - \alpha + 6\beta) - E_{\mathbf{k}}}{\beta a^2} = \frac{E_{\max} - E_{\mathbf{k}}}{\beta a^2} = \text{constant}. \quad (5.48)$$

This is an equation for a sphere. Thus, the constant energy surfaces are spherical with the corner points as centres. These spherical surfaces close to the corners of the Brillouin zone are shown in Fig. 5.9f. In between, the constant energy surfaces take different shapes as shown in Fig. 5.9b–e.

5.3.3 Fermi Surfaces of Some Real Crystals

In Sect. 5.3.1 and 5.3.2 we discussed the broad method of constructing Fermi surface for which we chose the simplest of lattices. The detailed treatment of Fermi surface determination for real lattices is beyond our scope. We shall merely quote some results.

5.3.3.1 Copper

The Fermi surface of copper is shown in Fig. 5.10. It is essentially spherical indicating an approximately free electron behaviour. However, in the (111) directions, the sphere projects out to touch the eight hexagonal faces of the first Brillouin zone. These projections are called ‘necks’ while the spherical part is called the belly.

5.3.3.2 Beryllium

The Fermi surface of hcp beryllium is shown in Fig. 5.11. It is in the form of a ‘coronet’ which encloses the unoccupied levels and two ‘cigars’ (only one is shown

Fig. 5.10 Fermi surface of copper showing the ‘necks’ (marked *N*); the *spherical* part is called the ‘belly’

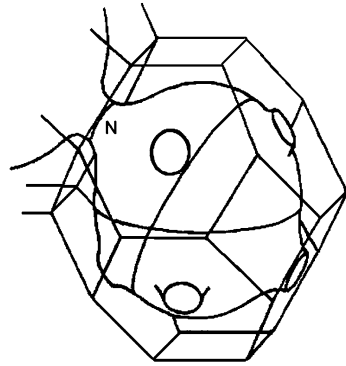
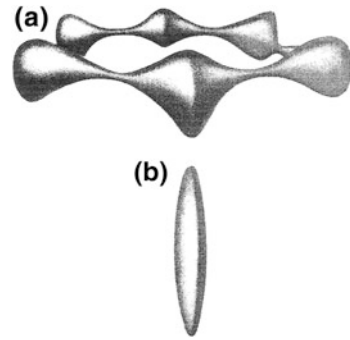


Fig. 5.11 Fermi surface of beryllium showing **a** the ‘coronet’ and **b** the ‘cigar’



in the figure) which contain electrons. Note the hexagonal symmetry of the Fermi surface.

5.3.3.3 Silicon and Germanium

Si and Ge both have the diamond structure and hence the same Brillouin zone. Yet there is a difference in the Fermi surfaces shown in Fig. 5.12. In both, the Fermi surface is ellipsoidal. However, in Si (Fig. 5.12a) there are six ellipsoids with their major axes along the (100) directions. On the other hand, in Ge (Fig. 5.12b), there are eight half-ellipsoids lying along the (111) directions.

Fermi surfaces of several crystals are discussed by Ashcroft and Mermin [3] and Singleton [4].

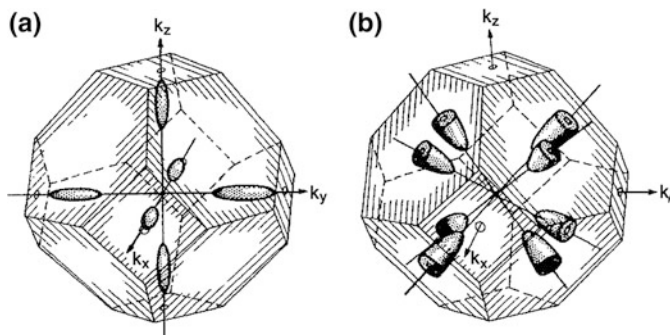


Fig. 5.12 Fermi surface of **a** silicon and **b** germanium

5.4 Examples of Band Structure

In the preceding chapter we discussed the principles of band structure calculations. Energy bands are determined from theoretical models. As we shall see, some energy band parameters can be determined experimentally. The actual determination of energy bands is an intricate process. In one of the earliest papers, Wigner and Seitz [11] mention that their calculations took “several afternoons spent on a manual desk calculator”. Kittel [7] remarks “few masters of band calculations learned their methods from books. Band calculation is an art learned from experience, developed in groups and needing access to computers”. We shall limit ourselves to a discussion of some already worked out examples of band structure calculations.

5.4.1 Aluminium

The structure and Brillouin zone of aluminium are shown in Fig. 5.13. Aluminium has the fcc lattice (Fig. 5.13a) and its first Brillouin zone is a truncated octahedron (Fig. 5.13b).

While presenting band structure, it is the practice to indicate energy values at some specific symmetry points and along some symmetry lines. These are shown in Fig. 5.13b. A notation for these points and lines was introduced by Bouckaert and Smoluchowski [12]. Here Γ is the zone centre. K , U , X , W and L are other points. Other symbols are Λ for the 111 axis, Δ for the 100 axis and Σ for the 110 axis.

The band structure of aluminium has been worked out by Heine [13], Behringer [14], Harrison [15] and Segall [16]; we shall discuss Segall’s results. Segall used the Green’s function method. The resulting energy bands are shown in Fig. 5.14. The following features are noticed.

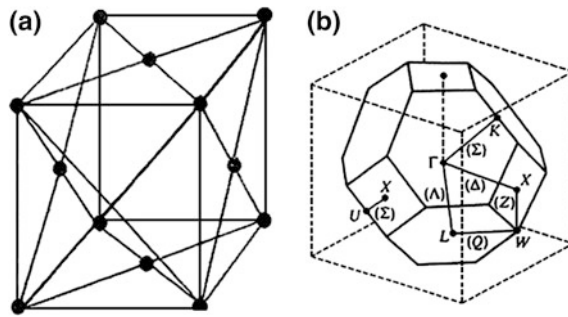


Fig. 5.13 **a** Crystal structure of aluminium, **b** Brillouin zone of Al showing symmetry points and symmetry lines

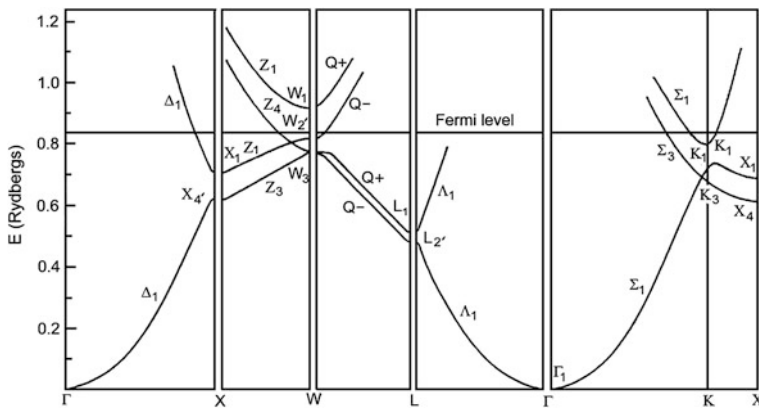


Fig. 5.14 Energy bands in aluminium

- (i) The gap at the first and second bands at X is small.
- (ii) First and second bands are degenerate at W .
- (iii) The Fermi level is slightly above the third band at W'_2 .
- (iv) The first band is completely filled, being below the Fermi level. The second and third bands are partially filled and the fourth band is empty.

Segall [16] plotted the conduction band energies E_k relative to the zone centre energy for the Δ , Λ and Σ directions against k (Fig. 5.15). Ignoring the gaps, he found that the points lie on a parabola with equation

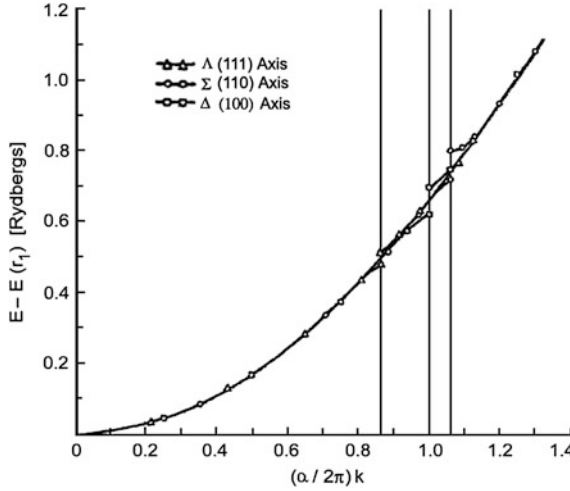


Fig. 5.15 Conduction band energies (relative to the zone centre energy) for Δ , Λ and Σ plotted against $(a/2\pi)k$

$$E_k = E_0 + \alpha k^2 \quad (5.49)$$

with $\alpha = 0.976$. The effective mass m^* (defined as $m^* = \hbar^2 / (d^2 E_k / dk^2)$) calculated from (5.49) is $1.025 m$. The values of the effective mass and the parabolic shape of the curve indicate a nearly free-electron behaviour.

5.4.2 Germanium

The crystal structure of germanium is shown Fig. 5.16. The sites of A and B are equivalent and are occupied with Ge atoms. Diamond, silicon and grey tin also have the same structure. The structure is based on an fcc lattice; hence, the Brillouin zone is the same as that shown in Fig. 5.13b. The band structure of Ge has been investigated by Phillips [17], Cardona and Pollock [18] and Herman et al. [19]. We shall discuss the results of Herman et al.

The energy bands obtained by Herman et al. are shown in Fig. 5.17. The following features may be noted.

- (i) The first band originating at Γ is parabolic.
- (ii) The first four bands Δ_1 , Δ_2 and Δ_5 (which is doubly degenerate) are completely filled. The two top-most valence bands converge at Γ'_{25} . They are called the heavy-hole and light-hole bands.
- (iii) The fifth and higher bands are empty.
- (iv) The band gaps are large compared to those in aluminium.

Table 5.1 Values of energy gap (E_g) and lattice constant a for crystals with diamond structure

Crystal	E_g [eV]	' a ' [$\text{\AA}$]
C	7	3.56
Si	1.2	5.43
Ge	0.8	5.66
Sn	–	6.46

- (v) The minimum energy gap (E_g) between Γ'_2 and Γ'_{25} is equal to 0.8 eV. In Table 5.1, the values of E_g for Ge and other related solids are compared with the corresponding lattice constants. Diamond has a large E_g and is an insulator. Ge and Si with smaller E_g are semiconductors. Sn (Grey), like aluminium, has negligible E_g and is metallic. It may be noted that the energy gap decreases as the lattice constant increases i.e. as the overlap of wave functions decreases (Chap. 4).

5.4.3 Gallium Arsenide

Gallium arsenide is the prototype of the II–VI and III–V compounds. They crystallize in the zinc-blende structure which is similar to the diamond structure (Fig. 5.16) with the difference that the A and B sites are occupied by Ga and As atoms. The first Brillouin zone for this structure is the same as for Al and Ge (Fig. 5.13b). The band structure for GaAs is shown in Fig. 5.18. The following features are observed.

- (i) Band 1 rises parabolically at Γ .
- (ii) Bands 2 and 3 are valence bands. As mentioned earlier, they are the light-hole and heavy-hole bands.
- (iii) Band 4 is the conduction band.

5.4.4 Sodium Chloride

Sodium chloride i.e. NaCl is one of the several alkali halides which crystallize with the structure shown in Fig. 5.19. The Na and Cl atoms exist in the crystal as ions. The NaCl lattice is made up of two interpenetrating fcc lattices. Its Brillouin zone, again, happens to be the same as that of Al (Fig. 5.13b). The band structure of NaCl has been studied by Clarke and Kliever [20], Kunz [21] and Page and Hygh [22]. We shall discuss the results of Page and Hygh.

Fig. 5.18 Energy bands in GaAs

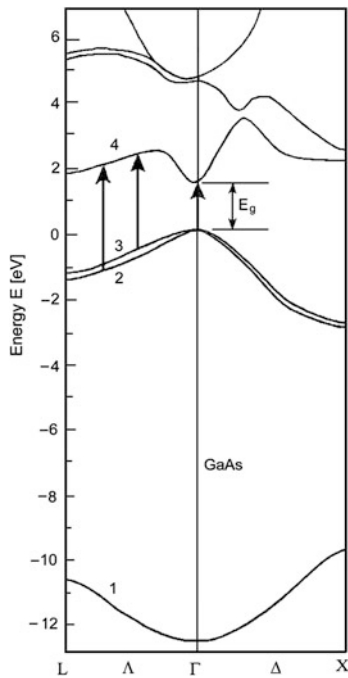
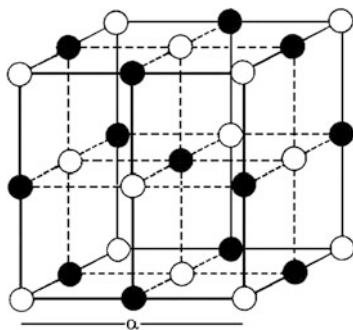


Fig. 5.19 Crystal structure of sodium chloride



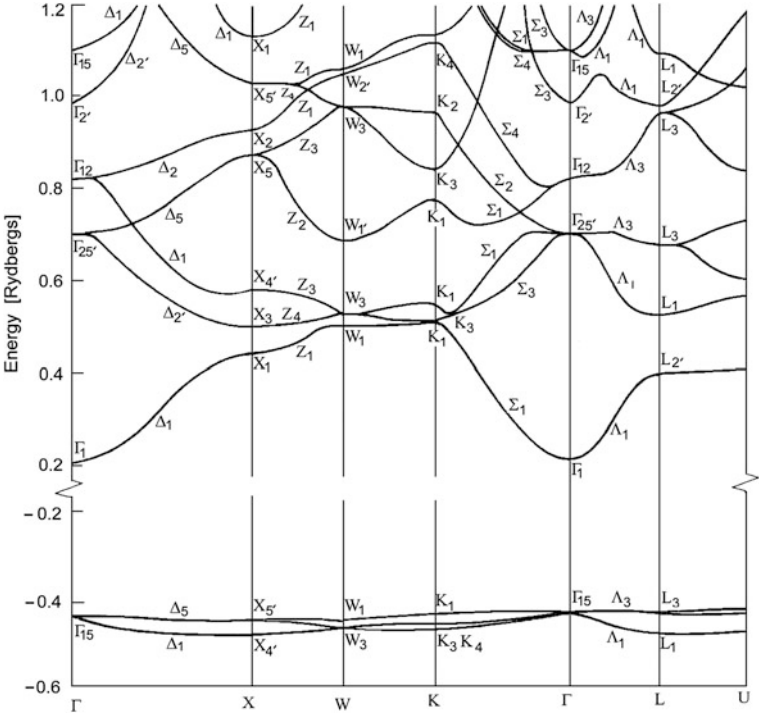


Fig. 5.20 Energy bands in NaCl

The energy bands of NaCl are shown Fig. 5.20. They have the following features.

- (i) The valence bands are fairly flat. They consist of the $3p^6$ state of Cl.
- (ii) The conduction band comprises $3s$ bands of Na, $3d$ band of Cl and $3p$ band of Na.
- (iii) The energy gap which corresponds to the $\Gamma_{15} \rightarrow \Gamma_1$ transition has a value 8.7 eV.

The effective masses calculated from $m^* = \hbar^2 / (d^2 E_k / dk^2)$ at different points of the conduction band are given in Table 5.2.

Table 5.2 Effective masses at the bottom of conduction band of NaCl

Symmetry point	m^* (in units of m)
Γ (towards X)	0.573
Γ (towards L)	0.712
Γ (towards K)	0.676

5.5 Effective Mass

In the preceding chapter we introduced the concept of the effective mass m^* . Based on the one-dimensional Kronig-Penney model, the expression for the effective mass turned out to be $m^* = \hbar^2 / (d^2 E_k / dk^2)$. For the free electron, the energy is $E = (\hbar^2 k^2 / 2m)$ and the constant energy surface is spherical. The effective mass m^* turns out to be just m , the free electron mass.

We shall now obtain a more general definition in three dimensions; the general definition is also applicable to non-spherical constant energy surfaces.

The angular velocity of the electron ω is given by

$$\omega = E_{\mathbf{k}} / \hbar. \quad (5.50)$$

The group velocity for the electron waves is given by

$$\mathbf{v} = \frac{\partial \omega}{\partial \mathbf{k}} = \frac{\partial}{\partial \mathbf{k}} (E_{\mathbf{k}} / \hbar) = \left(\frac{1}{\hbar} \right) \nabla_{\mathbf{k}} E_{\mathbf{k}} \quad (5.51)$$

Here $\nabla_{\mathbf{k}}$ is the gradient operator in the orthogonal k -space in which the coordinates are k_x , k_y , and k_z .

If an electric field $\mathbf{F}$ is applied, the force on the electron is $(-e\mathbf{F})$ and the time rate of change of energy is

$$d(E_{\mathbf{k}}) / dt = -e\mathbf{F} \cdot \mathbf{v} \quad (5.52)$$

An infinitesimal change of energy $dE_{\mathbf{k}}$ can be written as

$$dE_{\mathbf{k}} = \frac{dE_{\mathbf{k}}}{d\mathbf{k}} \cdot d\mathbf{k} = (\nabla_{\mathbf{k}} E_{\mathbf{k}}) \cdot d\mathbf{k}. \quad (5.53)$$

Combining (5.51), (5.52) and (5.53), we get

$$\hbar \frac{d\mathbf{k}}{dt} = -e\mathbf{F}. \quad (5.54)$$

The acceleration a is given by

$$\begin{aligned} a &= \frac{d\mathbf{v}}{dt} = \frac{1}{\hbar} \nabla_{\mathbf{k}} (dE_{\mathbf{k}} / dt) \\ &= \frac{1}{\hbar} \nabla_{\mathbf{k}} (-e\mathbf{F} \cdot \mathbf{v}) \\ &= \frac{1}{\hbar} \nabla_{\mathbf{k}} \left[(-e\mathbf{F}) \cdot \left(\frac{1}{\hbar} \right) \nabla_{\mathbf{k}} E_{\mathbf{k}} \right] \\ &= \left(\frac{1}{\hbar^2} \right) (\nabla_{\mathbf{k}} \nabla_{\mathbf{k}} E_{\mathbf{k}}) (-e\mathbf{F}). \end{aligned} \quad (5.55)$$

Since acceleration is force/mass, the term $(\hbar^2)(\nabla_k \nabla_k E_{\mathbf{k}})^{-1}$ has dimensions of mass; it is defined as the effective mass m^* . We may write the components of acceleration as

$$a_i = \left(\frac{1}{\hbar^2} \right) \sum_j \frac{\partial^2 E_{\mathbf{k}}}{\partial k_i \partial k_j} (-eF_j). \quad (5.56)$$

It is a rule in tensor analysis [8, 23] that when a quantity relates the components of a vector with the components of another vector, then that quantity is a second rank

tensor. In (5.56) the quantity $\left[\frac{1}{\hbar^2} \sum_j \frac{\partial^2 E_{\mathbf{k}}}{\partial k_i \partial k_j} \right]$ relates the components of vector a with the

components of the vector F . Hence $\left[\frac{1}{\hbar^2} \sum_j \frac{\partial^2 E_{\mathbf{k}}}{\partial k_i \partial k_j} \right]$ is a second rank tensor. From (5.55)

and (5.56), we see that the term $\left[\frac{1}{\hbar^2} \sum_j \frac{\partial^2 E_{\mathbf{k}}}{\partial k_i \partial k_j} \right]$ has dimensions of reciprocal mass. Hence

the tensor $\left[\frac{1}{\hbar^2} \sum_j \frac{\partial^2 E_{\mathbf{k}}}{\partial k_i \partial k_j} \right]$ is called the reciprocal mass tensor $\left[\frac{1}{m^*} \right]_{ij}$. It can be seen that

$$\left[\frac{1}{m^*} \right]_{ij} = \left[\frac{1}{\hbar^2} \sum_j \frac{\partial^2 E_{\mathbf{k}}}{\partial k_i \partial k_j} \right] = \left[\frac{1}{\hbar^2} \sum_j \frac{\partial^2 E_{\mathbf{k}}}{\partial k_j \partial k_i} \right] = \left[\frac{1}{m^*} \right]_{ji}. \quad (5.57)$$

Thus the reciprocal effective mass tensor is a symmetric tensor.

The tensor nature of effective mass means that the effective mass has different values in different directions which, in turn, means that electron motion is anisotropic. This is the consequence of the constant energy surface being non-spherical.

The full form of the reciprocal mass tensor is

$$\left[\frac{1}{m^*} \right]_{ij} = \begin{bmatrix} \frac{1}{m_{11}^*} & \frac{1}{m_{12}^*} & \frac{1}{m_{13}^*} \\ \frac{1}{m_{12}^*} & \frac{1}{m_{22}^*} & \frac{1}{m_{23}^*} \\ \frac{1}{m_{13}^*} & \frac{1}{m_{23}^*} & \frac{1}{m_{33}^*} \end{bmatrix}. \quad (5.58)$$

In solids with non-spherical constant-energy surfaces, the off-diagonal terms are large i.e. the field applied in one direction produces acceleration in another direction. It is even possible that some elements in the tensor are positive and some are negative. However, in many cases, there is some degree of symmetry in constant energy surface due to which the off-diagonal elements are equal to zero i.e. the tensor is a diagonal tensor. In such a case

Table 5.3 Conduction and hole effective masses (in units of m) of some semiconductors

Semiconductor	m_c^*	m_{lh}^*	m_{hh}^*
GaAs	0.067	0.082	0.45
InAs	0.023	0.025	0.41
InSb	0.014	0.016	0.40

Table 5.4 Anisotropic effective masses (in units of m) of some semiconductors

Semiconductor	$m_{ }^*$	$m_{\perp}^*$
Ge	1.64	0.082
Si	0.98	0.19

$$\left[\frac{1}{m^*} \right]_{ij} = \begin{bmatrix} \frac{1}{m_{11}^*} & \cdot & \cdot \\ \cdot & \frac{1}{m_{22}^*} & \cdot \\ \cdot & \cdot & \frac{1}{m_{33}^*} \end{bmatrix}. \quad (5.59)$$

In the case of a highly symmetric constant energy surface, $m_{11}^* = m_{22}^* = m_{33}^* = m^*$ and the effective mass is a scalar.

5.5.1 Types of Effective Masses

We have noted in Chap. 4 that electrons and holes have their respective effective masses. They are denoted as m_c^* and m_h^* respectively (c for conduction electrons and h for hole). Some authors refer to them as m_n^* and m_p^* (n for negative and p for positive). There are several other types, particularly in semiconductors. It was mentioned in Sect. 5.4 that in some semiconductors, there are two valence bands coinciding at the Γ point. These two bands have different curvatures which results in two different effective masses. These are denoted as m_{lh}^* and m_{hh}^* (lh for light hole and hh for heavy hole). Values of m_{lh}^* and m_{hh}^* for some crystals are given in Table 5.3.

The constant energy surfaces are highly anisotropic in some solids. Ge and Si are examples (Fig. 5.12) in which these surfaces are ellipsoidal. The effective masses along the two minor axes ($m_{\perp}^*$) will be the same while that along the major axis ($m_{||}^*$) will be different. In such cases, the reciprocal effective mass tensor is

$$\left[\frac{1}{m^*} \right] = \begin{bmatrix} \frac{1}{m_{\perp}^*} & \cdot & \cdot \\ \cdot & \frac{1}{m_{\perp}^*} & \cdot \\ \cdot & \cdot & \frac{1}{m_{||}^*} \end{bmatrix}. \quad (5.60)$$

Values of $m_{\perp}^*$ and $m_{||}^*$ for Ge and Si are given in Table 5.4.

Table 5.5 Values of effective mass (m^*/m) for some metals obtained from different properties

m^*/m	Li	Na	K	Rb	Cs
$(m^*/m)_{\text{Calc}}$	1.4	0.98	0.94	0.87	0.83
$(m^*/m)_{\text{Sp. heat}}$	2.3	1.3	1.2	1.3	1.51
$(m^*/m)_{\text{Susc.}}$	2.36	1.06	0.9	0.70	0.83

5.5.2 Comparison of Different Values of Effective Masses

The effective masses of electrons can be calculated from band structures; we shall call these values $(m^*/m)_{\text{calc}}$. We have seen in Chap. 4 that the experimental electronic specific heat and that calculated from free electron theory are in the ratio (m^*/m) ; we shall denote such values as $(m^*/m)_{\text{sp. heat}}$. Similarly the ratio of measured and calculated magnetic susceptibilities yield a value of effective mass which we shall denote as $(m^*/m)_{\text{susc.}}$. These values for some metals are given in Table 5.5. While these values from different origins generally agree in magnitude, there are some differences among them. Thus, the specific heat values are larger than the other values. For this reason, it is the practice to indicate the source of the m^*/m value. In the next section, we shall see that m^*/m values can be determined from cyclotron resonance experiments. The resulting m^*/m values are denoted as $(m^*/m)_{\text{CR}}$ as they may not agree exactly with other values.

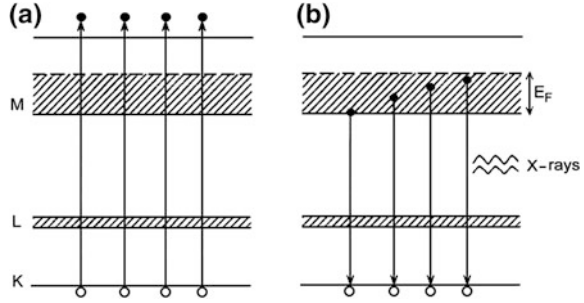
5.5.3 Physical Significance of the Effective Mass

The effective mass of electrons in most metals is in the range 0.5–2 times the free electron mass. The effective mass has nothing to do with the gravitational mass of the electron. If the effective mass is more or less than the free electron mass, it does not make the crystal heavier or lighter than the actual crystal [7]. The effective mass merely indicates that the motion of the electron in a periodic potential is much more complicated than that of a free electron. Whereas the motion of a free electron in an electric field is determined by the field alone, the motion of an electron in a periodic potential is determined by the electric field together with the periodic potential and on where the electron is located in the band. The joint effect of the field and the crystal potential results in replacing the free electron mass m by an effective mass m^* . Further, the tensor nature of m^* indicates the anisotropy of the constant energy surface (which is different from the crystal anisotropy).

5.6 Experiments on Band Structure

There are several experimental techniques to probe the band structures. Some of them yield information on band-structure related parameters like the Fermi energy and the effective masses. Others provide information about the Fermi surface. We shall discuss some of the methods.

Fig. 5.21 **a** Knocking out of electrons from K shell of Na; **b** filling up of electron vacancies in K shell by electrons from M band and emission of X-rays



5.6.1 Soft X-ray Emission

Let us recall the process of X-ray emission. When high energy electrons or other high energy particles strike an atom, they may succeed in knocking out electrons from the inner shells. To fill up the vacancies of electrons in the inner shells, electrons from outer shells jump into the inner shells. In this process, X-rays are emitted, their energy being equal to the difference in energies of the levels involved in the transition.

Let us consider the emission of X-rays from solid sodium. The energy bands in sodium are shown in Fig. 5.21a. The lowest level is the $1s$ level; In X-ray terminology, this is called the K-level. Above the K level are the filled $2s$, $2p$ levels. Together they constitute the L band. The top-most band is made up of overlapping conduction and valence bands containing $3s$ electrons; this is the M level. When bombarded by high energy electrons, some electrons from the K shell are knocked out leaving vacancies in the K band. Electrons from the conduction-valence M band jump into the K band (Fig. 5.21b) causing emission of X-rays. These X-rays are called soft X-rays. The photons have energies ~ 200 eV and wavelengths ~ 100 Å.

Let us note that the K and L bands are very narrow while the M band is a wide band. Further, the knocked out electrons leave the atom (Fig. 5.21a). On the other hand, The M electrons which jump into the vacancies, originate at different levels from the bottom to the top of the M-band (Fig. 5.21b). In other words, the emitted soft X-rays have a width equal to the width of the valence band i.e. the Fermi energy [7, 10]. As an example, the X-ray intensity versus photo energy curve obtained by Skinner [24] for lithium is shown in Fig. 5.22. The values of the Fermi energy for some metals obtained from soft-X-rays are compared in Table 5.6 with those calculated from the free electron theory.

Fig. 5.22 X-ray emission band of lithium

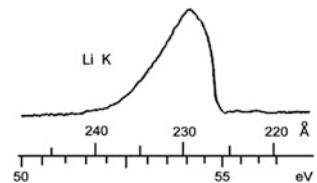


Table 5.6 Values of E_F of some metals

Metal	E_F [eV]	
	From soft X-rays	From free electron theory
Na	3.0	3.25
Li	4.2	4.74
Be	14	14.3
Al	12	11.7

The experimental method of soft-x-ray emission requires a source of high energy particles to impinge on a solid held in vacuum. The intensity can be measured with a suitable detector. The measurement of wavelength (photon energy) can be done by X-ray diffraction provided a crystal with a large lattice constant is available (note that these are long wavelength X-rays). Otherwise, a ruled grating has to be employed.

5.6.2 Cyclotron Resonance

It is known that a free electron in a magnetic field performs circular motion in a plane normal to the field. The angular frequency of such motion ω_c is given by

$$\omega_c = eH/mc \quad (5.61)$$

where e is the electron charge, H the magnetic field and m the mass of the free electron. The frequency ω_c is called the cyclotron frequency.

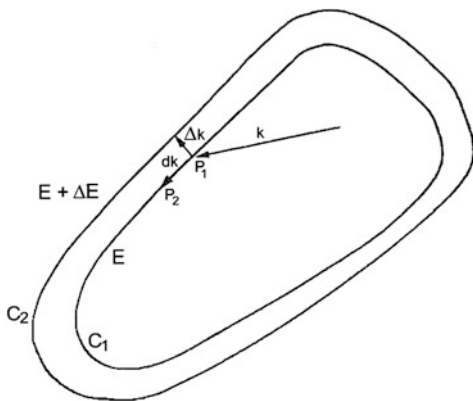
The concept of circular motion of an electron in a magnetic field was extended to electrons (and holes) in solids by Dorfman [25] and Dingle [26]. The expression for cyclotron frequency ω'_c of an electron in a solid is

$$\omega'_c = eH/m^*c \quad (5.62)$$

where m^* is the effective mass of the electron.

An interpretation of m^* in (5.62) can be attempted in terms of constant energy surfaces [4, 27]. Let a plane normal to the magnetic field intersect two nearby constant energy surfaces of energy E and $E + \Delta E$ (ΔE is assumed to be small); we shall call the closed curves of intersection C_1 and C_2 (Fig. 5.23). If $\mathbf{k}$ is the propagation vector of the electron at point P_1 on the curve C_1 , we may draw $\Delta\mathbf{k}$ normal at P_1 to touch curve C_2 . Let the electron travel from P_1 to P_2 along C_1 in time dt ; the line element P_1P_2 is dk .

Fig. 5.23 Intersection of plane perpendicular to the magnetic field with two constant energy surfaces



We have seen earlier (Sect. 5.5) that

$$\mathbf{v} = \frac{1}{\hbar} \frac{dE_{\mathbf{k}}}{d\mathbf{k}} \quad (5.63)$$

and

$$\hbar \frac{d\mathbf{k}}{dt} = \text{Force}. \quad (5.64)$$

In (5.63), we shall replace $dE_{\mathbf{k}}/d\mathbf{k}$ by $\Delta E/\Delta k$. Further, since the electron is in a magnetic field, the force in (5.64) is the Lorentz force $-\frac{e}{c}(\mathbf{v} \times \mathbf{H})$. With these substitutions, we get

$$\hbar \frac{dk}{dt} = \left(\frac{-eH}{\hbar c} \right) \left(\frac{\Delta E}{\Delta k} \right) \quad (5.65)$$

Let us now multiply both sides by dt and Δk and integrate round the orbit C_1 . Then,

$$\hbar \oint \Delta k dk = \left(\frac{-eH}{\hbar c} \right) \Delta E \oint dt. \quad (5.66)$$

In (5.66), $\oint dt$ is the period T to go round the orbit and $\oint \Delta k dk$ is the area $\Delta A_{1,2}$ between the two orbits C_1 and C_2 swept during a period. Thus, (5.66) becomes

$$\Delta A_{1,2} = \frac{eH}{\hbar^2 c} \Delta E T. \quad (5.67)$$

Since frequency $\omega'_c = 2\pi/T$, we have

$$\omega'_c = \left(\frac{2\pi eH}{\hbar^2 c} \right) \left(\frac{\Delta A_{1,2}}{\Delta E} \right)^{-1} \quad (5.68)$$

Comparing with (5.62) and noting that the Δ 's are infinitesimal, we have

$$m^* = \left(\frac{\hbar^2}{2\pi} \right) \left(\frac{\Delta A_{1,2}}{\Delta E} \right) = \left(\frac{\hbar^2}{2\pi} \right) \left(\frac{\partial A_{1,2}}{\partial E} \right) \quad (5.69)$$

We shall note the following:

- (i) The gradient $\left(\frac{\partial A_{1,2}}{\partial E} \right)$ is to be taken in a direction parallel to the magnetic plane i.e. in the plane C_1 .
- (ii) The results are valid for electrons for which $\frac{\partial A}{\partial E} > 0$ and for holes for which $\frac{\partial A}{\partial E} < 0$.
- (iii) The above discussion is in terms of k -space. It can be shown that an orbit in real space is similar to that in k -space but with a scaling factor $(\hbar c/eH)$.
- (iv) The effective mass m^* defined by (5.69) is conceptually different from effective masses like those from specific heats and the two need not agree.

The experimental arrangement for observing cyclotron resonance used by Galt et al. [28] is shown in Fig. 5.24. The sample is held in a cavity which is coupled to a wave guide. The power is introduced at one end of the wave guide and the absorption is measured by a detector at the other end. For good resolution, it is desirable that $\omega'_c \tau \gg 1$ (τ being the relaxation time). To ensure this, the experiment is done at low temperatures, high magnetic fields and at microwave frequencies. Typically, fields are ~ 10 – 20 T and the frequency is $\sim$ GHz. The sample is a very pure oriented single crystal.

The beauty of a cyclotron resonance technique lies in its ability to detect more than one type of carrier in a single experiment. As an example, the result of an experiment on Ge by Dresselhaus et al. [29] is shown in Fig. 5.25. The frequency was fixed at 24 GHz and the magnetic field was swept in each orientation of the crystal. Five peaks were observed. Of these, two correspond to light and heavy holes. The other three correspond to combinations of the effective masses $m_{\perp}^*$ and $m_{\parallel}^*$ of the electron in three orientations of the crystal. Similar results were obtained for Si. The effective masses obtained from this experiment are given in Table 5.7.

5.6.3 Anomalous Skin Effect

Electromagnetic radiation incident on a metal is mostly reflected. A small part of it penetrates into the metal but it gets absorbed as it travels until, finally, it dies out. Thus, the effective depth of penetration is limited to a value δ which is given by

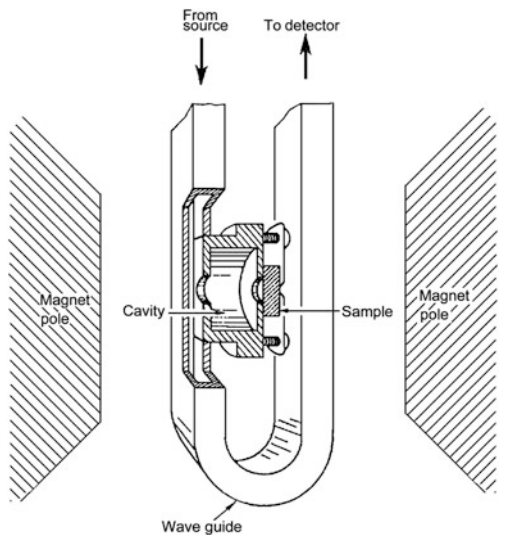


Fig. 5.24 Experimental set-up for the study of cyclotron resonance

Fig. 5.25 Absorption of radiation caused by cyclotron resonance in germanium

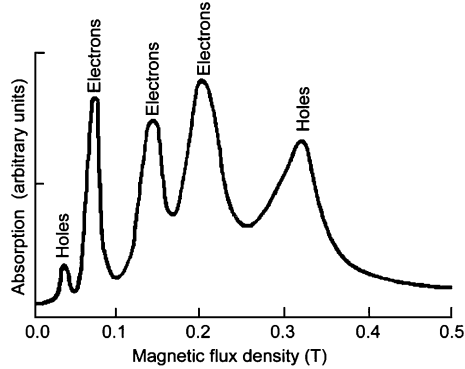


Table 5.7 Effective masses m^* (in units of m) from cyclotron resonance

Crystal	$m_{\perp}^*$	$m_{\parallel}^*$	m_{lh}^*	m_{hh}^*
Ge	0.082	1.64	0.044	0.3
Si	0.19	0.98	0.16	0.3

$$\delta = \left[\left(\frac{2\pi\omega}{c^2} \right) \sigma \right]^{-1/2} \tag{5.70}$$

where ω is the angular frequency of the radiation and σ the electrical conductivity of the metal. This is called the ‘normal skin effect’.

If λ is the mean free path of the electron the derivation of (5.70) assumes that $\lambda \ll \delta$. Typically, $\omega \approx 3 \times 10^{10} \text{ s}^{-1}$ and at room temperature $\sigma \approx 5 \times 10^5 \text{ } \Omega^{-1}$. With these values, the value of δ calculated from (5.70) is $\sim 10^{-4} \text{ cm}$. The value of λ at normal temperatures is $\sim 10^{-7} \text{ cm}$. Thus $\lambda \ll \delta$ which satisfies the condition assumed in deriving (5.70).

At low temperatures, σ is much larger. Further, if radiation at microscopic frequencies is employed, then δ calculated from (5.70) becomes comparable with λ and (5.70) is no longer valid. We shall denote the penetration depth in these conditions δ' . Pippard [30] studied this effect and showed that

$$\delta' = \left[\left(\frac{2\pi\omega}{c^2} \right) \left(\frac{ne^2}{mv_F} \right) \right]^{-1/2}. \quad (5.71)$$

Here, n is the concentration of electrons in unit volume, m the free electron mass and v_F the Fermi velocity. This is called the ‘anomalous skin effect’. The measurement of δ' yields information about v_F and, through it, about the Fermi surface. To determine δ' , the absorption of radiation has to be measured. The absorption is very small. Typically, at frequencies of $23,000 \text{ Mc s}^{-1}$, $6 \text{ } \mu\text{W}$ power is absorbed out of an incident power of 50 mW . An experimental method should be capable of measuring this small absorption.

The experimental set-up used by Pippard [31] is shown in Fig. 5.26. Here, A is the sample which is a metal single crystal with a polished surface. B is the rectangular waveguide. C is the vacuum enclosure immersed in liquid helium. D and D' are differential gas thermometers; D is connected to the sample and D' to the bath. E is one of the copper wires connecting the sample to D through F. G are tubes in the manometer and H the calibrating heating coil.

Polarized microwave energy is fed through the wave guides. It is mostly reflected and a small part is absorbed. This small absorption raises the temperature which is measured with the gas thermometers. Finally, from the absorption δ' is determined.

Pippard used a highly pure, flat and polished single crystal of copper. From his measurements, Pippard deduced the Fermi surface of copper (Fig. 5.27). This work is considered “the first successful experimental determination of the Fermi surface of a metal” [2].

5.6.4 Magnetoresistance

Magnetoresistance is the change in the electrical resistivity of a metal or semiconductor due to the application of magnetic field. It is expressed as the coefficient $[\rho(H) - \rho(0)]/[\rho(0)]$ where $\rho(H)$ is the resistivity at field H and $\rho(0)$ that at zero field.

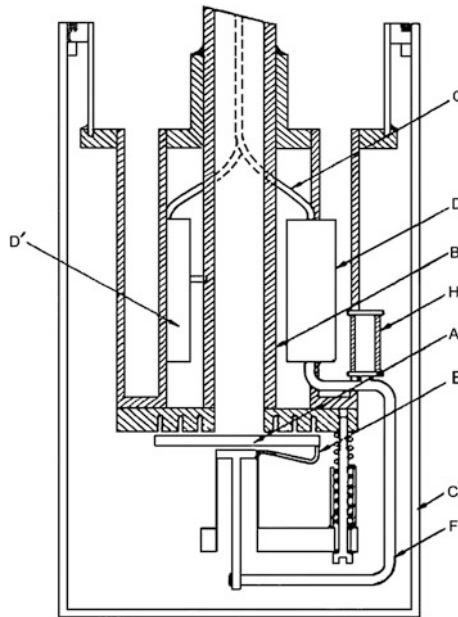
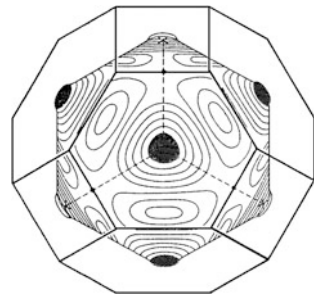


Fig. 5.26 Experimental set-up for study of anomalous skin effect

Fig. 5.27 Fermi surface of copper determined from anomalous skin effect



In Chap. 2 and 3, it was seen that the free electron theories predict a zero magnetoresistance while experiments indicate the existence of a finite magnetoresistance. This existence of magnetoresistance is due to the fact that electrons in a periodic potential behave differently from free electrons. The detailed treatment of magnetoresistance is beyond our scope. We shall limit ourselves to the results.

The intersection of a plane normal to a magnetic plane with the Fermi section could be a closed orbit or an open orbit (Fig. 5.28). In the case of a closed orbit, the average velocity of the electron in the plane of the orbit is zero. On the other hand, in an open orbit, the y -component of the velocity has a finite value. In the case of a closed orbit, the relevant theory [4] leads to

$$E_x = \frac{1}{\sigma_{xx} + (R^2 H^2 \sigma_{yy})^{-1}} J_x + \frac{RH}{1 + R^2 H^2 \sigma_{xx} \sigma_{yy}} J_y. \quad (5.72)$$

Here J_x and J_y are current densities, σ_{xx} , σ_{yy} the components of the conductivity tensor and R the Hall coefficient. It can be shown that both σ_{xx} and σ_{yy} vary as H^{-2} at high fields. Considering these variations, (5.72) leads to

$$\rho_{xx} = E_x/J_x = \text{constant at high fields}. \quad (5.73)$$

This means that ρ_{xx} saturates. On the other hand, for open orbits, the expression for ρ_{xx} is

$$\rho_{xx} = E_x/J_x \propto H^2. \quad (5.74)$$

From (5.73) and (5.74) we conclude that a field independent resistivity indicates presence of closed orbits whereas a resistivity proportional to H^2 indicates presence of open orbits.

Thus by measuring the magnetoresistance in different directions (by changing the orientation of the field with respect to the crystal) and studying its field-dependence, we get information about the existence of open and closed orbits; such information leads finally to the shape of the Fermi surface.

In Fig. 5.29, a radial plot of $[\rho(H) - \rho(0)]/[\rho(0)]$ for copper is shown [32]. The current flows in the [100] direction which is normal to the plane of the paper. The magnetic field is 18 kGs and its direction is varied from [001] to [010]. The sample is a very pure single crystal and the temperature is 4.2 K. The plot indicates existence of open orbits.

Fig. 5.28 Fermi surface orbits: **a** closed orbit; **b** open orbit

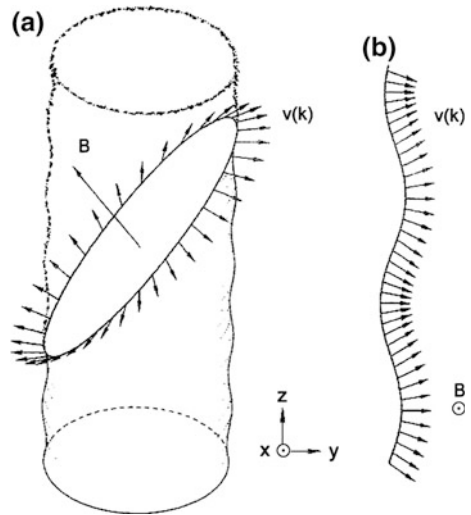
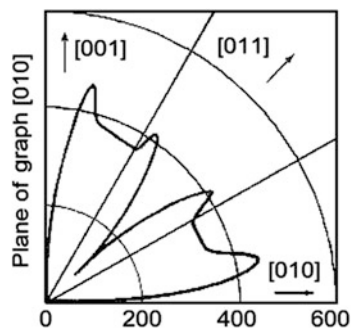


Fig. 5.29 Polar plot of magnetoresistance $[\rho(H) - \rho(0)]/\rho(0)$ for copper in different directions. Magnetic field of 18 kGs is in the plane of the paper



5.6.5 De Haas–van Alphen Effect

It was observed by de Haas and van Alphen [33] that at low temperatures the magnetic susceptibility (M/H) of bismuth measured as a function of the magnetic field showed oscillatory behaviour. This is shown in Fig. 5.30 where the susceptibility is plotted against the field H . It was soon realized that the oscillatory behaviour is revealed with greater clarity when the differential susceptibility (dM/dH) is plotted against ($1/H$).

The significance of this observation was pointed out by Onsager [34] who showed that the change in ($1/H$) in one oscillation of the susceptibility is given by

$$\Delta\left(\frac{1}{H}\right) = \left(\frac{2\pi e}{\hbar c}\right)\left(\frac{1}{A_e}\right) \quad (5.75)$$

where A_e is the area of an extremal cross-section of the Fermi surface in a plane normal to the applied field. In Fig. 5.31, some extremal cross-sections of a hypothetical Fermi surface are shown. Thus, if field (H_z) is applied along the z -axis, the extremal (maximum) orbits are (1) and (2) and the extremal (minimum) orbit is (3). If the magnetic field (H_y) is along y -axis, there is an extremal (maximum) orbit (4). By altering the direction of the magnetic field, areas of several extremal cross-sections can be measured. This information is useful in constructing a part or a whole of a Fermi surface.

The experiment is done on very pure single crystals at very low temperatures and at very high magnetic fields. These conditions are necessary to ensure good resolution and large amplitude. The general arrangement is shown in Fig. 5.32a. The specimen is enclosed in coaxial coils kept in a Dewar vessel and between pole pieces of a magnet. The two coils C and C_0 are connected in series and opposition so that their self-inductance is zero in the absence of the sample. But with the sample in place, when there is a burst of a high magnetic field (i.e. a time varying field), a voltage V is induced in the coil C which is proportional to dM/dt . We may write V as

Fig. 5.30 Plot of M/H against H for Bi single crystal along the binary axis

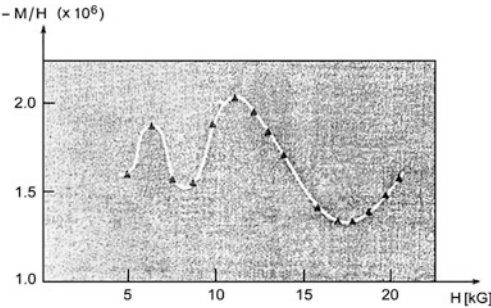


Fig. 5.31 Extremal orbits in a Fermi surface

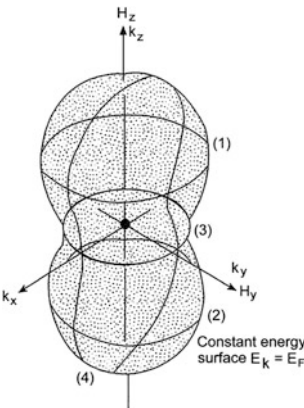


Fig. 5.32 **a** Experimental arrangement for observing de Haas-van Alphen effect; relative positions of magnet, coils and sample; **b** outer coil C_0 and inner coil C

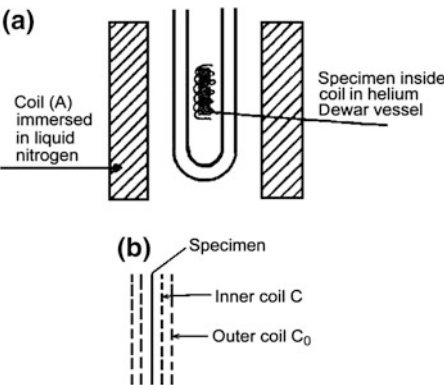


Fig. 5.33 a de Haas-van Alphen oscillations in silver; one low frequency oscillation is shown by *arrows*. The number of high frequency oscillations in one low-frequency oscillation yields the ratio $A(\text{belly})/A(\text{neck})$. The belly orbit and neck-orbit are shown in (b)

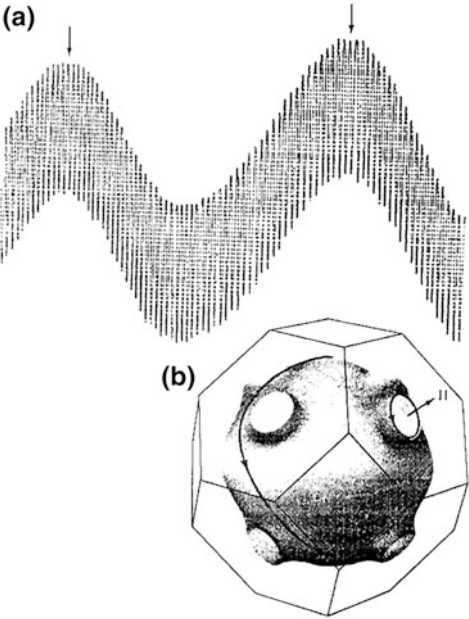


Table 5.8 Ratio of the areas of the ‘belly’ and ‘neck’ orbits in some metals

Metal	$A(\text{belly})/A(\text{neck})$
Cu	27
Ag	51
Au	29

$$V = \frac{dM}{dt} = \left(\frac{dM}{dH} \right) \left(\frac{dH}{dt} \right). \tag{5.76}$$

The voltage is fed to an oscilloscope. From the oscillation of V and knowledge of dH/dt , A_e can be estimated.

In Fig. 5.33, the de Haas-van Alphen oscillations in silver obtained by Joseph and Thorsen [35] are shown. The oscillations shown in Fig. 5.33a are a combination of two oscillations. The high frequency oscillations correspond to the larger ‘belly’ orbit (Fig. 5.33b) and the low frequency oscillations to the smaller ‘neck’ orbit. By counting the number of high frequency oscillations in a single low frequency oscillation, we directly get the ratio $A_{\text{belly}}/A_{\text{neck}}$ without even knowing the scale of the graph. Values of the ratio $(A_{\text{belly}}/A_{\text{neck}})$ for some metals are given in Table 5.8.

Table 5.9 Comparison of Sommerfeld Theory and Band Theory

Aspect	Sommerfeld theory	Band theory
Quantum numbers	$\mathbf{k}$	$\mathbf{k}$
Range of the quantum numbers	$\mathbf{k}$ extends over entire k -space	$\mathbf{k}$ extends over a Brillouin zone
Potential	Constant	Periodic
Wave function	Plane waves $\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}$	Modulated plane waves (Bloch function) $\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{\mathbf{k}}(\mathbf{r})$ is periodic
Energy	$E_{\mathbf{k}} = \frac{\hbar^2 k^2}{2m}$	$E_{\mathbf{k}}$ is structure dependent
Nature of energy levels	Discrete levels	Levels in the form of bands
Velocity	$\mathbf{v} = \frac{1}{\hbar} \frac{dE}{d\mathbf{k}} = \frac{\hbar \mathbf{k}}{m}$	$\mathbf{v} = \frac{1}{\hbar} \frac{\partial E_{\mathbf{k}}(\mathbf{r})}{\partial \mathbf{k}}$
Fermi sphere	Spherical for all crystals	Complicated surfaces varying with crystal structure
Mass of particles	Scalar mass m	Effective mass m^* which is of tensorial nature
Particles covered by the theory	Electrons	Electrons as well as holes

5.7 Comparison of Sommerfeld Theory and Band Theory

The quantum free electron theory (also called the Sommerfeld theory) is a great improvement over the Drude-Lorentz electron theory. Yet, it has some drawbacks, the most serious of them being its inability to account for the existence of semiconductors. The periodic potential theory (also called the band theory) was a further improvement over the Sommerfeld theory. In particular, it provided for the existence of forbidden energy gaps between bands and explained the behaviour of semiconductors. We shall conclude this two-part discussion of the band theory of solids with a comparison of features of these two models/theories (Table 5.9).

5.8 Problems

1. Construct the third Brillouin zone for the square lattice.
2. Prove that Bragg's law is satisfied at any point on the upper edge of the first Brillouin zone of the square lattice.
3. Using the results of the tight binding model, obtain the energy for the line joining the points $(\pi/a, 0)$ and $(0, \pi/a)$ in the first Brillouin zone of a square lattice.
4. Construct the first Brillouin zone for the simple cubic lattice.
5. The energy of the valence band in silicon is represented by

$$E_{\mathbf{k}} = Ak^2 \pm \left[B^2 k^4 + C^2 \left(k_x^2 k_y^2 + k_y^2 k_z^2 + k_z^2 k_x^2 \right) \right]^{1/2}.$$

The constants have values $A = -4.1$, $|B| = 1.6$ and $|C| = 3.3$. Estimate the effective masses of the holes.

6. Cyclotron resonance is observed in a solid at a frequency $f_c (= \omega_c/2\pi) = 24,000$ mc/s and field $H = 860$ oersteds. Estimate the effective mass of the electron.

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Chapter 6

Physics of Semiconductors

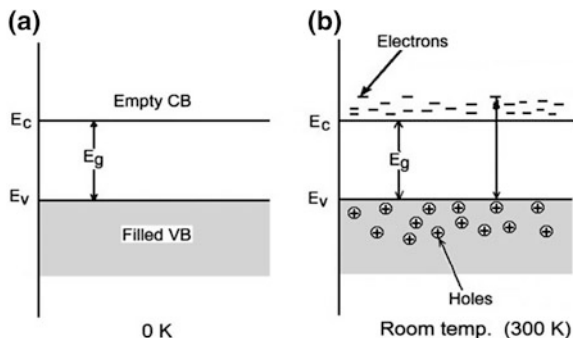
Abstract The most important feature of a semiconductor is that it has a small forbidden gap which enables an electron to jump from the valence band to the conduction band; this also gives rise to holes. There are different types of semiconductors. The behaviour of a semiconductor is complex, being dependent upon a number of factors like the energy gap, the effective masses, the mobilities, the concentration of donors and acceptors, the Fermi energy and the Fermi surface. Expressions are derived for the electrical conductivity of intrinsic and extrinsic semiconductors as a function of temperature. The temperature variation is exponential. Even a small concentration of chosen impurities brings about a large change in the conductivity. The conductivity of real crystals is anisotropic and tensorial in nature. Another important property of semiconductors is the Hall effect which has a different sign in *p*-type and *n*-type semiconductors. The dependence of the Hall effect on the shape of the constant energy surface is theoretically investigated. It is shown that the Hall effect and the magnetoresistance depend critically on the relaxation times. The experimental determination of mobility and its temperature dependence is discussed. The Haynes and Shockley experiment is described.

6.1 Introduction

A semiconductor is defined as a solid which has electrical conductivity intermediate between that of an insulator and a metal. More importantly, as temperature increases, the electrical conductivity of a semiconductor increases while that of a metal decreases. Such behaviour was, perhaps, observed long ago in many solids without realizing that the solid is a semiconductor. Thus, according to Ramakrishnan [1], Bose found that PbS is a good detector of microwaves without knowing that it is a semiconductor. Semiconductors developed an identity as a result of research during the second world war.

It is now known that semiconducting behaviour is due to a special band structure (Fig. 6.1a) in which, at $T = 0$, the valence band (VB) is completely filled, the

Fig. 6.1 Band structure of a semiconductor at **a** $T = 0$ and **b** $T > 0$



conduction band (CB) is completely empty and the two are separated by a narrow energy gap (E_g). The energy gap E_g is equal to the difference between the energy E_c at the bottom of the CB and the energy E_v at the top of the VB. At $T > 0$, the electrons at the top of the VB receive energy $\geq E_g$ and some of them are elevated into the lower region of the CB (Fig. 6.1b). Once they are in the CB, these electrons accept further energy and occupy higher levels in the CB; thus conduction takes place. The elevation of electrons from VB results in the creation of positively charged vacancies in the valence band; these are called 'holes' (Fig. 6.1b). The holes also contribute to conduction. As temperature increases (more energy becomes available), the concentration of electrons in CB and holes in the VB increases and hence the conductivity too increases. This increase in conductivity with temperature is exponential. A typical semilog plot of conductivity (σ) and temperature (T) is shown in Fig. 6.2; its slope is $E_g/2k_B$. Values of E_g for several semiconductors are given in Table 6.1. Semiconducting behaviour has been observed in a number of elements, inorganic compounds and even organics. However, elements with the diamond structure and compounds with the zinc blende structure have been most thoroughly investigated. Hence further discussion in this chapter will be mostly focused on these systems.

Fig. 6.2 Plot of $\log \sigma$ versus T^{-1} for an intrinsic semiconductor

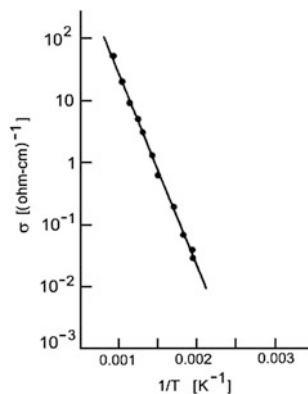


Table 6.1 Values of energy gap (E_g) for some semiconductors

Group	Crystal	E_g (eV)
IV	Si	1.1
	Ge	0.7
	α Sn	0.08
III–V	GaAs	1.4
	GaP	2.3
	GaSb	0.7
	InAs	0.4
	InP	1.3
	InSb	0.2
II–VI	CdS	2.6
	CdSe	1.7
	CdTe	1.5
	ZnS	3.6
	ZnSe	2.7
	ZnTe	2.3
IV–VI	PbS	0.4
	PbSe	0.3
	PbTe	0.3
Others	Ca ₂ Si	0.9
	Ca ₂ Sn	0.9
	Ca ₂ Pb	0.46
	AgI	2.8
	Ag ₂ Te	0.17
	Cu ₂ O	2.1
	Mg ₂ Si	0.7
	Mg ₂ Ge	0.7
	Mg ₂ Sn	0.3

Semiconductor science and technology now forms an independent discipline. The literature on the subject cannot be adequately enumerated. The physics of semiconductors has been discussed by McKelvey [2], Ashcroft and Mermin [3], Dekker [4], Levy [5] and Kittel [6] to name a few.

6.2 Types of Semiconductors

6.2.1 Intrinsic and Extrinsic Semiconductors

In Sect. 6.1 we have described how electrons are elevated into the CB due to thermal ionization of the upper electron energy levels in the VB and the consequent creation of holes. Pure semiconductors in which the electrical conduction is due to such thermally generated electrons and holes are called ‘intrinsic semiconductors’ and the conductivity due to this process is called ‘intrinsic conductivity’.

The conductivity of a semiconductor is substantially enhanced by the addition of chosen impurities even in small concentration. Let us consider the addition of a pentavalent impurity like phosphorous, arsenic or antimony to germanium. The valency of Ge is four. In the Ge structure (Fig. 6.3a) each Ge atom is covalently bonded to four neighbouring Ge atoms. Each pair of Ge atoms shares two electrons. If a pentavalent impurity, say As, is added to Ge, each impurity atom replaces one of the Ge atoms (Fig. 6.3b). It uses four of its electrons to bond with the neighbours. That leaves one surplus electron which is now only weakly attached to the impurity atom. This fifth electron is easily ionized by thermal energy and goes into the CB. On the other hand, if the impurity is a trivalent atom like B, Al, In, Ga, the three electrons of the impurity atom bond with three neighbours (Fig. 6.3c). The fourth incomplete bond is, in effect a hole in the system. In either case, the addition of impurities creates additional carriers (electrons or holes) in the host crystal resulting in larger conductivity. The larger the concentration of impurities the larger is the conductivity. Such semiconductors are called ‘impurity semiconductors’ or ‘extrinsic semiconductors’. If the additional carriers are electrons, the semiconductor is called ‘*n*-type’. If the additional carriers are holes, the semiconductor is called ‘*p*-type’. The impurities that give rise to additional electrons are called ‘donors’ and those that give rise to holes are called ‘acceptors.’

Let us now consider the binding energy (ionization energy) of the impurity atom vis-à-vis the host crystal. As an example, let us consider a pentavalent impurity in a

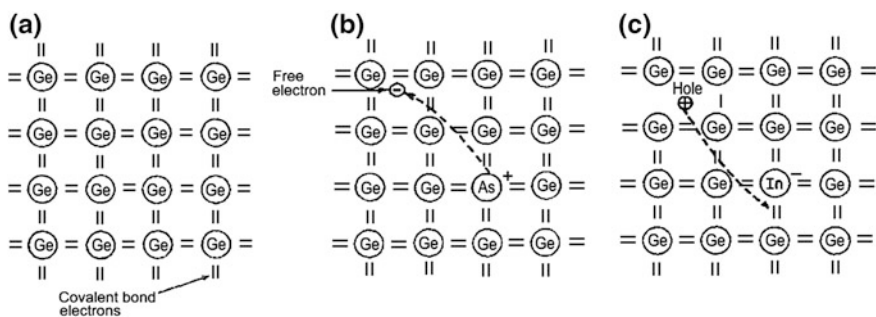


Fig. 6.3 Structure of Ge **a** pure **b** containing As^+ impurity (*n*-type) and **c** containing In^- impurity (*p*-type)

Ge crystal. We may imagine the fifth electron as rotating about the donor impurity which is now ionized. This is very much like an electron in a hydrogen atom. Using the Bohr theory of the hydrogen atom, the energy E' of the electron is

$$E' = -me^4/2\hbar^2n^2 \quad (6.1)$$

where the significance of the various quantities is well-known. Here, we have to make two modifications. Firstly, for an electron in an energy band, the mass m is to be replaced by the effective mass m_n^* . Secondly, the coulomb force in a medium is reduced by the dielectric constant ϵ of the host crystal; this results in a factor ϵ^2 in the denominator of (6.1). Thus the energy E' in (6.1) gets modified to E given by

$$E = m_n^*e^4/\epsilon^2\hbar^2n^2. \quad (6.2)$$

A similar calculation holds for the binding energy (ionization energy) of a hole created by a trivalent impurity. The ionization energies of some impurities in Si and Ge are given in Table 6.2. The donor and acceptor levels are shown in Fig. 6.4 vis-à-vis the energy bands in a semiconductor. The donor level is at a distance E_d (in terms of energy) below E_c (Fig. 6.4a) whereas the acceptor level is at a distance E_a above E_v (Fig. 6.4b). The donor energies as well as the acceptor energies are of the order of ~ 0.01 eV which is smaller than the thermal energy $k_B T$ at room temperature (~ 0.025 eV). Thus the electrons in donor levels are easily excited into the CB. Similarly, the electrons in VB are easily excited into the acceptor levels creating holes in the VB.

In Fig. 6.4, the energy bands of the donor and acceptor atoms are shown to lie between E_c and E_v . It was stated that an electron cannot have energy states in this region; hence the name ‘forbidden band’. This may appear contradictory. It may be clarified that the energy range from E_v to E_c is forbidden for the electrons of the host crystals and not for the impurity electrons. Application of the Bohr model also leads to the following expression for the radius r of an orbit of the impurity electron:

$$r = \epsilon n^2 \hbar^2 / e^2 m_n^*. \quad (6.3)$$

Table 6.2 Ionization energies (binding energies in eV) of donors and acceptors in Si and Ge

Impurity	Si	Ge	Impurity	Si	Ge
<i>Donors</i>			<i>Acceptors</i>		
P	0.044	0.012	B	0.045	0.010
As	0.049	0.013	Al	0.057	0.010
Sb	0.039	0.096	Ga	0.065	0.011
Bi	0.069		In	0.16	0.011

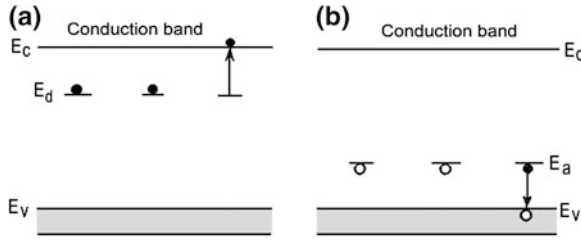


Fig. 6.4 Energy level diagram of a semiconductor containing **a** donor impurities at energy level E_d and **b** acceptor impurities at energy level E_a

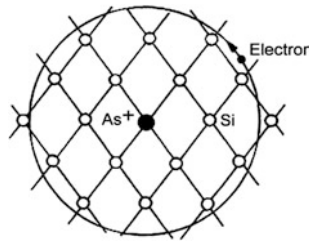


Fig. 6.5 The orbit of an electron around a donor impurity (As^+) in a semiconductor (Si)

The dielectric constant of Si is 11.7 and the effective mass of the electron in Si is $0.25 m$. With these values, the radius r of the first Bohr orbit calculated from (6.3) is $25 \text{ \AA}$. This is about 50 times the radius of the electron in the hydrogen atom which is $0.53 \text{ \AA}$. Further, this is about 20 times larger than the interatomic distance in Si which is $2.34 \text{ \AA}$. Thus, the orbit of the impurity electron includes several host atoms (Fig. 6.5).

6.2.2 Uniform and Nonuniform Semiconductors

If in a semiconductor the carriers (electrons/holes) are uniformly distributed, it is called a uniform semiconductor. On the other hand, if under some conditions, the carriers have a nonuniform distribution i.e. if the concentration of carriers is a function of coordinates, the semiconductor is called a nonuniform semiconductor. In a nonuniform semiconductor, it is implied that the concentration has a gradient. The existence of such concentration gradient gives rise to diffusion of carriers.

6.2.3 Direct Gap Semiconductors and Indirect Gap Semiconductors

Let us consider a semiconductor with a band structure shown in Fig. 6.6a. The minimum of CB is vertically above the maximum of VB i.e. both are at same $\mathbf{k}$. When an electron from the VB is excited into the CB the change in wave number $\Delta\mathbf{k} = 0$. If photons are employed to excite the electrons, the photon energy needed for such a transition is $\hbar\omega = E_g$. Such a semiconductor is called a ‘direct gap’ semiconductor. The group II-VI and group III-V compound semiconductors belong to this type.

Let us now consider a semiconductor with band structure shown in Fig. 6.6b. There are two minima in the CB with corresponding energy gaps E'_g and E_g . Energy gap E'_g corresponds to the transition from the maximum of the VB to the conduction band minimum which is vertically above the VB maximum. In this transition $\Delta\mathbf{k} = 0$. The other CB minimum has a $\mathbf{k}$ value different from that of the VB maximum. The transition of an electron to this second CB minimum involves change in energy as well as change in $\mathbf{k}$. The change in energy can be brought about by a photon but to bring about a change in $\mathbf{k}$ i.e. a change in momentum, a phonon is necessary. In Fig. 6.6b, $E'_g > E_g$. The first transition is possible provided a photon with energy E'_g is available. On the other hand, the second transition is possible only if a phonon is available to effect the change $\Delta\mathbf{k}$, in addition to energy E_g . Such semiconductors are called ‘indirect gap’ semiconductors e.g., Si and Ge.

The difference in direct gap semiconductors and indirect gap semiconductors is clearly seen in their optical behaviour. If photons are incident on a direct gap semiconductor, there will be no absorption of energy until the photon energy reaches a value $\hbar\omega = E_g$. For this and higher energies, the absorption coefficient α is large, the sudden increase taking place at $\hbar\omega = E_g$ (Fig. 6.7). In the indirect gap semiconductor, the increase in absorption takes place first at a frequency given by $\hbar\omega = E'_g$ and then at frequency $\hbar\omega = E_g$. Thus there are two absorption edges as seen in Fig. 6.8.

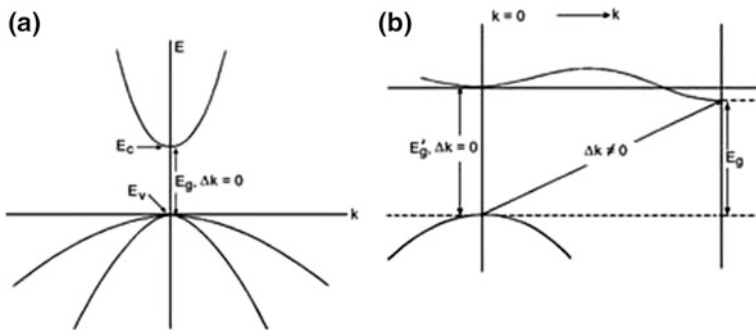


Fig. 6.6 Band structure of **a** a direct-gap semiconductor and **b** indirect gap semiconductor

Fig. 6.7 Optical absorption in a direct gap semiconductor (GaAs)

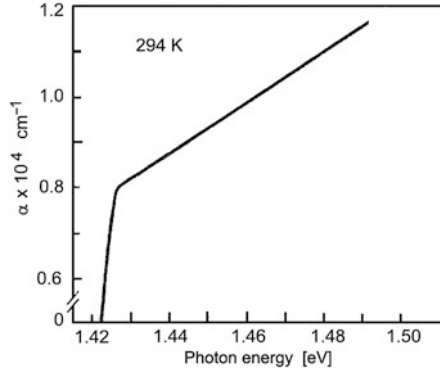
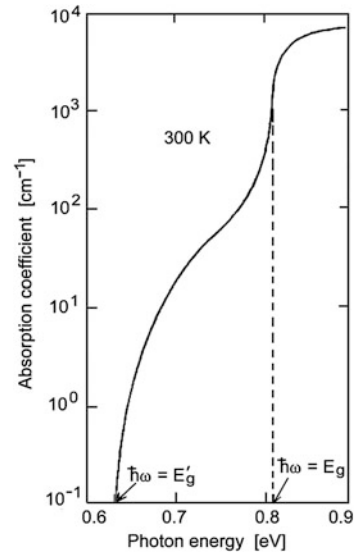


Fig. 6.8 Optical absorption in an indirect gap semiconductor (Ge)



6.3 General Physical Properties

We shall consider some general physical properties of semiconductors. As mentioned earlier, only semiconductors with the diamond structure and zinc blende structure will be considered.

6.3.1 Crystal Structure

The crystal structure of group IV elements is shown in Fig. 6.9a. As diamond has this structure, the structure of the whole group is called the diamond structure; the structure is fcc. The structure consists of two interpenetrating fcc lattices, each displaced with respect to the other by a quarter of a diagonal. The atomic coordinates are $(0, 0, 0)$, $(0, \frac{1}{2}, \frac{1}{2})$, $(\frac{1}{2}, 0, \frac{1}{2})$, $(\frac{1}{2}, \frac{1}{2}, 0)$, $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, $(\frac{1}{4}, \frac{3}{4}, \frac{3}{4})$, $(\frac{3}{4}, \frac{1}{4}, \frac{3}{4})$, and $(\frac{3}{4}, \frac{3}{4}, \frac{1}{4})$. The diamond structure has a centre of symmetry. Each atom has four nearest neighbours. The lattice constant of crystals with this structure are given in Table 6.3.

The crystal structure of cubic ZnS, known as the zinc blende structure, is shown in Fig. 6.9b. It consists of two interpenetrating fcc lattices, one occupied by only Zn atoms and the other by only S atoms. The two lattices are displaced with respect to the other by a quarter of a diagonal. The atomic coordinates are Zn: $(0, 0, 0)$, $(0, \frac{1}{2}, \frac{1}{2})$, $(\frac{1}{2}, 0, \frac{1}{2})$, $(\frac{1}{2}, \frac{1}{2}, 0)$ and S: $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, $(\frac{1}{4}, \frac{3}{4}, \frac{3}{4})$, $(\frac{3}{4}, \frac{1}{4}, \frac{3}{4})$, $(\frac{3}{4}, \frac{3}{4}, \frac{1}{4})$. Each Zn atom has four S nearest neighbours at the corners of a tetrahedron and vice versa. It can be seen that if all the atoms were identical, the structure is that of diamond. The zinc blende structure does not have a centre of symmetry. The lattice constants of some crystals with this structure are also given in Table 6.3.

Fig. 6.9 Crystal structure of **a** diamond and **b** zinc blende

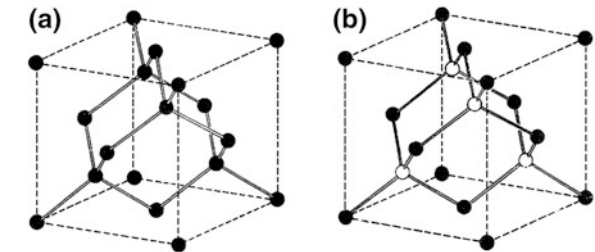


Table 6.3 Lattice constants (a) of some semiconductors

Structure	Crystal	a [Å]
Diamond	C	3.56
	Si	5.43
	Ge	5.65
	(α) Sn	6.46
Zinc blende	ZnS	5.41
	ZnSe	5.65
	GaP	5.45
	GaAs	5.65
	AlP	5.45
	AlAs	5.66
	InSb	6.46

6.3.2 Interatomic Binding

In the free state, the atoms of the group IV elements have two s electrons and two p electrons. When they become a part of the diamond structure, these four electrons form covalent bonds in pairs with the four neighbouring atoms. These bonds are very strong. The situation is similar in the zinc blende structure i.e. the Zn-S bonds are essentially covalent. But there is evidence that the electron sharing is not complete. Pauling's ionicity values and Szigeti's effective ionic charge values indicate that the bonds in the II-VI and III-V compounds are partially covalent (or partially ionic).

The existence of covalency in group IV elemental semiconductors and group II-VI, group III-V compound semiconductors is very important as the electrons in the valence band get excited by the breaking of these bonds.

6.3.3 Band Structure

The band structures of Ge, Si and GaAs have been discussed in detail in Chap. 5. Here, we shall briefly mention the main features. Figure 6.10a, b show the band structures of Si and Ge. In the light of the discussion in Sect. 6.2, it can be seen that both Si and Ge are indirect gap semiconductors. The minimum E_g (difference between the conduction band edge and valence band edge) has values 1.17 eV and 0.7 eV for Si and Ge but the transitions are at $\Delta\mathbf{k} \neq 0$. On the other hand, the band structure of GaAs (Fig. 6.10c) corresponds to a direct gap semiconductor. The value of E_g for $\Delta\mathbf{k} = 0$ is 1.4 eV.

The notable feature in all the three cases is that the highest bands in VB converge at $k = 0$ but are separate at $|k| > 0$. They correspond to heavy and light holes.

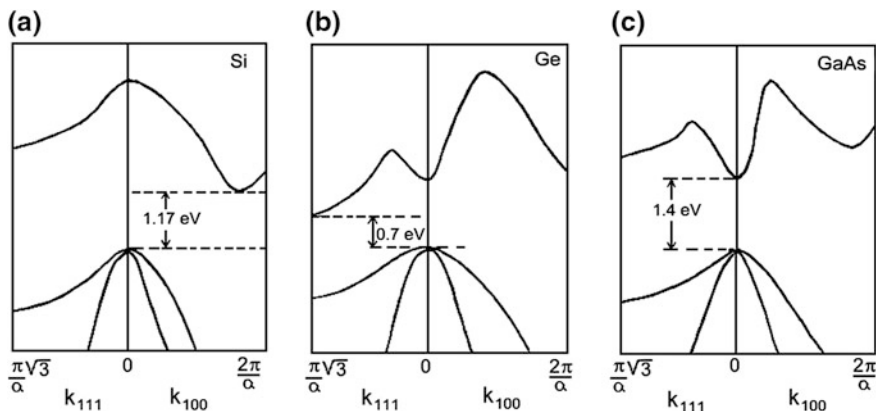


Fig. 6.10 Band structure of **a** Si, **b** Ge and **c** GaAs

Table 6.4 Effective masses of carriers in some semiconductors

Group	Crystal	Effective mass (in units of m)			
		Electrons (m_n^*)		Holes (m_p^*)	
III–V	GaAs	0.07		0.09	
	GaP	0.12		0.50	
	GaSb	0.20		0.39	
	InAs	0.03		0.02	
	InP	0.07		0.69	
	InSb	0.01		0.18	
II–VI	CdS	0.21		0.80	
	CdSe	0.13		0.45	
	CdTe	0.14		0.37	
	ZnS	0.40		5.41	
	ZnSe	0.10		0.60	
	ZnTe	0.10		0.60	
IV–VI	PbS	0.25		0.25	
	PbSe	0.33		0.34	
	PbTe	0.22		0.29	
IV		$m_\perp^*$	$m_\parallel^*$	m_{hh}^*	m_{lh}^*
	Si	0.97	0.19	0.5	0.16
	Ge	1.6	0.08	0.3	0.04

6.3.4 Effective Masses

The effective masses of several semiconductors are given in Table 6.4. The Fermi surfaces of Si and Ge have been discussed in Chap. 5. Due to the ellipsoidal shapes of the Fermi surface, there are two effective masses $m_\parallel^*$ and $m_\perp^*$ parallel and perpendicular to the long axes of the ellipsoids. Also, because of the splitting of the top-most level in the VB, there are two effective masses m_{hh}^* and m_{lh}^* corresponding to heavy holes and light holes. Values of these different types of effective masses are included in Table 6.4.

6.4 Electrical Conductivity of Semiconductors

In this section we shall consider the electrical conductivities of intrinsic and extrinsic semiconductors and also the anisotropy of conductivity.

6.4.1 Conductivity of Intrinsic Semiconductors

Let us consider an intrinsic semiconductor with the band structure shown in Fig. 6.11. At $T > 0$, such a semiconductor contains electrons in the CB and holes in the VB. The effect of an applied electric field E on an electron and a hole is shown in Fig. 6.12. Although the velocities of the two are in opposite directions, the current densities are in the same direction. In the following, we shall denote the concentrations of electrons and holes by n and p , their velocities by $\mathbf{v}_n$ and $\mathbf{v}_p$, the current densities by $\mathbf{J}_n$ and $\mathbf{J}_p$, their mobilities (velocity for unit field) by μ_n and μ_p and their contributions to the conductivity by σ_n and σ_p .

The current density due to electrons may be expressed as

$$\mathbf{J}_n = ne\mathbf{v}_n = ne\mu_n\mathbf{E}. \quad (6.4)$$

Since $\mathbf{J} = \sigma \mathbf{E}$, (6.4) leads to

$$\sigma_n = ne\mu_n. \quad (6.5)$$

Similarly, the conductivity due to holes may be written as

$$\sigma_p = pe\mu_p. \quad (6.6)$$

Fig. 6.11 Band structure of an intrinsic semiconductor

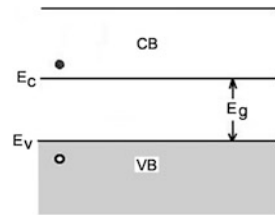
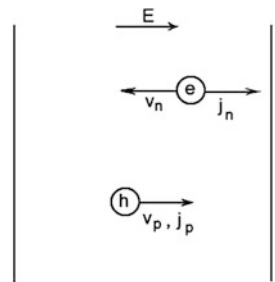


Fig. 6.12 Motion of electrons and holes in an electric field



The total conductivity σ is

$$\sigma = \sigma_n + \sigma_p = e(n\mu_n + p\mu_p). \quad (6.7)$$

For an intrinsic semiconductor, $n = p = n_i$. Hence (6.7) becomes

$$\sigma = en_i(\mu_n + \mu_p). \quad (6.8)$$

We shall now proceed to calculate the equilibrium concentration of electrons and holes in an intrinsic semiconductor.

From Chap. 3, we have

$$dn = D(E)f(E)dE. \quad (6.9)$$

where dn is the number of electrons in energy range from E to $E + dE$, $D(E)$ the density of states and $f(E)$ the Fermi-Dirac distribution function. We also have

$$D(E) = (4\pi/h^3)(2m)^{3/2}E^{1/2}. \quad (6.10)$$

and

$$f(E) = \frac{1}{\exp\left(\frac{E-E_F}{k_B T}\right) + 1}. \quad (6.11)$$

Substituting (6.10) and (6.11) in (6.9) and noting that for an electron in the CB the effective energy is not E but $E - E_c$, we get

$$dn = \left\{ (4\pi/h^3)(2m_n^*)^{3/2}(E - E_c)^{1/2} \right\} \left/ \left[\exp\left(\frac{E - E_F}{k_B T}\right) + 1 \right] \right\} dE. \quad (6.12)$$

Note that we have replaced m in (6.10) by m_n^* , the effective mass of the electron in the conduction band. The total number of electrons n in CB can be obtained from (6.12) by integrating between the limits $E = E_c$ to $E = \infty$. Thus,

$$n = (4\pi/h^3)(2m_n^*)^{3/2} \int_{E_c}^{\infty} \left\{ (E - E_c)^{1/2} \right\} \left/ \left[\exp\left(\frac{E - E_F}{k_B T}\right) + 1 \right] \right\} dE. \quad (6.13)$$

Since $k_B T \approx 0.025$ at room temperature, we shall ignore 1 in the square bracket. Thus

$$\left[\exp\left(\frac{E - E_F}{k_B T}\right) + 1 \right] \approx \exp\left(\frac{E - E_F}{k_B T}\right). \quad (6.14)$$

The significance of this approximation is that instead of the Fermi-Dirac expression, we are using the Maxwell-Boltzmann expression for $f(E)$. This makes things simpler. With this approximation, (6.13) becomes

$$\begin{aligned} n &= (4\pi/h^3)(2m_n^*)^{3/2} \int_{E_c}^{\infty} (E - E_c)^{1/2} \exp\left[-\left(\frac{E - E_F}{k_B T}\right)\right] dE \\ &= (4\pi/h^3)(2m_n^*)^{3/2} \exp\left(\frac{E_F - E_c}{k_B T}\right) \int_{E_c}^{\infty} (E - E_c)^{1/2} \exp\left[-\left(\frac{E - E_c}{k_B T}\right)\right] dE. \end{aligned} \quad (6.15)$$

Let us introduce a variable x given by

$$x = \frac{E - E_c}{k_B T}. \quad (6.16)$$

Then,

$$dE = k_B T dx \text{ and } x = 0 \text{ at } E = E_c. \quad (6.17)$$

With these substitutions, (6.15) becomes

$$\begin{aligned} n &= (4\pi/h^3)(2m_n^*)^{3/2} \exp\left(\frac{E_F - E_c}{k_B T}\right) \int_0^{\infty} x^{1/2} (k_B T)^{1/2} e^{-x} (k_B T) dx \\ &= (4\pi/h^3)(2m_n^* k_B T)^{3/2} \exp\left(\frac{E_F - E_c}{k_B T}\right) \int_0^{\infty} x^{1/2} e^{-x} dx. \end{aligned} \quad (6.18)$$

The integral in (6.18) is a standard integral with value $(\pi/4)^{1/2}$. Thus, (6.18) becomes

$$\begin{aligned} n &= 2 \left(\frac{2\pi m_n^* k_B T}{h^2} \right)^{3/2} \exp\left[-\left(\frac{E_c - E_F}{k_B T}\right)\right] \\ &= N_n \exp\left[-\left(\frac{E_c - E_F}{k_B T}\right)\right] \end{aligned} \quad (6.19)$$

where

$$N_n = 2 \left(\frac{2\pi m_n^* k_B T}{h^2} \right)^{3/2}. \quad (6.20)$$

Substituting an appropriate value for m_n^* , we get

$$N_n = 2.8 \times 10^{25} \left(\frac{T}{300} \right)^{3/2} \text{ m}^{-3} \quad \text{for Si.} \quad (6.21)$$

Let us consider the significance of N_n . From (6.11) and (6.14), the probability of occupancy of level E_c is

$$f(E_c) = 1 / \left[\exp \left(\frac{E_c - E_F}{k_B T} \right) + 1 \right] \approx \exp \left[- \left(\frac{E_c - E_F}{k_B T} \right) \right]. \quad (6.22)$$

We may rewrite (6.19) as

$$n(\text{at } E = E_c) = N_n f(E_c). \quad (6.23)$$

Comparing (6.23) with (6.9), we conclude that N_n is the effective density of states of electrons at the conduction band edge.

Let us now consider the concentration of holes in the VB. The procedure is very similar to that for electrons in the CB but with necessary differences. Firstly, if $f(E)$ was the probability of a state being occupied by an electron, the probability of a state not being occupied (i.e. for a hole to exist) is $[1 - f(E)]$. This probability is:

$$\begin{aligned} [1 - f(E)] &= 1 - \left\{ 1 / \left[1 + \exp \left(\frac{E - E_F}{k_B T} \right) \right] \right\} \\ &= \exp \left(\frac{E - E_F}{k_B T} \right) / \left[1 + \exp \left(\frac{E - E_F}{k_B T} \right) \right]. \end{aligned} \quad (6.24)$$

Noting that $E \ll E_F$ in the valence band, we shall ignore the exponential term in the denominator of (6.24) in comparison with 1. Then (6.24) becomes

$$[1 - f(E)] \approx \exp \left(\frac{E - E_F}{k_B T} \right). \quad (6.25)$$

Secondly, the effective energy of a hole in a VB is $(E_v - E)$. Thirdly, the effective mass of a hole is m_p^* .

The number of holes dp with energy range E to $E + dE$ can now be written as

$$\begin{aligned} dp &= D(E)[1 - f(E)]dE \\ &= (4\pi/h^3)(2m_p^*)^{3/2}(E_v - E)^{1/2} \exp \left(\frac{E - E_F}{k_B T} \right) dE. \end{aligned} \quad (6.26)$$

To obtain the total concentration of holes, p in the valence band, we shall integrate (6.26) between the limits $E = -\infty$ and $E = E_v$. Thus, we have

$$\begin{aligned}
p &= (4\pi/h^3)(2m_p^*)^{3/2} \int_{-\infty}^{E_v} (E_v - E)^{1/2} \exp\left(\frac{E - E_F}{k_B T}\right) dE \\
&= (4\pi/h^3)(2m_p^*)^{3/2} \exp\left(\frac{E_v - E_F}{k_B T}\right) \int_{-\infty}^{E_v} (E_v - E)^{1/2} \exp\left(\frac{E - E_v}{k_B T}\right) dE.
\end{aligned} \tag{6.27}$$

Let us introduce

$$x = \frac{E_v - E}{k_B T}. \tag{6.28}$$

Then,

$$dE = -k_B T dx \text{ and } x = 0 \text{ at } E = E_v. \tag{6.29}$$

Substituting (6.28) and (6.29) in (6.27), we get

$$p = (4\pi/h^3)(2m_p^* k_B T)^{3/2} \exp\left(\frac{E_v - E_F}{k_B T}\right) \int_0^{\infty} x^{1/2} e^{-x} dx. \tag{6.30}$$

As mentioned earlier, the value of the integral is $(\pi/4)^{1/2}$. Substituting this value in (6.30),

$$\begin{aligned}
p &= 2 \left(\frac{2\pi m_p^* k_B T}{h^2} \right)^{3/2} \exp\left[-\left(\frac{E_F - E_v}{k_B T}\right)\right] \\
&= N_p \exp\left[-\left(\frac{E_F - E_v}{k_B T}\right)\right]
\end{aligned} \tag{6.31}$$

where

$$N_p = 2 \left(\frac{2\pi m_p^* k_B T}{h^2} \right)^{3/2}. \tag{6.32}$$

For Si, (6.32) gives

$$N_p = 2.8 \times 10^{25} \left(\frac{T}{300} \right)^{3/2} \text{ m}^{-3}. \tag{6.33}$$

As in the case of N_n , it can be shown that N_p is the density of holes at the valence band edge. In (6.21) and (6.32) it is assumed that $m_n^* \approx m_p^*$.

In an intrinsic semiconductor $n = p$. Further, the product of the electron concentration and hole concentration is constant. Thus

$$np = \text{a constant} = n_i \quad (6.34)$$

This is a consequence of the law of mass action; n_i is called the intrinsic concentration. From (6.19), (6.31) and (6.34), we have

$$n_i^2 = np = \left\{ 2 \left(\frac{2\pi m_n^* k_B T}{h^2} \right)^{3/2} \exp \left[- \left(\frac{E_c - E_F}{k_B T} \right) \right] \right\} \left\{ 2 \left(\frac{2\pi m_p^* k_B T}{h^2} \right)^{3/2} \exp \left[- \left(\frac{E_F - E_v}{k_B T} \right) \right] \right\}$$

or

$$n_i = 2 \left(\frac{2\pi k_B T}{h^2} \right)^{3/2} (m_n^* m_p^*)^{3/4} \exp \left[- \left(\frac{E_g}{2k_B T} \right) \right] \quad (6.35)$$

where $E_g = (E_c - E_v)$ which is the energy gap.

Some numerical calculations are instructive. If we substitute for m_n^* , m_p^* and E_g for Ge in (6.35), we get $N_i \approx 2.4 \times 10^{19} \text{ m}^{-3}$. The electron concentration in Ge is $4.4 \times 10^{28} \text{ m}^{-3}$. Thus only about 5 bonds out of 10^{10} are broken to contribute to intrinsic conduction in Ge. This may be compared to metals where the number of conduction electrons is $\sim 10^{28}/\text{m}^3$.

Let us consider the intrinsic conductivity σ_i . According to (6.8),

$$\sigma_i = en_i(\mu_n + \mu_p) = 2e \left(\frac{2\pi k_B T}{h^2} \right)^{3/2} (m_n^* m_p^*)^{3/4} (\mu_n + \mu_p) \exp \left(\frac{-E_g}{2k_B T} \right). \quad (6.36)$$

The temperature variation of $(\mu_n + \mu_p)$ almost cancels the variation of $T^{3/2}$. Thus, σ_i essentially varies as $\exp(-E_g/2k_B T)$ and a plot of $\log \sigma_i$ against T^{-1} should be linear with a slope of $E_g/2k_B$. Such plots for some semiconductors are shown in Fig. 6.13; in the plot, instead of $\log \sigma$, the values of $\log \rho$ have been plotted where ρ is the resistivity ($=1/\sigma$).

Finally, let us consider the Fermi level E_F . In an intrinsic semiconductor $n = p$. Equating (6.19) and (6.31), we get

$$N_n \exp \left[- \left(\frac{E_c - E_F}{k_B T} \right) \right] = N_p \exp \left[- \left(\frac{E_F - E_v}{k_B T} \right) \right]. \quad (6.37)$$

Solving (6.37) for E_F , we get

$$E_F = \left(\frac{E_c + E_v}{2} \right) + \frac{3k_B T}{4} \log \left(\frac{m_p^*}{m_n^*} \right). \quad (6.38)$$

From (6.38), we note the following features which are shown in Fig. 6.14:

Fig. 6.13 Semi-log plot of resistivity versus T^{-1} for some intrinsic semiconductors

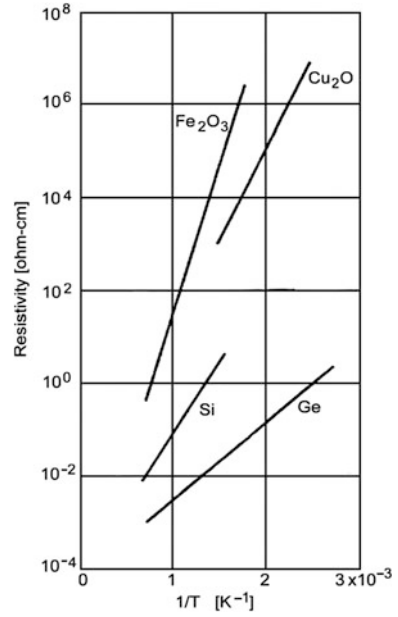
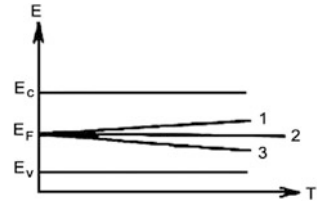


Fig. 6.14 Variation of Fermi level with temperature (not to scale) according to (6.38)



- (i) At $T = 0$, and if $m_n^* = m_p^*$ for other temperatures also, $E_F = (E_c + E_v)/2$, i.e. the Fermi level is located midway between the conduction band edge and the valence band edge (curve 2).
- (ii) At $T \neq 0$, E_F shows a very slight increase if $m_p^* > m_n^*$ (curve 1) and a very slight decrease if $m_p^* < m_n^*$ (curve 3).

It may be noted that some authors [7] choose E_v as the zero of the energy scale. Then, the first term in (6.38) becomes $(E_g/2)$.

Let us consider the Fermi level in a semiconductor from a conceptual point of view. In Chap. 3, we saw that the Fermi level in a metal is defined as (i) the energy for which the electron occupancy probability $f(E)$ has the value $1/2$ and also (ii) the highest energy level occupied by electrons at $T = 0$. Thus all levels below E_F are full and all levels above it are empty. In a semiconductor, the first definition still holds

but the second does not. In a metal E_F is at the edge of the valence band whereas in a semiconductor it lies between E_v and E_c , a region banned for electrons in a metal. E_F is still a level above the filled region and below the empty region, but it is well away from both E_c and E_v and its position is not uniquely defined. In fact, to quote Omar [7], in a semiconductor the Fermi level “is a theoretical concept and no electron need be present there”.

6.4.2 Conductivity of Extrinsic Semiconductors

A model of a semiconductor with donor impurities is shown in Fig. 6.15. As discussed in Sect. 6.2.1, the energy level E_d of the impurity is close to the energy E_c at the bottom of the conduction band and the Fermi level is located between E_c and E_d . At $T = 0$, all the donor levels (N_d) are occupied by electrons (Fig. 6.15a) whereas at $T > 0$, some donor levels (N_d^+) get ionized and the released electrons jump into the CB (Fig. 6.15b).

The electron concentration in the CB is still given by

$$n = N_n \exp \left[- \left(\frac{E_c - E_F}{k_B T} \right) \right]. \quad (6.19)$$

There are two contributions to the number of electrons in the CB: the electrons released by the donor atoms (N_d^+) and the electrons which have originated in the VB (their number being equal to the number of holes p). At this stage, the second contribution will be neglected. Then

$$n = N_d^+. \quad (6.39)$$

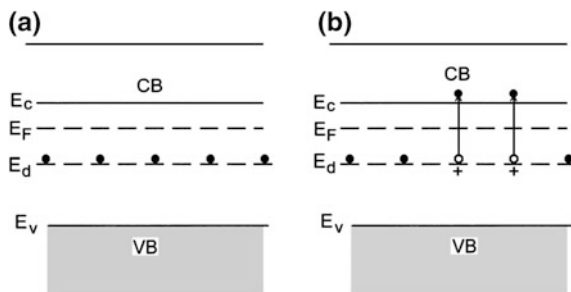


Fig. 6.15 Band structure of an n -type semiconductor **a** at $T = 0$, when the donor levels are occupied by impurity atoms and **b** at $T > 0$, when some of the donor atoms are ionized and their electrons are excited into the conduction band

From the law of mass action, we have

$$(N_d^+ p) = n_i^2 \text{ or } p = n_i^2 / N_d^+. \quad (6.40)$$

The concentration of ionized donors is

$$\begin{aligned} N_d^+ &= N_d [1 - f(E_d)] \\ &= N_d \left\{ 1 - \left[1 / 1 + \exp - \left(\frac{E_F - E_d}{k_B T} \right) \right] \right\} \\ &= N_d \left[\exp - \left(\frac{E_F - E_d}{k_B T} \right) / \left\{ 1 + \exp - \left(\frac{E_F - E_d}{k_B T} \right) \right\} \right]. \end{aligned} \quad (6.41)$$

Here we shall neglect the exponential term in comparison with unity. Then (6.41) becomes

$$N_d^+ = N_d \exp - \left(\frac{E_F - E_d}{k_B T} \right) \quad (6.42)$$

From (6.19), (6.39) and (6.42), we have

$$N_n \exp \left[- \left(\frac{E_c - E_F}{k_B T} \right) \right] = N_d \exp - \left(\frac{E_F - E_d}{k_B T} \right) \quad (6.43)$$

Solving (6.43) for E_F , we get,

$$E_F = \frac{E_c + E_d}{2} + \frac{k_B T}{2} \log \left(\frac{N_d}{N_n} \right). \quad (6.44)$$

We shall now note the following features from (6.44): (i) At $T = 0$, the Fermi level is midway between E_c and E_d ; (ii) at $T > 0$, E_F rises slowly as the temperature increases and (iii) (6.44) is valid only at small values of T ; it takes a complicated form at higher T [2].

Substituting (6.44) in (6.19) and reorganizing, we get

$$n = (N_n N_d)^{1/2} \exp \left(- \frac{\Delta E}{2 k_B T} \right) \quad (6.45)$$

where

$$\Delta E = E_c - E_d. \quad (6.46)$$

The conductivity of such a n -type semiconductor with a dopant concentration N_d is given by

$$\sigma_n = e\mu_n n = (e\mu_n)(N_n N_d)^{1/2} \exp\left(-\frac{\Delta E}{2k_B T}\right) \quad (6.47)$$

and

$$\log \sigma_n = \frac{1}{2} \log(e^2 \mu_n^2 N_n N_d) - \left(\frac{\Delta E}{2k_B}\right) T^{-1}. \quad (6.48)$$

Taking μ_n for Ge as $0.39 \text{ cm}^2/\text{V}\cdot\text{s}$ and the intrinsic carrier concentration as $7.2 \times 10^{-19} \text{ m}^{-3}$, the intrinsic conductivity σ_i is $6.68 \text{ } \Omega^{-1}\text{m}^{-1}$. If 1 ppm of a pentavalent impurity like P is added we get $n = 4.4 \times 10^{22} \text{ m}^{-3}$. Ignoring hole contribution, the extrinsic conductivity calculated from (6.47) is $2,760 \text{ } \Omega^{-1}\text{m}^{-1}$. Thus the conductivity has increased by a factor of 400.

A qualitative plot of $\log \sigma_n$ vs T^{-1} is shown in Fig. 6.16. It consists of three regions: region I from A to B, region II from B to C and region III from C to D. In region I (curve AB), the conductivity increases with increasing temperature, the conductivity is mainly due to the impurity electrons. Hence it is determined by N_d ; the curve in this region has a slope $\tan^{-1}(\Delta E/2k_B)$. At B, all the donor atoms have been ionized and N_d^+ takes the maximum value. As the temperature increases the conductivity cannot increase any more as N_d^+ cannot increase. In region II (curve BC), the temperature variation of σ is determined by the temperature variation of the mobility μ_n ; μ_n varies as $T^{-3/2}$. Thus, as temperature increases, σ decreases. By the time we reach region III, the only factor that can provide electrons is the intrinsic process which dominates over the other processes. The number of electrons originating in the VB is large and σ increases with increasing temperature. The slope of curve CD is $\tan^{-1}(E_g/2k_B)$. Figure 6.17 shows conductivity curves for a number of samples of Ge doped with increasing concentration of As impurity. The trends in the experimental curves agree with those in Fig. 6.16.

Fig. 6.16 Temperature variation of electrical conductivity of an *n*-type semiconductor according to (6.48)

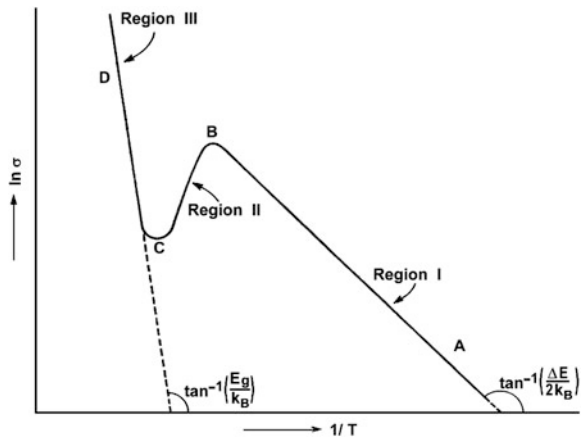
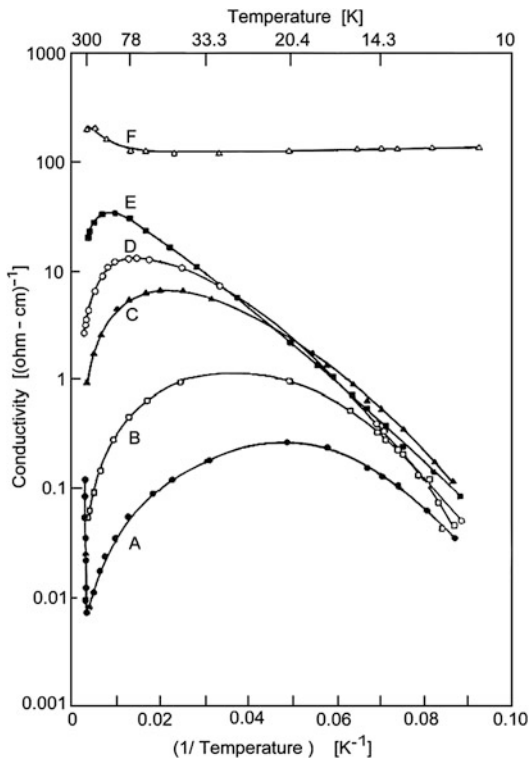


Fig. 6.17 Experimentally observed temperature variation of the electrical conductivity of germanium containing arsenic in different concentrations/cm³ (A = 10¹³, B = 10¹⁴, C = 10¹⁵, D = 10¹⁶, E = 10¹⁷, F = 10¹⁸)



Let us now consider a *p*-type semiconductor containing a trivalent impurity. The energy level diagram is shown in Fig. 6.18. The acceptor impurities occupy energy levels E_a above the valence band edge energy E_v . E_F is between E_a and E_v . At $T = 0$, the impurity levels are occupied by the impurity atoms (Fig. 6.18a) at $T > 0$, some electrons from the VB jump into the impurity levels which are now ionized (Fig. 6.18b); in this process, holes are created in the VB. Some holes are also present in the VB because of electrons getting excited to the CB. Let us denote the electron concentration in CB by n , hole concentration in VB by p , total acceptor concentration by N_a and concentration of ionized acceptor levels by N_a^- . We have

$$p = n + N_a^- \quad (6.49)$$

At low temperatures $N_a^- > n$. Hence (6.49) can be written as

$$p \approx N_a^-. \quad (6.50)$$

The concentration of ionized impurity levels is given by

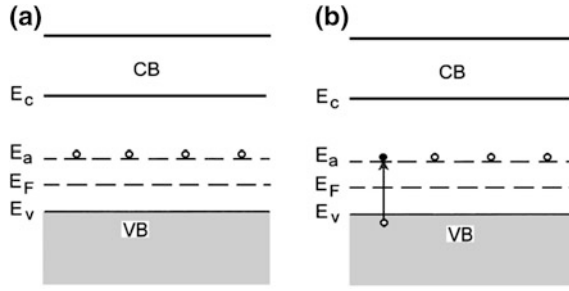


Fig. 6.18 Band structure of *p*-type semiconductor **a** at $T = 0$, when acceptor levels are occupied by impurity atoms and **b** at $T > 0$, when some of the acceptor atoms are ionized by electrons from the valence band causing holes in the VB

$$\begin{aligned} N_a^- &= N_a f(E_a) \\ &= N_a \left/ \left[1 + \exp\left(\frac{E_a - E_F}{k_B T}\right) \right] \right. \end{aligned} \quad (6.51)$$

Neglecting 1 in the denominator, (6.51) becomes

$$N_a^- = N_a \exp\left(\frac{E_F - E_a}{k_B T}\right). \quad (6.52)$$

The hole concentration in the VB is given by

$$p = N_p \exp\left[-\left(\frac{E_F - E_v}{k_B T}\right)\right]. \quad (6.31)$$

In view of (6.50), we shall equate (6.31) and (6.52) giving

$$N_p \exp\left[-\left(\frac{E_F - E_v}{k_B T}\right)\right] = N_a \exp\left(\frac{E_F - E_a}{k_B T}\right). \quad (6.53)$$

On reorganization, we get

$$E_F = \frac{E_a + E_v}{2} - \frac{k_B T}{2} \log\left(\frac{N_a}{N_p}\right). \quad (6.54)$$

The Fermi level is midway between E_a and E_v and shows a slight decrease as T increases.

Substituting (6.54) in (6.31) and reorganizing, we get

$$p = (N_a N_p)^{1/2} \exp\left(-\frac{\Delta E'}{2k_B T}\right). \quad (6.55)$$

where

$$\Delta E' = E_a - E_v. \quad (6.56)$$

Finally, we have the conductivity of a p -type semiconductor as

$$\sigma_p = e p \mu_p = (e \mu_p) (N_a N_p)^{1/2} \exp\left(-\frac{\Delta E'}{2k_B T}\right) \quad (6.57)$$

Thus, the temperature variation of σ_p is similar to that of σ_n .

6.4.3 Anisotropy of Conductivity

We have seen in Chaps. 2 and 3 that the isotropic electrical conductivity is given by

$$\sigma = (n e^2 \tau) \left(\frac{1}{m}\right) \quad (6.58)$$

where n is the electron concentration, τ the relaxation time and m the electron mass. In Chap. 4, we noted that $(1/m)$ is replaced by $[1/m^*]_{\alpha\beta}$ which is called the reciprocal effective mass tensor. Further, in Chap. 5 we noted that in some semiconductors like Si, the Fermi surface consists of ellipsoids and (6.58) is written as

$$[\sigma^{(i)}] = n^{(i)} e^2 \tau \left[\frac{1}{m^*} \right]_{\alpha\beta} \quad (6.59)$$

where (i) represents a part of the Fermi surface or a direction and $[\sigma^{(i)}]$ is called the conductivity tensor.

For the two ellipsoids whose major axes are along the x -axes, we may write (6.59) in the expanded form as

$$[\sigma^{(100)}] = n^{(100)} e^2 \tau \begin{bmatrix} \frac{1}{m_{\parallel}^*} & \cdot & \cdot \\ \cdot & \frac{1}{m_{\perp}^*} & \cdot \\ \cdot & \cdot & \frac{1}{m_{\perp}^*} \end{bmatrix} \quad (6.60)$$

where $m_{\parallel}^*$ is the effective mass along the major axis and $m_{\perp}^*$ that perpendicular to the major axis. In the same way, for the two ellipsoids with major axes along the y axis, we have

$$\left[\sigma^{(010)} \right] = n^{(010)} e^2 \tau \begin{bmatrix} \frac{1}{m_{\perp}^*} & \cdot & \cdot \\ \cdot & \frac{1}{m_{\parallel}^*} & \cdot \\ \cdot & \cdot & \frac{1}{m_{\perp}^*} \end{bmatrix} \quad (6.61)$$

And finally, for the ellipsoids with major axes along the z -axis, we have

$$\left[\sigma^{(001)} \right] = n^{(001)} e^2 \tau \begin{bmatrix} \frac{1}{m_{\perp}^*} & \cdot & \cdot \\ \cdot & \frac{1}{m_{\perp}^*} & \cdot \\ \cdot & \cdot & \frac{1}{m_{\parallel}^*} \end{bmatrix} \quad (6.62)$$

Since all ellipsoids are equivalent, the carrier densities are equal i.e. $n^{(100)} = n^{(\bar{1}00)} = n^{(010)} = n^{(0\bar{1}0)} = n^{(001)} = n^{(00\bar{1})} = n_0/6$ where n_0 is the overall carrier density. The total conductivity σ_0 is

$$[\sigma_0] = n_0 e^2 \tau \begin{bmatrix} \frac{1}{3} \left(\frac{2}{m_{\perp}^*} + \frac{1}{m_{\parallel}^*} \right) & \cdot & \cdot \\ \cdot & \frac{1}{3} \left(\frac{2}{m_{\perp}^*} + \frac{1}{m_{\parallel}^*} \right) & \cdot \\ \cdot & \cdot & \frac{1}{3} \left(\frac{2}{m_{\perp}^*} + \frac{1}{m_{\parallel}^*} \right) \end{bmatrix} \quad (6.63)$$

The scalar magnitude of the total conductivity is

$$\sigma_0 = \frac{n_0 e^2 \tau}{m_c^*} \quad (6.64)$$

where

$$\frac{1}{m_c^*} = \frac{1}{3} \left(\frac{2}{m_{\perp}^*} + \frac{1}{m_{\parallel}^*} \right) \quad (6.65)$$

m_c^* is called the conductivity effective mass.

6.5 Hall Effect in Semiconductors

In the free electron theory for metals (Chaps. 2 and 3), it was shown that the Hall coefficient R for metals is

$$R = -(1/nec) \quad (6.66)$$

where n is the electron density (number of electrons/unit volume); for metals R is negative. But it was observed in experiments that for some metals R is positive. In the band theory of solids (Chaps. 4 and 5), we saw that when an electron from the valence band gets excited into the conduction band, the electron vacancy is equivalent to a positively charged particle which is called a hole. It was pointed out that if only the holes are carriers, the Hall coefficient is

$$R = +(1/pec) \quad (6.67)$$

where p is the hole density. Note that R is positive in this case.

These derivations were based on several assumptions: (i) that there is a single velocity for all electrons, (ii) that the carriers are all electrons or all holes, (iii) that the constant energy surface is spherical and (iv) that the electrons have an isotropic mass. For a proper description of the Hall effect in semiconductors, it is necessary to remove at least some of these assumptions and obtain an expression for the Hall coefficient more rigorously. Detailed treatment of Hall effect in semiconductors is given by McKelvey [2] and Dekker [4].

To start with, we shall still assume that (i) the constant energy surfaces are spherical, (ii) the effective mass is isotropic and (iii) there is only one type of carrier (a hole) and remove these assumptions later. At this stage, we shall remove the assumption of a common velocity. This we shall do in two stages. Firstly, we shall bring in the concept of relaxation time and shall take the time average of velocity. Secondly, we shall bring in the concept of a velocity distribution i.e. the carriers have a range (distribution) of velocities; we should take the average of these velocities. For this purpose, we shall assume the Maxwell-Boltzmann velocity distribution.

Before proceeding further, we shall mention for the clarity of readers that we have used the symbol E for energy in the earlier sections. We shall continue to do that. But we shall also use $\mathbf{E}$ as the vector representing the electric field. This we have done to follow convention. Since the two quantities do not occur together, there should be no confusion.

6.5.1 Hall Effect in Semiconductors with Spherical Energy Surfaces

Let us consider a semiconductor with spherical energy surfaces. Let the semiconductor have only holes as carriers. Figure 6.19 shows the semiconductor sample to which an electric field $\mathbf{E}$ and magnetic field $\mathbf{H}$ are applied. Due to the electric field,

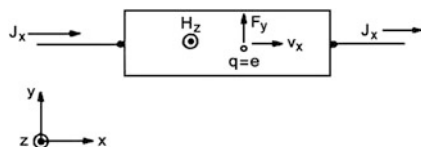


Fig. 6.19 Geometry of the Hall effect experiment with a *p*-type sample

a current density $\mathbf{J}$ develops in the sample. The relative orientation of $\mathbf{E}$, $\mathbf{H}$ and $\mathbf{J}$ (or their components) are shown in the figure.

The equation of motion of a hole may be written as

$$m_p^* \frac{d\mathbf{v}}{dt} = e\mathbf{E} + \frac{e}{c}(\mathbf{v} \times \mathbf{H}) \quad (6.68)$$

where m_p^* is the effective mass of the hole and $\mathbf{v}$ its velocity. The terms on the right hand side together represent the Lorentz force acting on the hole. In the context of Fig. 6.19, we see that the effective components which come into play are H_z , E_x , E_y and the velocity components v_x and v_y . Writing (6.68) in component form we have

$$\frac{dv_x}{dt} = \frac{eE_x}{m_p^*} + \omega_0 v_y \quad (6.69)$$

and

$$\frac{dv_y}{dt} = \frac{eE_y}{m_p^*} - \omega_0 v_x. \quad (6.70)$$

Here

$$\omega_0 = \frac{eH_z}{m_p^* c} \quad (6.71)$$

where ω_0 is the same as the frequency that occurs in cyclotron resonance (Chap. 5).

For convenience, we shall multiply (6.70) by i and add it to (6.69) thus giving us

$$\frac{dV}{dt} + i\omega_0 V = \frac{e\mathcal{E}}{m_p^*} \quad (6.72)$$

where V and $\mathcal{E}$ are complex quantities defined as

$$V = v_x + iv_y \text{ and } \mathcal{E} = E_x + iE_y \quad (6.73)$$

Let us multiply (6.72) by $\exp(i\omega_0 t)$. Then we get

$$\exp(i\omega_0 t) \frac{dV}{dt} + (i\omega_0) V \exp(i\omega_0 t) = \left(\frac{e\mathcal{E}}{m_p^*} \right) \exp(i\omega_0 t) \quad (6.74)$$

We can reorganize (6.74) as

$$d[V \exp(i\omega_0 t)] = \left(\frac{e\mathcal{E}}{m_p^*} \right) \exp(i\omega_0 t) dt. \quad (6.75)$$

By integration of (6.75), we get

$$V \exp(i\omega_0 t) = \left(\frac{e\mathcal{E}}{i\omega_0 m_p^*} \right) \exp(i\omega_0 t) + C. \quad (6.76)$$

where C is the constant of integration. Introducing $V_0 = v_{x0} + iv_{y0}$ as the value of V at $t = 0$, (6.76) gets modified to

$$V = V_0 \exp(-i\omega_0 t) + \left(\frac{e\mathcal{E}}{i\omega_0 m_p^*} \right) [1 - \exp(-i\omega_0 t)]. \quad (6.77)$$

As mentioned in the introductory remarks, this velocity function has to be subjected to two averaging procedures, firstly for the exponential distribution of relaxation times and secondly for the statistical distribution of velocities.

In the relaxation time approximation, if $P(t)$ is a property, its time average $\langle P \rangle$ is given by

$$\langle P \rangle = \int_0^\infty P(t) \exp(-t/\tau) dt \bigg/ \int_0^\infty \exp(-t/\tau) dt. \quad (6.78)$$

where τ is the relaxation time. Denoting the relaxation time of a hole by τ_p , the time average of V is given by

$$\begin{aligned} \langle V \rangle &= \int_0^\infty V \exp(-t/\tau_p) dt \bigg/ \int_0^\infty \exp(-t/\tau_p) dt \\ &= [1/(1 + i\omega_0 \tau_p)] [V_0 + (e\mathcal{E}\tau_p/m_p^*)]. \end{aligned} \quad (6.79)$$

We shall now carry out the averaging of $\langle V \rangle$ over a Maxwell-Boltzmann distribution of velocities. We shall note that such an average of the first term on the right hand side of (6.79) is always zero and we are left only with the second term. We shall now express $\langle V \rangle$ in the form of an imaginary and a real part as

$$\begin{aligned}
\langle V \rangle &= v_x + i v_y \\
&= \left(\frac{e}{m_p^*} \right) \left[\left(\tau_p - \frac{\omega_0^2 \tau_p^3}{1 + \omega_0^2 \tau_p^2} \right) E_x + \left(\frac{\omega_0 \tau_p^2}{1 + \omega_0^2 \tau_p^2} \right) E_y \right] \\
&\quad + i \frac{e}{m_p^*} \left[\frac{\tau_p}{1 + \omega_0^2 \tau_p^2} E_y - \frac{\omega_0 \tau_p^2}{1 + \omega_0^2 \tau_p^2} E_x \right].
\end{aligned} \tag{6.80}$$

If $Q(v)$ is a function of velocities, the average of $Q(v)$ over a Maxwell-Boltzmann velocity distribution is $\overline{Q(v)}$ which is given by

$$\overline{Q(v)} = \langle v^2 Q(v) \rangle / \langle v^2 \rangle. \tag{6.81}$$

Subjecting v_x and v_y to this procedure, we get their averages as

$$\bar{V} = \bar{v}_x + i \bar{v}_y \tag{6.82}$$

with

$$\bar{v}_x = \frac{e}{m_p^*} \left[\bar{\tau}_p E_x + \omega_0 \left(\frac{\overline{\tau_p^2}}{1 + \omega_0^2 \tau_p^2} \right) E_y - \omega_0^2 \left(\frac{\overline{\tau_p^3}}{1 + \omega_0^2 \tau_p^2} \right) E_x \right] \tag{6.83}$$

and

$$\bar{v}_y = \frac{e}{m_p^*} \left[\left(\frac{\overline{\tau_p}}{1 + \omega_0^2 \tau_p^2} \right) E_y - \omega_0 \left(\frac{\overline{\tau_p^2}}{1 + \omega_0^2 \tau_p^2} \right) E_x \right]. \tag{6.84}$$

The current density components are:

$$J_x = p e \bar{v}_x \quad \text{and} \quad J_y = p e \bar{v}_y \tag{6.85}$$

where p is the hole density. We should note that in the Hall experiment $J_y = 0$, i.e. $\bar{v}_y = 0$. Then (6.84) reduces to

$$E_y = \omega_0 \left[\left(\frac{\overline{\tau_p^2}}{1 + \omega_0^2 \tau_p^2} \right) / \left(\frac{\overline{\tau_p}}{1 + \omega_0^2 \tau_p^2} \right) \right] E_x. \tag{6.86}$$

Substituting (6.86) in (6.83), we get $\bar{v}_x$ in terms of E_x . Substituting this expression in (6.85), we get

$$J_x = \frac{p e^2}{m_p^*} \left\{ \bar{\tau}_p + \omega_0^2 \frac{[\overline{\tau_p^2 / (1 + \omega_0^2 \tau_p^2)}]^2}{[\overline{\tau_p / (1 + \omega_0^2 \tau_p^2)}]} - \frac{[\overline{\tau_p^3 / (1 + \omega_0^2 \tau_p^2)}]}{[\overline{\tau_p / (1 + \omega_0^2 \tau_p^2)}]} \right\} E_x. \tag{6.87}$$

For typical values of $H = 10,000$ oersteds, $\tau_p = 10^{-12}$ s and $m_p^* \approx m_0$, $\omega_0 \tau_p \approx 0.2$ and $\omega_0^2 \tau_p^2 \approx 0.04$ i.e. $\omega_0^2 \tau_p^2 \ll 1$. Hence we shall ignore $\omega_0^2 \tau_p^2$ in the term $(1 + \omega_0^2 \tau_p^2)$. With this approximation and substituting for ω_0 from (6.71), (6.87) reduces to

$$J_x = \frac{pe^2 \bar{\tau}_p}{m_p^*} \left\{ 1 - \left(\frac{e^2 H_z^2}{m_p^{*2} c^2} \right) \left[\frac{\bar{\tau}_p^3 \bar{\tau}_p - (\bar{\tau}_p^2)^2}{(\bar{\tau}_p)^2} \right] \right\} E_x. \quad (6.88)$$

With the approximation $\omega_0^2 \tau_p^2 \ll 1$, (6.86) becomes

$$E_y = \omega_0 \left(\bar{\tau}_p^2 / \bar{\tau}_p \right) E_x. \quad (6.89)$$

For the purposes of Hall effect, we shall neglect the second term in (6.88) and combining with (6.89), we finally get

$$E_y = \left(\frac{1}{pec} \right) \left(\frac{\bar{\tau}_p^2}{(\bar{\tau}_p)^2} \right) H_z J_x. \quad (6.90)$$

By definition, the Hall coefficient R is

$$R = E_y / H_z J_x. \quad (6.91)$$

Hence from (6.90), (6.91)

$$R = \left(\frac{1}{pec} \right) \left(\frac{\bar{\tau}_p^2}{(\bar{\tau}_p)^2} \right). \quad (6.92)$$

Writing the relaxation time terms in the form of the Maxwell-Boltzmann averaging expression, we get,

$$R = \left(\frac{1}{pec} \right) \left\{ \frac{\langle v^2 \rangle \langle v^2 \tau_p^2 \rangle}{\langle v^2 \tau_p \rangle^2} \right\}. \quad (6.93)$$

The value of the term in the floral bracket is $(3\pi/8)$. Hence, finally

$$R = \left(\frac{3\pi}{8} \right) \left(\frac{1}{pec} \right). \quad (6.94)$$

Similarly, if the carriers were only electrons, we will get

$$R = - \left(\frac{3\pi}{8} \right) \left(\frac{1}{nec} \right) \quad (6.95)$$

where n is the electron density. From (6.94) and (6.95) we note that the sign of R is different for p -type and n -type materials. Further, this rigorous treatment gives expressions close to those obtained from simpler treatment (Chap. 4).

We shall now remove the restriction that only holes are present and allow both electrons and holes to be present simultaneously with concentrations n and p respectively. We shall denote the various parameters as,

$$\begin{aligned} \text{Electrons: } & -e, v_{nx}, v_{ny}, t_n, m_n^*, \omega_{0n} = (eH_z/m_n^*c) \text{ and } \mu_n = -(e\bar{\tau}_n/m_n^*) \\ \text{Holes: } & +e, v_{px}, v_{py}, t_p, m_p^*, \omega_{0p} = (eH_z/m_p^*c) \text{ and } \mu_p = (e\bar{\tau}_p/m_p^*) \end{aligned}$$

Following the same procedure as above but allowing the presence of both electrons and holes, we get

$$\begin{aligned} J_x &= ne\bar{v}_{nx} + pe\bar{v}_{px} \\ &= \frac{ne^2}{m_n^*} [E_x\bar{\tau}_n - E_y\omega_{0n}\bar{\tau}_n^2 - E_x\omega_{0n}^2\bar{\tau}_n^3] + \frac{pe^2}{m_p^*} [E_x\bar{\tau}_p + E_y\omega_{0p}\bar{\tau}_p^2 - E_x\omega_{0p}^2\bar{\tau}_p^3] \end{aligned} \quad (6.96)$$

and

$$\begin{aligned} J_y &= ne\bar{v}_{ny} + pe\bar{v}_{py} \\ &= \frac{ne^2}{m_n^*} [E_y\bar{\tau}_n + E_x\omega_{0n}\bar{\tau}_n^2] + \frac{pe^2}{m_p^*} [E_y\bar{\tau}_p - E_x\omega_{0p}\bar{\tau}_p^2]. \end{aligned} \quad (6.97)$$

Expressing the effective masses in terms of mobilities, (6.96) and (6.97) become, on reorganization,

$$\begin{aligned} J_x &= ne\mu_n[E_x - E_y\omega_{0n}(\bar{\tau}_n^2/\bar{\tau}_n) - E_x\omega_{0n}^2(\bar{\tau}_n^3/\bar{\tau}_n)] \\ &\quad + pe\mu_p[E_x + E_y\omega_{0p}(\bar{\tau}_p^2/\bar{\tau}_p) - E_x\omega_{0p}^2(\bar{\tau}_p^3/\bar{\tau}_p)] \end{aligned} \quad (6.98)$$

and

$$J_y = E_x[n\mu_n\omega_{0n}(\bar{\tau}_n^2/\bar{\tau}_n) - p\mu_p\omega_{0p}(\bar{\tau}_p^2/\bar{\tau}_p)] + E_y(n\mu_n + p\mu_p). \quad (6.99)$$

Since, in a Hall experiment, $J_y = 0$, (6.99) becomes

$$E_y = \frac{[p\mu_p\omega_{0p}(\bar{\tau}_p^2/\bar{\tau}_p) - n\mu_n\omega_{0n}(\bar{\tau}_n^2/\bar{\tau}_n)]}{(n\mu_n + p\mu_p)} E_x. \quad (6.100)$$

Substituting (6.100) in (6.98) and on rearranging, we get

$$J_x = e(n\mu_n + p\mu_p) \left\{ 1 - \left[\frac{n\mu_n\omega_{0n}^2(\bar{\tau}_n^3/\bar{\tau}_n) + p\mu_p\omega_{0p}^2(\bar{\tau}_p^3/\bar{\tau}_p)}{n\mu_n + p\mu_p} \right] - \left[\frac{n\mu_n\omega_{0n}(\bar{\tau}_n^2/\bar{\tau}_n) - p\mu_p\omega_{0p}(\bar{\tau}_p^2/\bar{\tau}_p)}{n\mu_n + p\mu_p} \right]^2 \right\} E_x \quad (6.101)$$

We shall replace ω_{0n} and ω_{0p} by the definitions given earlier and, further, express the effective masses in terms of the mobilities. This gives

$$E_y = \left(\frac{1}{ec} \right) \frac{p[\bar{\tau}_p^2/(\bar{\tau}_p)^2] - (\mu_n^2/\mu_p^2)n[\bar{\tau}_n^2/(\bar{\tau}_n)^2]}{[(\mu_n/\mu_p)n + p]^2} H_z J_x. \quad (6.102)$$

For $[\bar{\tau}_n^2/(\bar{\tau}_n)^2]$ and $[\bar{\tau}_p^2/(\bar{\tau}_p)^2]$ we shall substitute the value $(3\pi/8)$. The Hall coefficient R is obtained from (6.102) as

$$R = \frac{E_y}{H_z J_x} = \left(\frac{3\pi}{8ec} \right) \frac{[p - (\mu_n^2/\mu_p^2)n]}{[(\mu_n/\mu_p)n + p]^2}. \quad (6.103)$$

We shall note the following features from (6.103):

- (i) R is positive for $p \gg n$ (p -type)
- (ii) R is negative for $p \ll n$ (n -type)
- (iii)

$$R = \left(\frac{3\pi}{8} \right) \left(\frac{1}{n_i ec} \right) \left(\frac{\mu_p - \mu_n}{\mu_p + \mu_n} \right) \quad \text{for } n = p = n_i (\text{intrinsic}). \quad (6.104)$$

It is interesting to note that just from the sign of R , one can determine whether a semiconductor is p -type or n -type. Further, from the magnitude of R , the concentration of charge carriers can be estimated. Combined with the value of the conductivity, mobilities can also be estimated.

6.5.2 Hall Effect in Semiconductors with Complex Energy Surfaces

We shall take the case of Si; its Fermi surface has six ellipsoids with their long axes oriented in the x , y , z directions (along the k_x , k_y , k_z directions, to be more precise). We have seen in Sect. 6.4.3 that for such a system, the effective reciprocal mass is a tensor $[\frac{1}{m^*}]_{\alpha\beta}$ and its principal elements are $\left(\frac{1}{m_{\perp}^*}\right)$ and $\left(\frac{1}{m_{\parallel}^*}\right)$. Let us assume that the semiconductor contains only holes.

The equation of motion (6.68) may be expanded and written in the matrix form as

$$\begin{bmatrix} \frac{dv_x}{dt} \\ \frac{dv_y}{dt} \\ \frac{dv_z}{dt} \end{bmatrix} = \begin{bmatrix} \frac{1}{m_{\parallel}^*} & 0 & 0 \\ 0 & \frac{1}{m_{\perp}^*} & 0 \\ 0 & 0 & \frac{1}{m_{\perp}^*} \end{bmatrix} \begin{bmatrix} eE_x + \frac{e}{c}(v_y H_z - v_z H_y) \\ eE_y + \frac{e}{c}(v_z H_x - v_x H_z) \\ eE_z + \frac{e}{c}(v_x H_y - v_y H_x) \end{bmatrix}. \quad (6.105)$$

The geometry of the experiment being the same as in Fig. 6.19, $H_x = H_y = 0$, $v_z = 0$ and $E_z = 0$. We may carry out the matrix multiplication and write the components as

$$\frac{dv_x}{dt} = \frac{eE_x}{m_{\parallel}^*} + \omega_{\parallel} v_y \quad (6.106)$$

and

$$\frac{dv_y}{dt} = \frac{eE_y}{m_{\perp}^*} - \omega_{\perp} v_x \quad (6.107)$$

where

$$\omega_{\parallel} = \frac{eH_z}{m_{\parallel}^*} \quad \text{and} \quad \omega_{\perp} = \frac{eH_z}{m_{\perp}^*} \quad (6.108)$$

Multiplying (6.106) by $\sqrt{\omega_{\perp}}$ and (6.107) by $i\sqrt{\omega_{\parallel}}$ and adding the two, we get

$$\frac{dZ}{dt} + i\sqrt{\omega_{\parallel}\omega_{\perp}}Z = e \left(\frac{\sqrt{\omega_{\perp}}}{m_{\parallel}^*} E_x + \frac{i\sqrt{\omega_{\parallel}}}{m_{\perp}^*} E_y \right) \quad (6.109)$$

where Z is a complex quantity defined by

$$Z = v_x \sqrt{\omega_{\perp}} + i v_y \sqrt{\omega_{\parallel}} \quad (6.110)$$

We may note the similarity between (6.72) and (6.109). Therefore, further procedure (like the double averaging) will be similar to that for solving (6.72) and we will get the x -component of velocities as

$$\begin{aligned} \bar{v}_x^{(100)} &= \bar{v}_x^{(\bar{1}00)} \\ &= e \left[\frac{\bar{\tau}}{m_{\parallel}^*} E_x + \frac{\omega_{\parallel}}{m_{\perp}^*} \left(\frac{\overline{\tau^2}}{1 + \omega_{\parallel}\omega_{\perp}\tau^2} \right) E_y - \frac{\omega_{\parallel}\omega_{\perp}}{m_{\parallel}^*} \left(\frac{\overline{\tau^3}}{1 + \omega_{\parallel}\omega_{\perp}\tau^2} \right) E_x \right] \end{aligned} \quad (6.111)$$

and

$$\bar{v}_y^{(100)} = \bar{v}_y^{(\bar{1}00)} = e \left[\frac{1}{m_{\perp}^*} \left(\frac{\tau}{1 + \omega_{\parallel} \omega_{\perp} \tau^2} \right) E_y - \frac{\omega_{\perp}}{m_{\parallel}^*} \left(\frac{\tau^2}{1 + \omega_{\parallel} \omega_{\perp} \tau^2} \right) E_x \right]. \quad (6.112)$$

Here (100) and $(\bar{1}00)$ refer to the two ellipsoids along k_x and $-k_x$ directions. Similar expressions are obtained for the velocities associated with the other two pairs of ellipsoids. These are added and using the fact that the current density in the y -direction is zero, we get

$$E_y = \left(\frac{eH_z}{c} \right) \left(\frac{\tau^2}{\bar{\tau}} \right) \frac{2m_{\parallel}^* + m_{\perp}^*}{m_{\perp}^{*2} m_{\parallel}^* \left(\frac{2}{m_{\perp}^*} + \frac{1}{m_{\parallel}^*} \right)} E_x. \quad (6.113)$$

From Sect. 6.4.3, we have

$$E_x = \left(\frac{J_x}{\sigma} \right) = \frac{3J_x}{pe^2 \bar{\tau} \left(\frac{2}{m_{\perp}^*} + \frac{1}{m_{\parallel}^*} \right)}. \quad (6.114)$$

Substituting in (6.113) and using the relation $R = E_y/H_z J_x$, we get

$$R = \left(\frac{1}{pec} \right) \left(\frac{\tau^2}{(\bar{\tau})^2} \right) \left[3m_{\parallel}^* (2m_{\perp}^* + m_{\parallel}^*) / (2m_{\parallel}^* + m_{\perp}^*)^2 \right]. \quad (6.115)$$

If instead of holes the carriers were electrons, we would get the same result but with a negative sign. Further, if the orientations of the electric field and magnetic field were different from those shown in Fig. 6.19, we would still get the same result. In other words, the Hall coefficient is isotropic at small magnetic fields.

6.6 Magnetoresistance

In a magnetic field, a moving charge carrier gets deflected from its straight line path. Its trajectory becomes curved and the mean distance traveled during successive collisions gets reduced; this increases the resistance. The increase in resistance in a magnetic field is called magnetoresistance. The magnetoresistance coefficient is defined as $\left(\frac{\Delta\rho}{\rho_0} \right) \frac{1}{H^2}$ where $\Delta\rho$ is the increase in resistivity, ρ_0 the resistivity in the absence of the field and H the magnetic field.

For a semiconductor containing holes and to which a magnetic field H_z and electric field E_x is applied, we have the current density

$$J_x = \frac{pe^2\bar{\tau}_p}{m_p^*} \left\{ 1 - \left(\frac{e^2 H_x^2}{m_p^{*2} c^2} \right) \left[\frac{\bar{\tau}_p^3 \bar{\tau}_p - \overline{(\tau_p^2)^2}}{(\bar{\tau}_p)^2} \right] \right\} E_x. \quad (6.88)$$

Equation (6.88) can be inverted and written as

$$E_x = \rho_0 J_x \left\{ 1 + \left(\frac{e^2 H_x^2}{m_p^{*2} c^2} \right) \left[\frac{\bar{\tau}_p^3 \bar{\tau}_p - \overline{(\tau_p^2)^2}}{(\bar{\tau}_p)^2} \right] \right\} = (\rho_0 + \Delta\rho) J_x \quad (6.116)$$

where $\rho_0 = (m_p^*/pe^2\bar{\tau}_p)$ is the resistivity in the absence of the magnetic field and the increase in resistivity is ρ_0 times the second term in the curly bracket. Thus, the magnetoresistance coefficient is

$$\left(\frac{\Delta\rho}{\rho_0 H^2} \right) = \left(\frac{e^2}{m_p^{*2} c^2} \right) \left[\frac{\bar{\tau}_p^3 \bar{\tau}_p - \overline{(\tau_p^2)^2}}{(\bar{\tau}_p)^2} \right]. \quad (6.117)$$

6.7 Mobility of Carriers

6.7.1 Definitions

The mobility (μ) of a carrier is defined as the drift velocity per unit electric field (E). Thus,

$$\mu = \frac{|v_D|}{E}. \quad (6.118)$$

If we substitute the free-electron theory expression for v_D in (6.118), we get

$$\mu = (e/m)\tau. \quad (6.119)$$

where τ is a relaxation time and m the free electron mass. Typically, $\tau = 2 \times 10^{-14}$ s. Substituting this value and the values for e and m in (6.119), we get $\mu \approx 10^4$ cm²/statvolt-s = 30 cm²/volt-s.

In semiconductors, there are two types of carriers, electrons and holes. They have their own mobilities μ_n and μ_p defined by

$$\mu_n = \frac{e\tau_n}{m_n^*} \quad \text{and} \quad \mu_p = \frac{e\tau_p}{m_p^*} \quad (6.120)$$

where τ_n and τ_p are the relaxation times and m_n^* and m_p^* the effective masses of electrons and holes respectively. It is the convention to treat both the mobilities as positive [6].

Let us consider a sample of a semiconductor which is n -type. Then substituting in (6.120) expressions for R and σ obtained earlier, we get

$$\mu_n = -cR\sigma \frac{(\bar{\tau}_n)^2}{\tau_n^2}. \quad (6.121)$$

Similarly for μ_p we get

$$\mu_p = cR\sigma \frac{(\bar{\tau}_p)^2}{\tau_p^2}. \quad (6.122)$$

μ_n and μ_p are called drift mobilities or conductivity mobilities whereas the term $\pm cR\sigma$ is called the Hall mobility μ_H . Thus,

$$\mu_{Hn} = -cR\sigma \text{ (} n\text{-type)} \quad \text{and} \quad \mu_{Hp} = cR\sigma \text{ (} p\text{-type)}. \quad (6.123)$$

The drift mobilities are then expressed as

$$\mu_n = \mu_{Hn} \frac{(\bar{\tau}_n)^2}{(\tau_n^2)} \quad \text{and} \quad \mu_p = \mu_{Hp} \frac{(\bar{\tau}_p)^2}{(\tau_p^2)}. \quad (6.124)$$

6.7.2 Experimental Determination of Mobilities

For semiconductors containing both types of carriers we have derived the following relation for the Hall coefficient:

$$R = \left(\frac{3\pi}{8ec} \right) \frac{[p - (\mu_n^2/\mu_p^2)n]}{[(\mu_n/\mu_p)n + p]^2}. \quad (6.103)$$

We shall use the law of mass action relation $np = n_i^2$ to eliminate either p or n from (6.103). The result is an equation for R which is a fourth degree function of n or p . Such an equation is difficult to solve analytically but can be solved by approximate numerical methods. The result is shown in Fig. 6.20. Another difficulty is that the curve indicates two roots (say p values) for each value of R . But it is possible to choose the correct root from experimental R values. If p and n values are determined and if the ratio (μ_n/μ_p) is known, it is possible to extract μ_n and μ_p values from experimental values of conductivity using the relation $\sigma = e(n\mu_n + p\mu_p)$. Values of the electron and hole mobilities are given in Table 6.5.

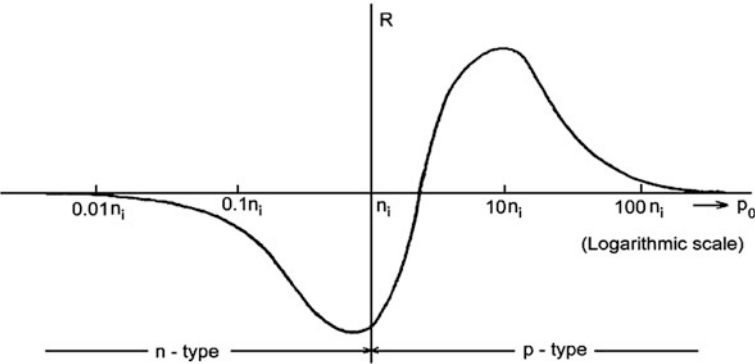


Fig. 6.20 Dependence of Hall coefficient on hole concentration

Table 6.5 Mobilities of electrons and holes in some semiconductors

Crystal	Mobility (cm ² /V-s)	
	Electrons	Holes
Si	1,600	400
Ge	3,800	1,800
InSb	77,000	1,250
InAs	23,000	~ 100
InP	3,400	650
GaSb	2,500	650
PbS	600	200
PbSe	900	700

6.7.3 Temperature Variation of Mobility

The conductivity and, hence, the mobility is influenced by the scattering of carriers. This scattering is temperature-dependent. There are several processes that cause scattering and there are elaborate theories for each scattering process. We shall discuss the most important processes viz., lattice scattering and impurity scattering.

6.7.3.1 Lattice Scattering

Electrons interact with phonons (quantized lattice vibrations). This interaction is called electron-phonon collision. As discussed in Chaps. 2 and 3, these collisions cause a deviation in the carrier trajectory; in effect, this determines the mobility. Seitz [8] worked out a detailed theory of lattice scattering. According to this theory, the mobility is

$$\mu = \left[\frac{2^{1/2} \times 6^{1/3}}{4\pi^{5/6}} \right] \left[\frac{N^{1/3} e \hbar^2 k_B^2 \theta_D^2 M}{m^{*5/2} C^2 (k_B T)^{3/2}} \right] \quad (6.125)$$

where θ_D is the Debye temperature and C is a constant with dimensions of energy. Further, M is the mass of an atom and N the number of atoms in unit volume.

6.7.3.2 Impurity Scattering

Electrons in an n -type semiconductor and holes in a p -type semiconductor get attracted/repulsed when they pass a charged impurity. The change in trajectory of an electron and hole due to Rutherford scattering is shown in Fig. 6.21 assuming the impurity to be positively charged. The effect of this type of scattering was investigated by Conwell and Weisskoff [9] who obtained

$$\mu = [2^{7/2} \epsilon^2 (k_B T)^{3/2} \log(1 + x^2)] / N_d \pi^{3/2} z^2 e^3 m^{*1/2} \quad (6.126)$$

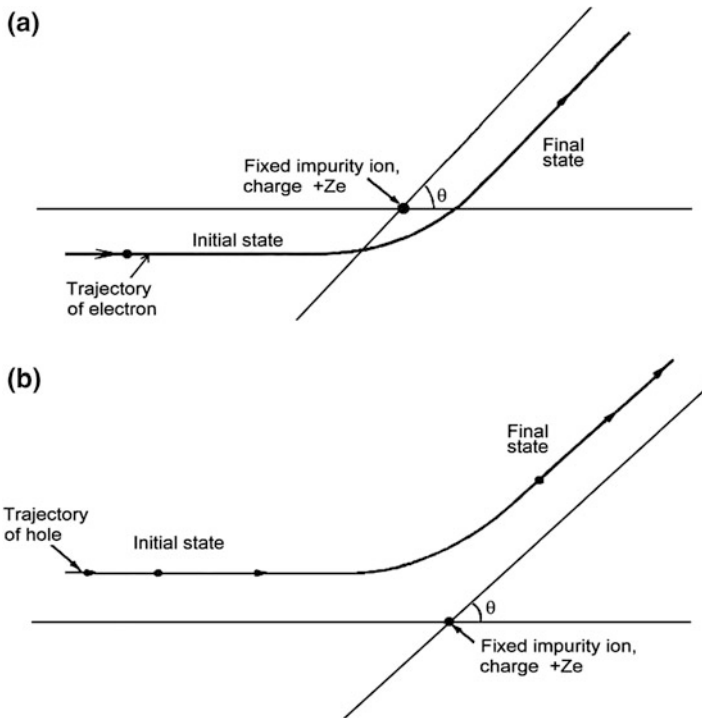
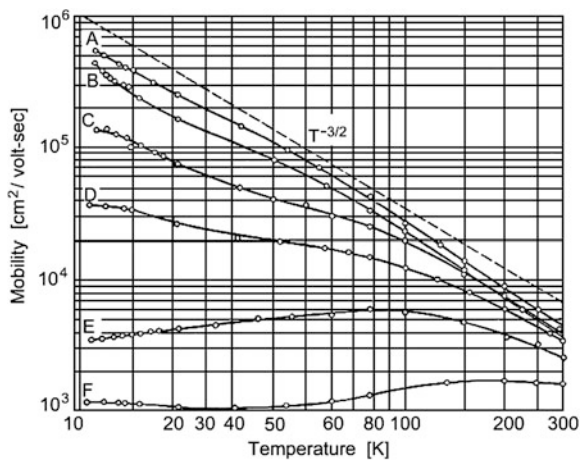


Fig. 6.21 Scattering of carriers by a positively charged impurity **a** electron carrier **b** hole carrier

Fig. 6.22 Temperature variation of mobility of some arsenic-doped germanium crystals; the *dashed line* represents a slope of $-3/2$; concentration of donors as in Fig. 6.17



where ϵ is the dielectric constant, N_d the concentration of donors, z the positive charge on the impurity (in electron units) and $x = 6\epsilon d k_B T / e^2$, d being half the distance between the impurity atoms. Note that μ varies as $T^{3/2}$ according to (6.126).

The experimental results from Conwell [10] on the temperature variation of mobility for a set of As-doped Ge are shown in Fig. 6.22 as $\log \mu$ versus $\log T$ -curves. The concentration of donors is $N_d = 10^{13} \text{ cm}^{-3}$ for curve A and increases progressively for the lower curves being $N_d = 10^{18} \text{ cm}^{-3}$ for curve F. For all curves except the two lowest curves, the slope is about $-3/2$ indicating lattice scattering. For curves E and F, the slope is positive at low temperature indicating impurity scattering.

6.8 Excess Carriers in Semiconductors

6.8.1 Creation of Excess Carriers

Let n be the equilibrium concentration of electrons in the conduction band of a semiconductor and p the equilibrium concentration of holes in the valence band. It is possible to increase the equilibrium concentrations by several methods viz., (i) by injecting carriers through a metal contact, (ii) by creation of electron-hole-pairs by irradiation with radiation having frequency $\hbar\omega \geq E_g$ or (iii) by bombardment with charged particles. If the host crystal has a high concentration of electrons (n -type), the electrons are called majority carriers and the holes created by any of the other methods are called minority carriers. The excess number of carriers, of course, disturbs the equilibrium, but it is soon restored by the recombination of holes and electrons.

Table 6.6 Lifetimes of holes on different samples of Ge

Sample numbers	Life-time (μs)	
	Ground surface	Etched surface
1	144	280
2	78	340
3	16	290
4	9	280

If Δn and Δp are the excess carrier concentrations, then $\Delta n = \Delta p$. The decay of carriers by recombination follows the laws:

$$(dn/dt) = -(1/\tau_n)\Delta n \quad \text{and} \quad (dp/dt) = -(1/\tau_p)\Delta p. \quad (6.127)$$

Here τ_n and τ_p are called the life-times of electrons and holes respectively. In this section the symbol τ represents the life-time and it is not to be confused with the relaxation times in earlier sections and chapters. The recombination process takes place in the surface as well as in the volume. Denoting the life-time for decay in surface by τ_s and that in the volume by τ_v , the overall lifetime τ is given by

$$(1/\tau) = (1/\tau_s) + (1/\tau_v). \quad (6.128)$$

The life-time τ varies with temperature. It is sensitive to the presence of impurities. Also, the life-time is affected by the presence of lattice defects; the life-times of some samples of Ge with ground and etched surfaces are given in Table 6.6.

6.8.2 Diffusion

The creation of excess carriers by any of the methods mentioned in the preceding section results in a pulse (Fig. 6.23a). It can be seen that the concentration of carriers decays with distance on either side of the peak. Let us consider what happens to the pulse with time. The carriers flow away with time; the pulse-height decreases and the pulse-width increases (Fig. 6.23b). If in addition, an electric field is applied, the pulse drifts as a whole (Fig. 6.23c).

In Sect. 6.2, we defined a non-uniform semiconductor as one in which the concentration of carriers is a function of coordinates. The concentration of excess carriers, as described above, is an example of non-uniform distribution. Whenever the concentration is non-uniform, there is a concentration gradient and particles move from a region of high concentration to a region of low concentration. This phenomenon is called diffusion. If (dn/dx) is the concentration gradient and J' the current density of particles (number of particles flowing in unit time across unit area of cross-section in a direction parallel to the gradient), then J' is proportional to the gradient or

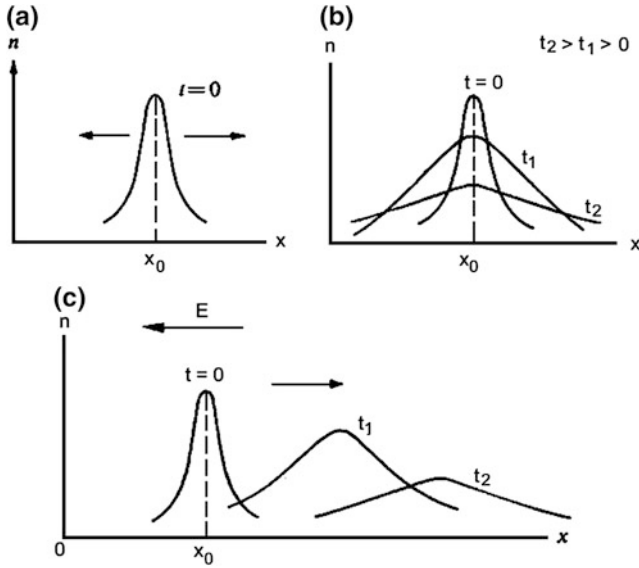


Fig. 6.23 Shape of a pulse of excess carriers **a** at time $t = 0$ **b** at different times t_1 and t_2 and **c** at different times in an electric field

$$J' = -D(dn/dx). \quad (6.129)$$

Equation (6.129) is the statement of Fick's law. The constant of proportionality D is called the diffusion constant. In (6.129), the negative sign is chosen to keep D positive. If the particles are charged (electrons or holes) and we multiply J' by the charge, it becomes the electric current density J . The electric current densities J_n and J_p for excess electrons and holes are

$$J_n = eD(dn/dx) \quad \text{and} \quad J_p = -eD(dp/dx). \quad (6.130)$$

Let us consider a semiconductor with only holes as carriers. Since there is only one type of carrier, we shall drop the subscript. If a gradient is present and also an electric field E , there are two contributions to J given by

$$J = -eD \frac{\partial p}{\partial x} + pe\mu E. \quad (6.131)$$

The first term is due to diffusion and the second due to drift. Now, we should note that the flow of carriers is a function of time also. The time rate of flow is given by

$$\left(\frac{\partial p}{\partial t} \right)_{\text{Flow}} = -\frac{1}{e} \frac{\partial J}{\partial x} = D \frac{\partial^2 p}{\partial x^2} - \mu \frac{\partial (pE)}{\partial x}. \quad (6.132)$$

This is known as the continuity equation. But we have also to include the variation in p due to recombination. We shall express it as

$$\left(\frac{\partial p}{\partial t}\right)_{\text{Recomb}} = -\frac{p - p_0}{\tau'} \quad (6.133)$$

where p_0 is the equilibrium concentration and τ' the recombination time. The total variation $\left(\frac{\partial p}{\partial t}\right)$ is

$$\left(\frac{\partial p}{\partial t}\right) = D \frac{\partial^2 p}{\partial x^2} - \mu \frac{\partial(pE)}{\partial x} - \frac{p - p_0}{\tau'}. \quad (6.134)$$

Equation (6.134) is called the diffusion equation. If we know the initial conditions, we can get p at any time or any position by solving (6.134). This, however, is not a straight forward problem. We shall however solve (6.134) for two simplified situations.

- (i) $(\partial p / \partial t) = 0$ and $E = 0$.

Equation (6.134) now takes the form

$$D \frac{\partial^2 p}{\partial x^2} - \frac{p - p_0}{\tau'} = 0. \quad (6.135)$$

This is a standard equation. Its solution is

$$p' = p - p_0 = A \exp[-x/(D\tau')^{1/2}]. \quad (6.136)$$

where A is a constant. The implication of (6.136) is that p' decays exponentially with x and practically vanishes at $x > (D\tau')^{1/2}$. This critical distance L_D is

$$L_D = (D\tau')^{1/2}. \quad (6.137)$$

and is called the diffusion length.

- (ii) $(\partial p / \partial t) = 0$ and $E \neq 0$

In this case (6.134) takes the form

$$\frac{\partial^2 p'}{\partial x^2} - \frac{\mu}{D} \frac{\partial p'}{\partial x} E - \frac{p'}{L_D^2} = 0. \quad (6.138)$$

Equation (6.138) is a standard equation. Its solution is given by

$$p' = A \exp[-(\gamma/L_D)x] \quad (6.139)$$

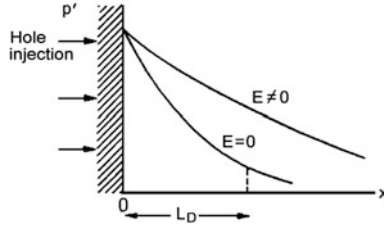


Fig. 6.24 Decay of excess holes with distance with $E = 0$ and $E \neq 0$

where

$$\gamma = \sqrt{1 + s^2} - s \quad \text{and} \quad s = \frac{\mu E L_D}{2D}. \quad (6.140)$$

The form of (6.136) and (6.139) is similar except that the diffusion length is now (L_D/γ) where γ is field dependent. The variation of p' with x for $E = 0$ and $E \neq 0$ is shown in Fig. 6.24.

6.8.3 Haynes and Shockley Experiment

Haynes and Shockley [11] devised an experiment that permitted direct measurement of mobilities of holes and electrons in a semiconductor.

The experimental arrangement is shown in Fig. 6.25. S is a rod of n -type germanium. The rod has metal contacts at the two ends. A field is applied across the rod by means of a battery such that the left end is positive. This field causes the flow of current I_B which flows from the left to the right i.e. electrons flow from right to left and holes from left to the right. There are two contacts at points A and B; A is called the emitter and B the collector. A pulse generator is connected in series with A and a recording system (oscilloscope) is connected in series with B. The point B is $-ve$ with respect to the sample.

At an instant t_1 , the pulse generator circuit is closed briefly. This results in injection of holes at A. This event is recorded on the oscilloscope as a sharp signal. These holes then travel from left to right. When they reach the collector B, their arrival at time t_2 is again recorded as a signal on the CRO. This signal is broad because the holes diffuse as they travel. In fact, by studying the width of the signal, information can be obtained about the diffusion constant.

If L is the distance between A and B and V the voltage, the drift velocity v_{Dp} of the holes is

$$v_{Dp} = L/(t_2 - t_1). \quad (6.141)$$

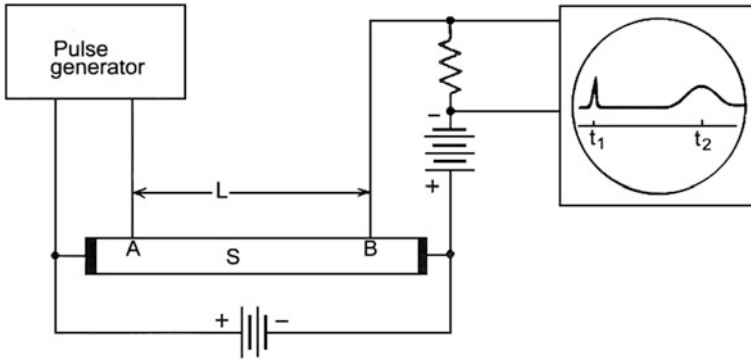


Fig. 6.25 The Haynes-Shockley experiment for direct measurement of mobility of minority carriers (holes)

But the drift velocity is also given by

$$v_{Dp} = \mu_p E = \mu_p (V/L) \quad (6.142)$$

where μ_p is the mobility of holes and V the voltage between A and B. From (6.141) and (6.142), we get

$$\mu_p = L^2 / V(t_2 - t_1). \quad (6.143)$$

In a typical experiment, $L \sim 20$ mm, $V \sim 10^3$ V/m and $(t_2 - t_1) \sim$ milliseconds. If we wish to measure the mobility of electrons, the voltages at A and B should be reversed and the sample should be p -type. Using this method, Haynes and Shockley obtained values $1,700 \text{ cm}^2/\text{Vs}$ and $3,600 \text{ cm}^2/\text{Vs}$ for the mobilities of holes and electrons respectively in Ge at 300 K.

6.9 Problems

- (1) An intrinsic semiconductor has conductivities $0.21 \text{ } \Omega^{-1} \text{ m}^{-1}$ at 293 K and $0.48 \text{ } \Omega^{-1} \text{ m}^{-1}$ at 305 K. Calculate E_g .
- (2) In an intrinsic semiconductor, the effective masses of holes and electrons are in the ratio $m_p^*/m_n^* = (1/6)$. If $E_g = 0.7$ eV, determine the position of the Fermi level at $T = 0$ and $T = 300$ K.
- (3) The Fermi level in an n -type semiconductor with a certain donor concentration is 0.5 eV below E_c . If the donor concentration is tripled, what is the new position of the Fermi level?

- (4) A semiconductor has an ellipsoidal energy surface with the equation

$$E(\mathbf{k}) = \hbar^2 \left[\frac{(k_x - k_z)^2}{2m_l^*} + \frac{(k_y^2 - k_z^2)}{2m_t^*} \right].$$

Show that the cyclotron frequency ω_0 is

$$\omega_0 = \left[\left(\frac{eH}{m_l^*} \right)^2 \cos^2 \theta + \left(\frac{e^2 H^2}{m_l^* m_t^*} \right) \sin^2 \theta \right]^{1/2}$$

where θ is the angle between H and the k_x -axis.

- (5) An n -type semiconductor has Hall coefficient $-7.3 \times 10^{-5} \text{ m}^3 \text{ C}^{-1}$ and conductivity $205 \text{ } \Omega^{-1} \text{ m}^{-1}$. Estimate the carrier density and carrier mobility.
- (6) The energy surface in the valence band of Si has the equation

$$E(\mathbf{k}) = -(4.1)k^2 \pm \left[(1.6)^2 k^4 + (3.3)^2 (k_x^2 k_y^2 + k_y^2 k_z^2 + k_z^2 k_x^2) \right]^{1/2}$$

where the numerical constants are in units of $\hbar^2/2m$. Calculate the values of the heavy-hole and light-hole effective masses.

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Chapter 7

Semiconductor Devices

Abstract Semiconductors have led to a number of devices with applications in science, technology and in day-to-day life. In this chapter, we discuss a few typical semiconductor devices in a qualitative manner. The first category of devices are the diodes. In the p - n junction diode, the presence of ionized donors and acceptors on either sides of the junction results in a potential barrier to the motion of the mobile electrons and holes across the junction. This motion is bias dependent. The relevant theory is worked out in detail. The rectifier action is discussed. In a Gunn diode the rectifier action is due to the creation of a negative differential conductivity region. In the tunnel diode the rectifier action is due to tunneling of the electrons and holes through the barrier. The second category of devices is the transistors. The working of the point contact transistor and junction transistor is discussed. Later versions like the field-effect transistor, the metal-oxide FET and the insulated bipolar transistor are also discussed. A few miscellaneous devices like the semiconductor solar cell, the semiconductor laser and the charge- coupled device are also described. Semiconductor devices are very sensitive to purity of material and perfection in fabrication. Crystal growth methods, purification by zone refinement and the junction fabrication techniques are discussed.

7.1 Introduction

The physics of semiconductors was discussed in the preceding chapter. In that chapter and also in earlier chapters, we confined ourselves to the physical principles of phenomena avoiding any discussion of their applications.

But semiconductors are different from other materials. The new concepts which had to be introduced to explain semiconductor behaviour are important but no less important are the devices that semiconductors led to. These devices have been hailed as “the best achievements of the human mind” [1] and are considered to have “revolutionized communication industry” [2]. A discussion of the physics of semiconductors is incomplete without a brief discussion of the devices based on

them. We shall therefore consider some representative devices. In doing so, our approach will be to discuss the physics involved without going into the circuitry nor too much into theory.

Pioneering semiconductor devices like the junction rectifier and the junction transistor are discussed by Kittel [3] and Dekker [4]. A few other devices are discussed by Omar [2] and Srivastava and Srinivasan [5]. Roy [1] deals exclusively with semiconductor devices.

7.2 Semiconductor Diodes

7.2.1 *p-n Junction Diode*

The simple *p-n* junction was designed by Shockley [6]. It was the first semiconductor device which found many applications. We shall first consider its structure and then discuss its most important application viz., as a rectifier.

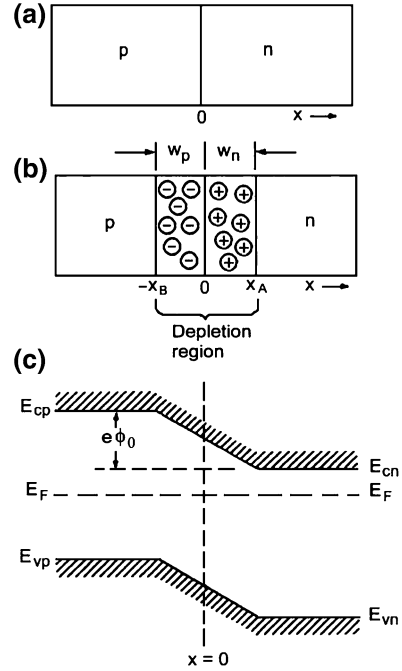
7.2.1.1 The Structure of a *p-n* Junction

A *p-n* junction diode is made up of a semiconductor like Si. One side of it is *p*-type and the other *n*-type as shown in Fig. 7.1a. We hasten to clarify that such a composite semiconductor is not made by simply placing a *p*-type sample and an *n*-type sample together in mechanical contact or even by welding or fusing them together. Instead the piece is grown as an integral whole with *p*-type impurities introduced on one side and *n*-type impurity on the other; the methods of fabricating such a unit will be discussed in Sect. 7.5. The concentration of impurity atoms changes from *p*-type to *n*-type in a narrow region which is called the *p-n* junction; the width of the junction is typically $\sim 1\ \mu$.

It is obvious that there is an excess of electrons on the *n*-side and an excess of holes on the *p*-side. The concentration of each (electrons and holes) has a strong gradient across the junction leading to diffusion. Electrons from the *n*-side diffuse to the *p*-side while the holes from the *p*-side diffuse to the *n*-side. This diffusion depletes the holes on the *p*-side and electrons on the *n*-side (Fig. 7.1b). The depletion of electrons on the *n*-side extends to a small region of width w_n , its edge being located at $x = x_A$. Similarly, the depletion of holes on the *p*-side extends to a small region of width w_p , its edge being located at $x = x_B$. These two regions are together called the depletion layer. The depletion layer (w_n) on the *n*-side has a net positive charge whereas that on the *p*-side (w_p) has a net negative charge. For this reason, the depletion layer is also called the space charge layer.

Because of their charged nature, the two layers give rise to an electric field; in the figure, the field is from the right to the left. The carrier concentration is uniform in regions away from the depletion layer. This field gives rise to a contact potential

Fig. 7.1 **a** A p - n junction, **b** structure of the depletion layer and **c** positions of the energy bands on either side of the junction and the contact potential ϕ_0



ϕ_0 and a potential barrier $e\phi_0$. The effect of the junction on the band energy levels is shown in Fig. 7.1c. The levels on the p -side are raised by an amount $e\phi_0$. The conduction band edges on the two sides of the junction are denoted by E_{cn} and E_{cp} . Similarly, the valence band edges are denoted by E_{vn} and E_{vp} respectively. E_F is the common Fermi level.

We shall now estimate the contact potential ϕ_0 . If n_{n0} and n_{p0} are the concentrations of electrons on the n -side and the p -side respectively in the equilibrium state, we have from Chap. 6,

$$n_{n0} = N_c \exp[-(E_{cn} - E_F)/k_B T] \quad (7.1)$$

and

$$n_{p0} = N_c \exp[-(E_{cp} - E_F)/k_B T]. \quad (7.2)$$

Dividing and reorganizing, we get

$$E_{cp} - E_{cn} = e\phi_0 = k_B T \log \frac{n_{n0}}{n_{p0}}. \quad (7.3)$$

From the law of mass action, we have

$$n_{n0}p_{n0} = n_{p0}p_{p0} = n_i^2 \quad (7.4)$$

where p_{n0} and p_{p0} are the equilibrium concentration of holes on the n and p sides and n_i is the intrinsic concentration. Substituting in (7.3), we get

$$e\phi_0 = k_B T \log \frac{n_{n0}p_{p0}}{n_i^2}. \quad (7.5)$$

If all the donors are ionized (i.e. electrons from donor levels have been excited into the conduction band) and also all the acceptors are ionized (i.e. all acceptor levels have been occupied by electrons from the valence band creating holes),

$$n_{n0} \approx N_d \quad (7.6)$$

and

$$p_{p0} \approx N_a. \quad (7.7)$$

Then, (7.5) becomes

$$e\phi_0 = k_B T \log \frac{N_d N_a}{n_i^2}. \quad (7.8)$$

Here N_d and N_a are concentrations of donor and acceptor impurities.

We shall now consider the band energy levels on the two sides of the junction when the voltage V is applied in two ways: forward bias (Fig. 7.2a) and reverse bias (Fig. 7.2b). The shifting of the bands due to the applied voltage in the two cases is shown in the same figure. In forward bias, the energy levels on the n -side are raised by an amount eV so that the potential barrier is reduced to $e(\phi_0 - V)$. On the other hand, in reverse bias, the levels on the n -side are lowered by an amount eV so that the potential barrier increases to $e(\phi_0 + V)$.

We shall now estimate the width of the depletion layer. If $e(\phi_0 + V) \gg k_B T$, the charge density ρ in the space charge region on each side of the junction is nearly constant and is equal to $-eN_a$ on the p -side and $+eN_d$ on the n -side. We may assume the charge density to be zero for $x < -x_B$ and $x > x_A$. The Poisson's equations for the two sides are

$$\frac{d^2 \phi_+}{dx^2} = -\frac{eN_d}{\epsilon} \quad \text{for } 0 < x < x_A, \quad (7.9)$$

$$\frac{d^2 \phi_-}{dx^2} = +\frac{eN_a}{\epsilon} \quad \text{for } -x_B < x < 0 \quad (7.10)$$

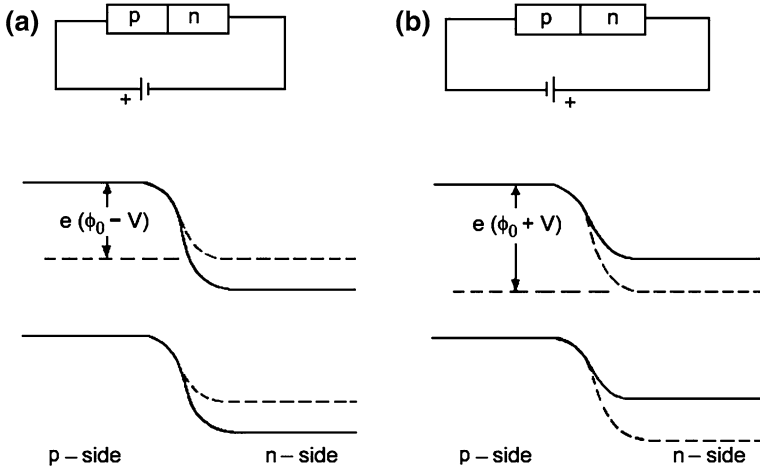


Fig. 7.2 Potential energy diagram for a p - n junction with **a** forward bias and **b** reverse bias; continuous curve without bias, dashed curve with bias

and

$$\frac{d^2\phi_+}{dx^2} = \frac{d^2\phi_-}{dx^2} = 0 \quad \text{for } x > x_A \quad \text{and} \quad x < -x_B. \quad (7.11)$$

Here ϵ is the dielectric constant. The subscripts $+$ and $-$ pertain to n and p sides respectively. Away from the depletion layer, the potential has a constant value (ϕ_0). In other words,

$$\phi_+(x) = \phi_0(x_A) = \text{constant} \quad \text{for } x > x_A \quad (7.12)$$

and

$$\phi_-(x) = \phi_0(-x_B) = \text{constant} \quad \text{for } x < -x_B \quad (7.13)$$

The total change in potential is $(\phi_0 - V)$. Thus

$$\phi_+(x_A) - \phi_-(-x_B) = (\phi_0 - V). \quad (7.14)$$

Also, the field is the negative of the derivative of the potential. Hence,

$$\left(\frac{d\phi_+}{dx} \right)_0 = - \left(\frac{d\phi_-}{dx} \right)_0 = \mathcal{E}_0 \quad (7.15)$$

where $\mathcal{E}_0$ is the field at the origin. Integrating (7.9) and (7.10) and applying the condition that the field is zero at $x = x_A$ and $x = -x_B$, we get

$$\mathcal{E}_+(x) = \frac{-eN_d}{\epsilon}(x_A - x) \quad \text{for } 0 < x < x_A \quad (7.16)$$

and

$$\mathcal{E}_-(x) = \frac{-eN_a}{\epsilon}(x + x_B) \quad \text{for } -x_B < x < 0 \quad (7.17)$$

Further at $x = 0$, the electric fields are equal. Thus,

$$N_d x_A = N_a x_B. \quad (7.18)$$

The meaning of (7.18) is that the number of positive charges on the n -side equals the number of negative charges on the p -side.

We choose the potential such that it is zero at $x = 0$. Then we combine (7.16) with (7.15). Integration of the resulting equation leads to

$$\phi_+(x) = \frac{eN_d}{2\epsilon}[x_A^2 - (x_A - x)^2] \quad \text{for } 0 < x < x_A \quad (7.19)$$

and

$$\phi_+(x) = \frac{eN_d}{2\epsilon}x_A^2 \quad \text{for } x \geq x_A \quad (7.20)$$

Following a similar procedure with (7.17), we get

$$\phi_-(x) = \frac{eN_a}{2\epsilon}[(x + x_B)^2 - x_B^2] \quad \text{for } -x_B < x < 0 \quad (7.21)$$

and

$$\phi_-(x) = -\frac{eN_a}{2\epsilon}x_B^2 \quad \text{for } x \leq -x_B \quad (7.22)$$

We also have

$$\phi_+(x_A) - \phi_-(-x_B) = (\phi_0 - V) = \frac{e}{2\epsilon}[N_d x_A^2 + N_a x_B^2]. \quad (7.23)$$

Substituting (7.18) in (7.23), we get

$$x_A = \left\{ \frac{2\epsilon(\phi_0 - V)}{e} \frac{N_a}{N_d(N_a + N_d)} \right\}^{1/2} \quad (7.24)$$

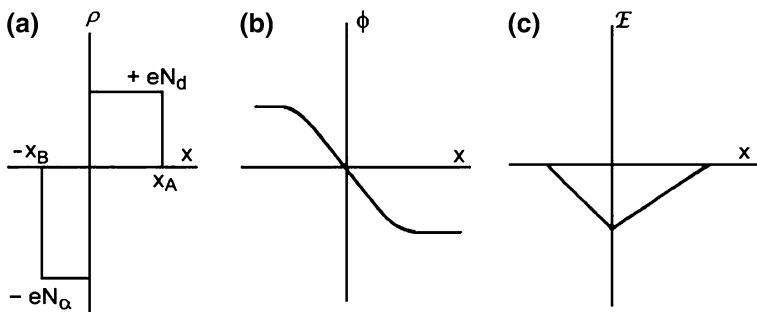


Fig. 7.3 Variation with x of **a** charge density ρ , **b** potential ϕ and **c** electric field ε

and

$$x_B = \left\{ \frac{2\epsilon(\phi_0 - V)}{e} \frac{N_d}{N_a(N_a + N_d)} \right\}^{1/2}. \quad (7.25)$$

From Fig. 7.1b, the width of the depletion layer is $x_A + x_B$. Thus,

$$\begin{aligned} \text{Width} = w_n + w_p = x_A + x_B &= \left\{ \frac{2\epsilon(\phi_0 - V)}{e} \left(\frac{1}{N_d} + \frac{1}{N_a} \right) \right\}^{1/2} \\ &= \left\{ \frac{2\epsilon\phi_0}{e} \left(\frac{1}{N_d} + \frac{1}{N_a} \right) \right\}^{1/2} \quad \text{for } V = 0. \end{aligned} \quad (7.26)$$

As mentioned earlier, this width is $\sim 1\mu$. The charge density ρ , the potential ϕ and the electric field ε are shown as functions of x in Fig. 7.3.

7.2.1.2 p-n Junction as a Rectifier

Let us consider the flow of carrier currents across the junction. In the equilibrium state when there is no external voltage, electrons flow from the n -side to the p -side. The resulting current may be denoted by I_{nr0} . Also, some electrons diffuse from p -side into n -side where they cause generation of electron-hole pairs. We shall denote the current due to this flow I_{ng0} . When there is equilibrium,

$$I_{nr0} + I_{ng0} = 0. \quad (7.27)$$

On application of the external voltage V in the forward direction, because of reduced potential barrier (Fig. 7.2a), the recombination of electrons and holes increases and so does the recombination current I_{nr} . I_{nr} is now given by

$$I_{nr} = I_{nr0} \exp(eV/k_B T). \quad (7.28)$$

The net electron current I_n is given by

$$I_n = I_{nr} + I_{ng} = I_{nr0} \exp(eV/k_B T) + I_{ng0} = -I_{ng0} [\exp(eV/k_B T) - 1]. \quad (7.29)$$

Equation (7.29) was obtained by substituting for I_{nr0} from (7.27); also, $I_{ng} = I_{ng0}$ since this current is not affected by the applied voltage. Similarly, we obtain the net hole current I_p as

$$I_p = I_{pg0} [\exp(eV/k_B T) - 1]. \quad (7.30)$$

The net current I is given by

$$I = I_p - I_n = (I_{ng0} + I_{pg0}) [\exp(eV/k_B T) - 1] = I_s [\exp(eV/k_B T) - 1]. \quad (7.31)$$

Here $I_s = (I_{ng0} + I_{pg0})$ is called the saturation current. If the voltage is applied in the reverse direction, V becomes negative in (7.31). Then the first term in the square bracket becomes negligible and we get

$$I = -I_s. \quad (7.32)$$

The rectification effect of a p - n junction is contained in (7.31) and (7.32). The voltage-current characteristics are shown in Fig. 7.4. Figure 7.4a shows the theoretical characteristics drawn according to (7.31) for forward bias and (7.32) for reverse bias. In the forward bias mode the current increases exponentially with voltage whereas in the reverse bias mode it remains constant i.e. independent of the voltage. Experimental I - V data from measurements on a p - n junction diode are shown in Fig. 7.4b. These are qualitatively similar to those in Fig. 7.4a. It may be

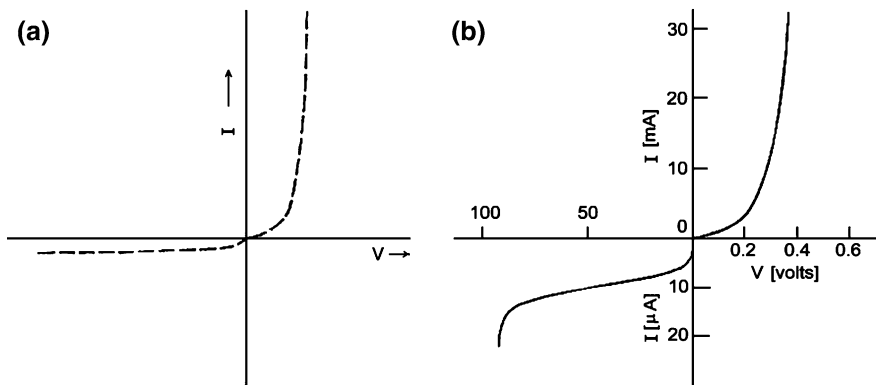


Fig. 7.4 I - V curves for a p - n junction rectifier **a** theoretical curve and **b** experimental curve

noted that the forward bias characteristic and the reverse bias characteristic are drawn on different scales; whereas the forward bias current is in mA, the reverse bias current is in μA .

7.2.2 Gunn Diode

The Gunn effect and the device based on it—Gunn diode—are possible only in some direct gap semiconductors like GaAs and InSb. Although the band structure of these solids has been discussed earlier, we shall note some features relevant to the Gunn effect.

The band structure of GaAs is shown in Fig. 7.5. There are two conduction band minima, one at the Γ point and the other at the X point. We shall refer to these two minima as the lower valley (LV) and the upper valley (UV). Values of important parameters like the effective mass (m^*), the mobility (μ), the density of states and energy gap (E_g) for LV and UV are given in Table 7.1.

The LV minimum is separated from the valence band (VB) by a forbidden gap of width 1.42 eV. The UV minimum is separated from the VB by a gap of width 1.78 eV. Thus the energy difference between the two minima is $\Delta E_g = 0.36$ eV. It may be noted that the effective mass of the electrons in the LV is much smaller than that in the UV. On the other hand, the mobility of electrons in the LV is much larger than that in the UV. Values of the density of states in the LV and UV are in the ratio 1:70.

Normally electrons from the top of the VB get thermally excited into the LV. However, when an electric field is applied, some of the electrons from the LV get transferred to the UV as shown in Fig. 7.6. This transfer is not simple and is

Fig. 7.5 Band structure of GaAs

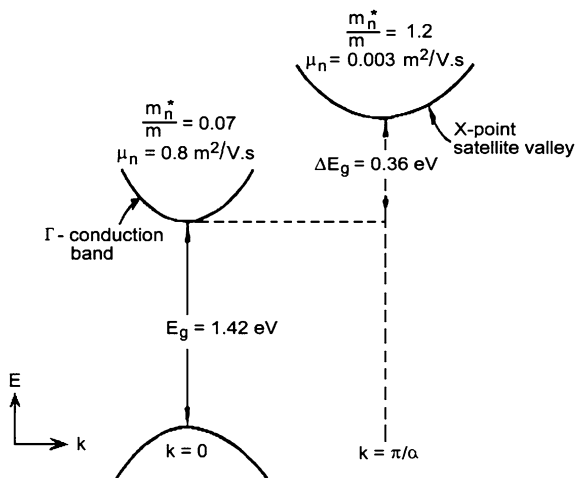
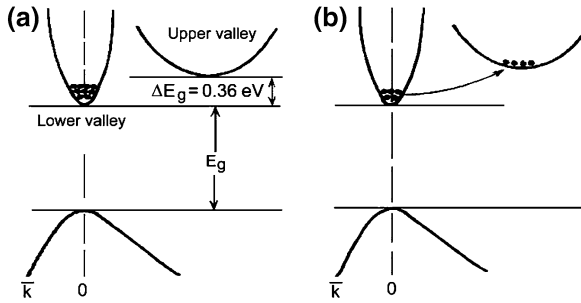


Table 7.1 Band structure parameters of GaAs

Parameter	Lower valley	Upper valley
Effective mass (in units of free electron mass)	0.07	1.2
Mobility ($\text{m}^2/\text{V.s}$)	0.8	0.003
Energy gap (eV)	1.42	1.78
Density of states (arbitrary units)	1	70

**Fig. 7.6** Transfer of electrons from lower valley to upper valley under applied electrical field **a** without field and **b** with field

dependent on the voltage in a complex manner. In fact, as we shall see, at certain voltages (~ 400 V/m), the semiconductor develops a negative conductance.

We shall adopt the following notation for the various parameters for the LV and UV: electron concentration (n_l , n_u), drift velocity (v_l , v_u), mobility (μ_l , μ_u) and electron energy at the conduction band edge (E_{cl} , E_{cu}). The average drift velocity v_D of the electrons is

$$v_D = \frac{n_l v_l + n_u v_u}{n_l + n_u} \quad (7.33)$$

By definition, $v_l = \mu_l \mathcal{E}$ and $v_u = \mu_u \mathcal{E}$ where $\mathcal{E}$ is the electric field. Substituting in (7.33) we have

$$v_D = \left[\mu_l + \left(\frac{n_u}{n_l} \right) \mu_u \right] \mathcal{E} / \left[1 + \left(\frac{n_u}{n_l} \right) \right]. \quad (7.34)$$

For n_u and n_l we have expressions similar to (7.1) and (7.2). Thus, we have

$$n_u = N_{cu} \exp[-(E_{cu} - E_F)/k_B T] \quad (7.35)$$

and

$$n_l = N_{cl} \exp[-(E_{cl} - E_F)/k_B T] \quad (7.36)$$

Dividing (7.35) by (7.36), we get

$$\begin{aligned} (n_u/n_l) &= (N_{cu}/N_{cl}) \exp[-(E_{cu} - E_{cl})/k_B T] \\ &= Z \exp(-\Delta E_g/k_B T) \end{aligned} \quad (7.37)$$

where $Z = (N_{cu}/N_{cl})$ and $\Delta E_g = E_{cu} - E_{cl}$. Substituting (7.37) in (7.34), we have

$$v_D = \left[\frac{\mu_l + \mu_u Z \exp(-\Delta E_g/k_B T_e)}{1 + Z \exp(-\Delta E_g/k_B T_e)} \right] \mathcal{E}. \quad (7.38)$$

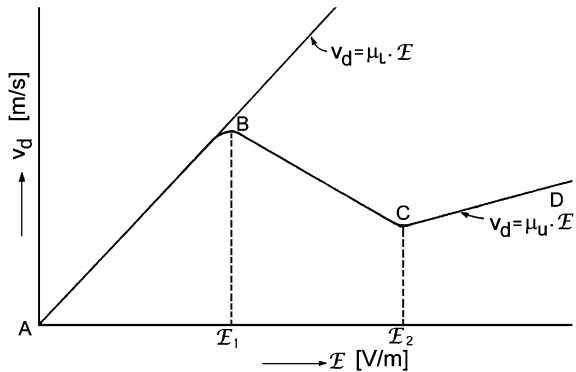
Note that in place of T , we have introduced the term T_e in (7.38). When the electron energy becomes very high, the electrons are no longer in equilibrium with the lattice. We then have to differentiate between the lattice temperature T and the electron temperature T_e . It can be shown [1] that

$$T_e = T + \frac{l e}{k_B} \mathcal{E} \quad (7.39)$$

where l is the mean free path of the electrons.

Without going into details, we shall note the implications of (7.38). For low electric fields, up to some value $\mathcal{E}_1$, $\Delta E_g \gg k_B T_e$ and (7.38) reduces to $v_D = \mu_l \mathcal{E}$. The v_D - $\mathcal{E}$ variation in this region is shown in Fig. 7.7 (part AB). Here the electrons move with mobility μ_l . For large electric fields, above some value $\mathcal{E}_2$, $\Delta E_g \ll k_B T_e$ and (7.38) reduces to $v_D = \mu_u \mathcal{E}$. This is shown in Fig. 7.7 (part CD of the curve). Lastly, for fields between $\mathcal{E}_1$ and $\mathcal{E}_2$, the v_D - $\mathcal{E}$ graph follows the route BC where it has negative slope. In this region, the mobility has some effective value

Fig. 7.7 Plots of drift velocity v_d against electric field $\mathcal{E}$ for a Gunn diode; BC is the NDC region



$\mu^* = -(dv_D/d\mathcal{E})$. Since mobility is directly related to the conductance, this region BC is called negative differential conductance (NDC) region. We shall see that the occurrence of the NDC region leads to a device.

Consider a thin bar of a semiconductor like GaAs biased as shown in Fig. 7.8. Let the voltage be such that the sample is in the NDC state. In such a situation, a space charge gets nucleated at the site of an impurity or any other imperfection in the bar. Such a tiny space charge-nucleus grows as a ‘domain’. Under the effect of the applied field, the domain not only grows but also drifts from the cathode to the anode (Fig. 7.9). Once the domain, which is essentially a pulse of electrons, reaches the anode, it delivers the charge at the anode. The bar now returns to its initial state and the process of formation of domain and its drift starts again. This, in effect, is the oscillation of the space charge. The frequency of oscillation ν_0 is given by

$$\nu_0 = v_D/L \quad (7.40)$$

where L is the length of the bar. In a typical Gunn diode, the sample length was 200 μ , the domain width 40 μ , the threshold voltage 13 V and the oscillation frequency 2.5 GHz.

The phenomenon of NDC was predicted by Ridley and Watkins [7] but the diode device based on it was designed by Gunn [8]. It is an excellent source of microwave radiation. It may be noted that the Gunn diode is a bulk device whose performance depends on the microscopic properties and not on any junction surface effects.

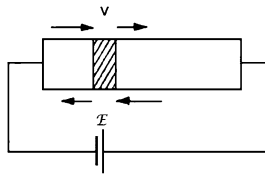


Fig. 7.8 A biased semiconductor bar

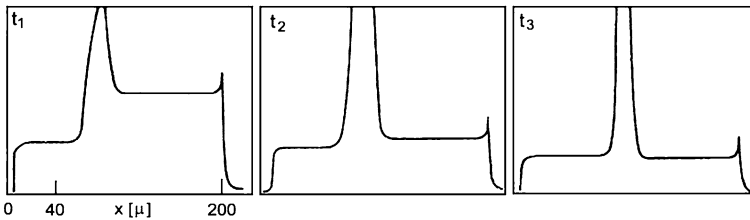


Fig. 7.9 Motion of a domain under an applied field at instants t_1, t_2, t_3 ($t_1 < t_2 < t_3$)

7.2.3 Tunnel Diode

According to classical mechanics, a particle incident on a barrier cannot pass through it. However, according to quantum mechanics, there is a finite probability of the particle passing through a potential barrier. The probability depends on the particle parameters and also on the potential parameters.

In a p - n junction, the strong field in the depletion layer acts as a barrier for the carriers on either side. Esaki [9] showed that under some conditions, the junction allows electrons to pass from one side to the other but not in the opposite direction. Thus the device works as a diode; it is called a tunnel diode. An elaborate quantum mechanical theory of the tunnel diode is discussed by Roy [1]. Here we shall discuss the working of a tunnel diode only qualitatively.

In the tunnel diode, the p and n regions are heavily doped ($\sim 10^{19} \text{ cm}^{-3}$). At this level of doping, the contact potential ϕ_0 is large, the depletion layer is narrow and the electric field is quite large ($\sim 900 \text{ kV/cm}$).

The energy levels on the two sides of the junction are shown in Fig. 7.10a. The Fermi level lies within the bands and is the same on both sides of the junction. Let us apply a reverse bias of V volts (Fig. 7.10b). The conduction band on the n -side is lowered by eV . The electrons on the p -side tunnel through the barrier and are transferred to the conduction band on the n -side giving rise to a current. This can happen only when the potential barrier is thin. Now, let us apply forward bias. This raises the levels on the n -side (Fig. 7.10c). The electrons on the n -side can now tunnel into the valence band on the p -side across the barrier. As the bias voltage V is increased, the current also increases. But beyond a certain voltage, the number of available states decreases (Fig. 7.10d) and the current starts decreasing reaching almost zero value. As the bias voltage increases further, the current again increases, not because of tunneling but because of diffusion of minority carriers across the junction.

The I - V characteristics of a tunnel diode are shown in Fig. 7.11. The current first increases, then decreases and, finally, increases again. There is similarity between these characteristics and those shown in Fig. 7.7 for the Gunn diode but the

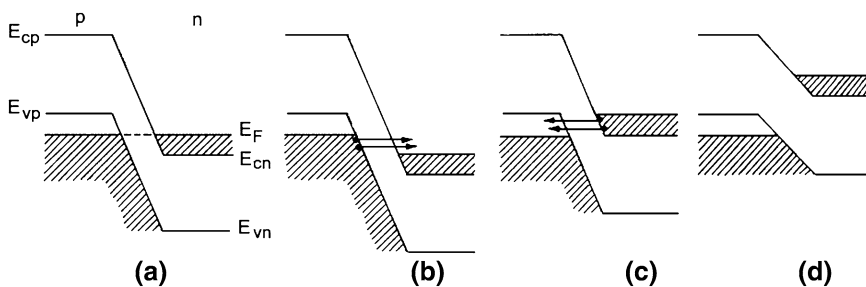


Fig. 7.10 Energy levels in a tunnel diode: **a** in equilibrium, **b** large current for reverse bias, **c** moderate current for small forward bias and **d** zero current for high forward bias

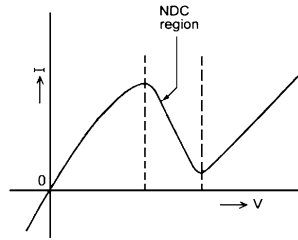


Fig. 7.11 I - V characteristics for a tunnel diode showing existence of NDC region

mechanisms are different. However, there is an NDC region in both; the role of the NDC region is to cause oscillations. Tunnel diodes are made from Si and GaAs and are useful as amplifiers and oscillators in the microwave range. Leo Esaki was awarded the Nobel Prize in 1973 for the invention of the tunnel diode.

7.3 Transistors

7.3.1 Point Contact Transistor

The design and working of the point contact transistor was proposed by Bardeen and Brattain [10]. The electrical arrangement is shown in Fig. 7.12. Two point contacts are placed close to each other with a spacing of ~ 0.01 cm on the surface of a germanium block. The left hand side contact is provided a positive voltage (forward bias) and is called the emitter. The contact on the right hand side is provided a negative voltage (reverse bias) and is called the collector. A large area, low resistance contact on the lower surface of the Ge-block is the third element of the circuit and is called the base electrode.

The flow of electrons and holes in the device is shown in Fig. 7.13. Under forward bias, the emitter releases the holes. The collector which is negatively biased attracts the holes towards itself. The nature of the metal contact is such that it offers a high resistance to electrons flowing from the metal contact to the semiconductor. Any change in the current in the emitter circuit brings about a larger change in the

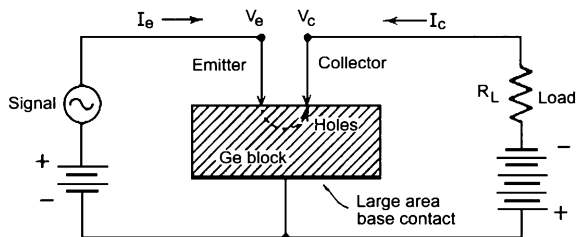
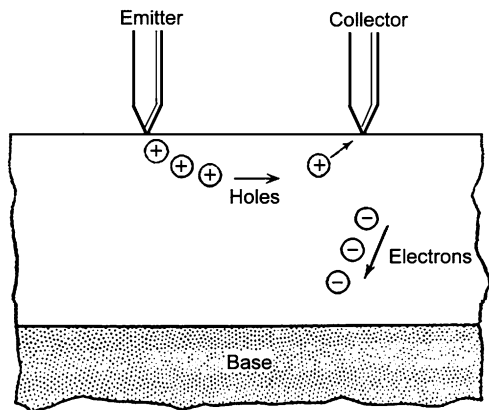


Fig. 7.12 Schematic circuit of a point contact transistor

Fig. 7.13 The motion of holes and electrons in a point contact transistor



current in the collector circuit. It follows that if an AC source is included in the emitter circuit, it will induce an AC signal of larger magnitude in the collector circuit. Thus the transistor acts as an amplifier. The amplification factor is given by

$$\alpha = -(\partial I_c / \partial I_e) \quad (7.41)$$

where I_c and I_e are currents in the collector and emitter circuits and V_c is the collector voltage. The numerical value of $\alpha \sim 2$. Although the design of the point contact transistor (also called Type A transistor) is simple, its detailed theory is quite complicated [10]. The characteristics of the point contact transistor are shown in Fig. 7.14.

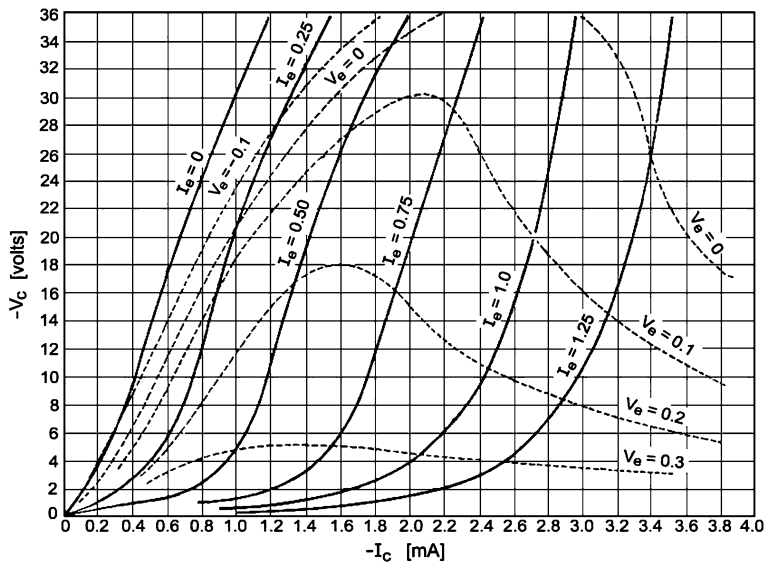


Fig. 7.14 Characteristics of a point contact transistor

J. Bardeen and W.H. Brattain shared the 1956 Nobel Prize for Physics for the invention of the point-contact transistor.

7.3.2 Junction Transistor

The idea of a junction transistor was proposed by Shockley et al. [11]. We shall describe the structure of an n - p - n transistor. The transistor consists of two n -type regions with a p -type region in between (Fig. 7.15a). Thus we have two n - p junctions J_e and J_c existing back to back and biased as shown in the figure. The n -type region on the left is called the emitter and the one on the right is called the collector. The central p -region is called the base electrode. The entire n - p - n assembly is a single piece of Ge in which the different regions are doped as required. In fact, the two n regions are also doped differently so that they have different conductivities. In a typical junction transistor the junction area is 0.3 cm^2 and the base layer thickness is 0.07 cm . The conductivities are: emitter, $100 (\text{ohm-cm})^{-1}$, base $1 (\text{ohm-cm})^{-1}$ and collector $0.1 (\text{ohm-cm})^{-1}$.

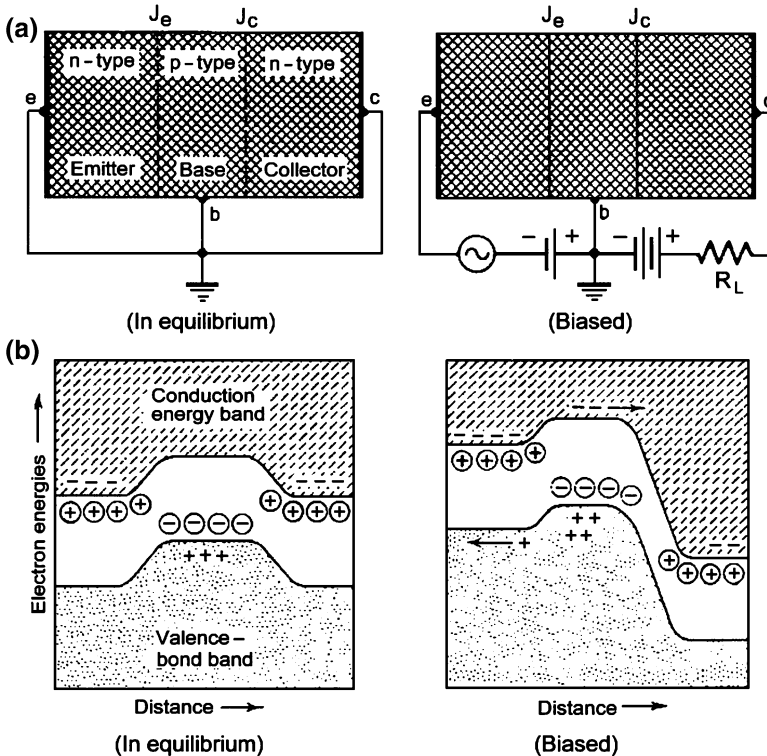


Fig. 7.15 n - p - n transistor **a** circuit in equilibrium and biased states, **b** energy levels in equilibrium and biased states

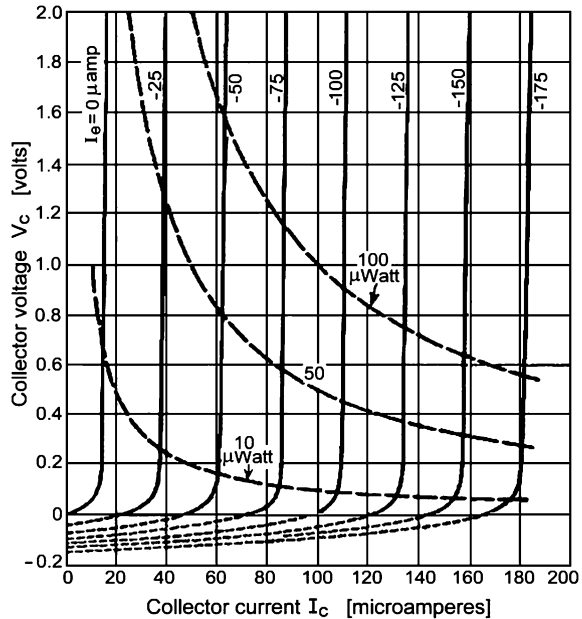
The detailed theory of the device is too long and complicated and will not be discussed here; it is given in the original paper by Shockley et al. [11]. We shall discuss the working only qualitatively.

When the n - p - n transistor is biased as shown in Fig. 7.15a, the energy level diagram across the two junctions is as shown in Fig. 7.15b. The J_c junction is reverse biased due to which the electrons in the collector region have low potential energy. The potential hill of the base region is too high for the electrons to climb and so they are held on the right hand side. Similarly the holes are held in the base region. In contrast, the electrons in the n -region on the emitter side experience a small potential hill. They easily climb it and pass into the base region. Once they enter this region, they diffuse and arrive at the junction on the right. Thus the flow of electrons from the left to right depends on the height of the base hill. This height can be adjusted by varying the voltage at the emitter keeping the base voltage constant. The voltage-current characteristics for an n - p - n transistor are shown in Fig. 7.16.

There is a similarity between the working of the n - p - n transistor and a triode valve: the emitter corresponds to the cathode, the base to the grid and the collector to the plate. Shockley et al. [11] have also considered the working of a p - n - p transistor and a p - n - p - n transistor.

W.B. Shockley received the 1956 Nobel Prize for physics for the discovery of the junction transistor.

Fig. 7.16 Current voltage characteristics of a n - p - n transistor



7.3.3 Field Effect Transistor

A field effect transistor (FET) is a device in which an electric field is used to control the conduction in a channel. We shall describe the configuration of a FET before deriving its I - V relations.

The FET structure is shown in Fig. 7.17. The main part is a single-crystal rod of n -type Si with ohmic contacts at its two ends. There are also two p^+ type contacts (G_1 and G_2) at opposite sides of the bar; these are called gates. The electrode S on the left is called the source and the electrode D on the right is called the drain. A small positive voltage V_d is applied to the drain D relative to S. Due to the voltage, a current flows through the channel. The reverse potential drop is not uniform but varies along the length of the bar being maximum at D and minimum at S. Let us analyze the performance of the FET at (i) small drain voltage and (ii) large drain voltage.

7.3.3.1 FET at Small Drain Voltage

In this case, the variation of voltage along the length of the rod may be ignored. The voltage is assumed to be constant and the channel cross section is also constant. Further, the channel is lightly doped. The depletion layer width is given by

$$a - b = [2\epsilon(\psi_0 + V_g) / eN_d]^{1/2} \quad (7.42)$$

where a and b are the channel dimensions as shown in the figure, ϵ the dielectric constant of the semiconductor, ψ_0 the junction potential, V_g the gate voltage and N_d the number of donors. At a certain gate voltage V_p , there is a pinch-off effect which

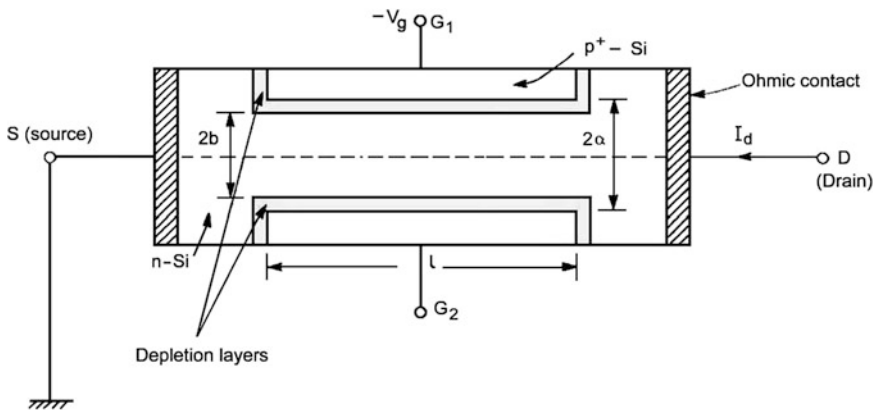


Fig. 7.17 Design of a field effect transistor

causes the channel thickness b to reduce to zero. We may now write (7.42) in a reorganized form as

$$V_p = [(eN_d a^2 / 2\epsilon) - \psi_0] \quad (7.43)$$

Substituting for $(eN_d a^2 / 2\epsilon)$ from (7.43) in (7.42), gives

$$b = a \left[1 - \left(\frac{\psi_0 + V_g}{\psi_0 + V_p} \right)^{1/2} \right]. \quad (7.44)$$

We may now write the channel conductance G at a gate voltage V_g as

$$G = \sigma_n \left(\frac{2bW}{l} \right) = \left(\frac{2e\mu_n N_d Wa}{l} \right) \left[1 - \left(\frac{\psi_0 + V_g}{\psi_0 + V_p} \right)^{1/2} \right] \quad (7.45)$$

where μ_n is the electron mobility, W the width of the channel perpendicular to the plane of the figure, σ_n the conductivity of the n -type channel and l the channel length. We can write the drain current I_d as

$$I_d = GV_d = \left(\frac{2e\mu_n N_d Wa}{l} \right) \left[1 - \left(\frac{\psi_0 + V_g}{\psi_0 + V_p} \right)^{1/2} \right] V_d. \quad (7.46)$$

For large V_g and V_p , ψ_0 can be ignored and (7.46) becomes

$$I_d = G_0 \left[1 - \left(\frac{V_g}{V_p} \right)^{1/2} \right] V_d \quad (7.47)$$

where $G_0 = (2e\mu_n N_d Wa / l)$.

The $I_d - V_d$ characteristics are shown in Fig. 7.18a and the transfer ($I_d - V_g$) characteristics are shown in Fig. 7.18b. For small gate voltage and drain voltage, I_d is a linear function of V_d .

7.3.3.2 FET at Large Drain Voltages

We shall now consider the case when the drain voltage is large. The potential drop is no longer constant and varies along the length of the rod. The transverse reverse bias across the gate and a point in the channel decreases as one moves from D to S. The depletion layer assumes the shape of a wedge converging towards the drain (Fig. 7.19). Let us consider an element between x and $x + \Delta x$ in the channel. The potential at x may be represented by $V(x)$. The reverse potential between the gate and the point at x becomes $[V_g + V(x)]$ and the voltage drop across the element is ΔV_x .

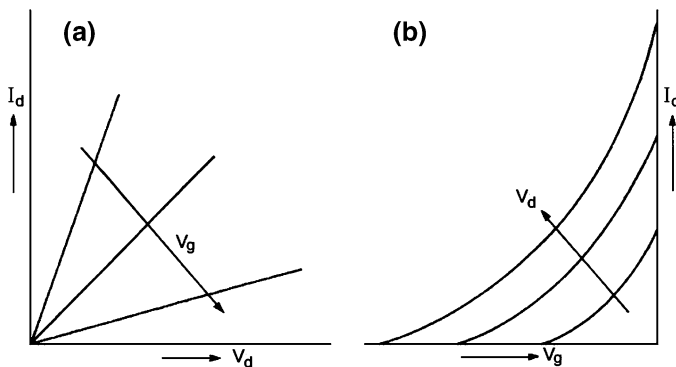
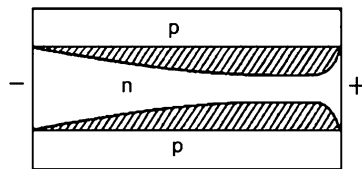


Fig. 7.18 Characteristics of a FET for small drain voltages: **a** drain characteristics and **b** transfer characteristics

Fig. 7.19 Wedge formation in a depletion layer



Following (7.46), we may express the current as

$$I_d = \left(\frac{2e\mu_n N_d W a}{l} \right) \left[1 - \left(\frac{V_x + V_g}{V_p} \right)^{1/2} \right] \Delta V_x. \quad (7.48)$$

Integrating over the length of the channel and reorganizing, we have

$$I_d = G_0 \left[V_d - \frac{2}{3} V_p \left(\frac{V_d + V_g}{V_p} \right)^{3/2} + \frac{2}{3} V_p \left(\frac{V_g}{V_p} \right)^{3/2} \right]. \quad (7.49)$$

By using the condition $(dI_d/dV_d) = 0$, we get

$$V_p = V_d + V_g. \quad (7.50)$$

The detailed analysis of (7.49) is complicated. We shall note the result: at low V_d values, the $I_d - V_d$ curve rises according to a power law but at large values of V_d , the curve becomes independent of V_d i.e. it saturates. The $I_d - V_d$ characteristics are shown in Fig. 7.20a and the transfer characteristics in Fig. 7.20b.

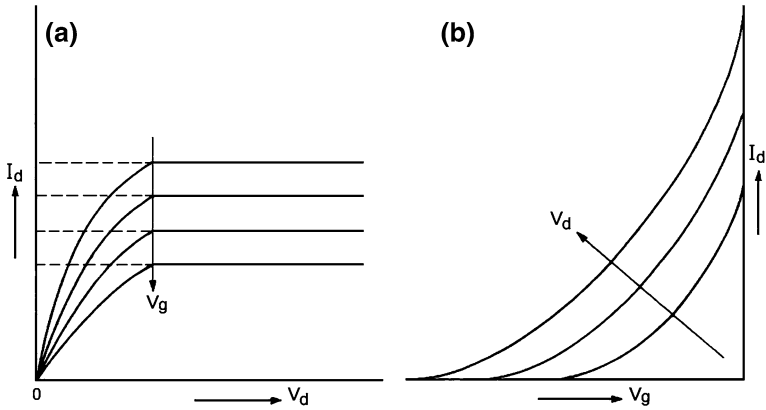


Fig. 7.20 Characteristics of a FET for large drain voltages: **a** drain characteristics and **b** transfer characteristics

7.3.4 MOSFET

MOSFET is the abbreviation for metal-oxide-semiconductor-field-effect-transistor. We shall first describe the configuration of the device and then consider the characteristics.

As seen in Fig. 7.21, a thin oxide (SiO_2) insulator layer intervenes between the metal electrodes and the semiconductor electrodes. The metal electrodes are termed the source (S), the gate (G) and the drain (D). The S and D electrodes connect with two n^+ semiconductor elements on either side of the device. The gate (G) electrode is located between the two n^+ elements. These n^+ elements are located at the upper part of a p -type sample. We shall see that a positive voltage applied to the gate electrode induces a conducting channel in the semiconductor substrate.

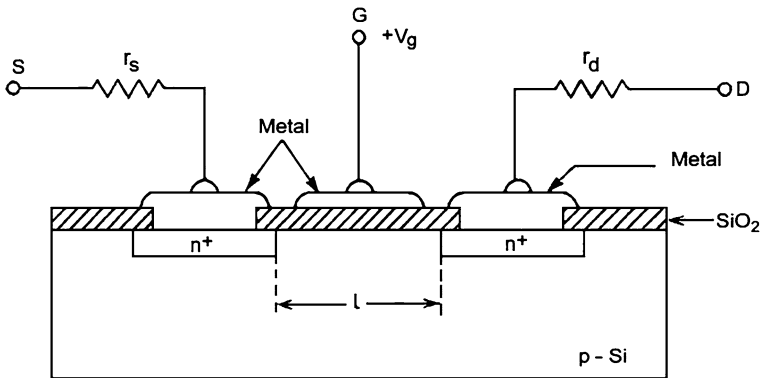


Fig. 7.21 Design of a MOSFET

In the absence of any gate voltage, no current flows through the channel because of the presence of a p - n and a n - p junction in series. On the application of a positive voltage to the gate, holes formed below the p -channel are driven towards the bulk semiconductor. This exposes the negative acceptor charges and electrons are drawn towards the interface with the bulk semiconductor. A layer of mobile electrons are formed connecting the source and the drain. If a suitable drain-source voltage is applied, electrical conduction becomes possible through the channel; increase in gate voltage (V_g) increases the channel current I_d .

We shall assume that the voltage at the gate is V_g . But the voltage at the channel is position-dependent; let us assume it to be $V(x)$ at the point x . Then the transverse voltage across the oxide insulator is $V_g - V(x)$; normally $V_g \gg V(x)$. The electric field $\mathcal{E}(x)$ at the point x is

$$\mathcal{E}(x) = [V_g - V(x)]/t \quad (7.51)$$

where t is the thickness of the oxide layer. The surface charge density $\rho_i(x)$ at the insulator-semiconductor interface is given by Gauss' law as

$$\rho_i(x) = \epsilon_i \mathcal{E}(x) \quad (7.52)$$

where ϵ_i is the interface dielectric constant of the insulator. ρ_i has contributions from both mobile and immobile acceptors. All the charges do not participate in the conduction and channel conduction begins only above a voltage V_T . We may express the channel current as

$$e\Delta n(x) = \epsilon_i [V_g - V(x) - V_T]/t \quad (7.53)$$

where $\Delta n(x)$ is the electron density per unit area of the insulator layer. The gate capacitance C_g is defined as

$$C_g = \epsilon_i l W / t \quad (7.54)$$

where W is the width of the gate electrode in a plane normal to the plane of the paper and l the channel length. Combining (7.53) and (7.54) to eliminate t , we get

$$e\Delta n(x) = (C_g / Wl) [V_g - V(x) - V_T]. \quad (7.55)$$

The conductance $G(x)$ of a length Δx of the channel of unit thickness inside p -Si is

$$G(x) = \sigma(x) \frac{Wl}{\Delta x} = \frac{e\mu_n \Delta n(x) W}{\Delta x}. \quad (7.56)$$

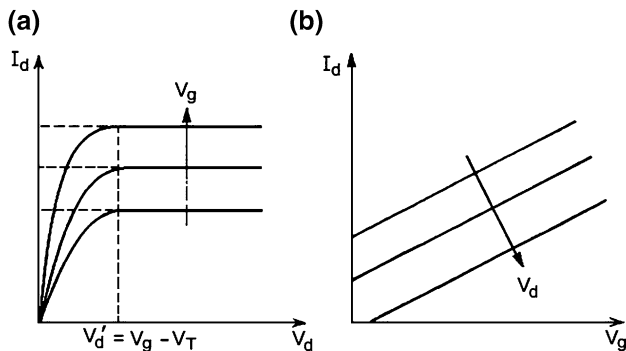


Fig. 7.22 Characteristics of a MOSFET: **a** drain characteristics and **b** transfer characteristics

Here μ_n is the electron mobility. Substituting for $e\Delta n(x)$ in (7.56) from (7.55), we have

$$G(x) = (\mu_n C_g / l) [V_g - V(x) - V_T] / \Delta x. \quad (7.57)$$

The channel current I_d under a drain voltage ΔV can be written as

$$I_d = G(x) \Delta V = \left(\frac{\mu_n C_g}{l} \right) [V_g - V(x) - V_T] \frac{\Delta V}{\Delta x}. \quad (7.58)$$

Integrating (7.58) over the length of the channel, we get

$$I_d = (\mu_n C_g / l^2) [V_g - V_T - (V_d / 2)] V_d \quad (7.59)$$

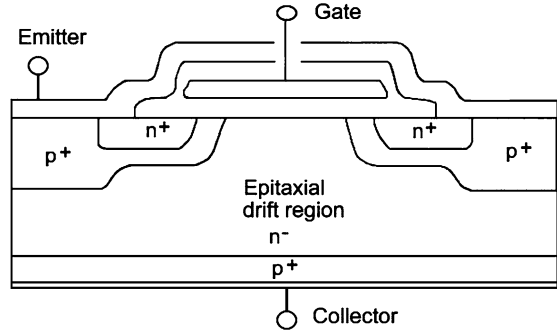
where V_d is the applied gate voltage.

The $I_d - V_d$ characteristics for different V_g values are shown in Fig. 7.22a. They are initially parabolic. If we put $(dI_d / dV_d) = 0$, we get $V_d' = V_g - V_T$. Thus the plot should saturate at the voltage V_d' . This is observed in Fig. 7.22a. From (7.59), the $I_d - V_g$ characteristics should be linear; this can be seen in Fig. 7.22b.

7.3.5 Insulated Gate Bipolar Transistor

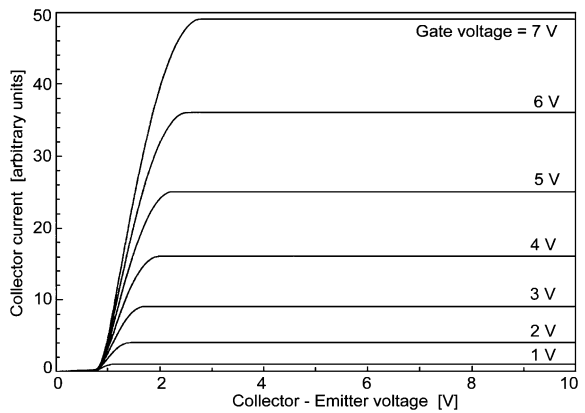
The insulated gate bipolar transistor (IGBT) was invented by Baliga [12, 13]. The structure of this Si-based device is shown in Fig. 7.23. It consists of four n - p - n - p layers controlled by a metal-oxide-semiconductor-gate. It can be considered a MOSFET with the drain region replaced by a p -type anode region.

Fig. 7.23 Design of the IGBT



The application of a gate voltage results in the formation of a channel of electrons under the gate terminal. Due to this, a current flows from the n^+ emitter through the p -body and the n^- drift region to the p^+ collector. This, in turn, causes a hole current to flow from the p^+ collector to the n -drift region; thus the device carries current. Due to the injection of holes from the p^+ collector to the n -drift region, there is a decrease in resistance and hence a voltage drop. The switching speed and the on-state voltage are two important parameters that determine the performance of the IGBT. Baliga optimized these two parameters by the technique of high energy electron irradiation. The emitter-collector voltage versus collector current characteristics at different gate voltages are shown in Fig. 7.24. IGBT has a high power gain and a low voltage drop. It can handle high voltages (~ 5 kV) and high currents (2000 Amps). Besides, it has the advantages of a small size and low weight, power loss and cost. These features make it a very popular device; it is used in switch-mode power supplies, traction motor control and induction heating.

Fig. 7.24 Characteristics of IGBT



7.4 A Few Other Devices

7.4.1 Semiconductor Solar Cell

If a p - n diode is in the open mode (open circuit) and the junction is irradiated by a source, electron-hole pairs are created which accumulate on the appropriate sides of the depletion layer. This results in an open circuit voltage across the p - n junction. In suitable conditions, the diode may deliver power to a load. When used in this manner, the junction diode is called a solar cell.

In a solar cell, since the p - n junction is to be exposed to radiation, the p - n junction is necessarily located close to the surface of the semiconductor (Fig. 7.25). If P_{in} is the incident power on the surface and R the surface reflection coefficient, the power absorbed is $P_{in}(1 - R)$. If $h\nu$ is the photon energy and n' the probability of the electron-hole creation, then the rate of electron-hole generation is $[n'P_{in}(1 - R)/h\nu]$. Let the cell be forward-biased by a voltage V . Then, as shown in Sec. 7.2.1, the current I flowing through the diode is

$$I = I_s[\exp(eV/k_B T) - 1] \quad (7.60)$$

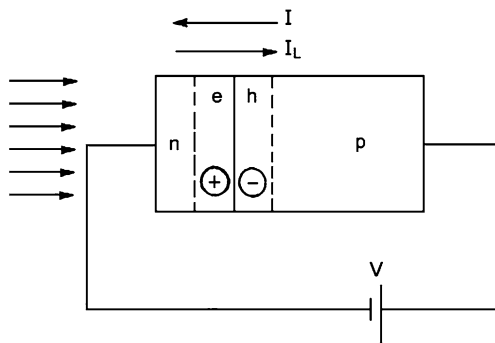
where I_s is the saturation current. When the cell is irradiated, electron-hole pairs are created in the depletion layer which drift due to the existence of the junction field. This results in a current I_L which flows in a direction opposite to the current I . Then, (7.60) gets modified to

$$I = I_s[\exp(eV/k_B T) - 1] - I_L. \quad (7.61)$$

The current I_L depends on the parameters P_{in} , ν , n' and R . At the open circuit voltage V_{oc} , the current $I = 0$. Thus, (7.61) can be reorganized as

$$V_{oc} = (k_B T/e) \log [1 + (I_L/I_s)]. \quad (7.62)$$

Fig. 7.25 Design of a solar cell



For weak intensity of radiation, $(I_L/I_s) \ll 1$. In such a case we can modify (7.62) as

$$V_{oc} \approx (k_B T/e)(I_L/I_s). \quad (7.63)$$

As the open circuit voltage V_{oc} is proportional to I_L , the cell is said to function in a linear mode. On the other hand, for larger radiation intensities, (7.62) reduces to

$$V_{oc} = (k_B T/e) \log (I_L/I_s). \quad (7.64)$$

We have plotted (7.61) in Fig. 7.26. It can be seen that in the fourth quadrant, the device has negative conductance. If we choose the working conditions in this quadrant, the cell can deliver power to an external load i.e. it can convert solar energy into electrical energy. Let us estimate the efficiency of such conversion.

The output power P may be expressed as

$$P = IV = I_s V [\exp(\beta V) - 1] - I_L V \quad (7.65)$$

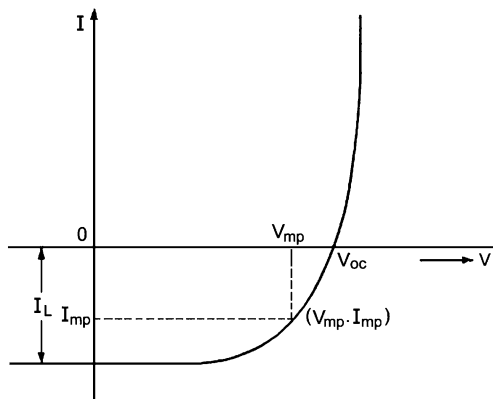
where $\beta = (e/k_B T)$. From (7.65), we can get the maximum power P_{mp} by putting $dP/dV = 0$. Then

$$\exp(\beta V_{mp}) = [1 + (I_L/I_s)] / (1 + \beta V_{mp}). \quad (7.66)$$

Here V_{mp} is the voltage at which the power is maximum. The corresponding current I_{mp} is

$$I_{mp} = I_s \exp(\beta V_{mp}) - I_s [1 + (I_L/I_s)]. \quad (7.67)$$

Fig. 7.26 I - V characteristics for a solar cell



By combining (7.66) and (7.67), we get

$$I_{mp} = -\beta I_s V_{mp} \exp(\beta V_{mp}). \quad (7.68)$$

The maximum power P_{mp} is

$$P_{mp} = V_{mp} I_{mp} = \beta I_s V_{mp}^2 \exp(\beta V_{mp}). \quad (7.69)$$

Substituting for $\exp(\beta V_{mp})$ from (7.66), we have

$$P_{mp} = \beta I_s V_{mp}^2 [1 + (I_L/I_s)] / (1 + \beta V_{mp}). \quad (7.70)$$

The efficiency of the solar cell η is defined as $\eta = P_{mp}/P_{in}$. Thus we have

$$\eta = P_{mp}/P_{in} = \frac{\beta I_s V_{mp}^2}{(1 + \beta V_{mp})} [1 + (I_L/I_s)] \frac{1}{P_{in}}. \quad (7.71)$$

Generally, $P_{in} \approx 1.35 \text{ kW/m}^2$. For a solar cell with Si, $\eta \approx 10\text{--}15\%$.

7.4.2 Semiconductor Laser

It is well-known that an electron from a higher energy level E_2 can jump to a lower level E_1 . In this process, a photon of frequency ω is emitted such that $\hbar\omega = E_2 - E_1$. There is an important constraint on such a transition viz., the population density of level E_2 should be greater than that of level E_1 . Generally, it is not so. In a laser, such a transition is made possible by bringing about a population inversion. This is a process in which electrons from level E_1 are ‘pumped’ into E_2 so that the electron population of E_2 becomes greater than that of E_1 .

Let us now see how laser action is possible in a p - n junction diode. The energy levels on the two sides of the p - n junction are shown in Fig. 7.27. The energy relation is

$$E_{FC} - E_{FV} \geq \hbar\omega. \quad (7.72)$$

To use the p - n junction as a laser, the p and n sides are heavily doped. The junction is forward biased. Electrons are injected from the right; they recombine with the holes which, in turn, are injected from the left. This results in population inversion and laser action. The active region is parallel to the junction face (Fig. 7.28) and the laser beam emerges from the side of the junction.

A detailed treatment of laser action in a junction diode is beyond the scope of this book. However, we shall note an important condition. The population inversion

Fig. 7.27 Energy levels in a highly doped p - n junction used as a laser

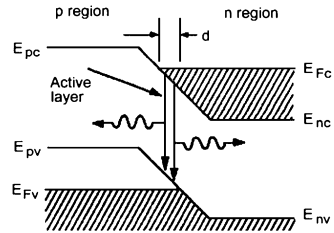
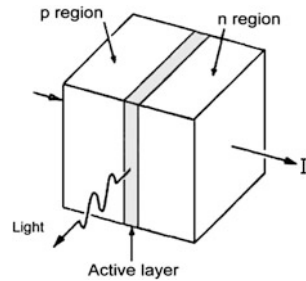


Fig. 7.28 Structure of a p - n junction laser



must be strong enough so that the gain is larger than the absorption due to several factors. This can be expressed by the equation

$$(I/d) \approx en/\tau'. \tag{7.73}$$

Here I is the junction current, d the width of the active layer, n the carrier concentration and τ' the carrier recombination time. Laser action is possible only in direct gap semiconductors like the II-VI and III-V compounds. Thus ZnS, CdTe and GaAs are good candidates but Ge and Si are not. The laser radiation emitted by a GaAs laser is shown in Fig. 7.29. Laser parameters for some semiconductor lasers are given in Table 7.2.

Fig. 7.29 Radiation from GaAs laser

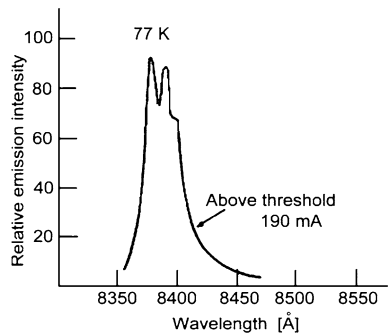


Table 7.2 Laser parameters of some semiconductors

Material	Photon energy (eV)	Wavelength (μ)
CdS	2.50	0.49
CdTe	1.58	0.78
Ga(As _{1-x} P _x)	1.41–1.95	0.88–0.63
GaAs	1.47	0.84
InP	1.37	0.90
GaSb	0.82	1.5
InSb	0.23	5.2

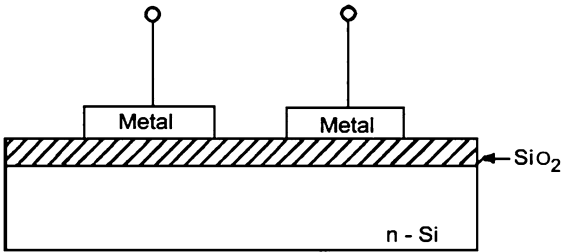
7.4.3 Charged Coupled Device

The configuration of a charged coupled device (CCD) is shown in Fig. 7.30. It consists of several (two in the figure) metal contacts grown over insulating oxide (SiO₂) layers. These layers, in turn, are deposited over a semiconductor (*n*-Si) substrate. When voltages are applied to the metal electrodes, depletion layers are formed in the semiconductor below the insulating layer. These depletion layers act as potential wells where minority carriers accumulate; they can be shifted from one potential well to the next by the application of suitable voltages to the metal contacts.

An important property of a CCD is its response to radiation. Irradiation of the metal contact results in creation of electron-hole pairs in the semiconductor substrate. The majority carriers are carried away by the voltage applied on the metal electrodes. On the other hand, the minority carriers remain stored in the potential well formed at the semiconductor-insulator interface. The density of such minority carriers is proportional to the intensity of illumination.

Finally, let us consider the effect of the voltages V_1, V_2, V_3 applied to the three metal contacts as shown in Fig. 7.31. In Fig. 7.31a, V_1 is greater than V_2 and V_3 ; the potential well is deepest under V_1 . If the voltages are as shown in Fig. 7.31b, electrons are stored in the well. After an interval of time, if the voltages are as shown Fig. 7.31c, i.e. V_2 is greater than V_1 and V_3 , the potential well is deepest under V_2 and the electrons are transferred from the first well to the second well. Similarly, if after a while, the voltages are as in Fig. 7.31d i.e. V_3 is greater than V_1

Fig. 7.30 Design of a CCD



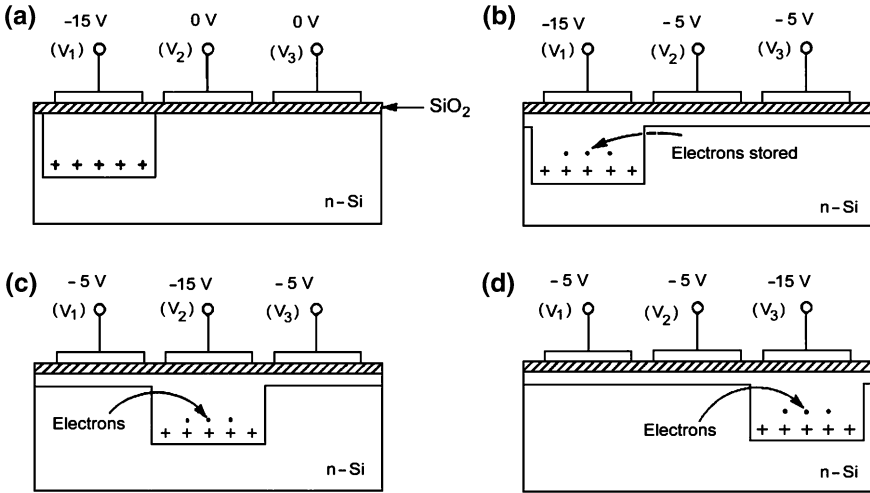


Fig. 7.31 Motion of charge in a CCD

and V_2 , the electrons in the second well are transferred from the second well to the third well under V_3 . In this way, the charge is carried from the left to the right and it is proportional to the intensity of radiation falling on the metal contacts. The CCD is an efficient detector for radiation and is used in the detection of i.r. and X-ray radiation.

7.5 Preparation of Device Material

“The discovery of the transistor in 1947–1948 was preceded by about 15, 20, 30 years of failed effort. People tried various semiconductors; they were either not pure enough or they could not make right junctions or, they did not have the right semiconductors”. This quote from Ramakrishnan [14] shows the importance of material purification in fabrication of devices. In this section, we shall discuss the following aspects of device material preparation: (i) purification, (ii) crystal growth and (iii) junction fabrication.

7.5.1 Material Purification

Semiconductor materials are purified by the technique called zone refinement introduced by Pfann [15–17]. The principle of the technique can be discussed with reference to Fig. 7.32.

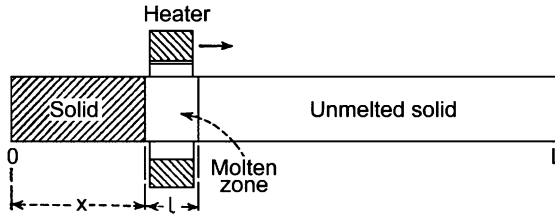


Fig. 7.32 Hot zone in a solid rod

The material is taken in the form of a bar (ingot) of length L . A narrow heater of length l is made to pass over the bar. The heater constitutes the hot zone. The traverse of the heater begins at $x = 0$. The material in a zone of length l melts. The heater passes in the x -direction. As the heater moves, the molten material behind it solidifies. The solid and the molten liquid are two solvent phases in which any impurity in the material acts as a solute. The solute has different solubilities in the solid and liquid phases of the solvent. Let us denote the equilibrium concentration of the solute in the solid and liquid phases by C_S and C_L . The ratio of these concentrations is called the segregation coefficient k_0 . Thus,

$$k_0 = C_S/C_L. \quad (7.74)$$

Figure 7.33 shows the constitutional diagrams for a solute-solvent combination. There are two possibilities: $k_0 < 1$ and $k_0 > 1$. In Fig. 7.33a, $k_0 < 1$ i.e. $C_S < C_L$; here the impurity lowers the melting point of the solvent. In Fig. 7.33b, $k_0 > 1$ i.e. $C_S > C_L$; here the impurity raises the melting point of the solvent.

We shall now examine qualitatively what happens in the bar as the molten zone, i.e. the heater, moves from the left to the right. Let us assume that the material to be

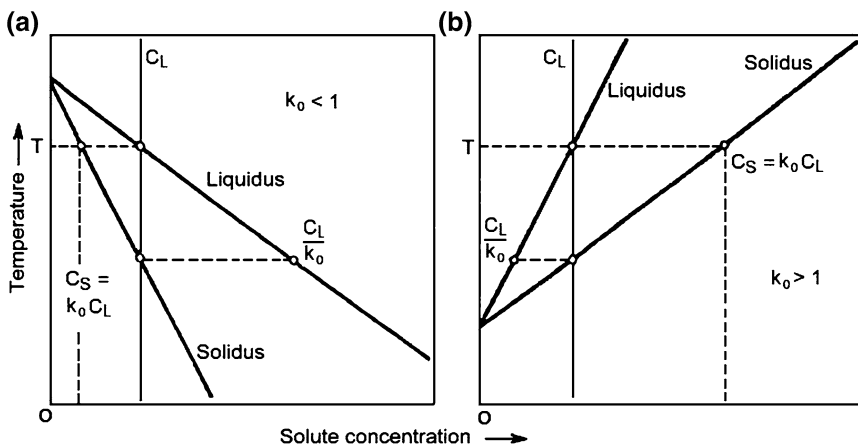


Fig. 7.33 Constitutional diagrams for a $k_0 < 1$ and b $k_0 > 1$

purified contains an impurity with $k_0 < 1$ which means that $C_S < C_L$. As the hot zone moves, the melt freezes and as it freezes, it rejects the extra amount of impurity ($C_S - C_L$) into the molten material in the new position of the heater. In other words, as the molten zone moves, it carries more and more of the impurity. When the hot zone has completed its journey (pass) from the left to the right, the left end of the rod has less impurity than it had initially. The more the number of passes, the lesser is the impurity concentration at the left end and the more is the concentration of impurity at the right end. This right end is cut-off at the end of the purification process. If the impurity has $k_0 > 1$, the purification process is reversed. The traveling hot zone leaves the impurity behind in the solidified mass. At the end of a pass, the left end of the bar has more of the impurity and the right end is purer. Thus, depending on the k_0 value of the impurity, the solid-liquid interface rejects some impurities and accepts others. The k_0 values of some impurities in Ge and Si are given in Table 7.3; most of them have $k_0 < 1$ except boron which has $k_0 > 1$ for Ge.

Let us examine the problem quantitatively. Equation (7.74) holds for slow freezing i.e. for slow velocities of the solid-liquid interface. At high velocities, the solid rejects the solute faster than it can diffuse into the bulk of the liquid. This results in a layer rich in the solute ahead of the interface. Now, the C_S/C_L ratio is not k_0 but k which is a function of the concentration of the solute in the layer and not in the bulk of the liquid. This effective distribution coefficient k is given by

$$k = k_0 / [k_0 + (1 - k_0) \exp(-\Omega)] \quad (7.75)$$

with

$$\Omega = f\delta/D. \quad (7.76)$$

Here f is the growth rate of the solid, δ the thickness of the liquid film and D the diffusion coefficient of the solute in the molten liquid. The parameter k is dimensionless. For small Ω , $k \rightarrow k_0$ and for large Ω , $k \rightarrow 1$; in general k lies between k_0 and 1.

Table 7.3 Segregation coefficients (k_0) for Ge and Si

Impurity	k_0 (Ge)	k_0 (Si)
P	0.12	0.04
As	0.04	0.07
Sb	0.003	0.002
B	20	0.68
Al	0.1	0.0016
Ga	0.1	0.004
In	0.001	0.0003

As the hot zone passes along the length of the bar, the solute concentration varies. Thus, we should express the concentration not just as C_S but as $C_S(x)$. The distance dependent concentration after one pass is given by

$$C_S(x) = C_0[1 - (1 - k) \exp(-kx/l)] \quad (7.77)$$

where C_0 is the initial concentration of the impurity at $x = 0$ and l the zone length. The solute concentration after a single pass is shown as a function of x in Fig. 7.34 for different values of k .

To improve purification, the heater is pushed to the left and the second pass starts; this process is repeated several times. For practical considerations, instead of n passes, a train of n heaters is employed (Fig. 7.35). a single pass of a train of

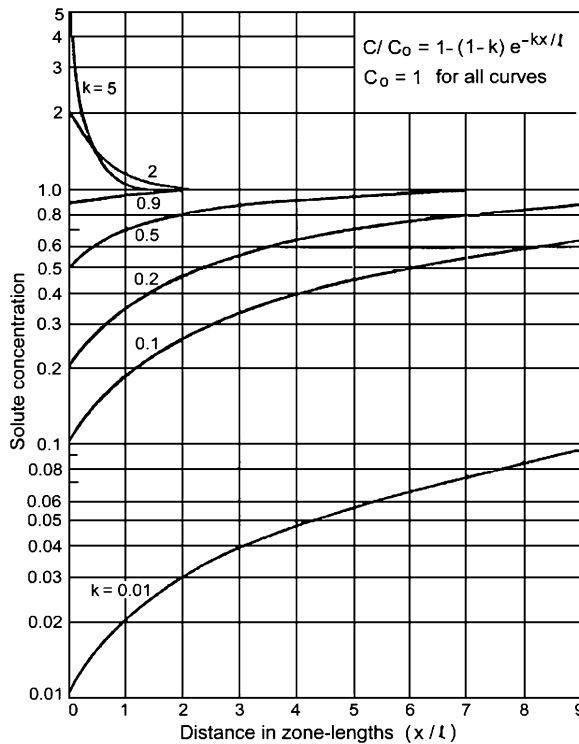


Fig. 7.34 Solute concentration as a function of (x/l) for single pass zone melting for different values of k

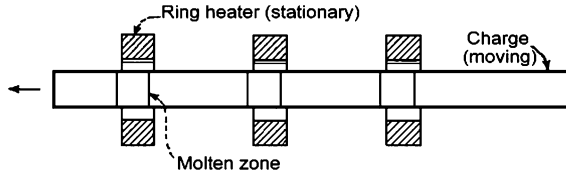


Fig. 7.35 A multiple-pass zone melting set-up

n heaters is equivalent to n passes of a single heater. After a number of passes, a final (ultimate) impurity distribution is reached. This ultimate distribution is given by

$$C_S(x)_{\text{ultimate}} = A \exp(Bx). \quad (7.78)$$

The constants A and B are obtained by solving the equations

$$k = BL / [\exp(BL) - 1] \quad (7.79)$$

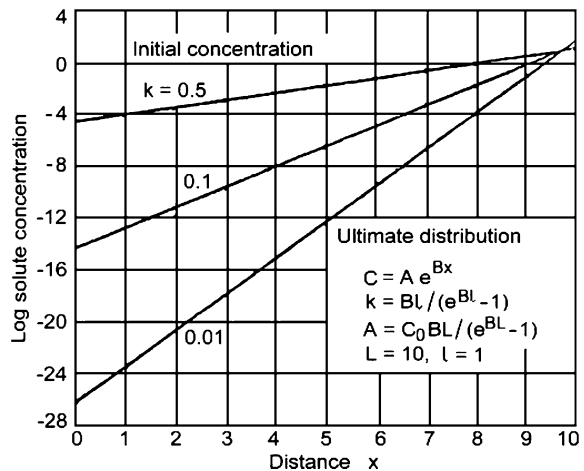
and

$$A = C_0 BL [\exp(BL) - 1] \quad (7.80)$$

Plots of the ultimate distribution for a system with $L/l = 10$ are shown in Fig. 7.36 for different values of k . It may be mentioned that for a set of values of k , l and L , a bar cannot be purified beyond the ultimate distribution.

Using experimental arrangements based on the above principles Pfann and Oslen [18] reduced the concentration of impurities in Ge to 1 in 10^{10} atoms of Ge.

Fig. 7.36 Ultimate distribution in zone melting



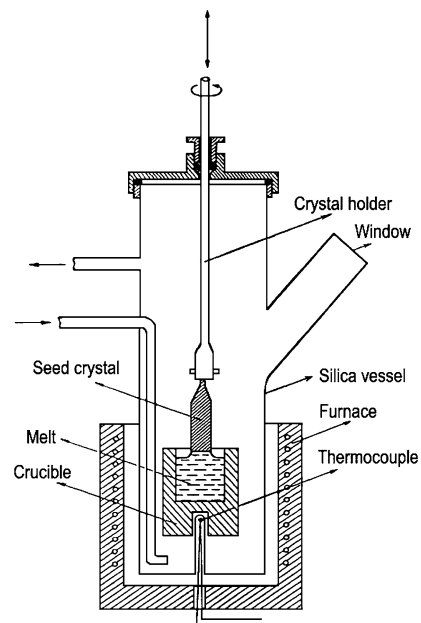
7.5.2 Crystal Growth

Single crystals of most semiconductors are grown by melting. The principle of crystal growth by melting is to melt the material and to cool the melt slowly. When the melt is close to freezing temperature a crystal nucleus is formed. In subsequent slow cooling, the atoms in the melt deposit themselves on the nucleus; the growth of the nucleus results in a single crystal. The two conditions that help the growth of a large single crystal are slow cooling and cooling through a temperature gradient. We shall now describe two commonly employed methods.

7.5.2.1 Czochralski Method

The experimental method is shown in Fig. 7.37. The growth vessel made of fused silica is placed in a furnace which has a large constant-temperature region. The furnace has a window through which the melt surface and the growing crystal can be observed. The material to be grown is taken in a crucible. The crucible is positioned in the constant temperature region of the furnace. The temperature is measured with a thermocouple. The crystal holder is a long hollow tube with a provision at the lower end to hold a seed crystal. Water can be circulated inside the tube part of the crystal holder. There is provision to raise or lower the crystal holder and also to rotate it. Also, inert gas can be introduced into the growth vessel.

Fig. 7.37 A Czochralski set-up for crystal growth



When the entire material has melted, the crystal holder along with the attached seed crystal is lowered in the furnace till the lower end of the seed touches the surface of the melt. After allowing a little time, the crystal holder is raised at a very slow speed (~ 1 mm/hr). The melt solidifies around the seed which can be seen through the window. After a single crystal of sufficient length has grown, the power to the furnace is slowly reduced so that the crystal cools slowly.

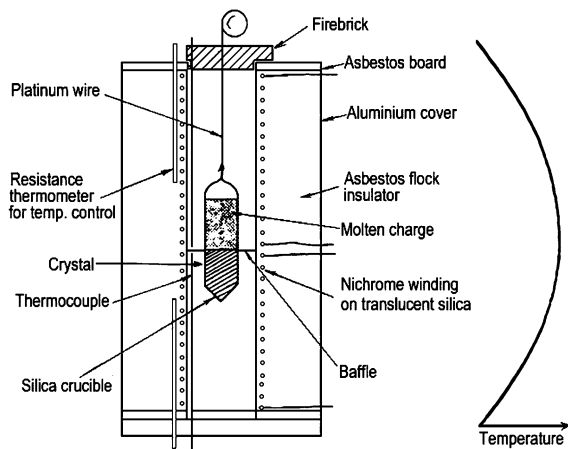
7.5.2.2 The Bridgman Method

This is a sealed container method. The experimental arrangement is shown in Fig. 7.38. The furnace is of translucent silica with nichrome windings. The number of winding turns are so adjusted that the furnace has a temperature gradient. There is a thermocouple to control the temperature of the furnace and another to measure the temperature of the growing crystal. The material to be grown is taken in a sealed silica tube. It has a tapering end and a hook at the top which links up the with a platinum wire. Initially the tube is located in a high temperature region such that the entire material in the tube melts. The tube is then lowered slowly. The tapered end of the tube is the first to enter the lower-temperature region in the furnace. Here, a part of the melt freezes and forms a tiny crystal nucleus. As the downward journey continues, more and more of the melt freezes around the crystal nucleus which finally results in a fairly long (2–5 cm) single crystal.

7.5.3 Fabrication of Junctions

A p - n junction is a piece of a semiconductor in which the impurity concentration changes from p -type to n -type in a narrow region. This change in concentration of

Fig. 7.38 A Bridgman set-up for crystal growth



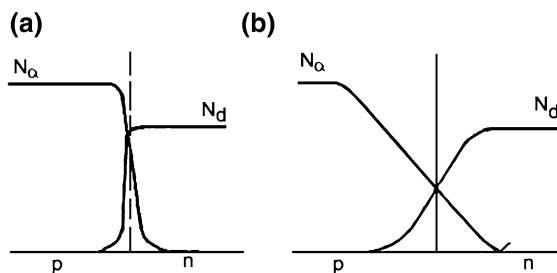


Fig. 7.39 Concentration of impurities across a junction: **a** abrupt junction, **b** graded junction; N_a , acceptor impurity; N_d , donor impurity

the two types may be ‘abrupt’ (Fig. 7.39a) or ‘graded’ (Fig. 7.39b). The number of junctions in a single piece also varies from device to device. Thus it may be a single p - n junction (Fig. 7.40) as in a junction diode or two junctions, n - p - n or p - n - p , as in a transistor (Fig. 7.40b) or three junctions (Fig. 7.40c) as in a semiconductor-controlled switch. These junctions cannot be made by placing p and n type samples adjacent to each other; rather, the variation in impurity concentration is effected in a single piece of the semiconductor by ingenious methods.

According to one method, the crystal growth starts in a Czochralski set-up with a charge that contains n -type impurity so that the crystal that grows is of n -type. At a certain stage of growth, p -type impurity is added to the molten charge. The crystal that grows subsequently is p -type; the resulting crystal includes a p - n junction.

In another method, a small quantity of a p -type impurity like In is placed on top a n -type Ge crystal. On heating, the In melts and dissolves some Ge. On slow cooling, the dissolved Ge forms a layer which is p -type.

In yet another method, a semiconductor sample is heated in a gaseous atmosphere of donors and acceptors at high temperature. The impurity now diffuses into the semiconductor. If the heating is with p -type impurity on one side and n -type impurity on the other, the resulting sample will contain a p - n junction. This is called the diffusion method.

Finally, we shall consider junction fabrication by zone refinement. Suppose the starting material is Ge with both boron and phosphorous being present as impurities. Note that boron is an acceptor and phosphorous a donor. The k_0 values are 20 and 0.12 for B and P respectively. If this charge is made to pass through a zone refinement unit, the two impurities are driven in opposite directions so that there will be a section where adjacent regions are rich in the two impurities; in other words, a p - n junction is formed. Figure 7.41 shows such p - n junctions formed in a Ge ingot.

Fig. 7.40 Types of junctions: **a** single junction, **b** double junction, **c** triple junction

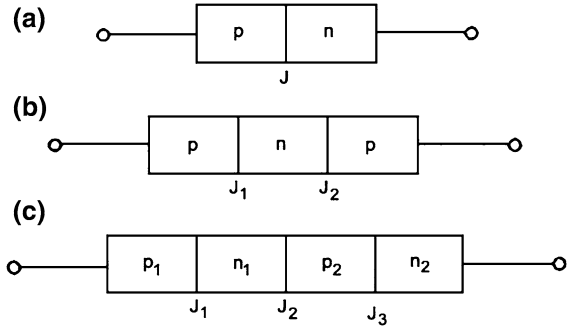
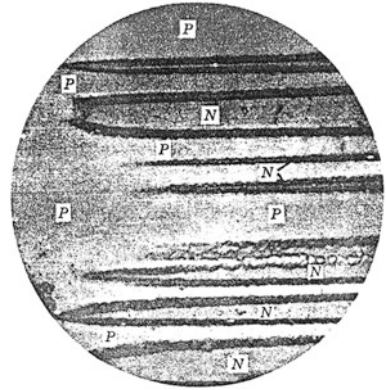


Fig. 7.41 *P-n* junction formation during zone refinement



7.6 Problems

- (1) A *p-n* junction is formed in Ge with boron and phosphorous as impurities. The dopants are 95 and 96 % ionised. Estimate the contact potential.
- (2) For the *p-n* junction in problem 1, estimate the width of the depletion layer (dielectric constant of Ge = 16).
- (3) Discuss the occurrence of NDC in a Gunn diode and explain how it leads to oscillations.
- (4) Presuming that the carrier mobilities in the upper valley and the lower valley μ_u and μ_l are proportional to $\mathcal{E}^{-1}$, derive the condition for the occurrence of NDC in GaAs.
- (5) Enumerate the factors that determine the efficiency of a semiconductor solar cell and discuss the role of each of them.

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Chapter 8

Magnetism I: Diamagnetism and Paramagnetism

Abstract This chapter deals with diamagnetism and paramagnetism. It starts with the definitions and experimental determination of the basic parameters of magnetism. According to the classical picture, the electron develops precessional motion in a magnetic field. Langevin calculated the induced magnetic moment due to this precession. The diamagnetic susceptibility derived from this moment is negative, small and independent of temperature. The same result is obtained from quantum mechanics using perturbation theory. Paramagnetism, on the other hand, is due to the presence of atoms or ions with a permanent magnetic moment. In a magnetic field, these magnetic dipoles tend to orient along the field; this orientation is governed by Maxwell-Boltzmann statistics. Langevin's classical theory leads to a susceptibility which is positive and follows the Curie law. According to quantum theory, the atomic magnets can take orientations permitted by space quantization. Quantum theory also leads to the Curie law but the susceptibility is now dependent on the spin. Theoretical predictions are compared with experimental results obtained on transition element compounds and rare earth element compounds. Paramagnetism in metals (Pauli paramagnetism) and Van Vleck paramagnetism are discussed. The production of low temperatures by adiabatic demagnetization of salts is also discussed.

8.1 Introduction

Man's acquaintance with magnetism is as old as civilization in itself. The discovery of the lodestone in times before Christ and Gilbert's invention of the magnetic compass in the 6th century were followed by the discoveries of principles and properties of magnetism in the 18th and 19th centuries by Coulomb, Oersted, Ampere and Savart. We should also remember the contributions of Faraday and Curie. Langevin was the first to give an atomistic interpretation of magnetism. The 20th century has seen an unprecedented burst of developments both in theories and applications of magnetism.

Various types of magnetism were mentioned in Chap. 1. This chapter deals with diamagnetism and paramagnetism. Diamagnetism is caused by atomic magnetic moments induced by the applied field. The diamagnetic susceptibility is small, negative and temperature-independent. Diamagnetism is universal; it exists in all materials, even in those which have other forms of magnetism. Paramagnetism, on the other hand, is caused by the presence of permanent magnetic dipoles which tend to orient themselves along the external magnetic field. Paramagnetic susceptibility is positive and is inversely dependent on the temperature.

Diamagnetism and paramagnetism are discussed by Levy [1], Wahab [2] and Omar [3] among others. Kubo and Nagamiya [4] and Ashcroft and Mermin [5] present detailed treatments.

8.2 Magnetic Parameters

The magnetic field, the magnetic induction, the magnetization, the magnetic permeability and the magnetic susceptibility are the most important magnetic parameters.

The basic magnetic force is the Coulomb force between two magnetic poles. If two magnetic poles each of strength m_1 and m_2 (in Weber) are at a distance r (in meters) from each other, a force F (in Newtons) acts between them. F (in Newtons) is given by

$$F = m_1 m_2 / \mu_0 r^2. \quad (8.1)$$

Here μ_0 is called the magnetic permeability of free space; its value is $4\pi \times 10^{-7}$ H/m (H and m represent Henry and meter respectively). Another definition of the magnetic field is the force exerted by an electric current on a magnetic pole. If we consider a current i (in amperes) flowing through a solenoid with n turns/ meter, the magnetic field H is

$$H = ni \quad (8.2)$$

The unit of magnetic field is ampere/meter (A/m). The unit Oersted (Oe) is also used ($1 \text{ A/m} = 4\pi \times 10^{-3} \text{ Oe}$; $1 \text{ Oe} = 79.6 \text{ A/m}$). A magnetic pole of strength m (web) placed in magnetic field of strength H (A/m) experiences a force F (N) given by

$$F = mH \quad (8.3)$$

Now let us consider a bar magnet of length l and poles of strength m and $-m$ at its ends placed in a uniform magnetic field H . It is acted upon by forces as indicated in Fig. 8.1a. If θ is the angle between the direction of the field H and the

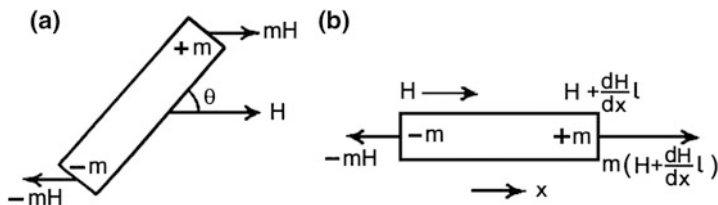


Fig. 8.1 **a** Action of forces on a bar magnet in a uniform magnetic field, **b** Action of forces on a bar magnet in a magnetic field with gradient

magnetization (line joining the poles), a couple acts on the magnet with moment L given by

$$L = -mlH \sin \theta. \quad (8.4)$$

Thus a uniform magnetic field has the effect of rotating a magnet; it cannot translate the magnet. On the other hand, if the field is nonuniform with a gradient dH/dx (Fig. 8.1b), the two poles experience forces $-mH$ and $m(H + \frac{dH}{dx}l)$. The net force F_x acting on the magnet is

$$F_x = ml \frac{dH}{dx}. \quad (8.5)$$

This force can bodily translate the magnet but cannot rotate it.

We note that in either case, the force acting on the magnet involves the quantity (ml) . This is called the magnetic moment; its units are Weber-meter [$1 \text{ wb-m} = (1/4\pi) \times 10^{10} \text{ gauss-cm}^3$]. A magnetic material may be considered to be made up of a large number of microscopic atomic magnets each with a magnetic moment μ_m . The total magnetic moment per unit volume is called the intensity of magnetization or just magnetization M . It is a vector with units wb/m^2 [$1 \text{ wb/m}^2 = (1/4\pi) \times 10^4 \text{ gauss} = 7.96 \times 10^2 \text{ gauss}$].

The magnetic susceptibility χ is defined as the magnetization per unit magnetic field. Thus,

$$\chi = M/H. \quad (8.6)$$

Since M and H have the same units (A/m), the susceptibility is dimensionless. If the magnetization M is for a gram molecule, the susceptibility is called the molar susceptibility χ_m . M and H are vectors. If we write (8.6) as $\mathbf{M} = \chi \mathbf{H}$, then χ is a property relating the components of two vectors. According to tensor algebra, a property that relates the components of two vectors is itself a second rank symmetric tensor [6, 7]. The susceptibility tensor is given by

$$[\chi_{ij}] = \begin{bmatrix} \chi_{11} & \chi_{12} & \chi_{13} \\ \chi_{12} & \chi_{22} & \chi_{23} \\ \chi_{13} & \chi_{23} & \chi_{33} \end{bmatrix}. \quad (8.7)$$

For an isotropic crystal, the off-diagonal components $[\chi_{ij}]$ become zero and the diagonal terms are equal. Thus

$$[\chi_{ij}]_{\text{isotropic}} = \begin{bmatrix} \chi & \cdot & \cdot \\ \cdot & \chi & \cdot \\ \cdot & \cdot & \chi \end{bmatrix}. \quad (8.8)$$

Hence for an isotropic material, the susceptibility is described by a single susceptibility χ i.e., the susceptibility is a scalar. Unless otherwise mentioned, we shall assume that all materials are isotropic with a single value for χ .

The magnetization of a material has its own effect apart from the effect of the applied field. To represent the combined effect of these two factors, we define the magnetic induction or the magnetic flux density $\mathbf{B}$ as

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}). \quad (8.9)$$

Since $\mathbf{M} = \chi \mathbf{H}$, we may write (8.9) as

$$\mathbf{B} = \mu_0(1 + \chi)\mathbf{H} = \mu\mathbf{H}. \quad (8.10)$$

Here, μ is called the absolute permeability of the medium (material). Like χ , μ also is a tensor. It is a scalar for isotropic materials.

For convenience, we introduce the relative permeability μ_r defined as

$$\mu_r = \mu/\mu_0 \quad (8.11)$$

From (8.10) we get

$$\mathbf{B} = \mu_0\mu_r\mathbf{H} \quad (8.12)$$

and

$$\mu_r = (1 + \chi) \quad (8.13)$$

For free space without any field, $M = 0$, $\chi = 0$, $\mu = \mu_0$ and $\mu_r = 1$; this leads to

$$\mathbf{B} = \mu_0\mathbf{H} \quad (8.14)$$

Finally, let us consider the potential energy associated with magnetization. The potential energy of a dipole of magnetic moment $\boldsymbol{\mu}$ in a magnetic field $\mathbf{H}$ is

$$U_{\text{pot.}} = -\boldsymbol{\mu} \cdot \mathbf{H} \quad (8.15)$$

Instead of a single dipole if we consider magnetization M , then the potential energy is

$$U_{\text{pot.}} = -\mathbf{M} \cdot \mathbf{H} = -MH \cos \theta \quad (8.16)$$

where θ is the angle between the two vectors.

We shall calculate the energy which a material develops if the magnetic field acting upon it is increased from 0 to H . From (8.16), the increase in energy dU per unit volume for an increase in field by dH is given by

$$dU = -MdH \quad (8.17)$$

Then

$$U = \int_0^U dU = - \int_0^H MdH = - \int_0^H \chi H dH = -\frac{1}{2}\chi H^2 \quad (8.18)$$

8.3 Experimental Methods

8.3.1 Production and Measurement of Magnetic Fields

For scientific purposes, magnetic fields are produced electrically. For small and moderate fields, coils with a few turns or solenoids are employed. For larger fields, electromagnets are used which again, are coils wound around cores. For a solenoid, there is a coil constant which is the proportionality factor between the field H and the coil current i ; this constant can be easily calculated from the dimensions of the solenoid and the number of turns. The magnetic field produced by an electromagnet can be measured by an induction method or by using a ‘search coil’. Nowadays, magnetic fields are measured from field-dependent effects like the Hall effect and the magnetoresistance using solid state probes like Ge for the former and Bi for the latter.

8.3.2 Measurement of Susceptibility

There are different approaches to the measurement of magnetization. Some of these are: measurement of force experienced by the magnetized material, measurement of voltage/current due to electromagnetic induction. We shall discuss some of the methods.

8.3.2.1 Guoy's Method

We have seen that in a nonuniform magnetic field a substance experiences a force which is proportional to its magnetization. The magnetization can therefore be measured either by directly measuring the force or by annulling the force.

The experimental set-up of Guoy's method is shown in Fig. 8.2. A long sample S is held between the pole pieces P_1, P_2 of an electromagnet. The sample is partially in the space between the pole pieces and partly outside so that it experiences the nonuniform magnetic field which is H_1 at the lower end of the sample and H_2 at the upper end. The sample experiences a force which is measured by a balance B which is linked to the sample.

If the cross-section of the sample is A and if the gradient of the field is dH_z/dz , then the force dF acting on a length dz of the sample is

$$dF = \text{grad } U = \text{grad} \left(-\frac{1}{2} \chi H^2 \right) dz A. \quad (8.19)$$

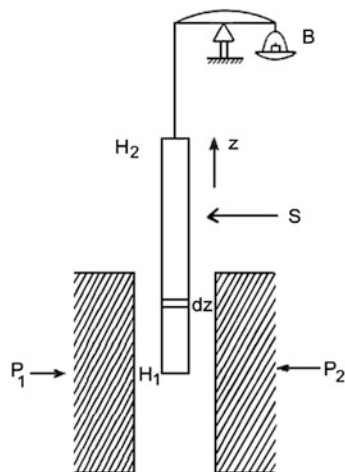
The total force F is

$$F = \int dF = - \int_{H_1}^{H_2} \frac{\chi A}{2} d(H^2) = -\frac{\chi A}{2} (H_2^2 - H_1^2) = \frac{\chi A}{2} (H_1^2 - H_2^2). \quad (8.20)$$

If the specimen is sufficiently long, $H_2 \approx 0$. Then (8.20) becomes

$$F = \frac{\chi A}{2} H_1^2. \quad (8.21)$$

Fig. 8.2 Guoy's method



As mentioned, F is measured with the balance. Knowing F and H , χ is calculated.

8.3.2.2 Sucksmith's Ring Balance

Another method is the ring balance designed by Sucksmith and Pearce [8] and shown in Fig. 8.3. Here the specimen is suspended from the lower end of an elastic ring R . When the magnetic field is applied, the sample experiences a force that deforms the ring. This deformation is indicated by the deflection of the image of a filament F by the mirrors M on the scale Sc . The deformation of the ring is a measure of the force acting on the specimen due to the applied magnetic field.

8.3.2.3 Horizontal Double Pendulum

The horizontal double pendulum designed by Lewis et al. [9] is a sensitive technique particularly suitable for the study of magnetic materials.

The set-up is shown in Fig. 8.4. It consists of a vernier eye-piece micrometer (A) to measure the displacement of the pendulum. The pendulum (B) itself consists of two fine glass threads which support the horizontal capillary boom; this holds the sample strip. The vertical suspension threads and the horizontal boom are protected in a thin glass-tube jacket. The electromagnet (C) consists of Cu-tube conductors. It generates a field of about 20,000 gauss with a current of 35 A. The pole pieces (D) are close to the horizontal boom.

The starting equation is

$$T = 2\pi(l/g)^{1/2} \quad (8.22)$$

Fig. 8.3 Sucksmith's ring balance

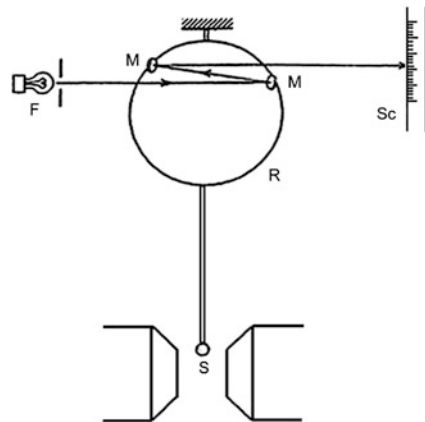
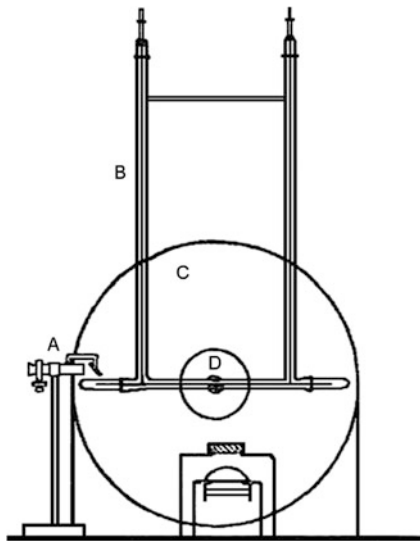


Fig. 8.4 Horizontal double pendulum



where T is the period and l the length of the pendulum. When the pendulum is displaced, the force F acting on it is

$$F = (mg/l)\delta \quad (8.23)$$

where δ is the displacement and m the mass of the composite pendulum. Combining (8.22) and (8.23), we get

$$F = (4\pi^2 m/T^2)\delta. \quad (8.24)$$

The displacement is calibrated against the force using a standard paramagnetic sample. The final working equation for the molar susceptibility χ_m is

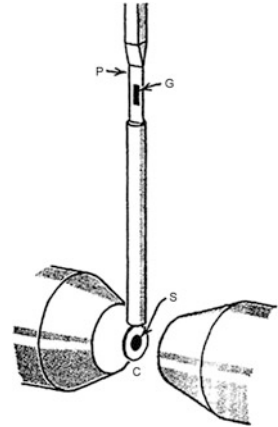
$$\chi_m = 2F/N_p P H^2 \quad (8.25)$$

where N_p is the concentration of paramagnetic ions in a gram-mole of the sample and P a correction factor.

8.3.2.4 Bozorth and Williams' Method

The set up designed by Bozorth and Williams [10] is shown in Fig. 8.5. It consists of a vertical pendulum (P) which is essentially a thin elastic metal strip. The specimen S is attached to the bottom of a pendulum. Around the specimen is a current-carrying coil C. When the magnetic field is applied, the force acting on the specimen stretches the metal strip. The extension of the strip is recorded with a

Fig. 8.5 Bozorth and Williams' method

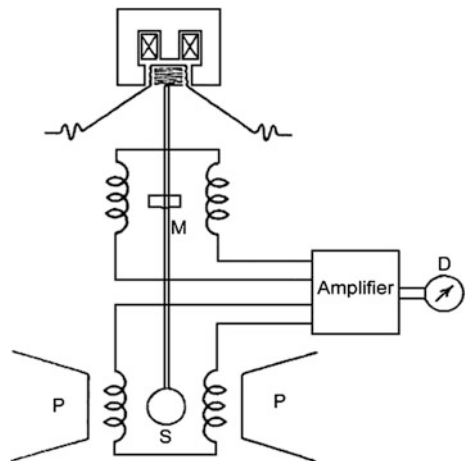


strain gauge *G* fixed on the strip. This extension is compensated by passing a suitable current through the coil such that the magnetic moment developed in the specimen by the magnetic field is cancelled by the magnetic moment produced by the coil.

8.3.2.5 Vibrating Specimen Magnetometer

The principle of this method developed by Foner [11] is shown in Fig. 8.6. The specimen *S* held between the pole pieces of an electromagnet (*P, P*) is forced to vibrate in a vertical direction. The vibration of the magnetized specimen induces an AC signal in a pair of coils placed on either side of the specimen. This is amplified and fed to a detector *D*. The pendulum also holds a standard magnet *M* at a point

Fig. 8.6 Vibrating specimen magnetometer



above the specimen and outside the magnetic field. As the specimen vibrates, the magnet also vibrates and generates an AC signal in the coils placed around it. This signal is also fed to the detector. The resultant signal is proportional to the magnetization of the specimen.

8.4 Diamagnetism

As mentioned earlier, diamagnetism is characterised by a small, negative and temperature-independent susceptibility. Diamagnetism is caused by the effect induced by an applied magnetic field on the orbital motion of the electrons. We shall see how it is explained theoretically.

8.4.1 Langevin's Classical Theory

We assume Bohr's model of the hydrogen atom with the electron revolving about the nucleus with angular frequency ω_0 . In the absence of any external magnetic field, the force F acting on the electron is the Coulomb force between the electron charge and the nuclear charge. The equation of motion is

$$F = m\omega_0^2 r = e^2/r^2 \quad (8.26)$$

with

$$\omega_0 = (e^2/mr^3)^{1/2}. \quad (8.27)$$

Here e is the electron charge, m the electron mass and r the radius of the electron orbit.

If a magnetic field is applied, an additional force ΔF acts on the electron; this is the Lorentz force which is given by

$$\Delta F = \frac{e}{c} (\mathbf{v} \times \mathbf{H}) \quad (8.28)$$

where $\mathbf{v}$ is the velocity of the electron. If the applied field H is in the z -direction and if the normal to the orbital plane is also in the z -direction, (8.28) becomes

$$\Delta F = \frac{e}{c} r\omega H. \quad (8.29)$$

We may note that ω is the altered frequency of the electron. The equation of motion now is

$$m\omega^2 r = \frac{e^2}{r^2} - \frac{e}{c} r \omega H \quad (8.30)$$

or

$$\omega^2 + (eH/mc)\omega - \omega_0^2 = 0 \quad (8.31)$$

The solution of (8.31) is

$$\omega = \pm \left[\omega_0^2 + \left(\frac{eH}{2mc} \right)^2 \right]^{1/2} - \left(\frac{eH}{2mc} \right). \quad (8.32)$$

For $\omega_0 \gg eH/2mc$, (8.32) can be approximated to

$$\omega = \pm \omega_0 - \left(\frac{eH}{2mc} \right). \quad (8.33)$$

Thus the angular frequency (ω_0) of the electron in the absence of the field changes by $(eH/2mc)$ in the presence of the field. The $\pm$ sign in (8.33) means that electrons with orbital angular momentum parallel to H are slowed down by $eH/2mc$ and those with orbital angular momentum antiparallel to H are speeded up. We may write (8.33) as

$$\omega = \pm \omega_0 - \omega_L \quad (8.34)$$

where

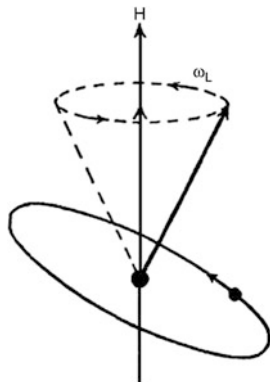
$$\omega_L = eH/2mc. \quad (8.35)$$

The change in frequency ω_L is called the Larmor frequency. It may be emphasized that it is not just a change in frequency. Rather, the orbit of the electron itself precesses about the direction of the magnetic field. The effect of the magnetic field on the orbital motion of the electron may be expressed in another way. In the presence of the magnetic field, the orbital angular momentum vector precesses about the magnetic field vector with an angular frequency ω_L (Fig. 8.7).

The motion of an electron in an orbit is equivalent to a current in a loop. In the absence of an applied magnetic field, these currents associated with several electrons get cancelled out with resultant zero current. But the current due to the precessional motion is in the same direction for all electrons; this current I is given by

$$I = qf/c \quad (8.36)$$

Fig. 8.7 Larmor precession in a magnetic field H



where q is the charge and f the frequency. Since $q = Ze$ and $f = \omega_L/2\pi$, (8.36) becomes

$$I = -\frac{Ze}{c} \frac{\omega_L}{2\pi} = -\left(\frac{Ze^2 H}{4\pi mc^2}\right). \quad (8.37)$$

Here Z is the total number of electrons in the atom. If A is the area of the orbit, the magnetic moment μ_i induced by this current is

$$\mu_i = IA. \quad (8.38)$$

If ρ is the radius of the orbit (instead of r),

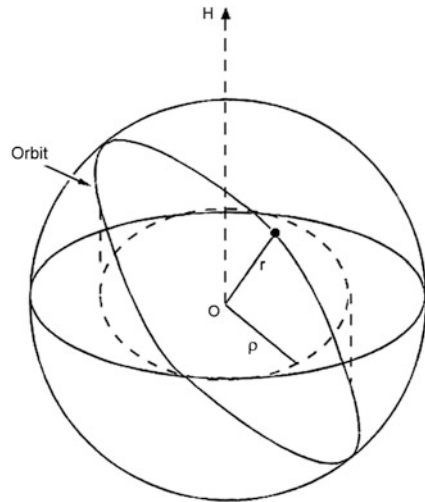
$$\mu_i = -\left(\frac{Ze^2 H}{4\pi mc^2}\right) \overline{\rho^2}. \quad (8.39)$$

Here we have used the mean square radius to take into account the variation of ρ from electron to electron.

If the plane of the circular electron orbit is not normal to the direction of the field, then we should consider the projection of the actual orbit on a plane normal to the field (Fig. 8.8). If x, y, z are the coordinates of the electron, r the radius of the orbit and ρ the radius of the projection, we have the following relations:

$$\begin{aligned} \overline{x^2} &= \overline{y^2} = \overline{z^2}, \\ \overline{x^2} + \overline{y^2} &= \overline{\rho^2}, \\ \overline{x^2} + \overline{y^2} + \overline{z^2} &= \overline{r^2}. \end{aligned} \quad (8.40)$$

Fig. 8.8 Projection of an inclined orbit on a plane normal to field H



From (8.40), we get

$$\overline{\rho^2} = \frac{2}{3} \overline{r^2}. \quad (8.41)$$

Substituting (8.41) in (8.39), we have

$$\mu_i = - \left(\frac{Ze^2 H}{6mc^2} \right) \overline{r^2} \quad (8.42)$$

and the susceptibility χ is

$$\chi = \frac{M}{H} = \frac{N\mu_i}{H} = - \left(\frac{Ze^2 N}{6mc^2} \right) \overline{r^2} \quad (8.43)$$

where N is the number of atoms in unit volume.

8.4.2 Quantum Mechanical Treatment

Following Kittel [12], we shall derive an expression for the diamagnetic susceptibility using quantum mechanics. To start with, we shall express the magnetic field $\mathbf{H}$ as the curl of a magnetic vector potential $\mathbf{A}$. Then

$$\mathbf{H} = \text{curl } \mathbf{A}. \quad (8.44)$$

In a magnetic field, the momentum $\mathbf{p}$ of a particle with charge e can be expressed as

$$\begin{aligned} \mathbf{p} &= \mathbf{p}_{\text{kin}} + \mathbf{p}_{\text{pot}} \\ &= m\dot{\mathbf{r}} + \frac{e}{c} \mathbf{A}. \end{aligned} \quad (8.45)$$

The kinetic energy T may be written as

$$T = \frac{1}{2} m \dot{\mathbf{r}}^2. \quad (8.46)$$

In view of (8.45), (8.46) may be written as

$$T = \frac{1}{2m} p^2 - \frac{e}{mc} \mathbf{p} \cdot \mathbf{A} + \frac{e^2}{2mc^2} A^2. \quad (8.47)$$

In the Schrödinger formulation the momentum operator is

$$\mathbf{p} = i\hbar \nabla. \quad (8.48)$$

The effect of the magnetic field is to add to the Hamiltonian a term $\mathcal{H}'$ given by

$$\mathcal{H}' = \frac{ie\hbar}{2mc} (\nabla \cdot \mathbf{A} + \mathbf{A} \cdot \nabla) + \frac{e^2}{2mc^2} A^2. \quad (8.49)$$

$\mathcal{H}'$ can be treated a small perturbation on the system. If the magnetic field H is uniform and acts in the z -direction, (8.44) can be expanded as

$$A_x = -\frac{1}{2} yH, A_y = -\frac{1}{2} xH \text{ and } A_z = 0, \quad (8.50)$$

Substituting in (8.49) and expanding, we get

$$\mathcal{H}' = \frac{ie\hbar H}{2mc} \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) + \frac{e^2 H^2}{8mc^2} (x^2 + y^2). \quad (8.51)$$

By definition, the angular momentum operator $\mathbf{L}$ is

$$\begin{aligned} \mathbf{L} &= -i\hbar \mathbf{r} \times \nabla \\ &= -i\hbar \left[\mathbf{i} \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right) + \mathbf{j} \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right) + \mathbf{k} \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) \right] \end{aligned} \quad (8.52)$$

We can see that the first term in (8.51) relates to the z -component of the angular momentum. It contributes to paramagnetism and is irrelevant to diamagnetism; hence, it can be ignored. The second term in (8.51) alone contributes to diamagnetism. From perturbation theory, the first order energy E' due to $\mathcal{H}'$ is given by

$$E' = \left(\frac{e^2 H^2}{12mc^2} \right) \bar{r}^2. \quad (8.53)$$

The magnetic moment μ_i is

$$\mu_i = -\frac{\partial E'}{\partial H} = -\left(\frac{e^2 H}{6mc^2} \right) \bar{r}^2. \quad (8.54)$$

The magnetization is $M = N\mu_i$ and the susceptibility χ is

$$\chi = -\left(\frac{Ze^2 N}{6mc^2} \right) \bar{r}^2 \quad (8.55)$$

which is the same as (8.43).

8.4.3 Comparison with Experimental Results

We see that the theories, both classical and quantum mechanical, predict a negative and temperature independent diamagnetic susceptibility in agreement with experimental observations. Besides, theory predicts a Z -dependence, Z being the total number of electrons of the atom/ion. For estimating the value of χ from (8.43, 8.55) we need the value of the probable mean square radius $\bar{r}^2$. This quantity can be evaluated only by quantum mechanical calculations. Assuming for the time being that $\bar{r}^2 \sim \text{\AA}^2$, we get $\chi \sim -10^{-6}$ units. It is small and thus theory conforms to the third aspect of χ .

Experimental values of χ for the rare gas atoms are given in Table 8.1. These values are small and negative and show the dependence on Z as predicted by theory;

Table 8.1 Diamagnetic susceptibilities of rare gas atoms

	χ (10^{-6} cm ³ /mole)	Z	χ/Z
He	-1.9	2	0.95
Ne	-7.2	10	0.72
A	-19.4	18	1.07
Kr	-28	36	0.78
Xe	-43	54	0.80

Table 8.2 Molar diamagnetic susceptibilities of ions in crystals ($10^{-6} \text{ cm}^3/\text{mole}$)

Ion	Experimental values	Theoretical values
F^-	-9.4	-8.1
Cl^-	-24.2	-25.2
Br^-	-34.5	-39.2
I^-	-50.6	-58.5
Li^+	-0.7	-0.7
Na^+	-6.1	-4.1
K^+	-14.6	-14.1
Rb^+	-22.0	-25.1
Cs^+	-35.1	-38.7
Mg^{2+}	-4.3	-3.1
Ca^{2+}	-10.7	-11.1
Sr^{2+}	-18.0	-21.0
Ba^{2+}	-29.0	-32.6

the ratio χ/Z is close to unity. Experimental and theoretical values for a number of ions in crystals are given Table 8.2; the agreement in the two sets is fair.

8.5 Paramagnetism

Paramagnetic materials are characterised by a small and positive susceptibility. Further, the susceptibility is temperature-dependent following a $\chi - T^{-1}$ relationship called the Curie law. Unlike diamagnetism which is due to induced magnetic moments, paramagnetism is due to the presence of permanent atomic magnetic moments.

8.5.1 Langevin's Classical Theory of Paramagnetism

As discussed earlier, according to the Bohr model, electrons of an atom rotate about the nucleus of the atom. This rotation is equivalent to a current I . If r is the radius of the circular orbit and ω_0 the angular frequency of the electron, the current I is given by

$$I = -\frac{e\omega_0}{2\pi c}. \quad (8.56)$$

If A is the area of the orbit, such a loop current produces a magnetic moment μ_m given by

$$\mu_m = IA = \left(-\frac{e\omega_0}{2\pi c}\right)\pi r^2 = -\frac{e\omega_0 r^2}{2c}. \quad (8.57)$$

The orbital angular momentum of the electron is $m\omega_0 r^2$. Hence,

$$\text{Magnetic moment/orbital angular momentum} = -\frac{e}{2mc} \quad (8.58)$$

The negative sign in (8.58) indicates that the magnetic moment points in a direction opposite to the orbital angular momentum.

Let us consider a substance containing N magnetic dipoles of magnetic moment μ_m in unit volume. In the absence of an external magnetic field, these dipoles are randomly oriented. If an external field is applied, the dipoles tend to align themselves parallel to the field. However, this tendency is opposed by the thermal agitation in the material. As a result only some, but not all, of the dipoles align themselves along the field. Different number of dipoles align themselves in different orientations. The method to find this number and the mean magnetic moment was developed by Langevin [13].

The potential energy U of a dipole in a magnetic field is

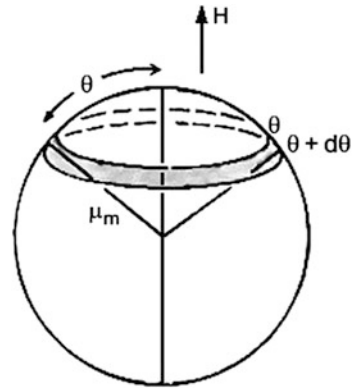
$$U = -\boldsymbol{\mu}_m \cdot \mathbf{H}. \quad (8.59)$$

If the magnetic moment makes angle θ with the field, (8.59) becomes

$$U = -\mu_m H \cos \theta. \quad (8.60)$$

Let dn be the number of dipoles oriented within the angular range θ to $\theta + d\theta$. We have to actually consider the dipoles within two shells (Fig. 8.9) with solid angles Ω and $\Omega + d\Omega$; $d\Omega$ is $2\pi \sin\theta d\theta$. According to Maxwell-Boltzmann statistics, dn is also proportional to $\exp(U/k_B T)$ i.e. $\exp(\mu_m H \cos \theta/k_B T)$. Thus the mean magnetic moment is

Fig. 8.9 Element of solid angle to obtain distribution of magnetic dipoles



$$\bar{\mu}_m = \frac{\int \mu \cos \theta \, dn}{\int dn} = \frac{\int_0^\pi (\mu_m \cos \theta) [\exp(\mu_m H \cos \theta / k_B T) 2\pi \sin \theta] d\theta}{\int_0^\pi [\exp(\mu_m H \cos \theta / k_B T) 2\pi \sin \theta] d\theta}. \quad (8.61)$$

Let us introduce $x = \mu_m H / k_B T$ and $y = \cos \theta$. Then,

$$\bar{\mu}_m = \mu_m \int_{-1}^{+1} y \exp(xy) dy \bigg/ \int_{-1}^{+1} \exp(xy) dy. \quad (8.62)$$

Using standard integrals, we get

$$\begin{aligned} \bar{\mu}_m &= \mu_m \left[\frac{\exp(x) + \exp(-x)}{\exp(x) - \exp(-x)} - \frac{1}{x} \right] \\ &= \mu_m [\coth x - (1/x)] = \mu_m L(x). \end{aligned} \quad (8.63)$$

The function $L(x)$ is called the Langevin function. The variation of $L(x)$ with (x) is shown in Fig. 8.10. It may be noted that for very large x (i.e. large fields or low temperatures) $L(x) \rightarrow 1$. This means that all magnetic dipoles align parallel to the field. For small values of x ,

$$\coth x = \frac{1}{x} + \frac{x}{3} + \frac{x^3}{15} + \dots \approx \frac{1}{x} + \frac{x}{3}$$

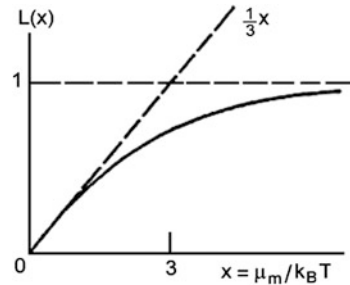
and

$$L(x) \approx x/3. \quad (8.64)$$

Hence, the magnetization M for unit volume is

$$\begin{aligned} M &= N\mu_m(x/3) \\ &= \frac{1}{3} N\mu_m \left(\frac{\mu_m H}{k_B T} \right) = \left(\frac{N\mu_m^2}{3k_B} \right) \frac{H}{T} \end{aligned} \quad (8.65)$$

Fig. 8.10 The Langevin function $L(x)$ versus x



The paramagnetic susceptibility χ is

$$\chi = \frac{M}{H} = \left(\frac{N\mu_m^2}{3k_B} \right) \frac{1}{T} = \frac{C}{T}. \quad (8.66)$$

Thus, the theory accounts for Curie law with $C = N\mu_m^2/3k_B$.

8.5.2 Quantum Theory of Paramagnetism

The quantum theory of paramagnetism differs from the classical theory in the following respects: (i) It redefines the permanent magnetic moment in terms of the orbital angular momentum and the spin angular momentum, (ii) it recognises that orbital and spin angular momenta are quantized, (iii) it recognises that the magnetic momentum is also quantized and (iv) it takes into account the phenomenon of space quantization according to which the angular momentum vectors $\mathbf{L}$, $\mathbf{S}$, $\mathbf{J}$ assume different orientations in a magnetic field and each precesses around the field.

The quantum theory of paramagnetism is based on the vector model which has been discussed in detail in chap. 11. We shall briefly recapitulate the main points. According to the model, the electronic state of an atom is described by the orbital angular momentum quantum number L , the spin angular quantum number S , their resultant quantum number J and their associated magnetic quantum numbers m_L , m_S , m_J (which are projections of L , S , and J on a given direction). However any four of them are sufficient to describe the electronic state of the atom. The quantised unit of the magnetic moment is $(e\hbar/2mc)$ in the cgs system ($e\hbar/2m$ in the SI system). This is called the Bohr magneton μ_B . Its value is 0.9273×10^{-20} erg/oersted. The atomic properties dependent on the quantum numbers are summarised in Table 8.3. In the table, the quantity g is the Lande's splitting factor defined as

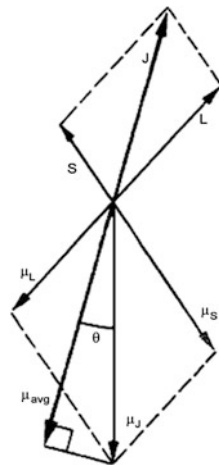
$$g = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}. \quad (8.67)$$

Its values are 1 and 2 for $S = 0$ and $L = 0$ respectively. Note also that $\frac{\mu_S}{S} = 2\frac{\mu_L}{L}$, i. e. the gyromagnetic ratio for spin angular momentum is double that for the orbital angular momentum. This has an important effect viz., while the magnetic moment due to $\mathbf{L}$ is in line with $\mathbf{L}$ and that due to $\mathbf{S}$ is in line with $\mathbf{S}$, the magnetic moment

Table 8.3 Quantum numbers and associated atomic properties

Quantum numbers	Angular momentum	Magnetic moment
L	$\sqrt{L(L+1)} \hbar$	$-\sqrt{L(L+1)} \mu_B$
S	$\sqrt{S(S+1)} \hbar$	$-2\sqrt{S(S+1)} \mu_B$
J	$\sqrt{J(J+1)} \hbar$	$-g\sqrt{J(J+1)} \mu_B$

Fig. 8.11 Orientation of L , S , J and the associated magnetic moments



due to $\mathbf{J}$ is not in line with $\mathbf{J}$ (Fig. 8.11). Hence the effective quantum number in the presence of a magnetic field is not J but M_J which is the projection of J on the field direction. Thus the effective magnetic moment is

$$\mu_J = -g\mu_B M_J. \quad (8.68)$$

Since g is a function of J and since J can assume $(2J + 1)$ orientations in the presence of the field, hence M_J too can assume $(2J + 1)$ values in the field; in spectroscopic terms this causes a splitting of energy levels.

The procedure to obtain χ is similar to that in the classical theory with two differences. Firstly, the energy of the atomic magnet is

$$U = -M_J g \mu_B H. \quad (8.69)$$

Secondly, to obtain the mean magnetic moment, instead of integration, we shall resort to summation for all permitted orientations of the magnetic moment. Thus the mean magnetic moment is

$$\bar{\mu}_m = \frac{\sum_{-J}^{+J} M_J g \mu_B \exp(M_J g \mu_B H / k_B T)}{\sum_{-J}^{+J} \exp(M_J g \mu_B H / k_B T)}. \quad (8.70)$$

We shall discuss the implication of (8.70) in two regimes:

8.5.2.1 At Low Magnetic Fields/High Temperatures

In this regime, $M_J g \mu_B H / k_B T \ll 1$. We can then make the approximation

$$\exp(M_J g \mu_B H / k_B T) \approx [1 + (M_J g \mu_B H / k_B T)]. \quad (8.71)$$

Thus the magnetization M can be written as,

$$M = N \bar{\mu}_m = \frac{N \sum_{-J}^{+J} M_J g \mu_B [1 + (M_J g \mu_B H / k_B T)]}{\sum_{-J}^{+J} [1 + (M_J g \mu_B H / k_B T)]}. \quad (8.72)$$

Carrying out the summation, we get for the paramagnetic susceptibility,

$$\chi = \frac{M}{H} = \left[\left(\frac{N g^2 \mu_B^2}{3 k_B} \right) J(J+1) \right] \frac{1}{T}. \quad (8.73)$$

If we compare (8.72) with (8.66), we can draw two conclusions. Firstly (8.70) represents Curie law with $C = (N g^2 \mu_B^2 / 3 k_B) J(J+1)$. Secondly, μ_m is replaced by μ_J the total magnetic moment associated with J as

$$\begin{aligned} \mu_J^2 &= g^2 J(J+1) \mu_B^2. \\ &= p_{\text{eff}}^2 \mu_B^2 \end{aligned} \quad (8.74)$$

p_{eff} is called the effective number of Bohr magnetons.

8.5.2.2 At High Fields and Low Temperatures

Now, $(M_J g \mu_B H / k_B T)$ is no longer less than 1. We cannot make the approximation (8.71) and we have to use (8.70) as it is. The mathematical procedure to reduce (8.70) to a closed form is long and complicated. We shall simply assume the result:

$$M = N g J \mu_B B_J(x) \quad (8.75)$$

with

$$x = (g \mu_B J)(H / k_B T). \quad (8.76)$$

The function $B_J(x)$ is defined as

$$B_J(x) = \frac{2J+1}{2J} \coth\left(\frac{2J+1}{2J}x\right) - \frac{1}{2J} \coth\left(\frac{x}{2J}\right) \quad (8.77)$$

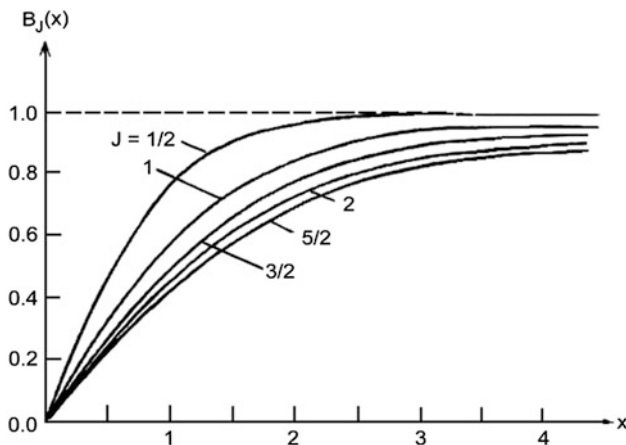


Fig. 8.12 The Brillouin function $B_J(x)$ versus x for different J values

and is known as the Brillouin function. A plot of $B_J(x)$ for different values of J is shown in Fig. 8.12. For $x \gg 1$, the value of $\coth$ of any function $\rightarrow 1$ and hence $B_J(x) \rightarrow 1$. Then we have

$$M = Ng\mu_B \quad (8.78)$$

This means that at high fields, all atomic magnets align themselves along the field; in other words, there is saturation.

In order to calculate either χ or p_{eff} from (8.73, 8.74), we need the knowledge of the J value of atoms/ions. These J values are estimated from Hund's rules together with Pauli's exclusion principle. The Hund's rules are:

- (i) Orbital angular momenta add up to give maximum L .
- (ii) Spin momenta add up to give maximum S .
- (iii) The value of J is $L - S$ if the shell is less than half filled, $L + S$ if it is more than half filled and 0 if it is just half filled i.e., $L = 0$ and $J = S$.

Finally, we shall note that filled electron shells do not contribute to the magnetic moment of the atom. Thus, paramagnetism is displayed only by atoms with incomplete shells. This condition is satisfied by ions of rare group elements (incomplete $4f$ shells) and ions of iron group (incomplete $3d$ shell).

8.5.3 Comparison with Experiment

8.5.3.1 Comparison of Diamagnetic and Paramagnetic Susceptibilities

The theoretical ratio of χ_{para} and χ_{dia} is

$$\frac{\chi_{para}}{\chi_{dia}} = -\frac{2J(J+1)}{Zk_B T} \frac{\hbar^2}{m r^2} \quad (8.79)$$

At room temperature this ratio is ~ 500 [5]. The experimental value of this ratio is also of the same order. Thus the paramagnetic susceptibility completely masks the diamagnetic susceptibility.

8.5.3.2 Curie Law

The classical theory as well as the quantum theory of paramagnetism predict a Curie law behaviour i.e., $\chi = C/T$. In Fig. 8.13, the magnetic susceptibility of the paramagnetic salt $\text{CuSO}_4 \cdot \text{K}_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ is plotted against the reciprocal temperature ($1/T$). The straight line plot is a demonstration of the Curie law.

8.5.3.3 Variation of Susceptibility with (H/T)

We have seen that the dependence of the paramagnetic susceptibility on the combined effect of the magnetic field and the temperature (H/T) is given by the Brillouin function. Plots of the susceptibility of some salts containing ions of rare earths and iron group elements are shown in Fig. 8.14 from data by Henry [14]. These plots exactly fit the theoretical curves for Brillouin functions corresponding

Fig. 8.13 Plot of magnetic susceptibility χ of $\text{CuSO}_4 \cdot \text{K}_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ against ($1/T$)

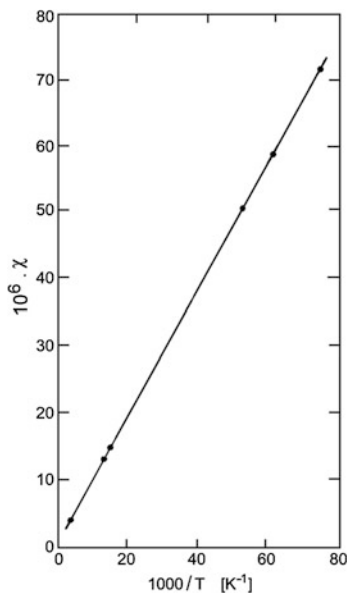
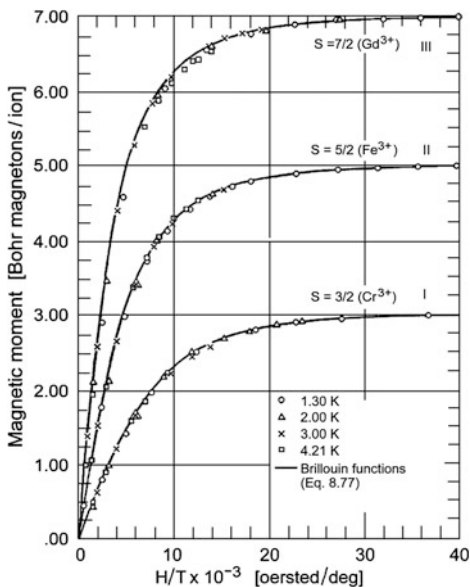


Fig. 8.14 Plot of magnetic moment versus H/T for spherical samples of (I) potassium chromium alum, (II) ferric ammonium alum, and (III) gadolinium sulphate octahydrate



to appropriate J values. In these cases, the orbital angular momentum is quenched, i.e. $J = S$. The S values are shown in the figure.

8.5.3.4 Effective Number of Bohr Magnetons

From the experimental values of χ and a knowledge of the J values, the effective number of Bohr magnetons can be calculated from (8.74). These p_{eff} values provide interesting information.

The experimental p_{eff} values for the rare earth ions are given in Table 8.4 and also shown in Fig. 8.15 along with values calculated from (8.74). There is good agreement between the two sets of values except in the case of Sm^{3+} and Eu^{3+} . Van Vleck and Frank [15] have pointed out that the discrepancy in these two cases is due to the effect of the higher states of the L - S multiplet on the effective J value.

The p_{eff} values for the iron group ions are also shown in Fig. 8.15 and given in Table 8.5. Here there are large differences between the experimental and calculated values. It is observed that the agreement improves dramatically if we replace J in (8.74) by S giving

$$p_{\text{eff}} = 2\sqrt{S(S+1)} \quad (8.80)$$

The values of p_{eff} calculated from (8.80) are also given in Table 8.5 and are shown in Fig. 8.15. It is as if the orbital angular momentum does not contribute to the paramagnetism of these ions. This effect is called the ‘quenching of the orbital angular momentum’.

Table 8.4 Effective magneton numbers for trivalent rare earth ions

Ion	Configuration	Basic level	$p(\text{calc}) = g[J(J + 1)]^{1/2}$	$p(\text{exp})$ approx.
Ce ³⁺	4 <i>f</i> ¹ 5 <i>s</i> ² 6 <i>p</i> ⁶	² F _{5/2}	2.54	2.4
Pr ³⁺	4 <i>f</i> ² 5 <i>s</i> ² 6 <i>p</i> ⁶	³ H ₄	3.58	3.5
Nd ³⁺	4 <i>f</i> ³ 5 <i>s</i> ² 6 <i>p</i> ⁶	⁴ I _{9/2}	3.62	3.5
Pm ³⁺	4 <i>f</i> ⁴ 5 <i>s</i> ² 6 <i>p</i> ⁶	⁵ I ₄	2.68	–
Sm ³⁺	4 <i>f</i> ⁵ 5 <i>s</i> ² 6 <i>p</i> ⁶	⁶ H _{5/2}	0.84	1.5
Eu ³⁺	4 <i>f</i> ⁶ 5 <i>s</i> ² 6 <i>p</i> ⁶	⁷ F ₀	0	3.4
Gd ³⁺	4 <i>f</i> ⁷ 5 <i>s</i> ² 6 <i>p</i> ⁶	⁸ S _{7/2}	7.94	8.0
Tb ³⁺	4 <i>f</i> ⁸ 5 <i>s</i> ² 6 <i>p</i> ⁶	⁷ F ₆	9.72	9.5
Dy ³⁺	4 <i>f</i> ⁹ 5 <i>s</i> ² 6 <i>p</i> ⁶	⁶ H _{15/2}	10.63	10.6
Ho ³⁺	4 <i>f</i> ¹⁰ 5 <i>s</i> ² 6 <i>p</i> ⁶	⁵ I ₈	10.60	10.4
Er ³⁺	4 <i>f</i> ¹¹ 5 <i>s</i> ² 6 <i>p</i> ⁶	⁴ I _{15/2}	9.59	9.5
Tm ³⁺	4 <i>f</i> ¹² 5 <i>s</i> ² 6 <i>p</i> ⁶	³ H ₆	7.57	7.3
Yb ³⁺	4 <i>f</i> ¹³ 5 <i>s</i> ² 6 <i>p</i> ⁶	² F _{7/2}	4.54	4.5

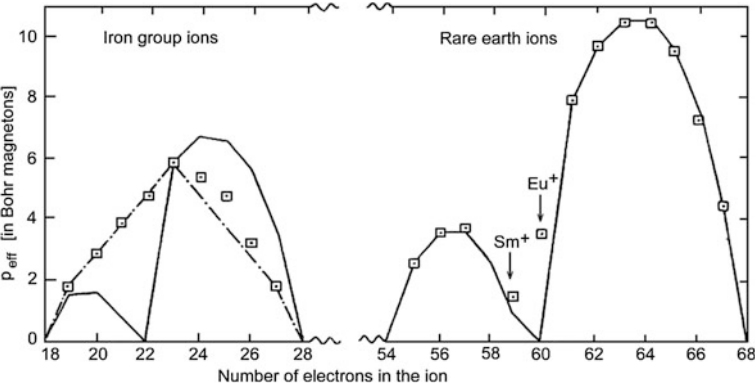


Fig. 8.15 Effective number of Bohr magnetons in iron group ions and rare earth ions; *continuous curves from (8.74); dashed curve from (8.80)*

8.6 Pauli Paramagnetism

Metals show a feeble paramagnetism. If we apply the Langevin theory of paramagnetism to a metal, we would find that the susceptibility is $\sim 10^{-4}/\text{cm}^3$ varying inversely with temperature. But experimentally, a much smaller susceptibility $\sim 10^{-6}/\text{cm}^3$ is observed; further this susceptibility is independent of temperature. An explanation of this paramagnetism of metals was provided by Pauli [16] in terms of the spin of electrons; hence this paramagnetism is called Pauli paramagnetism and the corresponding susceptibility is called the Pauli susceptibility χ_{Pauli} .

Table 8.5 Effective magneton numbers for iron group ions

Ion	Configuration	Basic level	$p \text{ (calc)} = g[J(J+1)]^{1/2}$	$p \text{ (calc)} = 2[S(S+1)]^{1/2}$	$p \text{ (exp)}$
Ti ³⁺ , V ⁴⁺	3d ¹	² D _{3/2}	1.55	1.73	1.8
V ³⁺	3d ²	³ F ₂	1.63	2.83	2.8
Cr ³⁺ , V ²⁺	3d ³	⁴ F _{3/2}	0.77	3.87	3.8
Mn ³⁺ , Cr ²⁺	3d ⁴	⁵ D ₀	0	4.90	4.9
Fe ³⁺ , Mn ²⁺	3d ⁵	⁶ S _{5/2}	5.92	5.92	5.9
Fe ²⁺	3d ⁶	⁵ D ₄	6.70	4.90	5.4
Co ²⁺	3d ⁷	⁴ F _{9/2}	6.54	3.87	4.8
Ni ²⁺	3d ⁸	³ F ₄	5.59	2.83	3.2
Cu ²⁺	3d ⁹	² D _{5/2}	3.55	1.73	1.9

According to the Pauli exclusion principle, only two electrons can be associated with a given electronic state having quantum numbers n , l and m_l . These will have a parallel spin and an antiparallel spin with reference to a fixed direction like the direction of a magnetic field. We may represent the density of state curves as the sum of two parts, one for parallel spins and the other for antiparallel spins, each having half the number of states permitted by Fermi-Dirac statistics (Chap. 3). These curves are shown in Fig. 8.16a. Each electron in one part is equally likely to occupy the other part. For convenience, we assume one part to be occupied entirely by parallel spins and the other by antiparallel spins. When a magnetic field H is applied, the parallel spin electrons will experience an energy shift of $-\mu_B H$. Similarly, the antiparallel electrons will experience an energy shift of $+\mu_B H$. The density of states curve for antiparallel spins is thus shifted relative to that for parallel spins by $2\mu_B H$. This is shown in Fig. 8.16b. Here $\mu_B H \ll E_{F0}$ where E_{F0} is the Fermi energy at $T = 0$. The Fermi levels in the two parts are also relatively shifted. This is an unstable situation. To regain equilibrium, electrons from the top of the right hand side are transferred to empty levels on the left hand side until the Fermi levels on the two sides are level (Fig. 8.16c).

Pauli derived an expression for the susceptibility using free electron theory and Fermi-Dirac statistics. Thus we have

$$N(E)dE = \int_0^{\infty} D(E)f(E)dE, \quad (8.81)$$

where

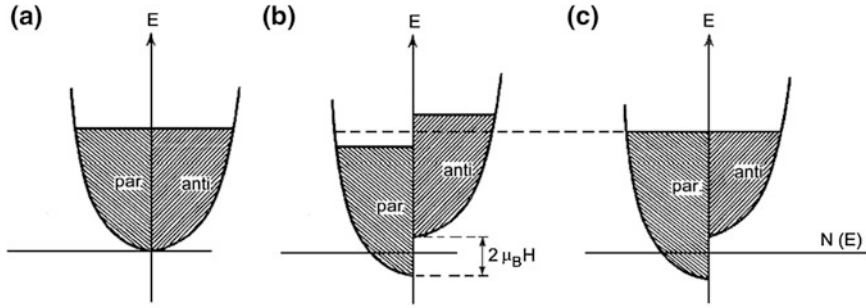


Fig. 8.16 Density of state curves for free electrons **a** In absence of field, **b** In presence of field (non equilibrium), **c** In presence of field (equilibrium)

$$D(E)dE = \frac{2\pi V}{h^3} (2m)^{3/2} E^{3/2} dE \quad (8.82)$$

and

$$f(E) = 1 / \left[\exp \left(\frac{E - E_F}{k_B T} \right) + 1 \right]. \quad (8.83)$$

Further, he made the approximation

$$D(E \pm \mu_B H) = D(E) \left[1 \pm \frac{\mu_B H}{2k_B T} \right]. \quad (8.84)$$

This was justified since $\mu_B H \ll 2k_B T$. The magnetization M is the magnetic moment per unit volume due to the difference in magnetic moments of electrons in the two parts (Fig. 8.16). We have

$$M = \frac{\mu_B}{V} \int_0^\infty \left[\frac{1}{2} D(E + \mu_B H) - \frac{1}{2} D(E - \mu_B H) \right] f(E) dE. \quad (8.85)$$

Using (8.81–8.84) in (8.85), the expression for the Pauli susceptibility is

$$\chi_{\text{Pauli}} = M/H = \left[\frac{3n}{2k_B T_F} \mu_B^2 \right] \quad (8.86)$$

where T_F is the Fermi temperature. If typical values of T_F are substituted we get $\chi_{\text{Pauli}} \sim 10^{-6}/\text{cm}^3$. There are two similarities in the Langevin susceptibility and the Pauli susceptibility. Firstly, both are of ‘minute’ magnitude [5]. Secondly, dimensionally, both have a temperature term in the denominator; however, in the

Table 8.6 Susceptibility of the alkali metals ($\chi \times 10^6$ per gram)

		Li	Na	K	Rb	Cs
I.	Calculated spin susceptibility from (8.86)	1.5	0.66	0.60	0.31	0.24
II.	As corrected for effective mass	2.1	0.64	0.56	0.27	0.20
III.	Observed bulk susceptibility	3.54	0.70	0.54	0.22	0.20

Langevin picture the temperature is variable whereas it is constant in the Pauli expression.

Values of χ obtained from experiments and χ_{Pauli} calculated from (8.86) are given in Table 8.6. While there is order of magnitude agreement between the two sets, there are substantial differences also. An explanation for this difference is to be found if we use the concept of effective mass (m^*/m) from the band theory. While comparing the results of free electron theory and band theory (Chap. 5), it was pointed out that wherever mass m occurs in the former, it should be replaced by the effective mass m^* . Now, the Fermi temperature is defined as

$$T_F = \frac{E_F}{k_B} = \frac{\hbar^2}{2mk_B} \left(\frac{3\pi^2 N}{V} \right). \quad (8.87)$$

Thus, the χ_{Pauli} values calculated from (8.86) should be multiplied by m^*/m , or

$$(\chi_{\text{Pauli}})_{\text{corrected}} = \chi_{\text{Pauli}} (m^*/m). \quad (8.88)$$

In Table 8.6 we have included these mass corrected values of χ_{Pauli} using the values of m^*/m given by Kittel [12]. The agreement between experimental values and $(\chi_{\text{Pauli}})_{\text{corrected}}$ is much better.

8.7 Adiabatic Demagnetization

Adiabatic demagnetization is a method of producing low temperatures employing paramagnetic materials. This process was proposed independently by Debye [17] and Giauque [18].

The principle of the method can be understood from Fig. 8.17. The variation of entropy S of a paramagnet salt with temperature T is denoted by curve AB; there is no magnetic field ($H = 0$). Let the point B correspond to temperature T_2 . A magnetic field H is applied to the system. Some of the spins in the paramagnetic salt align themselves along the field. Alignment of the spins means an increase in order and that, according to the second law of thermodynamics, means a decrease in entropy to a value represented by point B'. During this process, the specimen is in good thermal contact with the surroundings i.e. $\Delta T = 0$ and the process of magnetization is isothermal. The point B lies on another S - T curve A'B' similar to AB. At this

Fig. 8.17 Entropy-temperature plot for adiabatic demagnetization

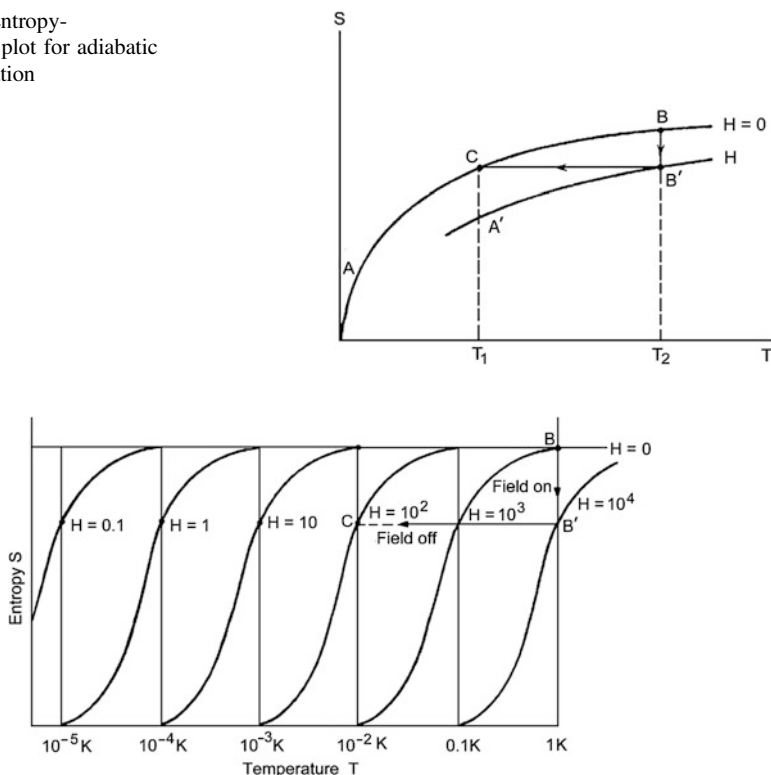


Fig. 8.18 Successive cycles of cooling by adiabatic demagnetization

stage the magnetic field is withdrawn i.e. the system is demagnetized. The system is isolated so that the spins remain oriented and the system moves to the state represented by the point C on curve AB. The ordered state of the spins represented by B is equivalent to their being at a lower temperature represented by the point C; the point C pertains to a lower temperature. The path B'C is an adiabatic path since $\Delta S = 0$. Thus the process may be described as isothermal magnetization of a paramagnetic salt (along path BB') followed by adiabatic demagnetization (along path B'C); in the process, the temperature has decreased from T_2 to T_1 . By repeating the process (Fig. 8.18), it has been possible to reach temperatures as low as $\sim 10^{-3}$ – 10^{-5} K. The salt used in the pioneering experiment was iron ammonium alum $[\text{FeNH}_4(\text{SO}_4)_3 \cdot 12\text{H}_2\text{O}]$.

We shall now show that the entropy indeed decreases along the path BB' due to the application of the field. From thermodynamics we have

$$TdS = dE + MdH. \quad (8.89)$$

Since E is a function of T and H , (8.89) can be written as

$$TdS = \left(\frac{\partial E}{\partial T}\right)dT + \left(\frac{\partial E}{\partial H}\right)dH + MdH. \quad (8.90)$$

Since $dT = 0$ along the path BB' , (8.90) may be written as

$$\left(\frac{\partial S}{\partial H}\right)_T = \frac{1}{T} \left[\left(\frac{\partial E}{\partial H}\right)_T + M \right]. \quad (8.91)$$

From one of Maxwell's equations, we have

$$\left(\frac{\partial S}{\partial H}\right)_T = \left(\frac{\partial M}{\partial T}\right)_H. \quad (8.92)$$

Thus, we may write

$$dS = \left(\frac{\partial M}{\partial T}\right)_H dH. \quad (8.93)$$

Substituting $M = N\mu_B^2 P_{eff}^2 H / 3k_B T$ and integrating, we get

$$S = S_{H=0} - (N\mu_B^2 P_{eff}^2 H^2) / 6k_B T^2. \quad (8.94)$$

This is the decrease in entropy due to demagnetization.

8.8 Miscellaneous Effects in Diamagnetism and Paramagnetism

For completeness we shall discuss, briefly and qualitatively, two contributions to magnetic properties; although magnitude-wise they are small effects, they involve important concepts.

8.8.1 Van Vleck Paramagnetism

A detailed quantum mechanical treatment of the magnetic susceptibility of ions with shells which are one electron short of being half filled leads to

$$\chi = -\frac{N}{V} \left[\frac{e^2}{4mc^2} \langle 0 | \sum_i (x_i^2 + y_i^2) | 0 \rangle - 2\mu_B^2 \sum_n \frac{|\langle 0 | (\mathbf{L}_z + g\mathbf{S}_z) | n \rangle|^2}{E_n - E_0} \right]. \quad (8.95)$$

We shall not deal with (8.95) beyond discussing the significance of the two terms. The first term is nothing but the Larmor expression for diamagnetic susceptibility. The second term arises out of the special electronic structure of the shell. This is a positive term; thus it is a paramagnetic effect. This paramagnetic correction to the Larmor diamagnetic susceptibility is called Van Vleck paramagnetism [19]. The overall magnetic behaviour of a material containing ions with a shell one ion short of being half filled is thus determined by the difference between the Larmor diamagnetism and the Van Vleck paramagnetism.

8.8.2 Landau Diamagnetism

We have seen that conduction electrons endow metals with a temperature-independent paramagnetic susceptibility—the Pauli susceptibility χ_{Pauli} . Landau showed that a diamagnetic effect arises from a coupling of the field with the orbital motion of the electron. There is a complicated oscillatory dependence of M and H at very low temperature and high field in very high purity samples. There is a net non-vanishing magnetization antiparallel to H which is due to the orbital electronic motion induced by the field over and above the Larmor mechanism. This is called Landau diamagnetism and the associated susceptibility is denoted by χ_{Landau} . It was shown by Peierls [20] that

$$(\chi_{\text{Landau}}) = -\frac{1}{3} \chi_{\text{Pauli}}. \quad (8.96)$$

Thus the observed experimental susceptibility is the sum of the Larmor, Pauli and Landau contributions; it is quite complicated to determine any one of them individually.

8.9 Problems

1. Evaluate the Larmor frequency in Hertz for an electron in a field of 1 weber/m².
2. Consider an electron in a circular orbit around a hydrogen atom with radius r and angular frequency ω . Express ω in terms of e , m and r . Now apply a magnetic field H normal to the orbital plane. Show that the angular frequency is

shifted by $\Delta\omega = eH/mc$. What should be the field to make $\Delta\omega$ one percent of the orbital frequency of the electron in a He atom?

3. The diamagnetic susceptibility of Cu is -0.2×10^{-6} . The density of copper is 8.93 g/cm^3 and the atomic weight is 63.5. Calculate the average radius of the Cu ion.
4. A paramagnetic salt contains 10^{28} ions/ m^3 with a magnetic moment $1 \mu_B$. Calculate the paramagnetic susceptibility and also the magnetization produced in a uniform magnetic field of 10^6 A/m at room temperature.
5. Show that the Brillouin function approaches the Langevin function for $J \rightarrow \infty$.
6. The number of unpaired electrons in Dy^{3+} is 5. Calculate the paramagnetic susceptibility for a salt containing 1 kg mole of Dy^{3+} ions at 300 K.
7. Consider an ion with a partially filled shell of angular momentum J and Z additional electrons in fixed shells, show that the ratio χ_{para} to χ_{dia} is

$$\chi_{\text{para}}/\chi_{\text{dia}} = \frac{2J(J+1)}{Zk_B T} \left(\frac{\hbar^2}{mr^2} \right).$$

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Chapter 9

Magnetism II: Ferromagnetism, Antiferromagnetism and Ferrimagnetism

Abstract In crystals containing atoms with nonzero spin, the spins are normally in a random state. They assume an ordered arrangement below a critical temperature. Depending upon the type of this order, the crystals develop ferromagnetism, antiferromagnetism or ferrimagnetism. The magnetic behavior of these three types is discussed in this chapter. Ferromagnetic crystals are characterized by a susceptibility that follows the Curie-Weiss law and existence of spontaneous magnetization. Weiss attributed ferromagnetism to the presence of an internal magnetic field and obtained the Curie-Weiss law. To account for spontaneous magnetization, Weiss assumed the presence of domains. A quantum mechanical interpretation was given by Heisenberg who introduced the concept of the exchange interaction between ordered spins. Ferromagnetism occurs when equal spins assume parallel alignment. When equal spins adopt an antiparallel alignment, it results in antiferromagnetism. These materials have a characteristic susceptibility which has a maximum at a certain temperature. Neel interpreted antiferromagnetism by extending the idea of exchange interaction. A third type of magnetism, ferrimagnetism, results by the antiparallel alignment of unequal spins. Neel's theory of ferrimagnetism is also discussed. Ferromagnets and ferrimagnets have the properties of hysteresis and domain structure. Their role in practical applications is discussed.

9.1 Introduction

In the preceding chapter, we considered two types of magnetism viz., diamagnetism and paramagnetism. The first is the effect induced by a magnetic field and the second is due to the presence of permanent atomic magnets which are free to rotate under the influence of a magnetic field.

The present chapter deals with magnetic phenomena arising out of interactions of electron spins. In their book, Ashcroft and Mermin [1] rightly titled the corresponding chapter as “Magnetic Ordering”. There are three types of ordering of spins: parallel equal spins, antiparallel equal spins and antiparallel unequal spins.

They give rise to ferromagnetism, antiferromagnetism and ferrimagnetism respectively. Each type of magnetism exists below a critical temperature. Above the critical temperature, each type becomes paramagnetic with the susceptibility following a modified Curie law. Ferromagnetism and ferrimagnetism display the property of spontaneous magnetization and the associated phenomenon of domain structure.

Many concepts which are introduced while dealing with ferromagnetism play an important role in magnetism as a whole. Hence, ferromagnetism is discussed in considerable detail. Our treatment is based essentially on the molecular field theory as it brings out the main features in a simple manner. Other advanced approaches are touched upon only briefly. The idea of ‘domains’ was introduced by Weiss [2] in his theory of ferromagnetism but they were later found to exist in ferrimagnetic materials also. Hence the topic of domains is taken up after the discussion on ferrimagnetism.

Treatments of these aspects of magnetism are included in books on solid state physics by Kubo and Nagamiya [3] and Ashcroft and Mermin [1] and specialized books on magnetism by Chikazumi and Charap [4], Morrish [5], Crangle [6] and Yosida [7].

9.2 Ferromagnetism

9.2.1 General

The characteristic properties of a ferromagnetic substance are: (i) a positive susceptibility, (ii) spontaneous magnetization, (iii) existence of a critical temperature at which the spontaneous magnetization vanishes, (iv) a modified Curie law for the susceptibility above the critical temperature and (v) existence of domains.

The property of ferromagnetism is displayed by the transition elements Fe, Ni, Co, the rare earth elements Gd and Dy and also by some alloys and compounds.

9.2.2 The Weiss Theory of Ferromagnetism

The first theory of ferromagnetism was proposed by Weiss [2]. He made the assumption that a ferromagnetic substance has spontaneous magnetization, i.e., a magnetization which exists even in the absence of an external magnetic field. But specimens do exist which do not have any magnetization. Weiss reconciled this contradiction by introducing the concept of ‘domains’. Domains are small regions (Fig. 9.1). Each region has a spontaneous magnetization in a certain direction. The direction of magnetization differs from domain to domain so that the net magnetization of the specimen is zero.

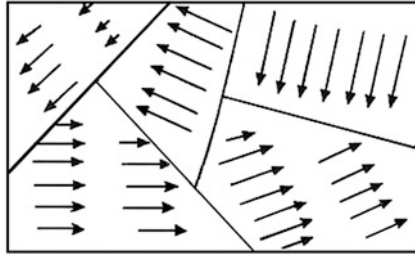


Fig. 9.1 Domains in a ferromagnetic crystal. Each domain has a magnetization but the net magnetization of the sample is zero

The presence of spontaneous magnetization means that even in the absence of an external field, the atomic magnets are aligned in one direction. There must be something which regiments the atomic magnets. For this, Weiss introduced the concept of an internal field—the molecular field. Weiss assumed that the molecular field H_m is proportional to the magnetization M . Thus

$$\mathbf{H}_m = N_W \mathbf{M}. \quad (9.1)$$

Following Morrish [5], we shall denote the constant of proportionality by N_W . It is called the molecular field constant or the Weiss constant.

The magnetization M is given by (see Chap. 8)

$$M = Ng\mu_B JB_J(x) \quad (9.2)$$

where N is the number of atoms in unit volume, g the Lande splitting factor, μ_B the Bohr magneton, J the total angular momentum quantum number (resultant of orbital angular momentum L and spin angular momentum S). $B_J(x)$ is the Brillouin function defined by

$$B_J(x) = \frac{2J+1}{2J} \coth \frac{2J+1}{2J} x - \frac{1}{2J} \coth \frac{x}{2J} \quad (9.3)$$

with

$$x = g\mu_B JH/k_B T. \quad (9.4)$$

In (9.4) H is the magnetic field and T the temperature. For a ferromagnetic substance H should be replaced by $(H + N_W M)$. Thus (9.4) becomes

$$x = (g\mu_B J/k_B T)(H + N_W M). \quad (9.5)$$

At $T \rightarrow 0$, $x \rightarrow \infty$ and $B_J(x) \rightarrow 1$. Hence (9.2) reduces to

$$M(0) = Ng\mu_B J \quad (9.6)$$

where $M(0)$ is the magnetization at $T = 0$; it may be noted that this is also the maximum possible magnetization. From (9.2), we get

$$\frac{M(T)}{M(0)} = B_J(x). \quad (9.7)$$

For spontaneous magnetization $H = 0$. Substituting in (9.5) and combining with (9.6), we get

$$\frac{M(T)}{M(0)} = \left(\frac{k_B T}{N N_W g^2 \mu_B^2 J^2} \right) x. \quad (9.8)$$

In (9.7) and (9.8), we have replaced M by $M(T)$ to drive home the temperature dependence of the spontaneous magnetization.

At a given temperature the spontaneous magnetization has to satisfy (9.7) and (9.8). To obtain the magnetization, a graphical procedure is adopted (Fig. 9.2). First, for a chosen value of J , a curve is drawn for (9.7) i.e., for $B_J(x)$ as a function of x . Next, a set of straight lines are drawn to represent (9.8). For a chosen value of T , we look for the point of intersection of the corresponding line with the curve for $B_J(x)$. This point of intersection (for example, the point P) on the curve represents the value of $M(T)/M(0)$ for the chosen value of T . If we consider the other straight lines with smaller slope (lines more inclined towards the right), we will get $M(T)/M(0)$ for higher x i.e. lower T . On the other hand, if we consider straight lines with a larger slope i.e., for higher T , we reach a temperature (T_f) for which the straight line is tangential to the $B_J(x)$ curve at the origin. This is the highest temperature for which $M(T)/M(0)$ has a real value. For larger slope, i.e., at $T > T_f$, the straight line does not intersect the curve and $M(T)/M(0)$ does not have any value. This critical temperature at which the spontaneous magnetization, and hence

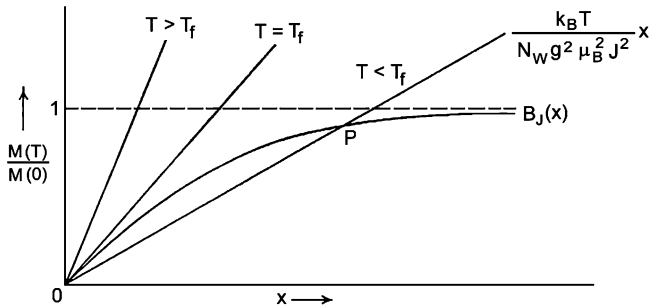


Fig. 9.2 Graphical method for determining the spontaneous magnetization at different temperatures

ferromagnetism, vanishes is denoted by T_f . It is called the ferromagnetic Curie temperature.

We shall now establish a connection between the critical temperature T_f and the Weiss constant N_W . For $x \ll 1$, i.e., close to the origin in Fig. 9.2, (9.3) approximates to

$$B_J(x) \approx \frac{J+1}{3J} x. \quad (9.9)$$

At the origin ($x = 0$), the tangent to the curve and the straight line for $T = T_f$ coincide. Thus

$$\text{Slope of the tangent at } (x = 0) = \frac{J+1}{3J} \quad (9.10)$$

and

$$\text{Slope of the straight line for } (T = T_f) = \frac{k_B T_f}{N N_W g^2 \mu_B^2 J^2}. \quad (9.11)$$

Equating the two, we get

$$T_f = \left[\frac{J(J+1)N g^2 \mu_B^2}{3k_B} \right] N_W. \quad (9.12)$$

Thus the critical temperature T_f is proportional to the Weiss constant N_W .

Let us revert to the variation of $M(T)/M(0)$ with temperature. Combining (9.8) and (9.11), we get

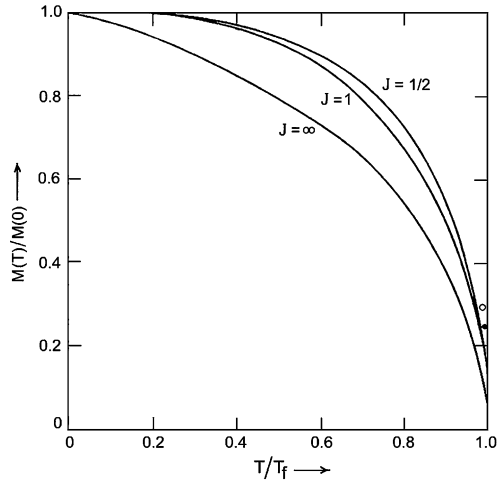
$$\frac{M(T)}{M(0)} = \left(\frac{J+1}{3J} \right) \left(\frac{T}{T_f} \right) x. \quad (9.13)$$

For a chosen value of J , a plot of $M(T)/M(0)$ versus (T/T_f) is a universal curve. Such curves for different values of J are shown in Fig. 9.3. From these curves, the value of $M(T)/M(0)$ at different temperatures can be read off.

We shall now consider the temperature variation of the susceptibility χ above the critical temperature. As spontaneous magnetization ceases at $T = T_f$, magnetization is possible at $T > T_f$ only if an external field is applied. If the applied field is low, it may produce magnetization which is much less than the saturation magnetization. We can then use the approximate form of $B_J(x)$ which leads to

$$M = \frac{Ng\mu_B(J+1)}{3} x \quad (9.14)$$

Fig. 9.3 Theoretical plot of spontaneous magnetization $M(T)/M(0)$ as a function of T/T_f for different assumed values of J



where x is given by

$$x = \frac{Jg\mu_B}{k_B T} (h + N_W M). \quad (9.15)$$

Combining (9.12) and (9.13), we can express M in terms of H . Then the susceptibility χ is

$$\chi = \frac{M}{H} = \frac{C}{T - \theta} \quad (9.16)$$

where

$$C = \frac{Ng^2\mu_B^2 J(J+1)}{3k_B} \quad (9.17)$$

and

$$\theta = \frac{Ng^2\mu_B^2 J(J+1)}{3k_B} N_W = CN_W. \quad (9.18)$$

Equation (9.16) resembles the Curie law in paramagnetism with the difference that T is replaced by the reduced temperature $(T - \theta)$; (9.16) is called the Curie-Weiss law. θ is called the paramagnetic Curie temperature. From the definition of T_f (9.12) and θ (9.18), we see that $\theta = T_f$. We continue to refer to C as the Curie constant.

9.2.3 Experimental Results

We shall first compare experimental results on magnetization with theory. Methods of determination of magnetization have been discussed in Chap. 8. There is a difficulty in the measurement of spontaneous magnetization. Because of the presence of domains (Sect. 9.5), it is not possible to measure the absolute or true magnetization. A part of the applied field is used up in removing domain structure and in orienting a single domain along the direction of the field. This difficulty is, to some extent, overcome by making measurements at different magnetic fields. For magnetic fields from about 10^3 to 3×10^4 Oe, the measured magnetization M_H is found to empirically fit the equation

$$M_H = M_S(T) \left[1 - \frac{a}{H} - \frac{b}{H^2} \right] + CH. \quad (9.19)$$

The experimental results of $M(T)/M(0)$ as a function of T/T_f for Ni are shown in Fig. 9.4. The data points are evenly distributed around the theoretical curve for $J = S = \frac{1}{2}$ shown in Fig. 9.3. The experimental results on iron and cobalt (not shown in the figure) also lie on the same curve.

It was pointed out that the saturation magnetization $M(0)$ equals $Ng\mu_B J$. Since the ferromagnetic metals behave as if the orbital angular momentum is quenched, $J = S = \frac{1}{2}$. For the electron $g = 2$. Hence $M(0) = N\mu_B$. Then the number of electrons associated with an atom should be $M(0)/N\mu_B$. This is called the effective number of electrons p_{eff} . The values of p_{eff} are given in Table 9.1. Although the number of electrons associated with an atom is an integer, it is found that p_{eff} is non-integral.

We shall now consider the temperature variation of susceptibility in the region $T > T_f$. Weiss's theory predicts that χ will vary according to the Curie-Weiss law (9.16). Plots of χ^{-1} versus T are thus expected to be linear. Such plots for some ferromagnetic metals are shown in Fig. 9.5. Values of C and θ obtained from these plots are given in Table 9.1. A closer look at the $\chi^{-1} - T$ plot shows that it has a

Fig. 9.4 Experimental data on spontaneous magnetization of nickel; values of $M(T)/M(0)$ are plotted against (T/T_f) ; continuous curve is the theoretical curve from Fig. 9.3 for $S = \frac{1}{2}$

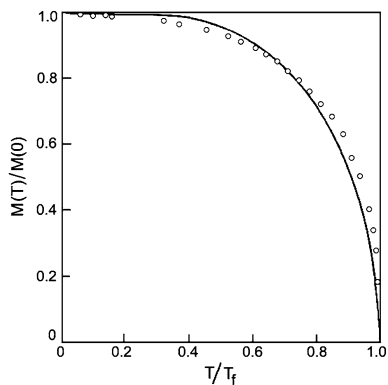


Table 9.1 Ferromagnetic properties of some metals

Metal	Saturation magnetization (cgs) (0 K)	P_{eff}	T_f (K)	θ (K)	C (cgs units/ mole)
Fe	1,752	2.22	1,043	1,093	1.26
Co	1,446	1.716	1,400	1,428	1.22
Ni	510	0.61	631	650	0.32
Gd	1,980	7.10	289	302	7.8

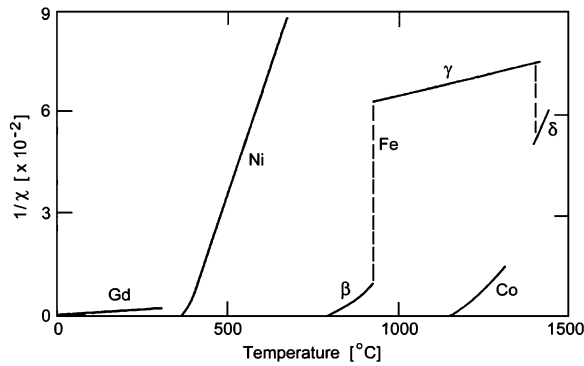


Fig. 9.5 Linear plots of $1/\chi$ versus temperature for some ferromagnetic metals illustrating the Curie-Weiss law

curvature in the region close to T_f . If we extrapolate the linear part of the plot (Fig. 9.6), it has θ as the intercept on the T -axis. Thus, we find $\theta \neq T_f$; instead $\theta > T_f$. To distinguish them, T_f is called the ferromagnetic Curie point and θ the paramagnetic Curie point.

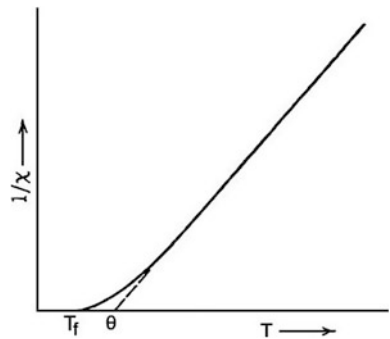


Fig. 9.6 Plot of $(1/\chi)$ versus temperature showing curvature near $T = T_f$; extrapolation of linear part yields value of paramagnetic Curie temperature θ

9.2.4 The Heisenberg Model

9.2.4.1 Magnitude of the Internal Field

We have seen that Weiss assumed the existence of internal molecular field H_m in a phenomenological spirit. This field was supposed to be responsible for the parallel alignment of the atomic magnetic dipoles in a ferromagnetic material. This alignment is destroyed at the Curie temperature T_f by the thermal agitation. Obviously at $T = T_f$, the thermal energy equals the energy due to spontaneous magnetization. Thus,

$$\mu_B H_m \approx k_B T_f. \quad (9.20)$$

Typically, $T_f \approx 10^3$ K. Substituting in (9.20), we get

$$H_m = k_B T_f / \mu_B = 10^7 \text{ Oe}. \quad (9.21)$$

This is a large value. If ' a ' is the interatomic distance, the field produced by the dipole-dipole interaction is $\approx \mu_B / a^3 = 10^3$ Oe; the internal field is larger than this dipole-dipole field. In fact, the internal field is larger than the magnetic fields produced in laboratories. Therefore, it is necessary to know the origin of such a large field which is responsible for spontaneous magnetization.

9.2.4.2 Heisenberg's Exchange Interaction

An understanding of the origin of the internal molecular field was provided by Heisenberg [8]. Using quantum mechanics, he proposed an exchange interaction between two electrons. His theory was based on the Heitler-London treatment of the hydrogen molecule.

To start with, we shall consider two non-interacting electrons attached to two neighbouring atoms (Fig. 9.7). We shall denote the two nuclei by A and B, the two electrons by 1 and 2 and the distance between them by r_{AB} , r_{12} , r_{A2} and r_{B1} . The Schrödinger equation for the two atoms are

$$\left[-\frac{\hbar^2}{2m} \nabla_1^2 + V(q_1) \right] \psi = E_A \psi \quad (9.22)$$

and

$$\left[-\frac{\hbar^2}{2m} \nabla_2^2 + V(q_2) \right] \psi = E_B \psi. \quad (9.23)$$

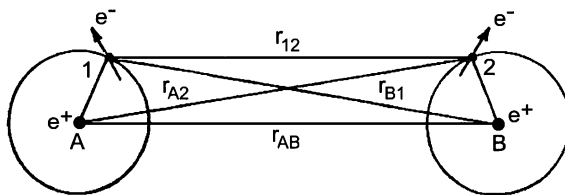


Fig. 9.7 The hydrogen molecule

The solutions of (9.22) are $\psi_A(1)$ when electron 1 is attached to atom A and $\psi_A(2)$ when the electron 2 is attached to atom A. Similarly, $\psi_B(1)$ is the solution of (9.23) when electron 1 is attached to atom B and $\psi_B(2)$ the solution of (9.23) when electron 2 is attached to atom B. If the two atoms are treated as a single system with no interaction between them, the Schrödinger equation may be written as

$$-\frac{\hbar^2}{2m} [\nabla_1^2 + \nabla_2^2 + V(q_1) + V(q_2)] \psi = E\psi. \quad (9.24)$$

The solutions of (9.24) are $\psi_A(1)\psi_B(2)$ and $\psi_A(2)\psi_B(1)$ with $E = E_A + E_B$ for both electrons.

Now let us assume that there is an interaction between the two electrons. This interaction, called the exchange interaction, may be represented by an extra potential V_{AB} in the Schrödinger equation. The term V_{AB} is given by

$$V_{AB} = e^2 \left[\frac{1}{r_{AB}} + \frac{1}{r_{12}} - \frac{1}{r_{B1}} - \frac{1}{r_{A2}} \right]. \quad (9.25)$$

V_{AB} acts as a perturbation on the 2-electron system. Using the perturbation theory, we get the energy E of the system as

$$E = K \pm J_e \quad (9.26)$$

where K is a constant. K and J_e are given by

$$K = \int \psi_A^*(1)\psi_B^*(2) V_{AB} \psi_A(1)\psi_B(2) d\tau_1 d\tau_2 \quad (9.27)$$

and

$$J_e = \int \psi_A^*(1)\psi_B^*(2) V_{AB} \psi_A(2)\psi_B(1) d\tau_1 d\tau_2 \quad (9.28)$$

where $d\tau_1$ and $d\tau_2$ are volume elements. K is the Coulomb interaction energy and J_e the exchange integral. It may be noted that the interaction between the electrons is through their spins. The sign of J_e is plus when the spins are parallel and it is minus

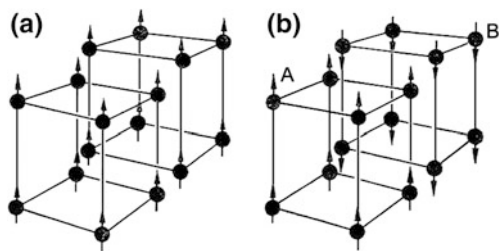


Fig. 9.8 Spin alignments in the Heisenberg model: **a** parallel spins, **b** antiparallel spins

Table 9.2 Values of the reduced interatomic distance r'_{AB} for some metals

Metal	Cr	Mn	Fe	Co	Ni
r'_{AB}	2.60	2.94	3.26	3.64	3.94

when they are antiparallel. When J_e is positive, $(K - J_e) < (K + J_e)$ and the system is stable; this is the ferromagnetic state. On the other hand, when J_e is negative, the system is in the antiferromagnetic state. Systems with parallel spins and antiparallel spins envisaged in the Heisenberg model are shown in Fig. 9.8.

Since only a few materials are ferromagnetic, it means that J_e is positive only for those materials. Slater [9] and Bethe [10] studied this problem and found that the value and sign of J_e depends critically on the value of the reduced interatomic distance r'_{AB} ; r'_{AB} is defined as r_{AB}/r_0 where r_{AB} is the interatomic distance and r_0 is the orbital radius. Values of r'_{AB} for some transition metal elements are given in Table 9.2; these values range from 2.6 to 4. Values of J_e for the transition elements are shown in Fig. 9.9 as a function of r'_{AB} . The data points lie on a smooth curve which rises from the negative J_e region to the positive J_e region. The curve crosses the r'_{AB} axis at $r'_{AB} \approx 3$ and, after reaching a maximum, tails off at ~ 4.5 . It can be seen that Fe, Ni and Co which have $r'_{AB} \geq 3$ have positive J_e and are ferromagnetic. On the other hand, Mn and Cr with $r'_{AB} < 3$ have negative J_e and are not ferromagnetic. The question then arises as to whether a solid containing Mn ions would be ferromagnetic if its r'_{AB} value is >3 . Indeed MnAs and MnSb with $r'_{AB} > 3$ are found to be ferromagnetic. If the spins of the two atoms are denoted by vectors $\mathbf{S}_1$ and $\mathbf{S}_2$, (9.26) can be expressed as

$$E = K - 2J_e \mathbf{S}_1 \cdot \mathbf{S}_2 \tag{9.29}$$

The Coulomb energy term K is irrelevant to the problem of ferromagnetism, hence it can be ignored. We can now write (9.29) in the form of a Hamiltonian:

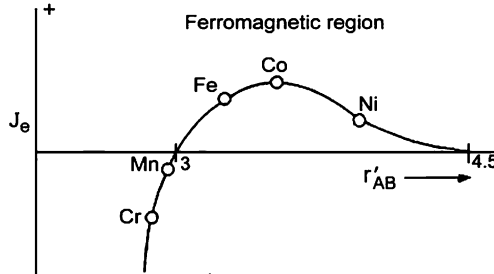


Fig. 9.9 The Slater-Heisenberg plot between r'_{AB} and J_e

$$\mathcal{H} = -2J_e \mathbf{S}_1 \cdot \mathbf{S}_2. \quad (9.30)$$

We shall now extend the treatment to the whole crystal. It is convenient to denote the spin values as $\mathbf{S}_i$ and $\mathbf{S}_j$ instead of $\mathbf{S}_1$ and $\mathbf{S}_2$. Taking $\mathbf{S}_i$ as the spin of a reference atom, the Hamiltonian for the crystal is

$$\mathcal{H} = -2J_e \mathbf{S}_i \cdot \mathbf{S}_j. \quad (9.31)$$

This is called the two-electron Hamiltonian. By solving (9.31), it is possible to obtain results on ferromagnetism but the procedure is long and complicated.

9.2.4.3 The Ising Model

A simpler procedure was suggested by Ising [11]. He expanded the scalar product in (9.31) as

$$\mathcal{H} = -2J_e \sum_j [S_{xi}S_{xj} + S_{yi}S_{yj} + S_{zi}S_{zj}]. \quad (9.32)$$

Next, it is assumed that the first two terms in (9.32) can be neglected. This reduces (9.32) to

$$\mathcal{H} = -2J_e \sum_j S_{zi}S_{zj}. \quad (9.33)$$

The reduced Hamiltonian (9.33) is called the Ising Hamiltonian. We shall further assume (i) that J_e is the same for all the nearest neighbours of atom i , (ii) that J_e is negligible for neighbours other than the nearest, (iii) that the spin S_{zj} can be replaced by its time average $\langle S_{zj} \rangle$ which is given by $M/g\mu_B N$ and (iv) that the number of nearest neighbours is Z . With these assumptions and substitutions (9.33) becomes

$$\mathcal{H} = -2Z \frac{S_{zi} J_e M}{Ng\mu_B}. \quad (9.34)$$

On the other hand, the Hamiltonian in the Weiss field is

$$\mathcal{H} = -gS_{zi}\mu_B N_W M. \quad (9.35)$$

Equating (9.34) and (9.35), we have

$$N_W = 2ZJ_e / Ng^2\mu_B^2. \quad (9.36)$$

Combining (9.36) and (9.12), we can write

$$T_f = 2ZJ_e S(S+1)/3k_B. \quad (9.37)$$

Applying (9.36) and (9.37) to a simple cubic lattice with $Z = 6$ and $S = 1/2$, we get

$$(J_e/k_B T_f) = 1/3. \quad (9.38)$$

Thus, the Heisenberg model replaces Weiss' internal field which was phenomenologically introduced by a quantum mechanical exchange interaction between the spins of electrons. Stoner connected the Weiss parameter N_W and the critical temperature T_f with the exchange integral J_e .

9.2.5 Other Methods

The characteristic ferromagnetic properties were satisfactorily explained by the phenomenological theory of Weiss and later by the quantum mechanical exchange theory of Heisenberg. We shall briefly discuss a few other methods.

9.2.5.1 The Series Expansion Method

If a system containing electron pairs is in a magnetic field H , the Heisenberg Hamiltonian gets modified to

$$\mathcal{H} = -2J_e \sum_j \mathbf{S}_i \cdot \mathbf{S}_j - g\mu_B \mathbf{H} \cdot \sum_k \mathbf{S}_k. \quad (9.39)$$

The first summation is over all neighbouring pairs and the second summation includes all atoms in the crystal. It can be seen that the first term is the familiar Heisenberg exchange energy and the second term is the interaction energy of the spins and the magnetic field. The procedure in this method is to start with the

quantum mechanical partition function $\mathfrak{Z} = \text{tr} \exp[-(\mathcal{H}/k_B T)]$, where tr stands for trace of the matrix. It may be noted that in quantum mechanics the spin-operators are Pauli matrices. Once the partition function is known, magnetization and susceptibility can be obtained from the relation

$$M = Nk_B T \frac{\partial \mathfrak{Z}}{\partial H}. \quad (9.40)$$

These calculations, though lengthy, essentially relate to terms like $\overline{(\sum \mathbf{S}_i - \mathbf{S}_j)^n}$. Following this procedure, Brown and Luttinger [12] obtained the susceptibility as

$$\chi = \left[\frac{Ng^2 \mu_B^2 S(S+1)}{3k_B T} \right] \left[\sum_{n=0}^{\infty} \frac{a^n}{t^n} \right] \quad (9.41)$$

where t is $(k_B T/J_e)$. Brown and Luttinger have calculated the a -coefficients up to the sixth term for various values of S and for different structures. The interesting aspect of (9.41) is that the first term in the series leads to the Curie law, the first two terms together lead to the Curie-Weiss law and the first three terms lead to a $\chi^{-1} - T$ curve with the observed curvature near $T = T_f$.

9.2.5.2 Bethe-Peierls-Weiss Method

Bethe [13] and Peierls [14] developed an order-disorder theory for alloys. The ferromagnetic-paramagnetic transition is also an order-disorder transition. The theory of Bethe and Peierls was extended to ferromagnetism by Weiss [15]; incidentally, this Weiss is not the originator of the internal field theory.

Let us consider a cluster of atoms including a central atom i with Z number of nearest neighbours. The central atom interacts with its nearest neighbour through a Heisenberg exchange interaction. The atoms outside the cluster interact with the Z neighbours of the central atom as if there is an effective magnetic field H_e . This field operates only below the Curie point T_f . When an external field H is applied, the Hamiltonian of the cluster is given by

$$\mathcal{H} = -2J_e \mathbf{S}_i \cdot \sum_{j=1}^Z \mathbf{S}_j - g\mu_B \mathbf{H}_e \cdot \sum_{j=1}^Z \mathbf{S}_j - g\mu_B \mathbf{H} \cdot \mathbf{S}_i. \quad (9.42)$$

Here, the external field is assumed to interact only with the central atom. Making the Stoner-Ising approximation, (9.42) is simplified to

$$\mathcal{H} = -2J_e S_{zi} \sum S_{zj} - g\mu_B H_e \sum S_{zj} - g\mu_B H S_{zi}. \quad (9.43)$$

After considerable mathematics Weiss [15] obtained the average atomic magnetic moment μ as

$$\mu = -\mu_B \frac{Z}{Z-1} \left(\frac{g\mu_B H_e}{2k_B T} \right). \quad (9.44)$$

The $\chi^{-1} - T$ curve obtained from (9.44) predicts a curvature at $T \approx T_f$ in agreement with experimental observation. The difference between T_f and θ is about 5 %.

9.2.5.3 Stoner's Theory

The models so far discussed assume that the electrons are localised at the atoms. The properties of metals and alloys are described by the band theory of solids in which mobile electrons and holes are the main actors. Since ferromagnetic materials are mostly metals and alloys, Stoner [16] applied the band theory to investigate ferromagnetism. Stoner's theory is also known as the collective electron theory or the band theory of ferromagnetism.

Stoner built his theory on the following assumptions.

- (i) He included only the role of the $3d$ band electrons and ignored the $4s$ band.
- (ii) He assumed the $3d$ band to be parabolic near the Fermi level i.e., the energy is given by

$$E = \frac{\hbar^2 k^2}{2m^*} \quad (9.45)$$

and the density of states by

$$D(E)dE = KE^{1/2}dE. \quad (9.46)$$

The symbols are explained in Chaps. 3 and 4.

- (iii) Electrons and holes obey Fermi-Dirac statistics.
- (iv) The exchange interaction between electrons with parallel spins and anti-parallel spins is given by the Weiss expression

$$E_1 = \pm N_W M \mu_B. \quad (9.47)$$

At this stage it is convenient to introduce ξ and θ' defined as

$$\xi = M/N\mu_B \text{ and } k_B\theta' = N_W N \mu_B^2. \quad (9.48)$$

Then the energy E_1 is $\pm Nk_B\theta'\xi$ where N is the number of electrons or holes/unit volume.

The magnetization M is the product of μ_B and the difference in the number of parallel magnetic moments and antiparallel magnetic moments. Thus, we have

$$M = \int \{F[E(k) - N_W N \mu_B^2] - F[E(k) + N_W N \mu_B^2]\} D(E) dE \quad (9.49)$$

where

$$E(k) = \frac{\hbar^2 k^2}{2m^*} \pm N_W M \mu_B \quad (9.50)$$

and $F(E)$ is the Fermi-Dirac distribution function (Chap. 3). If a magnetic field H is present, (9.50) becomes

$$E(k) = \frac{\hbar^2 k^2}{2m^*} \pm N_W M \mu_B \pm \mu_B H. \quad (9.51)$$

Substituting (9.50) in (9.49) and putting $T = 0$, the magnetization M takes the form

$$\begin{aligned} M(0) &= \left[\frac{\mu_B}{(2\pi)^2} \right] \times \left[\int_0^{E_F + N_W N \mu_B^2} \left(\frac{2m^*}{\hbar^2} \right)^{3/2} E^{1/2} dE - \int_0^{E_F - N_W N \mu_B^2} \left(\frac{2m^*}{\hbar^2} \right)^{3/2} E^{1/2} dE \right] \\ &= \left(\frac{\mu_B}{6\pi^2} \right) \left(\frac{2m^*}{\hbar^2} \right)^{3/2} \left[(E_F + N_W N \mu_B^2)^{3/2} - (E_F - N_W N \mu_B^2)^{3/2} \right]. \end{aligned} \quad (9.52)$$

From (9.52) we have the following results:

- The magnetization $M(0)$ depends on the relative magnitude of the exchange energy and the Fermi energy i.e., on the parameter $k_B \theta' / E_F$. For $k_B \theta' / E_F > 2^{-1/3}$, $\xi(0) = M(0) / N \mu_B = 1$ and for $k_B \theta' / E_F < 2/3$, $\xi(0) = M(0) / N \mu_B = 0$. For $2/3 < k_B \theta' / E_F < 2^{-1/3}$, $\xi(0)$ lies between 0 and 1. Since the effective number of electrons can be calculated from $M(0)$, we can see why this number is non-integral.
- The calculation of $M(T)$ (or $\xi(0)$) is even more complicated. But it has been done by Stoner. The values of $M(T) / M(0) = \xi(T) / \xi(0)$ calculated for different values of $(k_B \theta' / E_F)$ are shown in Fig. 9.10.
- The susceptibilities (rather $1/\chi$) calculated from theory are also shown in the same figure.

Thus Stoner's band model of ferromagnetism leads to reasonable values of magnetization and susceptibility and, more importantly, to non-integral values of the effective number of electrons.

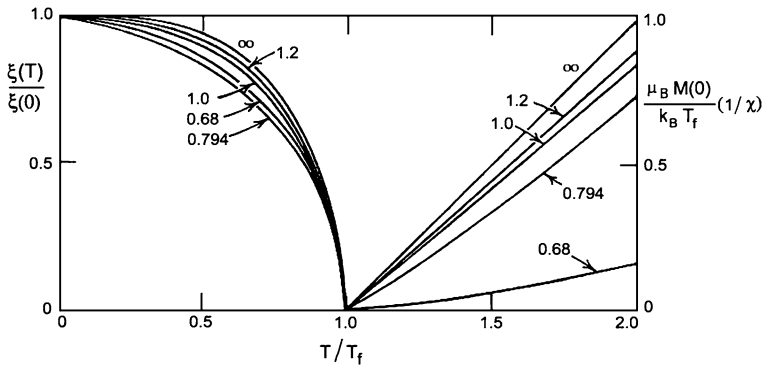


Fig. 9.10 Plot of magnetization (lhs) and reciprocal susceptibility (rhs) against T/T_f

9.2.5.4 Spin Wave Model

We shall discuss the concept of a spin wave qualitatively. Consider a ferromagnetic sample of unit volume at $T = 0$. Let us apply a small magnetic field in the negative z direction. Being at $T = 0$, the system is in the ground state. With $S = \frac{1}{2}$ and $g = 2$, the magnetic moment is $M = N|\mu_B|$ per unit volume. We shall now increase the temperature slightly. With this thermal energy, the spin of a particular atom will be reversed. The Heisenberg exchange energy will try to invert the reversed spin. However, the reversal takes place gradually over a large number of atoms as shown in Fig. 9.11. In other words, the reversal of the spin takes place by the propagation of a spin wave. This concept was introduced by Bloch [17] and the spin wave is called a Bloch wave. The spin wave can have only certain wavelengths.

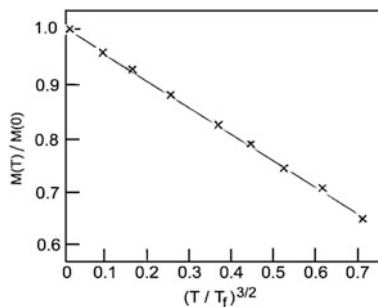
If the temperature is further increased, two spins may be reversed; then there may be two spin waves travelling in the crystal. If they meet at some point, they may get scattered. Or, the spin reversal may take place on two adjacent atoms; the reversed spins are now bound together. If more than two spins undergo reversal, there will be more spin waves and more collisions. Bloch ignored these possibilities and considered each reversal to give rise to a single spin wave. The energy of the system due to this mechanism is the sum of the energies of all the independent waves. Such superposition is possible if the number of waves i.e., the number of spin reversals is small. This can happen only at temperatures below T_f . Spin waves have properties possessed by photons and phonons and they are called magnons.

The quantum mechanical treatment of spin waves is too complicated; it has been discussed by Morrish [5] and Ashcroft and Mermin [1]. We shall merely mention



Fig. 9.11 A spin wave propagating along a chain of atoms

Fig. 9.12 $M(T)/M(0)$ plotted against $(T/T_f)^{3/2}$



its most important result viz., the temperature dependence of magnetization. This is given by

$$M(T) = M(0) \left\{ 1 - A \left(\frac{k_B T}{J_e} \right)^{3/2} \right\} \text{ for } T \leq T_f. \quad (9.53)$$

Here A is a constant ≈ 0.1174 for the simple cubic lattice. Thus magnetization varies as $T^{3/2}$. Hence, (9.53) is known as Bloch's $T^{3/2}$ law. Experimental data on $M(T)$ for Gd is shown in Fig. 9.12. It is consistent with (9.53).

9.3 Antiferromagnetism

9.3.1 General

Manganese oxide (MnO) is a cubic crystal with NaCl structure. Its susceptibility was measured by Bizette et al. [18]. They observed that MnO has a unique temperature variation very different from paramagnetics and ferromagnetics. As shown in Fig. 9.13a, the average susceptibility χ_{PC} (measured on a polycrystalline sample) initially increases with increasing temperature up to a certain temperature T_N . At T_N the trend changes abruptly and the average χ starts decreasing with increasing temperature. The single crystal susceptibilities $\chi_{\parallel}$ and $\chi_{\perp}$ are also shown. These are parallel or perpendicular to what is called the 'easy' or 'preferred' direction of magnetization. The temperature variation of $\chi_{\parallel}$ is qualitatively similar to that of the polycrystalline χ_{PC} . On the other hand $\chi_{\perp}$ remains almost independent of temperature till T_N . Thereafter, it varies like average χ . The reciprocal susceptibilities are shown in Fig. 9.13b. This type of temperature variation of susceptibility is now taken as a characteristic of a new magnetic property called antiferromagnetism.

The theory of antiferromagnetism is based on the assumption that the spins have an antiparallel arrangement. One set of spins may have all spins up and the other set may have spins equal to those in the first set but all down. A variety of spin arrangements are possible like triangular, spiral and canted. Since the spins in

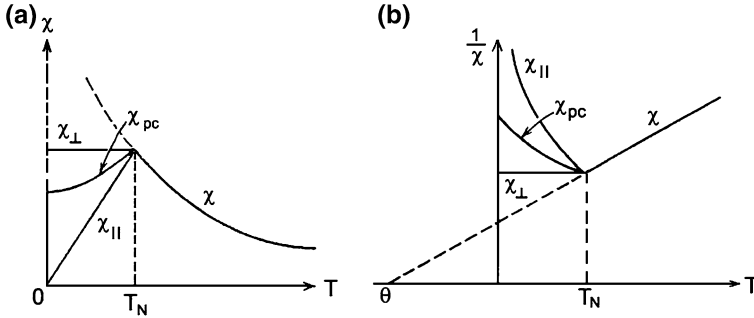


Fig. 9.13 Typical temperature variation of the susceptibility of an antiferromagnetic substance: **a** susceptibility, **b** reciprocal susceptibility

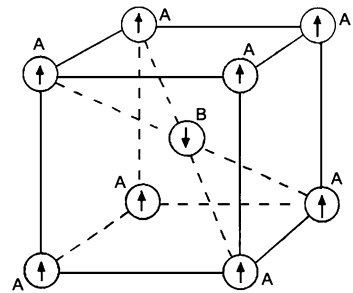
opposite directions are equal in magnitude, there is no spontaneous magnetization and no domain structure. We may mention that neutron diffraction experiments support the idea of such arrangement of spins.

We shall consider the molecular theory of antiferromagnetism and compare theoretical predictions with experimental results.

9.3.2 Molecular Field Theory of Antiferromagnetism

The molecular theory of antiferromagnetism was developed by Neel [19], Bitter [20] and Van Vleck [21]. For convenience, we shall consider the bcc lattice with atoms at each corner and the body-centre (Fig. 9.14). The spins of the atoms at the corner are antiparallel to those of the atoms at the body-centres. Though these atoms are the same, we shall denote the corner atoms as *A*-atoms and those at the body-centre as *B*-atoms to differentiate their spin alignments. In fact, the two sets of atoms constitute *A* and *B* sublattices and the treatment is called the two-sublattice model.

Fig. 9.14 Antiparallel spin alignment in the *A* and *B* lattices of a bcc antiferromagnetic crystal



The molecular fields at the sublattices A and B are

$$\begin{aligned}\mathbf{H}_{mA} &= -N_{AA}\mathbf{M}_A - N_{AB}\mathbf{M}_B, \\ \mathbf{H}_{mB} &= -N_{AB}\mathbf{M}_A - N_{BB}\mathbf{M}_B.\end{aligned}\tag{9.54}$$

Here M_A and M_B are the magnetizations of the two lattices and the N 's are various Weiss coefficients. It can be seen that

$$N_{AA} = N_{BB} = N_{ii} \text{ and } N_{AB} = N_{BA}.\tag{9.55}$$

N_{ii} and N_{AB} are positive. If there is an external field H acting on the system, the total field acting on each system is

$$\begin{aligned}\mathbf{H}_A &= \mathbf{H} - N_{ii}\mathbf{M}_A - N_{AB}\mathbf{M}_B, \\ \mathbf{H}_B &= \mathbf{H} - N_{AB}\mathbf{M}_A - N_{ii}\mathbf{M}_B.\end{aligned}\tag{9.56}$$

When the system is in equilibrium, the magnetization of the A sublattice is given by

$$M_A = \frac{1}{2}N g \mu_B S B_S(x_A).\tag{9.57}$$

with

$$x_A = \frac{Sg\mu_B}{k_B T} H_A.\tag{9.58}$$

$B_S(x_A)$ is the Brillouin function defined as

$$B_S(x_A) = \frac{2S+1}{2S} \coth \frac{2S+1}{2S} x_A - \frac{1}{2S} \coth \frac{x_A}{2S}.\tag{9.59}$$

Similarly the magnetization of the B -lattice is given by

$$M_B = \frac{1}{2}N g \mu_B S B_S(x_B)\tag{9.60}$$

with

$$x_B = \frac{Sg\mu_B}{k_B T} H_B.\tag{9.61}$$

$B_S(x_B)$ is the Brillouin function defined as

$$B_S(x_B) = \frac{2S+1}{2S} \coth \frac{2S+1}{2S} x_B - \frac{1}{2S} \coth \frac{x_B}{2S}. \quad (9.62)$$

In (9.57) and (9.60), N is the number of atoms/unit volume having a permanent dipole moment due to spin S . It may be noted that (9.57) and (9.60) are the same as (9.2) with J replaced by S .

We shall now discuss how the theory leads to an expression for the magnetization and the susceptibility as a function of temperature. For convenience, we shall consider three different temperature regions: $T > T_N$, $T = T_N$ and $T < T_N$.

At $T > T_N$

For high temperatures i.e., for small values of x_A and x_B , (9.57) and (9.60) can be approximated to

$$M_A = \frac{N g^2 \mu_B^2 S(S+1)}{6k_B T} H_A \quad (9.63)$$

and

$$M_B = \frac{N g^2 \mu_B^2 S(S+1)}{6k_B T} H_B. \quad (9.64)$$

At $T > T_N$, the system is in the paramagnetic region. The vectors $\mathbf{H}$, $\mathbf{M}_A$ and $\mathbf{M}_B$ are parallel and we may write

$$H_A = H - N_{ii}M_A - N_{AB}M_B, \quad (9.65)$$

and

$$H_B = H - N_{AB}M_A - N_{ii}M_B. \quad (9.66)$$

Substituting (9.65), (9.66) in (9.63) and (9.64), we get

$$M_A = \frac{N g^2 \mu_B^2 S(S+1)}{6k_B T} (H - N_{ii}M_A - N_{AB}M_B) \quad (9.67)$$

and

$$M_B = \frac{N g^2 \mu_B^2 S(S+1)}{6k_B T} (H - N_{AB}M_A - N_{ii}M_B). \quad (9.68)$$

Adding (9.67) and (9.68) and on reorganising, we get the total magnetization

$$M = M_A + M_B = 2 \left\{ \frac{N g^2 \mu_B^2 S(S+1)}{6k_B T} \right\} H / \left[1 + \left\{ \frac{N g^2 \mu_B^2 S(S+1)}{6k_B T} \right\} (N_{ii} + N_{AB}) \right]. \quad (9.69)$$

Finally, from (9.69), we get the susceptibility χ as

$$\chi = \frac{M}{H} = \frac{C}{T + \theta} \quad (9.70)$$

where

$$C = \left\{ \frac{N g^2 \mu_B^2 S(S+1)}{3k_B} \right\} \text{ and } \theta = \frac{1}{2} C (N_{ii} + N_{AB}). \quad (9.71)$$

Since N_{ii} and N_{AB} are positive, θ is positive. Equation (9.70) is a modified Curie-Weiss law for antiferromagnetics. Variation of the susceptibility with temperature for the region $T > T_N$ is shown in Fig. 9.15.

At $T = T_N$

In the vicinity of T_N , (9.67) and (9.68) are valid. With $H = 0$, these equations become

$$M_A = \frac{C}{2T} (-N_{ii} M_A - N_{AB} M_B) \quad (9.72)$$

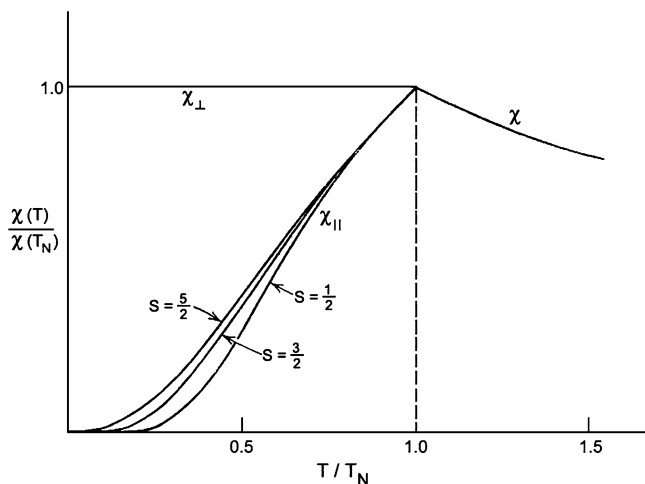


Fig. 9.15 Temperature variation of the susceptibility of an antiferromagnetic substance predicted from theory

and

$$M_B = \frac{C}{2T}(-N_{AB}M_A - N_{ii}M_B). \quad (9.73)$$

Solutions for M_A and M_B are possible only if the determinant of coefficients of M_A and M_B vanishes. This will lead us to

$$T_N = \frac{C}{2T}(N_{AB} - N_{ii}). \quad (9.74)$$

Substituting for C from (9.71), we get

$$\frac{\theta}{T_N} = \frac{N_{AB} + N_{ii}}{N_{AB} - N_{ii}}. \quad (9.75)$$

Equation (9.75) is a relation between θ and T_N in terms of the interaction coefficients. If $N_{ii} = 0$, $(\theta/T_N) = 1$. On the other hand, if $N_{ii} \neq 0$, but is less than N_{AB} , $\theta > T_N$. If $N_{ii} \rightarrow N_{AB}$, it may appear that θ/T_N will take a very large value. But this is not so. If θ/T_N increases beyond a certain value, the sublattice structure destabilises and the magnetic structure will itself change. A more detailed analysis shows that θ/T_N has values in the range 1–3 for bcc and rutile lattices and between 1 and 5 for the fcc lattice.

At $T < T_N$

The anisotropy in magnetic behaviour of a single crystal will be discussed later. In anticipation, we shall assume that magnetization in a crystal is preferred (easy) in some direction. Hence magnetization in a direction (say the easy direction) will be different when an external field is applied parallel or perpendicular to that direction.

First we shall consider the case when the field H is applied parallel to the easy direction (Fig. 9.16a). Then H is parallel to the magnetization of the A sublattice and antiparallel to the magnetization of the B sublattice. Then (9.58) and (9.56) give us

$$x_A = \frac{g\mu_B S}{k_B T}(H - N_{ii}M_A + N_{AB}M_B). \quad (9.76)$$

Similarly, from (9.61) and (9.56), we get

$$x_B = \frac{g\mu_B S}{k_B T}(-H + N_{AB}M_A - N_{ii}M_B). \quad (9.77)$$

But when $H = 0$, $M_A = -M_B = M_0$ and $x_A = x_B = x_0$ with

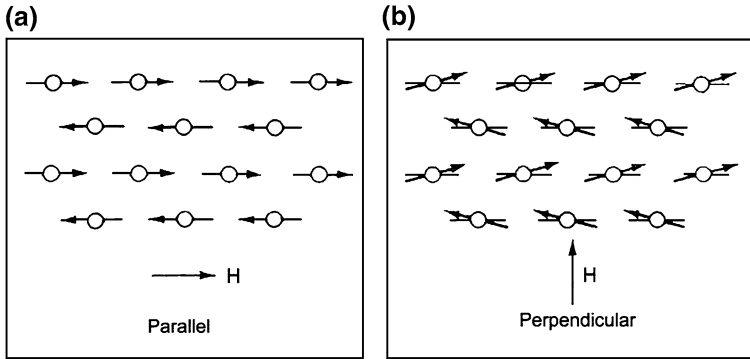


Fig. 9.16 Effect of applied field on the spins: **a** H parallel to spin, **b** H perpendicular to spin

$$x_0 = \frac{g\mu_B S}{k_B T} [(N_{AB} - N_{ii})M_0]. \quad (9.78)$$

We may expand a Brillouin function in a Taylor series in H and as an approximation retain only the first term. Then

$$B_S(x_A) = B_S(x_0) + \frac{g\mu_B S}{k_B T} [H + N_{ii}(M_0 - M_A) + N_{AB}(M_B - M_0)] B'_S(x_0) \quad (9.79)$$

and

$$B_S(x_B) = B_S(x_0) - \frac{g\mu_B S}{k_B T} [H + N_{AB}(M_0 - M_A) + N_{ii}(M_B - M_0)] B'_S(x_0). \quad (9.80)$$

In (9.79) and (9.80), $B'_S(x_0)$ is the derivative of $B_S(x_0)$ with respect to the argument. Solving (9.79) and (9.80) for M_A and M_B , we get the magnetization of the system induced by the field as $(M_A - M_B)$ from which we can get the susceptibility $\chi_{\parallel}$ in the easy direction as

$$\chi_{\parallel} = \frac{M_A - M_B}{H} = \frac{N\mu_B^2 g^2 S^2 B'_S(x_0)}{[k_B T + \frac{1}{2}(N_{ii} + N_{AB})\mu_B^2 g^2 S^2 N B'_S(x_0)]}. \quad (9.81)$$

According to (9.81), $\chi_{\parallel}$ is a function of S , T and $B'_S(x_0)$. We see that $\chi_{\parallel} \rightarrow 0$ as $T \rightarrow 0$ and at $T = T_N$, $\chi_{\parallel}$ equals the value given by the modified Curie-Weiss law (9.70). Curves for $\chi_{\parallel}$ calculated from (9.81) for different assumed values of S are shown in Fig. 9.15.

Let us now consider the case where the external field H is perpendicular to the easy axis (Fig. 9.16b). The field now exerts a torque on the dipole tending to rotate it such that it aligns along the field. This tendency is opposed by the molecular field. As a result the dipole assumes an intermediate position making angle ϕ with the

easy direction. At equilibrium the total torque (resultant of torque due to external field and torque due to molecular field) on each dipole will be zero. Thus

$$|\mathbf{M}_A \times (\mathbf{H} + \mathbf{H}_{mA})| = 0. \quad (9.82)$$

Using (9.54), we have

$$|\mathbf{M}_A \times \mathbf{H}| - N_{AB} \mathbf{M}_A \times \mathbf{M}_B = 0. \quad (9.83)$$

Or,

$$M_A H \cos \phi - N_{AB} M_A M_B \sin 2\phi = 0. \quad (9.84)$$

We can simplify (9.84) as

$$2M_B \sin \phi = H/N_{AB}. \quad (9.85)$$

Remembering that $M_A = M_B$, the total magnetization along the field direction is

$$M = (M_A + M_B) \sin \phi = 2M_B \sin \phi = H/N_{AB}. \quad (9.86)$$

Hence the susceptibility normal to the easy direction $\chi_{\perp}$ is

$$\chi_{\perp} = M/H = 1/N_{AB}. \quad (9.87)$$

The point to be noted is that $\chi_{\perp}$ is independent of T and of N_{ii} . The susceptibility $\chi = 1/N_{AB}$ at $T = T_N$. Finally, the isotropic susceptibility χ_{PC} for a polycrystalline sample is given by

$$\chi_{PC} = \frac{\chi_{\parallel} + 2\chi_{\perp}}{\chi}. \quad (9.88)$$

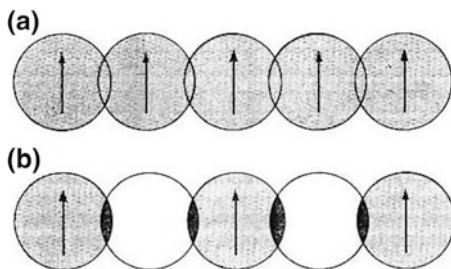
The temperature variation of $\chi_{\parallel}$, $\chi_{\perp}$ and χ_{PC} before $T = T_N$ and of χ at $T > T_N$ is shown in Fig. 9.15.

9.3.3 The Origin of Antiferromagnetism

The antiferromagnetism in MnO which is a prototype of an antiferromagnet is due to the spins of the neighbouring Mn ions. But their separation is $\sim 4.43 \text{ \AA}$. According to Fig. 9.9, at this separation, the Heisenberg exchange interaction is weak and ferromagnetic. Hence the origin of antiferromagnetism cannot be the direct Heisenberg exchange due to overlapping of Mn ions (Fig. 9.17a).

A different mechanism was introduced by Kramers [22] and later refined by Anderson [23] and Van Vleck [24]. It is called superexchange. It involves the

Fig. 9.17 Antiferromagnetic interaction: **a** direct exchange, **b** superexchange



indirect exchange between magnetic ions like Mn through their overlapping with a common nonmagnetic ion like oxygen (Fig. 9.17b). We shall now elaborate the concept of superexchange.

Let us consider two Mn^{2+} ions and a single O^{2-} ion arranged on a line (Fig. 9.18a). The ground state of the system consists of one electron on each Mn^{2+} ion in d_1 states and two electrons on the O^{2-} in the p state. The axis of the p -orbitals which form a dumb-bell coincides with the line joining the two Mn^{2+} ions. There is a finite probability that one of the oxygen electrons is transferred to a Mn ion (Fig. 9.18b). This is an excited state in which there will be a coupling between the electrons in the d_1 and d_1' states of the Mn ion and between the remaining electron of the oxygen ion and one d_2 electron on the other Mn ion. It may be noted that this coupling is due to the spin of the Mn ion. Other possible electron transfers are shown in Fig. 9.18c, d. This picture explains the interaction between two neighbouring Mn ions through the participation of the oxygen ions. The quantum mechanical calculation of the superexchange interaction involves use of third order perturbation theory which is beyond our scope. It is found that qualitatively the superexchange Hamiltonian is similar in form to the direct exchange Hamiltonian.

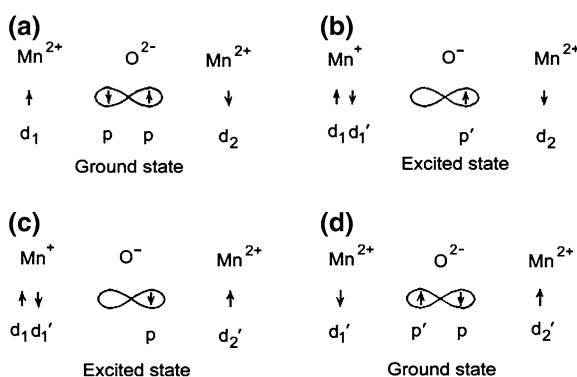


Fig. 9.18 Details of the superexchange mechanism in MnO

9.3.4 Experimental Results

9.3.4.1 Evidence of Spin Ordering

Direct evidence of spin ordering can be obtained from neutron diffraction experiments. The details of this technique will be discussed in the next chapter; here we shall only cite the results on MnO and MnF₂ which are the earliest antiferromagnetic systems to have been investigated.

MnO has the cubic NaCl structure. The crystal structure and spin orientation are shown in Fig. 9.19. The spins of the Mn ions in a given (111) plane are all in one direction whereas those in the next (111) plane are in the opposite (antiparallel) direction.

MnF₂ has the tetragonal rutile structure. The crystal structure and spin orientation is shown in Fig. 9.20. The spins are arranged parallel to the *c*-direction but the direction of spins is opposite in alternate planes.

9.3.4.2 Temperature Variation of Susceptibility

Experimental results obtained by Bizette et al. [18] on the temperature variation of the susceptibility of a powder sample of MnO are shown in Fig. 9.21. As the temperature rises from very low temperatures the susceptibility increases up to $T = 122$ K which may be identified as the critical temperature or Neel temperature T_N . The experimental data obtained by Bizette and Tsai [25] on a single crystal of MnF₂ are shown in Fig. 9.22. It can be seen that $\chi_{\parallel}$ increases with temperature up to the Neel temperature. $T_N = 74$ K. On the other hand $\chi_{\perp}$ is nearly constant with

Fig. 9.19 Crystal structure and spin ordering in MnO; circles with plus and minus signs represent Mn²⁺ ions with spins parallel and antiparallel to the easy direction. The dark spheres represent oxygen ions

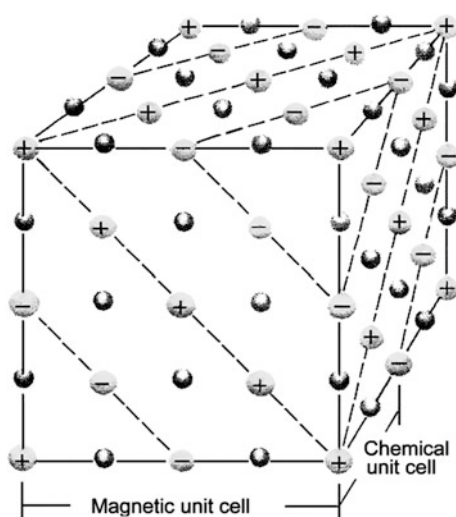


Fig. 9.20 Crystal structure and spin ordering in MnF_2 ; dark circles are fluorine ions

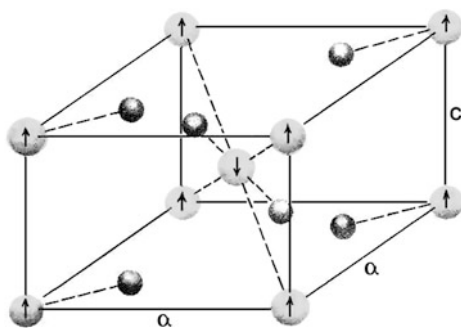


Fig. 9.21 Susceptibility of MnO measured on a powder sample

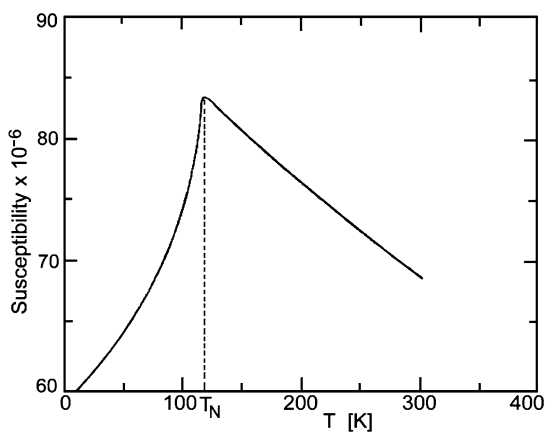
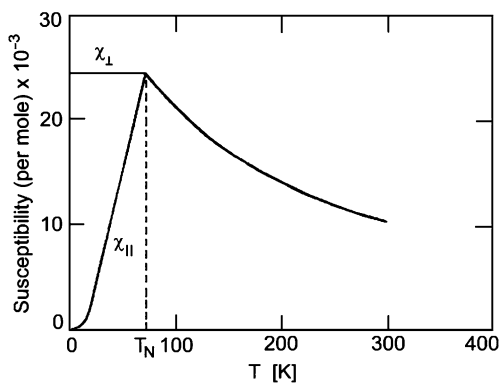


Fig. 9.22 Susceptibility of MnF_2 measured on a single crystal



respect to temperature. Above T_N , there is a single susceptibility χ which decreases with increasing temperature in a modified Curie-Weiss fashion. These observations are in accordance with the molecular field theory.

9.3.4.3 Analysis of Antiferromagnetic Parameters

Values of the antiferromagnetic parameters T_N , θ , θ/T_N , C and $\chi(0)/\chi(T_N)$ of some antiferromagnetic materials are given in Table 9.3. Theory predicts a value of 0.67 for $\chi_P(0)/\chi_P(T_N)$. The experimental values are close to this value. However, MnS and MnO₂ have values which are much larger whereas FeCl₂ and FeCO₃ have values which are much smaller.

For the ratio θ/T_N , theory predicts values in the range 1–3 for the bcc and rutile structures and between 1 and 5 for fcc structures. Most of the experimental values are within these ranges. Surprisingly, θ is negative for some systems; this implies a negative value for the interaction coefficient N_{ii} .

The Curie constant C , obtained by fitting the modified Curie-Weiss law to experimental data on χ above T_N , is useful in estimating values of the effective number of Bohr magnetons p_{eff} . For MnO, we get $p_{\text{eff}} = 5.95$. This is to be compared with the theoretical value of 5.92.

Table 9.3 Parameters for some antiferromagnetic compounds

Compound	Crystal structure	T_N (K)	θ (K)	θ/T_N	C_{mole}	$\chi(0)/\chi(T_N)$
MnO	fcc	122	610	5.0	4.40	0.69
FeO	fcc	185	570	3.1	6.24	0.77
CoO	fcc	291	280	0.96	3.0	–
NiO	fcc	515	–	–	–	0.67
MnS	fcc	165	528	3.2	4.30	0.82
MnF ₂	Rutile	74	113	1.5	4.08	0.75
FeF ₂	Rutile	85	117	1.4	3.9	0.72
CoF ₂	Rutile	40	53	1.3	3.3	–
NiF ₂	Rutile	78	116	1.5	1.5	–
MnO ₂	Rutile	86	–	–	–	0.93
Cr ₂ O ₃	Rhombohedral	307	1,070	3.5	2.56	0.76
α -Fe ₂ O ₃	Rhombohedral	950	2,000	2.1	4.4	–
FeS	Hex. layer	613	857	1.4	3.44	–
FeCl ₂	Hex. layer	24	–48	–2.0	3.59	<0.2
CoCl ₂	Hex. layer	25	–38.1	–1.5	3.46	~0.6
NiCl ₂	Hex. layer	50	–68.2	–1.4	1.36	–

9.4 Ferrimagnetism

9.4.1 General

We see that in ferromagnetism, the spins are all equal and parallel even in zero external magnetic fields and there is spontaneous magnetization. In antiferromagnetism, the spins are equal but antiparallel so that there is no spontaneous magnetization. There is a third possibility; the spins may be antiparallel but unequal. Some such possible spin arrangements are shown in Fig. 9.23. In Fig. 9.23a there is one spin pointed downward in sublattice *A* and two spins pointed up in sublattice *B*; thus there is spontaneous magnetization. In Fig. 9.23b, there is one spin down in *A* and one spin up in *B*, but the two spins are unequal resulting in spontaneous magnetization. Other spin arrangements resulting in non-zero field are shown in Fig. 9.23c–g.

The magnetic behaviour of such a system was theoretically analysed by Neel [26] based essentially on the assumption of a Weiss-like field. The spontaneous magnetization of such a system starts with a large value at very low temperatures but reduces to zero at a critical temperature called T_{FN} . On the other hand, the reciprocal susceptibility starts with a large value at $T > T_{\text{FN}}$; it reduces with decreasing temperature linearly but takes a sudden plunge to zero at $T \approx T_{\text{FN}}$. This behaviour is called ferrimagnetism. Ferrimagnetism was first observed in materials called ferrites which have spinel structure.

Let us consider the ferrites. A typical member of this family is magnetite with the formula Fe_3O_4 ($\text{FeO} \cdot \text{Fe}_2\text{O}_3$). By replacing the divalent Fe ion in FeO with other divalent ions (like Mn^{2+} , Co^{2+} , Ni^{2+} , Cu^{2+} , Zn^{2+} or Cd^{2+}) we get the various members

Fig. 9.23 Examples of possible ferromagnetic spin arrangements

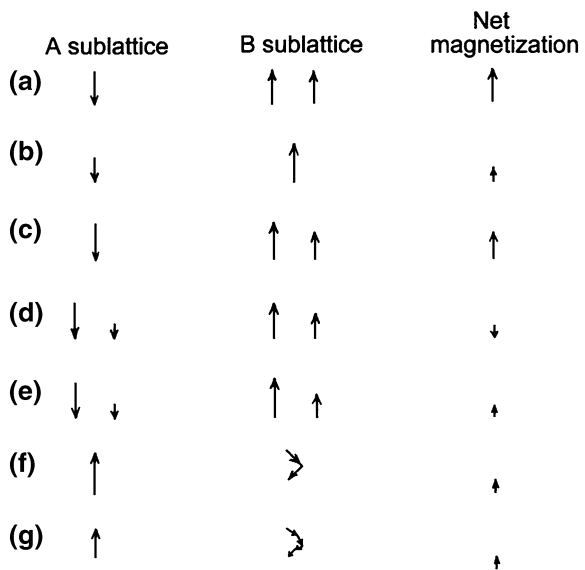
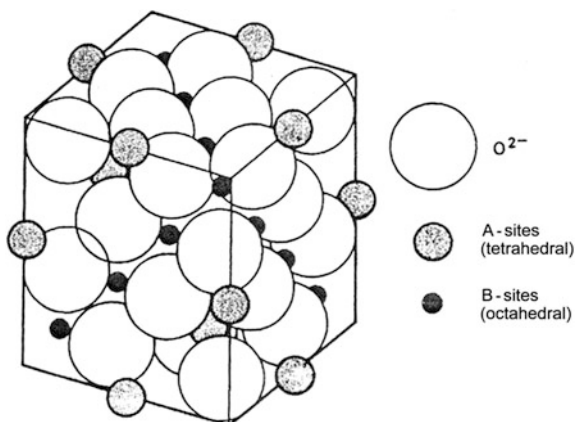
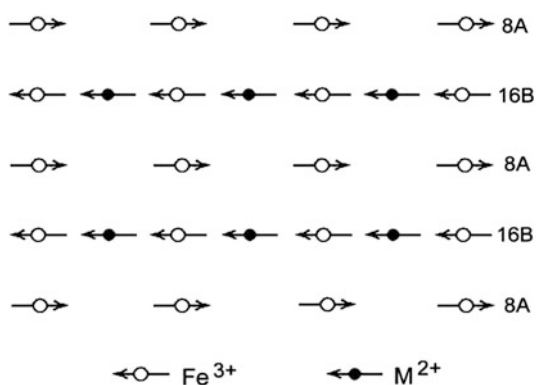


Fig. 9.24 The spinel structure

of the ferrite family. Ferrites have the spinel structure shown in Fig. 9.24. Here the empty circles denote oxygen. The black and grey circles represent metal ions. The oxygen ions with their large radius ($1.82 \text{ \AA}$) form an fcc lattice. In this fcc lattice, the metal ions occupy two types of interstitial sites. The eight grey circles occupy the tetrahedral sites (called 'A' sites). The sixteen black circles occupy the octahedral sites (called 'B' sites). From valence considerations, we should expect the M^{2+} ions on A sites and the Fe^{3+} ions on the B sites. This is called a normal spinel. But in reality, half the Fe^{3+} ions are located at A sites. The remaining half of the Fe^{3+} ions and all the M^{2+} ions occupy the B sites. This is called the inverse spinel structure. The anti-parallel arrangement of spins is as shown in Fig. 9.25. It can be seen that the net magnetization in zero field is due to the magnetization of the M^{2+} ions on the B sites.

Fig. 9.25 Arrangement of spins of the Fe^{3+} ions and the M^{2+} ions in spinel structure

9.4.2 Neel's Theory of Ferrimagnetism

Let us consider two antiparallel sublattices A and B with unequal spins. The expressions for the internal fields $\mathbf{H}_{mA}$ and $\mathbf{H}_{mB}$ are the same as in antiferromagnetism. Thus, we have

$$\mathbf{H}_{mA} = -N_{AA}\mathbf{M}_A - N_{AB}\mathbf{M}_B \quad (9.89)$$

and

$$\mathbf{H}_{mB} = -N_{BA}\mathbf{M}_A - N_{BB}\mathbf{M}_B. \quad (9.90)$$

The field coefficients $N_{AB} = N_{BA}$. In antiferromagnetism we had $N_{AA} = N_{BB}$. But the two sublattices are not crystallographically equivalent; so $N_{AA} \neq N_{BB}$. Also $\mathbf{M}_A \neq \mathbf{M}_B$. If an external field $\mathbf{H}$ is applied, the net fields $\mathbf{H}_A$ and $\mathbf{H}_B$ at the two sublattices are

$$\mathbf{H}_A = \mathbf{H} - N_{AA}\mathbf{M}_A - N_{AB}\mathbf{M}_B \quad (9.91)$$

and

$$\mathbf{H}_B = \mathbf{H} - N_{AB}\mathbf{M}_A - N_{BB}\mathbf{M}_B. \quad (9.92)$$

N_{AA} , N_{BB} and N_{AB} are positive. It is convenient to express N_{AA} and N_{BB} in terms of N_{AB} as

$$N_{AA} = \alpha N_{AB} \text{ and } N_{BB} = \beta N_{AB} \quad (9.93)$$

where α and β are constants.

We now write the magnetizations of the two sublattices as

$$M_A = \sum_i N_i g \mu_B S_i B_{S_i}(x_A) \quad (9.94)$$

where

$$x_A = \frac{S_i g \mu_B}{k_B T} H_A. \quad (9.95)$$

The Brillouin function is

$$B_{S_i}(x_A) = \frac{2S_i + 1}{2S_i} \coth \frac{2S_i + 1}{2S_i} x_A - \frac{1}{2S_i} \coth \frac{x_A}{2S_i}. \quad (9.96)$$

Similarly we have

$$M_B = \sum_j N_j g \mu_B S_j B_{S_j}(x_B) \quad (9.97)$$

where

$$x_B = \frac{S_j g \mu_B}{k_B T} H_B. \quad (9.98)$$

The Brillouin function is

$$B_{S_j}(x_B) = \frac{2S_j + 1}{2S_j} \coth \frac{2S_j + 1}{2S_j} x_B - \frac{1}{2S_j} \coth \frac{x_B}{2S_j}. \quad (9.99)$$

Here N_i and N_j are the numbers of atoms per unit volume with spin quantum numbers S_i and S_j respectively. We shall discuss the implications of the above equations in different temperature regions viz., $T > T_{\text{FN}}$, $T = T_{\text{FN}}$ and $T < T_{\text{FN}}$.

At $T > T_{\text{FN}}$

At the critical temperature, spontaneous magnetization vanishes and paramagnetism sets in. We shall call this temperature T_{FN} (to distinguish it from the antiferromagnetic Neel temperature T_N). At T_{FN} and above, the material becomes paramagnetic and we may write the magnetization in the Curie-form as

$$M_A = \frac{C_A}{T} H_A \text{ and } M_B = \frac{C_B}{T} H_B \quad (9.100)$$

where C_A and C_B are the Curie constants for the two sublattices. These are given by

$$C_A = \sum_i \frac{N_i g^2 \mu_B^2 S_i(S_i + 1)}{3k_B} \text{ and } C_B = \sum_j \frac{N_j g^2 \mu_B^2 S_j(S_j + 1)}{3k_B}. \quad (9.101)$$

The magnetic fields (9.91) and (9.92) may be written in the scalar form as

$$H_A = H - N_{AA}M_A - N_{AB}M_B \quad (9.102)$$

and

$$H_B = H - N_{AB}M_A - N_{BB}M_B. \quad (9.103)$$

Combining (9.100) and (9.102), we get

$$(T + C_A N_{AA})M_A + C_A N_{AB}M_B = C_A H \quad (9.104)$$

Similarly, combining (9.100) and (9.103), we get

$$C_B N_{AB} M_A + (T + C_B N_{BB}) M_B = C_B H \quad (9.105)$$

Solving (9.104) and (9.105), we get

$$M_A = \left[\frac{C_A(T + C_B N_{BB}) - C_A C_B N_{AB}}{(T + C_A N_{AA})(T + C_B N_{BB}) - C_A C_B N_{AB}^2} \right] H \quad (9.106)$$

and

$$M_B = \left[\frac{C_B(T + C_A N_{AA}) - C_A C_B N_{AB}}{(T + C_A N_{AA})(T + C_B N_{BB}) - C_A C_B N_{AB}^2} \right] H. \quad (9.107)$$

With some mathematics (9.106), (9.107) can be combined and reorganised to give

$$\frac{1}{\chi} = \frac{H}{M_A + M_B} = \frac{T}{C} - \frac{1}{\chi_0} - \frac{\sigma}{T - \theta'}, \quad (9.108)$$

where

$$\left. \begin{aligned} C &= C_A + C_B, \\ \frac{1}{\chi_0} &= -\frac{1}{C^2} (C_A^2 N_{AA} + C_B^2 N_{BB} + 2C_A C_B N_{AB}), \\ \sigma &= \frac{C_A C_B}{C^3} \left\{ C_A^2 (N_{AA} - N_{AB})^2 + C_B^2 (N_{BB} - N_{AB})^2 - 2C_A C_B [N_{AB}^2 - (N_{AA} + N_{BB})N_{AB} + N_{AA}N_{BB}] \right\} \\ \theta' &= -\frac{C_A C_B}{C} (N_{AA} + N_{BB} - 2N_{AB}) \end{aligned} \right\}. \quad (9.109)$$

A schematic plot of $1/\chi$ versus T drawn according to (9.108) is shown in Fig. 9.26. It is a hyperbola with $(1/\chi) \rightarrow 0$ as $T \rightarrow T_{\text{FN}}$. At very high temperatures, the third term in (9.108) can be neglected. Then (9.108) simplifies to

$$\frac{1}{\chi} = \frac{T}{C} - \frac{1}{\chi_0}. \quad (9.110)$$

Thus the high temperature part in Fig. 9.26 is linear which when extrapolated to $(1/\chi) = 0$ yields an intercept $(-C/\chi_0)$ on the T -axis; we may call it θ . Equation (9.110) can now be expressed as

$$\chi = \frac{C}{T + \theta}. \quad (9.111)$$

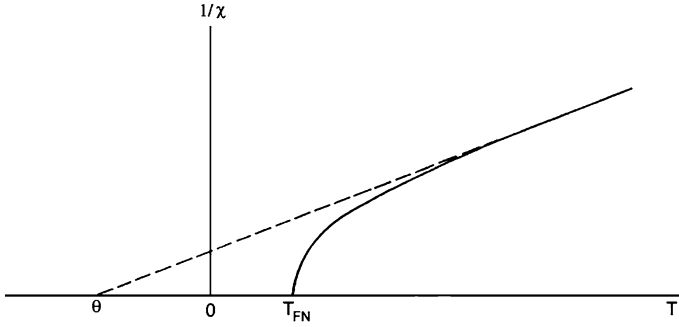


Fig. 9.26 Reciprocal susceptibility of a ferrimagnet above T_{FN}

The parameter θ is called the paramagnetic Curie temperature. One can notice a similarity between (9.111) and (9.70) for an antiferromagnetic system. However, the curve in Fig. 9.26 differs from those for ferromagnetic and antiferromagnetic systems in having a severe curvature close to $T = T_{FN}$ where it plunges to $(1/\chi) = 0$.

At $T = T_{FN}$

In (9.104), (9.105), we shall put $H = 0$, and $T = T_{FN}$. Then we get

$$(T_{FN} + C_A N_{AA})M_A + C_A N_{AB}M_B = 0 \quad (9.112)$$

and

$$C_B N_{AB}M_A + (T_{FN} + C_B N_{BB})M_B = 0. \quad (9.113)$$

Equations (9.112) and (9.113) can be solved only if the determinant of the coefficients of M_A and M_B vanishes. In that case, we get

$$T_{FN} = -\frac{1}{2}(C_A N_{AA} + C_B N_{BB}) + \frac{1}{2} \left[(C_A N_{AA} - C_B N_{BB})^2 + 4C_A C_B N_{AB}^2 \right]^{1/2} \quad (9.114)$$

At $T < T_{FN}$

In this region, there is ferrimagnetic ordering. The sublattices are antiparallel. We shall replace S_i by S_A and S_j by S_B . Further we shall assume $N_{AB} > N_{AA}, N_{BB}$. Then the sublattice magnetizations (9.94) and (9.97) reduce to

$$M_A(T) = M_A(0)B_{SA}(x_A) \text{ and } M_B(T) = M_B(0)B_{SB}(x_B) \quad (9.115)$$

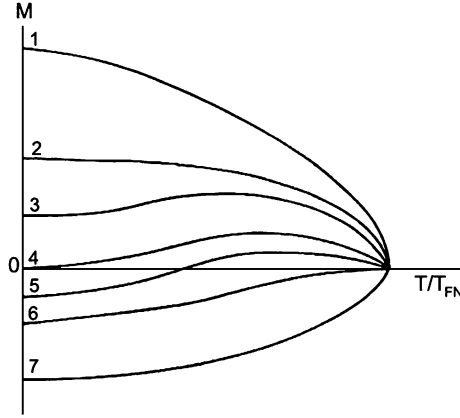


Fig. 9.27 Temperature variation of net magnetization M for various values of $M_A(0)/M_B(0)$; curves 1, 2, 3 with $M_A(0)/M_B(0) < 1$, curve 4 with $M_A(0)/M_B(0) = 1$, curves 5, 6, 7 with $M_A(0)/M_B(0) > 1$

and

$$x_A = \frac{S_A g \mu_B}{k_B T} (-N_{AA} M_A + N_{AB} M_B) \quad (9.116)$$

and

$$x_B = \frac{S_B g \mu_B}{k_B T} (N_{AB} M_A - N_{BB} M_B). \quad (9.117)$$

Note the signs in (9.116) and (9.117) which are in keeping with the antiparallel spin orientation in the two sublattices.

The evaluation of the net magnetization $M(T) = [M_A(T) - M_B(T)]$ from (9.115) is not straightforward. In his original paper, Neel [26] described a graphical procedure. Now-a-days, it can be obtained directly using a computer. The plots of $M(T)$ versus (T/T_{FN}) for different assumed values of $M_A(0)/M_B(0)$ are shown in Fig. 9.27. All plots converge to $M(T) = 0$ at $T = T_{FN}$.

9.4.3 Experimental Results

The temperature variation of spontaneous magnetization $M(T)$ and of reciprocal susceptibility $1/\chi$ for two ferrites Fe_3O_4 and NiFe_2O_4 are shown in Fig. 9.28. The sets of curves are in agreement with (9.108) and (9.115). Magnetization $M(T)$ and reciprocal susceptibility $1/\chi$ both plunge to zero at $T = T_{FN}$. It is a coincidence that both materials have the same T_{FN} .

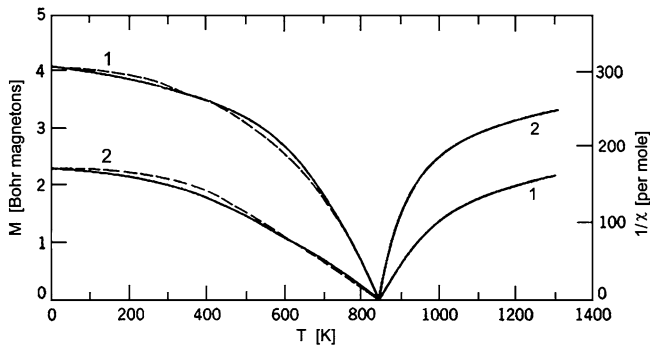


Fig. 9.28 Temperature variation of M (lhs) and reciprocal susceptibility $1/\chi$ (rhs) for (1) Fe_3O_4 and (2) NiFe_2O_4 ; *solid lines* are experimental curves and *dashed lines* are theoretical curves

Table 9.4 Ferrimagnetic parameters for some simple spinels (magnetic moments are in Bohr magnetons per formula unit for 0 K)

Spinel	Moment A site	Moment B site	Net moment		T_{FN} (K)
			Theoretical	Experimental	
$\text{Fe}^{3+}[\text{Fe}^{2+}\text{Fe}^{3+}]\text{O}_4$	5	4 + 5	4	4.1	858
$\text{Fe}[\text{NiFe}]\text{O}_4$	5	2 + 5	2	2.3	858
$\text{Fe}[\text{CuFe}]\text{O}_4$	5	1 + 5	1	1.3	728
$\text{Fe}[\text{CoFe}]\text{O}_4$	5	3 + 5	3	3.7	793

The ferrimagnetic parameters of some spinels are given in Table 9.4. The experimental values of the net magnetic moment are close to the predicted values. Some deviations are possible due to the uncertainty in the value of S because of incomplete quenching of orbital angular momentum or due to the mixing of the two ions at the B sites.

Louis Neel, a Frenchman, has made fundamental contributions to the theories of antiferromagnetism and ferrimagnetism; he was awarded the Nobel Prize for physics in 1970.

9.5 Domains and Related Topics

Both ferromagnetic and ferrimagnetic materials have a domain structure i.e. they consist of a number of spontaneously magnetized regions with the direction of magnetization being the same within a region but varying from domain to domain.

The following discussion applies equally to ferromagnetic and ferrimagnetic substances.

9.5.1 The Concept of Domains

The concept of spontaneous magnetization was introduced by Weiss [2] according to which some substances have a magnetization even in the absence of an external magnetic field. In such a sample all magnetic dipoles are arranged in a single direction. Such a sample is shown in Fig. 9.29a. But very often we come across samples of the same material which have no magnetization. How is this possible? Weiss explained this contradiction by invoking the idea of domains. Consider the possibility of a sample with two regions (Fig. 9.29b) each being magnetized but the two being magnetized in opposite directions. The net magnetization of the sample is zero. The same will be the case with a sample having four regions with direction of magnetizations as shown in Fig. 9.29c. In the limit, a sample may consist of several small regions (Fig. 9.29d) each spontaneously magnetized but the direction of magnetization varying from region to region, the net magnetization of the sample being zero. Such regions are called ‘domains’.

Let us consider the energetics of domain formation. A single-domain sample is shown in Fig. 9.30a. It has free magnetic poles at the ends and their lines of force are spread considerably in the space around the sample. The magnetic energy $(1/8\pi) \int H^2 dV$ associated with these lines of force will have a substantial value. If another sample has domains with directions of magnetization as shown in Fig. 9.30b, the spread of the lines of force is considerably less than in the first case and the energy integral will have a lesser value. If, further, there are triangular prism domains at the two ends of the sample (Fig. 9.30c), the lines of force do not spread

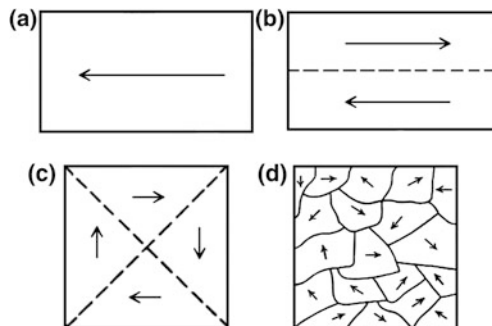


Fig. 9.29 Development of domain structure: **a** a uniformly magnetized sample, **b** a sample with two oppositely magnetized regions with net zero magnetization, **c** a sample with four magnetized regions with net zero magnetization, **d** a sample with several magnetized regions (domains) with net zero magnetization

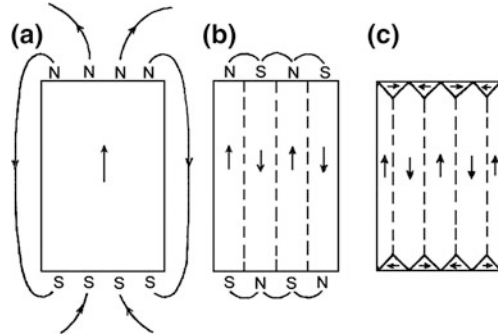


Fig. 9.30 Energetics of domain formation

into space at all; these prism domains are called ‘closure domains’. It is obvious that the magnetic energy in this case is zero. Hence domains reduce the magnetic energy and therefore domain formation is favoured from energy considerations.

9.5.2 Observation of Domains

Domains were first observed by Bitter [27] and independently by Hamos and Theissen [28]. They applied a drop of colloidal suspension of fine ferromagnetic particles on the surface of a ferromagnetic crystal placed on an electromagnet (Fig. 9.31). The suspension consisted of Fe_3O_4 particles precipitated from a mixture

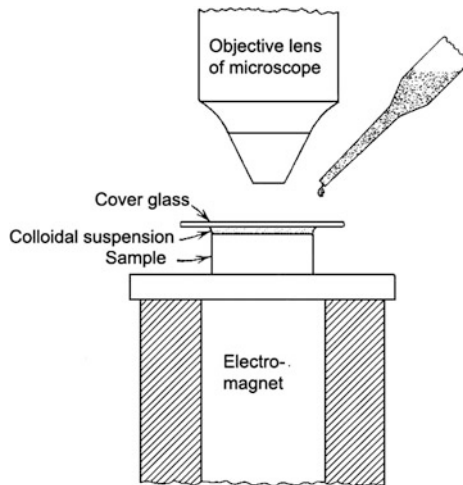
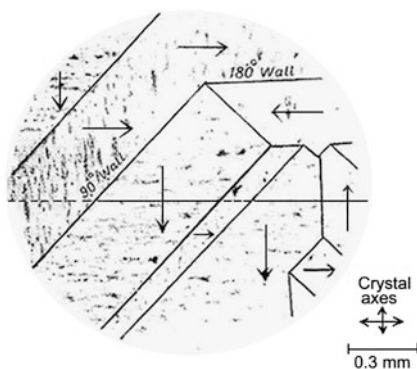


Fig. 9.31 Bitter’s method

Fig. 9.32 Domain pattern obtained by Bitter's method



of $\text{FeCl}_2 \cdot 2\text{H}_2\text{O}$, $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ and NaOH . The magnetite precipitate is then added to a soap solution and the mixture is boiled. The colloidal Fe_3O_4 particles are attracted to the magnetic free poles along the domain walls. Thus the domain walls are revealed as boundaries of a domain. The domain structure formed on the surface of a sample can be observed with a reflection microscope. A typical domain pattern is shown in Fig. 9.32. The field of view consists of a number of regions enclosing parallel striations. The direction of spontaneous magnetization in a domain is normal to the striations. The magnetization direction varies from domain to domain. Electropolishing of the surface and use of electron microscope ensures better definition and resolution.

Another method is the magnetic Kerr effect. This is the rotation of the plane of polarisation when a beam of plane-polarised light undergoes reflection at the surface of a magnetized sample. The arrangement is shown in Fig. 9.33a and the path of the beam is shown in Fig. 9.33b. The incident beam passes through a Nicol prism and is reflected down with a half-silvered mirror. On reflection at the sample, the plane of polarisation is rotated by angle θ . This reflected beam now passes through the half-silvered mirror and a second Nicol prism. The beam is analysed and recorded through a microscope assembly. The optical pattern of domains observed by Roberts and Bean [29] on a MnBi crystal is shown in Fig. 9.34.

The shapes of domains differ with the domain structure. The schematic of a closure domain is shown in Fig. 9.35a and the corresponding observed domain in Fig. 9.35b. In Fig. 9.36 the schematic of a 'tree' shaped domain and the observed pattern are shown. Domain patterns differ from one crystal to another. Also, they differ from one crystal plane to another in the same crystal; these features are shown in Fig. 9.37.

Fig. 9.33 **a** Magnetic Kerr effect **b** optics of the magnetic Kerr effect

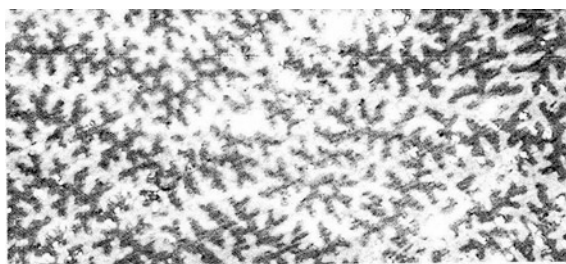
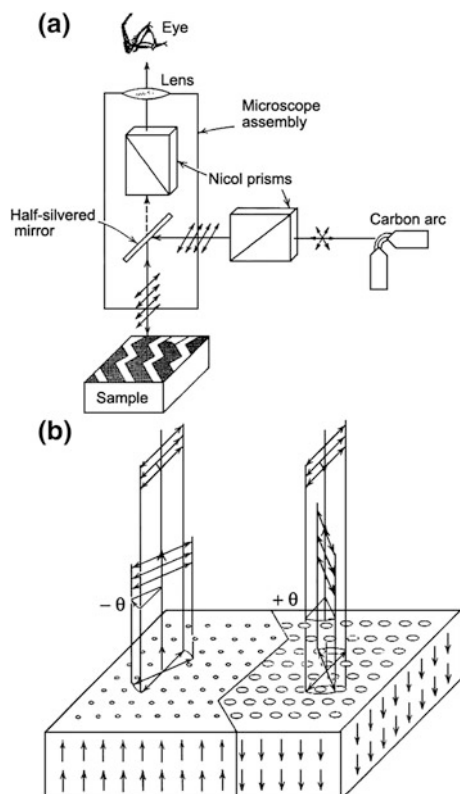


Fig. 9.34 Domain pattern obtained by the magnetic Kerr effect method

9.5.3 Magneto-Crystalline Anisotropy

So far we have treated magnetization as isotropic. Experiments have shown that it is not so and that, in a crystal, there are preferred directions of magnetization. This is called magneto-crystalline anisotropy. Plots of M_H for some directions as a function of applied field H are shown in Fig. 9.38. Here M_H is the magnetization produced in

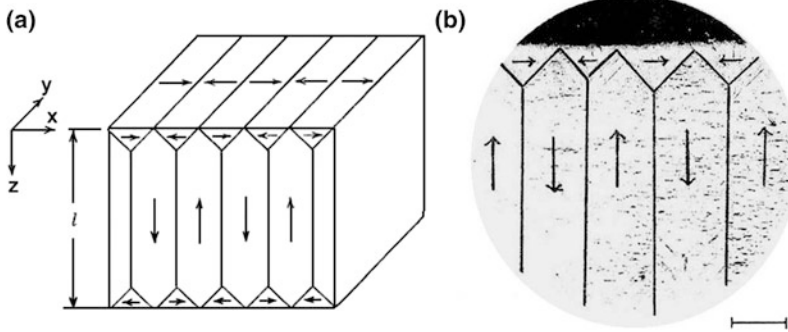


Fig. 9.35 **a** Closure domains, **b** domain pattern

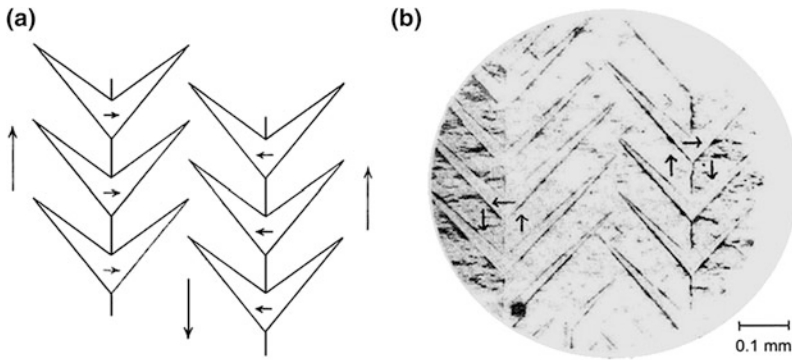


Fig. 9.36 **a** Tree typed domains, **b** domain patterns

the direction of the field. It is seen that to produce the same magnitude in different directions, different amounts of fields are required. The direction in which a given value of magnetization can be produced with the least field is called the ‘easy’ or ‘preferred’ direction. The direction in which maximum field is required to produce a given magnetization is called the ‘hard’ direction.

The anisotropy in magnetization is related to the anisotropy in the ‘magneto-crystalline energy’. This is defined as the work required to produce magnetization in a certain direction compared to that in the easy direction. In thermodynamic sense, this is a free energy which we shall denote by F_K . For a cubic crystal the magneto-crystalline energy is given by

$$F_K = K_1(\alpha_1^2\alpha_2^2 + \alpha_2^2\alpha_3^2 + \alpha_3^2\alpha_1^2) + K_2\alpha_1^2\alpha_2^2\alpha_3^2 \quad (9.118)$$

where $\alpha_1, \alpha_2, \alpha_3$ are direction cosines of the direction K with respect to the crystal axes. The corresponding expression for a hexagonal crystal is

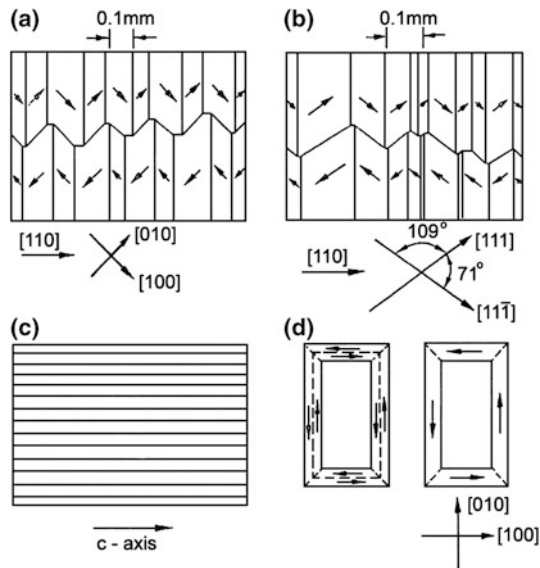


Fig. 9.37 Domain patterns **a** on (001) plane of Si-Fe crystal, **b** ($1\bar{1}0$) plane of Ni, **c** (001) plane of Co, **d** (100) plane of Si-Fe

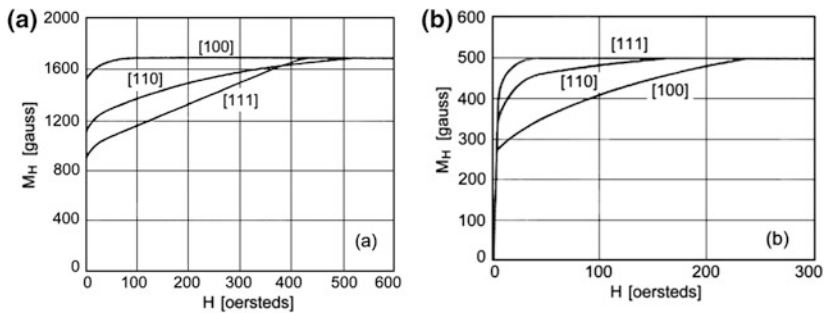


Fig. 9.38 Magnetization curves of **a** Fe and **b** Ni single crystals in different directions (indicated by their Miller indices)

$$F_K = K'_1 \sin^2 \theta + K'_2 \sin^4 \theta \quad (9.119)$$

where θ is the angle made by the direction of magnetization with the c -axis. Values of the anisotropy constants K_1 , K_2 , K'_1 and K'_2 are given in Table 9.5.

Table 9.5 Anisotropy constants at $T = 293$ K

Metal	K_1 (ergs/cm ³)	K_1' (ergs/cm ³)	K_2 (ergs/cm ³)	K_2' (ergs/cm ³)
Fe	4.6×10^5		1.5×10^5	
Ni	-5×10^4			
Co		4.1×10^6		1.0×10^6

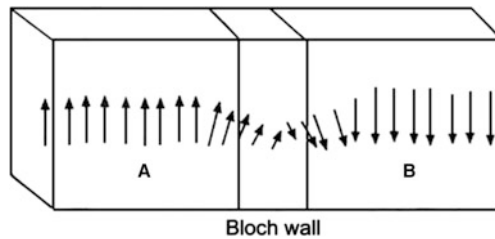
9.5.4 Domain Wall

The boundary between two domains is called the domain wall. We shall consider two adjacent domains which have magnetization in opposite directions i.e. the spins in the two domains make an angle π . Bloch [17] pointed out that in going across a domain boundary, the spin direction does not change abruptly; rather, it changes gradually as shown in Fig. 9.39.

Let us consider why it is so, first qualitatively and then quantitatively. The Heisenberg exchange energy between two neighbouring spins is $-2J_e \mathbf{S}_1 \cdot \mathbf{S}_2$. If the spins are parallel i.e., the angle between them is zero, the exchange energy is $-2J_e S^2$. If the two spins make an angle ϕ , the exchange energy is $-2J_e S^2 \cos \phi \approx -2J_e S^2 (1 - \frac{\phi^2}{2})$. Thus when the angle between the two spins changes from 0 to ϕ , the exchange energy increases by $J_e S^2 \phi^2$. Instead of considering two adjacent electrons, let us consider a chain of $(N + 1)$ electrons where the spins at the two ends make an angle ϕ_0 . Then $\phi = \phi_0/N$ and the net increase in exchange energy for the chain is

$$(E_{\text{exch.}})_{\text{chain}} = N J_e S^2 \phi^2 = N J_e S^2 \left(\frac{\phi_0}{N} \right)^2 = \frac{J_e S^2 \phi_0^2}{N}. \quad (9.120)$$

If $\phi_0 = \pi$ and $N = 100$, the increase in exchange energy is 100 times less than if adjacent spins had abruptly changed direction by π . Hence the domain boundary is a wall of some thickness rather than a sharp boundary.

**Fig. 9.39** Change in orientation of spins across the boundary between two domains: A and B

Let us consider the problem quantitatively. We start with an element of the wall of unit area and thickness Na , N being the number of spins in a line and a being the interatomic spacing. The total wall energy/cm² (σ_W) is

$$\sigma_W = \sigma_{\text{exch.}} + \sigma_{\text{anis.}} \quad (9.121)$$

Here $\sigma_{\text{exch.}}$ is the exchange energy and $\sigma_{\text{anis.}}$ the magneto-crystalline anisotropy energy. The exchange energy is obtained by multiplying the energy in (9.120) by $1/a^2$ which is the number of rows of spins across 1 cm² of the wall. $\sigma_{\text{anis.}}$ is K times the volume (Na) of the wall. Thus, (9.121) may be written as

$$\sigma_W = (J_e S^2 \phi_0^2 / Na^2) + KNa. \quad (9.122)$$

The equilibrium value is obtained from the condition $\frac{d\sigma_W}{dN} = 0$. This leads to

$$N = (J_e S^2 \phi_0^2 / Ka^3)^{1/2}. \quad (9.123)$$

Let us apply (9.123) for iron for which we use approximate values: $J_e = (k_B T_f)/3$, $\phi_0 = \pi$, $S = \frac{1}{2}$ and $K = 10^5$ ergs/cm³. This leads to $N = 300$ and thickness $t = Na \approx 1000$ Å. We can substitute this value of N in (9.123) to get σ_W ; it is ~ 1 erg/cm². It may be noted that this is an approximate treatment.

9.5.5 Magnetostriction

So far we have not considered whether magnetization has any effect on the dimensions of the magnetized sample. Experimentally, it is observed that indeed there is such an effect which is called ‘magnetostriction’.

Linear magnetostriction is denoted by a quantity λ called the magnetostriction constant which is defined as

$$\lambda = \delta l / l. \quad (9.124)$$

Here δl is the change in the length l caused by magnetostriction. The magnetostriction in different directions in an iron crystal is shown in Fig. 9.40. In view of the direction-dependence, the magnetostriction in different directions is indicated by attaching the Miller indices to the magnetostriction constants e.g. λ_{100} , λ_{111} etc.

In a cubic crystal, the anisotropy in magnetostriction can be described by

$$\frac{\delta l}{l} = \frac{3}{2} \lambda_{100} (\alpha_1^2 \beta_1^2 + \alpha_2^2 \beta_2^2 + \alpha_3^2 \beta_3^2)^{-1/3} + 3 \lambda_{111} (\alpha_1 \alpha_2 \beta_1 \beta_2 + \alpha_2 \alpha_3 \beta_2 \beta_3 + \alpha_3 \alpha_1 \beta_3 \beta_1) \quad (9.125)$$

Fig. 9.40 Magnetostriction in iron crystal

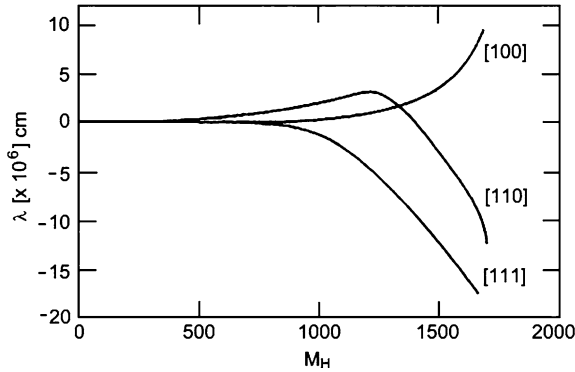


Table 9.6 Magnetostriction constants of some metals

Substance	Magnetostriction constants ($\times 10^{-6}$)	
	λ_{100}	λ_{111}
Fe	20.7	-21.2
Ni	-45.9	-24.3
40 at% Co-Fe	146.6	8.7
50 at% Co-Fe	119.3	41.3
70 at% Co-Fe	81.3	70.0

where $\alpha_1, \alpha_2, \alpha_3$ are the direction cosines of the direction of magnetostriction and $\beta_1, \beta_2, \beta_3$ those of the direction in which δl is measured. Values of λ_{100} and λ_{111} for some ferromagnetic crystals are given in Table 9.6.

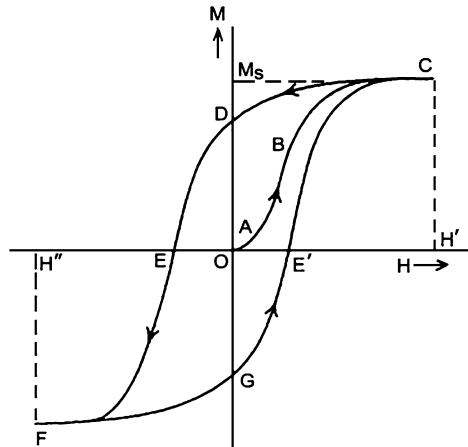
A simple qualitative interpretation of magnetostriction is that when magnetization changes the dimensions of a sample, its elastic strain energy increases but its magneto-crystalline anisotropy energy decreases. Equilibrium is achieved at a strain for which the increase in elastic strain energy is balanced by the decrease in magneto-crystalline energy.

9.5.6 Hysteresis

In Chap. 8, we have stated the relation between magnetization M and field H as $M = \chi H$ where χ is the susceptibility. This relation is reversible; if we increase H , M increases and if we decrease H , M also decreases. But in ferromagnetic and ferrimagnetic materials, the relation between M and H is not reversible.

Figure 9.41 is a plot of magnetic field H against the magnetization M for a ferromagnetic material. We start from the origin O. As the field increases the magnetization increases up to some point A. This part of the curve is reversible. As

Fig. 9.41 The M - H plot
(hysteresis loop)



the magnetic field increases further, the increase in M is faster up to the point B. Thereafter the magnetization tends to saturate i.e. an increase in the field does not lead to an increase in magnetization any longer. The curve becomes flat at the point C for which the field is H' . The magnetization at C is called the saturation magnetization M_{sat} .

If the magnetic field is now slowly withdrawn, the magnetization does not decrease along the path CBAO. Instead, the decrease in magnetization with decreasing field is now slower and the magnetization takes the path CD. The point D is of significance. Even when H has been reduced to zero, some magnetization (represented by OD) still remains. This magnetization for $H = 0$ is called remanent magnetization or remanence M_r . How can the sample be demagnetized? For this a field has to be applied in the negative direction; the sample is demagnetized at the field OE; this is called the coercive field H_c .

The susceptibility at any stage of magnetization is the slope of the M - H curve along the curve OABC. The susceptibility at the point A is called the initial susceptibility (χ_0), that at point B is called the incremental susceptibility (χ_i) and that close to C the maximum susceptibility (χ_{max}). We can define corresponding permeabilities from the relation $\mu = (\chi + 1)$.

Suppose we continue to increase the negative field beyond the point E, M now takes negative values i.e. the direction of magnetization is reversed. This negative magnetization increases along the path EF at which stage it attains saturation; the field now is H'' . If we now change the field from H'' to H' through O, the magnetization takes the path FGE'C. Thus when the magnetic field changes from $0 \rightarrow H' \rightarrow 0 \rightarrow H'' \rightarrow 0 \rightarrow H'$, the magnetization traces the closed path E'CDEFGE'. The phenomenon of the magnetization not returning to 0 when field is withdrawn is called hysteresis and the closed curve is called the hysteresis loop. It can be shown that during one cycle of magnetization, the system consumes heat equal to $W = \oint H dM$ which is the area of the hysteresis loop.

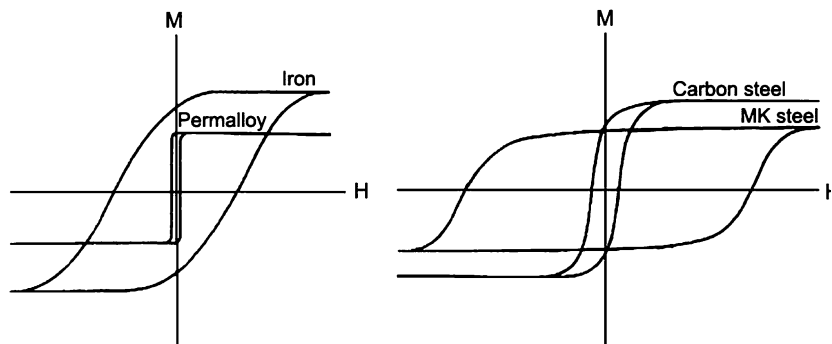


Fig. 9.42 Hysteresis loops of different materials

The shape and size of the hysteresis loop varies from material to material. Some examples are shown in Fig. 9.42. Also the value of the coercive field H_c varies over a wide range. For technical applications magnetic materials are classified as (a) soft materials and (b) hard materials or permanent magnets. The requirements of soft materials are high maximum permeability and low loss. The properties of some soft materials are given in Table 9.7. For permanent magnets, the requirements are large remanence and large coercive field. Properties of some permanent magnetic materials are given in Table 9.8.

It may be mentioned that magnets made of alloys of rare earths have superior properties. In particular, Sm and Nd magnets are popular. They have very high remanence and high coercivity. Rare earth magnets can lift steel spheres thousands of times their own weight. Sm magnets have a high T_c and are useful in high temperature applications.

Table 9.7 Properties of some soft magnetic materials

Material	Maximum permeability	Saturation flux density (Gs)	Coercive force H_c (oersteds)
Cold-rolled steel	2,000	21,000	1.8
Iron	5,000	21,500	1.0
Purified iron	180,000	21,500	0.05
4 % silicon-iron	7,000	19,700	0.5
Hipernik	70,000	16,000	0.05
78 permalloy	100,000	10,700	0.05
Mu metal	100,000	6,500	0.05
Supermalloy	800,000	8,000	0.002
Permendur	5,000	24,500	2.0

Table 9.8 Properties of some hard magnetic materials

Material	Coercive force H_c (oersteds)	Remanence (Gs)	Hysteresis loss ($\times 10^{-6}$ ergs)
Carbon steel	50	10,000	0.20
Tungsten steel	70	10,300	0.32
36 % cobalt steel	240	9,500	0.97
Remalloy or comol	250	10,500	1.1
Alnico II (sintered)	520	6,900	1.4
Alnico V	550	12,500	4.5
Alnico VI	750	10,000	3.5
Vicalloy II (wire)	510	10,000	3.5
Cunife (wire)	550	5,400	1.5
Vectolite	1,000	1,600	0.60
Platinum-cobalt	2,600	4,500	3.8
Hyflux	390	6,000	0.97

There is a close connection between the domains and the hysteresis curve. When a magnetic field is applied, the magnetization in individual domains increases. This happens in two processes shown in Fig. 9.43. The first process is the displacement of the Bloch walls due to which the domain grows (Fig. 9.43a). The second process (Fig. 9.43b) is the rotation of a domain which causes an increase in the magnetization in the direction of the field.

In fact, both these processes operate in different regions of the field as the specimen is magnetized. This is shown in Fig. 9.44. In the low field region, magnetization proceeds by reversible boundary displacement. In the medium field region, the magnetization is by irreversible boundary displacement. Finally, in the high field region, the magnetization is due to rotation of domains. Here, the magnetization is complete and saturation sets in.

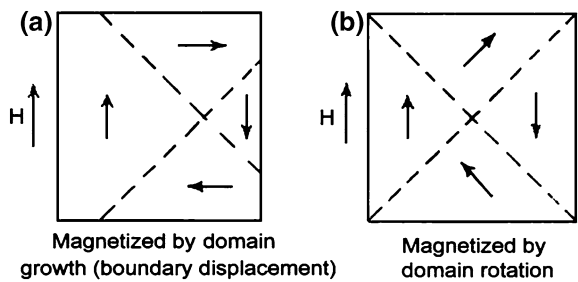


Fig. 9.43 Magnetization by **a** domain wall displacement **b** domain rotation

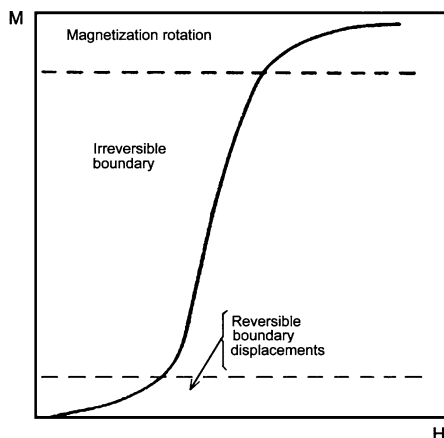


Fig. 9.44 Typical magnetization curve illustrating role of domains in different regions of the curve

9.5.7 Magnetic Bubbles

Consider a thin film about $20\ \mu\text{m}$ thick of a ferromagnetic material with its easy axis normal to the film surface. In such a sample, domain structure can be observed in transmission. In the demagnetized state, the film contains equal areas with magnetization up and down (Fig. 9.45a). When a small steady field is applied, it makes the unfavourable areas shrink i.e. if the field is up, areas with downward magnetization shrink (Fig. 9.45b).

When marked anisotropy is absent, the domain walls move freely (Fig. 9.46a). Applying a steady field normal to the film makes the domains thinner (Fig. 9.46b). If now a strong field is applied over and above the steady field, the negative domains break into cylindrical domains which look like dots (Fig. 9.46c). These are called bubbles. They are about $1\ \mu\text{m}$ in dimensions. Repeated application of the pulsed field produces more bubbles (Fig. 9.46d).

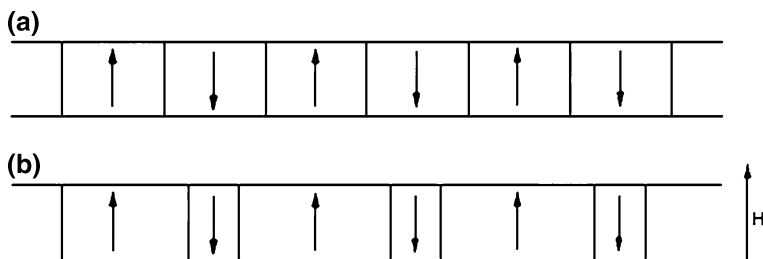
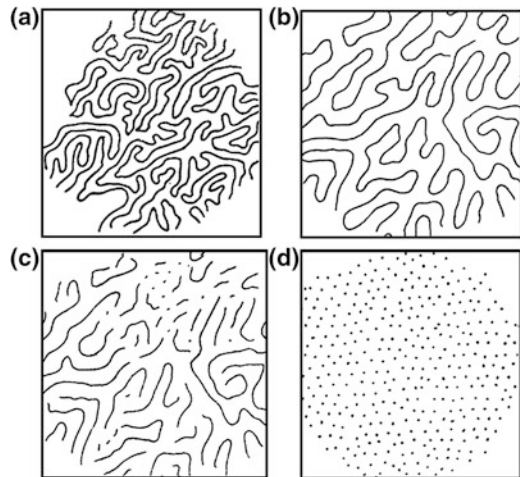


Fig. 9.45 Domains in a thin film of a ferro (or ferri) magnetic material: **a** without field, **b** with field

Fig. 9.46 Development of magnetic bubbles: **a** in no field, **b** in steady field, **c** after one superposed pulse, **d** after several pulses



The bubbles are magnetic dipoles. They repel one another and can be moved about on the sample by the application of a field gradient. They have several applications e.g. in tape recorders and logic operations. The preferred materials for magnetic bubbles are rare earth garnets.

9.6 Problems

1. A ferromagnetic system with $J = S = \frac{1}{2}$ is placed in a magnetic field $H = 5 \times 10^4$ amp-m. Calculate the fraction of spins parallel to the field and also the average component of the atomic magnetic moment in the direction of the field at $T = 300$ K.
2. Starting with (9.81) for an antiferromagnetic system, calculate $\chi_{\parallel}$ (a) at $T = 0$, (b) $T > T_N$, (c) $T = T_N$ for $S = \frac{1}{2}$ and $S = \infty$.
3. Taking the effective fields on the two sublattices in an antiferromagnetic system are $H_A = H - N_{ii}M_A - N_{AB}M_B$ and $H_B = H - N_{AB}M_A - N_{ii}M_B$, show that
$$\frac{\theta}{T_N} = \frac{N_{AB} + N_{ii}}{N_{AB} - N_{ii}}.$$
4. Assuming a two sublattice model for a ferrimagnetic system, discuss the variation of the reciprocal susceptibility at temperatures above and below the Neel temperature.
5. Discuss Stoner's theory of ferromagnetism qualitatively and show how it accounts for nonintegral values of the effective number of Bohr magnetons.

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Chapter 10

Magnetism III: Magnetic Symmetry and Magnetic Structures

Abstract We discuss magnetic symmetry and determination of magnetic structures in this chapter. The antiparallel alignment of spins in antiferromagnetic and ferrimagnetic crystals is not always linear. The spins assume a variety of orientations resulting in exotic magnetic structures. The chapter begins with the symmetry aspects of spin ordering. The concepts of magnetic symmetry, magnetic symmetry elements, magnetic point groups and magnetic space groups are explained. The preferred tool for the experimental determination of magnetic structures is neutron diffraction. Neutron diffraction theory is discussed and neutron diffraction instrumentation is described in detail. Neutron diffraction studies started with the antiferromagnetic manganese oxide and manganese fluoride. Studies of more complicated magnetic structures are also discussed. An important class of magnetic structures is the helical structure. Its stability is discussed. Magnetic structures of rare earth elements are described.

10.1 Introduction

We have seen in the preceding chapter that in some materials, the spin magnetic moments assume ordered positions in a certain temperature range. There are three types of ordering: (i) equal parallel spins, (ii) equal antiparallel spins and (iii) unequal antiparallel spins. These spin arrangements are shown in Fig. 10.1. These orderings give rise to three types of magnetic materials: ferromagnetic, antiferromagnetic and ferrimagnetic. This is only a broad classification. Within the latter two groups, the spins assume a variety of orientations. As an introduction to the exotic field of magnetic structures, several spin arrangements are shown in Figs. 10.2 and 10.3.

In this chapter we introduce magnetic symmetry, magnetic point groups and magnetic space groups. The neutron diffraction method of determining magnetic structures is described in detail. Several examples of solved magnetic structures are discussed.

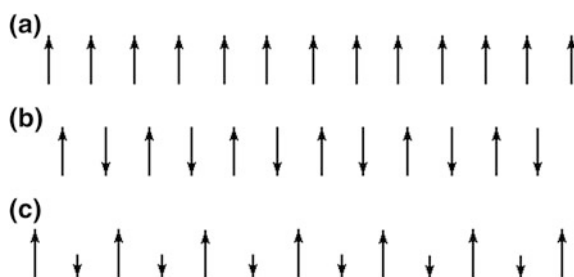


Fig. 10.1 Linear arrays of spins illustrating **a** Ferromagnetic, **b** Antiferromagnetic and **c** Ferrimagnetic orderings

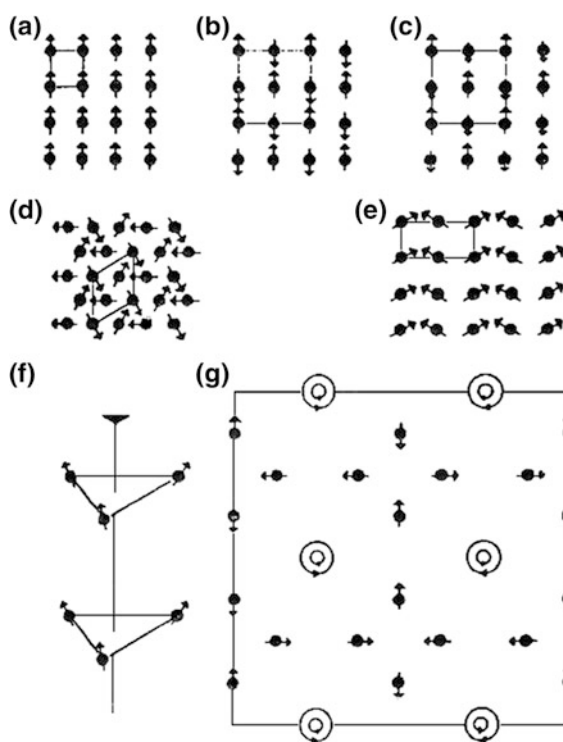
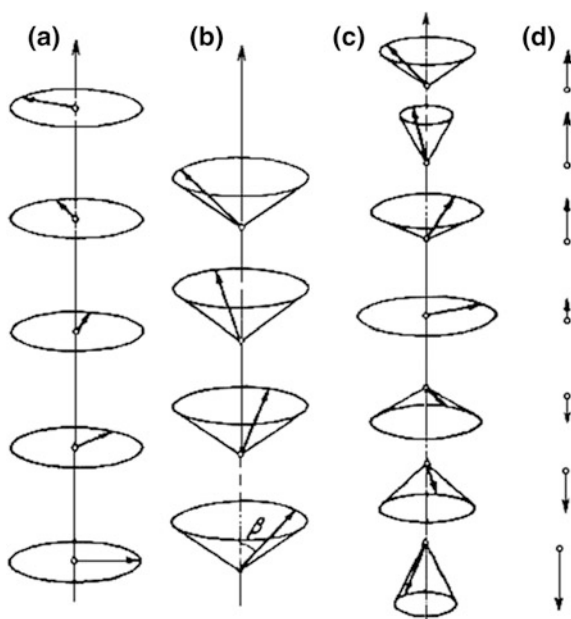


Fig. 10.2 Types of magnetic structures: **a** Ferromagnet, **b** Antiferromagnet, **c** Ferrimagnet, **d** Triangular structure, **e** Weak ferromagnet, **f** Umbrella structure, **g** Three dimensional (multiaxial) structure

The topic of magnetic symmetry is treated by Bhagavantam [1], Prandl [2], Cracknell [3] and Sirotnin and Shaskalskaya [4]. Oles et al. [5] have compiled solved magnetic structures.

Fig. 10.3 Types of helicoidal structures: **a** Simple spiral, **b** Ferromagnetic (*conical*) spiral, **c** Complex spiral, **d** Static longitudinal spin wave



10.2 Magnetic Symmetry

10.2.1 General

A detailed treatment of magnetic symmetry involves use of group theory which is beyond our scope. We shall therefore develop the concept of magnetic symmetry qualitatively and only note the results of rigorous treatments.

Before taking up magnetic symmetry, let us briefly consider the principles of conventional crystal symmetry. A crystal has symmetry elements inversion, rotation axes, mirror planes and rotation-reflection axes. Independent combinations of these symmetry elements are called point groups; there are 32 point groups. Crystals (assumed to be infinite in extent) also have translational symmetry elements, glide planes and screw axes. Independent combinations of point group symmetry elements and translational symmetry elements are called space groups; there are 230 space groups.

Some crystals contain atoms with non-zero spin. These atoms have an associated spin magnetic moment. If these magnetic moments are ordered in a definite manner, the crystal will have additional symmetry elements. Following Bhagavantham [1], we shall consider as an example the system shown in Fig. 10.4. The arrows indicate the sense of spins. In Fig. 10.4a, if the spins are ignored, the system has X_3 as a 2-fold rotation axis and X_2X_3 as a mirror plane. But if the spins are considered, X_2X_3

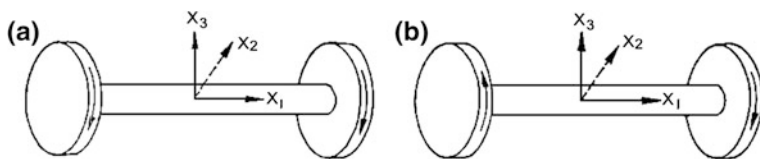


Fig. 10.4 Illustration of magnetic symmetry operation; **a** Spins down, **b** Antiparallel spins

is still a reflection plane but X_3 is no longer a rotation axis. Again, in Fig. 10.4b, if spins are ignored, X_3 is a rotation axis and X_2X_3 is a reflecting plane. But in this case, if the spins are considered X_3 is still a rotation axis but X_2X_3 is not a reflection plane. The important conclusion is that the symmetry elements of a system with spins are different from those of a system without spins.

In the case discussed above, on rotation (in Fig. 10.4a) and on reflection (in Fig. 10.4b), the up-spins become down-spins and vice versa. The system can be brought into congruence with itself by once again reversing the spins, up to down and down to up. This composite operation is called complementary operation and is denoted by symbol R in the Zheludev notation. Shubnikov, on the other hand, introduced magnetic symmetry in a different way. He regarded objects as having elements with black and white colours. An operation that changes black to white and white to black is equivalent to a symmetry operation which reverses spins. The absence of a change in colour is indicated by gray colour. In Shubnikov's system, the operation is denoted by $1'$. In the Shubnikov notation, $1'$ is called the 'time inversion' operator. Applying $1'$ twice will reproduce the original colours.

10.2.2 Magnetic Point Groups

The principle of deriving magnetic point groups is to combine the symmetric operator $1'$ with the ordinary (conventional) point group symmetry operations. The result is a magnetic point group. Thus, addition of $1'$ to the point group 4 results in magnetic point groups 4, $4'$ and $4 \times A$; A is antisymmetric group comprising operations 1 and $1'$. These are shown in Fig. 10.5. Similarly, point group m results

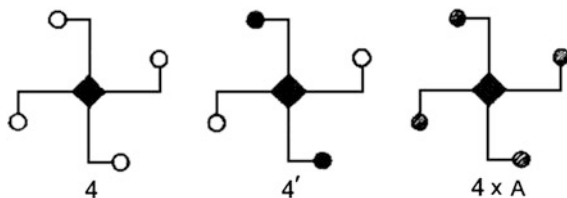


Fig. 10.5 Operation $1'$ acting on point group 4

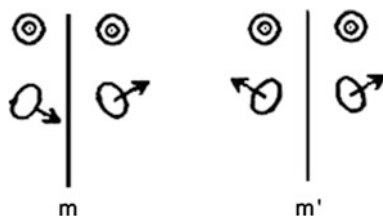


Fig. 10.6 Operation $1'$ acting on point group m

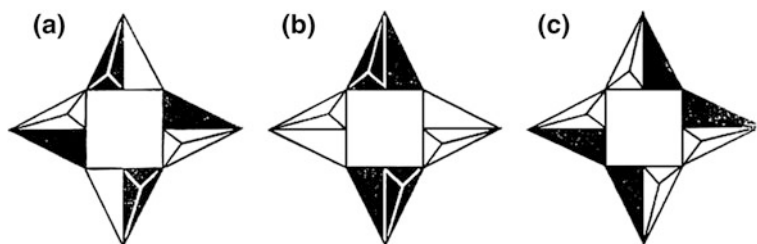


Fig. 10.7 Operation $1'$ acting on point group $\bar{4}2m$; **a** $\bar{4}2m$, **b** $\bar{4}2'm'$ and **c** $\bar{4}'2'm$

in magnetic point groups m and m' (Fig. 10.6). Sirotnin and Shaskalskaya [4] have given the example of conventional point group $\bar{4}2m$ which on application of $1'$ results in $\bar{4}2'm'$, $\bar{4}'2m'$ and $\bar{4}'2'm$ (Fig. 10.7). The total number of magnetic point groups is 90 as against the 32 conventional point groups. A complete list of magnetic point groups is given in Table 10.1.

10.2.3 Magnetic Space Groups

The combination of conventional space group symmetry elements with the magnetic symmetry element $1'$ results in magnetic space groups. The magnetic space groups were first derived and listed by Belov et al. [6]. They are 1421 in number as against the 230 conventional space groups. This means that each conventional space group has several magnetic space groups associated with it. These are listed in Table 10.2 for the space group $D_{2h}^{17}/Cmcm$. The magnetic space groups of some magnetic crystals are given in Table 10.3.

Before proceeding further, we shall summarize the results of conventional (or ordinary) crystallography and magnetic crystallography (Table 10.4).

Table 10.1 List of the 90 magnetic point groups

1					
$\bar{1}$	$\bar{1}'$				
2	$2'$				
m	m'				
$2/m$	$2'/m$	$2/m'$	$2'/m'$		
222	$2'2'2'$				
$mm2$	$m'm2'$	$m'm'2$			
mmm	$m'mm'$	$m'm'm$	$m'm'm'$		
4	$4'$				
$\bar{4}$	$\bar{4}'$				
$4/m$	$4'/m$	$4/m'$	$4'/m'$		
422	$4'22'$	$42'2'$			
$4mm$	$4'm'm$	$4\ m'm'$			
$\bar{4}2m$	$\bar{4}'2'm$	$\bar{4}'2m'$	$\bar{4}2'm'$		
$4/mmm$	$4/m'mmm'$	$4'/mmm'm$	$4'/m'm'm$	$4/mmm'm'$	$4/m'm'm'm'$
3					
$\bar{3}$	$\bar{3}'$				
32	$32'$				
$3m$	$3m'$				
$\bar{3}m$	$\bar{3}'m$	$\bar{3}'m'$	$\bar{3}m'$		
6	$6'$				
$\bar{6}$	$\bar{6}'$				
$6/m$	$6'/m$	$6/m'$	$6'/m'$		
622	$6'2'2'$	$62'2'$			
$6mm$	$6'm'm$	$6m'm'$			
$\bar{6}m2$	$\bar{6}'m'2$	$\bar{6}'m2'$	$\bar{6}m'2'$		
$6/mmm$	$6/m'mmm'$	$6'/mmm'm$	$6'/m'm'm$	$6/mmm'm'$	$6/m'm'm'm'$
23					
$m\bar{3}$	$m'3$				
432	$4'32'$				
$\bar{4}3m$	$\bar{4}'3'm'$				
$m\bar{3}m$	$m'3m$	$m3m'$	$m'3m'$		

Table 10.2 The family of magnetic space groups belonging to $D_{2h}^{17}/Cmcm$

$Cmcm$	$C_p mcm$
$Cm'cm$	$C_p m'cm$
$Cmc'm$	$C_p mc'm$
$Cmcm'$	$C_p mcm'$
$Cm'c'm$	$C_p m'c'm$
$Cmc'm'$	$C_p mc'm'$
$Cm'cm'$	$C_p m'cm'$
$Cm'c'm'$	$C_p m'c'm'$

Table 10.3 Chemical and magnetic space groups of some crystals

Crystal	Space group		References
	Chemical	Magnetic	
Fe	$Im3'm$	$I4/m'm'm'$	Sirotn and Shaskalskaya [4]
MnO	$Fm3 m$	$C_{2c} 2/m$	Prandl [2]
MnF ₂	$P4/mnm1'$	$P4'_2/mnm'$	Sirotn and Shaskalskaya [4]
FeSO ₄	$Cmcm$	$C_p m'cm'$	Prandl [2]

Table 10.4 Results of conventional and magnetic crystallography

Feature	Conventional crystallography	Magnetic crystallography
Primitive lattices	7	7
Bravais lattices	14	36
Point groups	32	90
Space groups	230	1421

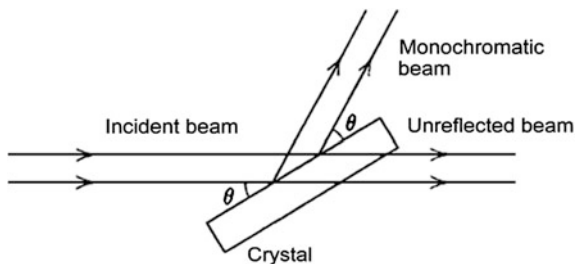
10.3 Neutron Diffraction

10.3.1 General

The development of X-ray diffraction method paved the way for unraveling the structure of crystals. X-rays are scattered by atomic electrons. Therefore, what X-ray diffraction finally reveals is the electron density distribution and the atomic positions. This information is called the crystal structure or, sometimes, the chemical crystal structure.

De Broglie introduced the concept of matter waves having a wavelength $\lambda = h/mv$ where h is Planck's constant, m the particle mass and v the particle velocity. This idea was confirmed by the discovery of the diffraction of electrons by crystals. Here again the electron waves are scattered by the atomic electrons. Consequently, electron diffraction, like X-ray diffraction, yields electron density, atomic positions and crystal structure.

Fig. 10.8 Basic principle of neutron diffraction



In course of time, the diffraction of neutrons by crystals was also discovered. Unlike X-rays, the neutron waves are scattered by the atomic nuclei. This scattering takes place by two mechanisms: (i) nuclear scattering which is due to the collisions between the neutrons and the nuclei and (ii) magnetic scattering which is due to interaction of the magnetic moment of the neutron with that of the atom. The former provides information about the nuclear structure which is essentially the same as the chemical structure provided by X-rays. The latter on the other hand provides information about the magnetic structure which is the symmetric arrangement of ordered spins.

The basic principle of diffraction of neutrons is the same as that of diffraction of X-rays. This is shown in Fig. 10.8. A beam of neutrons is incident on a crystal. Depending on the wavelength λ of the neutron waves and the interplanar spacing d of the crystal, the neutron waves are diffracted at an angle θ such that Bragg's law $n\lambda = 2d \sin \theta$ is satisfied; n is an integer.

10.3.2 Neutron Diffractometer

The basic components of the experimental arrangement for recording a neutron diffraction pattern are shown in Fig. 10.9. They are (a) the source, (b) the collimator, (c) the monochromator, (d) the sample holder and (e) the detector. We shall now discuss the details of these components.

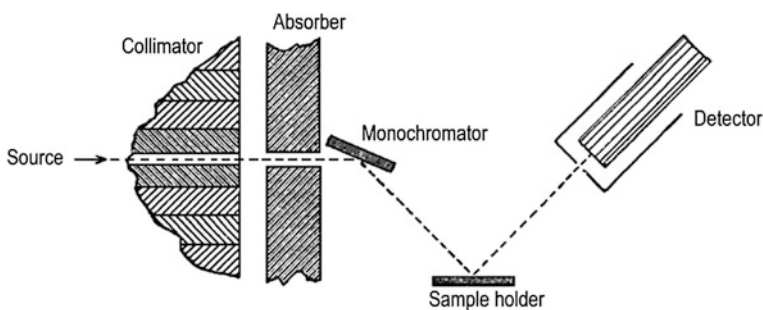


Fig. 10.9 Basic components of a neutron diffractometer

- (a) In pioneering work, the source of neutrons was a Ra-Be source. However, it is now completely superseded by a nuclear reactor.
- (b) Most materials are almost completely transparent to neutrons because of the uncharged nature of the neutrons. Lead, however, is a comparably good absorber. Hence collimation is achieved by long thick tubes or blocks of lead with a channel for the neutrons.
- (c) The wavelength of neutrons depends on the velocity through the De Broglie relation $\lambda = h/mv$. The neutrons produced in a reactor have a velocity distribution. In the reactor, the neutrons undergo collisions with the atoms of what is called a moderator in which their velocity gets reduced. Finally, they achieve equilibrium appropriate to the temperature of the moderator. This determines the mean velocity of the neutrons. For a temperature of ~ 320 K of the moderator, the mean velocity is such that the wavelength is ~ 1.4 Å; this velocity is suitable for diffraction of neutrons. Such neutrons are called thermal neutrons. But even thermal neutrons have a velocity spread. Hence, in neutron diffraction, monochromatization means selecting neutrons with some desired velocity.

In early experiments, a device called chopper (Fig. 10.10a) was used. It consisted of two discs made of an absorber each having a window; the two windows are in a line so that neutrons entering through one window can pass through the other. The discs are mounted on a shaft which can be rotated. Neutrons entering through the first window can pass through the second window only if their velocity is equal to l/t where l is the distance between the two discs and t the period of rotation of the chopper. Neutrons with other velocities are absorbed by the second disc.

However, nowadays, the phenomenon of diffraction of neutrons is itself used for monochromatization. If a crystal with its d value predetermined by X-rays is used, it will diffract only the wavelength that satisfies Bragg's law (Fig. 10.10b). Single crystals of Cu, Pb, Zn, Ge and pyrolytic graphite are employed for the purpose [7].

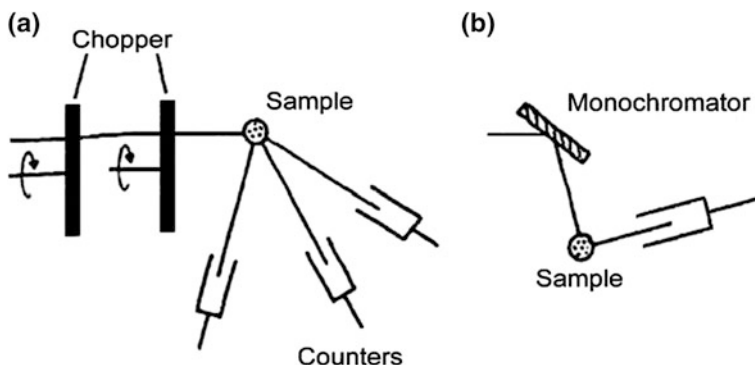


Fig. 10.10 Types of monochromators; **a** Chopper, **b** Crystal diffraction

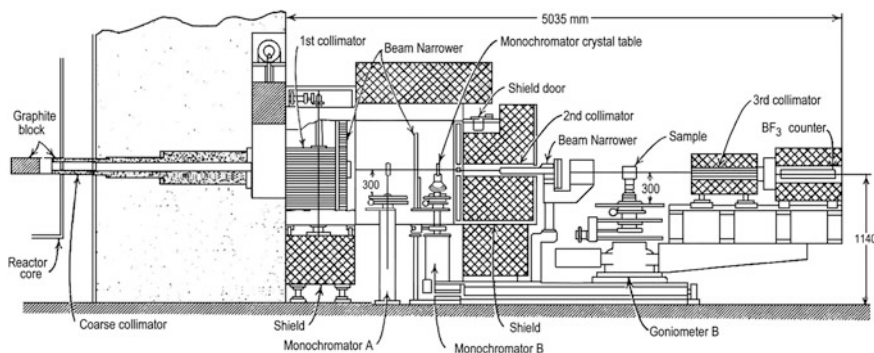


Fig. 10.11 Details of a neutron diffractometer

- (d) If the material is in the form of powder, the sample is in the form of a vanadium capsule containing the powder or a compressed cylinder. In view of the low absorption by the diffracting material, the size of the sample is large compared with that in X-ray diffraction. Typical size is 2.5 cm in diameter and 5 cm in length. Single crystal samples are mounted on a goniometer which has provisions for orienting the crystal.
- (e) The detector is generally a BF_3 counter mounted on a rotatable table. The actual equipment is much more elaborate. A lateral view is shown in Fig. 10.11. The figure is self-explanatory. Multiple collimation and multiple monochromatization is employed. Heavy shielding (cross-hatched regions in the figure) is used to avoid multiple scattering of neutrons in space. Note also the dimensions; neutron diffractometer occupies quite some space.

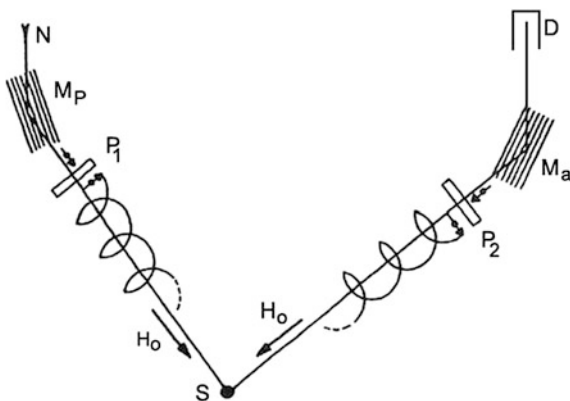
10.3.3 Polarized Neutrons

For some studies, a beam of polarized neutrons is used. The principle of a polarized neutron spectrometer is shown in Fig. 10.12. The incident beam of unpolarized neutrons is diffracted by a crystal which is magnetized in the vertical direction. The diffracted beam is partially polarized. If the neutrons are polarized in the same direction as the magnetized crystal (called the polarizer) the differential cross-section $\left(\frac{d\sigma}{\Omega}\right)_1$ is

$$\left(\frac{d\sigma}{\Omega}\right)_1 = (C - D)^2 \quad (10.1)$$

where C and D are the amplitudes of the nuclear scattering and magnetic scattering. On the other hand, if the neutron polarization is antiparallel to the magnetization of the polarizer, the differential cross-section $\left(\frac{d\sigma}{\Omega}\right)_2$ is

Fig. 10.12 Polarized neutron diffractometer. N unpolarized incident beam; M_p , M_a polarizer and analyzer, P_1 , P_2 $\pi/2$ spin-turn coils, S Sample, H_0 guide field, D detector



$$\left(\frac{d\sigma}{\Omega}\right)_2 = (C + D)^2 \quad (10.2)$$

If the polarizer crystal is chosen such that $C = D$, $\left(\frac{d\sigma}{\Omega}\right)_1$ vanishes and we get only a negatively polarized neutron beam. It is found that this condition is satisfied by the (111) plane of magnetite (Fe_2O_3) and the (200) plane of Fe-Co crystal.

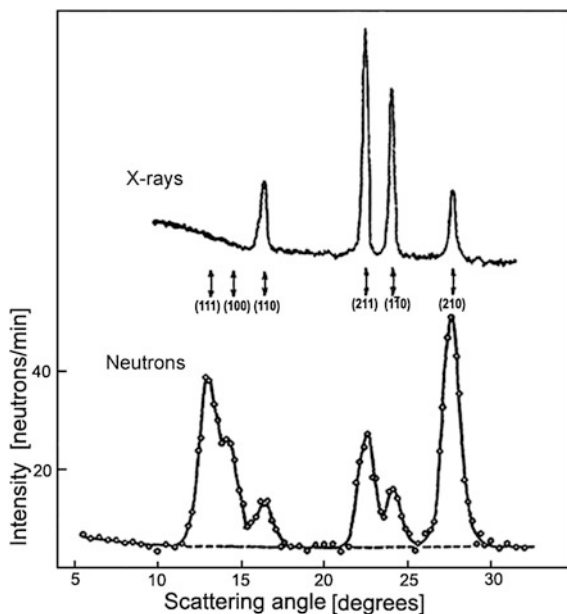
The diffracted beam then passes through a magnetic field parallel to the polarization of the neutrons. To reverse the polarization of the beam, the magnetic field of a r.f. field is employed which ‘flips’ the neutron. By using both the beams, nuclear scattering is almost completely eliminated and only magnetic scattering prevails.

The advantages in using polarized neutrons and also their applications are discussed by Hayter [8] and Satya Murthy and Paranjape [9].

10.3.4 Analysis of Neutron Diffraction Data

X-ray and neutron diffraction patterns for the powder samples of a magnetic material are shown in Fig. 10.13. Some reflections in the X-ray pattern are repeated in the neutron diffraction pattern. These represent the chemical crystal structure of the sample. The reflections in the X-ray pattern are caused by the scattering of X-rays by electrons and those in the neutron diffraction pattern by nuclear scattering. The extra lines occurring in the neutron diffraction pattern are due to the magnetic scattering by atoms with non-zero spins which are in an ordered state (the magnetic structure). In the discussion that follows, our emphasis will be on these extra reflections.

Fig. 10.13 X-ray and neutron diffraction patterns of a magnetic material



10.3.4.1 Lattice Constants

From the Bragg angles (θ) and the Miller indices (h, k, l) of the reflections, the lattice constants can be calculated from the relations:

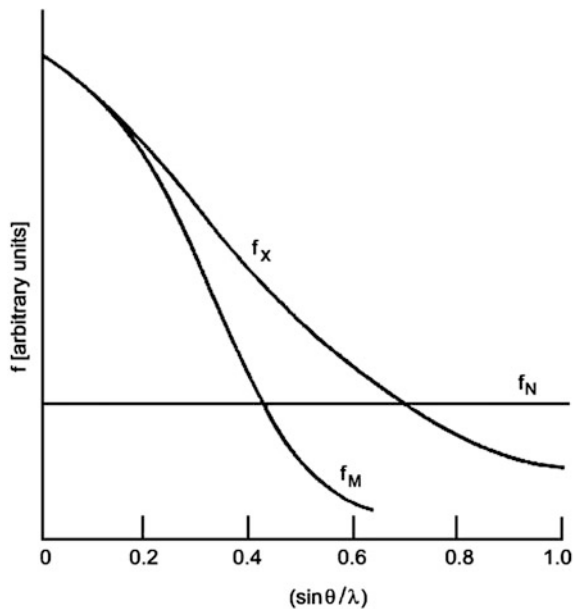
$$\left. \begin{aligned} \text{For cubic crystals : } \sin^2 \theta &= \left(\frac{\lambda^2}{4a^2} \right) (h^2 + k^2 + l^2). \\ \text{For tetragonal crystals : } \sin^2 \theta &= \left(\frac{\lambda^2}{4a^2} \right) (h^2 + k^2) + \left(\frac{\lambda^2}{4c^2} \right) l^2. \\ \text{For hexagonal crystals : } \sin^2 \theta &= \left(\frac{\lambda^2}{3a^2} \right) (h^2 + hk + k^2) + \left(\frac{\lambda^2}{4c^2} \right) l^2. \end{aligned} \right\} \quad (10.3)$$

Here a is the lattice constant for a cubic crystal and a and c are the lattice constants for tetragonal and hexagonal crystals. Similar relations for other crystal systems can be found in text books on X-ray diffraction.

10.3.4.2 Intensities

Intensities of the diffraction beams are the most important input for the determination of the structure. In order to understand the mechanism of neutron diffraction, it is useful to start with the mechanism of X-ray diffraction. X-rays are scattered by the electrons in an atom. The resultant of the amplitudes of waves scattered by all

Fig. 10.14 X-ray and nuclear and magnetic form factors of an atom as a function of $(\sin \theta/\lambda)$



the electrons in the atom is called the atomic scattering factor or the X-ray form factor f_X . In a similar way the neutrons are scattered by the nucleus. The form factor due to neutron-nucleus interaction is called the nuclear form factor and is denoted by f_N . The neutron has spin and an associated magnetic moment. Atoms with unpaired electrons (non-zero spin) also have a magnetic moment. The interaction between neutron magnetic moment and the atomic magnetic moment results in scattering which is called magnetic scattering. The form factor for this kind of scattering is denoted by f_M . Values of f_X , f_N and f_M are plotted against $(\sin \theta/\lambda)$ in Fig. 10.14; here θ is the Bragg angle. It may be noted that f_N is constant. Both f_X and f_M decrease with $(\sin \theta/\lambda)$; the decrease with $(\sin \theta/\lambda)$ is faster for f_M .

We shall now consider the concept of 'structure factor'. This is the ratio of the resultant amplitude of waves scattered by all atoms in the unit cell to the amplitude of the wave scattered by a single electron. For X-rays, the structure factor for the plane with Miller indices hkl is

$$F_X(hkl) = \sum_j f_{Xj} \exp i\phi_j. \quad (10.4)$$

Here ϕ_j is the phase; it is given by

$$\phi_j = 2\pi(hu_j + kv_j + lw_j) \quad (10.5)$$

where u_j , v_j , w_j are the coordinates of the j th atom in the unit cell. The coordinate is the actual Cartesian coordinate expressed as a fraction of the lattice constant.

Similarly, when neutron waves are diffracted, the nuclear structure factor has the form

$$F_N(hkl) = \sum_j f_{Nj} \exp 2\pi i(hu_j + kv_j + lw_j) \quad (10.6)$$

where f_{Nj} is the nuclear form factor for the j th nucleus. If the unit cell contains some atoms with non-zero spin and if the spins are in an ordered state, the neutrons are magnetically scattered and the magnetic structure factor for these waves is

$$F_M(hkl) = \sum_j f_{Mj} \exp 2\pi i(hu_j + kv_j + lw_j) \quad (10.7)$$

where u_j , v_j and w_j , are the coordinates of the j 'th atom with non-zero spin. Note that in (10.7), f_{Mj} can take positive or negative values depending on whether the spin of that atom is in the up-state or down-state.

Finally, the integrated intensity I_{hkl} of a Bragg reflection of scattered neutrons is given by

$$I_{hkl} = kI_0 A_{hkl} J_{hkl} e^{-2W} |F_{hkl}|^2 / \sin 2\theta \quad (10.8)$$

in which kI_0 is an instrumental constant, A_{hkl} the absorption correction, J_{hkl} the multiplicity, e^{-2W} the Debye-Waller correction for thermal motion and F_{hkl} is the structure factor [10]. In structure determination $|F_{hkl}|^2$ is taken as the intensity.

In neutron diffraction, if both nuclear scattering and magnetic scattering contribute to a reflection, the structure factor squared is given [11] as

$$|F_{hkl}|^2 = |F_N(hkl)|^2 + q^2 |F_M(hkl)|^2 \quad (10.9)$$

Here $q^2 = \sin^2 \theta$ where θ is the angle between the normal to the reflecting plane and the direction of magnetization of the sample. Thus $0 \leq q^2 \leq 1$. Special values of q^2 are 1 when the directions are perpendicular and 0 when they are parallel; $q^2 = 2/3$ for an unmagnetized (paramagnetic) sample. In the literature on neutron diffraction, $|F_N(hkl)|$ and $|F_M(hkl)|$ are denoted by C and D , a practice introduced by Halpern and Johnson [11].

In determining magnetic structures, the intensities are used in different ways; we shall discuss some of these.

- (i) In some structures, select reflections have zero intensity in the non-magnetic state and non-zero intensity in the magnetic state. These are extra reflections which appear in the neutron diffraction pattern of the magnetic sample. The mere presence or absence of these reflections throws light on the type of magnetic order.

As an example, let us consider a bcc lattice with identical atoms at the corner (0, 0, 0) and at the body-centre ($\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$). In the non-magnetic state, when only nuclear scattering contributes, the structure factor is

$$\begin{aligned} F_N(hkl) &= f_N[\exp 2\pi i(h \cdot 0 + k \cdot 0 + l \cdot 0) + \exp 2\pi i(h \cdot \frac{1}{2} + k \cdot \frac{1}{2} + l \cdot \frac{1}{2})] \\ &= f_N[1 + \exp \pi i(h + k + l)] \end{aligned} \quad (10.10)$$

Thus $F_N = 0$ for $(h + k + l)$ odd and $F_N \neq 0$ for $(h + k + l)$ even. Hence, the (100) reflection will be absent but (200) will be present. If the substance is in the ferromagnetic state, then the structure factor is similar with f_M substituting f_N i.e.

$$F_M(hkl) = f_M[1 + \exp \pi i(h + k + l)] \quad (10.11)$$

Now again, the (100) reflection will be absent and the (200) reflection will be present. On the other hand, if the substance is antiferromagnetic, the form factor of the waves from the corner atom and the body-centre will be different (say f_{M1} and f_{M2}) and the structure factor is

$$F_M(hkl) = f_{M1} + f_{M2} \exp \pi i(h + k + l). \quad (10.12)$$

It can be seen that the structure factor and hence the intensity for (100) is no longer zero. The presence of the (100) reflection leads to the conclusion that the substance is antiferromagnetic.

- (ii) If a clear indication of magnetic structure is not available from intensities of individual reflections, then, as done in X-ray structure determination, a spin arrangement is assumed and theoretical intensities are calculated from (10.6)–(10.9). The calculated intensities are compared with observed intensities. If there is over-all agreement, the assumed structure is taken as correct. Otherwise another structure is assumed and the process is repeated until we reach the correct structure.
- (iii) In X-ray diffraction, a Fourier synthesis of the intensities leads to an electron density distribution map from which atomic positions can be located. From the Fourier synthesis of the intensities of a magnetic crystal, information can be obtained about the spin orientations.

10.4 Examples of Magnetic Structures

10.4.1 General

Experimental results on the temperature variation of magnetic susceptibility clearly distinguish the various magnetic types. Some information on magnetic structures is provided by properties like magnetic anisotropy and magnetostriction. X-ray

diffraction provides information regarding the basic chemical structure. Clinching evidence about the magnetic structure i.e. the symmetric orderly alignment of spins is provided by neutron diffraction. The principle and practice of neutron diffraction has been discussed in the preceding section. Magnetic structures of a large number of crystals have been reported; we shall discuss a few of them.

10.4.2 Ferromagnetic Structures

The magnetic structures of iron and cobalt have been studied by Shull et al. [12]. Fe has the bcc structure (Fig. 10.15a). It has a Curie temperature of 1,043 K and so it is ferromagnetic at room temperature. The neutron diffraction pattern of Fe powder at room temperature is shown in Fig. 10.16. The reflections can be indexed on the basis of the bcc structure assuming the same lattice constant as derived from the X-ray diffraction pattern. There are no extra reflections. Assuming that the reflections are due to nuclear scattering, their intensities can be calculated from the theoretical nuclear form factors. The observed intensities are systematically larger than the calculated intensities. Obviously there is a strong magnetic contribution. From the difference between the observed intensities and the calculated intensities, the magnetic form factors can be estimated. These form factors are shown in Fig. 10.17 as a function of $(\sin \theta/\lambda)$ along with the nuclear form factor. The experimental magnetic form factor shows the same variation with $(\sin \theta/\lambda)$ as is observed in the theoretical magnetic form factor (Fig. 10.14), thus confirming that Fe is in the ferromagnetic state. The spin orientation in Fe is shown in Fig. 10.15b. It has been mentioned that from a Fourier synthesis of the intensities, the spin density distribution map (sddm) can be constructed. The sddm for Fe is shown in Fig. 10.18. It is seen that the spin density spreads more along [100] direction than along the [110] direction [13].

The results on Co are similar to those on Fe.

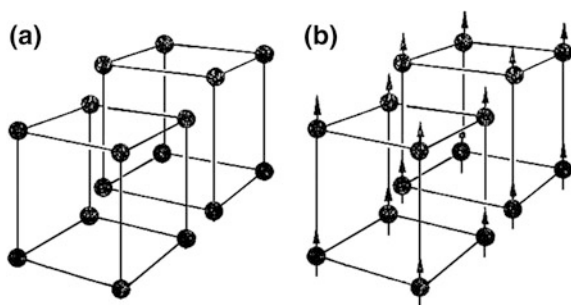


Fig. 10.15 Body-centred cubic structure of iron; **a** In paramagnetic state without ordered spins, **b** In ferromagnetic state with parallel ordering of spins

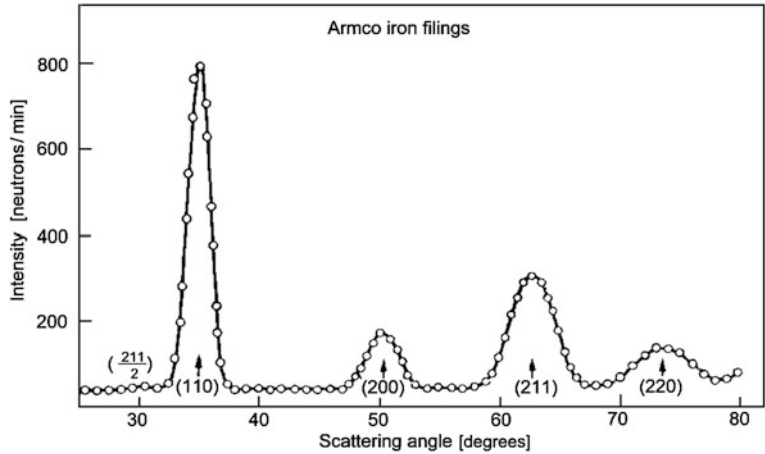


Fig. 10.16 Neutron diffraction pattern of ferromagnetic iron

Fig. 10.17 Magnetic scattering form factor (expressed as the differential cross-section) of iron as a function of $(\sin \theta/\lambda)$; *dashed curve* nuclear scattering form factor

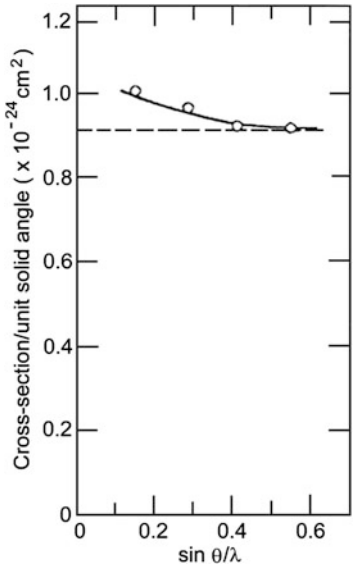
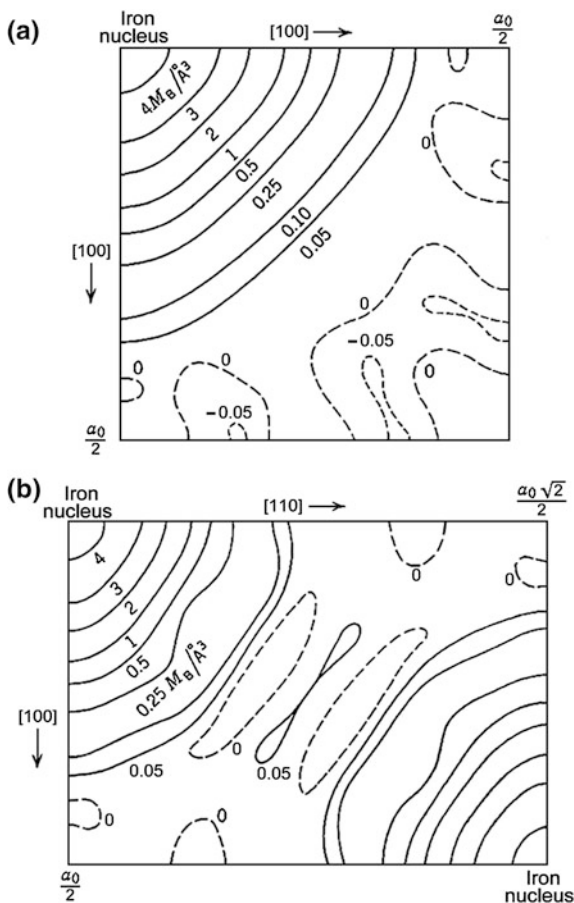


Fig. 10.18 Spin density distribution of iron **a** On (001) plane, **b** On (110) plane



10.4.3 Antiferromagnetic Structures

10.4.3.1 Transition Elements

Like Fe and Co, the transition metals chromium and α -manganese also have bcc structure (Fig. 10.19a). Magnetic susceptibility measurements show that they are antiferromagnetic with $T_N = 475$ and 100 K respectively.

Shull and Wilkinson [14] carried out neutron diffraction studies on Cr. The neutron diffraction pattern recorded at 20 K is shown in Fig. 10.20. The (110), (200), (211) reflections are due to nuclear scattering. It was pointed out in Sect. 10.3 that in antiferromagnetic bcc lattices, the (100) reflection should appear. This reflection is seen in the pattern. Further the intensity of the (100) reflection (Fig. 10.21) decreases with increasing temperature, finally approaching zero at $T \approx 475$ K. Also, the shape of the temperature versus intensity curve resembles a

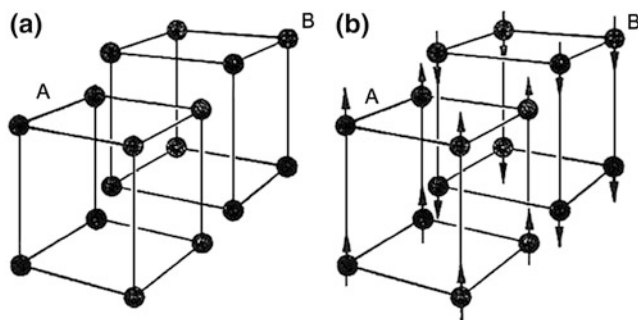


Fig. 10.19 Body-centred cubic structure of Cr and α -Mn; **a** In paramagnetic state without ordered spins, **b** In ferromagnetic state with antiparallel ordering of spins

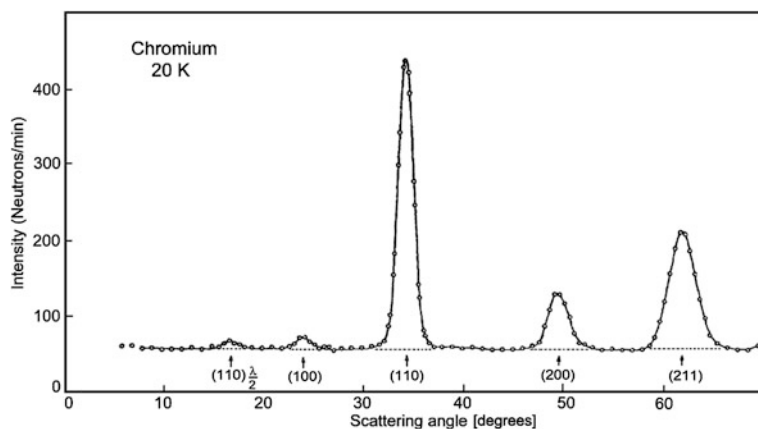


Fig. 10.20 Neutron diffraction pattern for Cr taken at 20 K; note the presence of the (100) magnetic reflection

Brillouin function. These features clearly indicate that the (100) reflection is due to antiferromagnetism in Cr. The spin arrangement is shown in Fig. 10.19b.

Shull and Wilkinson [14] also studied α -Mn. The neutron diffraction patterns recorded at 295 and 20 K are shown in Fig. 10.22. The reflections in the 295 K pattern are due to nuclear scattering. The pattern at 20 K also shows these reflections along with extra reflections due to antiferromagnetism. The intensities of the extra reflections decrease with increasing temperature and approach zero value at $T \approx 100$ K showing that they are related to antiferromagnetism. However, the (100) reflection which is seen in the Cr pattern in Fig. 10.20 does not show up in the 20 K pattern (Fig. 10.22). This suggests that the antiferromagnetism in Mn is weaker than that in Cr.

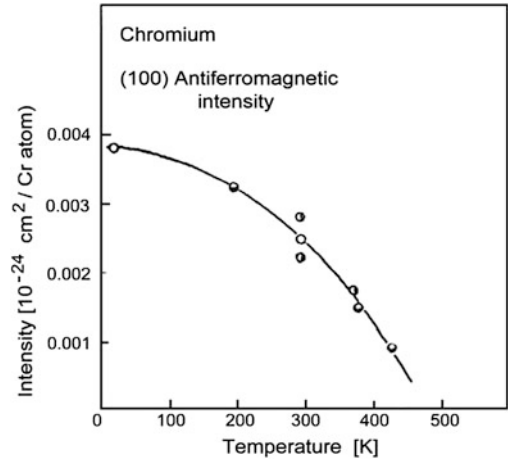


Fig. 10.21 Temperature variation of the intensity of the (100) magnetic reflection

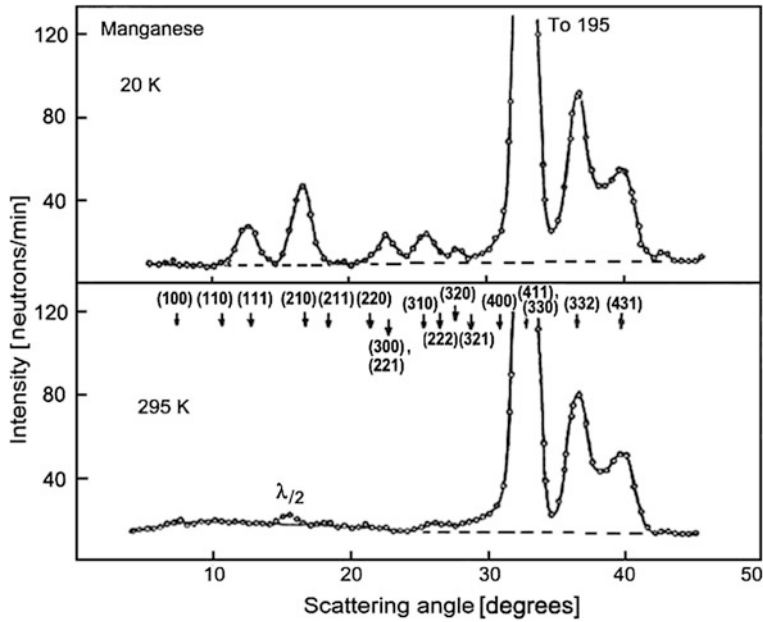


Fig. 10.22 Neutron diffraction patterns of α -Mn at 20 and 295 K

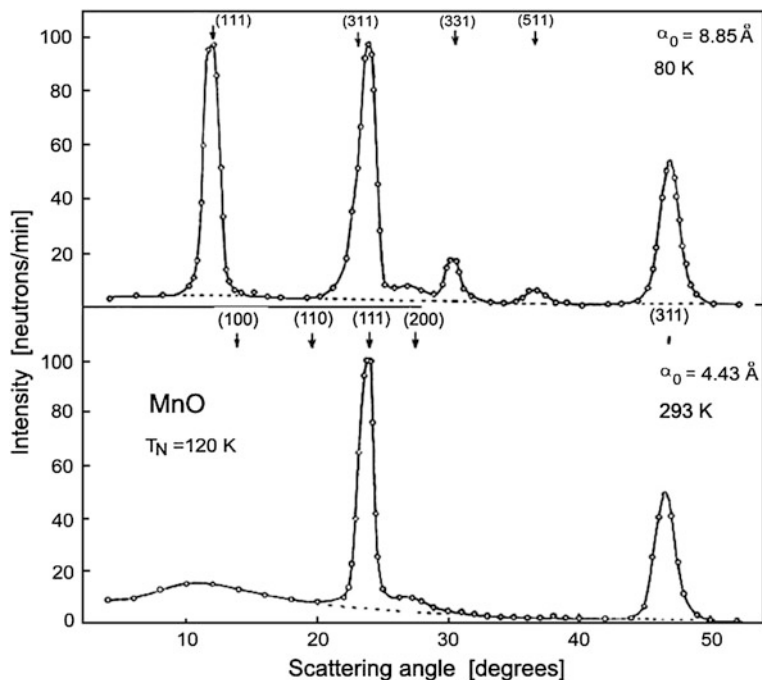


Fig. 10.23 Neutron diffraction patterns for MnO above and below the Neel temperature

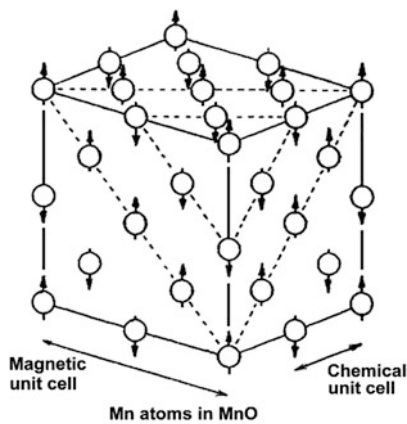
10.4.3.2 Transition Metal Oxides

Manganese oxide (MnO) is cubic with NaCl structure and a lattice constant $a = 4.45 \text{ \AA}$. It is antiferromagnetic with $T_N = 120 \text{ K}$. MnO was the first ever magnetic structure to be explored with neutron diffraction.

The neutron diffraction patterns of MnO recorded by Shull and Smart [15] at 293 K (i.e. above T_N) and at 80 K (i.e. below T_N) are shown in Fig. 10.23. The room temperature pattern is that expected from a crystal with NaCl structure having $a = 4.45 \text{ \AA}$. The X-ray pattern of MnO (not shown here) has strong hkl -all-even reflections and weak hkl -all-odd reflections. In contrast, the neutron diffraction pattern has weak hkl -all-even reflections and strong hkl -all-odd reflections. This is because the signs of nuclear scattering amplitudes are of opposite sign for the Mn and O ions.

Both the neutron diffraction patterns shown in Fig. 10.23 conform to the NaCl structure. However, there is a dramatic difference between the two. Whereas the room temperature (RT) pattern can be indexed with a lattice constant of $4.423 \text{ \AA}$, the low temperature (LT) pattern can be indexed with a lattice constant of $8.85 \text{ \AA}$. It is therefore concluded that while the RT pattern is caused by nuclear scattering, the LT pattern is caused by magnetic scattering from the spin moments of Mn ions in antiparallel orientation. The chemical unit cell and magnetic unit cell of MnO are

Fig. 10.24 Magnetic structure of MnO; note the magnetic unit cell edge which is twice the chemical unit cell edge



shown in Fig. 10.24. In the chemical lattice alternate points are occupied by Mn and O ions. In the magnetic lattice alternate points are occupied by spin-up and spin-down ions of Mn. The Mn–Mn distance in the chemical lattice is 4.42 Å. The Mn $\uparrow$ –Mn $\downarrow$ distance in the magnetic lattice is 8.85 Å (the arrow indicates spin-up).

The temperature variation of the intensity of (111) reflection of MnO is shown in Fig. 10.25. The intensity decreases with increasing temperature and tends to zero as the temperature tends to T_N . Further, the temperature versus intensity curve has the shape of a Brillouin function. This shows that the (111) reflection is due to the antiferromagnetic state. It may be mentioned that whereas the space group of MnO is $Fm3m$, its magnetic space group is $C_{2c}2/m$.

Shull et al. [16] investigated the orientations of the spins. They assumed three models: (a) magnetic moments aligned along [100] direction as shown in Fig. 10.24 with $q^2 = 2/3$, (b) magnetic moments perpendicular to the (111) sheets with $q^2 = 2/3$ for (111) reflection, 32/33 for (311) and 32/57 for (331) reflections and (c) magnetic moments aligned arbitrarily in the (111) sheets with $q^2 = 1$ for the (111) reflections and various values for the other reflections. The intensities of the magnetic reflections for these three models are given in Table 10.5 along with observed intensities. It is seen that only model (a) gives intensities agreeing with observed intensities.

Shull et al. [14] also studied FeO, NiO and CoO. Their neutron diffraction patterns are similar to that for MnO in the antiferromagnetic state. Thus their magnetic structure is similar to the one shown in Fig. 10.24.

Clifford Shull's experiment on MnO opened up the field of magnetic structures. He was awarded the Nobel Prize for physics in 1994.

Fig. 10.25 Temperature variation of the intensity of the magnetic reflection (111) of MnO

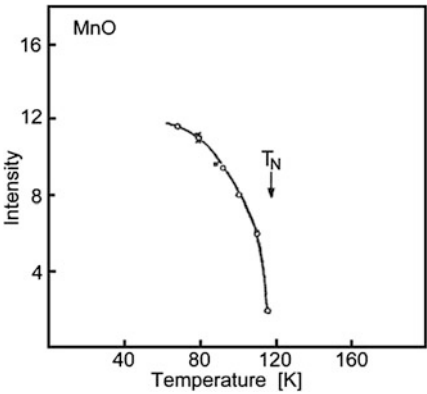


Table 10.5 Observed and calculated neutron diffraction intensities for MnO in antiferromagnetic state

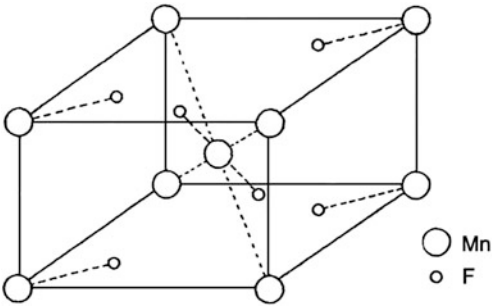
	Calculated for various models			Observed
	(a)	(b)	(c)	
(111)	1,038	0	1,560	1,072
(311)	460	675	–	308
(331)	129	109	–	132
(511), (333)	54	24	–	70

10.4.3.3 MnF₂ and Isomorphous Crystals

MnF₂ has the tetragonal rutile structure shown in Fig. 10.26. Magnetic susceptibility measurements show that MnF₂ is antiferromagnetic with $T_N = 75$ K.

The magnetic structure of MnF₂ was studied by Erickson [17]. The neutron diffraction patterns obtained by him at RT and at 23 K (LT) are shown in Fig. 10.27. The pattern at RT consists of reflections due to nuclear scattering by the chemical unit cell. The pattern at LT shows the same reflections (as in the RT

Fig. 10.26 Crystal structure of MnF₂



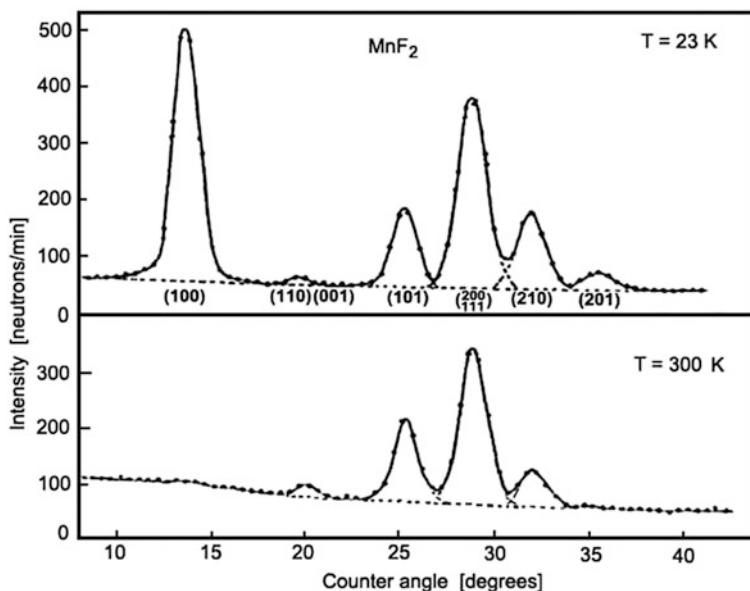
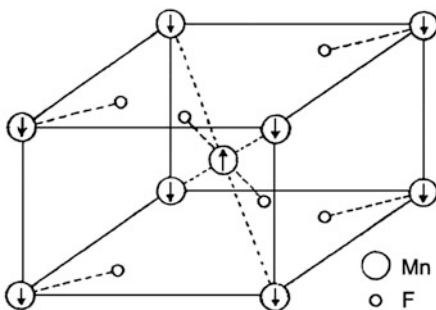


Fig. 10.27 Neutron diffraction patterns for MnF_2 in the paramagnetic state (300 K) and in the antiferromagnetic state (23 K)

pattern) and, in addition, some new reflections. The presence of the former shows that the chemical unit cell at LT is the same as that at RT. The presence of the latter reflections has to be attributed to the magnetic superlattice. If the magnetic lattice is the same as the chemical lattice, the new reflections can be indexed as (100), (111), (210) and (201). The chemical unit cell has two Mn^{2+} ions at (000) and $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. The spin moments of these two ions are antiparallel. The absence of the (001) reflection indicates that the spin moments are aligned along the c -axis.

The magnetic structure of MnF_2 is shown in Fig. 10.28. The environment of each Mn^{2+} ion can be described as two parallel neighbours at 3.31 Å along the c -axis, eight antiparallel neighbours at 3.81 Å, four parallel neighbours at 4.87 Å and

Fig. 10.28 Magnetic structure of MnF_2



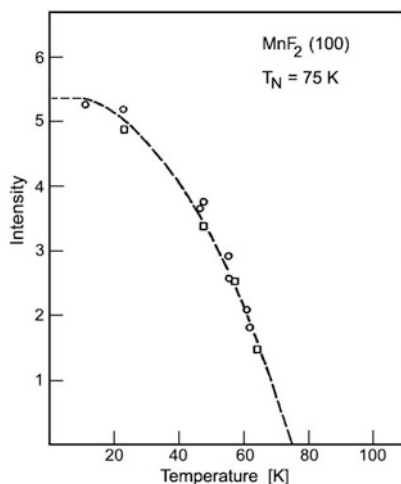


Fig. 10.29 Temperature variation of the (100) magnetic reflection of MnF_2

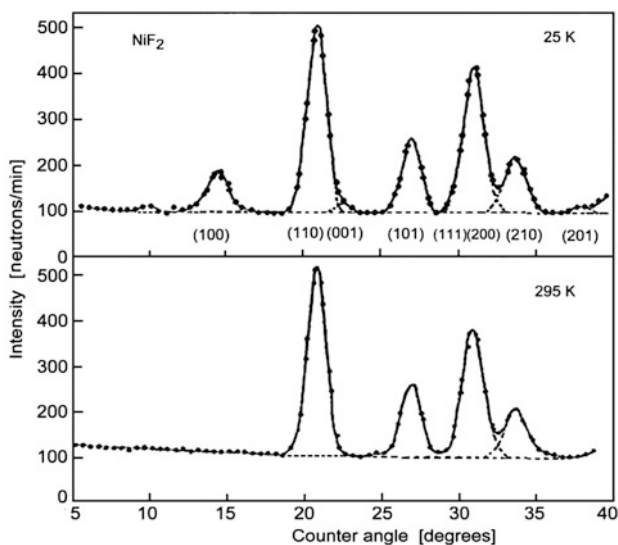


Fig. 10.30 Neutron diffraction patterns for NiF_2 at 295 and 25 K; note presence of magnetic reflection (001) in the low temperature pattern

eight antiparallel neighbours at $6.03 \text{ \AA}$. Thus, the spin ordering in MnF_2 is antiferromagnetic.

The temperature variation of the intensity of the (100) reflection is shown in Fig. 10.29. It decreases with increasing temperature and tends to zero at T_N . This confirms that the origin of the extra lines is in the antiferromagnetism of MnF_2 .

Isomorphous compounds CaF_2 and FeF_2 have been studied by Erickson [17] and have shown a similar magnetic structure as MnF_2 . In NiF_2 , which is otherwise isomorphous to MnF_2 , the (001) reflection (absent in MnF_2) appears with an unmistakable intensity (Fig. 10.30). This indicates that the spins are not exactly parallel to the c -direction but make an angle of 10° to the axis.

10.4.3.4 Fe_2O_3 (Hematite)

$\alpha\text{-Fe}_2\text{O}_3$ (Hematite) is trigonal with $a = 5.42 \text{ \AA}$ and $\alpha = 55^\circ 17'$. There are two molecules in the unit cell (Fig. 10.31). Note the Fe positions; these Fe ions lie in sheets in the (111) plane (or normal to the [111] direction). Magnetic susceptibility measurements by Morin [18] show that $\alpha\text{-Fe}_2\text{O}_3$ is antiferromagnetic with $T_N = 950 \text{ K}$.

Shull et al. [16] recorded X-ray diffraction pattern and neutron diffraction pattern of Fe_2O_3 at room temperature. These patterns are shown in Fig. 10.32. The reflections present in the X-ray pattern are also present in the neutron diffraction pattern indicating that these reflections are due to nuclear scattering. In addition, the neutron diffraction pattern contains lines which can be indexed as (111) and (100); since these are not present in the X-ray pattern, these must be due to magnetic scattering. In fact, as we shall see, much information can be extracted from these reflections.

To decide the magnetic structure of $\alpha\text{-Fe}_2\text{O}_3$, Shull et al. [16] calculated the intensities of the (111) and (100) reflections from different models. Designating the four Fe ions as A, B, C, D and a spin-up by + and a spin-down by –, the three assumed models are: (a) $+-+ -$, (b) $++--$ and (c) $+ - + -$. The calculated structure factors for the (111) and (100) reflections are zero for model (c); hence it is ruled out. The calculated intensities will depend not only on models (a) and (b) but also on the assumed orientations of the spins. Here, again, there are three possibilities. These are:

- I. The spin moments are directed along the unit cell edges.
- II. The moments are directed along the space diagonal of the unit cell and are thus perpendicular to the (111) Fe sheets.
- III. The moments are in the (111) sheets and directed towards one of the three nearest neighbours in the sheet.

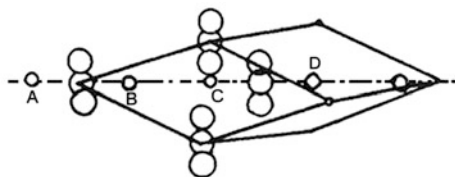


Fig. 10.31 Chemical structure of $\alpha\text{-Fe}_2\text{O}_3$; *small circles* represent Fe ions while *large circles* represent oxygen ions

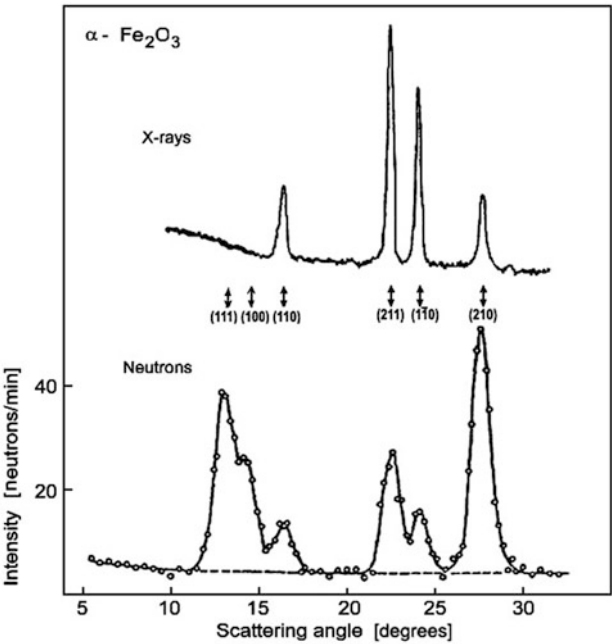


Fig. 10.32 X-ray and neutron diffraction patterns of $\alpha\text{-Fe}_2\text{O}_3$; innermost two reflections in the neutron diffraction pattern are antiferromagnetic in origin and are absent in the X-ray pattern

Table 10.6 Comparison between observed and calculated F^2 values for various magnetic structure models of $\alpha\text{-Fe}_2\text{O}_3$

$F^2 \rightarrow$	Calculated						Observed	
	Model (a)			Model (b)				
$hkl\downarrow$	(I)	(II)	(III)	(I)	(II)	(III)	300 K	80 K
(111)	1.25	0	4.3	0.23	0	0.81	4.9	<0.05
(100)	1.40	1.59	0.96	2.32	2.64	1.59	0.91	1.37

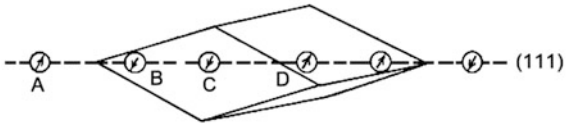


Fig. 10.33 Magnetic structure of $\alpha\text{-Fe}_2\text{O}_3$; only the iron ions are shown

Intensities were calculated for the two reflections at 300 and 80 K. They are given in Table 10.6 along with observed intensities. It is seen that RT data favours model (a) with spin orientation III whereas LT data favours model (a) with spin orientation II (Fig. 10.33).

10.4.3.5 LiFePO₄

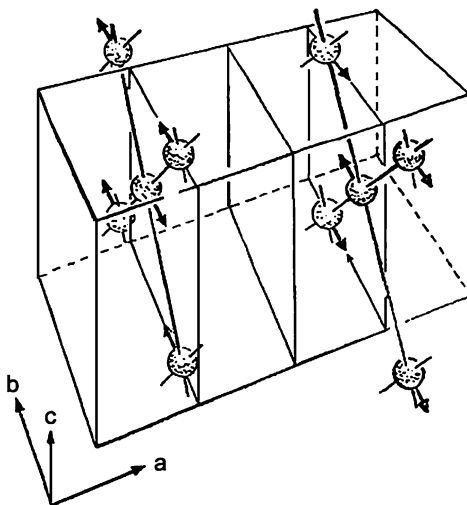
Magnetic susceptibility measurements, X-ray structure determination and magnetic structure determination of LiFePO₄ was carried out by Santoro and Newnham [19].

Magnetic susceptibility measurements made over the range 20–300 K showed that LiFePO₄ is antiferromagnetic with $T_N \approx 50$ K. X-ray powder diffraction studies at RT revealed that LiFePO₄ is orthorhombic with $a = 10.31$ Å, $b = 5.997$ Å and $c = 4.686$ Å; there are four molecules in the unit cell. Intensity analysis showed that LiFePO₄ has the olivine structure with space group $Pnma$.

Neutron diffraction patterns were recorded at 300 K (i.e. above T_N) and at 9 K (i.e. below T_N). The patterns at LT had reflections common with the X-ray pattern. The intensities calculated with X-ray-determined parameters agreed with observed intensities showing that there is no change in the chemical unit cell.

In the olivine structure, the Fe²⁺ ions have coordinates (0.28, $\frac{1}{4}$, 0.98), (0.72, $\frac{3}{4}$, 0.02), (0.22, $\frac{3}{4}$, 0.48) and (0.78, $\frac{1}{4}$, 0.52). Antiferromagnetic spin orientations $+- -+$ are assigned to the Fe ions. If the spins are assumed to be collinear with a or c , a large intensity is predicted for the (010) reflection. But this reflection is not observed in the recorded pattern. It was concluded that the spins are collinear with the b direction. With this assignment of spin, there is good agreement between calculated and observed intensities particularly for the strong (101) and (201) reflections. While the space group of the chemical structure is $Pnma$, the space group of the magnetic structure is $Pnma'$. The magnetic structure is shown in Fig. 10.34.

Fig. 10.34 Magnetic structure of LiFePO₄; only the iron ions are shown



10.4.3.6 CrSe, An Umbrella Structure

The magnetic structures that have been discussed so far were all collinear i.e. half the spins were oriented in one direction and the other half in the opposite direction. We shall now consider the magnetic structure of CrSe which is not collinear but has a novel spin ordering.

CrSe has the hexagonal NiAs structure (Fig. 10.35). The unit cell dimensions measured with X-rays are $a_0 = 3.674 \text{ \AA}$ and $c_0 = 6.001 \text{ \AA}$. The space group is $P6_3/mmc$. From susceptibility, it is concluded that CrSe is antiferromagnetic with $T_N = 300 \text{ K}$.

The magnetic structure of CrSe was investigated by Corliss et al. [20]. They recorded the neutron diffractions patterns at RT and 77 K (Fig. 10.36). The RT pattern could be indexed on the basis of the NiAs structure using nuclear scattering amplitudes. The 77 K pattern could be indexed only on the basis of a unit cell with

Fig. 10.35 Chemical structure of CrSe

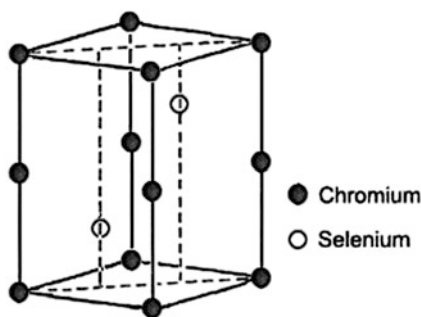
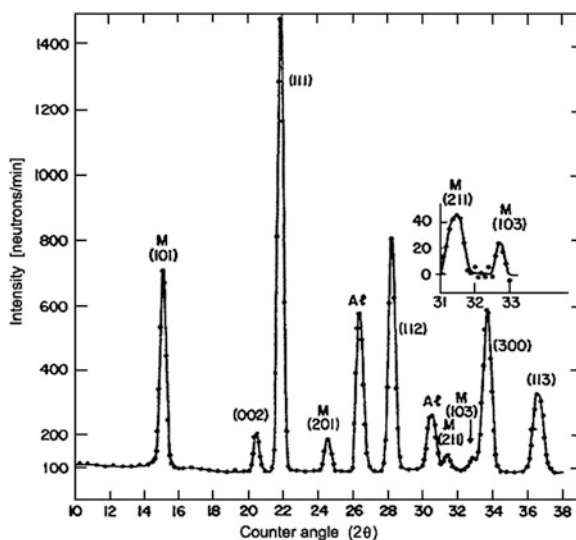


Fig. 10.36 Neutron diffraction pattern for CrSe at 77 K. The indexing is based on an enlarged cell. Magnetic reflections are indicated by symbol *M*. Inset shows enlargement of (211) and (103) reflections



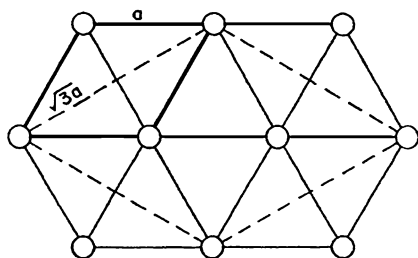
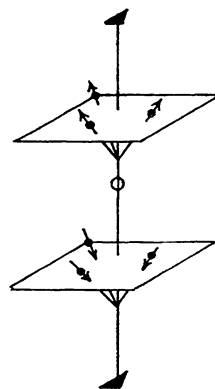


Fig. 10.37 Projection of unit cell of CrSe on basal plane. Chemical cell is shown by *dark lines* and magnetic cell by *dashed lines*

Table 10.7 Comparison of observed and calculated intensities for CrSe

hkl	$I_{\text{calc.}}$		$I_{\text{obs.}}$
	Collinear model	Umbrella model	
101	1,858	1,858	1,858
201	639	337	342
211	514	235	216
103	26	160	164

Fig. 10.38 Umbrella model for spin configuration in CrSe; the spins in each level have a 3-fold symmetry



$a = \sqrt{3}a_0$ and $c = \sqrt{3}c_0$ (Fig. 10.37). To interpret the magnetic reflections in the low temperature pattern, a collinear spin structure was assumed. Table 10.7 shows that the intensities calculated with this model do not agree with observed intensities. A model was then assumed in which the spins were not restricted to being parallel or antiparallel to a single direction. In this structure, planes parallel to the basal plane contain three chromium atoms whose spins form an ‘umbrella’ structure with three-fold symmetry. The intensities calculated from this model are in better agreement with observed intensities. The Shubnikov space group of this structure is $P\bar{6}2'm'$ (Fig. 10.38).

10.4.3.7 Helical Structures

Let us first describe a helical structure. Consider a hexagonal crystal consisting of layers perpendicular to the c -axis (Fig. 10.39). The spin moments in the top layers are shown to be along the a -direction. All other spins in this layer will be aligned in the same direction. Consider the next layer. The spins in this layer are parallel to each other but this direction makes angle α_0 with the spin direction in the upper layer. As we go down the c -axis, the direction of spin moments change from one layer to the next by the same angle α_0 . Such a structure is called a ‘helical’ structure. The angle of rotation of the spin from layer to layer is called the ‘turn angle’.

Let us consider the stability of a helical structure. We shall start with the top-most layer in Fig. 10.39. The spins in this layer are all in parallel alignment (ferromagnetic order). We shall denote the exchange integral within the layer by J_0 . We shall denote the exchange integrals for the interaction between the spin in the top layer and a spin in the next layer by J_1 , that with the next by J_2 and so on. The net exchange energy $(E_{\text{ex}})_0$ for the top layer is

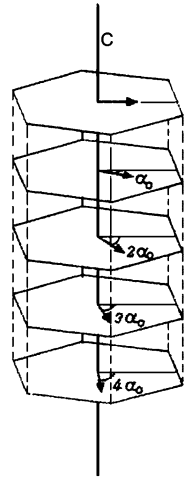
$$(E_{\text{ex}})_0 = -2S^2[J_0 + 2J_1 \cos \alpha_0 + 2J_2 \cos 2\alpha_0 + \dots] \quad (10.13)$$

where S is the spin. For N layers, the energy is

$$E_{\text{ex}} = -2S^2N[J_0 + 2J_1 \cos \alpha_0 + 2J_2 \cos 2\alpha_0 + \dots]. \quad (10.14)$$

It can be shown that this energy has a minimum for some value $(\alpha_0)_{\text{min}}$ of the turn angle i.e. the helical structure is stable when the turn angle is $(\alpha_0)_{\text{min}}$. Enz [21] showed that for a 3-layer system

Fig. 10.39 Helical spin configuration in a hexagonal crystal



$$(\alpha_0)_{\min} = -J_1/4J_2. \quad (10.15)$$

In general, $(\alpha_0)_{\min}$ has values other than 0 and π .

We shall now discuss the first helical structure that was observed viz., the magnetic structure of MnAu_2 . MnAu_2 is antiferromagnetic with $T_N \approx 363$ K. The chemical crystal structure of MnAu_2 is shown in Fig. 10.40a wherein the Mn positions are shown. X-ray studies showed that MnAu_2 has the tetragonal body-centred arrangement of Mn atoms with $c/a = 2.6$. Each Mn atom has 2 Au atoms at the same distance above and below the Mn atom. The Mn atoms are in layers perpendicular to the c -direction, the interlayer distance being $c/2$.

The neutron diffraction patterns of MnAu_2 recorded by Herpin and Meriel [22] at 363 and 293 K are shown in Fig. 10.41a, b. The (002) and (101) peaks observed at 363 K are due to chemical (nonmagnetic) state due to nuclear scattering. These peaks are also present in the LT pattern. But there are two extra peaks on either side of (002). Also, the (101) peaks appear to have broadened. By taking the difference of a and b, the broad peak (101) also shows up two peaks on either side of (101) as can be seen in Fig. 10.41c. The presence of these peaks could be explained by a helical arrangement of spin moments along the c -axis with a turn angle of $\alpha_0 = 51^\circ$ per layer (or 102° /unit cell) along c . This spin arrangement is shown in Fig. 10.40b.

Helical magnetic structures have been observed in MnO_2 , CrSb , CuCr_2O_2 and MnCr_2O_4 .

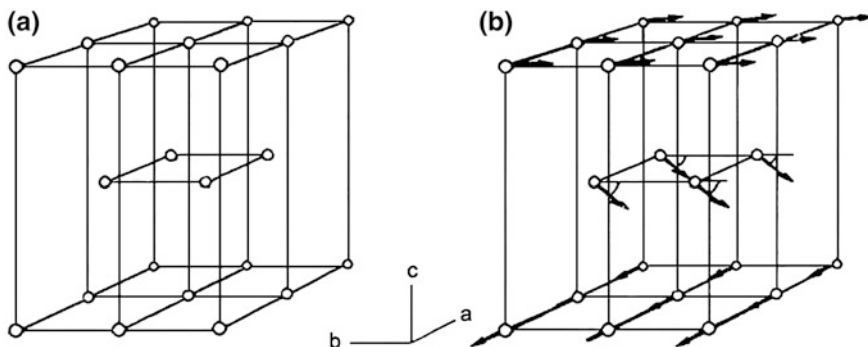
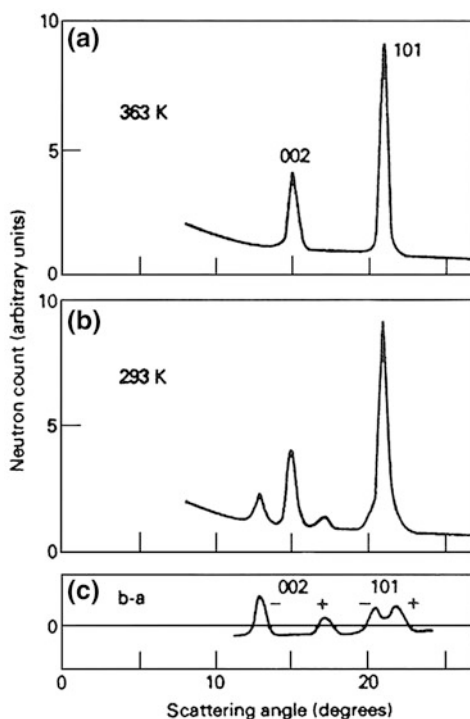


Fig. 10.40 **a** Chemical structure of MnAu_2 , **b** Helical magnetic structure of MnAu_2 (only Mn positions are shown)

Fig. 10.41 Neutron diffraction patterns for MnAu_2
a At 363 K and **b** At 293 K,
c Difference of **(b)** and **(a)**



10.4.4 Ferrimagnetic Crystals

10.4.4.1 Magnetite

Magnetite (Fe_3O_4) is a ferrite with the spinel structure. One fourth of the unit cell is shown in Fig. 10.42. Half the Fe^{3+} ions are located at the tetrahedral sites. The octahedral sites are occupied by all the Fe^{2+} ions and the remaining half of the Fe^{3+} ions. Magnetic susceptibility measurements show that Fe_3O_4 is ferrimagnetic with $T_{\text{FN}} = 858 \text{ K}$.

X-ray diffraction and neutron diffraction patterns at RT recorded by Shull et al. [16] are shown in Fig. 10.43. The chemical structure is the same in both patterns. From the nuclear positions known from X-ray structure determination, we can calculate the nuclear scattering contribution to intensities using nuclear scattering amplitudes. Similarly, based on an assumed model of spins, we can calculate the contribution of magnetic scattering using appropriate magnetic form factors. The sum of the two contributions is the calculated intensity for a reflection.

The ferrimagnetic structure assumed for Fe_3O_4 is shown in Fig. 10.44. The Fe ions in the tetrahedral positions have their spin moments pointed upwards whereas those in the octahedral sites have spin moments pointed downwards. These being unequal, the substance behaves like a ferrimagnet. The calculated and observed

Fig. 10.42 Chemical structure of Fe_3O_4 (one fourth of unit cell)

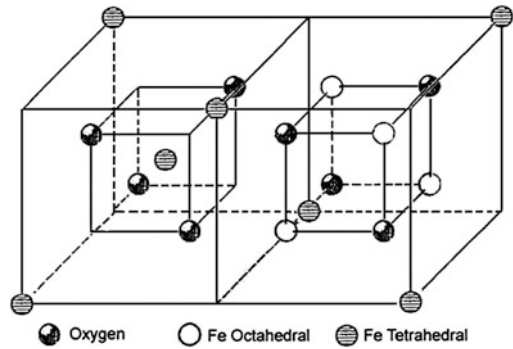


Fig. 10.43 X-ray and neutron diffraction patterns for Fe_3O_4

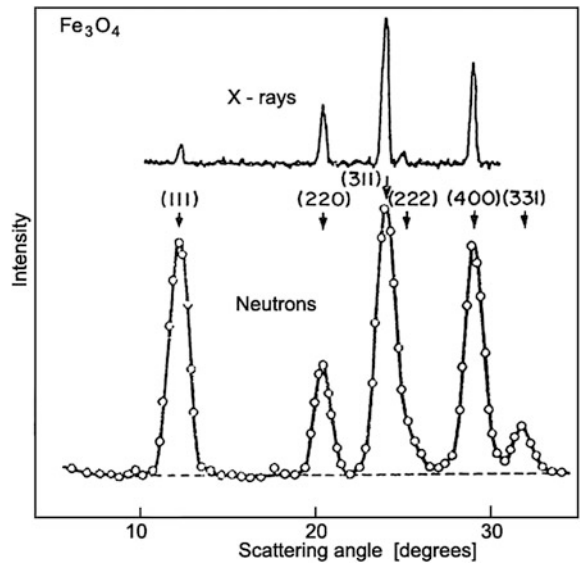


Fig. 10.44 Magnetic structure of Fe_3O_4 (one fourth of unit cell)

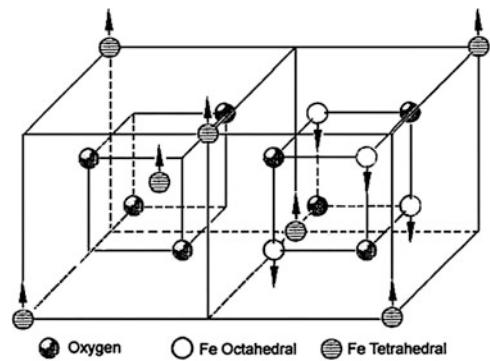


Table 10.8 Comparison between observed and calculated intensities for Fe_3O_4

Reflection	Calculated intensities			Observed intensity
	Magnetic	Nuclear	Total	
(111)	902	32	934	860
(220)	125	218	343	360
(311) } (222) }	112	948	1060	1070
(400)	116	649	765	780
(331)	94	116	110	135
(422) } (333) } (511) }	20	670	690	700
(440) } (531) }	32	1692	1724	1730

intensities are given in Table 10.8. The agreement is satisfactory. This model was adopted after other models were found to give unsatisfactory intensities.

10.4.4.2 Copper Chromite

Copper chromite (CuCr_2O_4) is tetragonal with a slightly distorted spinel structure (Fig. 10.45). The divalent Cu ions are at the A sites and the trivalent Cr ions are at the B sites. Magnetic susceptibility measurements show that the compound is ferrimagnetic with a Neel temperature $T_{\text{FN}} = 135$ K and a magnetic moment of $\sim 0.4 \mu_{\text{B}}$ /molecule.

The neutron diffraction patterns obtained by Prince [23] at RT and LN temperature are shown in Fig. 10.46. The low temperature pattern shows the (101) and (002) reflections which are purely magnetic. The magnetic reflections as well as the nuclear reflections can be indexed on the basis of the chemical cell. On the other hand, the (002) reflection which is present with unmistakable intensity is not permitted by the space group of the chemical cell because of the d -glide operation. The appearance of this reflection can be explained as due to nonparallel spins of Cr ions related by the glide operation. There is no contribution of magnetic scattering to the (110) and (112) reflections which means that Cr ions in the same sheet have parallel spins. These facts indicate that the Cr ions in any (001) sheet are parallel and those in adjacent sheets make angles with each other. The $\sim 0.5 \mu_{\text{B}}$ magnetic moment of a molecule can be explained by assuming one of the two possible arrangements of the Cr ions shown in Fig. 10.47, two ions having moment of $3 \mu_{\text{B}}$ and one ion having a moment of $1 \mu_{\text{B}}$. Thus the spin configuration in copper chromite crystal is ferrimagnetic.

Fig. 10.45 CuCr_2O_4 structure projected on plane perpendicular to $[001]$

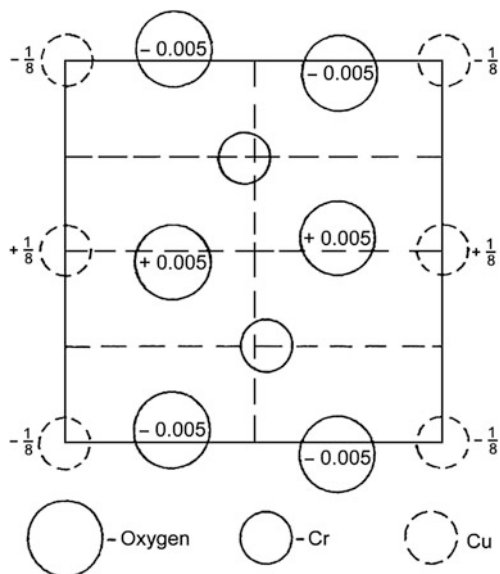
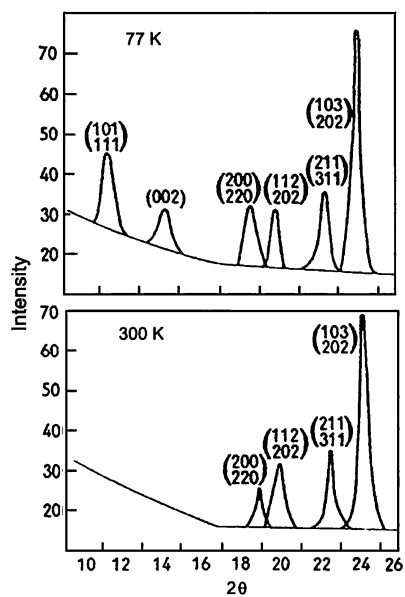


Fig. 10.46 Neutron diffraction patterns for CuCr_2O_4 at 300 and 77 K



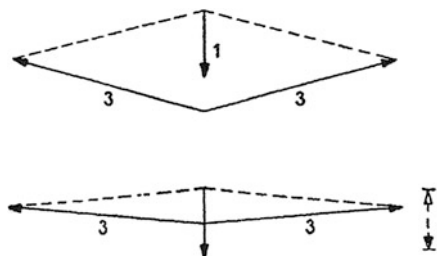
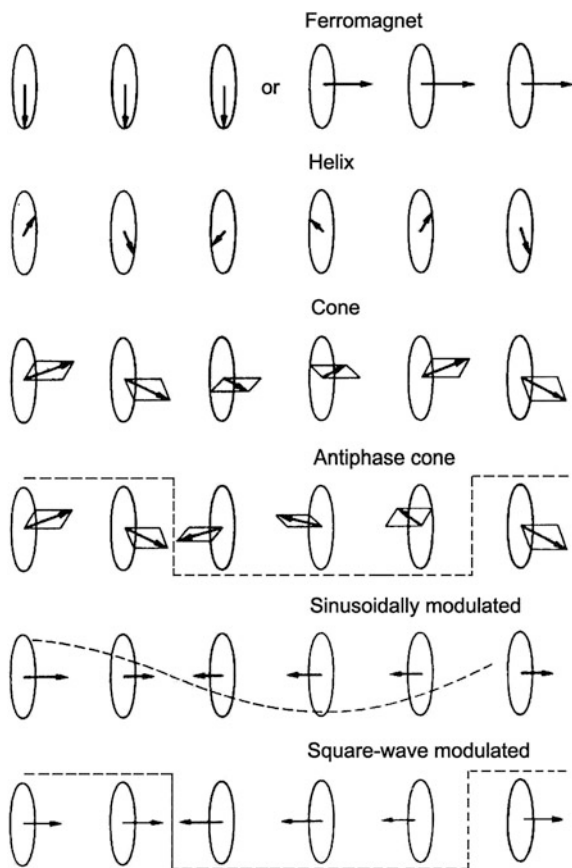


Fig. 10.47 Two possible ways of arranging moments of Cr ions to yield observed moment of $0.5 \mu_B/\text{molecule}$

10.4.5 Rare Earth Metals

Although the rare earth metals have the same electronic structure (unpaired electrons in the $4f$ shell) they display a variety of exotic magnetic structures. At a given

Fig. 10.48 Variety of magnetic structures found in rare earths



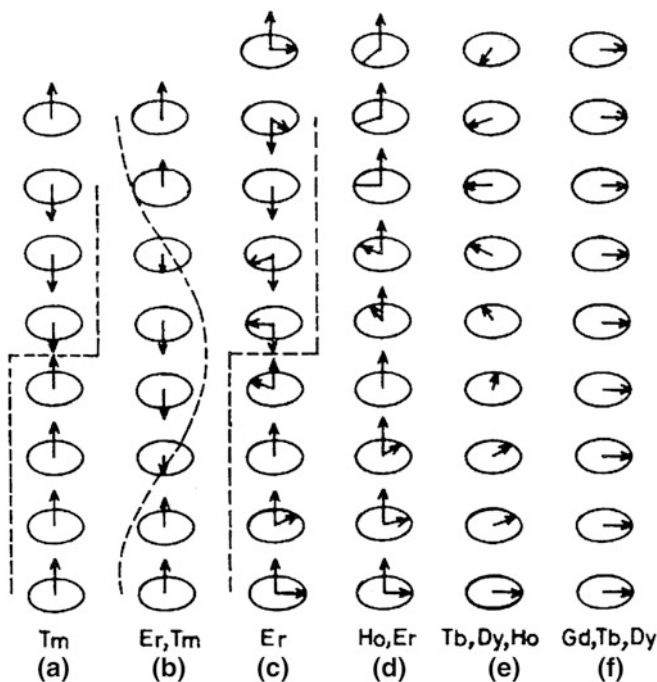


Fig. 10.49 Magnetic structures of specific rare earth metals

temperature, different rare earths have different magnetic structures. Further, a given rare earth metal has different magnetic structures in different temperature ranges. A detailed discussion of the magnetic structures of rare earth metals is given by Koehler [24], Kubo and Nagamiya [25] and Sinha [26]. Since we are now well-acquainted with the procedure of determining the magnetic structure, we shall only summarize the results on magnetic structures of rare earth metals.

The different types of magnetic structures found in the rare earth metals are shown in Fig. 10.48. The magnetic structures assumed by specific rare earth metals are shown in Fig. 10.49. The observations on magnetic structures of rare earth metals are summarized in Table 10.9.

Tm (Fig. 10.49a) has a seven-layer repeat distance (3 layers of north-pointing moments and 4 layers of south-pointing moments). An oscillatory moment is observed in Er (Fig. 10.49b) in which the z -components of the moments in a hexagonal layer are parallel to each other and the magnitude of this component varies sinusoidally. At 53 K, Er displays type c (Fig. 10.49), where several layers with moments pointing up are followed by an equal number of layers with moments pointing down. Ho and Er also display type d (Fig. 10.49) in which the perpendicular component of the moment shows a helical variation. Tb, Dy and Ho display the helical structure shown in Fig. 10.49e. At different temperature ranges, Gd, Tb and Dy transform to the ferromagnetic structure (Fig. 10.49f).

Table 10.9 The magnetism and crystal structure of rare-earth metals (μ is magnetic moment)

Metal	Crystal structure	Magnetism	Remarks
Ce	Hexagonal ABAC type ($c/2a = 1.63$); face-centred cubic	A hexagonal ABAC type is antiferromagnetic with $T_N = 12.5$ K	Highly dense nonmagnetic face-centred cubic structure (no f electrons)
Pr	Hexagonal ABAC type ($c/2a = 1.62$)	Antiferromagnetic with $T_N = 25$ K. Sine structure in b_1 direction	Either the spins on the BC planes are ordered, or those on the A, A planes
Nd	Hexagonal ABAC type ($c/2a = 1.61$)	Antiferromagnetic with $T_N = 19$ K. Sine structure in b_1 direction	Above 7.5 K only the B, C-plane spins are ordered; below 7.5 K those in or those on A, A are ordered also
Gd	Hexagonal close-packed ($c/a = 1.59$), the coefficient of expansion c below T_c is large if negative, whereas a is small if positive	Ferromagnetic, $T_c = 289$ K 289–245 K: $\mu // c$ axis; 245–225 K: continuous change from $\mu // c$ to $\mu \perp c$ 225–165 K: $\mu \perp c$ axis; 165–0 K: continuous change from $\mu \perp c$ to angle $\angle(\mu, c) = 34^\circ$	
Tb	Hexagonal close-packed ($c/a = 1.58$)	Spiral structure from $T_N = 228$ to 221 K and $\mu \perp c$; ferromagnetic below 221 K; $\mu // b$ to μ (0 K) = $9.34 \mu_B$	
Dy	Hexagonal close-packed ($c/a = 1.57$), $c/a = 1.59$ at low temperatures	Spiral structure from $T_N = 179$ to 85 K, $\mu \perp c$; ferromagnetic below 85 K; $\mu // a, \mu$ (0 K) = $10.0 \mu_B$	
Ho	Hexagonal close-packed ($c/a = 1.57$)	Spiral structure from $T_N = 132$ to 20 K, $\mu \perp c$; conic structure below 20 K; moment in the c axis direction 117μ ; rotational component perpendicular to the axis $10.2 \mu_B$	
Er	Hexagonal close-packed ($c/a = 1.57$), at low temperatures $c/a = 1.575$	Sine structure parallel to the c axis from $T_N = 85$ to 53 K; components parallel to and perpendicular with the c axis vibrate from 53 K to 20 K; conic structure below 20 K; $\mu // c = 8, \mu$ (0 K) = $\mu_B, \mu \perp c = 4 \mu_B$	
Tm	Hexagonal close-packed ($c/a = 1.57$)	Sine structure parallel to the c axis below $T_N = 56$ K; gradually start vibrating circularly from 40 K down, then ferrimagnetic with +++++ as the period; each atom has μ (0 K) = $7 \mu_B$	

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Chapter 11

Magnetic Resonance

Abstract Magnetic resonance (NMR and ESR) based on the Stern–Gerlach experiment and the Rabi experiment has been extensively applied to the solid state. After a brief introduction, general theoretical principles are built around four concepts, namely, Larmor precession, magnetization, complex susceptibility and spin Hamiltonian. Experimental techniques of NMR, namely, continuous wave (CW) NMR, free induction decay, pulsed solid state NMR and the magic angle spinning technique are discussed focusing on determination of spin-lattice relaxation time studies. Superconducting phase transition, Knight shift measurements, NMR diffraction and imaging are discussed. This is followed by ESR theory and a complete analysis of ESR spectrum, CW and pulsed ESR experimental techniques and two case studies namely microsymmetry and crystal field effects in barium titanate and microwave penetration depth determination in high-temperature superconductors. Two current trends in ESR namely Si quantum computer and electrically detected magnetic resonance in Si thin film solar cells are briefly discussed.

11.1 Introduction

11.1.1 General

This chapter focuses on the basics, theory, experimental aspects and applications of nuclear magnetic resonance (NMR) and electron spin resonance (ESR). When applied to crystalline solids that possess magnetic nuclei and unpaired electron spins, the techniques of magnetic resonance yield nuclear and electronic spin density distribution besides elucidating a host of solid state phenomena.

11.1.2 Spins of Atoms, Electrons and Nuclei

In chapters on magnetism, we considered the response of electrons and ions in solids to static magnetic fields. In this chapter we shall consider the response of electrons, magnetic ions as well as nuclei to applied alternating magnetic fields.

The general relationship [1] between the magnetic moment μ and the angular momentum J of an atom is

$$\mu = \gamma \hbar J. \quad (11.1)$$

Here γ is called the gyromagnetic ratio. Since we are going to deal with electrons and also nuclei we shall use the subscript 'e' for quantities in (11.1) to refer to electrons. Thus (11.1) may be written as

$$\mu_e = \gamma_e \hbar J. \quad (11.2)$$

Equation (11.2) may also be expressed as

$$\mu_e = -g_e \beta_e J. \quad (11.3)$$

We have from (11.2) and (11.3)

$$g_e \beta_e = -\gamma_e \hbar \quad (11.4)$$

where β_e is the Bohr magneton defined as

$$\beta_e = e \hbar / 2mc. \quad (11.5)$$

It can be easily recognized as the magnetic moment of the electron in the $n = 1$ orbit in Bohr's atomic theory. Magnetic moments are conveniently expressed in units of β_e whose value is 9.2741×10^{-21} erg/gauss. g_e in (11.3) is another important quantity called Lande's level splitting factor, defined as

$$g_e = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)} \quad (11.6)$$

where L is the orbital angular momentum and S the spin angular momentum. g_e takes values 1 for $S = 0$ and 2 for $L = 0$ but has other values given by (11.6) for $L \neq 0$, $S \neq 0$. It may be recalled (Chap. 1) that the actual value of J in (11.3) is $\sqrt{J(J+1)}$. The projection of J in the direction of a magnetic field H is M_J . Because of space quantization M_J has the set of $(2J+1)$ values $-J, -(J-1), \dots, (J-1), J$. The magnetic moment in the direction of the magnetic field is

$$(\mu_e)_J = -g_e \beta_e M_J. \quad (11.7)$$

The nucleus also has a spin I which bestows a magnetic moment μ_n on the system. For the nuclear magnet quite analogous to the electron magnet, the nuclear magnetic moment μ_n is given by

$$\mu_n = \gamma_n \hbar I \quad (11.8)$$

where γ_n is the nuclear gyromagnetic ratio. Analogous to (11.3) for the electron we have for the nucleus

$$\mu_n = g_n \beta_n I \quad (11.9)$$

where β_n is the nuclear magneton defined as the ratio $e\hbar/2M_p c$ where M_p = proton mass. β_n has the value 5.0509×10^{-24} erg/gauss. The value of β_n/β_e is 1830.1, the same as the value of M_p/m . The splitting factor for nuclear spin is g_n and its value is 5.9. Again, the actual value of I in (11.8, 11.9) is $\sqrt{I(I+1)}$.

Thus we have defined the following parameters for the electron magnet and the nuclear magnet:

Electron: $\mu_e, \beta_e, \gamma_e, g_e$

Nucleus: $\mu_n, \beta_n, \gamma_n, g_n$.

The magnetic moments of some nuclei of relevance to nuclear magnetic resonance are given in Table 11.1.

The gyromagnetic ratio forms the basis of the two following experiments that set the stage for magnetic resonance of the solid state.

11.1.3 Two Molecular Beam Magnetic Resonance Experiments

11.1.3.1 Stern–Gerlach Experiment

This pioneering molecular beam magnetic resonance experiment [2, 3] paved the way for magnetic resonance experiments which rely on the concept of precession of atomic spin (electron or nuclear) in an externally applied magnetic field. In this experiment which firmly established the spatial quantization of intrinsic spin angular momentum, a beam of atomic silver particles are passed through a non-uniform magnetic field created by a magnet with a very sharp edge on one of the pole tips (Fig. 11.1). When the beam of atoms was directed along this sharp edge the atoms would feel a vertical force in the inhomogeneous field. An atom, with its magnetic moment horizontal, would experience no force and go straight past the magnet. An atom with its magnetic moment pointed upward would experience an upward force toward the sharp edge. On the other hand, an atom with its magnetic moment pointed downward would experience a downward force. Stern and Gerlach found two distinct spots on the glass plate detector: the silver atoms had formed two beams. From a measurement of the deflection of the beams, the strength of the magnetic moment of silver could be deduced. A modern discussion of the experiment in the context of quantum measurement is given in [4]. Indeed this experiment enables us to see the relation between the postulate of dynamics of

Table 11.1 Magnetic moments and resonance frequencies for selected NMR nuclei

Nucleus	Spin	Magnetic moment (in units of β_n)	NMR frequency (MHz) for 1 T magnetic field
Neutron	1/2	−1.9135	29.1
^1H	1/2	2.7935	42.6
^7Li	3/2	3.2571	16.5
^{17}O	5/2	−1.893	5.772
^{19}F	1/2	2.627	40.08
^{23}Na	3/2	2.2178	11.3
^{25}Mg	5/2	−0.855	2.605
^{27}Al	5/2	3.6419	11.1
^{29}Si	1/2	0.555	8.45
^{31}P	1/2	1.131	17.2349
^{39}K	3/2	0.391	1.986
^{55}Mn	5/2	3.468	10.50
^{59}Co	7/2	4.616	10.05
^{63}Cu	3/2	2.2266	11.3
^{65}Cu	3/2	2.3850	12.1
^{35}Cl	3/2	0.8222	4.2
^{87}Rb	3/2	2.7506	13.93
^{89}Y	1/2	−0.137	2.086
^{91}Zr	5/2	−1.30284	3.9725

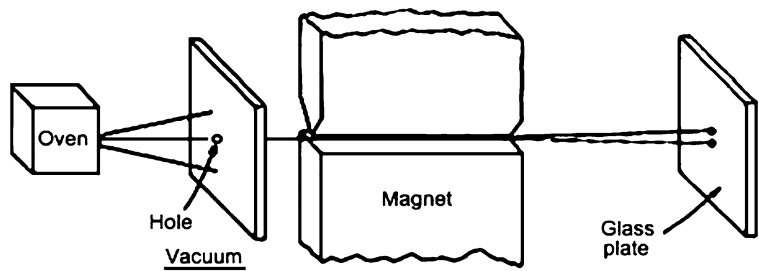


Fig. 11.1 Stern-Gerlach experiment (schematic) demonstrating spatial quantization of electron spin and enabling the first quantum measurement of magnetic moment

Schrödinger equation and the postulate of quantum measurement. Practically speaking, the apparatus of Stern and Gerlach is actually a quantum mechanical state selector, or, beam polarizer. The separated beams emerging from the region of the field gradient are polarized, each characterized by the component of the total

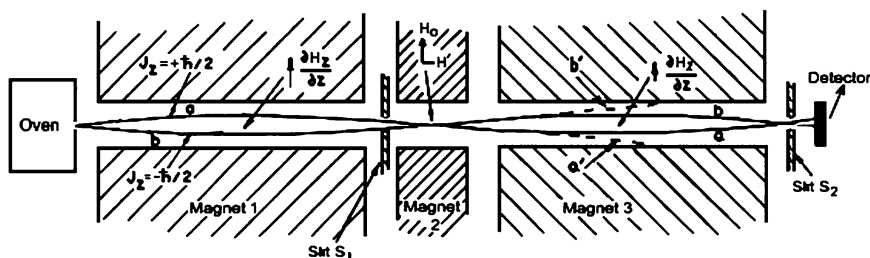


Fig. 11.2 The Rabi molecular beam resonance apparatus for measuring magnetic moments

J vector (m_j). The apparatus may be reversed functionally so that it works as an analyzer. It motivated Rabi and colleagues to perform the following experiment.

11.1.3.2 Rabi Experiment

In their trail-blazing experiment Rabi et al. [5] showed that electromagnetic waves with the right frequency could flip magnetically aligned nuclei from one state to the other. Photons at this special nuclear resonance frequency had just the right energy to boost nuclei from the lower energy state to the higher energy state. They experimented on beams of nuclei passing through magnetic fields as depicted in Fig. 11.2.

Magnet 1 has a field with a strong and positive field gradient while magnet 2 has a uniform field. Magnet 3 is just like magnet 1 but with the field inverted that is a negative field gradient. Atoms coming out of the oven with $J_z = +h/4\pi$ go along curve a and those with $J_z = -h/4\pi$ go along curve b and only these will get past slit S_1 . They go straight through magnet 2 and magnet 3. The downward gradient of this magnet reverses the trajectories a and b . They go on to the detector through slit S_2 . A set of coils in the gap of magnet 2 producing a small horizontal magnetic field H' oscillating with a frequency ω adds to the strong constant vertical field H_0 . If $\omega = \omega_p$ the precession frequency of atoms in the field H_0 , the alternating field will cause some of the atoms passing by to make transitions from one J_z to the other. A detector set up to measure the current of arriving atoms as a function of ω will register a decrease when ω is “in resonance” with ω_p . Knowing ω_p the g -value of the atom could be deduced [6]. By this method nuclear magnetic moments can be measured accurately by observing the spin resonance of the nucleus or neutrons, that is, transitions between Zeeman levels. This formed the basis of magnetic resonance experiments of solids [7].

These two experiments established the ideas of spin and intrinsic angular momentum of quantum particles. More importantly, they led to three major discovery experiments to be briefly discussed below.

11.1.4 Discovery Experiments

In a major discovery Purcell et al. [8] reported the nuclear magnetic resonance (NMR) absorption in a sample of paraffin wax while Bloch et al. [9, 10] detected a similar absorption in liquid water by the method of nuclear induction. The existence of the quantum state and the reality of precession are the two discoveries of nuclear magnetic resonance.

Nuclear magnetic resonance is the absorption and re-emission of rf radiation by nuclei in magnetic field. Nuclear magnetic resonance has found many applications in physics, chemistry and medicine. The 1952 Nobel prize in physics was awarded jointly to Felix Bloch and Edward Purcell for the experimental discovery of nuclear magnetic resonance.

Zavoisky [11] discovered the electron paramagnetic resonance absorption of Cu^{2+} in $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ and Mn^{2+} in MnSO_4 , in 1944, pioneering the study of ESR in solids.

We now discuss general theoretical principles that govern magnetic resonance phenomena in solids.

11.2 General Theoretical Principles

Precession and nutation are the intrinsic properties which the classical mechanical geomagnetic earth [12] shares with the quantum mechanical magnetic atom, nucleus and the more fundamental electron and neutron. Precession ensures dynamical stability as in a gyroscope. Precession of a wobbling top introduces the idea of a precession frequency and Larmor precession and relaxation which is the basis of magnetic resonance.

The basic experimental technique for producing transitions between m_j states of isolated spins which interact only with high frequency and DC magnetic fields as done by Rabi was discussed in the last section. The need to do magnetic resonance experiments on macroscopic quantities connected with magnetic moments in solids with spin densities $\sim 10^{22}$ per cc interacting with the heat bath or 'lattice' naturally leads us to develop generally four basic theoretical ideas: (i) Larmor precession coming from classical and quantum mechanics, (ii) macroscopic magnetization coming from statistical mechanics, (iii) complex susceptibility coming from electromagnetic theory and (iv) spin Hamiltonian which is intrinsic to magnetic resonance method.

Magnetic systems (Fe, Co, Ni; Li, Na, Cu; V, Nb, Sn, Pb among others) that possess both magnetic moments and angular momentum exhibit the phenomenon of magnetic resonance. Resonance implies that we are in tune with a natural frequency of the magnetic system, corresponding to the frequency of the gyroscopic precession of the magnetic moment in an external static magnetic field. As a method to probe the solid state, resonance method permits accession of precise and detailed local structural information not obtainable otherwise. We now discuss the four concepts one by one.

11.2.1 Larmor Precession

Historically speaking, Landau and Lifshitz in 1935 predicted the existence of ferromagnetic resonance of the Larmor precession, which was independently verified in experiments by Griffiths and Zavoiskii in 1946. It is the precession of the magnetic moments of electrons, atomic nuclei and atoms about an external magnetic field. The magnetic field applied exerts a torque on the magnetic moment given by

$$\mathbf{\Gamma} = \boldsymbol{\mu} \times \mathbf{H}_0 = \gamma \mathbf{J} \times \mathbf{H}_0 \quad (11.10)$$

where μ is the magnetic dipole moment, $\mathbf{J}$ the angular momentum vector, and $\mathbf{H}_0$ is the external magnetic field (Fig. 11.3). $\mathbf{\Gamma}$ being always normal to the μ - $\mathbf{H}_0$ plane changes only the direction of the moment vector as the latter precesses around $\mathbf{H}_0$ with a frequency

$$\omega_0 = \gamma H_0 \quad (11.11)$$

where γ is the gyromagnetic ratio for the nucleus or the electron given by $\gamma_n = g_n e / 2m_n c$ and $\gamma_e = g_e e / 2m_e c$ respectively, g_n and g_e being spectroscopic splitting factors, m_n and m_e being masses of nucleus and electron respectively, and c the velocity of light. The most important role of Larmor precession is the following: the oscillatory magnetic moment normal to H_0 , produced by Larmor precession interacts with a small oscillatory magnetic field $H_1 \cos \omega t$, also normal to H_0 , and

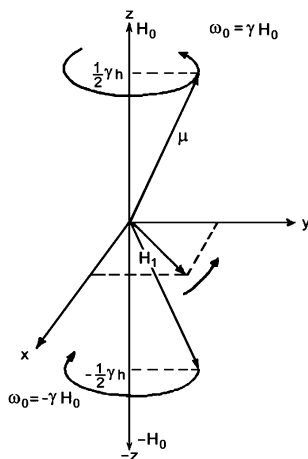


Fig. 11.3 Larmor precession of a nuclear or electron spin magnetic moment in an applied magnetic field H_0 ($\omega_0 = \gamma H_0$) along z -axis. An oscillatory magnetic field H_1 at radio or microwave frequency applied normal to H_0 ‘flips’ the moment changing the sense of precession and causing resonance absorption

changes the direction of the component of μ along $\mathbf{H}_0$ by 180° thereby changing the energy of the nuclear or electronic dipole and causing resonance absorption in the solid sample.

In NMR and ESR the Larmor precession is shown experimentally through resonance absorption. Let us illustrate it using the H-atom which consists of a proton magnet ($I = 1/2$) and an electron magnet ($S = 1/2$). The corresponding spin-degeneracies $m_I = \pm 1/2$ and $m_S = \pm 1/2$ are lifted by an externally applied DC magnetic field. Resonance absorption of rf or microwave radiation takes place when the sample bathed in a rf or microwave magnetic field is made to satisfy the resonance condition (11.11). The resonance frequency is

$$\nu \text{ [MHz]} = 4.258 H_0 \text{ [in kilogauss]} = 42.58 H_0 \text{ [in Tesla]} \quad (11.12a)$$

for the proton and

$$\nu \text{ [GHz]} = 2.80 H_0 \text{ [in kilogauss]} = 28.0 H_0 \text{ [in Tesla]} \quad (11.12b)$$

for the electron spin. The quantum mechanical selection rules are $\Delta m_I = \pm 1$ (NMR) and $\Delta m_S = \pm 1$ (ESR). Values of NMR frequencies for some nuclei are given in Table 11.1.

Next we discuss the statistical basis of macroscopic magnetization essentially on the lines of Schumacher [13].

11.2.2 Macroscopic Magnetization

In a solid where a large number of identical magnetic moments interact with a heat bath or ‘lattice’, what are the basic considerations that govern the relative populations of the different m_I levels? Take the case of nuclear magnetic moments in the system and assume that the density of the system is low enough so that we can apply Boltzmann statistics. Consider N spin $1/2$ nuclei in a magnetic field H_0 applied in the z direction. The populations of the $m_I = \pm 1/2$ states are N_+ and N_- with $N_+ + N_- = N$ (Fig. 11.4). The nuclei have translational degrees of freedom—kinetic and potential energy. The Zeeman energy of each spin is

$$E = -\gamma_n \hbar H_0 m_I. \quad (11.13)$$

For a general spin I the magnetic energy or Zeeman energy of the spin system is given by

$$E = \sum_{m_I=-I}^{+I} E(m_I) N(m_I) \quad (11.14)$$

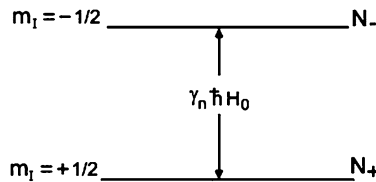


Fig. 11.4 Energy levels of a spin $\frac{1}{2}$ in a magnetic field H_0 . Ordering of levels occurs for $\gamma_n > 0$

where $N(m_l)$ is the number of spins in a given m_l state at a temperature T . It is given by the exact expression

$$N(m_l) = \frac{N \exp(-\hbar \gamma_n m_l H_0 / k_B T)}{\sum_{m_l=-I}^{+I} \exp(-\hbar \gamma_n m_l H_0 / k_B T)}. \quad (11.15)$$

The denominator of (11.15) represents the ‘sum over states’ or the partition function of statistical mechanics. In the high temperature approximation when $\gamma_n \hbar H_0 \ll k_B T$, $N(m_l)$ can be usefully approximated as

$$N(m_l) = \frac{N}{2I+1} \exp(-\hbar \gamma_n m_l H_0 / k_B T). \quad (11.16)$$

Substituting (11.16) in (11.14) we obtain

$$E \cong \frac{N}{2I+1} \sum_{m_l=-I}^{+I} \gamma_n \hbar m_l H_0 \left(1 - \frac{\gamma_n \hbar H_0 m_l}{k_B T} \right) = -\frac{N \gamma_n^2}{2I+1} \frac{(\hbar H_0)^2}{k_B T} \sum_{m_l=-I}^{+I} m_l^2.$$

The summation yielding $I(I+1)(2I+1)/3$, the Zeeman energy is

$$E = N \gamma_n^2 \hbar^2 I(I+1) H_0^2 / 3 k_B T. \quad (11.17)$$

The classical equivalent of (11.17) is $E = -N \mu^2 H_0^2 / 3 k_B T$ so that the quantum mechanical equivalent of μ^2 is $\gamma_n^2 \hbar^2 I(I+1)$ and the factor of 3 in the denominator of (11.17) is the quantum average of m_l^2 just as it is the average of $\cos^2 \theta$ (θ being the angle between the moment and H_0) in the classical calculation of E in the high temperature limit. The magnetic moment per unit volume is $E/V = -\mathbf{M}_0 \cdot \mathbf{H}_0$. Thus from (11.17)

$$\mathbf{M}_0 = \frac{N \gamma_n^2 \hbar^2 I(I+1)}{3 k_B T} \mathbf{H}_0. \quad (11.18)$$

Now we can ask the question: Suppose the excess spin population $n = N_+ - N_-$ is in a non-equilibrium condition to begin with. Then how does it approach thermal

equilibrium with the ‘lattice’ with the equilibrium value n_0 ? The equation to be solved is

$$\frac{dn}{dt} = \frac{(n_0 - n)}{T_1} \quad (11.19)$$

subject to the initial condition $n(0) = 0$. In other words, if the field were switched on suddenly at $t = 0$ after having always been zero before, then

$$n = n_0[1 - \exp(-t/T_1)] \quad (11.20)$$

where T_1 is the spin-lattice relaxation time—the time constant of the approach of the spin system to thermal equilibrium of the lattice. M_0 value can be obtained in another way through n_0 :

$$M_0 = n_0\mu_z/V \quad (11.21)$$

where $n_0 = N \tanh(\gamma_n H_0/2k_B T)$ in the high temperature limit gives the same value for M_0 . The static magnetic susceptibility χ_0 is

$$M_0 = \chi_0 H_0 \quad (11.22)$$

leading to the formula

$$\chi_0 = (N/V)\gamma_n^2 \hbar^2 I(I+1)/3k_B T. \quad (11.23)$$

A further question is: how does the spin system—now a macroscopic sample—respond to alternating (rf) fields? The answer to this question leads us to the concept of complex susceptibilities.

11.2.3 Complex Susceptibility Through Bloch Equations of Motion

In order to describe magnetic resonance experiments in liquids and liquid-like solids Bloch proposed phenomenological equations of motion for the macroscopic magnetization vector $\mathbf{M}$. The idea is to express the interaction of the magnetization with (i) the static and alternating external fields and (ii) with the lattice and to write a term that expresses the interaction of magnetic moments with each other and with the internal magnetic fields in the sample.

The three Bloch equations accommodate the effect of interaction of the moments undergoing resonance and other magnetic moments in the sample via dipole–dipole interaction or through quantum mechanical exchange interaction through a single parameter that affects only the transverse magnetization namely the components M_x

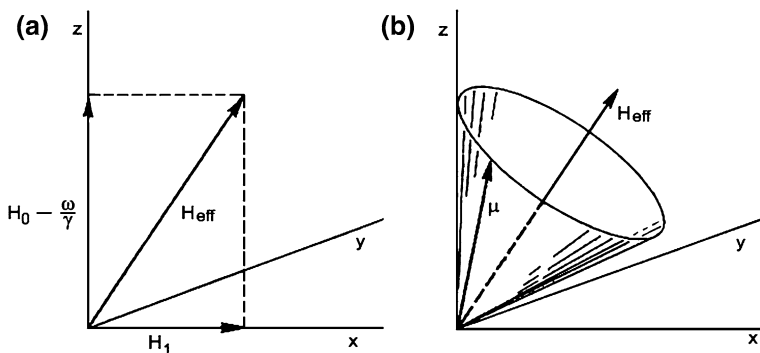


Fig. 11.5 **a** Effective field H_{eff} in relation to the static field H_0 and the time-varying field H_1 ; **b** motion of the moment μ in the rotating coordinate system describing a cone around H_{eff}

and M_y . Bloch equations describe a periodic motion (Fig. 11.5) of the magnetization occurring in the rotating frame in which the magnetic moments precess, tracing out a cone about an effective field H_{eff} , inclined with respect to the applied field.

The internal interactions of spins with each other are given by the partial differential equation

$$\partial M_{x,y} / \partial t = -M_{x,y} / T_2 \quad (11.24)$$

defining the parameter T_2 as the transverse or spin-spin relaxation time. According to (11.24) the magnetization decays exponentially to zero. Spin lattice relaxation time T_1 diminishes $M_{x,y}$. Field inhomogeneity adds to the rate at which $M_{x,y}$ diminishes so that $T_2 < T_1$. Combine (11.24) with torque and relaxation equations to get the Bloch equations

$$dM_z/dt = [(M_0 - M_z)/T_1] + \gamma_n(\mathbf{M} \times \mathbf{H})_z, \quad (11.25a)$$

$$dM_{x,y}/dt = -[M_{x,y}/T_2] + \gamma_n(\mathbf{M} \times \mathbf{H})_{x,y} \quad (11.25b)$$

where $\mathbf{H} = \mathbf{k}H_0 + \mathbf{i}H_1(t)$.

A pulsed H_1 gives free induction (or exponential) decay of M_x as the transient response of the Bloch equations. Thus,

$$M_x(t) = M_{x0} \cos \omega_0 t \exp(-t/T_2). \quad (11.26)$$

A continuous H_1 yields the steady-state solution in terms of real and imaginary parts $\chi'(\omega)$ and $\chi''(\omega)$ of the complex susceptibility of the nuclear spin system (sample). In the rotating frame the rf field is $\mathbf{H} = \mathbf{k}H_0 + \mathbf{i}H_1(t)$ but in the laboratory frame

$$\mathbf{H}_1(t) = \mathbf{i}2H_1 \cos \omega t \quad (11.27)$$

the transformation being $\omega = -\omega\mathbf{k}$. Ignoring the counter-rotating component the equations for magnetization are

$$dM_z/dt = -\gamma_n M_y H_1 + [(M_0 - M_z)/T_1], \quad (11.28)$$

$$dM_x/dt = -\gamma_n M_y [H_0 - (\omega/\gamma_n)] - (M_x/T_2), \quad (11.29)$$

$$dM_y/dt = -\gamma_n \{M_z H_1 - M_x [H_0 - (\omega/\gamma_n)]\} - (M_y/T_2). \quad (11.30)$$

In the steady state, the left hand sides of the above equations are zero. From the first of the three equations $(M_0 - M_z)$ is proportional to $M_y H_1$. From the definitions of χ' and χ'' , M_y is proportional to H_1 so $(M_0 - M_z) \sim H_1^2$. In the linear approximation to H_1 , we can replace M_z by M_0 in the second and third equations above. Let $M_+ = M_x + iM_y$. Here $i = \sqrt{-1}$. Then

$$dM_+/dt = -M_+ \{ (1/T_1) + i\gamma_n [H_0 - (\omega/\gamma_n)] \} + i\gamma_n M_0 H_1. \quad (11.31)$$

The steady-state solution to the above equation is

$$M_+ = \frac{i\gamma_n M_0 H_1}{(1/T_2) + i\gamma_n [H_0 - (\omega/\gamma_n)]}. \quad (11.32)$$

Let us define $\omega_0 = \gamma_n H_0$, substitute $M_0 = \chi_0 H_0$ and then separate the real and imaginary parts to get

$$M_x = \chi_0 \omega_0 T_2 \frac{(\omega_0 - \omega)}{1 + (\omega - \omega_0)^2 T_2^2} H_1, \quad (11.33)$$

$$M_y = \chi_0 \omega_0 T_2 \frac{1}{1 + (\omega - \omega_0)^2 T_2^2} H_1. \quad (11.34)$$

But

$$M_x = (\chi' \cos \omega t + \chi'' \sin \omega t) 2H_1. \quad (11.35)$$

So

$$\chi' = \frac{\chi_0 \omega_0 T_2}{2} \frac{(\omega_0 - \omega) T_2}{1 + (\omega - \omega_0)^2 T_2^2} \quad (11.36)$$

and

$$\chi'' = \frac{\chi_0 \omega_0 T_2}{2} \frac{1}{1 + (\omega - \omega_0)^2 T_2^2}. \quad (11.37)$$

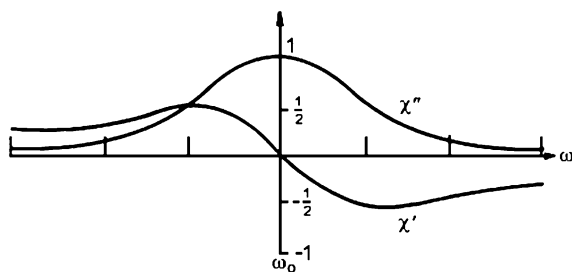


Fig. 11.6 The real (χ') and imaginary (χ'') parts of complex susceptibility based on the solutions of Bloch equations. The full width at half maximum is $2/T_2$. Dispersion (χ') peaks are at $\omega = \omega_0 \pm 1/T_2$

Plots of χ' and χ'' are shown in Fig. 11.6.

Next we discuss the unique concept of magnetic resonance, namely, the spin Hamiltonian—the vital link between theory and experiment.

11.2.4 Spin Hamiltonian

The nuclear spin Hamiltonian is given by

$$\mathcal{H} = -\gamma_n H_0 I_z + \frac{e^2 q Q}{[4I(I+1)][3I_{x3}^2 - I^2 + \eta(I_{x1}^2 - I_{x2}^2)]} + \gamma_n \sigma H_0 I_z \quad (11.38)$$

where H_0 is the external magnetic field, I_z the nuclear spin projection on the field H_0 , I the nuclear spin, γ_n is nuclear gyromagnetic ratio, $e^2 q Q$ is the quadrupole coupling constant, η the asymmetry parameter and σ is the chemical shift tensor. Much information can be derived by the use of the spin Hamiltonian, particularly about the anisotropic chemical shift.

What is chemical shift? The effective field at the nucleus of an atom H_{eff} is somewhat reduced relative to the applied field H_0 because of the shielding provided by the enveloping electron cloud which is characterized by a coefficient σ such that

$$H_{\text{eff}} = H_0(1 - \sigma). \quad (11.39)$$

σ depends on the density of electrons and the shielding is diamagnetic in nature, basically a consequence of Lenz's law as applied to the electrons circulating about an atom or molecule. Larmor precession of the electron orbit induces a small field that opposes the applied field. In molecules and solids bonding constraints alter the course of induced currents. A given element will have a different shielding constant in a different chemical environment. Thus the magnetic resonance of a nucleus in a

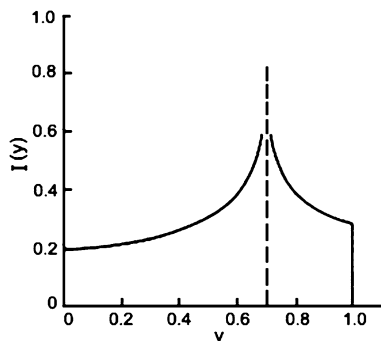


Fig. 11.7 Schematic line shape of a completely anisotropic NMR spectrum of a polycrystalline sample with three principal frequencies. y is the dimension-less frequency and $I(y)$ is the normalized NMR intensity. This anisotropy may come from chemical shift or quadrupole interactions anisotropy [14]

given external field may occur at different frequencies in different chemical compounds. This is called the chemical shift which naturally contains information about the chemical structure of a solid material.

Polycrystalline solids give broad-line NMR spectra (Fig. 11.7) the nature of which is determined by the effects of (a) asymmetric quadrupole interaction (exhibited by nuclei with $I > 1/2$ interacting with the electric field gradient) and (b) asymmetric chemical shift. The parameters characterizing these effects are directly connected with the local symmetries and electronic structure (charge and spin distributions in the sample).

11.3 Experimental Techniques of NMR

We discuss very briefly the continuous wave (CW) and pulse NMR techniques followed by methods for analyzing quadrupole split single crystal NMR spectra and also for determination of relaxation times.

11.3.1 Continuous Wave NMR

The basic principle [15] of this early technique is based on the Q -Meter detection of NMR signals essentially generated by electromagnetic induction (Fig. 11.8). In the Q -Meter NMR detector, the coil L is used as a resonant circuit by placing a capacitor C_0 in parallel with L_0 such that the Larmor frequency $\gamma_n H_0$ is the same as the resonant frequency $\omega_0 = 1/(L_0 C_0)^{1/2}$. In the figure the oscillator and large resistor R form a constant current generator of current I_0 . Parallel resonant circuit

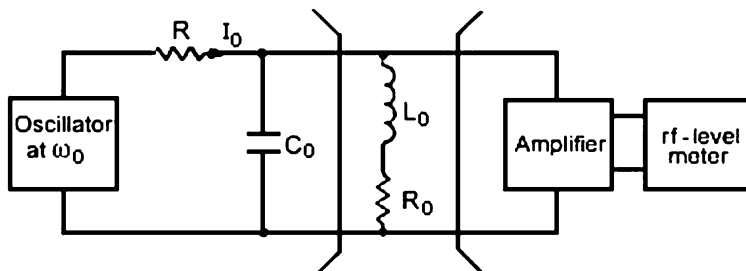


Fig. 11.8 Detecting NMR through Q-meter

tuned to ω_0 presents a real impedance $Z_0 (= Q\omega_0 L_0)$ at resonance. The voltage across it $V_0 = I_0 Z_0$ is amplified and its magnitude detected by the peak-reading voltmeter. As we go through resonance by changing H_0 , the real part of complex inductance changes fractionally by an amount $4\pi \chi'' Q$ so that the voltage at the amplifier input changes by $I_0 4\pi \chi'' Q^2 \omega L_0$.

11.3.2 Pulse NMR

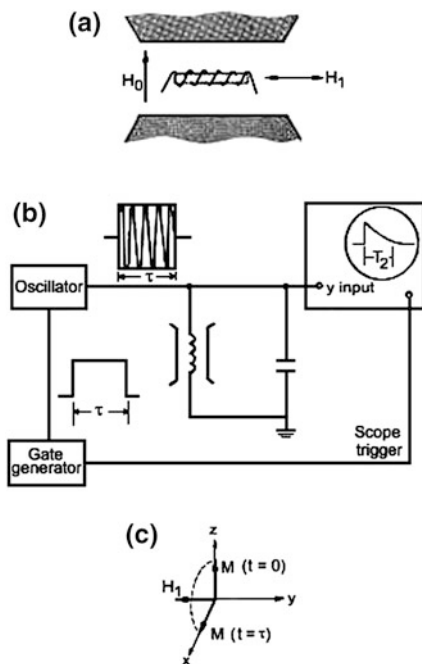
In Fourier transform NMR, the transformed free induction decay (FID) is the observable NMR signal generated by non-equilibrium nuclear spin magnetization precessing about the magnetic field (conventionally along z). This non-equilibrium magnetization can be induced, generally by applying a pulse of resonant radio-frequency close to the Larmor frequency of the nuclear spins. If the magnetization vector has a non-zero component in the xy plane, then the precessing magnetization will induce a corresponding oscillating voltage in a detection coil surrounding the sample. This time-domain signal is typically digitized and then Fourier transformed in order to obtain a frequency spectrum of the NMR signal i.e. the NMR spectrum. The duration of the NMR signal is ultimately limited by T_2 relaxation, but mutual interference of the different NMR frequencies present also causes the signal to be damped more quickly. When NMR frequencies are well-resolved, as is typically the case in the NMR of samples in solution, the overall decay of the FID is relaxation-limited and the FID is approximately exponential (with a time constant T_2 or more accurately T_2^*). FID durations will then be of the order of seconds for nuclei such as ^1H . If NMR line shapes are not relaxation-limited (as is commonly the case in solid-state NMR), the NMR signal will generally decay much more quickly e.g. microseconds for ^1H NMR.

11.3.2.1 Free Induction Decay Principle NMR

Put simply, free induction decay of nuclear magnetization is analogous to the transient response of the simple LCR circuit to which an ac input voltage is applied momentarily and switched off; the AC output voltage of the LCR circuit rapidly decays to zero with a decreasing amplitude. The envelope of the decaying amplitude is usually an exponential decay, like that of a damped harmonic oscillator. The Fourier transformation of this kind of decay gives the NMR spectrum of the sample in a coil which was empty in the LCR circuit.

How is free induction decay of an NMR sample recorded? Refer to Fig. 11.9. A coil produces a 90° pulse (or a transverse rf pulse of magnitude H_1) in the rotating frame (angular velocity $\omega_0 = \gamma_n H_0$) acting for a time τ such that $\gamma_n H_1 \tau = \pi/2$. It also serves to pick up the signal caused by the rotating magnetization immediately after the action of the $\pi/2$ pulse. If the coil volume is unity, cross section area A and has n turns, then the induced voltage immediately after $\pi/2$ pulse is turned on will be $V_0 = -n/c \eta d\phi/dt = -1/c 4\pi A n \eta dM/dt = -1/c 4\pi n \eta A \omega M_0$ where $\Phi (= BA)$ is flux linking the coil, $B = 4\pi M$ and η is the sample filling factor ($0 < \eta < 1$) which accounts for the incomplete flux linkage between sample and coil. $|V| \sim$ several mV if the coil is part of a resonant circuit of reasonable Q . Here M_0 is the initial (equilibrium) precessing magnetization fully turned over to the xy plane of the laboratory. The transverse magnetization decays exponentially with a time constant

Fig. 11.9 Observing nuclear free induction decay (schematic): **a** geometry of DC and AC magnetic fields H_0 and H_1 , **b** electronic circuit schematic and **c** H_1 and M vectors in the rotating frame [13]



T_2 , $T_2 \gg 2\pi/\omega_0$ and the signal is contained within a slowly varying envelope. The envelope decays as $V_0 \exp(-t/T_2)$ known as the free induction decay.

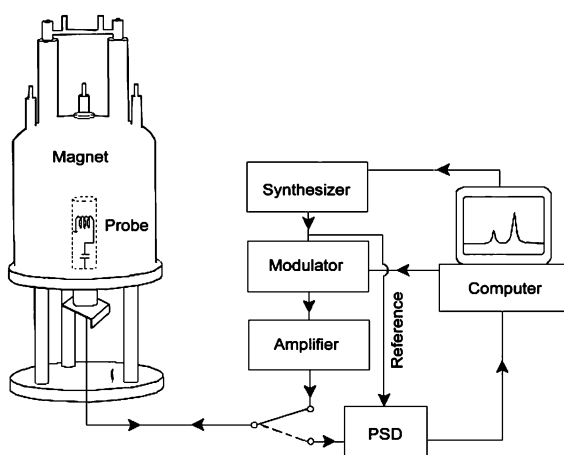
11.3.2.2 The Complete Pulsed Solid State NMR Spectrometer

In this spectrometer (Fig. 11.10) the sample is located in the coil of a rf resonant circuit-part of the probe. Sample and probe sit at the center of a superconducting magnet. Sample nuclei are excited by rf pulses generated in a synthesizer with a modulator and a high power amplifier. The pulse sequences are controlled by a computer. Relatively very weak sample response signal is directed through an rf switch to a phase sensitive detector (PSD) whose reference signal is derived from the synthesizer. Finally the computer displays the NMR spectrum of the sample.

11.3.2.3 Magic Angle Spinning (MAS) Technique

^1H NMR spectrum of water molecules which require 10 picoseconds for complete reorientation is a sharp line ~ 0.1 Hz wide while the line from a static sample of solid ice is 100 kHz. Broadening of solid state NMR spectra due to dipolar, chemical shift and quadrupolar anisotropy (for $I > 1/2$ nuclei) hide valuable information about structure and dynamics. The magic angle spinning (MAS) technique [16] shown in Fig. 11.11 overcomes this difficulty. Macroscopic rotation of the solid sample at the magic angle θ_m of $54^\circ 44' 8.4''$ averages all dipolar interaction $(3 \cos^2 \theta - 1)$ terms in the dipolar interaction Hamiltonian, θ is the angle between the direction of H_0 and the vector joining two nuclei (protons for example). Through MAS of a powder sample the average positions of the nuclei i and j and

Fig. 11.10 A typical pulse FT NMR spectrometer [16]



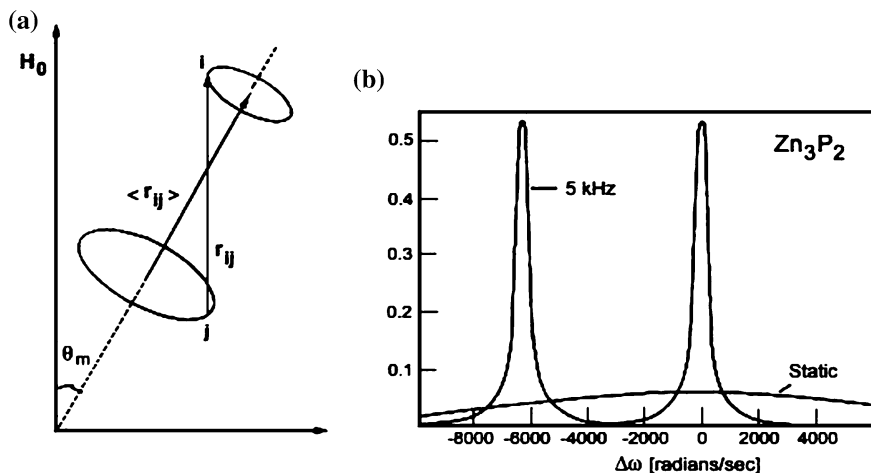


Fig. 11.11 a Magic angle spinning (MAS) b demonstrating the MAS advantage; ^{31}P NMR spectrum of rotating and static sample of polycrystalline zinc phosphate Zn_3P_2 . The two lines arise from crystallographically inequivalent P sites in the tetragonal unit cell [16]

the average internuclear vector $\langle r_{ij} \rangle$ are made to lie along that axis. MAS thus, narrows the overall NMR response by resolving from different nuclear sites. Thus the broadening for each line is less than the frequency difference between them. MAS imposes an average axial symmetry in an otherwise asymmetric environment. Chemical shift of a non-rotating single crystal depends on the orientation of the crystal with respect to an external magnetic field. While sample rotation effectively removes the dipolar broadening of NMR lines, it also averages out the chemical shift anisotropy to the isotropic value—value equal to the trace of the chemical shift tensor. Even a polycrystalline sample rotating under magic angle gives a set of narrow lines corresponding to the inequivalent sites of the nucleus being observed. Sample rotation rate must be comparable (in Hz) with the width of the spectrum of the static sample, only then are the satellite spectra detached from the narrowed central line and separately resolved. MAS is routinely used in solid state NMR.

11.3.3 Analysis of NMR Spectra

The NMR Hamiltonian ($\mathcal{H}_{\text{NMR}}$) used to analyze results of experimental NMR spectra such as those schematized by Fig. 11.12 is

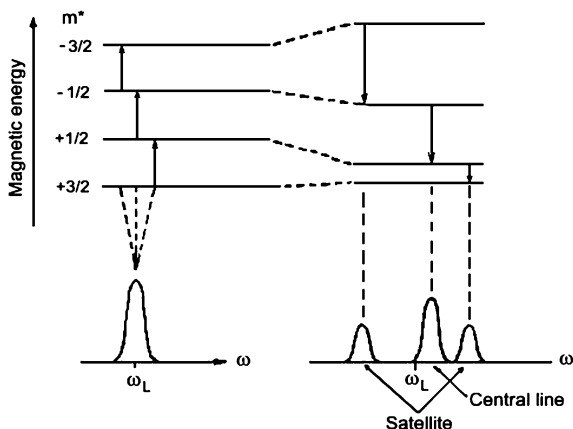
$$\mathcal{H}_{\text{NMR}} = \mathcal{H}_Z + \mathcal{H}_Q \quad (11.40)$$

where $\mathcal{H}_Z$ is the nuclear Zeeman term and $\mathcal{H}_Q$ describes the nuclear electric quadrupole interaction.

Fig. 11.12 Energy levels of a nucleus with $I = 3/2$

(a) without and (b) with quadrupolar interaction.

Arrows indicate NMR transitions induced by a radio frequency field. The resulting NMR spectrum is shown at the bottom



In a typical FT pulse NMR experiment [17] NMR signals of say ^{39}K in the KHSO_4 single crystal are measured for a fixed static magnetic field (typically 9.4 T) and the central radio frequency at 18.672 MHz. A sequence of one 90° pulse, 800 scans and a repetition time of 1 s were used.

Rotating the crystal about three mutually perpendicular axes, normal to the three mutually perpendicular planes of the crystal (identified previously) the angular variation of NMR spectra may be obtained (Fig. 11.13a). The angular rotation patterns of the NMR spectrum measured at room temperature when the sample is rotated in the ab - and ac planes are shown in Fig. 11.13b. These patterns actually represent the angular dependence of the second-order quadrupole shifts of the ^{39}K resonance in a KHSO_4 single crystal. Data fits to the strongly anisotropic symmetric second rank electric field gradient (EFG) tensors. Crystal rotation about crystallographic axes brings out the existence of two pairs of magnetically equivalent but crystallographically inequivalent nuclei one each from K(1) and K(2). The four NMR lines observed thus represent frequency shifts of these two groups- smaller shifts from K(1) and the larger shifts from K(2) nuclei in the crystal structure. Figure 11.13c depicts the microscopic environments of K(1) and K(2) consistent with the NMR observations. The quadrupolar parameters for ^{39}K in KHSO_4 for K (1) and K(2) are: $e^2qQ/h = 1.06, 1.45$ MHz and $\eta = 0.58, 0.85$ respectively. Thus the quadrupole parameters associated with ions surrounding the resonant nucleus in a single crystal could be unequivocally obtained from an analysis of NMR spectra.

11.3.4 Determination of Spin-Lattice Relaxation Time (T_1)

Pulse FT NMR spectrometers (Varian 200, Bruker 400) are usually employed to measure NMR signals from single crystal samples as a function of temperature. Two techniques employed to extract spin lattice relaxation time T_1 are (a) inversion

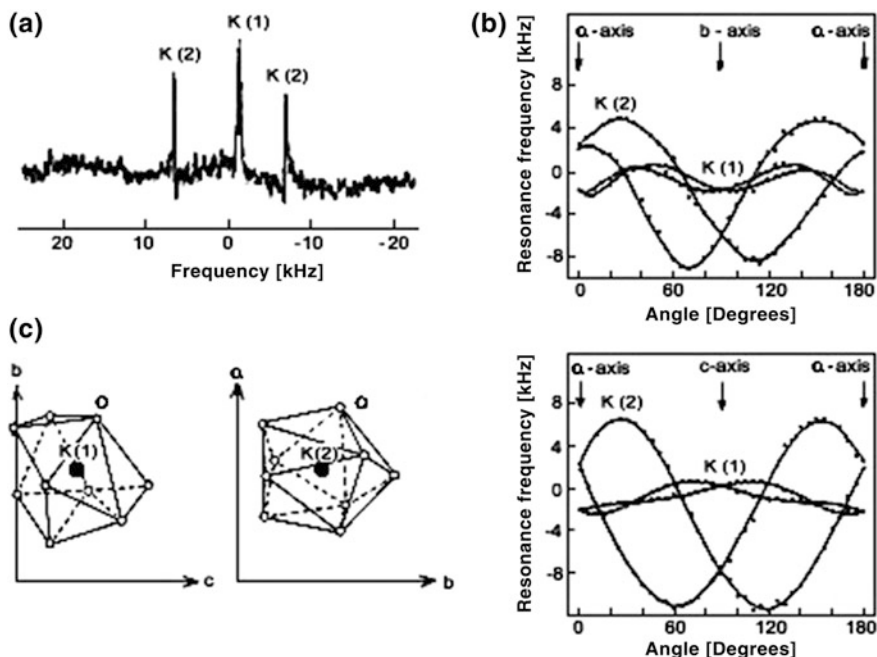


Fig. 11.13 **a** Resolved pulse NMR spectrum of ^{29}K in KHSO_4 when the static magnetic field H_0 is parallel to $(a + 30^\circ)$ -axis in the ac plane of the single crystal. The zero of the NMR frequency axis is the ^{39}K nuclear resonance frequency 18.672 MHz. The four lines with similar intensities arise from the KHSO_4 crystal structure microsymmetry. **b** Rotation pattern of ^{39}K NMR in KHSO_4 measured in the ab - and ca -planes at room temperature **c** Surroundings of K ions in KHSO_4 : around K(1) in projection along a -axis and around K(2) in projection along the c -axis

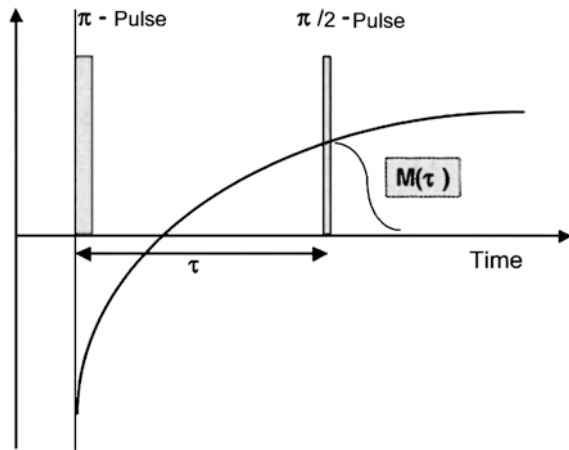
recovery (Fig. 11.14b) saturation recovery, adopting pulse sequences $\pi - t - \pi/2$ and $\pi/2 - t - \pi/2$ respectively.

Application of a pulse of the rf field H_1 with a duration t_p makes the magnetization precess at an angle given by $\gamma_n H_1 t_p$. Choose the pulse length so as to invert (π -pulse) or tilt the magnetization into the x - y plane ($\pi/2$ pulse). Precession of magnetization in the xy plane will induce a voltage in the coil producing the free-induction decay (FID) signal. After initial π -pulse $M_z(t)$ can be monitored by the amplitude of FID after a $\pi/2$ reading pulse at the evolution time t which is varied in the experiment.

The inversion recovery traces of say ^1H ($I = 1/2$) nuclei follow an exponential function. In such a case T_1 can be determined by fitting the traces with the following equation [18]:

$$[S(\infty) - S(t)]/2S(\infty) = \exp(-Wt) \quad (11.41)$$

Fig. 11.14 Principle of T_1 measurement by inversion recovery using $\pi - \tau - \pi/2$ pulse sequence



where $S(t)$ is the nuclear magnetization at time t and W is the transition probability corresponding to $\Delta m_I = \pm 1$. The relaxation times are given by

$$T_1 = 1/W. \quad (11.42)$$

In a typical experiment the experimental values corresponding to the logarithm of LHS of (11.41) are plotted against time (t) and the slope of this plot directly gives T_1 . Recovery traces at several temperatures fitted to (11.40) gives the temperature dependence of T_1 . The saturation recovery method is applicable to quadrupolar nuclei (e.g. ^7Li ($I = 3/2$)) where one is interested in finding out how the quadrupolar relaxation processes enable the nuclear spin system to recover after its magnetization has been saturated. A temperature-dependent relaxation time is indicative of fluctuations in the electric field gradient (EFG) tensor that are driven by thermally activated motion. When the central line is at saturation, the saturation recovery functions for the ^7Li transitions are given by [17] as

$$[S(\infty) - S(t)]/S(\infty) = 0.5[\exp(-2W_1t) + \exp(-2W_2t)] \quad (11.43)$$

where W_1 and W_2 ($n = 1, 2$) are the transition probabilities corresponding to the $\Delta m_I = \pm 1$ and $\Delta m_I = \pm 2$ transitions respectively. The spin-lattice relaxation time is given by [18] as

$$1/T_1 = (2/5)(W_1 + 4W_2). \quad (11.44)$$

A typical T_1 versus T plot for ^1H NMR in $\text{K}_4\text{LiH}_3(\text{SO}_4)_4$ crystals is shown in Fig. 11.15 with the inset showing the variation of LHS of (11.41) with delay time at 300 K [19].

A typical temperature dependence of T_1 for ^7Li NMR in $\text{K}_4\text{LiH}_3(\text{SO}_4)_4$ crystals together with the corresponding saturation recovery traces is presented in Fig. 11.16 [19].

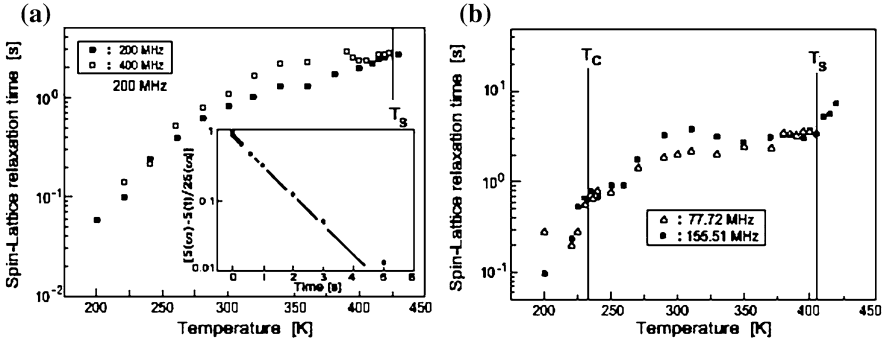


Fig. 11.15 Temperature dependence of T_1 for ^1H in $\text{K}_4\text{LiH}_3(\text{SO}_4)_4$ crystals. *Inset* variation in $S(\infty) - S(t)/2S(\infty)$ with delay time at 300 K. **b** ^7Li in $(\text{NH}_4)_4\text{LiH}_3(\text{SO}_4)_4$ crystals

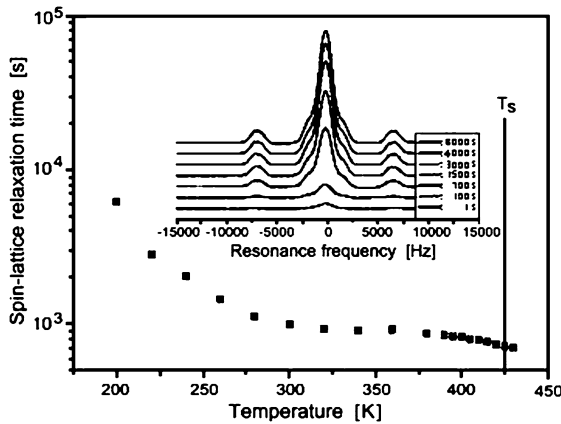


Fig. 11.16 Temperature dependence of the spin-lattice relaxation time for ^7Li in $\text{K}_4\text{LiH}_3(\text{SO}_4)_4$ crystals *Inset* shows saturation recovery trace for ^7Li as a function of delay time at 300 K. T_1 is not affected by the phase transition at $T_s = 425$ K

11.4 Case Studies in NMR

NMR is an important and well-established probe in solid state physics in static fields, but its use in pulsed field magnets is currently being developed.

In NMR one exploits the magnetic moment $\mu_n = \gamma_n \hbar I$ of atomic nuclei with spin I . Due to its angular momentum the nucleus precesses about an applied magnetic field H_0 with the so called Larmor frequency $\omega = \gamma_n H_0$. In typical NMR experiments one measures the change of the Larmor frequency (the NMR shift) or the nuclear relaxation that are both influenced by the chemical or electronic properties of the solid material under study. Short radio-frequency (rf) pulses in a tank circuit's rf coil provide a field H_1 that oscillates with the carrier frequency $\omega_c - \omega$ along an axis perpendicular to the external magnetic field H_0 .

We discuss briefly three case studies.

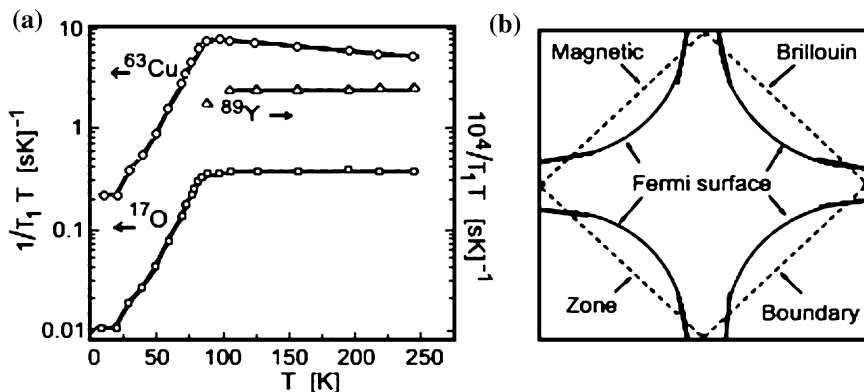


Fig. 11.17 **a** Spin-lattice relaxation rate versus temperature for $\text{YBa}_2\text{Cu}_3\text{O}_7$ ($T_c = 90$ K), ^{89}Y relaxation rate is (shown on the scale at *right*) a factor 10^3 – 10^4 lower than those of ^{63}Cu and ^{17}O and **b** a model of a Fermi surface for high T_c cuprates

11.4.1 NMR of the Superconducting Phase Transition

In high T_c superconductors the superconducting properties are unusual in that superconductivity arises from d -wave pairing instead of the s -wave pairing found in BCS superconductors. Also the normal state ($T > T_c$) properties such as electrical resistivity of the high T_c compounds deviate from that of a normal metal. For example Fermi surface apparently exists only in those directions in k -space where magnetic excitations do not interfere. NMR data on these materials have successfully been described by a semi-phenomenological susceptibility of a homogeneous spin fluid made up of a mixture of almost localized magnetic and nearly free electron contributions. The temperature dependence of relaxation for $\text{YBa}_2\text{Cu}_3\text{O}_7$ (Fig. 11.17a) is T -independent on Y and O sites while strongly T -dependent on Cu sites. Figure 11.17b shows a model for the 2D Fermi surface which covers only a part of the Brillouin zone [20].

11.4.2 Knight Shift

The shift in the value of external field at which the nuclear resonance occurs in the metal, compared to that of a salt, is the Knight shift [21, 22]. In a metal the atomic valence electron is the one donated by each atom to the collection of itinerant conduction electrons so that the S -electron wave function near the nucleus in the metal is not the same as in an atom. So the hyperfine interaction between the electron and the nucleus $a(S)\mathbf{I} \cdot \mathbf{S}$ is modified by a factor x_i (< 1) so that

$$\mathcal{H} = a(S)x_i\mathbf{I}_i \cdot \mathbf{S}_j \quad (11.45)$$

is the interaction between the i th nucleus and the j th electron. The information buried in x_i may be measured. This interaction is responsible for (1) the Knight shift, (2) nuclear relaxation and the Korringa relation, (3) Overhauser effect, and (4) the electron resonance shift (“day shift”). We shall focus only on the Knight shift and refer to [13] for a lucid discussion of other effects.

Consider the interaction energy of a nucleus of spin I and the gyromagnetic ratio γ_n . Then

$$U = (-\gamma_n \hbar H_0 + a \langle S_z \rangle) I_z. \quad (11.46)$$

Here the first term represents the interaction with the applied magnetic field H_0 and the second term the average hyperfine interaction of the nucleus with the conduction electrons. Note that the second interaction is observed in ESR also and arises from the so-called Fermi contact interaction. The average value of the conduction electron spin $\langle S_z \rangle$ connects to the Pauli spin susceptibility χ_s of the conduction electrons whose concentration is N through the relation

$$M_z = g_e N \beta_e \langle S_z \rangle = \chi_s H_0. \quad (11.47)$$

Thus the interaction could be written as

$$U = \left(-\gamma_n \hbar + \frac{a \chi_s}{g_e N \beta_e} \right) H_0 I_z = -\gamma_n \hbar H_0 \left(1 + \frac{\Delta H}{H_0} \right) I_z \quad (11.48)$$

The Knight shift K is then defined as

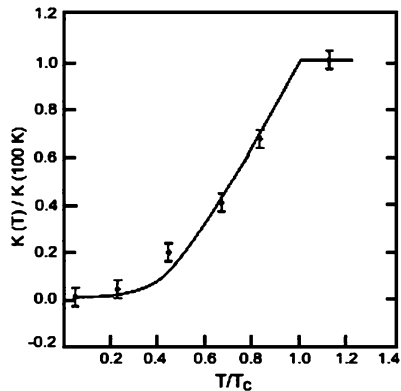
$$K = -\frac{\Delta H}{H_0} = \frac{a \chi_s}{g_e N \beta_e \gamma_n \hbar}. \quad (11.49)$$

K arises because the metal apparently sets up a small magnetic field in addition to the externally applied field. K is (a) associated with the electron spin paramagnetism of the metal, (b) nearly temperature independent and (c) proportional to the applied magnetic field. Typically 0.1 to 1 % of the total field arises from the interaction of nuclear moments with electronic moments reduced by ‘up-down’ arrangement; the electronic moments are several thousand times greater than nuclear moments.

Electron spin magnetization is concentrated close to the nuclear spins. Thus relative to the wave function of free electrons (ψ_{free}), the wave function of the conduction electrons is highly concentrated about positively charged nuclei (ψ_{nucleus}) and the shift is increased by a factor proportional to the electron density at the nucleus. Thus

$$\Delta H = (8\pi/3) \chi_s H_0 \langle |\psi_{\text{nucleus}}|^2 / |\psi_{\text{free}}|^2 \rangle. \quad (11.50)$$

Fig. 11.18 Normalized Knight shift of ^{63}Cu in the chains of a cuprate superconductor as a function of normalized temperature T/T_c [23]



$\langle |\psi_{\text{nucleus}}| \rangle^2$ represents an average over the Fermi surface of the metal because only electrons near the Fermi surface can have unpaired spins. This equation is reasonable and a rigorous consequence of energy band approximation and applies for a cubic metal. For a noncubic metal K becomes anisotropic.

An example of K measurement is connected with the superconducting transition observed in the cuprate superconductor where ^{63}Cu shift is followed through the T_c (Fig. 11.18). The vanishingly small shift seen for $T/T_c < 0.2$ is what is expected for a BCS superconductor with spin pairing at low temperatures. Thus NMR provides a unique perspective of the phenomenon of HTSC through Knight shift [23].

11.4.3 NMR Diffraction

Magnetic resonance imaging is based on the concept of NMR diffraction quite similar to the concept of crystal diffraction and reciprocal lattice in solid state physics. It is thus analogous to X-ray and optical diffraction. This concept developed by the Mansfield group [24, 25] linking MRI to conventional optics leads to a k -space analysis of imaging as briefly described below.

A reciprocal vector in k -space is defined by

$$\mathbf{k} = (1/2\pi)\gamma_n \mathbf{G}t \quad (11.51)$$

where $\mathbf{G}$ is the magnetic field gradient and t is the duration of a magnetic field gradient applied to the sample. t is involved because magnetic resonance data are acquired in the time domain. The time domain signal $S(t)$ written in k -space format is

$$S(k) = \int_{-\infty}^{+\infty} d^3r \rho(r) \exp(i\gamma_n H_0 + i\mathbf{k} \cdot \mathbf{r}) \quad (11.52)$$

where $\rho(r)$ represents the nuclear spin density within the volume element at r .

Equation (11.52) is analogous to that of a plane wave scattered from some point P (Fig. 11.19) displaced by r from the origin O [26]. The wavy line represents the total accumulated phase γH_0 of the ‘scattered’ wave at the origin while the straight lines represent the additional phase accumulation due to the scattering in the gradient at P. Thus the ‘MRI wave’ has a wavelength $\lambda = 2\pi/k$ which determines the spatial resolution obtainable in MRI and not the wavelength of the radio-frequency radiation at the Larmor frequency. Consequently MRI is able to resolve spatial features far smaller than would be possible from normal scattering of the radio-frequency electromagnetic wave.

For a clearer link between MRI and scattering, consider a static distribution of spins forming an orthorhombic lattice with unit cell dimensions a, b, c . The displacement vector $\mathbf{r}_{lmn}(q)$ from the origin to the q th spin site in the (l, m, n) unit cell is given by

$$\mathbf{r}_{lmn}(q) = (l + u_q)\mathbf{a} + (m + v_q)\mathbf{b} + (n + w_q)\mathbf{c}. \quad (11.53)$$

where $\mathbf{a} = a\mathbf{i}$, $\mathbf{b} = b\mathbf{j}$ and $\mathbf{c} = c\mathbf{k}$. ($\mathbf{i}, \mathbf{j}, \mathbf{k}$) are unit vectors along the three directions, u_q, v_q and w_q are fractions of the primitive cell dimensions and l, m, n are integers. The NMR signal can now be rewritten as a discrete scattering equation

$$S = \sum_{lmn} a_{lmn} \exp[2\pi(le + mf + ng)] \sum_q f_q \exp 2\pi i(u_q e + v_q f + w_q g) \quad (11.54)$$

where $e = ak_x/2\pi$, $f = bk_y/2\pi$, $g = ck_z/2\pi$, and $a_{lmn} = 0$, the dimensionless e, f, g being the Miller indices at time t . The term involving the summation over q ,

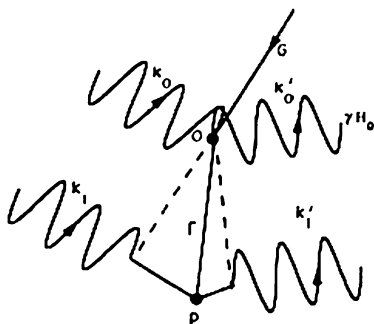


Fig. 11.19 Illustrating NMR diffraction. NMR ‘wave’ from point P displaced by r from origin. Wavy line total phase of scattered wave at the origin. Straight line additional phase accumulation from scattering in gradient G at P, k_1'

which corresponds to signal contributions from within the unit cell may be recognized as analogous to the geometrical structure factor S_k in electron or neutron scattering. The NMR scattering cross section $f_q = 0$ or 1 according to whether resonant spins are absent from or are present within the cell. Equation (11.54) gives plane-wave diffraction from a regular lattice. Thus the NMR signal $S(\mathbf{k})$ yields a diffraction pattern that, for regular structures, may be used to determine directly the lattice constants, just as in X-ray diffraction studies. More significantly, NMR diffraction measurements carry both phase and amplitude information, implying that the inverse Fourier transformation of the NMR diffraction pattern yields directly the spatial distribution of ensembles of spins.

The fact that signal and spin density are conjugate is the fundamental relationship of MRI. In the k -space formalism the NMR signal $S(k_z)$ from a three-dimensional object encoded using a gradient along the z direction may be written as

$$S(k_z) = \int_{-\infty}^{+\infty} \left[\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \rho(x, y, z) dx dy \right] \exp(2\pi i k_z z) dz \quad (11.55)$$

where the term in square brackets is called the spectrum of the sample in the presence of the gradient. The spectrum $P_{xy}(z)$ is the one-dimensional (1D) projection of the three-dimensional (3D) nuclear spin density onto the direction of the field gradient, or, the projection profile. Note that all nuclei in a plane perpendicular to the field gradient will experience the same field and contribute to the signal amplitude at the same frequency. The projection therefore becomes the Fourier transformation of the signal $S(k_z)$:

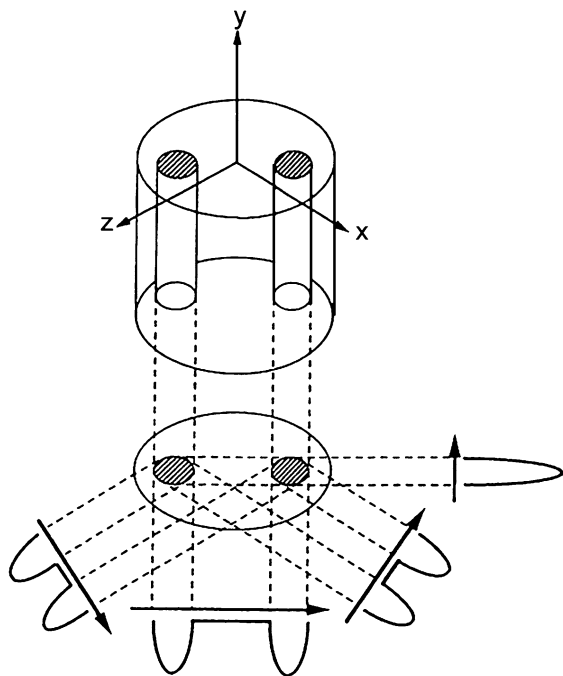
$$p_{xy}(z) = \int_{-\infty}^{+\infty} S(k_z) \exp(-2\pi i k_z z) dk_z. \quad (11.56)$$

From an operational point of view (11.51) shows that k -space may be traversed by moving either in time or in gradient magnitude, the direction of the traverse invariably determined by the direction of $\mathbf{G}$. The manner in which the sampled points are distributed in k -space is known as a raster, based on Cartesian and polar coordinates. Figure 11.20 shows Lauterbur's [27, 28] first practical demonstration of Nuclear MRI using projection-reconstruction technique.

11.5 ESR Theory

Both NMR and ESR detect magnetic dipole transitions between the ground state m_I and m_S levels of a nucleus (ex. H-atom proton, $m_I = \pm 1/2$) and an electron spin (H atom electron, $m_S = \pm 1/2$) placed in a solid, the levels being separated by an external static magnetic field. While nuclear spin, nuclear magnetic moment and nuclear

Fig. 11.20 Illustration of Lauterbur's projection-reconstruction method to create a magnetic resonance image of two tubes of water. *Arrow:* direction of magnetic field gradient used to encode the NMR signal with spatial information. *Shapes below:* 1D projections; Dotted line, back projection and 2D-NMR image of tubes



electric quadrupole moment are practically unaffected by the surroundings, for electronic systems the much greater physical size and the much smaller excitation energies makes the electron spin system strongly dependent on the surroundings (see Sect. 11.7 for details). Kittel [29] and Dekker [30] have discussed magnetic resonance from two different perspectives that provide valuable insights into the phenomena. A paramagnetic atom placed in a crystal may have angular momentum, magnetic moment and electric quadrupole moment quite different from those of its free atom. Thus sodium atom is different from sodium metal and sodium chloride in that strong, weak and no electron spin resonance are observed as expected. To observe ESR in insulating NaCl, one needs to dope it with, say, Mn impurity or irradiate with X-rays to create *F*-centres. However there is no free atom analogue for the *F*-center.

ESR is exhibited by materials with iron group atoms (e.g. Cu or Mn) or rare-earths (e.g. Gd) having incomplete *3d* and *4f* subshell, simple metals with conduction electrons, ferromagnets (e.g. Co) and ferrimagnets (Fe_3O_4) and imperfections in semiconductors and insulators capable of trapping electrons and holes.

The theory of ESR is discussed in terms of the 'electron Hamiltonian' [31, 32] which unlike the nuclear Hamiltonian discussed earlier is mainly made up of (i) electron kinetic energy, (ii) electron potential energy, (iii) spin-orbit coupling, (iv) coupling of electronic spins with each other in systems with $S > \frac{1}{2}$, the so-called electronic splitting or zero-field splitting and (v) spin and orbital magnetic moments coupled to an externally applied magnetic field, (vi) hyperfine energy due to the magnetic interactions between the unpaired electron and several nuclei, (vii) nuclear

electric quadrupole moment interaction with the crystalline electric field gradients, and, (viii) nuclear spin-electronic spin/orbital moment coupling.

11.5.1 The ESR Hamiltonian

Including the unpaired electron spin (S)-nuclear spin (I) hyperfine interaction, the nuclear Zeeman interaction and the nuclear electric quadrupole interaction of the spin system the ESR spin Hamiltonian may be written as

$$\mathcal{H}_{\text{ESR}} = \beta_e \mathbf{S} \cdot g_e \mathbf{H} + \mathbf{S} \cdot \mathbf{A} \mathbf{I} - \beta_n g_n \mathbf{I} \cdot \mathbf{H} + \mathbf{I} \cdot \mathbf{Q} \mathbf{I} \quad (11.57)$$

where g_n and β_n are the nuclear g -factor and nuclear Bohr magneton respectively, A and Q are the hyperfine and quadrupole interaction tensors, symmetric and diagonalizable. The g - and A -tensors and their variations with respect to temperature, pressure and concentration of paramagnetic impurities in a diamagnetic host crystal or a composite or a porous medium could be related to physico-chemical phenomena and microstructure. The above equation—the spin Hamiltonian—is generally a polynomial in S_{eff} [31]—the effective angular momentum where $2S_{\text{eff}} + 1$ equals the total number of electronic states whose properties are being described and which may be used to determine the energy levels of the spin system and thus deduce the transition frequencies. $\mathcal{H}_{\text{ESR}}$ is thus the link between theory and experiment.

Consider a spin half nucleus containing an unpaired electron ($I = S = 1/2$) with both electron Zeeman and magnetic hyperfine interactions being isotropic. Taking the static magnetic field along z -axis (11.57) becomes

$$\mathcal{H}_{\text{ESR}} = g_e \beta_e H S_z + a S_z I_z + g_n \beta_n H I_z \quad (11.58)$$

where a is the isotropic hyperfine coupling constant (measured in Hz). The four possible combinations of m_I and m_S , $(I + 1/2, +1/2)$, $(I + 1/2, -1/2)$, $(I - 1/2, +1/2)$ and $(I - 1/2, -1/2)$, define the four spin states whose energies are given by

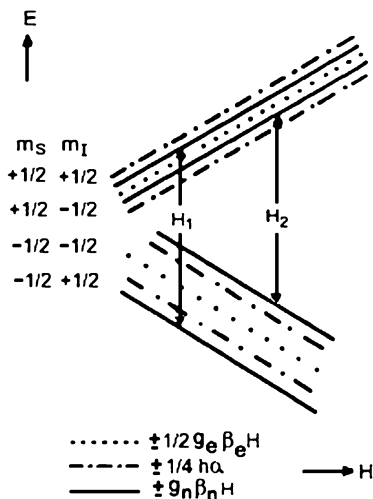
$$E = \langle m_S m_I | \mathcal{H}_{\text{ESR}} | m_S m_I \rangle = m_S g_e \beta_e H + m_S m_I h a - m_I g_n \beta_n H. \quad (11.59)$$

Figure 11.21 shows these levels as a function of H . The ESR selection rule $\Delta m_S = \pm 1$ corresponds to a change of $\hbar$ in the spin angular momentum. During this transition and in the presence of hyperfine structure (when ESR will split into two or in general $(2I + 1)$ line structure) m_I must not change so that the total angular momentum of the nuclear spin system and the electron spin system is conserved. This implies the operation of the extended selection rule $\Delta m_s = \pm 1$, $\Delta m_I = 0$ so that the ESR transitions are observed when

$$h\nu = g_e \beta_e H_0 \pm (1/2) h a \quad (11.60)$$

giving an isotropic hyperfine interaction $A_{\text{iso}} = h a / g_e \beta_e$.

Fig. 11.21 Energy levels and transitions for a spin system with $S = I = 1/2$ as a function of applied magnetic field. Electron Zeeman (.....), hyperfine (---) and nuclear Zeeman (—) effects are shown. Two ESR transitions at H_1 and H_2 are observed



ESR measures g_e and hyperfine coupling constants which is the intrinsic magnetic field of the electron H_e produced by the intrinsic nuclear spin angular momentum I so that

$$h\nu = (H_0 + H_e)g_e\beta_e \text{ or } H = h\nu/g_e\beta_e = H_0 \pm H_e = H_0 - am_I \quad (11.61)$$

The nuclear Zeeman energies do not normally affect the transitions so that ESR does not measure g_n . Thus the unequivocal identification of the particular nucleus is generally impossible from ESR. This ambiguity may however be resolved by a simultaneous establishment of ESR ($\Delta m_s = \pm 1$, $\Delta m_I = 0$) and NMR (resonant absorption of radio frequency radiation by magnetic nuclei ($\Delta m_s = 0$, $\Delta m_I = \pm 1$) in the electron nuclear double resonance (ENDOR) technique [33].

11.5.2 ESR Spectrum and Its Analysis

The continuous wave X- or Q-band ESR spectrum is generally a plot of the first derivative of the imaginary part of the complex magnetic susceptibility of the paramagnetic system with respect to the applied DC magnetic field. The spectrum may consist of a single line or a group of lines and could vary with changes in temperature or the crystallographic orientation of the specimen in the DC magnetic field for crystal rotations about a given axis (a , b , c). Four basic quantities or parameters are derivable from the spectrum: (i) g -factor or g -tensor, (ii) hyperfine splitting parameter or A tensor, (iii) line shape and line width and (iv) double-integrated intensity. Of the four the last one is readily available in modern spectrometers with in-built software.

An analysis of the four quantities reveal the following information about the system: (a) nature of the paramagnetic center or complex i.e., electron (*F*-center in alkali halides) or hole-type (*V*-center in *X*-irradiated alkali halides), (b) the precise identity of the complex/center in terms of its nearest neighbours and local geometry/symmetry, (c) the distribution of unpaired spin density over the neighbours, and (d) the bulk paramagnetic susceptibility of the sample without the need for diamagnetic corrections.

11.5.2.1 The *g*-Factor

A free spin system $S = 1/2$ with a single line spectrum due to $m_S = -1/2 \rightarrow +1/2$ transition in resonance with the applied microwave magnetic field (normal to the DC field H_0) satisfies the condition $h\nu = g_e \beta_e H_0$. With ν directly known, and H_0 known from the resonance spectrum, g_e can be found. Values of g_e determined from resonance experiments show a variation from the free electron value 2.0023.

The electron Zeeman interaction responsible for the ESR spectrum is the leading term of (11.57) which with reference to a Cartesian coordinate frame x, y, z with electron magnetization parallel to the applied magnetic field having components H_x, H_y, H_z , becomes

$$\mathcal{H}_Z = \beta_e (g_{xx} S_x H_x + g_{yy} S_y H_y + g_{zz} S_z H_z). \quad (11.62)$$

For paramagnetic systems in which the ground and excited states are mixed by spin (*S*)-orbit (*L*) interaction, the elements of the *g*-tensor connecting the pseudo-vector *S* and the *H*-vector are

$$g_{ij} = 2\delta_{ij} - 2\lambda \frac{\langle n|L_i|0\rangle \langle 0|L_j|n\rangle}{E_n - E_0} \quad (11.63)$$

where $\delta_{ij} = 1$ if $i = j$ and 0 otherwise. $|0\rangle$ is the orbital singlet ground state of energy E_0 , $|n\rangle$ is the excited state of energy E_n and λ is the spin-orbit interaction which varies widely for 3*d*-group ions; for Ti^{3+} it is 154 cm^{-1} while for Cu^{++} it is -852 cm^{-1} .

11.5.2.2 The Hyperfine Coupling Constant *A*

The interaction between the electronic and nuclear magnetic dipoles produces this coupling constant which in terms of operators *S* and *I* reads

$$\mathcal{H}_{hf} = S A I, \quad (11.64)$$

the second term in (11.57) where A is a symmetric second rank tensor like g . In a reference frame xyz in which A is diagonal

$$\mathcal{H}_{hf} = A_{xx}S_x I_x + A_{yy}S_y I_y + A_{zz}S_z I_z, \quad (11.65)$$

which for the frequent case of axial symmetry becomes

$$\mathcal{H}_{hf}(\text{axial}) = A_{||}S_z I_z + A_{\perp}(S_x I_x + S_y I_y), \quad (11.66)$$

where z -axis is the axis of symmetry of the complex containing the unpaired electron. The A -tensor has two parts: (i) isotropic contribution (A_{iso}) due to the s -electron density found at the corresponding magnetic nucleus, (ii) the anisotropic contribution due to the spin and orbital dipole-dipole interactions with the nuclei. $A_{\text{iso}} = (A_{||} + 2A_{\perp})/3$ is due to the Fermi-contact interaction given by

$$\mathcal{H}_{\text{contact}} = (8\pi/3)g_e \beta_e g_n \beta_n |\psi(0)|^2 S I, \quad (11.67)$$

(a term important for NMR of metals too) from which

$$A_{\text{iso}} = (8\pi/3)g_e \beta_e g_n \beta_n |\psi(0)|^2 = 4.24 g_n |\psi(0)|^2 (\text{mT}), \text{ using } \beta_n = \beta_e/1836 \quad (11.68)$$

where $|\psi(0)|^2$ is the unpaired electron density at the nucleus. For a wave function

$$\psi = a|s\rangle + \sqrt{1-a^2}|p\rangle \quad (11.69)$$

and

$$A_{\text{iso}}(\text{expt})/A_{\text{iso}}(\text{theory}) = a^2. \quad (11.70)$$

A_{iso} (theory) has been computed for all the magnetic atoms from H to Bi [34, 35]. In the case of paramagnetic ions such as $\text{Cu}^{2+}(3d^9)$ which give a large value of A_{iso} despite apparently small spin density at the nucleus, Fermi contact interaction cannot account for A_{iso} . In such cases the inner (core) s -electrons (e.g. $3s$ in the case of Cu) are polarized by the valence electrons and these ‘core polarized’ electrons have a sizeable spin density at the $^{65,63}\text{Cu}$ nucleus to give a large A_{iso} .

The anisotropic part of the hyperfine interaction A_{aniso} is due to the dipole-dipole interaction between the electron and the nuclear dipoles

$$\mathcal{H}_{\text{dipolar}} = g_e g_n \beta_e \beta_n \left[\frac{\mathbf{S} \cdot \mathbf{I}}{r^3} - \frac{3(\mathbf{S} \cdot \mathbf{r})(\mathbf{I} \cdot \mathbf{r})}{r^5} \right]. \quad (11.71)$$

In terms of the angle θ between the vector r joining the electron and nucleus and the magnetic field

$$\mathcal{H}_{\text{dipolar}} = g_e g_n \beta_e \beta_n [(3 \cos^2 \theta - 1)/r^3] I_z S_z. \quad (11.72)$$

In terms of an equivalent magnetic field this dipolar interaction gives rise to the anisotropic part of the hyperfine interaction

$$A_{\text{aniso}} = \beta_n g_n \left\langle \frac{3 \cos^2 \theta - 1}{r^3} \right\rangle I_z \quad (11.73)$$

spatially averaged over all of the unpaired electron wave function. Thus $A_{\text{aniso}} \sim \langle 1/r^3 \rangle$. A_{aniso} helps fix the geometry of the paramagnetic complex in a crystal.

11.5.2.3 ESR Line Shape and Line Width

The shape of an ESR absorption or first derivative signal depends on (a) whether the unpaired electron is located in a metal or semiconductor or an insulator (b) the immediate neighbourhood of the electron with which it interacts electrostatically or electromagnetically. The strength of these interactions give the resonance line a nonzero line width. According to the uncertainty relation $\Delta E \Delta t \geq \hbar$. If $\Delta t = T_1$, the spin-lattice relaxation time, then the magnitude of T_1 decides the ‘lifetime’ broadening and line width. The line width is related to the line intensity through the line shape function because the computation of the line width depends on the line shape function-implying an intimate relation between the three observables.

Individual ESR signals have two familiar shapes [36]—the Lorentzian and the Gaussian shapes—due to homogeneous broadening and inhomogeneous broadening mechanisms respectively. In the first mechanism all dipoles experience the same static, time average magnetic fields but different instantaneous magnetic fields. In the second mechanism the dipole is subjected to slightly different effective magnetic fields such that at a given time only a small fraction is in resonance as the external magnetic field is swept through the line leading to a superposition of a large number of slightly staggered individual ‘spin packets’ or an envelope.

An inhomogeneous magnetic field or anisotropic interactions as seen in g - and A - tensors in ceramics or unresolved hyperfine structure gives Gaussian shapes. Metal clusters (and colloidal nanoparticles) and conduction electrons in metals give Lorentzian and highly asymmetric Dysonian shapes [37] respectively, while defects and free radicals in insulators give Gaussian lines. F -centers in RbBr crystals give an inhomogeneously broadened 290 G wide (Gaussian) ESR line [29]. Lorentzian lines are broader than those broadened by T_1 . A relaxation time T_2 relates to the width of the normalized ESR line in the absence of saturation as

$$1/T_2 = (\text{shape constant}) \times (\text{gyromagnetic ratio}) \times (\text{line width at half height}), \quad (11.74)$$

the shape constant being 1 for Lorentzian and $\sqrt{\pi e \ln 2}$ for Gaussian lines. $1/T_2$ is a linear function of line width and includes ‘lifetime’ (or T_1) broadening and other homogeneous mechanisms characterized by T_2' so that

$$(1/T_2) = (1/T_2') + (1/2T_1) \quad (11.75)$$

where T_2' is the spin-spin relaxation time ($\ll T_1$ for free radicals). Lines are also broadened/narrowed by exchange interactions between paramagnetic centers in concentrated spin systems. T_1 and T_2' often decide the temperatures at which ESR would become observable.

11.5.2.4 ESR Intensity

The integrated/double integrated intensity of the ESR absorption/first derivative signal is proportional to the density of paramagnetic centers in the sample and to the net paramagnetic susceptibility of the sample.

If (a) ESR line shape is Lorentzian, (b) Curie law ($\chi \sim 1/T$) applies, (c) applied microwave power does not saturate the ESR signal and (d) the signal to noise ratio is 1, then ($N_{\min}$) the ESR detection sensitivity or the minimum number of paramagnetic centers detectable in ESR is given by

$$N_{\min} = [3V_c k_B T_s \Gamma / 2\pi g_e^2 \beta_e^2 S(S+1) H_r Q_u'] [F k_B T_d b / P_0]^{1/2} \quad (11.76)$$

where V_c is the volume of rectangular cavity operated in TE_{102} mode; k_B the Boltzmann constant, T_s the sample temperature, Γ the half width at half maximum intensity (in Gauss) of the absorption line, H_r the magnetic field at the center of the absorption line. Q_u' is the effective unloaded Q -factor of ESR cavity $= Q_c Q_u / (Q_c + Q_u)$ where Q_c is the dielectric Q , Q_u the unloaded Q , T_d the detector temperature, b the bandwidth of the detecting-amplifying system, P_0 the microwave power (ergs/s) incident on the cavity and F the non-thermal noise figure = 1 for an ideal spectrometer. Wertz and Bolton [38] estimate the value 10^{11} for $N_{\min}$.

Assuming identical geometry for sample and standard, so that H_1 is the same for both, and non-saturating conditions, the concentration $[x]$ of the paramagnetic species relative to that of standard $[\text{std}]$ is

$$[x] = [\text{std}] A_x G_{\text{std}} M_s g_{\text{std}}^2 [S(S+1)]_{\text{std}} / A_s G_x M_x g_x^2 [S(S+1)] \quad (11.77)$$

where A 's are the appropriate areas under resonance curve, G 's the receiver gains of spectrometer, M 's the modulation amplitudes for sample and standard and g_{std} the g -value of the standard sample used for calibration. The concentration of paramagnetic spins in the sample, say N , is fundamentally related to the paramagnetic

susceptibility χ_{EPR} to emphasize its deduction from EPR experiment through the equation

$$\chi_{\text{EPR}} = Ng_e^2 \beta_e^2 J(J+1)/3k_B T \quad (11.78)$$

where $J = L + S$ and other symbols are familiar. An application of this procedure is discussed in [8]. In a recent work on K-doped graphite powder, the ESR signal intensity when calibrated against an intensity reference yields an accurate characterization allowing the density of states to be obtained [39].

11.5.3 *g-Tensor and A-Tensor Analysis*

Single crystals containing ESR active species give spectra whose analysis gives the principal values of the g - and A -tensors. This task begins with the identification of three mutually perpendicular directions in the crystal in a manner similar to the NMR spectra analysis of quadrupole coupling parameters. These could be [100], [010] and [001] in an orthorhombic crystal. These need not always correspond to crystallographic axes. ESR spectra are usually recorded for crystal rotations about each of these three axes.

Three sets of spectra arising from crystal rotations about x , y and z axes with yz , zx and xy planes in the DC magnetic field need to be analyzed after identifying groups of peaks corresponding to chemically distinct centres or physically inequivalent but chemically equivalent centres, reduced plots are made of g_{ij}^2 or $(g^4 A^2)_{ij}$ ($ij = xy, yz, zx$) versus θ . Thereafter each of the observed plots is fitted by the method of least squares to

$$g_{ij}^2(\theta) = U + V \cos 2\theta + W \sin 2\theta \quad (11.79)$$

(with a similar equation for $g^4 A^2$) with

$$U = (g_{ii}^2 + g_{jj}^2)/2, \quad V = (g_{ii}^2 - g_{jj}^2)/2, \quad W = g_{ij}^2. \quad (11.80)$$

Then the elements of the g^2 (or hyperfine tensor) are found and the matrix is symmetrized and diagonalized. The diagonalization can be carried out by solving the equation $|G - \lambda I| = 0$, where G is the matrix described below, λ are the eigen values required and I is the unit matrix. Alternatively, we can use Schonland's method [40]. This consists in accurately locating the maxima and minima using the relation $\tan 2\theta = W/V$ and forming the matrix elements as follows:

$$\{g^2\} = G = \begin{bmatrix} G_{xx} & G_{xy} & G_{xz} \\ G_{yx} & G_{yy} & G_{yz} \\ G_{zx} & G_{zy} & G_{zz} \end{bmatrix}. \quad (11.81)$$

Here

$$\begin{aligned} G_{xx} &= \alpha_y + \alpha_z - \alpha_x, \\ G_{yy} &= \alpha_z + \alpha_x - \alpha_y, \\ G_{zz} &= \alpha_x + \alpha_y - \alpha_z. \end{aligned} \quad (11.82a)$$

$$G_{yz} = \pm \sqrt{(\delta_x - \alpha_y - \alpha_z)(\delta_x - \alpha_y + \alpha_z)}, \quad (11.82b)$$

with G_{zx} and G_{xy} written out in cyclic manner. α 's and δ 's are determined using the accurately located $g_{\max}^2$ and $g_{\min}^2$ for each of the three rotations as follows:

$$\alpha_i = [g_{\max}^2(i) + g_{\min}^2(i)]/2, \text{ and } \delta_i = [g_{\max}^2(i) - g_{\min}^2(i)]/2, \quad i = x, y, z. \quad (11.83)$$

If the paramagnetic centre is found at n_1 sites for rotations about x -axis, n_2 sites for rotations about y -axis and n_3 sites for rotations about z -axis, then one has to sort out the correct matrix elements out of a possible $n_1 n_2 n_3$ combinations of maxima and minima. The choice can be narrowed down by seeking coincidences in the g_{ij}^2 values (for example g_{xx}^2 occurs twice- for rotations about y and z axes [41]). Finally the diagonalization of the matrix is effected using Jacobi matrices so that

$$\begin{bmatrix} l_{xx} & l_{xy} & l_{xz} \\ l_{xy} & l_{yy} & l_{yz} \\ l_{xz} & l_{yz} & l_{zz} \end{bmatrix} \begin{bmatrix} g_{xx}^2 & g_{xy}^2 & g_{xz}^2 \\ g_{xy}^2 & g_{yy}^2 & g_{yz}^2 \\ g_{xz}^2 & g_{yz}^2 & g_{zz}^2 \end{bmatrix} \begin{bmatrix} l_{xx} & l_{xy} & l_{xz} \\ l_{xy} & l_{yy} & l_{yz} \\ l_{xz} & l_{yz} & l_{zz} \end{bmatrix}^{-1} = \begin{bmatrix} g_{xx}^2 & 0 & 0 \\ 0 & g_{yy}^2 & 0 \\ 0 & 0 & g_{zz}^2 \end{bmatrix}. \quad (11.84)$$

Taking square roots of the diagonal elements gives principal g -values. The direction cosine matrix (l -matrix) gives the direction cosines of $g_{xx}(l_{xx}, l_{xy}, l_{xz})$, $g_{yy}(l_{xy}, l_{yy}, l_{yz})$ and $g_{zz}(l_{xz}, l_{yz}, l_{zz})$. The paramagnetic center is eventually located by the help of principal g - and A values and their direction cosines.

The analysis of single crystal ESR spectra proceeds exactly similar to that of NMR spectra discussed earlier. For a good case study of analysis of all the spin Hamiltonian parameters see the work on Cu^{++} impurity in Na_2SO_4 single crystals [42].

11.5.4 Polycrystalline ESR Spectra

ESR spectrum of a paramagnetic powder sample (composed of randomly oriented small crystallites) is the envelope of the spectra from all possible orientations of the paramagnetic center or radical species with respect to the external magnetic field. To analyze such spectra is to extract the principal values of g - and hyperfine

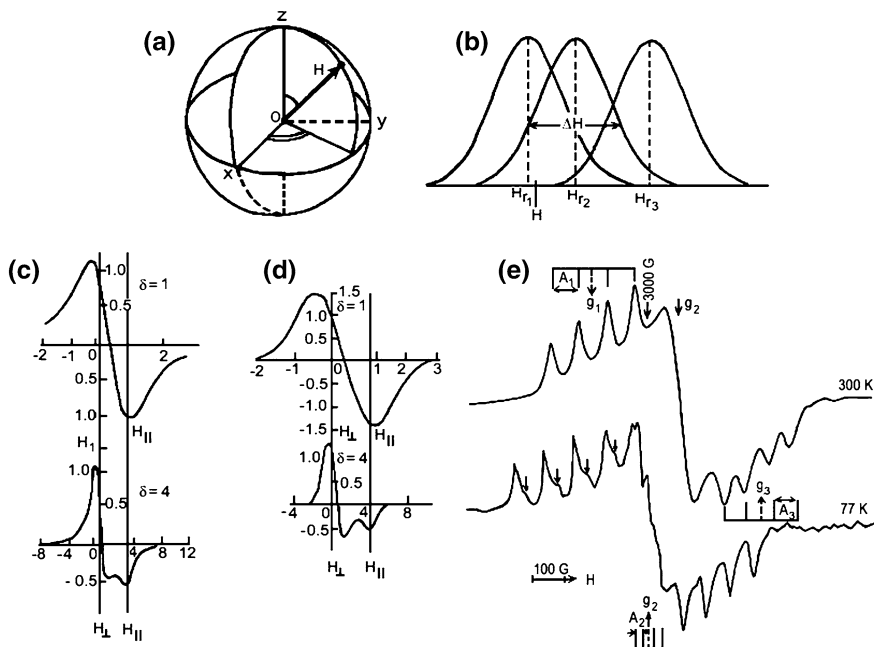


Fig. 11.22 Analysis of polycrystalline ESR spectral line shapes: **a** coordinate system showing H vector in a unit sphere, **b** contributions from three different crystallites to the absorption at the magnetic field H . The magnitudes of $H_{r2}-H_{r1}$ and $H_{r3}-H_{r2}$, the successive differences in resonance fields are relative to line width ΔH of second crystallite with resonance field H_{r2} , **c** theoretical shapes of asymmetric ESR lines based on Lorentzian form of individual lines, **d** theoretical shapes of asymmetric ESR lines based on Gaussian form of individual lines; comparison with experimental spectra enable extraction of reasonably accurate g -parameters and **e** a typical powder pattern with orthorhombic symmetry for g and A observed in $\text{KHSO}_4:\text{Cu}^{2+}$ at 300 K and 77 K [41]

tensors. In the Lebedev procedure [43] a unit sphere is considered in which the coordinates of the ESR center is fixed while $\mathbf{H}$ is a vector whose coordinates are θ and ϕ (Fig. 11.22). Every orientation of $\mathbf{H}$ on the sphere is equally probable. The number of centers with a magnetic field orientation between θ and $\theta + \Delta\theta$ and between ϕ and $\phi + \Delta\phi$ are given by the solid angle

$$\Delta\Omega = \sin\theta \Delta\theta \Delta\phi. \quad (11.85)$$

But only a fraction of these radicals would contribute to the microwave energy absorption at a magnetic field between H and $H + dH$, this fraction depending on the individual line width ΔH and the distance from resonance $H_r - H$. This fraction is defined by a function $f(H_r - H, \Delta H)$, where H_r the resonance field of the individual line itself is a function of orientation (Fig. 11.22b). The aim is to determine the total energy absorption at the magnetic field H . The intensity of this absorption (I) at a given magnetic field is the product of the function over all solid angles, because all orientations contribute to the absorption, in principle:

$$\begin{aligned}
I &= (1/4\pi) \sum_i f(H_r - H, \Delta H) \Delta\Omega_i \\
&= (1/4\pi) \sum_{j=0}^{2\pi} \sum_{i=0}^{\pi} f(H_r - H, \Delta H) \sin\theta_i \Delta\theta_i \Delta\phi_j \\
&= (1/4\pi) \int_{j=0}^{2\pi} \int_{i=0}^{\pi} f(H_r - H, \Delta H) \sin\theta \, d\theta \, d\phi
\end{aligned} \tag{11.86}$$

where the summations are replaced by the integrals and $1/4\pi$ is the normalized factor. The derivative of absorption (usually measured) is

$$dI/dH = (1/4\pi) \int_0^{2\pi} \int_0^{\pi} f'(H_r - H, \Delta H) \sin\theta \, d\theta \, d\phi \tag{11.87}$$

The specific form of the function $f'(H_r - H, \Delta H)$ depends on the shape of the individual derivative curves. For a Lorentzian absorption curve

$$f'(H_r - H, \Delta H) = \frac{16}{\pi(\sqrt{3}\Delta H)^3} (H_r - H) \left[1 + \frac{4}{3} \left(\frac{(H_r - H)^2}{\Delta H} \right) \right]^{-2} \tag{11.88}$$

where H is the swept magnetic field and ΔH the magnetic field separation between the maximum and the minimum in the derivative curve, assumed to be constant, although it can be orientation dependent. The spin Hamiltonian for the given situation decides the form taken by H_r . The expected line shapes of asymmetric ESR spectra based on Lorentzian and Gaussian shapes for individual lines are shown in Fig. 11.22c,d respectively while a typical experimental ESR line shape for orthorhombic symmetry in g - and A - tensors is shown in Fig. 11.22e.

ESR spectra of polycrystalline samples are related to those of their monocrystalline counterparts in an important way. The turning points in the angular dependence of resonance fields of the latter determine the peak positions of the former [44]. The derivatives of the resonant positions with respect to the spin Hamiltonian parameters including the orientations of the magnetic field may be used to calculate the peak positions and intensities of the ESR spectrum without the need to compute the complete powder spectrum. We briefly discuss ferro/antiferromagnetic resonance in the following section.

11.5.5 Ferromagnetic and Antiferromagnetic Resonance

Ferromagnets and antiferromagnets are fundamentally important and technologically applicable materials which crystallize in a variety of structures in the cubic

family. Iron, cobalt, nickel among transition metals and gadolinium and dysprosium among rare earths are examples of ferromagnets while chromium and manganese are antiferromagnetic metals. Among compounds MnO , NiO , CrO_2 and $\alpha\text{-Fe}_2\text{O}_3$ and MnF_2 are antiferromagnetic. Magnetite (Fe_3O_4) and nickel ferrite (NiFe_2O_4) are famous examples of ferrimagnetic materials. These latter compounds have ‘sublattices’ in their spinel structural framework occupied exclusively by divalent or trivalent cation leading to uncompensated magnetization which makes them ferrimagnetic.

Resonance absorption at high frequencies occurs in ferromagnets, ferrimagnets and in antiferromagnets, known as ferromagnetic resonance (FMR) and antiferromagnetic resonance (AFMR) respectively. For FMR the resonance condition is

$$h\nu = g_e\beta_e[(H_0 + \alpha\mu_0 M)(H_0 + \beta\mu_0 M)]^{1/2} \quad (11.89)$$

where h is Planck’s constant, g_e the spectroscopic splitting factor, β_e the Bohr magneton, ν the microwave frequency, H_0 the magnetic field at resonance, μ_0 the free space permeability and M the magnetization of the sample. α and β depend on field orientation and sample shape; $\alpha = \beta = 0$ for spherically shaped samples. Ferromagnetic Fe, Co and Ni have g_e -values 2.10, 2.18 and 2.21 while ferrimagnetic yttrium iron garnet and Ni ferrite have g_e ’s 2.11 and 2.18 respectively.

To observe AFMR one requires absorption of considerably more energetic photons than FMR- in the range 10^{11} – 10^{12} Hz. Why? There are two reasons: (i) large energy is stored in the antiferromagnetic solid as a result of magnetic anisotropy (H_A), (ii) a molecular field (exchange field, H_E) is exerted by each magnetic sublattice on the other. In contrast to FMR this exchange field makes nonzero contribution to the effective field. Therefore AFMR condition is

$$h\nu = g_e\beta_e\{[H_A(2H_E + H_A)]^{1/2} \pm (1 - \alpha)H_{\text{ext}}\} \quad (11.90)$$

where H_{ext} is the externally applied field and α is the demagnetizing factor appropriate for external field. A tends to zero as T approaches the Neel temperature (T_N) at which the antiferromagnet transforms to a paramagnet. As an estimate of the H_{ext} required to observe AFMR, consider MnF_2 ($T_N = 67.7$ K); for this case $H_A = 0.88$ T and $H_{\text{ext}} = 56$ T so that the value of $[H_A(2H_E + H_A)]^{1/2}$ is 9.67 T which only a superconducting magnet can supply [45]. Faraday rotation isolator, a mandatory component in all microwave equipment is essentially based on ferromagnetic resonance.

11.6 Experimental Techniques in ESR

In this section, we briefly discuss spectrometers used in the ESR study of solids namely (i) CW ESR spectrometer using a reflection cavity and a transmitter-receiver dual cavity and (ii) Pulsed ESR spectrometer.

11.6.1 Continuous Wave ESR Spectrometer

The continuous wave ESR experiment is a steady-state experiment in which continuous irradiation of the unpaired electron with microwaves of X-band frequencies (8–12 GHz, 3.75–2.5 cm waves) or Q-band frequencies (33–50 GHz, 9–6 mm waves) keeps the system in equilibrium. The CW spectrometer uses an uninterrupted monochromatic ‘beam’ of microwaves generated by a klystron, a suitable microwave guiding system, a cylindrical or rectangular metallic cavity to house the sample and a detecting system to monitor the spectrum and finally an electronic system to record the ESR spectrum.

The frequency of the klystron may be locked to the cavity frequency through auxiliary electronic circuitry and measured by comparison with a quartz crystal standard in the frequency counter.

A linearly varied static magnetic field derived from an electromagnet with the pole pieces symmetrically flanking the cavity, or, a superconducting magnet separates the Kramers’ doublets that is, two energy levels that differ only in spin orientation, and prepares the sample for the resonance experiment.

The schematic of the typical spectrometer for X-band work is shown in Fig. 11.23. The cavity may be tuned to the incoming microwaves and concentrates the radiation in its standing wave pattern. The sample is placed in the cavity in a region corresponding to the maximum microwave field. The level of power in the cavity can be ascertained by tapping a fraction of it to a crystal (Si rectifier) detector which converts the microwave radiation into a DC signal.

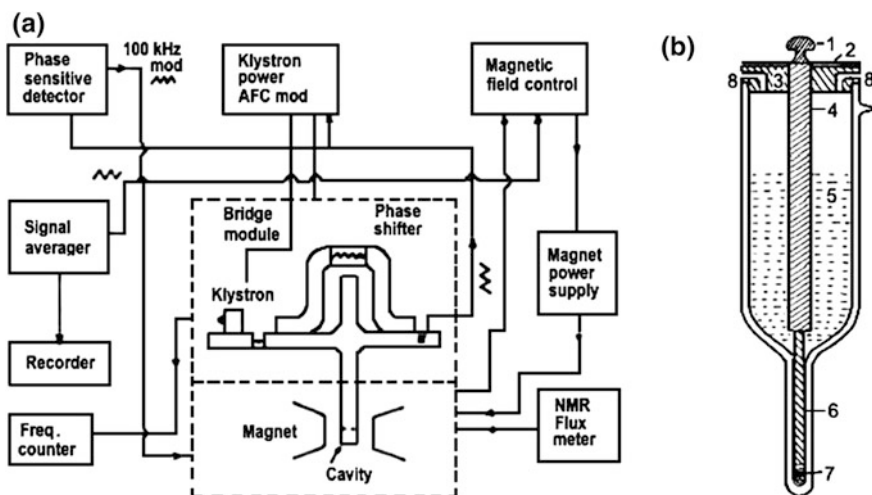


Fig. 11.23 **a** Continuous wave ESR spectrometer for X-band work. **b** Bath-type cold-finger ‘cryostat’ for single crystal ESR work at 77 K. 1-rotation head (aluminum); 2-cursor on perspex disc; 3-perspex disc graduated 0–360°; 4-perspex rod; 5-liquid nitrogen; 6-Teflon crystal holder; 7-crystal stuck with silicone grease; 8-escape for nitrogen vapours [41]

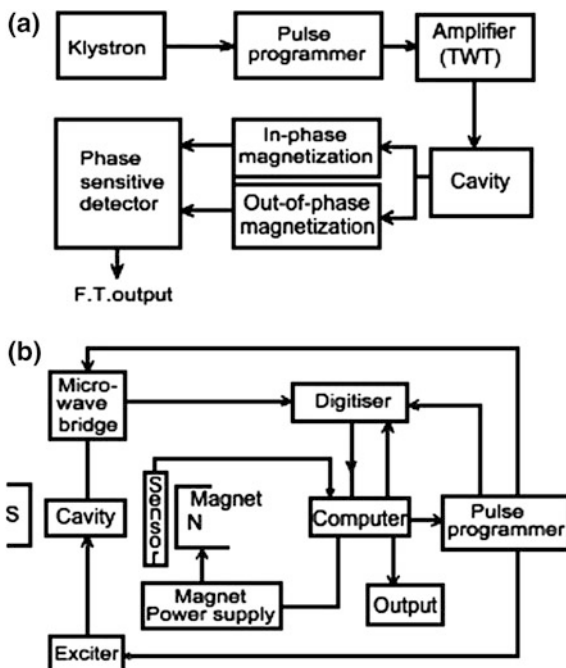
A small alternating magnetic field modulation superimposed on the static field enables the field to be swept through resonance twice during each cycle. The output signal from the crystal detector will be modulated with twice the frequency of modulation. For high sensitivity the modulation frequency should be high compared to audio-frequency. There is a trade-off between this frequency and the excess noise produced by the detecting system. A broad minimum around 30 MHz corresponds to maximum sensitivity but the excess crystal noise ($1/f$ noise) at 100 kHz is very small. This fact plus simplicity of design makes 100 kHz the standard modulation frequency of most commercial instruments. A pair of modulation coils are embedded in the wall of the cavity whose walls are thin enough to let the 100 kHz modulation without attenuation but thick enough to be a short for microwave frequencies. The static magnetic field for X-band work is ~ 0.33 T while for Q-band (~ 8 mm) work one needs fields ~ 1.3 T. Work at shorter wavelengths, say 4 mm and 2 mm, requires the use of superconducting magnets which produce fields ~ 5 T. The magnetic field should be uniform over the whole volume of the specimen to resolve spectra to better than 25 mG or $2.5 \mu\text{T}$. A Hall-effect probe located in the magnet gap stabilizes and controls the field. A nuclear magnetic resonance flux meter may also be used for such stabilization and field measurement. Resonance field positions are usually accurate to five significant figures.

The 'super heterodyne detection' involves mixing of the signal reflected from the cavity with the output of a local oscillator (klystron operating at 39 MHz above or below that of signal klystron). After detection the 100 kHz signal is sent through a narrow band amplifier to a phase sensitive detector. The voltage output of the phase sensitive detector may be changed to a pulse train by a voltage-to-frequency converter, counted and stored in the channels of a signal averager which enables repetitive scans to be made through the spectrum to improve signal to noise ratio. Finally the output may be plotted as a curve or digitized using an analog-to-digital converter and computer interfacing electronics. The ESR signal (s) thus obtained from the CW instrument is proportional to the first derivative of the absorption line contour. To avoid signal dispersion the amplitude of the modulation sweep must be less than one half the line width of the spectral line.

11.6.2 The Pulsed or Fourier Transform (FT) ESR Spectrometer

This system (Fig. 11.24a) detects the resonance in the time domain (as in NMR discussed earlier) by observing the recovery of magnetization of the spin system from a condition of extreme non-equilibrium to which it was deliberately driven previously. Microwave pulses are applied to the sample over selected finite intervals of time, so that the orientation of the precessing magnetic moment about the external magnetic field (Larmor precession) can be turned by chosen angles about

Fig. 11.24 **a** Pulsed FT ESR spectrometer schematic, **b** a general time-domain ESR spectrometer in schematic



the x , y , or z axes (with z axis kept parallel to the external magnetic field). To this end, it is possible to use a single pulse or more pulses with chosen intervals of time in between, recording a signal (referred to as an “echo signal”) that is proportional to the magnetization of the sample at appropriate intervals following application of the last pulse, a condition of extreme non-equilibrium to which it was deliberately driven previously.

In a pulsed ESR spectrometer microwave radiation from a high power (1–2 kW) klystron is converted into microwave pulses and shaped by a pulse gate. These pulses are then amplified by a traveling wave tube (TWT) amplifier and sent to the microwave cavity situated between the pole pieces of an electromagnet. The in-phase and out-of-phase components of the magnetization recovery signal are obtained simultaneously using separate detection channels. Finally these two signals are combined and recorded/displayed as the FT output.

A general time-domain spectrometer [46] used for relaxation experiments, where T_1 could be measured in the pulsed mode and the time-dependence of ESR signal upon pulsed laser excitation of the sample could be monitored, is shown in Fig. 11.24b. The computer selects the type of pulse sequence necessary and the pulse programmer generates such a pulse sequence. The computer also controls the digitizer which accepts after amplification the ESR signal from the microwave bridge and starts digitizing the signal at the falling edge of the excitation pulse. After recording the signal the digitizer transfers the digitized signal to the computer

which in turn processes and outputs it suitably. The computer also controls the magnetic field. It can either scan the magnetic field (for CW operation) or hold it at a desired value (for pulsed operation). Kevan and Bowman [47] and Mishra [48] describe and discuss more about this technique which is a ‘multifrequency’ technique.

11.7 Case Studies in ESR

ESR has been applied to a wide variety of solid state materials as shown in Table 11.2. In this section we focus on a few important ones among these. In all the applications spin Hamiltonian parameters are derived from the experiments as detailed in Sect. 11.5 to obtain the information sought.

11.7.1 Microsymmetry-Crystal Field Effect

Cubic perovskite structures such as barium titanate, lead titanate and lead zirconate (Fig. 11.25a) are capable of hosting transition metal ions such as Mn, Fe, Cu. Cu is a good system to study the effect of microsymmetry or local symmetry/structure on the EPR spectra and also investigate metastable phase formation and phase transitions thereof. Let us illustrate this with the polycrystalline EPR spectra of Cu^{2+} introduced at the Ti-site in BaTiO_3 stabilized in hexagonal and tetragonal environments of oxygens [49]. Figure 11.25b shows the room temperature Q -band (43 GHz frequency) EPR spectrum of $\text{BaTiO}_3 + 0.02 \text{ BaO} + 0.005 \text{ CuO}$ sintered at 1,400 °C. Generally the spectra are reminiscent of a Cu^{2+} ion in axial symmetry surroundings that could arise from a distorted octahedron of oxygen ions in the BaTiO_3 structure. There are two sets of EPR features which are shown magnified representing Cu^{2+} in two distinct local symmetries. Each of these two symmetries—tetragonal and hexagonal—could be exclusively stabilized through control of concentration of Cu and sintering temperature with $g_{\parallel}$ (tetragonal) $>$ $g_{\parallel}$ (hexagonal) (Fig. 11.25c). Each of these two spectra may be described by an axial spin-Hamiltonian given by

$$\mathcal{H} = g_{\parallel} \beta_e H_z S_z + g_{\perp} \beta_e (H_x S_x + H_y S_y) + A_{\parallel} S_z I_z + A_{\perp} (S_x I_x + S_y I_y) \quad (11.91)$$

with $S = 1/2$ and $I = 3/2$. $g_{\parallel}$ and $g_{\perp}$, $A_{\parallel}$ and $A_{\perp}$ are the principal axis components of the g and A tensors respectively. Table 11.3 gives the spin Hamiltonian parameters determined by simulation.

Table 11.2 Applications of ESR in solid state physics

Type of material		Information sought
I	Metals and metallic materials	
A	Ferromagnetic metals and alloys	Magnetic phase identification, magnetization, damping and magneto-striction parameters
	(a) Crystalline	
	(b) Glassy	
B	Superconductors	Metal-insulator transitions, phase purity in ceramics, microwave impedance and penetration depth
II	Semiconductors	
	(a) Crystalline	Defect type location, concentration and geometry
	(b) Amorphous	Annealing behaviour under plastic deformation, irradiation and implementation
III	Insulators	
A	<i>Inorganic</i>	
	(a) Monocrystalline	Microscopic structure and electronic information
	(b) Polycrystalline	
	(c) Glassy	
B	<i>Organic</i>	
	(a) Monocrystalline	Free radical production, identification and detection, kinetics, motion of spin labels
	(b) Polycrystalline	
C	<i>Polymers</i>	
	(a) Conducting	Behaviour under stress, irradiation and chemical doping. Defect concentration, correlation to quasi-particle motion and conduction mechanism
	(b) Nonconducting	Conformation, dynamics and polymerization mechanism
D	<i>Ferroelectrics, optoelectronics and superionics</i>	Phase transitions, defect orientations, thermal and optical stability
E	<i>Catalysts</i>	Nature and number of surface species and its bonding to support

Both principal g values are greater than free electron g of 2.0023 and $g_{||} > g_{\perp}$ for both spectra as expected for ions with $3d^9$ configuration at distorted octahedral sites. The five-fold degenerate $3d$ orbitals are split by an octahedral crystal field into a triplet T_{2g} and a doublet E_g . A tetragonal distortion caused by the Jahn-Teller effect (JTE) splits the deeper lying state E_g of the $3d^9$ configuration further leading to an orbital singlet ground state. Thus the measured spectra of Cu^{2+} in BaTiO_3 can only be obtained if the Cu^{2+} ions are in octahedral environments elongated along one of the cubic axis leaving the bound state $d_{x^2-y^2}^2$ orbitally non-degenerate. Cu^{2+} ions are in octahedral environments and exhibit axial symmetry and hence they are incorporated at Ti-sites in the BaTiO_3 lattice confirming the assumptions derived from

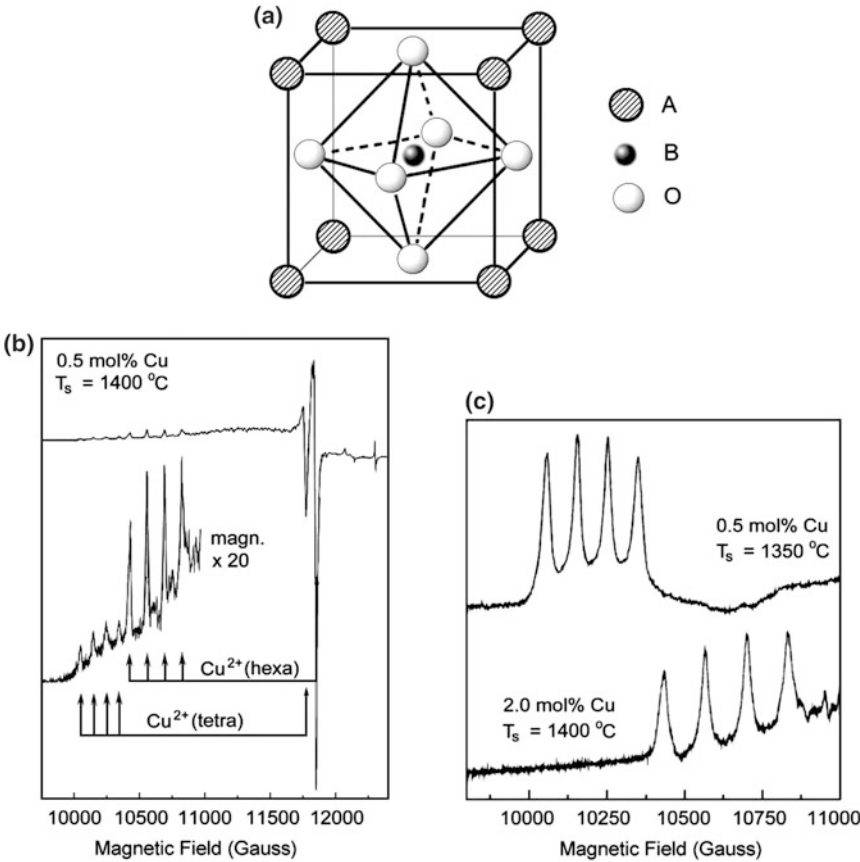


Fig. 11.25 **a** The perovskite structure (ABO_3). In $BaTiO_3:Cu$, Cu impurity occupies B or Ti site. **b** Room temperature EPR spectrum of a sample with a nominal composition of $BaTiO_3 + 0.02 BaO + 0.005 CuO$ sintered at $1,400\text{ }^\circ C$ in the Q -band (34.0 GHz). Two different spectra of Cu^{2+} in two distinct microsymmetries in the $BaTiO_3$ structure—hexagonal and tetragonal—are seen with easily identified parallel and perpendicular components of g tensor. **c** Low-field part of the room temperature EPR spectra (Q -band) of single phase tetragonal (*top*) and hexagonal $BaTiO_3$ [49]

Table 11.3 Cu^{2+} in tetragonal (T) and hexagonal (H) $BaTiO_3$: principal g and A values [49]

	$g_{ }$	$g_{\perp}$	$A_{ }$ (10^{-4} cm^{-1})	$A_{\perp}$ (10^{-4} cm^{-1})
<i>T phase</i>				
40 K	2.3852(6)	2.0663(6)	110(2)	8(4)
300 K	2.3811(6)	2.0645(6)	108(2)	8(4)
<i>H phase</i>				
40 K	2.2848(6)	2.0518(5)	151(2)	13(4)
300 K	2.2841(6)	2.0508(5)	140(2)	12(4)

considerations of ionic radii and other experimental data. Doping of Cu^{2+} at Ti^{2+} would create oxygen vacancies in the BaTiO_3 lattice. These could lie in the first coordination sphere of the copper complex perpendicular to the σ -bonds and thus stabilize the local tetragonal symmetry. Cu also influences the high-temperature cubic-hexagonal phase transformation at high temperature under two conditions: (i) existence of a sufficiently high density of oxygen vacancies and (ii) the driving force should be an initiator of the phase transformation. Dynamical JTE could drive the transition which means that the degeneracy of the d^1 configuration of Ti^{3+} in undoped BaTiO_3 would be eliminated by a distortion of the regular oxygen octahedron of the cubic phase leading to the two distorted octahedra of the hexagonal structure. Degenerated states occur both in higher single electron E_g level ($\text{Cu}^{2+}(3d^9)$, also $\text{Mn}^{3+}(3d^4)$) and in the lower single electron T_{2g} level $\text{Ti}^{3+}(3d^1)$ leading to strong and weak JTE respectively so that Cu^{2+} should cause a strong JTE and a decrease in cubic-hexagonal transition temperature. Hexagonal phase is present in BaTiO_3 doped with 2 mol % Cu implying that O^{2-} vacancies and JTE active acceptor ions are high enough and the ionic mobilities are adequate.

11.7.2 Superconductors

Microwave penetration depth (λ) is an important parameter that characterizes a high temperature superconductor (e.g. $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_8$ or 2212 with $T_c = 82$ K) in

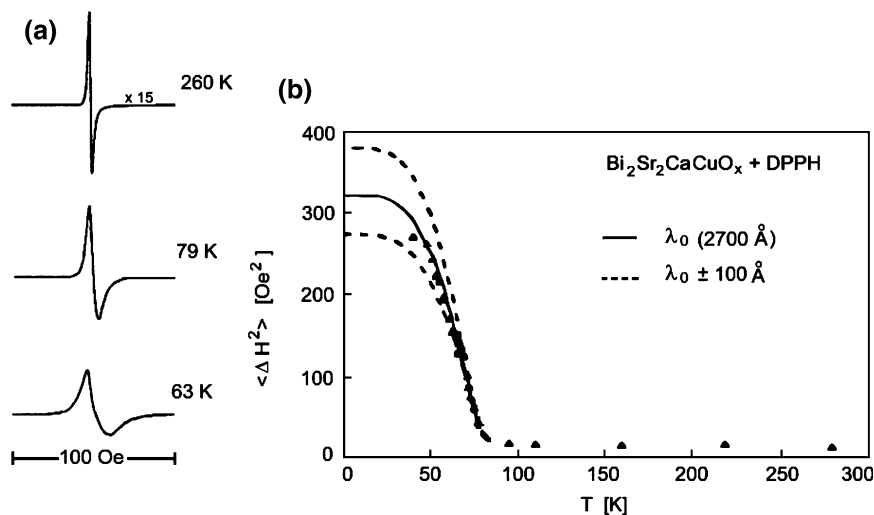


Fig. 11.26 **a** ESR spectra of DPPH adsorbed on Bi-based 2212 ($T_c = 82$ K) superconductor for determination of microwave penetration depth. Significant line broadening and asymmetry at $T < T_c$ is due to flux lattice formation, **b** temperature dependence of line width of DPPH ESR signal. *Solid line* theoretical fit (see text) which yields a λ_0 of 270 nm with $T_c = 84$ K. *Dashed lines* Fits with 270 ± 10 nm which shows that λ_0 deduced is accurate to better than 3.7 % [50]

connection with the formation of the so-called flux-line lattice and applications thereof. DPPH may be easily adsorbed on the surface of a superconductor pellet by immersing it in a benzene solution of DPPH. Rakvin et al. [50] have studied the temperature-dependent ESR spectra of DPPH-adsorbed on 2212 in order to determine λ_0 . Figure 11.26a displays typical spectra wherein at $T \leq T_c$ the ESR line is inhomogeneously broadened due to the formation of the flux-line lattice. From the temperature-dependent line width the magnetic penetration depth λ_0 at $T = 0^\circ\text{K}$ is determined. The variation of λ_0 follows the law $[1-(T/T_c)^4]^{1/2}$ which could be used to fit experimental line-width data (Fig. 11.26b). Significantly, experiments on single crystal samples show that the main contribution to line-broadening comes from the flux lattice and not from granularity and screening effects.

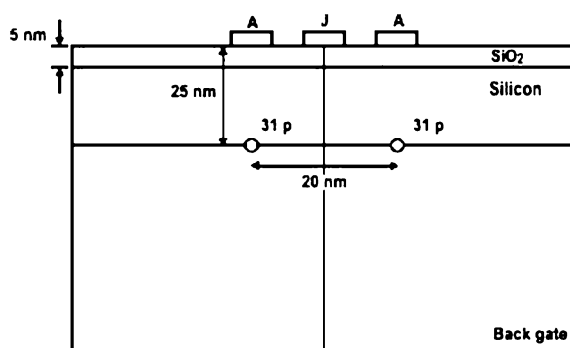
11.8 Current Trends and Developments

In this section we briefly discuss two new magnetic resonance techniques and experiments that have implications for areas such as single spin detection and quantum computation besides device characterization.

11.8.1 Si Quantum Computer

If a single phosphorus atom is placed in the vicinity of a silicon transistor and the atom is irradiated by microwaves, one can write quantum information on the spin of its nucleus. This is the idea of the quantum computer. Kane proposed the idea in 1998 [51] conceptualized in Fig. 11.27. The potential for quantum computing comes from harnessing the laws of quantum physics that allow the spin state of an electron or an atom's nucleus to achieve “superposition”—existing in more than one state at a time. A classical computer bit can exist either as a 1 or a 0, but a ‘quantum bit’ or qubit is capable of existing in multiple states at the same time. Kane suggested harnessing the nuclear spin of phosphorus atoms (^{31}P , $I = 1/2$) embedded in

Fig. 11.27 Principle of Kane's quantum computer



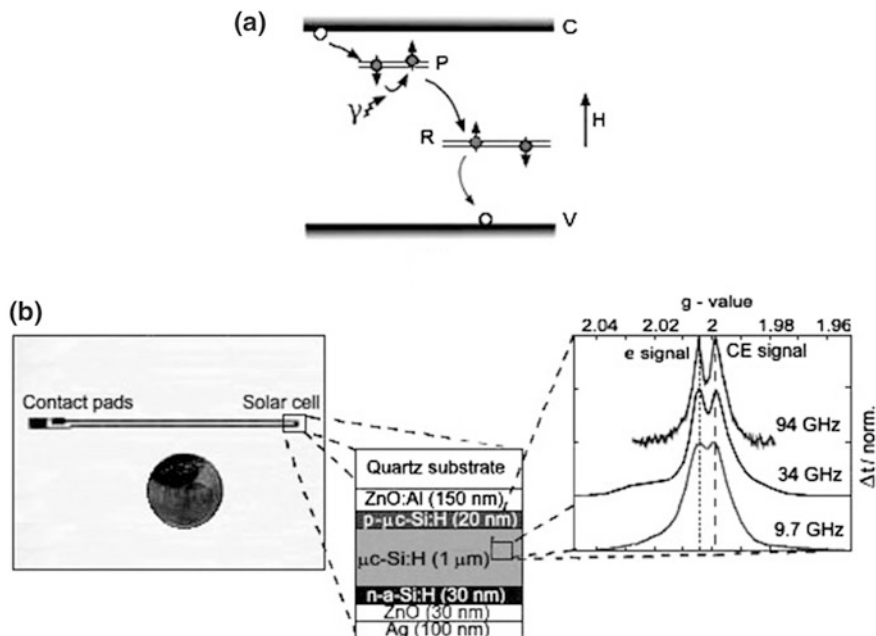


Fig. 11.28 The EDMR mechanism. **a** A donor impurity (P) sits just below the conduction band (C). An acceptor (R) sits between the donor and the valence band making a recombination pathway for the donor electron (small filled circle) to recombine with a hole in the valence band (V). A photon (γ) of a specific frequency can flip spins against the magnetic field (H). **b** Pulsed multi-frequency electrically detected magnetic resonance (EDMR) at X-, Q- and W-Band (9.7, 34, and 94 GHz)

a silicon crystal as a qubit. Si-based quantum computing also offers long coherence times for electron and nuclear spins. That means the electron spin states and nuclear spin states acting as qubits could hold on to their information for long periods of time, something that other quantum computing schemes have struggled with.

Nuclear spins alone will not interact significantly with other nuclear spins 20 nm away. Although nuclear spin is useful to perform single-qubit operations, to make a quantum computer, two-qubit operations are also required. This is the role of electron spin in this design. Under A -gate control (Fig. 11.28), the spin is transferred from the nucleus to the donor electron. Then, a potential is applied to the J gate, drawing adjacent donor electrons into a common region, greatly enhancing the interaction between the neighbouring spins. By controlling the J gate voltage, two-qubit operations are possible. In a significant experimental development, the first step towards the realization of Si quantum computer has been taken [52].

11.8.2 EDMR of Silicon Thin Film Solar Cell

Electrically detected magnetic resonance (EDMR) [53] is a materials characterization technique that improves upon ESR. It involves measuring the change in the electrical resistance of a sample when exposed to certain microwave frequencies. It can be used to identify very small numbers (down to a few hundred atoms) of impurities in semiconductors and devices. It has been applied to ion-implanted Si:P nanostructures [54] and quantum dots [55]. The mechanism and implementation of EDMR are given in Fig. 11.28.

Pulsed multi-frequency electrically detected magnetic resonance (EDMR) at *X*-, *Q*- and *W*-Band (9.7, 34, and 94 GHz) was applied to investigate paramagnetic centers in microcrystalline silicon thin-film solar cells under illumination. The EDMR spectra are decomposed into resonances of conduction band tail states (*e* states) and phosphorus donor states (*P* states) from the amorphous layer and localized states near the conduction band (*CE* states) in the microcrystalline layer. The *e*-resonance has a symmetric profile at all three frequencies, whereas the *CE* resonance reveals an asymmetry especially at *W*-band. This is suggested to be due to a size distribution of Si crystallites in the microcrystalline material. A gain in spectral resolution for the *e* and *CE* resonances at high fields and frequencies demonstrates the advantages of high-field EDMR for investigating devices of disordered Si. The microwave frequency independence of the EDMR spectra indicates that a spin-dependent process independent of thermal spin-polarization is responsible for the EDMR signals observed at *X*-, *Q*- and *W*-bands.

11.9 Summary and Outlook

In this chapter we have introduced the two basic experiments of Stern and Gerlach and Rabi et al. which laid the foundations of magnetic resonance. Four general concepts: Larmor precession, magnetization, complex susceptibility and spin Hamiltonian were developed as a platform to discuss theory, experimental techniques and applications of NMR and ESR. The analysis of single crystal and powder NMR and ESR spectra proceed along similar lines to yield chemical shift, Knight shift and quadrupole coupling constants as well as *g*-tensor and hyperfine tensor. Bloch equations provide a sound basis for performing nuclear and electron relaxation experiments using pulsed magnetic fields. The applications discussed provide a motivation to look for new ideas to apply the methods to solve solid state and materials physics problems. Finally emerging applications such as high field NMR and high frequency ESR as well as the silicon quantum computer ensure that the technique will inspire a whole new generation of students in solid state physics.

11.10 Problems

1. (a) Verify that the Larmor frequency of ^{85}Rb is 700 kHz/G. (b) Assuming a static magnetic field of 1 Tesla compute the frequency required to perform ^{85}Rb NMR. (c) The sensitivity of the NMR absorption depends on the difference in population of the two Zeeman levels of the nucleus (n_1, n_2 at 300 K) whose degeneracy is lifted by the static magnetic field. This population difference follows the Boltzmann distribution. For NMR absorption to be detected at 10 MHz, what is the difference in population?
2. The envelope of the free induction decay function $F(t)$ following a 90 degree pulse is proportional to the FT of the absorption line shape $g(u)$. Defining $u = \nu - \nu_0$ show that for a symmetrical line

$$F(t) = A \int_{-\infty}^{\infty} g(u) \cos ut \, du$$

where A is a constant. (a) Expand the right side of the $F(t)$ equation in a power series in t and show that the even moments of the line shape (note: second moment is the line width) given by the formula

$$\langle \Delta \nu^{2n} \rangle = (-1)^n (d^{2n} F(t) / dt^{2n})_{t=0} / F(0).$$

(b) Abragam's [18] two-parameter formula for the free induction decay of a dipolar solid is $F(t) = \exp\{-(a^2 t^2 / 2)\} (\sin bt) / bt$. Show that the line shape $g(u)$ that corresponds to this form of $F(t)$ results from the superposition of Gaussian curves of root mean square half-width a , with a rectangular envelope of width $2b$. (c) From the result of (a) and the expansion of $F(t)$ given in (b) show that the parameters a and b are related to the second and fourth moments $\langle \Delta \nu^2 \rangle$ and $\langle \Delta \nu^4 \rangle$ by $a^2 + (1/3)b^2$ and $3a^4 + 2a^2 b^2 + (1/5)b^4$ respectively.

3. The population difference n in the two level system of spins with populations N_+ and N_- is perturbed by an rf field. Show that $dn/dt = 0$.
4. Derive the principal g -values of the G matrix (Sect. 11.5) by solving the eigen value problem $|G - \lambda I| = 0$ where λ 's are the eigen values and I is the unit matrix. Compare your result to those obtained by the Schonland's method.
5. ESR of F -centers has been observed in irradiated (a) RbCl crystals and (b) KCl crystals. Reason and discuss as to why the spectrum is well-resolved in (a), but featureless in (b). What information does the spectrum of RbCl give?
6. Cu^{2+} and Mn^{2+} can be introduced as impurities into MgO crystal with NaCl structure, the impurity substituting Mg^{2+} site with six O^{2-} ions as nearest neighbours (octahedral symmetry). However, at room temperature while Mn^{2+} gives an EPR, Cu^{2+} does not give an EPR spectrum. Why? What must one do to observe Cu^{2+} EPR in MgO? (Refer to [31]).

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Chapter 12

Fast Ion Conduction

Abstract Fast ion conduction refers to the rapid movement of ions via diffusion of a large number of in-built defects in certain ionic crystals leading to ionic conductivities in the range $1\text{--}10\ \Omega^{-1}\text{cm}$ around ambient temperature. Such conductivity due to fast diffusion of Ag^+ , Li^+ , F^- , O^{2-} is possible only in a few special crystal structures supported by a robust sublattice such as Γ in AgI , O^{2-} in LiCoO_2 , PO_4^{3-} in LiFePO_4 , Pb^{2+} in PbF_2 and Zr^{4+} in Y-doped ZrO_2 . Beginning from a traditional thermodynamical-statistical basis of Schottky and Frenkel defects, the nature of fast ion conduction is discussed via specific examples. A fundamental model for fast ion conductor is discussed which predicts solid-like ion motion at low temperatures that changes to liquid-like motion at high temperatures. Tubandt's method for ion transport number measurement, AC conductivity and tracer diffusion coefficient experiments are described. A few applications, like the all-solid-state battery, fuel cells, chemical sensors and nanoscale memory devices are discussed.

12.1 Introduction

Fast ion conduction is a phenomenon dating back to Michael Faraday who first noted high solid state conductivity. In 1833 he reported that Ag_2S conducted electricity in the solid state and observed that hot lead fluoride also conducted electricity via F^- ions [1]. The chemistry Nobel Laureate Walther Nernst observed ($\sim 1,900$) that mixed oxides of ZrO_2 and Y_2O_3 glowed white hot when a current was passed through them at high temperatures. This unusual behavior was attributed to oxygen ion conduction forming the basis for the Nernst glower lamp. However, it was in 1914 that the rare and unique electrical property of AgI was discovered by Tubandt and Lorenz [2] when they observed that above $147\ ^\circ\text{C}$ AgI exhibited an unusually high ionic conductivity of $1.3\ \Omega^{-1}\text{cm}^{-1}$. This is larger by several orders than the conductivity of normal ionic conductors ($\sim 10^{-8}\text{--}10^{-5}\ \Omega^{-1}\text{cm}^{-1}$) and comparable with that of liquid electrolytes ($\sim 10^{-3}\text{--}1\ \Omega^{-1}\text{cm}^{-1}$). AgI

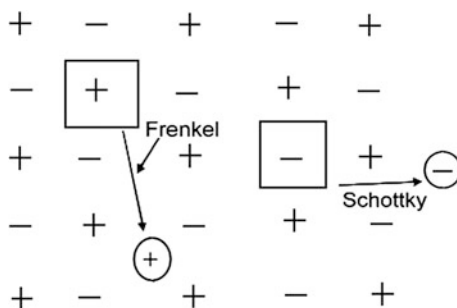
is representative of a new class of solids called solid electrolytes or ‘fast ion conductors’. This chapter is devoted to a brief study of fast ion conductors. In the general context of solid state physics fast ion conductors support extensive structural disorder based on Frenkel defects and possess ‘liquid-solid duality’ and thus challenge basic solid state concepts of well-defined Brillouin zone and optical phonons [3].

12.2 Nature of Ionic Conduction

Crystalline solids may be broadly divided into (i) metallic and (ii) nonmetallic solids. Nonmetallic solids include semiconductors and ionic solids. Ionic solids are composed of oppositely charged ions—cations (with + charge) and anions (with - charge)—arranged alternately on the points of a specific crystal lattice. Ion formation (example, Na^+ and Cl^- ions in the face-centered cubic lattice) arises from the tendency of atoms to become stable by attaining closed-shell configuration.

While ionic solids are perfect insulators at absolute zero, they become imperfect at temperatures greater than $T = 0$ K. An imperfect ionic crystal contains defects mainly arising from lattice vibrations which start as soon as $T > 0$ K. These defects are charged and atomistic and play a very crucial role in ion conduction. Conductivity by the motion of ions via these defects—mainly vacancy and the interstitial—is called ionic conductivity. Cation vacancies and an equal number of anion vacancies (to maintain electrical neutrality of the ionic crystal) are Schottky defects. Cation interstitials and their associated cation vacancies are cation Frenkel defects (anion Frenkel defects also arise). They are illustrated in Fig. 12.1. A Schottky defect (in say NaCl) is formed by the movement of a negative (Cl^-) or positive (Na^+) ion from the interior to the surface of the crystal to produce cation and anion vacancies individually or in pairs with a formation energy of 2.07 eV. A Frenkel defect is formed (say in AgCl) by a cation moving to an interstitial site within the crystal (with a formation energy of 1.35 eV) so that the Ag^+ vacancy and the Ag^+

Fig. 12.1 Schottky and Frenkel defects in ionic crystals



interstitial constitute a Frenkel defect pair. Movement of Na^+ and Ag^+ ions causes ion conduction in NaCl and AgCl respectively. A tendency for fast ion conduction is indicated by the fact that just below the melting point the concentration of Frenkel defects in AgCl is 0.6 %, that is, 1 in 200 Ag^+ ions have moved into interstitial locations.

The number n_S of Schottky defect pairs that can form at a temperature T is given by

$$n_S \approx N \exp(-E_S/2k_B T) \quad (12.1)$$

where N is the total number of lattice sites, E_S the energy of formation of a Schottky pair and k_B the Boltzmann constant [4]. The number of Frenkel defects n_F which is much smaller than the number of lattice sites N and the number of interstitial sites N' is given by

$$n_F \approx (NN')^{1/2} \exp(-E_F/2k_B T) \quad (12.2)$$

where E_F is the energy necessary to remove an atom from a lattice site to an interstitial position to form a Frenkel pair [2].

While the production of Schottky defects lowers the density of the crystal through an increase in volume but not mass, creation of Frenkel defects does not change density in view of zero volume change. Schottky defects are important for normal ionic conduction while Frenkel defects are important for fast ion conduction.

In general, the ionic conductivity of a binary compound (NaCl or AgCl) is given by

$$\sigma = n(Ze)\mu \quad (12.3)$$

where n is the number of predominant mobile ions/defects, Ze is the charge on the mobile ion and μ the mobility of the ion. σ is a function of temperature and a measurement of the temperature dependence enables determination of defect formation and migration energies in a given crystal. Note that ion mobility is temperature dependent and μ and σ are obtainable from ion diffusivity (D) by using the Einstein relation [3].

$$k_B T \mu = ZeD \quad (12.4)$$

The density of vacancies is strongly temperature dependent and increases rapidly with increase in T , with an upper limit ≥ 0.01 mol % for most solids. The vacancies cause an expansion of the crystal lattice—one of the reasons for the thermal expansion of solids [5].

Equation (12.3) is basically Ohm's law, $J = \sigma E$, valid for all solids including metals. However unlike in metals and semiconductors, the mobile ions in ionic

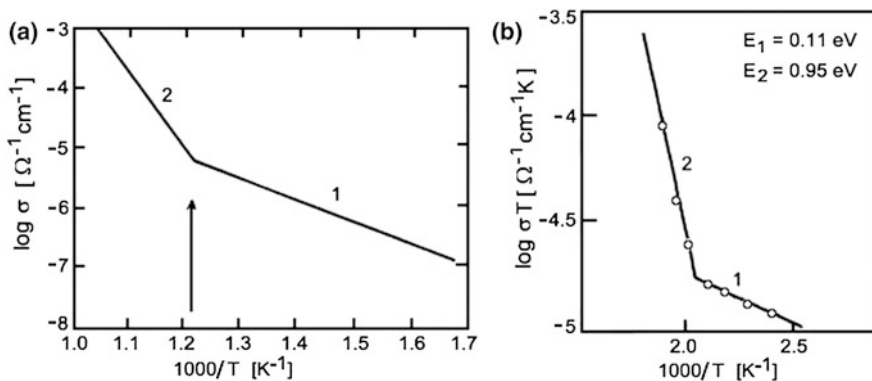


Fig. 12.2 **a** Typical conductivity plot for NaCl. Arrow break separating intrinsic and extrinsic regions. **b** Conductivity plot of a fast ion conductor LiNaSO₄ obtained by deep quenching from 900 °C to room temperature

crystals must be physically transported across two electrodes from cathode to anode so that both charge and mass transport is involved.

The defects could be intrinsic (thermally generated) or extrinsic—created by doping the ionic crystal with aliovalent impurities (say Ca⁺⁺ in NaCl). The temperature dependence of ionic conductivity $\sigma(T)$ is given by

$$\sigma(T) = (\sigma_{01}/T) \exp(-E_1/k_B T) + (\sigma_{02}/T) \exp(-E_2/k_B T), \quad (12.5)$$

where (σ_{01}/T) and (σ_{02}/T) are pre-exponential factors that are related to ion jump (diffusion) mechanisms. E_1 is the thermal activation energy for defect migration and E_2 for combined migration and creation.

Experimental data usually plotted as $\log \sigma T$ or $\log \sigma$ versus $1,000/T$ (the Arrhenius plot) ideally exhibits two linear segments separated by a knee (Fig. 12.2a). The slope of the low temperature plot is $E_1/k_B = E_m$ where E_m is the migration energy. The slope of the high temperature plot is $E_2/k_B = E_m + E_F/2$ where E_F is the defect formation energy. From the values of the two slopes, E_m and E_F can be estimated separately. Figure 12.2a is the conductivity plot for NaCl and Fig. 12.2b that for melt-quenched LiNaSO₄. The conductivity of the latter is higher than that of the former by an order of 2–3. The former is a normal ionic conductor while the latter is a fast ion conductor [6]. For more details see Lidiard [7] and Hayes and Stoneham [8].

It is to be noted that in a typical ionic crystal the ionic conductivity, either by chemical doping or thermal activation, is not substantial and thus may be termed ‘slow ion conduction’. This is basically due to the close-packed crystal structure and rather low defect concentration besides discreteness of ion hops. However it is desirable to overcome these difficulties to enhance the conductivity. In the next section we consider fast ion conduction.

12.3 Fast Ion Conduction

Fast ion conduction is a process quite different from the normal ion conduction. In materials exhibiting fast ion conduction there is no need to create point defects thermally and the migration energy for defects is of the order of $k_B T$. The ionic conductivity of a fast ion conductor is in the range $10^{-3} \Omega^{-1} \text{ cm}^{-1}$ – $10 \Omega^{-1} \text{ cm}^{-1}$ at ambient temperature. In this section we shall discuss some general aspects of fast ion conduction.

12.3.1 General Characteristics

The Frenkel defect is of much interest in fast ion conduction. Silver halides—unlike alkali halides—possess Frenkel defects. Thus AgCl, which has a lower energy of formation of Frenkel defect compared to larger Schottky defect formation energy of NaCl, already exhibits an enhanced conductivity relative to NaCl. AgBr has a very interesting conductivity behavior especially at temperatures near T_{melting} with its Frenkel defect concentration showing an abrupt increase. This increase is almost like a virtual phase transition to a fast ion conducting phase except that such a transition is interrupted by melting.

While AgCl and AgBr have the same structure as NaCl and NaBr, AgI does not have an fcc structure (unlike NaI), but a zincblende or wurtzite structure instead. Let us point out the difference between a normal ion conductor and a fast ion conductor through NaI and AgI. Figure 12.3 compares the latter two structures from the point of view of ion conduction process through Schottky and Frenkel defects respectively.

The essential requirements for a fast ion conductor are the following:

1. A high concentration of ionic charge carriers with a small ionic radius ($<0.15 \text{ nm}$).
2. A high concentration of vacancies or interstitial ions.
3. A low activation energy for ion hopping.
4. An appreciable degree of non-stoichiometry.
5. An unusually small entropy and attempt frequency of ion hopping so that the lattice is undisturbed during ion transport.

A generally common feature of fast ion conductors is the possession of a somewhat continuous open path for the diffusion of interstitial ions. An open structure arising from the marginally covalent Ag-I bond such as in AgI (Fig. 12.3b) ensures that all these requirements are satisfied. The partially covalent Ag-I bond in AgI creates easy paths for migration of Ag⁺ ions literally paving the way for fast ion conduction. In fact, structures with ‘liquid-like molten sublattice’ [9, 10] have mobile ion/defect concentrations of $\sim 10^{22}$ per cc so that all the above criteria are satisfied, resulting in a conductivity of $\sim 0.1 \times 10^{-3} \Omega^{-1} \text{ cm}^{-1}$ at $T > 420 \text{ K}$.

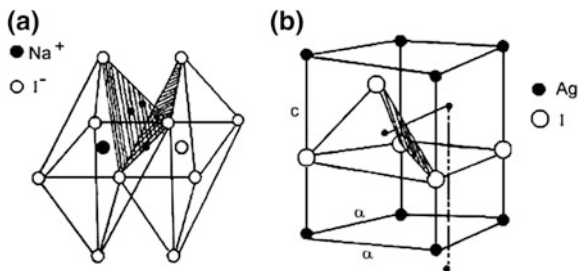


Fig. 12.3 Defects in **a** fcc NaI with Γ octahedra around Na^+ and **b** wurtzite AgI with Γ tetrahedral around Ag^+ . NaI is a normal ion conductor while hexagonal AgI is a fast ion conductor above 147 °C when it transforms to bcc phase through a massive first order phase transition. Note conduction path in **b**

Figure 12.4 compares the conductivity profiles of NaI and AgI. The conductivity plots of $\beta\text{-PbF}_2$ and K β -alumina (to be discussed later) are also included in Fig. 12.4b.

In a fast ion conductor the crystal structure ensures a large concentration of mobile ions and defects so that no formation energy is involved. Thus (12.5) reduces to

$$\sigma = \sigma_{01} \exp(-E_m/2k_B T). \quad (12.6)$$

E_m is very small, <0.1 eV (0.05 eV for bcc AgI), so that

$$\sigma = \sigma_0 \exp(-E_m/2k_B T) \quad (12.7)$$

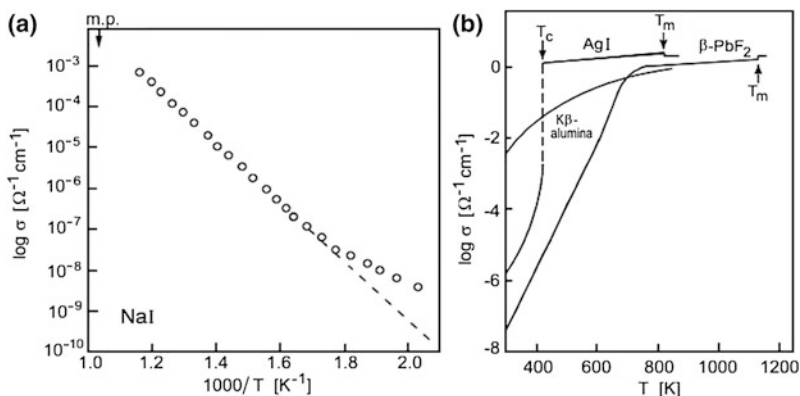


Fig. 12.4 Ionic conductivity of **a** NaI normal ion conductor and **b** AgI, a fast ion conductor. In **b** conductivity of $\beta\text{-PbF}_2$ and K β -alumina are also included for further discussion

and the Arrhenius plot is a single straight line with a small slope; σ_0 is the pre-exponential factor. Note that (12.7) is qualitatively different from (12.6) (see Sect. 12.3.1). Equation (12.6) was derived from the theory of NaCl-type conductors. It is important to remember that here the defect formation energy is an important component. While making the transition to ‘fast’ ion conductors as is done here by way of (12.7) it is to be realized that there is no defect formation here. So $E_F = 0$. Thus both (12.6) and (12.7) have to be written although they appear redundant.

12.3.2 Classification of Fast Ion Conductors

The conductivity profiles of AgI, PbF₂ and K- β alumina (Fig. 12.4b) forms the basis of classification of fast ion conductors. Boyce and Huberman [11] have proposed this classification based on the manner in which the fast ion conducting state is realized in a particular compound. There are essentially three types.

Type I: These become fast ion conductors above the temperature of a first order structural phase transition. The wurtzite phase of AgI or the β -phase transforms to the bcc or α -phase when σ abruptly jumps by three orders of magnitude at the first order phase transition temperature viz., 420 K. For $T > 419$ K the number of mobile Ag⁺ ions becomes $\sim 10^{22}$ per cc through a process called ‘sublattice melting’ which implies that practically all Ag⁺ ions in AgI contribute to ion transport.

Type II: This group of compounds attain high conductivity following a gradual and continuous disordering process within the same crystallographic phase. Below the phase transition temperature T_c illustrated by β -PbF₂ in Fig. 12.4(b) the number of conducting defects increase rapidly before saturating at $T > T_c$. Specific heat anomaly and lattice expansion accompany ionic conductivity changes.

Type III: In this type of materials no clear phase transition occurs. However high levels of ionic conduction are attained through increased mobility of a fixed number of thermally activated defects. Arrhenius plots of such materials are linear. Sodium- or potassium β -alumina is an example.

Tables 12.1 and 12.2 list important fast ion conductors along with their room temperature structures and phase transition temperatures. In the next subsection we briefly discuss important structures of fast ion conductors.

12.3.3 Structural Varieties

Materials that exhibit unusually fast ionic conduction do so because of special characteristics related to their crystal structures. Therefore we briefly discuss the most important crystal structures.

Table 12.1 Fast ion conductors (Type I)

Compound	Structure (low temp)	Phase transition temperature [K]	Entropy [J/K mol]	Ionic conductivity (σ) [$\Omega^{-1} \text{ cm}^{-1}$]	Temperature [K]
RbAg ₄ I ₅	(Rhombohedral)	122		0.25	300
	(Cubic)	209			
AgI	Wurtzite (hexagonal) (β)	419 (bcc, α)	14.5	1.3	419
Ag ₂ S	Monoclinic	830 (melting)	11.3		
		452 (bcc)	9.3	4×10^{-5}	293
Ag ₂ Se	Orthorhombic	1,115 (melting)	12.6	4.1	473
		406 (bcc)	10	3.1	473
Ag ₂ Te	Monoclinic	1,170 (melting)			
		423 (fcc)		6×10^{-4}	351
Ag ₃ SI	Rhombohedral (γ)	1,075 (melting)			
		157 ($pm\bar{3}m$, β)		1.0	523
CuBr	Zincblende	519 ($Im\bar{3}m$, α)			
		664 (bcc)	9.0	0.017	588
CuI	Zincblende	761 (melting)	12.6		
		642 (hep)	1.8		
Cu ₂ S	Orthorhombic	(fcc)		0.09	723
		878 (melting)			
		376 (hexagonal)	9.9	Chalcocite 52-0.07, T_c (371–381)	
		708 (fcc)	1.8	Cu _{2-x} S 0.07-2400 ($x = 0-0.2$)	
		1,402 (melting)		(mixed electronic-ionic conductor)	

(continued)

Table 12.1 (continued)

Compound	Structure (low temp)	Phase transition temperature [K]	Entropy [J/K mol]	Ionic conductivity (σ) [$\Omega^{-1} \text{ cm}^{-1}$]	Temperature [K]
Cu ₂ Se	Tetragonal	395 (fcc)		0.11	423
		1,113 (melting)			
		1,428 (melting)	19.6		
Li ₂ SO ₄	Monoclinic	848 (fcc)		~ 1	1,073
		1,130 (melting)			
Li ₂ O	Antifluorite	1,350 (superionic phase)		0.001	1,273
		1,705 (melting)			

Table 12.2 Fast ion conductors (Type II and Type III)

Compound	Structure (low temp)	Phase transition tem- perature [K]		Ionic conductivity (σ) [$\Omega^{-1}\text{cm}^{-1}$]	Temperature [K]
		T_{σ} saturation ^a	T_{melting}		
<i>Type II</i>					
PbF ₂	Fluorite	703	1095	5	~ 873
CaF ₂	Fluorite	1,423	1691	0.0001–45	713–1,373
SrF ₂	Fluorite	1,453	1673	0.00014–1.4	853–1,253
BaF ₂	Fluorite	1,233	1593	0.0001–25	703–1,353
SrCl ₂	Fluorite	993	1148	0.2	1,000
LaF ₃	Tysonite	–	1766	0.003–1.2	403–943
ZrO ₂ -10 mol % Y ₂ O ₃	Fluorite			0.3–50	873–1,673
<i>Type III</i>					
Li - β alumina				0.0013	298
Na- β alumina				0.014	298
Na ₂ ZrSi ₄ O ₁₁				6.0×10^{-5}	573

^aTemperature at which conductivity saturates

12.3.4 *RbAg₄I₅*

The crystal structure of RbAg₄I₅ determined through single crystal X-ray diffraction data has revealed a tetramolecular cubic unit cell with a lattice constant of 1.124 nm. Structure refinement fixes the space group as *P*4₁3 [12]. The arrangement of Γ^- ions is such that it provides fifty six iodide tetrahedral sites per unit cell, These sites share faces in such a manner as to provide diffusion paths for the silver ions. Furthermore two low temperature phases exist (Table 12.1).

12.3.5 *α -AgI*

In α -AgI a bcc arrangement of Γ^- ions exists (Fig. 12.5a) within which Ag⁺ ions move interstitially. Ag⁺ ions are so mobile that it is hard to determine their position by X-ray diffraction. Presumably they move rapidly among sites with threefold and fourfold coordination within the relatively static iodine structure in the unit cell. Just two Ag⁺ ions have the freedom to occupy as many as 42 positions in the unit cell proposed by Stroock [9]. Interestingly the density of this fast ionic conducting

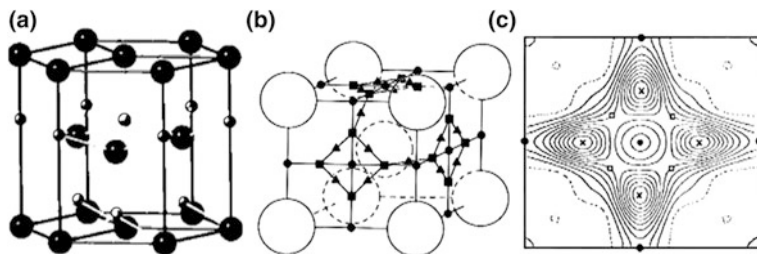


Fig. 12.5 **a** Structure of Wurtzite or β -AgI. Small spheres: Ag, big spheres: I, **b** Stroock model for the crystal structure of α -AgI (bcc), big circles: I^- . 42 sites (shown as black triangles, squares and circles) are available for occupation by 2 Ag^+ ions in the unit cell. **c** Diffusion path for Ag^+ ion for α -AgI in the $(x, y, 0)$ projection from neutron scattering experiment at 160 °C. X: tetrahedral sites, (●): octahedral sites, □: trigonal sites providing unhindered passage

phase is higher than that of the ambient temperature wurtzite phase. The lattice parameter of α -AgI is roughly about one half of that of the much better conducting $RbAg_4I_5$ at 0.506 nm which indicates the role of free volume on ionic conductivity.

In the three dimensional iodine sublattice of α -AgI (Fig. 12.5b) there exists an open tunnel space in the cube edge direction. As an ion proceeds along the tunnel it moves through an array of anions. The latter may be described as a series of alternating north-south and east-west dumb bell pairs that define a series of 2D apertures through which the mobile ions move. This type of aperture is quite rare because it permits a wide variety of paths to accommodate the relative magnitudes of attractive and repulsive forces as the mobile ion moves through the tunnel. The crucial point is that the competing effects of polarization and repulsive terms are out of phase and give rise to a flatter potential for ions of intermediate size such as Ag^+ rather than that of smaller size as Li^+ or bigger size as Na^+ leading to exceptional ion mobility in this structure. Figure 12.5c shows the diffusion path for Ag^+ ion in for α -AgI in the $(x, y, 0)$ projection deduced from neutron scattering experiment at 160 °C of Cava and co-workers [13].

12.3.6 Na- β -Alumina

Na- β -alumina is a two-dimensional fast Na^+ ion conductor with the general formula $Na_2O.nAl_2O_3$ ($5 \leq n \leq 11$) and is a byproduct of the glass-making industry. Its unusual crystal structure is built from close-packed layers of oxygen ions stacked in three dimensions. Figure 12.6b (left) shows the 10-layer unit cell of Na- β alumina consisting of the block-like structure of spinel-type separated by conducting (mirror) planes normal to the c -axis of the hexagonal unit cell. Figure 12.6b (right) shows the exclusive layer in which Na^+ ions reside. Every fifth such layer is oxygen deficient which makes Na^+ more mobile because Na^+ ions have a smaller radius

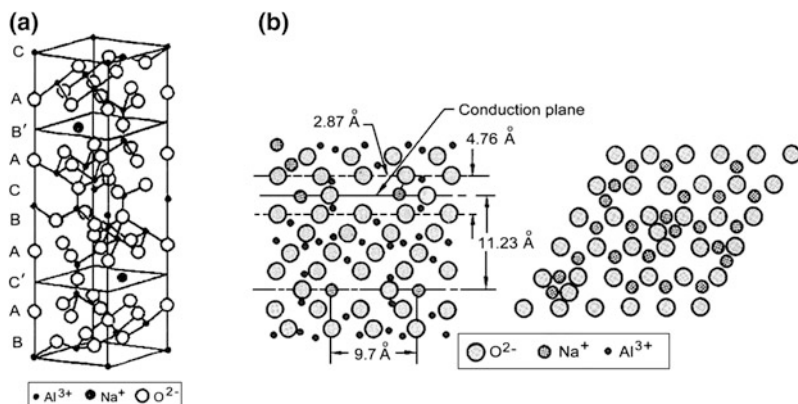


Fig. 12.6 **a** The 10-layered unit cell of two-dimensional fast ion conductor Na-β alumina. **b** *Left* tightly packed Al–O spinel-type blocks held together by Al–O–Al bonds. The conduction plane occurs in between these blocks in which Na⁺ ions diffuse fast. *Right* The same plane viewed with 90° rotation relative to the left figure showing sparsely populated Na⁺ which migrate through vacancies at high temperature. Note the absence of Al³⁺ ions in this plane

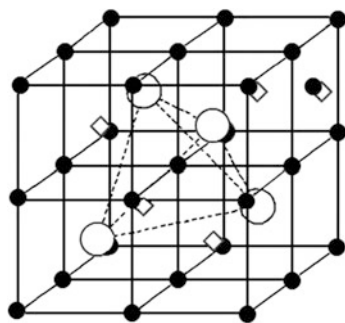
than O²⁻ ions. Such planes are loosely packed and contain equal numbers of Na⁺ and O²⁻ ions. Na⁺ diffusion is possible only along these planes separated by 1.123 nm. In between these alkali ion planes is a block with four close packed oxygen layers (i.e., 16 O atoms) and 11 Al atoms arranged appropriately. Seven Al atoms reside in octahedral sites while the remaining four Al's occupy tetrahedral voids, resulting in a crystal packing similar to the Mg spinel MgAl₂O₄. It is interesting that the lattice parameters of Na β alumina, $a = 0.5594$ nm, $c = 2.253$ nm, is very little different from Li, K, Rb, Tl and Ag-β aluminas. An unusual structural perspective of Na⁺ diffusion in these different aluminas is that the activation energy for tracer diffusion shows a parabolic variation with cation size, the minimum occurring for Na- and Ag-compounds.

12.3.7 Fluorite and Antifluorite

Figure 12.7 displays the essentially fcc lattice of the fluorite structure adopted by a considerable number of fast ion conductors (Table 12.2) including PbF₂ and ZrO₂.

Features of this structure based on mineral fluorite or CaF₂ include (a) slightly bigger cations in comparison to other structures, (b) cations making up a framework fcc sublattice while anions occupy the tetrahedral sites thereby created, a situation tailor-made for fast anion transport via anion Frenkel defects. Co-ordination number for cations is 8 and for anions it is 4 and their positions are (000); (1/4, 1/4, 1/4) and (3/4, 3/4, 3/4) respectively. The antifluorite structure adopted by Li₂O and Na₂S

Fig. 12.7 Fluorite or CaF_2 structure shared by PbF_2 and cubic polymorph of ZrO_2 stabilized by CaO or Y_2O_3 with anion Frenkel defects. Filled circles are anions and open circles are cations. Diamonds are sites for anion interstitials



among others is also a fcc packed structure but this time of anions. All tetrahedral voids are filled by cations which makes fast ion conduction possible only at high temperatures in Li_2O .

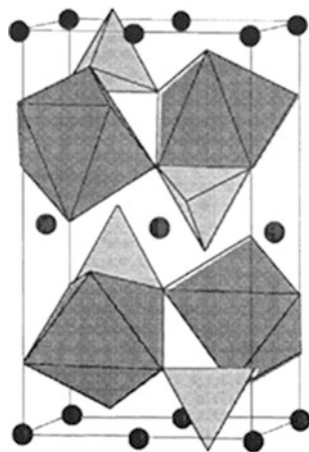
12.3.8 Olivine-Based LiFePO_4 Structure

This is built on oxygen close-packing with Li^+ and Fe^{2+} occupying one-half of the octahedral sites while P occupies one eighth of the available tetrahedral sites. This peculiar distribution of both Li^+ and Fe^{2+} within octahedral sites generates LiO_6 and FeO_6 octahedra as shown in Fig. 12.8 This connectivity has a special impact on both ionic and electronic conductivity of LiFePO_4 .

The phosphorus atom of the phosphate (PO_4) forms with the four oxygen atoms an edge-shared tetrahedron with phosphorus in the center. A zig zag three-dimensional framework is formed by FeO_6 octahedra sharing common-O corners with PO_4 tetrahedra [14] stabilizing Fe in the divalent state but not in the trivalent state as confirmed by Mossbauer spectroscopy [15]. Lithium ions reside within the octahedral channels in a zigzag structure. In the lattice, FeO_6 octahedra are connected by sharing the corners of the bc face. LiO_6 groups form a linear chain of edge-shared octahedra parallel to the b -axis. A FeO_6 octahedron shares edges with two LiO_6 octahedra and one PO_4 tetrahedron. In crystallography, this structure is thought to be the $Pmnb$ space group of the orthorhombic crystal system. The lattice constants are: $a = 0.6008$ nm, $b = 1.0334$ nm, and $c = 0.4693$ nm. The volume of the unit lattice is 0.2914 nm³. The application of this important material as a Li ion battery cathode is discussed in Sect. 12.5.

For a more comprehensive discussion of crystal structures see Hull [16].

Fig. 12.8 Crystal structure of Li^+ conductor orthorhombic LiFePO_4 based on oxygen hexagonal close-packing. Plane of paper represents the bc plane. Black dots O^{2-} ions. FeO_6 and LiO_6 octahedra are shown in dark and light shades respectively



12.4 Electrical Conductivity

12.4.1 Basic Theory

When an ionic solid is placed in an electrical circuit, a force is experienced by the ions due to the a voltage developed across it. This force causes the motion of cations or anions via defects. Thus electricity is conducted through the solid and a current flows in the external circuit. This conductivity may be related to the mobile ion diffusion coefficient if we account for the force exerted on the ions by the electric field. The three basic mechanisms of ion conduction are shown in Fig. 12.9.

The Einstein relation (12.4) connects ion mobility (conductivity) and diffusivity or diffusion coefficient. Now we discuss a simple atomistic model of vacancy/interstitial transport [17]. This model is based on the concept of random walk of vacancies and interstitials via isolated jumps between lattice sites in the crystal structure. In this model the diffusion coefficient D of a particle species is given by

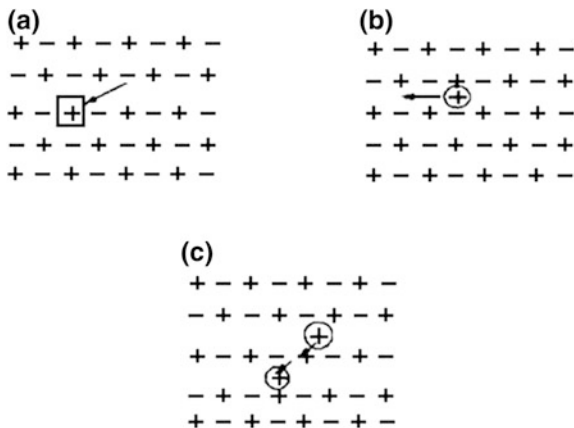
$$D = v(d^2)g \quad (12.8)$$

where v is the average jump frequency, d the distance moved during a single jump and g the geometric factor given by $(1/g) = \text{number of possible directions from a given site}$ ($g = 1/6$ for cubic close packed structures and $1/2$ for jumps in one dimension). v may be written as

$$v = \beta v_0 \exp(-\Delta G/RT) \quad (12.9)$$

where β is the fraction of the species free to move, v_0 the attempt frequency, ΔG is the free energy of migration, R the gas constant, and $\exp(-\Delta G/RT)$ the probability of a successful jump.

Fig. 12.9 Basic mechanisms of ion conduction **a** vacancy mechanism **b** interstitial mechanism **c** interstitialcy mechanism



Dividing ΔG into entropy ΔS and enthalpy ΔH components which relate to defect energies,

$$\Delta G = \Delta H - T\Delta S \quad (12.10)$$

and substituting in (12.8) we get

$$D(T) = g d^2 \beta v_0 \exp(-\Delta S/R) \exp(-\Delta H/RT). \quad (12.11)$$

β is defect concentration when transport occurs by defect motion as defined earlier. For normal highly pure ionic conductors the defect concentration β is dominated by the formation of intrinsic defect pairs (except in NaI and NaBr) so that an additional temperature dependence is required:

$$\beta = \exp(-\Delta G_F/RT), \quad (12.12)$$

where ΔG_F is the free energy of formation of the intrinsic defect pair (cation-anion vacancy or cation/anion vacancy and the corresponding interstitial). β is quite small at low T which constrains diffusivity. The slope of a plot of $\ln D$ versus $1/T$ is $-(\Delta H + \Delta H_F)/R$ where ΔH_F is the enthalpy of defect pair formation.

In fast ion conductors, essentially all of the ions in the mobile ion sublattice (Ag^+ ions in αAgI or F^- ions in PbF_2) may be considered mobile. Therefore β is (a) large and (b) effectively independent of T . In this case the slope of the $\ln D$ versus $1/T$ is $(-\Delta H/R)$. Now suppose both charge and mass are transported by the same species. Then ionic conductivity σ is related to D , particle diffusivity, by the Nernst-Einstein relation

$$\sigma = DCz^2F^2/RT \quad (12.13)$$

where C is the concentration of mobile species (not defects), z the charge of mobile species and F the Faraday constant. In view of (12.12) and (12.10) one may expect linear $\log \sigma T$ versus $1/T$ relation:

$$\sigma = (\sigma_0/T) \exp(-\Delta H/RT) \quad (12.14)$$

where σ_0 = extrapolated intercept at $1/T = 0$ on the $\ln \sigma T$ versus $1/T$ plot. Combining (12.11) and (12.13) we get

$$\sigma_0 = C\beta d^2 v_0 g(z^2 F^2/R) \exp(-\Delta S/R). \quad (12.15)$$

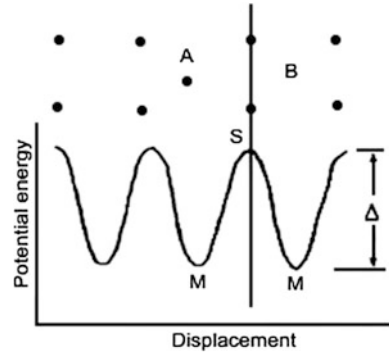
Interestingly fast ionic conductors have a much lower value of σ_0 than do normal ionic conductors. In fact lower by five orders of magnitude. Which term in (12.15) is responsible for this? It is not C , d , g and z . Note that g is much larger in fast ion conductors than in normal ionic conductors in which defect concentration is essentially determined by intrinsic thermally activated defect pair formation. Then consider β , v_0 and ΔS . β is expected to be independent of temperature in fast ion conductors. Therefore it is not β that makes the pre-exponential factor of FIC much smaller than in normal conductors. What about ΔS then? It is not presently known for species migration in FICs. Reasonably, (1) ΔS may be unusually small. Because of structural accommodation for easy uninterrupted motion of ions, the lattice vibrational frequencies of the surrounding crystal structure do not change significantly; (2) the attempt frequency v_0 is also unusually low if mobile ions sit in relatively shallow potential wells- again due to structural peculiarities. Therefore theoretical approaches to fast ion conduction [18] must consider the above factors besides low activation energy (as low as 2–3 kcal/mole) and low pre-exponential factors $\sigma_0 \sim 100 \Omega^{-1} \text{cm}^{-1}$.

12.4.2 A Fundamental Model for Fast Ion Conduction

Although (12.15) gives a basic feel for the fast ion conductor, it is important to remember (Fig. 12.4b, Tables 12.1 and 12.2) that fast ion conduction occurs only in a few special structures in Type I and Type II materials discussed above and that too through a massive and abrupt, or a gradual, slow ion to fast ion conducting phase transition. Thus to fully appreciate fast ion conduction phenomenon, it is necessary to build models based on structures of both low and high temperature phases. We shall consider briefly a fundamental model that is intuitive and thus provides an insight into the phenomenon. A basic formula is derived for the frequency-dependent ionic conductivity that could be compared with the experimental results.

A fast ion conductor is crudely comparable to a metal. The concentration of mobile ions (say Ag^+ in AgI) compares favourably to the conduction electron concentration in a metal. Just as electrons in a metal are regarded as a Fermi gas or a liquid, ions in a fast ion conductor may be looked upon as a gas/liquid trapped in a

Fig. 12.10 Interstitial diffusion in a 2-dimensional lattice



framework solid. Most significantly the fast ion conductor responds to both time and spatially-varying fields. Huberman and Sen [19, 20] proposed the model of an ionic liquid in a lattice. The ions are treated as hopping between sites while performing finite amplitude oscillations around the potential minima provided by the lattice. The detailed lattice vibrational spectrum is ignored. This is like adopting the long-wavelength (or low frequency) approximation in solid state physics. The aim of this model is to calculate the temperature dependence of the ionic conductivity over several decades of frequency, that is the temperature dependence of AC conductivity. In this model, the mobile ions have two basic degrees of freedom: (i) an oscillatory motion around the local site potential minima with a characteristic frequency ω_D and (ii) a random walk process by which the mobile ions can diffuse throughout the lattice. Focus on the time τ_0 spent by the ion in the potential minima offered by the lattice (Fig. 12.10). Note that in order to go from A to B the ion has to pass from one potential minimum (M) to another by crossing the barrier at saddle point (S). Δ represents the necessary potential energy. τ_0 is given by

$$\tau_0 = r_0^2 / \{2dD(T)\} \quad (12.16)$$

where d is the dimensionality and $D(T)$ the temperature-dependent diffusion coefficient. The time τ_1 spent in free flight of the mobile ion from one site to another is obtained from the equation

$$\tau_1 = r_0(M/k_B T)^{1/2} \quad (12.17)$$

where r_0 is the jump distance, M the ion mass and k_B the Boltzmann constant. Note that both τ 's are strong functions of temperature so that a small change in temperature will make the system go over from a solid like behavior ($\tau_1/\tau_0 = 0$) to gas-like behavior ($\tau_1/\tau_0 = 1$). In the intermediate 'liquid-like regime' the system exhibits a rich structure in the response function.

Suppose the system is probed experimentally within a time interval τ that is short compared to the 'residence' time τ_0 . The ions will respond as if they are oscillators bound to a finite harmonic potential. On the other hand, for probing times longer

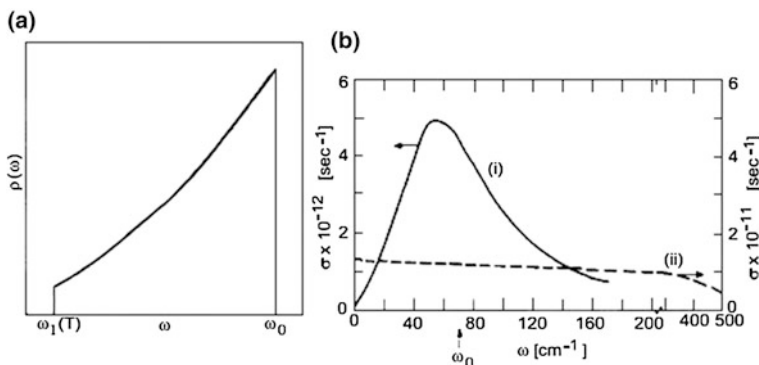


Fig. 12.11 **a** Model density of states $\rho(\omega)$ as a function of ion hopping frequency ω in the ‘ionic liquid in a lattice’ model of fast ion conductor. ω_0 is a function of T . **b** Calculated frequency dependence of ionic conductivity σ at (i) 300 K and (ii) 2,000 K with appropriate scaling of the ordinate

than τ_0 the response will be essentially that of a diffusing ion. There is no long range order for the mobile ions. So the spectrum may be described by a Debye density of states above a temperature-dependent cut-off frequency $\omega_0 = (\tau_0)^{-1}$ while below ω_0 the density of states is diffusive (Fig. 12.11a).

To calculate the full frequency-dependent conductivity, consider the equation of motion for the mobile ion in the form

$$\frac{d^2u}{dt^2} + 2\Gamma \frac{du}{dt} + \omega^2 u = f(t) \quad (12.18)$$

where u is the ionic coordinate, Γ the damping coefficient and $f(t)$ is a random or stochastic force with a Gaussian distribution such that

$$\langle f(t) \rangle = 0 \quad (12.19)$$

and

$$\langle f(0)f(t) \rangle = 2D\delta(t). \quad (12.20)$$

Angular brackets denote thermal average and the LHS of (12.20) represents the correlation of the ion motion at time t to its motion at $t = 0$. D is given by a macroscopically determined diffusion coefficient to be discussed in Sect. 12.5.3; so the ionic mobility in this model is essentially macroscopic.

Below the cut-off frequency the diffusive motion of the ions is described by the following equation of motion:

$$\frac{d^2u}{dt^2} + \gamma \frac{du}{dt} = f(t) \quad (12.21)$$

where the coefficient γ is to be determined by the zero frequency behaviour. For a 3D fast ion conductor such as RbAg_4I_5 or $\alpha\text{-AgI}$ there are many available sites per mobile ion; so that one can neglect ion-ion correlations. Thus conductivity $\sigma(\omega)$ can be calculated in terms of mobile ion velocity auto correlation function $\langle v(0)v(t) \rangle$ from

$$\sigma(\omega) = [(Ze)^2 n \pi / 2 \hbar \omega] [1 - \exp(-\hbar \omega / k_B T)] \int_{-\infty}^{+\infty} \langle v(0)v(t) \rangle \exp(-i\omega t) dt \quad (12.22)$$

where Ze is the ionic charge and n the number density of ionic charge carriers; $\langle v(0)v(t) \rangle$ is determined from (12.18) and (12.21) with the density of states shown in Fig. 12.11. Equations (12.21) and (12.22) emphasize the importance of ion-ion interactions characteristic of fast ion conductors and its effect on the frequency dependent conductivity. In the DC limit ($\omega \rightarrow 0$) the diffusion coefficient is given by

$$D = (1/d) \text{ Real part of } \int_0^\omega \langle v(0)v(t) \rangle dt. \quad (12.23)$$

Therefore knowing r_0 (defined earlier) one can get γ as

$$\gamma = (k_B T / MD)(\omega_0 / \omega_D)^3 \quad (12.24)$$

where D is the diffusion coefficient in the zero-frequency limit. γ is the damping coefficient considering the ion motion as that of a damped harmonic oscillator subjected to a random force. The zero frequency behaviour of conductivity is nothing but σ_{dc} . Finally $\sigma(\omega)$ is given by

$$\sigma(\omega) = n(Ze)^2 / M [(\omega_0 / \omega_D)^3 \{ \gamma / (\gamma^2 + \omega^2) \} + (3\Gamma \omega^2 / \omega_D^2) F(\omega, \Gamma)] \quad (12.25)$$

where

$$F(\omega, \Gamma) = \int_{\omega_0^2}^{\omega_D^2} x dx / [(x - \omega^2)^2 + 4\Gamma^2 \omega^2 x]. \quad (12.26)$$

Note that (12.25) is made up of two terms: (i) A Lorentzian function centered around ω and width γ and (ii) a broad resonance around ω_D corresponding to a single phonon density of states (Fig. 12.11b). In the limit $\omega \rightarrow 0$ the conductivity becomes

$$\sigma(\omega = 0) = [n(Ze)^2/k_B T] D_0 e^{-U/k_B T} \quad (12.27)$$

which, with U as the activation energy for diffusion, is the Nernst-Einstein relation written in the form

$$\sigma(T) = n(T)(Ze)^2 D(T)/k_B T. \quad (12.28)$$

The major conclusions from this model are: (i) The DC electrical transport properties of a fast ion conductor become purely diffusive which is the well-known Arrhenius behavior. (ii) In the limit $\omega \rightarrow \infty$ the conductivity goes to zero as ω^{-3} . (iii) In the $\tau = 0$ limit, (12.25) predicts the true Debye spectrum with the low temperature specific heat behaving as T^3 , and, finally, (iv) as the temperature increases the cut-off frequency ω_0 moves towards the Debye frequency; For temperatures close to the hopping activation energy $F(\omega, \Gamma) \rightarrow 0$ as $\omega_0 \rightarrow \omega_D$ and the conductivity becomes like that of a Drude metal (curve 2 in Fig. 12.11b). Huberman-Sen model thus predicts correctly that the response of the fast ion conducting system shifts from solid-like at low temperatures to liquid-like at moderate temperatures characterized by a hopping diffusion to a classical gas response as the cut-off frequency approaches the Debye frequency.

More sophisticated models have taken into account ion-ion interactions (neglected in this simple but effective approach) to reproduce the experimental temperature dependence of conductivity in AgI and PbF_2 [18]. It is very interesting to note that single-particle two-dimensional Brownian motion based on the model discussed above has been experimentally observed in the high-frequency ionic conductivity of the fast ion conductor Ag_3SI [21].

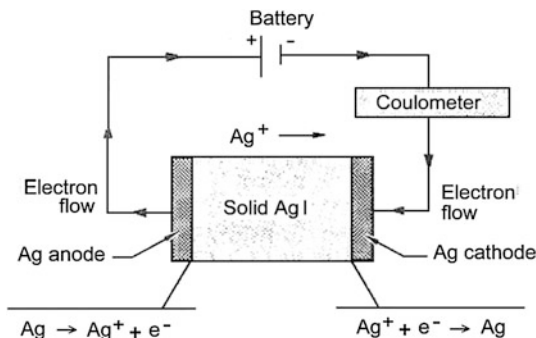
12.5 Experimental Methods

In this section three experimental methods to characterize fast ion conductors are briefly described. (i) Transport number method of Tubandt, (ii) AC conductivity and (iii) tracer diffusion technique for diffusivity.

12.5.1 Tubandt Method

The first conclusive evidence for the transport of charge through solids by the flow of ions was provided by Tubandt in 1910 as summarized in [22]. Figure 12.12 shows the schematic of the set up. A bulk sample of AgI was heated to 150 °C. The silver electrodes were weighed before and after the passage of a measurable quantity of current. The amount of Ag gained by the cathode exactly equaled that

Fig. 12.12 Schematic of Tubandt's pioneering experiment on ion transport in solid AgI



lost by the anode. The total number of atoms transferred was equal to the current registered by the coulometer.

12.5.2 AC Ionic Conductivity Measurements

Ionic and fast ionic conductors usually have negligible electronic conduction so that the entire conductivity comes from the hopping motion of ions. Thus measurement of AC electrical conductivity is important for materials and devices thereof. Furthermore the DC conductivity (σ_{dc}) which is derivable from AC conductivity is related to the charge or diffusivity D_σ through the equation

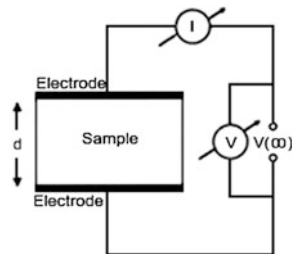
$$D_\sigma = \sigma_{dc} k_B T / N_{ion} q^2 \quad (12.29)$$

where N_{ion} is the density of mobile ions and q the electrical charge per ion. Therefore conductivity measurement becomes an indispensable complement of tracer diffusion to be discussed subsequently.

Measurements of ionic conductivity are carried out with an AC bridge because ionic and fast ionic conductors exhibit polarization effects at the electrodes. Figure 12.13 shows an experimental set up. The measurements are made with two identical electrodes applied to the faces of the sample in the form of a rectangular parallelepiped. The general procedure is to apply an electrical signal of angular frequency ω ($\omega = 2\pi\nu$) in the form of a known voltage or current. The frequencies range from 50 Hz to several MHz. A low cost AC conductivity meter based on phase-sensitive detection may be fabricated [23].

Essentially one requires a signal generator, an ammeter and a voltmeter for current and voltage measurements and the phase shift between the current $I(\omega)$ and voltage $V(\omega)$. The experiment involves the measurement of the complex impedance of the sample defined as

Fig. 12.13 AC conductivity of ionic/fast ionic conductor by the AC bridge method



$$Z(\omega) = V(\omega)/I(\omega). \quad (12.30)$$

$Z(\omega)$ has two components: the real Z' and the imaginary Z'' . Hence

$$Z(\omega) = Z' - iZ''. \quad (12.31)$$

The complex conductivity $\sigma(\omega)$ also has two components:

$$\sigma(\omega) = \sigma'(\omega) + i\sigma''(\omega). \quad (12.32)$$

Conductivity is related to complex impedance by

$$\sigma(\omega) = [1/Z(\omega)](d/A) \quad (12.33)$$

where d is the sample thickness and A the area of cross-section of the sample. A plot of Z' versus Z'' with ω as a variable parameter gives the so-called Cole-Cole plot which is a semicircle. The point at which the semicircle intersects the real axis defines the intrinsic resistance of the sample in a simple Debye relaxation model of a parallel resistance-capacitance circuit. From this value σ_{dc} may be deduced. By doing the experiment at different frequencies one can generate the σ_{dc} versus $1/T$ plot and get the relevant activation energy. One can also work in the complex admittance formalism instead of complex impedance since one is reciprocal of the other [24].

12.5.3 Tracer Diffusion Measurements

One of the most important direct methods for ion diffusivity or diffusion coefficient determination is the tracer diffusion method (Fig. 12.14). This is a four-step technique (Fig. 12.14a). In step 1 the radioactive tracer (Na^{24} for NaCl [25]) is deposited onto the strain-free surface of the sample using evaporation or dripping of a liquid solution. Step 2 consists of thermal diffusion anneal carried out at temperature T for some diffusion time t by placing the sample in an encapsulated quartz ampoule under vacuum or argon atmosphere. If a thin layer of M tracer atoms/cm²

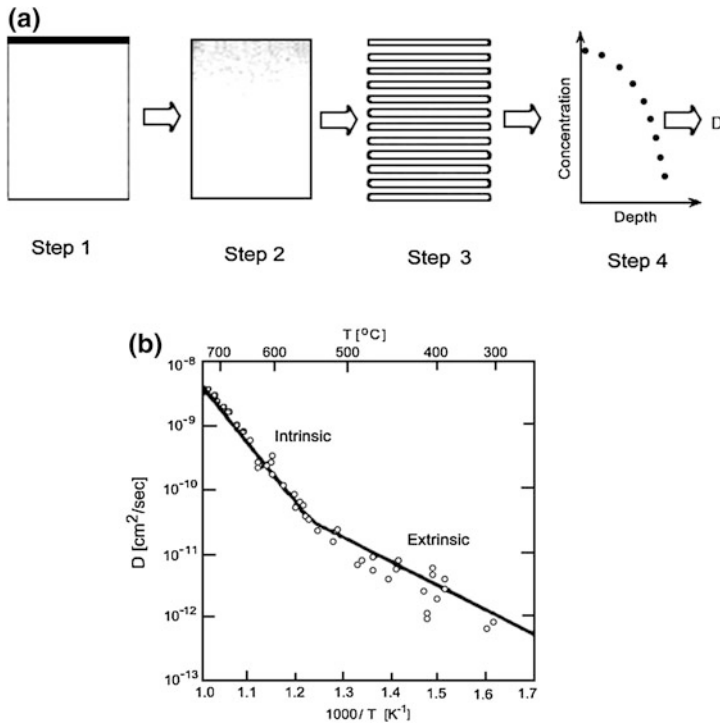


Fig. 12.14 Tracer diffusion method for ion diffusivity: **a** technique in four steps, **b** typical plot of D versus $1,000/T$ for NaCl

has been deposited at the surface $x = 0$ of a semi-infinite sample. Then the tracer concentration after a ‘diffusion anneal’ involving $C(x,t)$ has a space-time dependence [26] given by

$$C(x,t) = [M/(\pi Dt)^{1/2}] \exp(-x^2/4Dt). \quad (12.34)$$

Equation (12.34) represents the thin film solution of Fick’s second law of diffusion or the diffusion equation

$$\partial C/\partial t = \nabla \cdot (D \nabla C). \quad (12.35)$$

Equation (12.35) relates the time-rate of change of mobile ion concentration to the spatial second differential coefficient of the same concentration, D occurring as a parameter dependent on the crystal structure of the solid sample. Note that diffusion from a thin film source follows a Gaussian distribution of concentration with respect to the $t = 0$ concentration. Note that Fick’s first law is contained in the rhs of (12.35) namely diffusion flux or the number of particles crossing unit area is

$$\mathbf{J} = -D\nabla C. \quad (12.36)$$

Step 3 is a major step of analysis of diffusion profile (actually the concentration-depth profile) by adopting the method of serial sectioning of the sample using lathes and microtomes and subsequent determination of the amount of tracer per section. The sections must be parallel. For radioactive tracers the specific activity per section $A(x)$ is given by

$$A(x) = kC(x). \quad (12.37)$$

$A(x)$ is obtained from the section mass and the count rate. The count rate refers to γ - or β -particles emitted in the nuclear reaction involving the radio tracer. The count rate must be corrected for background. In step 4, a plot is made between concentration and the depth of the sample sectioned referred to as depth profile. A plot of $\log A(x)$ versus penetration distance squared is a straight line whose slope is $1/4Dt$. From this slope and the diffusion time t tracer diffusivity D is determined. Figure 12.14b gives a typical diffusivity plot for Na^{24} diffusion in NaCl .

12.5.4 Conductivity Optimization

After identifying a FIC it is essential to find a composition that exhibits optimum conductivity so that it could be applied as a solid electrolyte in the design and fabrication of electrochemical devices. We briefly discuss two examples of optimization: (i) $\text{LiI-Al}_2\text{O}_3$ and (ii) CaO-ZrO_2 .

12.5.5 $\text{LiI-Al}_2\text{O}_3$

Liang [27] studied the conduction characteristics of the polycrystalline lithium iodide containing aluminum oxide. It was found that the incorporation of aluminum oxide substantially increased the conductivity of lithium iodide. Lithium iodide containing 33–45 mol per cent aluminum oxide exhibited conductivities in the order of $10^{-5} \Omega^{-1}\text{cm}^{-1}$ at $25 \pm 2^\circ\text{C}$. No substantial amount of aluminum oxide was found soluble in the lithium iodide under the experimental condition. The increase in conductivity is not due the classical doping mechanism but due to the formation of a composite or a dispersed or heterogeneously doped solid electrolyte in which interfacial ion transport dominates. The electronic conductivity of the lithium iodide-aluminum oxide solid was negligible compared to the ionic conductivity. Therefore, it is suitable as a solid electrolyte in solid-state battery systems. The pacemaker battery is discussed in Sect. 12.4.

12.5.6 CaO-Stabilised-ZrO₂

Pure ZrO₂ is monoclinic below 1,150 °C transforming to the tetragonal phase above this temperature. However CaO (also Y₂O₃) stabilizes the fluorite structure. Ca²⁺ in CaO and Y³⁺ in Y₂O₃ will introduce one and one and a half moles of anion vacancies when substituted for an equivalent mole of ZrO₂. Generally substitution of CaO and Y₂O₃ is in the range of 8–20 mol % creating anion vacancies from 4 and 10 %. These vacancies are responsible for the very high rates of oxygen diffusion. At 1,000 °C, ZrO₂: CaO has an ionic conductivity of $\sim 55 \times 10^{-3} \Omega^{-1} \text{cm}^{-1}$ and an activation energy of $\sim 1.1 \text{ eV}$ [22]. The sensor application based on CaO-ZrO₂ is discussed in Sect. 12.6.

12.6 Applications

Conversion of chemical energy stored in a fast ion conductor in the form of mobile ions, into useable electrical or mechanical energy is the basis of all applications of fast ion conductors. To do that one needs a conversion mechanism or a transducer. The purpose is to get energy that can be stored and in portable form for ready use wherever required. These transducers effect conversion between electrical and chemical quantities namely energies, potentials and fluxes.

Some practical applications of fast ion conductors are given in Table 12.3 and discussed in the following sections.

Table 12.3 Practical applications of fast ion conductors^a

Use	Specific needs	Examples
Ion monitor	High discrimination, low electronic conduction (σ_{el})	ZrO ₂ /Y ₂ O ₃ as an O ₂ monitor
Fuel cell electrode	High temperature stability, $\sigma_{\text{el}} \geq 100 \Omega^{-1} \text{cm}^{-1}$	PrCoO ₃ ; fuel gas is the other electrode
Fuel cell electrolyte	Impermeable to fuel gas; $\sigma_{\text{el}} \leq 10 \times 10^{-3} \Omega^{-1} \text{cm}^{-1}$	ZrO ₂ /Y ₂ O ₃ for H ₂ combustion
Battery anode	Electropositive material; no reactions or solubility in electrolyte; High σ_{el}	Li and other metals
Battery cathode	Electronegative material; High σ_{el}	NiS, FeS ₂ , Li _x CoO ₃ , PbI ₂ , I ₂ + P2VP ^b
Battery electrolyte	Low leakage current; chemical inert ness, low σ_{el}	β -alumina for the Na/S battery; Al ₂ O ₃ :Li for pacemaker, LiI for pocket calculator, also Li ₃ N, LiPON ^c

^a In all cases an ionic conductivity $\geq 1 \Omega^{-1} \text{cm}^{-1}$ is desirable

^b Poly 2-Vinyl pyridine

^c Lithium phosphorus oxynitride

12.6.1 Solid State Batteries

The major applications of fast ion conductors are to the all solid state battery [28, 29]. The familiar car battery or the ‘lead-acid battery’ which provides the basic understanding for the solid state battery is discussed first. This also illustrates the concept of electrochemical potential and potential steps. In the lead-acid battery the anode is metallic lead (Pb) while the cathode is a layer of PbO_2 formed on a Pb substrate. These two electrodes are immersed in dilute sulfuric acid (H_2SO_4) liquid electrolyte (Fig. 12.15aA) which is partially ionized into H^+ ions (protons) and SO_4^{2-} ions.

During the discharge process both Pb and the PbO_2 are converted to lead sulfate PbSO_4 via the two reactions:

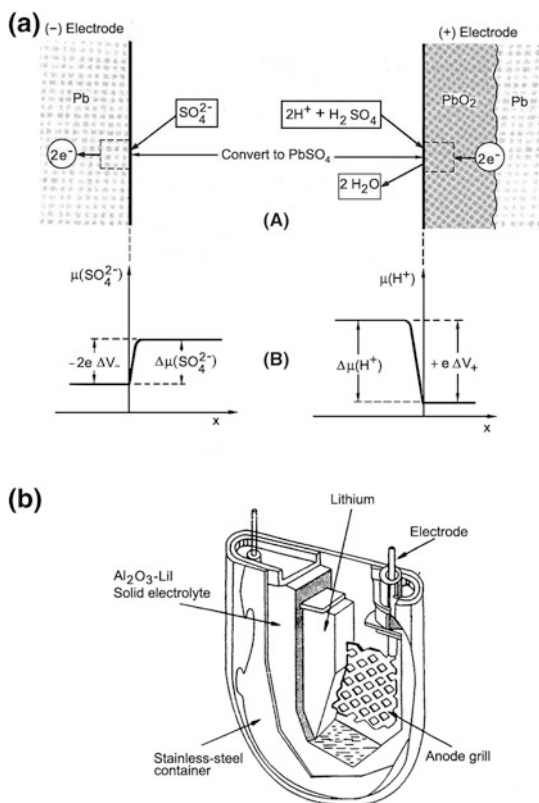
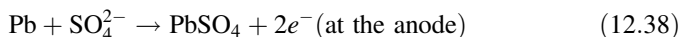
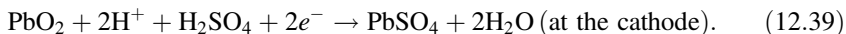


Fig. 12.15 **a** Lead-acid battery (car battery): (A) components and reactions, (B) electrochemical potentials at the anode-electrolyte and cathode-electrolyte interfaces [30]. **b** The pacemaker battery with $\text{Al}_2\text{O}_3\cdot\text{LiI}$ as the solid electrolyte

and



Equation (12.38) makes the cathode a sink for SO_4^{2-} ions keeping the internal chemical potential $\mu(\text{SO}_4^{2-})$ of the sulfate ions at the surface of the anode lower than that inside the electrolyte (Fig. 12.15aB). Likewise (12.39) makes the cathode a sink for protons, keeping the internal chemical potential $\mu(\text{H}^+)$ of the protons lower at the surface of the cathode than inside the electrolyte. The chemical potential gradients drive the ions towards the electrodes; also they drive the electrical currents during the discharge process.

Suppose the battery terminals are not connected. Then the electrons are depleted from the cathode and accumulate in the anode thus charging both. Consequently electrochemical potential steps develop at the electrode-electrolyte interfaces. These are steps of exactly the correct magnitude to equalize the chemical potential steps and to stop the diffusion of ions which does not allow the chemical reactions to proceed further. An external current allowed to flow enables reactions to resume. Electron flow through the electrolyte is almost nil because the electrolyte has only a negligible electron concentration.

During the charging process reactions opposite to (12.38) and (12.39) take place. Because now an external applied voltage generates the potential steps at the electrode surface of such magnitude as to reverse the sign of the total chemical potential gradients and hence the direction of ion flow.

Let ΔV_- and ΔV_+ be the differences in the electrostatic potential of the anode and the cathode relative to the common electrolyte. Sulfate ions carry two negative charges so that diffusion will stop when

$$-2q\Delta V_- = \Delta\mu(\text{SO}_4^{2-}). \quad (12.40)$$

Diffusion of protons will stop when

$$+q\Delta V_+ = \Delta\mu(\text{H}^+). \quad (12.41)$$

The two potentials ΔV_- and ΔV_+ are called half-cell EMFs and have magnitudes -0.4 V and $+1.6$ V respectively. Thus the total electrostatic potential developed across the battery to stop the diffusion reaction is

$$\Delta V = \Delta V_+ - \Delta V_- = 2.0 \text{ volt} \quad (12.42)$$

which is the open-circuit voltage or EMF of the battery. It drives the electrons from the $(-)$ terminal to $(+)$ terminal, when the two are connected. Note that the only effective electron flow path is through the external connection between the electrodes.

The three participants in the electrochemical reactions in a lithium-ion battery are the anode, cathode and the solid electrolyte. Figure 12.15b shows the primary battery used as a pacemaker for heart patients [31]. LiFePO_4 is currently being developed as a cathode for rechargeable Li^+ -ion batteries. In these secondary or rechargeable batteries Li^+ ions move through the electrolyte from cathode to anode during discharge and from anode to cathode during recharging in a reversible process called intercalation.

The lithium cell used in common consumer applications uses metallic lithium as anode and manganese dioxide as cathode, with a salt of lithium dissolved in an organic solvent as the electrolyte. Another type of lithium cell having a large energy density is the lithium-thionyl chloride cell used in commercial/industrial applications, or are installed into devices where the consumer does not replace them. This cell contains a liquid mixture of thionyl chloride (SOCl_2) and lithium tetrachloroaluminate (LiAlCl_4), which act as the cathode and electrolyte respectively. A porous carbon material is used as a cathode current collector which receives electrons from the external circuit.

12.6.2 Fuel Cell

A fuel cell converts chemical energy into electrical energy. In fact a fuel cell is an electrolysis cell operated in the reverse with hydrogen and oxygen as fuels. Thus electrolysis is basic to fuel cell technology. It is both isothermal and isobaric. We can consider the cell to be a dilute sulfuric acid electrolyte in which are immersed two platinum electrodes. Through reactions similar to those considered above, water is split into hydrogen and oxygen:



Effective work necessary to decompose 1 mol of water is related to the molar Gibbs free energies of the reactants:

$$W' = \Delta G = G(\text{H}_2\text{O}) - G(\text{H}_2) - \frac{1}{2}G(\text{O}_2) \quad (12.44)$$

ΔG is -237 kJ per mole at room temperature.

In electrolysis this work is done by a current I that flows under an external voltage V_0 . If t is the time required to split one mole of water and Q the total charge flowing through the cells so that

$$W' = QV_0. \quad (12.45)$$

When sulphate ions move to the cathode to split water releasing molecular oxygen gas and two electrons,

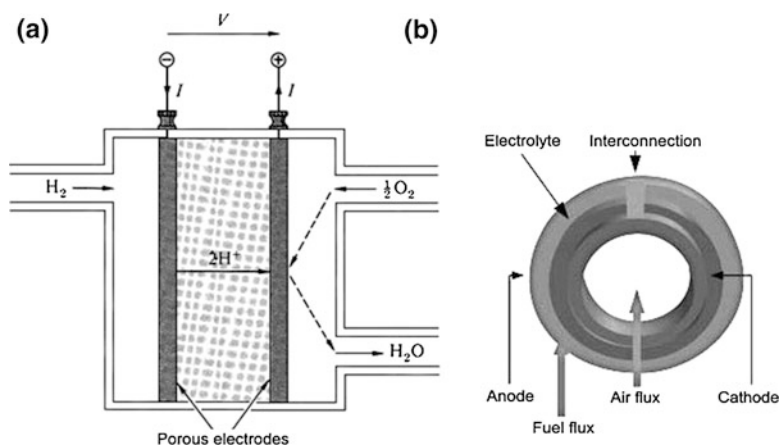


Fig. 12.16 **a** Conventional fuel cell, **b** solid oxide fuel cell

$$Q = -2N_A e = -1.93 \times 10^5 \text{ Coulomb} \quad (12.46)$$

where N_A is the Avogadro number. Combining (12.44–12.46) we get the condition for electrolysis which requires a minimum voltage

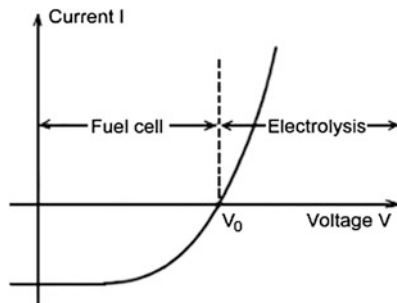
$$V_0 = -\Delta G / 2N_A e \quad (12.47)$$

or 1.229 volts. In order to maintain a finite current flow a voltage larger than V_0 needs to be applied. This is because V_0 alone just reduces to zero the potential barrier between the systems on either side of the reaction (12.47). When $V > V_0$ excess power $(V - V_0)I$ will be dissipated as heat in the electrolyte.

If $V < V_0$ the reaction (12.43) will proceed from right to left provided hydrogen gas is available at the cathode and oxygen gas at the anode. To make it happen the electrodes must be fabricated as porous sponges through which hydrogen and oxygen are forced under pressure (Fig. 12.16a). This type of device produces a voltage V_0 between the electrodes. If the electrodes are connected external current will flow. This arrangement is the so-called hydrogen-oxygen fuel cell; an important power source. Thus an electrochemical cell can operate in two ranges with current-voltage characteristics shown in Fig. 12.17.

A solid oxide fuel cell (Fig. 12.16b) is made up of four layers including an air flux layer, the other three being porous ceramics namely the anode, the cathode and the electrolyte. The fourth one is the air flux. The electrolyte is typically 3–8 mol % yttria stabilized zirconia. Oxygen transport (as O^{2-}) occurs through the electrolyte. A single cell consisting of these four layers stacked together is typically only a few millimeters thick. Hundreds of these cells are then connected in series to form what most people refer to as an “SOFC stack”. The ceramics used in SOFCs become electrically and ionically active when they reach very high temperatures. Thus the

Fig. 12.17 The I - V curves of an electrolyte or fuel cell with the corresponding operating voltages



stacks have to run at temperatures ranging from 500 to 1,000 °C. At such high temperatures reduction of oxygen into oxygen ions occurs at the cathode. These ions can then diffuse through the solid oxide electrolyte to the anode where they can electrochemically oxidize the fuel. In this reaction, a water byproduct is given off as well as two electrons. These electrons then flow through an external circuit where they can do work. The cycle then repeats as those electrons enter the cathode material again.

12.6.3 Chemical Sensors

Fast ion conductors can be used as chemical sensors for ions in solution or for trace gases. The principle is illustrated in Fig. 12.18a for a CaO-stabilized zirconia (CSZ) sensor. CSZ is shown as alternate layers of O^{2-} ions beginning from top interleaved with Zr^{4+} ion layers randomly occupied by Ca^{2+} ions (hatched circles). The sensor operates at $\sim 1,000$ °C. The oxygen in the zirconia lattice and oxygen outside would equilibrate when the oxygen pressures on both sides are equal. If the oxygen (partial) pressures on one side differs from that on the other, oxygen ions flow through zirconia generating an EMF according to the equation

$$E = \left(\frac{RT}{|z|F} \right) \ln(P_1/P_2) \quad (P_1 > P_2) \quad (12.48)$$

where R is the gas constant, T the temperature in K, z the ion charge, F the Faraday constant and P_1 , P_2 the partial pressures as shown in Fig. 12.14a. The electrode EMF is proportional to the ion concentration.

Figure 12.18b displays an automobile exhaust sensor. The configuration is: the inner side of the sensor is air at pressure P_1 /ZrO₂-Y₂O₃ (YSZ)/Pt: exhaust gas. At the interface 1 the reaction is $O_2 + 4e^- \rightarrow O^{2-}$ while at the interface 2 it is $O^{2-} \rightarrow O_2 + 2e^-$. The heater maintains YSZ at a suitably high temperature required for sensing.

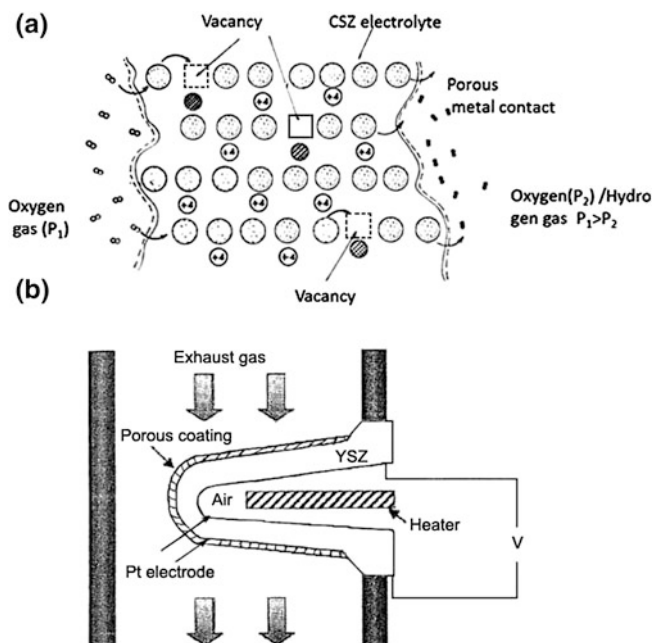


Fig. 12.18 a Sensor principle, b automobile exhaust sensor

12.6.4 Nanoscale Memory Device

A nanoscale nonvolatile memory device based on Ag^+ fast ionic conductor RbAg_4I_5 has been fabricated directly on a Si substrate with lateral size scaled down to 100 nm (Fig. 12.19) [32]. By applying voltages with different polarity the device can be switched between high- and low-resistance states. The ratio between high and low resistances could be as high as 1,000. The low resistance state shows rectifying diode characteristics. The configured states remain nonvolatile after switching and the device could be switched repeatedly for nearly 1,000 times. These results point to the possibility of direct integration of nanoscale nonvolatile memory device with Si-based circuit and has potential for high-density memory applications.

12.7 Summary and Outlook

Beginning from a traditional thermodynamic-statistical picture of Schottky and Frenkel defects we have discussed in this chapter the nature of fast ion conduction with the help of specific examples. Basic experimental methods for investigation of fast ion diffusion have been outlined. A few applications have been mentioned. It is

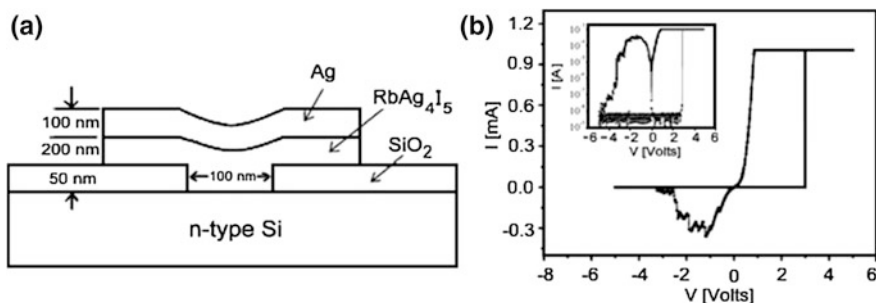


Fig. 12.19 **a** Nanoscale nonvolatile memory device on *n*-type Si substrate. **b** *I*-*V* curve showing switching characteristic; Insert *I*-*V* curve with current shown on logarithmic scale

clear that fast ion conduction is possible only in a few special crystal structures which are supported by a robust sublattice (I^- in AgI and Pb^{2+} in PbF_2). At some well defined temperature, by a mechanism that is as yet baffling, the conducting ion sublattice either suddenly ‘melts’ or else disorders extensively providing $\sim 10^{22}$ mobile ions per cc for conduction when the compound becomes a solid electrolyte. The mechanism of fast ion conduction, despite much theoretical effort, is an open problem, of course driven by electrochemical power conversion prospects for electrical vehicles and micro- and nano-ionic devices.

12.8 Problems

- From Fig. 12.2 estimates the Schottky defect pair formation energy (E_F) and the defect migration energy (E_m) for NaCl.
 - Derive the formula for n_F , the concentration of Frenkel defects in an ionic conductor at a temperature T .
- Compute the maximum ionic conductivity that can be obtained in the point defect model of an ionic crystal.
- Make a plot of (12.34) taking C/M on the y-axis and $x/\sqrt{Dt}$ on the x-axis. Take diffusion length $2\sqrt{Dt}$ as a parameter having values 0.5, $1/\sqrt{2}$, 2 and 4 to generate a family of curves. Interpret the result.
- Derive the real and imaginary parts of the impedance of an ionic conductor, Z' and Z'' , modeled as a parallel combination of an ohmic resistance R and a capacitance parallel to it. Represent the impedance on a complex Z' - Z'' plane to get the semi-circular plot called Cole-Cole plot. This semicircle passes through the origin for $\omega \rightarrow \infty$, through $(R, 0)$ for $\omega = 0$. Indicate, on the plot, the point for which $\omega =$ time constant of the circuit. What is the meaning of $(R, 0)$ with respect to the sample?

5. The wurtzite structure of β -AgI and the zinc blende structure of γ -AgI, are the two low-temperature polymorphs of AgI. Discuss the similarities and differences between these two structures and that of the fast ionic conducting phase α -AgI with I-based bcc structure. Why do you think the conductivity of the Ag^+ ions σ increases in the order $\sigma_\alpha > \sigma_\gamma > \sigma_\beta$?

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Chapter 13

Superconductivity

Abstract This chapter deals with superconductivity. Several experimental observations like the Meissner effect, the critical field, the existence of type I and type II superconductors, the isotope effect and the determination of thermal conductivity and specific heat in the normal and superconducting states are discussed. The thermodynamics of superconductors is discussed in detail. The values of quantities like the free energy, the entropy and the specific heat in the normal and superconducting states are worked out. The concept of the energy gap in the electron energy spectrum is explained. Some of the theories of superconductivity like London's theory, the Bardeen–Cooper–Schreiffer theory and the Ginzburg–Landau theory are discussed. The concepts of penetration depth, flux quantization, coherence length and Cooper pairs are explained. The connection between thermal vibrations and superconductivity is emphasized. Tunneling through barriers is an important aspect in superconductivity. Both tunneling of electrons (normal tunneling) and Cooper pairs (Josephson tunneling) are discussed. The AC and DC Josephson effects are discussed. Quantum interference effect and its application (SQUID) are described. In the 1980s, superconductivity was observed in some compounds with transition temperatures above the liquid nitrogen temperature. The structure and properties of some HT superconducting compounds are discussed.

13.1 Introduction

We have noted in earlier chapters that electrical conductivity of solids has a wide range of values; 10^{-22} – 10^{-14} for insulators, 10^{-9} – 10^2 for semiconductors and 10^4 – 10^8 for metals (all in units of $\text{ohm}^{-1} \text{ cm}^{-1}$). In fact the range is still wider, going up to almost infinity in the case of superconductors. But this occurs only at extremely low temperatures. The discovery of superconductors became possible only with the availability of liquid helium. Superconductivity has led to the discovery of several new phenomena, elaborate theories and a number of applications.

In this chapter, we shall start with the discovery of superconductivity and then describe some properties of superconductors. This is followed by a discussion of the main theories. Some applications of superconductivity are discussed. For completeness, we have included a section on high temperature superconductors.

There is a vast literature on superconductivity. Apart from discussions in text books by Kubo and Nagamiya [1], Blakemore [2] and Ashcroft and Mermin [3], Kittel [4], there are books dedicated to superconductivity [5–7].

13.2 Discovery of Superconductivity

Kammerlingh Onnes succeeded in the liquefaction of helium in 1908. This opened up a new low temperature range for studying properties of materials. Onnes took up a study of the electrical resistance of metals at liquid helium temperatures.

Onnes [8] discovered superconductivity while studying the electrical resistance of mercury. The equipment used by Onnes is shown in Fig. 13.1. Purified mercury is introduced in vacuo in a series of U-tubes (A–B) of diameter 0.05 mm. This system of U-tubes is placed in a specially designed cryostat containing liquid

Fig. 13.1 Apparatus used by Kammerlingh Onnes in the discovery of superconductivity

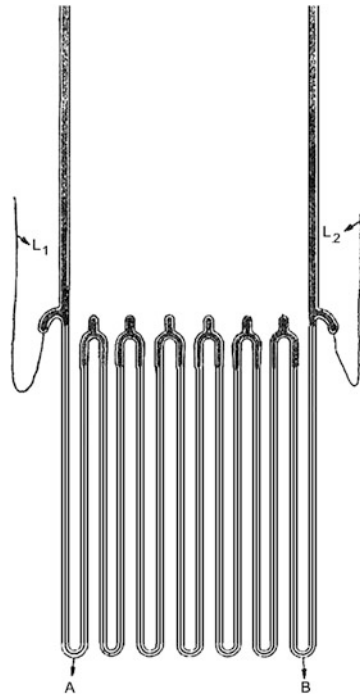
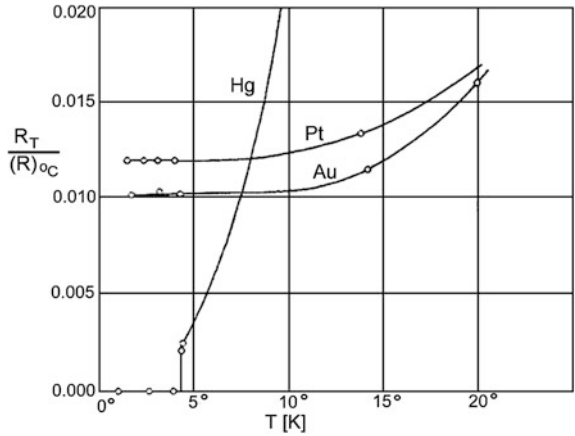


Fig. 13.2 Temperature variation of fractional resistance $[R_T/(R)_{0C}]$ of mercury, platinum and gold at very low temperatures

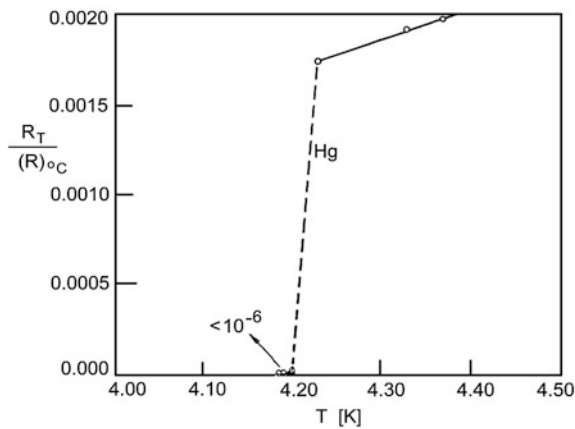


helium where the mercury freezes and becomes a long wire. The ends of the mercury wire are connected to platinum wires L_1 and L_2 to facilitate passing of current and the measurement of potential difference.

The results obtained by Onnes on Hg and a few other metals are shown in Fig. 13.2. The figure shows the electrical resistance (expressed as a fraction of its value at 0 °C) at different temperatures. It can be seen that the resistance of Pt and Au decreases smoothly as the temperature is lowered from about 20 K down to ~ 1.5 K. The resistance of mercury also decreases smoothly (though more steeply than in Pt and Au) till about 5 K but at a temperature of 4.2 K, it abruptly drops almost to zero. In the words of Onnes [9], the resistance which was 1/500 at 4.2 K dropped to ‘a thousand millionth part’ of that at normal temperature.

That the fall in resistance was abrupt and sharp is seen more clearly in Fig. 13.3. In terms of conductivity, the conductivity has reached a value which is a thousand million times that at normal temperature. Onnes concluded that mercury has entered a new state—the state of superconductivity.

Fig. 13.3 Abrupt fall in the fractional resistance $[R_T/(R)_{0C}]$ of mercury



The temperature at which superconductivity appears was called the ‘transition point’ by Onnes. It is generally referred to as the transition temperature T_c . Below T_c , a metal is in the superconducting state and above T_c it behaves like a normal metal.

13.3 Occurrence

Mercury was the first metal in which superconductivity was observed by Kammerlingh Onnes. In the same series of experiments he observed superconductivity in Sn ($T_c = 3.72$ K) and Pb ($T_c = 7.19$ K). Onnes’ discovery of superconductivity led to a study of superconductivity by many other research groups and laboratories and since then superconductivity has been observed in several other metals. A list (not exhaustive) of superconducting metals along with their transition temperatures is given in Table 13.1. Among elements which are superconductors at ambient pressures, niobium has the highest transition temperature ($T_c = 9.25$ K).

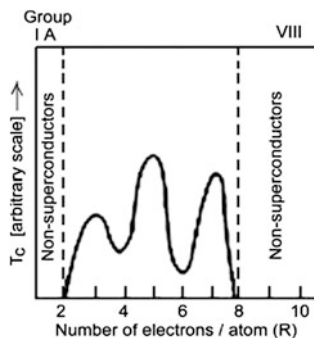
Mathias [10] proposed the following empirical rules for the appearance of superconductivity:

- (i) Superconductivity occurs in metallic elements in which the number of valence electrons per atom (R) has values between 2 and 8 (Fig. 13.4).
- (ii) Elements with odd R have high T_c values and those with even R values have low T_c values.
- (iii) Some elements which are not superconductors in the pure form become superconductors in the alloyed form.
- (iv) Maximum T_c has been observed for alloys and compounds having R close to 5 and 7.

Table 13.1 Values of T_c for some elements which are superconductors at atmospheric pressure

Element	T_c (K)
Niobium	9.25
Lead	7.19
Tantalum	4.47
Mercury	4.15
Tin (grey)	3.72
Indium	3.41
Tellurium	2.38
Aluminium	1.17
Gallium	1.08
Cadmium	0.56
Titanium	0.4

Fig. 13.4 Plot of T_c against number of valence electrons (Z) for some metals



- (v) Transition metals with $R < 2$ do not exhibit superconductivity. Also, monovalent metals at ambient pressures and noble metals do not show superconductivity. Ferromagnetic metals and rare earth elements except lanthanum do not display superconductivity at ambient pressures.
- (vi) In general, superconductivity occurs in materials with large resistivity ρ . Empirically, materials with $R\rho > 10^6$ show superconductivity.

A few other observations are:

- (i) The sharpness of transition temperature depends on the state of purity of the sample. In some cases it can occur within a temperature of 0.001 K.
- (ii) Presence of traces of paramagnetic elements lowers the transition temperature or even destroys the superconductivity; for instance a few parts per million of Fe destroys superconductivity in molybdenum. One atomic percent of gadolinium lowers the transition temperature of lanthanum from 5.6 to 0.6 K.

The development of high pressure techniques has made it possible to study materials at high pressures. The earlier piston-displacement methods can be used only up to a pressure of 25 GPa. But the invention of the diamond anvil techniques has extended the pressure range to 260 GPa. Several elements have been scanned for possible superconductivity at these high pressures and, indeed many of them have revealed the existence of superconductivity. The elements which are non-superconducting at normal pressures but become superconducting at higher pressures are listed in Table 13.2 along with the pressure and T_c [11]. The onset of superconductivity in these elements could be due to different reasons. Two important mechanisms are crystal structure transition and a transition in electronic structure. Bismuth is an example of the first type. It is non-superconducting in the trigonal structure at normal pressure but becomes a superconductor with $T_c = 8.7$ K at a pressure of 9 GPa where it has a fcc structure. The onset of superconductivity in the alkaline earth metals Ca, Sr, Ba is an example of the second type; these metals undergo an s - d electronic transition at high pressure.

Table 13.2 Values of T_c for some elements which are superconducting at a higher pressure

Element	Pressure (GPa)	T_c (K)
Si	12.0	8.5
Ge	11.5	5.4
Sb	36.0	8.5
Bi	9.0	8.7
Fe	21.0	2.0
Ca	150	4.0
Sr	50.0	4.0
Ba	20.0	5.0
Cs	8.0	1.7
Ce	5.0	1.7

Table 13.3 summarizes information on elemental superconductors at ambient pressures and high pressures.

The systems covered in Tables 13.1 and 13.2 have $T_c \leq 10$ K. In view of possible applications, there has been a search for superconductors with higher T_c . A number of superconducting compounds are listed in Table 13.4. Some of them have $T_c > 10$ K. Among these Nb_3Ge has the highest T_c of 23 K (which is between liquid hydrogen and liquid nitrogen temperatures).

The subject of superconductivity is now a century old. During this period superconductivity has been observed “in most of the natural elements, alloys, thousands of inorganic compounds and even organic compounds and in the double-stranded molecule of DNA” [11]. Detailed discussion of all these systems is beyond our scope.

To complete this discussion, we mention the discovery by Bednorz and Muller [12] of superconductivity in $La_{2-x}M_xCuO_4$ ($M = Ba$ or Sr) ceramic at $T_c \approx 40$ K. The ceramic $YBa_2Cu_3O_7$ was soon found to have a $T_c \approx 93$ K (which is above the liquid nitrogen temperature). We shall discuss this aspect in a later section.

13.4 Properties of Superconductors

When a metal becomes a superconductor some properties such as thermal properties, magnetic properties and thermodynamic properties show some changes. Changes in thermodynamic properties depend on whether the transition takes place with or without the presence of magnetic field. We discuss the thermal and magnetic properties in this section and the thermodynamic properties in Sect. 13.5.

Table 13.3 Summary of information on elemental superconductors

		Ambient pressure superconductor		High pressure superconductor		
		T_c (K) $T_c^{\max}$ (K) P (GPa)		$T_c^{\max}$ (K) P (GPa)		
H						He
Li	0.0004 14 30	Be 0.026				Ne
Na		Mg				Ar
K		Ca	25 161	Sc	0.39 19.6 106	B
Rb		Sr	7 50	Y	0.546 19.5 115	Al
Cs	1.3 1.2	Ba	5 18	La-Lu	0.12 4.483 8.6 62	Si
Fr		Ra		Insert Ac-Lr		Ge
						Ga
						Zn
						Cu
						Ni
						Co
						Fe
						Mn
						Cr
						V
						Ti
						Sc
						Ca
						Sr
						Rb
						K
						Na
						Li
						H
						He
						Ne
						Ar
						Kr
						Xe
						Rn
						At
						Po
						Bi
						Pb
						Tl
						Hg- α
						Cd
						In
						Sn
						Sb
						Te
						I
						Br
						O
						F
						Lu
						Yb
						Tm
						Er
						Ho
						Dy
						Tb
						Gd
						Eu
						Sm
						Pm
						Nd
						Pr
						Ce
						La-fcc
						La-Lu
						Ac
						Th
						Pa
						U
						Np
						Pu
						Am
						Cm
						Bk
						Cf
						Es
						Fm
						Md
						No
						Lr

Table 13.4 Values of T_c for some superconducting compounds

Compound	T_c [K]
Nb ₃ Ge	23.20
Nb ₃ Ga	20.30
Nb ₃ Al	18.60
Nb ₃ Sn	18.05
NbN	17.30
ZrN	13.70
NiBi	4.25
InSb	1.90

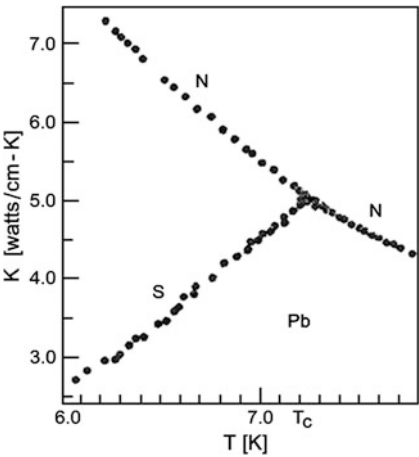
13.4.1 Thermal Properties

13.4.1.1 Thermal Conductivity

Thermal conductivity of tin is found to change drastically below the transition temperature and the value approaches zero, approximately varying as the cube of the transition temperature.

For lead, the variation in thermal conductivity with temperature is shown in Fig. 13.5. Curve N shows the thermal conductivity in the normal state and the curve S in the superconducting state. It can be seen that below T_c the thermal conductivity (K) shows a sharp decrease with decreasing temperature. On the other hand the curve for the normal state shows a continuous and smooth decrease with increasing temperature.

Fig. 13.5 Thermal conductivity of Pb as a function of temperature. Below T_c the thermal conductivity is plotted on the lower curve. The data points on the *upper curve* below T_c were obtained by application of magnetic field; they represent the normal state of Pb



Normally good electrical conductors are good heat conductors. The superconductors are good electrical conductors but are poor thermal conductors. This indicates that when a normal conductor changes to a superconductor at T_c , some fraction of electrons do not transport heat (entropy). This is the characteristic property of superconducting electrons.

13.4.1.2 Thermoelectric Effect

An electrical circuit of two superconductors or one superconductor and the other normal conductor shows none of the thermoelectric effects. The absence of Peltier effect indicates that the electrons in the superconducting phase do not carry any entropy. This shows that entropy in the superconducting state is less than the entropy in the normal state (see Sect. 13.5).

13.4.2 Magnetic Properties

13.4.2.1 Critical Field

Superconductivity may be destroyed by application of a strong magnetic field. The field may be either an externally applied field or produced by a current flowing inside the metal. The critical value of the applied field required to destroy superconductivity is denoted by H_c . The critical field H_c is a function of temperature. The variation of critical field as a function of temperature $H_c(T)$ may be represented approximately by a parabola with the following equation:

$$H_c(T) = H_0 \left[1 - \left(\frac{T}{T_c} \right)^2 \right] \quad (13.1)$$

where H_0 is the field necessary to destroy superconductivity at absolute zero.

The temperature variation of the critical field for a typical superconductor is shown in Fig. 13.6. At any point of the curve, the material is in the normal state above H_c and in the superconducting state below H_c . The temperature variation of H_c for some selected elements is shown in Fig. 13.7. The value of H_c for some elements is shown in Table 13.5.

13.4.2.2 Meissner Effect

Meissner and Ochsenfeld [13] discovered that when a magnetic field is applied to a superconductor at a temperature $< T_c$, the magnetic field is completely expelled (Fig. 13.8a). They measured the magnetic flux distribution outside the samples of tin

Fig. 13.6 Schematic plot of the critical magnetic field as a function of temperature

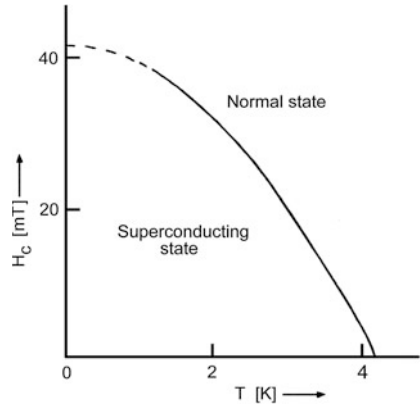


Fig. 13.7 Temperature variation of critical field for some superconducting elements

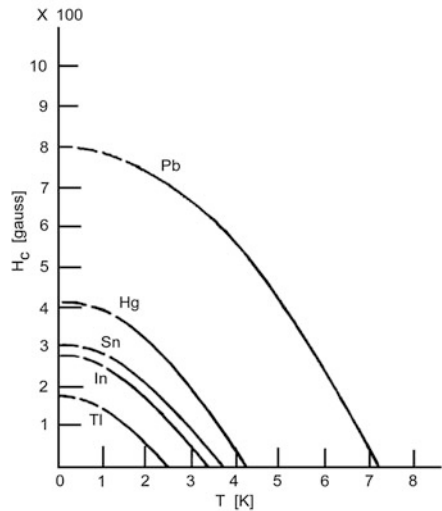


Table 13.5 Critical magnetic field (H_c) at 0 K for some elements

Element	H_c (gauss)
Nb	2060
Pb	803
Ta	830
La	1100
Zr	47
Ti	171
Ga	51
Hg	417

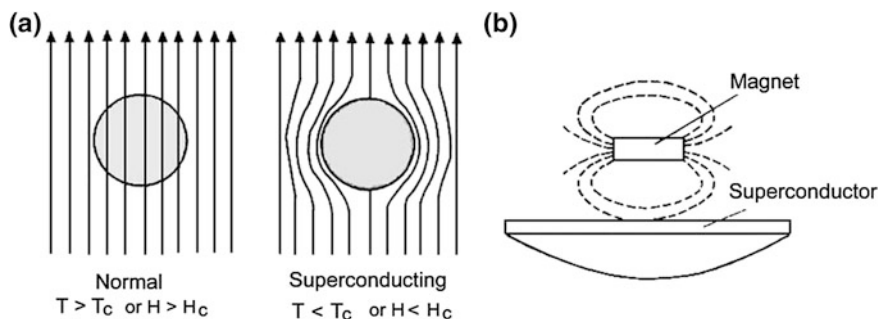
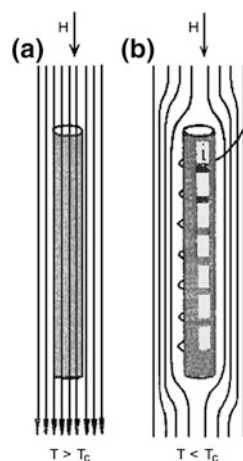


Fig. 13.8 **a** The Meissner effect and **b** principle of levitation

Fig. 13.9 Flux distribution around a superconducting rod at **(a)** $T > T_c$ and **(b)** $T < T_c$; note the flux exclusion due to induction current i



and lead placed in a magnetic field at a temperature below their transition temperature. Their experiments indicated that the magnetic behaviour for a perfect conductor and a superconductor differ. This could not be explained solely by taking into account that a superconductor is a resistanceless material. Their findings indicated that in addition to being a perfect conductor, a superconductor is also a perfect diamagnet i.e. $B/H = 0$. The applied field is completely cancelled by the induced magnetization in the specimen which opposes the applied field. This phenomenon is known as Meissner effect. The Meissner effect was first demonstrated by Arkadyev [14] by the celebrated ‘suspended magnet’ experiment (Fig. 13.8b). The magnet is lifted when the lines of force of the magnet are repelled by the surface of the superconductor. This phenomenon is known as magnetic levitation.

Consider a sample of a superconductor in the form of a long rod and apply a magnetic field parallel to the long axis as shown in Fig. 13.9a. When the sample is cooled to a temperature below T_c the flux is excluded. This is because of the induction current in the sample (Fig. 13.9b).

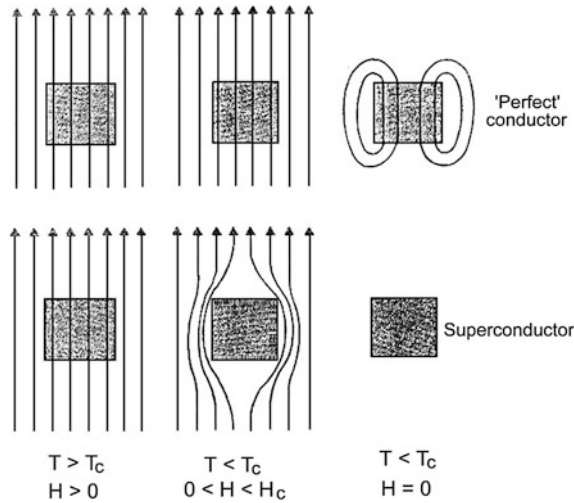


Fig. 13.10 Difference between flux distribution in a hypothetical perfect conductor and a superconductor

We now compare the flux configuration in a ‘perfect conductor’ and a superconductor in the form of solid blocks (Fig. 13.10). We shall first examine a perfect conductor (first row). Initially at $T > T_c$ and at $T < T_c$ in the presence of the field the sample acquires induced current but when the field $H = 0$ the induced field remains frozen in the sample. In the second row for the superconductor, initially it is at $T > T_c$ (normal state) in the presence of the field. When $T < T_c$ and field is below H_c , it expels the flux, since the metal is superconducting. Finally, the termination of the field induces an opposing surface current and leaves the superconductor field-free and current-free. It is to be noted that the transition from a normal to a ‘perfect conductor’ is not reversible, while in the case of a superconductor the transition between superconducting and normal state is reversible.

We now examine the flux distribution in a superconducting ring (Fig. 13.11). In the first row, when $T > T_c$ and the magnetic field is on, the flux penetrates everywhere. When the sample is cooled to $T < T_c$ and the field is still on, the flux is excluded from the cross-section of the ring by the induction current in the loop. Finally, when magnetic field is withdrawn, the flux inside the ring is trapped because the flux cannot pass through the superconductor. A large current is induced by the field which maintains the trapped flux. This current persists even after the field is withdrawn as the resistance is zero. This is called the persistent current. On the other hand, in the second row, to start with the ring is at $T > T_c$ and at zero field. It is next cooled to $T < T_c$ and the field is still zero. Finally, when the ring is at $T < T_c$ and the field is switched on, the flux is excluded from the ring as well as the hole since the flux lines are unable to cross the superconductor. Thus, in this case, in contrast to the case of the solid block, the flux configuration depends on whether the field is applied before or after the sample is cooled. In the Meissner effect, this

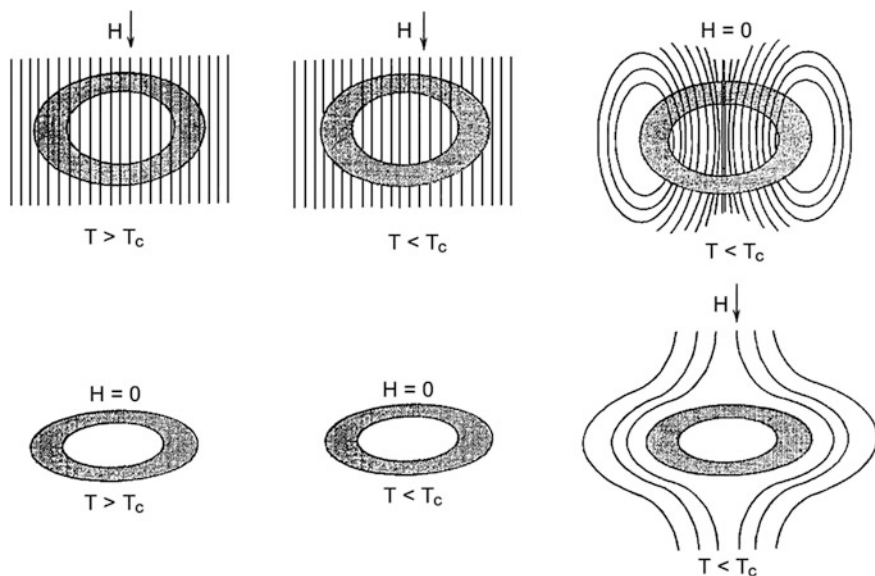


Fig. 13.11 Flux distribution around a superconducting ring with cooling and application of field in different sequences

exclusion of the magnetic flux from the interior of the sample plays an important role. Unlike the electric field which cannot penetrate a sample, a magnetic field can penetrate a superconductor within a range of 10^{-6} – 10^{-4} cm.

13.4.3 Type I and Type II Superconductors

We consider the equation for magnetic induction B in terms of the applied field H and magnetization M as

$$B = H + 4\pi M. \quad (13.2)$$

When $B = 0$, we have

$$H = -4\pi M. \quad (13.3)$$

Curves showing magnetization versus applied magnetic field are shown in Fig. 13.12. Figure 13.12a corresponds to a superconductor showing complete Meissner effect and obeying (13.2). Above the critical field H_c the specimen is a normal conductor. It is to be noted that the negative value of M represents diamagnetism. Pure specimens of many materials show this behaviour and are called type I superconductors. They are also called ‘soft superconductors’.

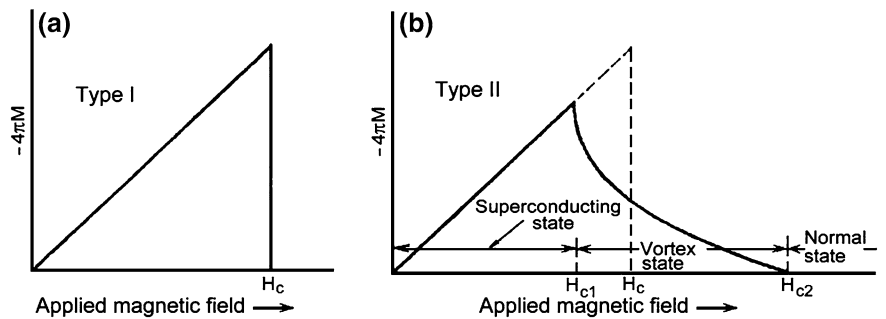


Fig. 13.12 Curves showing magnetization ($-4\pi M$) versus applied magnetic field for (a) type I and (b) type II superconductors

Table 13.6 Critical magnetic field H_{c2} at 0 K for some type II superconductors

Superconductor	H_{c2} (10^4 gauss)
Nb ₃ Ge	38
Nb ₃ Al	17.5
Nb ₃ Ga	3.4
NbN	47.0
La ₃ In	13.4
NiBi	4.2
InSb	1.9
V ₃ Si	25

There are some materials exhibiting the magnetization curve shown in Fig. 13.12b. They are termed type II superconductors or ‘hard superconductors’. While pure samples of many materials are of type I, alloys and transition metals with high resistivity values in the normal state tend to be of type II. The electrical properties of superconductors of type II differ from those of type I in that they have a vortex state in between a superconducting and a normal state.

In type II superconductors the magnetic flux is excluded completely from the sample for comparatively weak fields up to H_{c1} . Above H_{c1} in the region up to H_{c2} there exists a partial penetration of flux. It is said to be in the vortex state. Above H_{c2} for high fields (>100 times H_c) the flux penetrates completely and the superconductivity vanishes.

Values of the critical field H_c of type I superconductors are too low for any technical application. But type II materials with high values of H_{c2} are of technical importance. Values of H_{c2} for some materials are listed in Table 13.6.

Type I may be changed to type II by addition of alloying elements. They have high electrical resistivity and short mean free path; hence they favour superconductivity. Lead is a superconductor of type I in its pure form. But with addition of 2 wt% of indium, it changes to type II. The magnetization curves with increasing

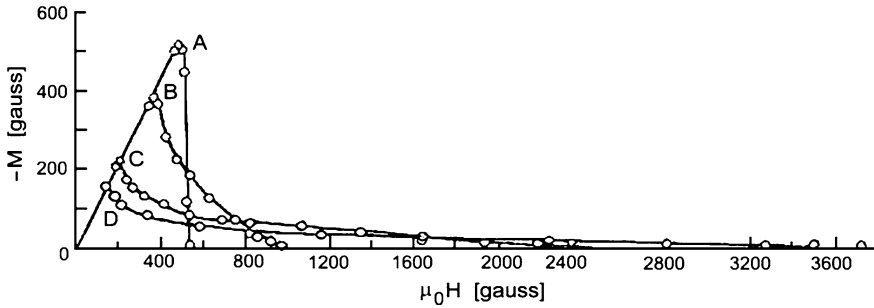
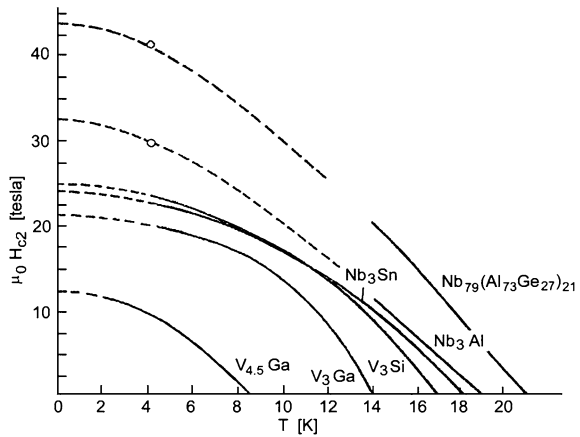


Fig. 13.13 Magnetization curves at 4.2 K for lead and lead-indium alloys: A, Pb; B, Pb-2.08 wt% In; C, Pb-8.23 wt% In; D, Pb-20.4 wt% In

Fig. 13.14 Plot of H_{c2} versus temperature for some type II superconductors



percentage of indium are shown in Fig. 13.13 [15]. It is to be noted that the areas under the curves are the same for type I (A) and type II (B, C, D) samples. Other characteristics like the transition temperature, the energy gap and the intrinsic electronic structure remain unaltered. The dependence of the critical magnetic field H_{c2} on temperature for some type II superconductors was studied by Foner et al. [16]; their results are shown in Fig. 13.14.

13.4.4 Isotope Effect

The variation of critical temperature with isotopic mass M is termed isotope effect. For instance for mercury T_c changes from 4.185 to 4.146 K for a change in the isotopic mass from 199.5 to 203.4 atomic mass units. The critical temperature of different isotopes of a metallic element varies with isotopic mass approximately as

the inverse square root of the atomic mass. This dependence of temperature on the atomic mass provides direct evidence that the ionic motion plays an important role in establishing superconductivity. The involvement of phonons in superconductivity was first proposed by Frohlich [17] and evidence in support is provided by isotope effect.

The variation of T_c with isotopic mass M may be expressed as

$$T_c = \frac{1}{M^\alpha} \quad (13.4)$$

where $\alpha \approx 0.5$. The Debye temperature θ_D also has a similar mass dependence ($\theta_{\text{Debye}} \propto M^{-1/2}$). This is another indication of the involvement of lattice vibrations in superconductivity.

It was observed in due course that $\alpha = 0.5$ is applicable to s - p metals. For some metals such as transition metal superconductors, the values range from 0 to 0.3. However the importance of isotope effect in intuitively suggesting the concept of electron-phonon interaction cannot be ignored.

13.5 Thermodynamics of Superconducting Transition

The transition from normal to superconducting state may be expressed in terms of thermodynamic quantities as in the case of transition from liquid to vapour phase. This will allow us to obtain expressions for entropy, latent heat and specific heat in terms of critical field and temperature.

We consider a type I superconductor with complete Meissner effect so that, the flux B is zero, i.e. $M = -H/4\pi$.

The Gibb's free energy G per unit volume is given by

$$G = U - TS - HM \quad (13.5)$$

where U is the internal energy, T the temperature, S the entropy, H the magnetic field and M the magnetization. In this equation, pV in the standard equation is replaced by $-HM$ as we are dealing with a magnetic system. Differentiating (13.5) and using $dU = TdS + HdM$ from the second law of thermodynamics, we have at constant temperature

$$dG = -MdH = \frac{H}{4\pi}dH. \quad (13.6)$$

For the superconducting state, the Gibb's free energy $G_s(H)$ for an increase of field from zero to H is

$$\int_0^H dG = \int_0^H \frac{H}{4\pi} dH = \frac{H^2}{8\pi} \quad (13.7)$$

or,

$$G_s(H) - G_s(0) = H^2/8\pi. \quad (13.8)$$

At $H = H_c$ the critical field, the normal and superconducting states are at equilibrium at a given temperature T . Hence

$$G_n(T, H_c) = G_s(T, H_c) = G_s(0) + H_c^2/8\pi. \quad (13.9)$$

The variation of free energy recorded for aluminium as a function of temperature is shown in Fig. 13.15a. It can be seen that for temperatures $T < T_c$, G_n is higher than G_s . The curves merge at the transition temperature.

Fig. 13.15 **a** Experimental values of G_n and G_s for Al at low temperature. G_s is measured at zero magnetic field and G_n below T_c by applying magnetic field; **b** free energies G_n and G_s versus applied magnetic field

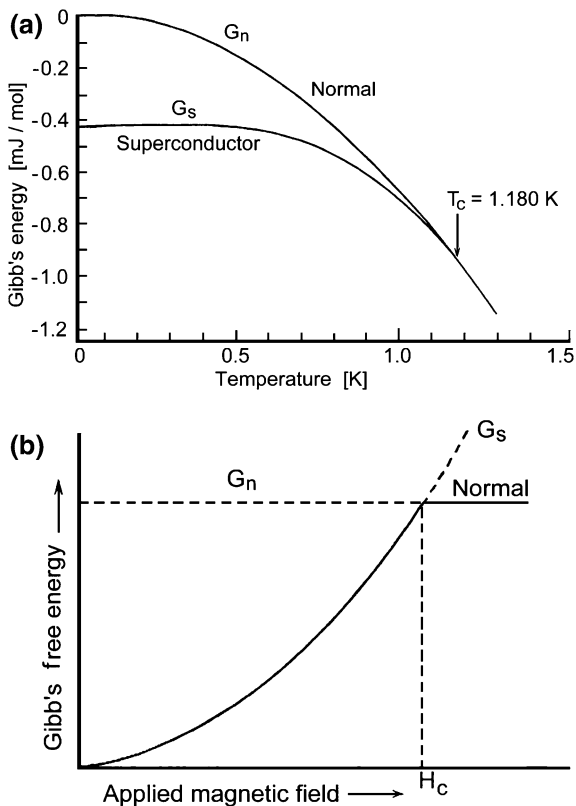
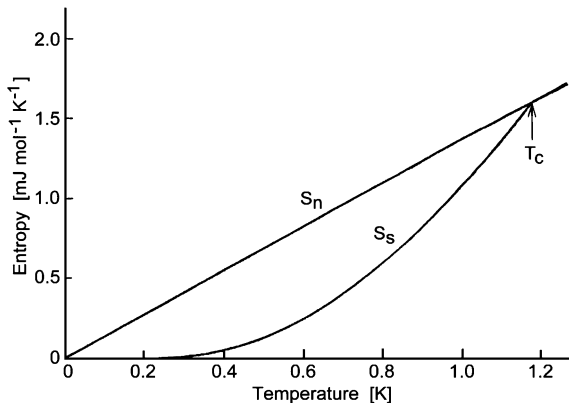


Fig. 13.16 Plot of entropy versus temperature for Al in normal (S_n) and superconducting (S_s) states



The variation of free energy for the normal and superconducting states is shown as a function of applied field in Fig. 13.15b. Up to the critical field the free energy for the superconducting phase is lower than that of the normal state. At the critical field the free energies are equal. Beyond the critical field, the free energy of the superconducting phase shows an increase. For the normal metal, the free energy is independent of the applied field.

Entropy Entropy is defined as $S = -(\partial G/\partial T)_p$. Hence, using (13.9) we get

$$\left(\frac{\partial G_n}{\partial T}\right)_p - \left(\frac{\partial G_s}{\partial T}\right)_p = -(1/8\pi)2H_c \left(\frac{\partial H_c}{\partial T}\right)_p$$

or,

$$S_n - S_s = -\frac{H_c}{4\pi} \frac{\partial H_c}{\partial T}. \quad (13.10)$$

As $(\partial H_c/\partial T)_p$ is always a negative quantity, $S_n - S_s$ is positive. In the presence of a magnetic field, the entropy in the normal state is greater than that in the superconducting state. Figure 13.16 shows the experimental results for aluminum. The lower entropy in the superconducting state shows that electrons in the superconducting state are in an orderly state.

Latent Heat The latent heat

$$Q = T(S_n - S_s) = -T \frac{H_c}{4\pi} \frac{\partial H_c}{\partial T}. \quad (13.11)$$

At $T = T_c$, $H_c = 0$. Hence the latent heat is zero at the transition temperature.

13.5.1 Specific Heat

The specific heat per unit volume C is given by $dQ/dT = TdS/dT$. Then the difference between specific heats in normal and superconducting states is given by $C_n - C_s = T \frac{\partial}{\partial T} (S_n - S_s)$. Using (13.11), we get

$$C_n - C_s = -\left(\frac{T}{4\pi}\right) \left[H_c \frac{\partial^2 H_c}{\partial T^2} + \left(\frac{\partial H_c}{\partial T} \right)^2 \right]. \quad (13.12)$$

At the transition temperature, we have $T = T_c$ and $H_c = 0$. Then (13.12) becomes

$$C_s - C_n = \Delta C = \left(\frac{T_c}{4\pi} \right) \left(\frac{\partial H_c}{\partial T} \right)^2. \quad (13.13)$$

Equation (13.13) is known as Rutgers's law; it has been verified experimentally [5]. Combining (13.13) with (13.1), we get

$$C_s - C_n = \left(\frac{3}{2\pi} \right) \left(\frac{H_0^2}{T_c} \right) \left[\left(\frac{T}{T_c} \right)^3 - \frac{1}{3} \left(\frac{T}{T_c} \right) \right]. \quad (13.14)$$

Both C_s and C_n consist of two components, the lattice component (with index l) and an electronic component (with index e). We may express C_s and C_n as $C_s = C_{ls} + C_{es}$ and $C_n = C_{ln} + C_{en}$. Assuming that the lattice components are the same in the normal state and the superconducting state, we may rewrite (13.14) as

$$C_s - C_n = C_{es} - C_{en} = \left(\frac{3}{2\pi} \right) \left(\frac{H_0^2}{T_c} \right) \left[\left(\frac{T}{T_c} \right)^3 - \frac{1}{3} \left(\frac{T}{T_c} \right) \right]. \quad (13.15)$$

From the free electron theory (Chap. 3), it is known that

$$C_{en} = \gamma T. \quad (13.16)$$

Comparing (13.15) and (13.16), we get

$$\gamma = \left(\frac{1}{2\pi} \right) \left(\frac{H_0^2}{T_c^2} \right). \quad (13.17)$$

Obviously, C_{es} is the T^3 term in (13.15). Thus we have

$$C_{es} = 3\gamma T_c \left(\frac{T}{T_c} \right)^3. \quad (13.18)$$

Experimental results agree with (13.18) except at very low temperatures.

From experiments at very low temperatures on the superconducting phase, it is found that the electronic contribution C_{es} decreases exponentially. Taking this into account, (13.18) may be modified as

$$\frac{C_{es}}{\gamma T_c} = a \exp(-bT_c/T) \quad (13.19)$$

where a and b are constants. Further, (13.19) may be written as

$$\frac{C_{es}}{\gamma T_c} = a \exp(-\Delta/k_B T) \quad (13.20)$$

where Δ is called the energy gap parameter; we shall discuss this parameter in a separate section.

A second order transition is characterized by a nil latent heat at the transition point and a discontinuity in the specific heat at the transition point. From (13.11) and (13.13), we see that the superconducting transition fulfills these characteristics and, hence, is a second order transition.

We have discussed theoretically several aspects of the specific heats in the normal and superconducting states. We shall now compare these theoretical results with experimental observations:

- (i) A comparison of ΔC from (13.12) with experimental data on tin from the work of Keesom and Laer [18] is shown in Fig. 13.17. The dashed curve represents values from (13.12). It can be seen that the agreement is fair.
- (ii) Figure 13.18 shows data on specific heat of gallium in the normal and superconducting states [19]. Data for the normal state below T_c was obtained by application of a magnetic field of 200 Gauss. A sudden drop in specific heat is noticed at T_c .

Fig. 13.17 Temperature variation of ΔC for tin; smooth curve represents experimental data [18]; dashed curve from (13.12)

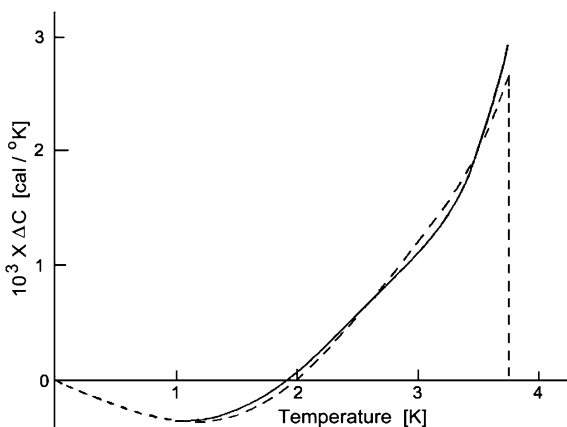


Fig. 13.18 Temperature variation of specific heat (C_p) of gallium in the normal and superconducting states [19]; values for the normal state below T_c were obtained by applying a magnetic field

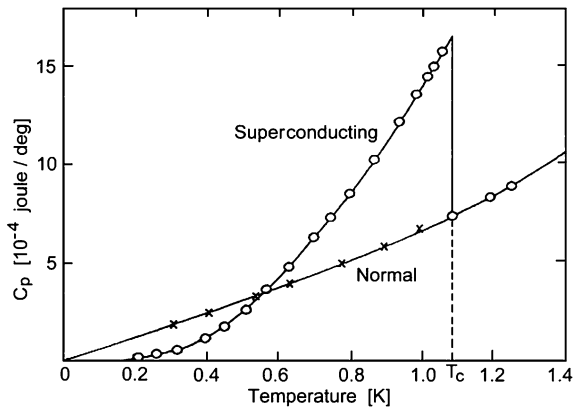
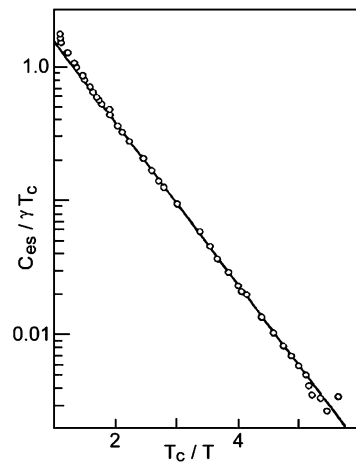


Fig. 13.19 Temperature variation of the electronic specific heat C_{es} of gallium below T_c ; the plot has slope $(\Delta/k_B T)$ in conformity with (13.20)



- (iii) A semi-log plot of the electronic specific heat C_{es} of gallium in the superconducting phase against T_c/T is shown in Fig. 13.19. The straight line plot is in conformity with (13.20).
- (iv) Figure 13.20 shows data on specific heat of NbSe_2 in the normal and superconducting phases [20]. Plot of C/T against T^2 yields γ as intercept. The figure explains how values of C_{en} , C_{es} and $\Delta C/T_c$ can be obtained. C_{ph} is the lattice specific heat which is calculated from a known value of the Debye temperature.

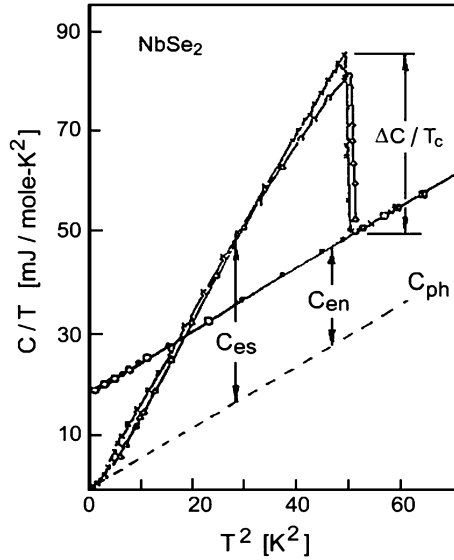


Fig. 13.20 The normal electron contribution (C_{en}), superconducting electron contribution (C_{es}) and phonon contribution (C_{ph}) to the specific heat of superconductor NbSe_2

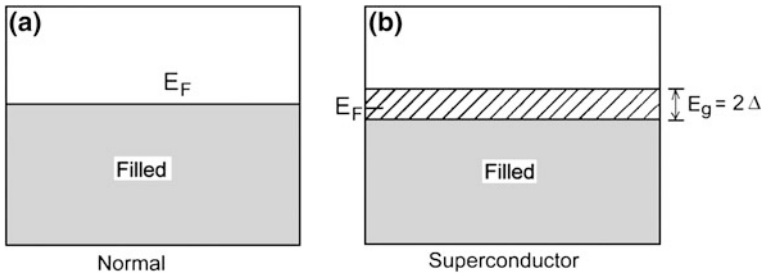


Fig. 13.21 **a** Filled conduction band in a normal metal; **b** the energy gap $E_g (=2\Delta)$ at the Fermi level E_F in a superconductor

13.5.2 Energy Gap

There is an energy gap in the energy spectrum of the electrons in a superconductor. This represents the energy required by the superconducting electrons to change over to normal electrons. The position of the energy gap (E_g) vis-à-vis the Fermi energy (E_F) and the filled band is shown in Fig. 13.21. The energy gap E_g is much smaller than the energy gap in a semiconductor. In a semiconductor the origin of the energy gap is in the electron-lattice interaction. This interaction keeps the electrons in an ordered state with respect to the Fermi gas.

Table 13.7 Values of energy gap E_g by different methods for some superconductors

Superconductor	E_g in units of $k_B T_c$		
	From specific heat	From infrared absorption	From electron tunneling
Sn	3.6	3.6	4
In	3.9	4.1	4
Pb		4.1	4.2
Hg (α)			4.6
Al			3.4
Nb			3.8
V			3.4
Zn			3.2
Cd			3.2
Ta			3.6

The energy gap in a superconductor is temperature dependent. It decreases to zero as the temperature approaches the critical temperature T_c . The energy gap E_g is twice the energy gap parameter Δ which was introduced in the expression for C_{es} (13.20).

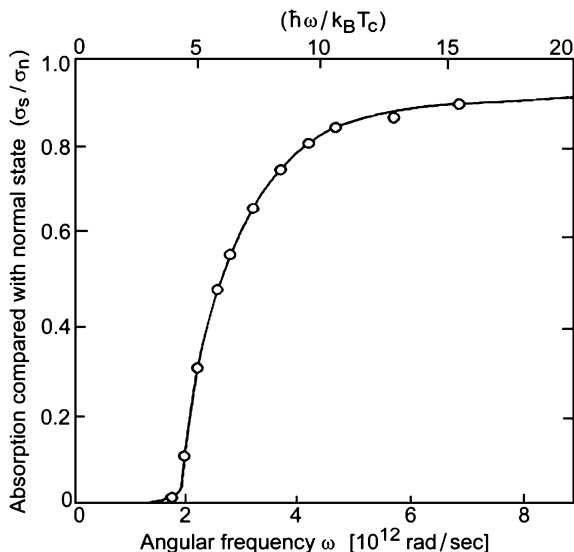
The energy gap can be determined from the experimental values of C_{es} . It can also be determined from other phenomena like radiation absorption and quantum tunneling which will be discussed later. The values of E_g for some superconducting elements are given in Table 13.7.

13.5.3 Absorption of High Frequency Electromagnetic Radiation

Superconductors have zero DC resistance at T_c . But the AC resistance is not zero though it is small. The absorption of electromagnetic radiation is studied in terms of complex resistance or complex conductivity. The absorption of electromagnetic radiation by superconductors was studied by Biondi et al. [21], Glover and Tinkham [22, 23] and by Ginsberg and Tinkham [24]. Although their experimental techniques were slightly different, the results are the same, viz., the absorption of radiation by a superconductor suddenly increases at a critical frequency ω_c which lies in the far infrared or microwave region (10^{10} – 10^{12} Hz).

Ginsberg and Tinkham made measurements of absorption by thin films in the frequency range $\omega = 10^{12}$ rad/s. Their results for absorption are given as the ratio σ_s/σ_n , σ_s and σ_n being the complex conductivities in the superconducting and normal states. They studied lead, tin and indium. Their results for indium are shown in Fig. 13.22. It can be seen that the absorption which is low increases suddenly at $\omega_c \approx 2 \times 10^{12}$ rad/s.

Fig. 13.22 Frequency variation of the absorption of electromagnetic radiation by an indium film in the superconducting state



This frequency is associated with an energy $\hbar\omega_c = 4k_B T_c$. We can interpret this absorption as selective absorption of energy by the superconducting electrons to cross over the energy gap E_g in order to become normal electrons. The optical effect is that the indium thin film which was semi-transparent at frequencies less than ω_c becomes opaque at ω_c and higher frequencies. This frequency gives the energy gap ($E_g = \hbar\omega_c$). The results obtained by Ginsberg and Tinkham are given in Table 13.7. The energy gap values obtained from absorption are in agreement with those from specific heats. It is to be noted that whereas absorption experiments give values of $E_g = 2\Delta$, the specific heat experiments give values $E_g = \Delta$.

13.6 Theories of Superconductivity

The first theoretical approach to account for superconductivity was proposed 21 years after its discovery by Gorter and Casimir [25, 26]. The theory, known as ‘the two-fluid model’, is analogous to the two-fluid theory of liquid helium. In the superconducting state the electrons consist of two types, normal electrons and superconducting electrons. We shall denote their densities or concentrations by n_n and n_s respectively. The ratio of n_n and n_s changes with temperature. The superconducting electrons do not carry any entropy and flow with zero resistance, while the normal electrons behave like the electrons in any normal metal. Above the transition temperature all the electrons are normal and as the temperature reaches 0 K, all are of the superconducting type. Thermodynamical analysis has shown that the free energy and the entropy in the superconducting phase are lesser than those of the normal phase in the temperature range $0 - T_c$ (Sect. 13.5).

The two-fluid theory could account for the experimentally observed parabolic behaviour for the critical field and T^3 law for the specific heat, but could not account for several other characteristic properties mentioned in Sect. 13.4.

In the following sections we discuss Londons' modified two-fluid theory [27] followed by an account of the microscopic theory proposed by Bardeen, Cooper and Schrieffer [28]. Finally, the theory by Ginzburg and Landau [29] will be discussed in brief.

13.6.1 London Equations

London and London [27] modified the two-fluid theory of Gorter and Casimir [25] to take into account the perfect diamagnetic property of the superconducting state. The normal and super currents are assumed to flow parallel to each other in a superconductor.

The total current density $\mathbf{J}$ is given by

$$\mathbf{J} = \mathbf{J}_s + \mathbf{J}_n \quad (13.21)$$

where $\mathbf{J}_s$ is the superconducting electron current density and $\mathbf{J}_n$ the normal current density. Since the supercurrent $\mathbf{J}_s$ flows without any resistance it is assumed to carry the entire induced current. The contribution from normal current is ignored in the present treatment.

$\mathbf{J}_s$ is the flow of current due to electrons of mass m , charge e and density n_s . Under the influence of an electric field E , the equation of motion for the super-electrons is given by

$$m \frac{d\mathbf{v}_s}{dt} = -e\mathbf{E} \quad (13.22)$$

where $\mathbf{v}_s$ is the velocity of the superelectrons. The current density is

$$\mathbf{J}_s = -n_s e \mathbf{v}_s. \quad (13.23)$$

Using (13.22) and (13.23), we get

$$\frac{d\mathbf{J}_s}{dt} = \frac{n_s e^2}{m} \mathbf{E}. \quad (13.24)$$

Equation (13.24) is Londons' first equation. In this equation if the electric field $\mathbf{E} = 0$, the current $\mathbf{J}_s$ may still be finite. In comparison, for normal current flow

given by $\mathbf{J}_n = \sigma \mathbf{E}$ (where σ is the conductivity), no current flow is possible for $\mathbf{E} = 0$. The electric field $\mathbf{E}$ and the magnetic induction $\mathbf{B}$ are related by Maxwell's relation

$$\text{Curl } \mathbf{E} = -\frac{1}{c} \frac{d\mathbf{B}}{dt} \quad (13.25)$$

where c is the velocity of light. Substituting for $\mathbf{E}$ from (13.24) in (13.25), we have

$$\text{Curl} \left[\left(\frac{mc}{n_s e^2} \right) \frac{d\mathbf{J}_s}{dt} \right] = -\frac{d\mathbf{B}}{dt}. \quad (13.26)$$

Integrating (13.26) with respect to time gives

$$\text{Curl} \left[\left(\frac{mc}{n_s e^2} \right) \mathbf{J}_s \right] = -\mathbf{B}. \quad (13.27)$$

F. London and H. London assumed that the constant of integration is zero. This also implies an assumption that in the bulk of the superconductor, where $\mathbf{B} = 0$, the current flow is irrotational. Substituting Λ for $(mc/n_s e^2)$, (13.27) may be written as

$$\text{Curl} (\Lambda \mathbf{J}_s) = -\mathbf{B}. \quad (13.28)$$

Equation (13.28) is known as Londons' second equation. If this equation is added to Maxwell's equations they describe together the superconductor characteristics. We consider Maxwell's two equations

$$\text{Curl } \mathbf{B} = (4\pi/c) \mathbf{J} \quad (13.29)$$

and

$$\text{div } \mathbf{B} = 0. \quad (13.30)$$

Using the identity,

$$\text{Curl } \text{Curl } \mathbf{B} = \text{grad } \text{div } \mathbf{B} - \nabla^2 \mathbf{B}, \quad (13.31)$$

(13.29) and (13.30) lead to

$$\nabla^2 \mathbf{B} = -\text{Curl } \text{Curl } \mathbf{B} = -\text{Curl} (4\pi/c) \mathbf{J}_s. \quad (13.32)$$

Substituting from (13.28) we get

$$\nabla^2 \mathbf{B} = \frac{4\pi}{c\Lambda} \mathbf{B}. \quad (13.33)$$

Writing $4\pi/c\Lambda = 1/\lambda^2$ we get

$$\nabla^2 \mathbf{B} = \frac{1}{\lambda^2} \mathbf{B}. \quad (13.34)$$

where

$$\lambda = \left(\frac{mc^2}{4\pi n_s e^2} \right)^{1/2}. \quad (13.35)$$

λ is called the London penetration depth. The solution of the one-dimensional form of (13.34) is

$$B(x) = B(0)e^{-x/\lambda}. \quad (13.36)$$

Equation (13.36) shows that the solution of the type $B(x) = \text{constant}$ cannot exist. The graphical representation of (13.36) is shown in Fig. 13.23. Equation (13.36) indicates that the flux density decreases exponentially inside the superconductor falling to $1/e$ of its initial value at a distance λ , the penetration depth. Equation (13.36) shows that if $B(0)$ is the field at the plane boundary at $x = 0$, i.e. at the yz plane, the field $B(x) \rightarrow 0$ as $x \rightarrow \infty$ indicating that flux inside the bulk of the superconductor is zero. Hence (13.36) accounts for Meissner effect.

The penetration depth λ varies with temperature according to the relation

$$\lambda(T) = \lambda(0) \left(1 - \frac{T^4}{T_c^4} \right)^{-1/2}. \quad (13.37)$$

Here $\lambda(0)$ is the penetration depth at $T = 0$. $\lambda(0)$ can be estimated from (13.35) by replacing n_s with n_n . As an example, $\lambda(0) = 520 \text{ \AA}$ for Hg.

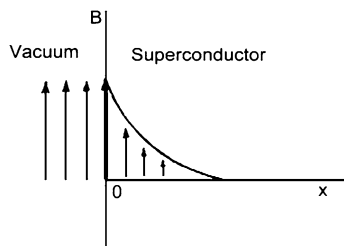


Fig. 13.23 Exponential decrease of magnetic flux (B) inside a superconductor placed in a magnetic field according to (13.36)

We now consider the effect of alternating current in a superconductor. The alternating current for the superconducting electrons is given by

$$\mathbf{J}_s = \mathbf{J}_{s0} e^{i\omega t}. \quad (13.38)$$

Using (13.24), we get

$$\mathbf{J}_s = -\frac{in_s e^2}{\omega m} \mathbf{E}. \quad (13.39)$$

Combining (13.35) with (13.39) we get

$$\mathbf{J}_s = -\frac{ic^2}{4\pi\lambda^2\omega} \mathbf{E}. \quad (13.40)$$

For normal electrons $\mathbf{J}_n = \sigma \mathbf{E}$, and since $\mathbf{J} = \mathbf{J}_n + \mathbf{J}_s$, we have

$$\mathbf{J} = \left(\sigma - \frac{ic^2}{4\pi\lambda^2\omega} \right) \mathbf{E}. \quad (13.41)$$

For high frequencies, (13.41) predicts that the metal behaves like a normal metal. Hence for optical frequencies, no difference is detected between the normal and superconducting states. However, a difference is observed at lower frequencies.

According to Londons' equations, the penetration depth is a fundamental length characteristic of a superconductor. During experimental work on superconductors with high frequency field, Pippard [30] observed that λ is not a simple constant. It shows anisotropy, changes with mean free path of electrons and is temperature dependent.

13.6.2 Coherence Length

Pippard extended Londons' theory by introducing a parameter ξ characterizing the superconductor. ξ is called the 'coherence length' and is defined as the distance within which the ratio of superconducting electrons to normal electrons cannot change drastically in a spatially varying magnetic field. It is also a measure of the minimum spatial extent of a transition layer between a normal conductor and a superconductor [4].

The Londons' equation (13.28) can be expressed in terms of the vector potential $\mathbf{A}$ of the local magnetic field as

$$\text{Curl}(\mathbf{A} \mathbf{J}_s) = -\mathbf{B} = \text{Curl} \mathbf{A}. \quad (13.42)$$

Then

$$\text{Curl } \mathbf{J}_s = -\frac{c}{4\pi\lambda^2} \text{Curl } \mathbf{A} \quad (13.43)$$

Equation (13.43) shows that the variation in $\mathbf{A}$ and $\mathbf{J}_s$ is similar. This is valid for fields which are uniform. For fields which vary rapidly in space, the average of $\mathbf{A}$ over a coherence length should be taken. This requires extra kinetic energy. We now proceed to obtain an expression for the increase in energy due to modulation following Kittel [4].

The wave equation for a plane wave is

$$\psi(x) = e^{ikx} \quad (13.44)$$

where k is the propagation vector. The wave function for a strongly modulated wave $\varphi(x)$ will be the sum of the modulated and plane waves. Thus,

$$\varphi(x) = e^{-1/2}(e^{i(k+q)x} + e^{ikx}) \quad (13.45)$$

where q is a modulating vector.

We obtain the kinetic energy using the momentum operator

$$p_k = m\dot{x} \quad (13.46)$$

and

$$\text{kinetic energy operator} = \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \right). \quad (13.47)$$

For unmodulated wave:

The probability density is

$$\psi^* \psi = e^{-ikx} e^{ikx} = 1 \quad (13.48)$$

Kinetic energy is

$$\begin{aligned} \int \psi^* \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \right) \psi dx &= \int e^{-ikx} \left(-\frac{\hbar^2}{2m} \right) \frac{\partial^2 e^{ikx}}{\partial x^2} dx \\ &= \frac{\hbar^2 k^2}{2m} \end{aligned} \quad (13.49)$$

For modulated wave:

The probability density is

$$\begin{aligned}
 \varphi * \varphi &= \frac{1}{2} \left(e^{-i(k+q)x} + e^{-ikx} \right) \left(e^{i(k+q)x} + e^{ikx} \right) \\
 &= \frac{1}{2} (2 + e^{iqx} + e^{-iqx}) \\
 &= 1 + \cos qx.
 \end{aligned} \tag{13.50}$$

Kinetic energy is

$$\begin{aligned}
 \int \varphi * \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \right) \varphi dx &= \frac{1}{2} \left(\frac{\hbar^2}{2m} \right) [(k+q)^2 + k^2] \\
 &= \frac{\hbar^2}{2m} k^2 + \frac{\hbar^2}{2m} kq + \frac{\hbar^2}{2m} q^2.
 \end{aligned} \tag{13.51}$$

Neglecting the q^2 term, the extra energy required to modulate is given by the second term of (13.51) as $\frac{\hbar^2}{2m} kq$. If this energy exceeds the energy gap, superconductivity will be destroyed. The critical value q_0 of the modulated wave is given by

$$\left(\frac{\hbar^2}{2m} \right) k_F q_0 = E_g \tag{13.52}$$

where k_F is the value of k at the Fermi surface. The intrinsic coherence length ξ_0 is given by $1/q_0$; hence we have

$$\xi_0 = \frac{\hbar^2 k_F}{2m E_g}. \tag{13.53}$$

In terms of the electron velocity at the Fermi surface v_F , where $v_F = \frac{\hbar k_F}{m}$, we have

$$\xi_0 = \frac{\hbar v_F}{2E_g} = \frac{\hbar v_F}{k_B T_c}. \tag{13.54}$$

A similar result was obtained by Faber and Pippard [31]; ξ_0 is given by

$$\xi_0 = a \frac{\hbar v_F}{k_B T_c} \tag{13.55}$$

where a is a constant obtained empirically. BCS theory [28] gives

$$\xi_0 = 0.18 \frac{\hbar v_F}{k_B T_c}. \tag{13.56}$$

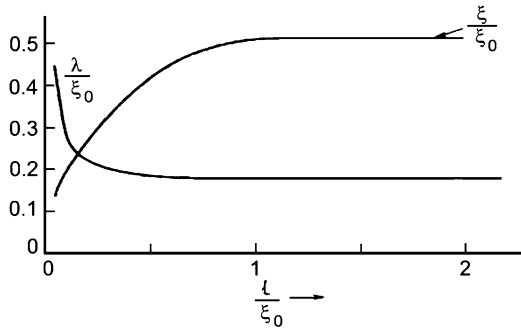


Fig. 13.24 Variation of the coherence length ζ and the penetration depth λ with the mean free path l (all parameters are expressed as fractions of the intrinsic coherence length ξ_0)

The coherence length ζ and the penetration depth λ are functions of mean free path l of conduction electrons in the normal state. The coherence length ζ is related to the intrinsic coherence length ξ_0 and the mean free path l as

$$\frac{1}{\zeta} = \frac{1}{\xi_0} + \frac{1}{l}. \quad (13.57)$$

These functions are shown in Fig. 13.24.

13.6.3 BCS Theory of Superconductivity

Nearly half a century after the discovery of superconductivity, Bardeen, Cooper and Schreiffer [28] proposed a comprehensive, microscopic theory (since popularly known as the BCS theory). The theory involves advanced mathematical techniques beyond our scope. Hence a qualitative treatment is provided.

Frohlich [17] and Bardeen [32] suggested independently that absorption and emission of phonons by electrons in a phonon field could result in an attractive interaction responsible for superconductivity. Frohlich, while dealing with work concerning polaron states in ionic crystals, suggested that the electron-phonon interaction plays a role in superconductivity. Further he suggested that if electron-phonon interactions are important, T_c should be proportional to the Debye temperature θ_D for isotopes of a given element. As θ_D is known to be proportional to $M^{-1/2}$ where M is the isotopic mass, T_c should also vary as $M^{-1/2}$. Subsequent experimental results on T_c of isotopes of several elements indeed confirmed the $T_c - M^{-1/2}$ relation and thus lent support to Frohlich's suggestion of the important role of electron-phonon interaction in superconductivity.

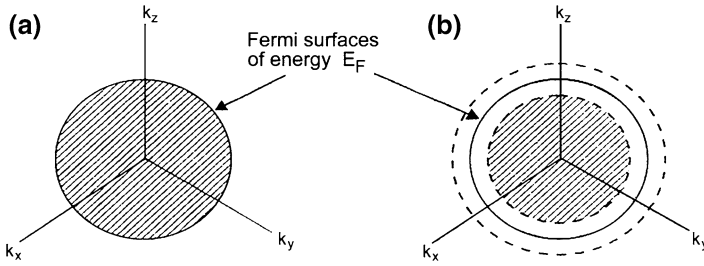


Fig. 13.25 **a** Fermi level in a normal metal with all states below E_F being occupied; **b** the BCS ground state

According to London [33], since superconductivity is manifested as a quantum phenomenon like superfluidity, the magnetic flux should be quantized in units of $2\pi\hbar c/e$. Subsequent experiments by Deaver and Fairbank [34] showed that the flux is indeed quantized, but the quantized unit of flux has half the value i.e. $\pi\hbar c/e$. This implies that the superconducting carriers have a charge $-2e$ which agrees with the concept of Cooper pairs. Cooper [35] proposed that states involving pairs of electrons with attractive interaction are possible. Further, the two electrons moving in a field of phonons could interact mutually by exchange of phonons called virtual phonons. A virtual phonon is one which according to the uncertainty principle does not last long enough to be observed before it is absorbed. These electrons are called Cooper pairs. It is important to note that the BCS theory does not use a boson condensation process to explain superconductivity. The electron pairs form the superconducting carriers and unpaired electrons form the normal carriers.

The effect of attractive interaction on all the electrons in the superconductor at zero temperature is that they are described by the BCS ground state rather than the occupied Fermi level of the normal state (Fig. 13.25a, b).

The BCS ground state is made up of Cooper pairs in which if one electron state is unoccupied, so is the other. It differs from the normal state as some electron states just outside the normal Fermi surface are occupied, some just inside are unoccupied (Fig. 13.26). In a BCS state, the orbitals are occupied in pairs. If an orbital with wave vector $\mathbf{k}$ and spin up $\uparrow$ is occupied, then the orbital with wave vector $-\mathbf{k}$ and spin down $\downarrow$ is also occupied. Also if $\mathbf{k}_1\uparrow$ is vacant, $-\mathbf{k}_1\downarrow$ is also vacant. The Cooper pairs have zero spin and have many characteristics of bosons. The electron-phonon-electron interaction, which could make the coulombic repulsion between the two electrons smaller, favours superconductivity (Fig. 13.27). A Cooper pair ($\mathbf{k}_1\uparrow-\mathbf{k}_1\downarrow$) can go to any other unoccupied position of a pair ($\mathbf{k}_2\uparrow-\mathbf{k}_2\downarrow$) by the exchange of a virtual phonon q .

The Cooper pair is a bound state. The energy difference between the paired state and the unpaired state is the energy gap E_g ; E_g is twice the ‘energy gap parameter Δ ’ i.e. $E_g = 2\Delta$. An energy E_g is required to break a Cooper pair resulting in the

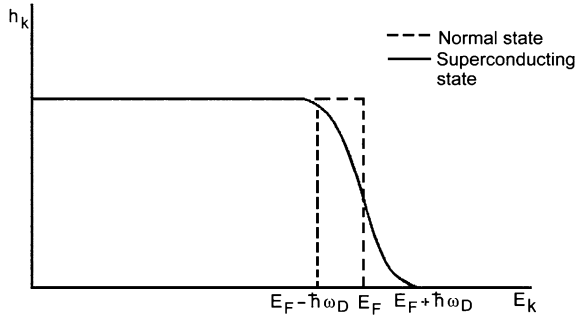
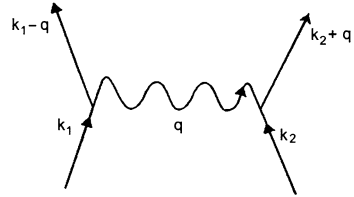


Fig. 13.26 Occupation probability h_k of electrons in a superconducting BCS state; some electrons from the normal state in energy range $(E_F - \hbar\omega_D)$ to E_F are now occupying energy states in the range E_F to $(E_F + \hbar\omega_D)$

Fig. 13.27 Exchange of virtual phonons between electrons



creation of two quasi-particle excitations. The BCS theory gives the following relation for 2Δ at 0 K.

$$\frac{2\Delta(0)}{k_B T_c} = 3.52. \quad (13.58)$$

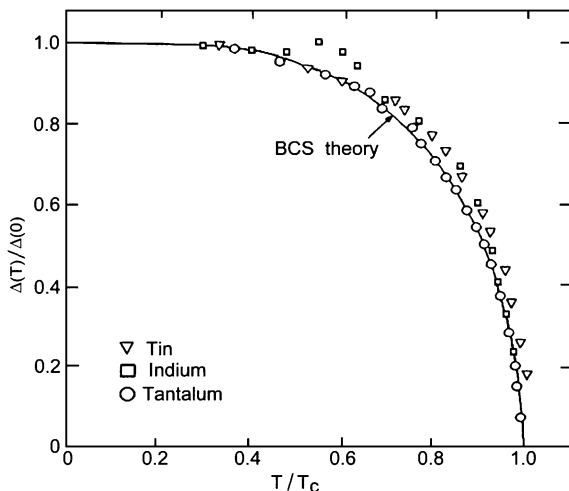
This is a universal relation in the sense that it does not contain any phenomenological quantities. Values of E_g determined by experimental methods for several elements are given in Table 13.7. Most of them are consistent with (13.58) although in the case of Pb and Hg there is significant deviation.

The BCS theory predicts a temperature dependent energy gap. In the vicinity of T_c , the temperature variation is given by

$$\frac{\Delta(T)}{\Delta_0} = 1.76 \left(1 - \frac{T}{T_c} \right)^{1/2}. \quad (13.59)$$

It follows from (13.59) that $\Delta(T) \rightarrow 0$ as $T \rightarrow T_c$. Another interpretation of (13.59) would be that the fraction of superconducting electrons decreases with increasing temperature until at $T = T_c$, all electrons are normal electrons. Experimental results on temperature variation of $\Delta(T)$ for some superconductors are shown in Fig. 13.28. These results are consistent with the BCS theory (13.59).

Fig. 13.28 Temperature dependence of the superconducting energy gap; experimental results are compared with the BCS theory



The most important concept introduced in the BCS theory is the concept of a bound state of two electrons—the Cooper pairs. From this concept several observed properties of superconductors like the critical field, specific heat, spectroscopic absorption, Meissner effect and even the London equation follows in a natural way.

13.6.4 Ginzburg–Landau (GL) Theory

Ginzburg and Landau [29] proposed a theory for superconductivity which is related to Landau's theory of second order phase transitions. They describe the superconducting state in terms of an order parameter $\psi(r)$ which vanishes above T_c and whose magnitude measures the degree of superconducting order at any position. ψ is interpreted as the wave function of the condensed super electrons and is related to the local density of superconducting electrons n_s as

$$n_s = |\psi(r)|^2. \quad (13.60)$$

GL theory in full form is too complicated to present here. Instead we consider the theory in a phenomenological way to highlight the important assumptions and results.

- (i) The phase transition is characterized by the order parameter which is identified with the macroscopic wave function ψ . ψ is a function which varies with space.
- (ii) The order parameter vanishes above T_c and its magnitude measures the degree of superconducting order below T_c .

- (iii) The order parameter can be viewed as a one-particle wave function describing the position of centre of mass of an electron pair. A single wave function is assumed as all the electron pairs are in the same two-electron state.
- (iv) The free energy is written as a function of ψ and vector potential $\mathbf{A}$. By minimizing the free energy, ψ and the supercurrent are obtained.

The theory accounts for all the parameters of a superconductor: penetration depth, coherence length and flux quantization.

The basic postulate of the GL theory is that if ψ is small and varies slowly in space, the free energy density f for a particle can be expanded in a series as a function of ψ (where $\psi = |\psi|e^{i\phi}$) as

$$f = f_n + \alpha|\psi|^2 + \frac{\beta}{2}|\psi|^4 + \frac{1}{2m^*} \left| \left(\frac{\hbar}{i} \nabla - \frac{e^*}{c} \mathbf{A} \right) \psi \right|^2 + \frac{h^2}{8\pi}. \quad (13.61)$$

Here, f_n is the free energy density in the normal phase, α and β phenomenological parameters, $\hbar^2/8\pi$ the magnetic energy density, m^* and e^* are the mass and charge of the particle.

The following differential equations for ψ and the current J are derived using variational principles:

$$\alpha\psi + \beta|\psi|^2\psi + \frac{1}{2m^*} \left(\frac{\hbar}{i} \nabla - \frac{e^*}{c} \mathbf{A} \right)^2 \psi = 0 \quad (13.62)$$

and

$$\mathbf{J} = \frac{e^*}{m^*} \left(\hbar \nabla \phi - \frac{e^*}{c} \mathbf{A} \right) |\psi|^2. \quad (13.63)$$

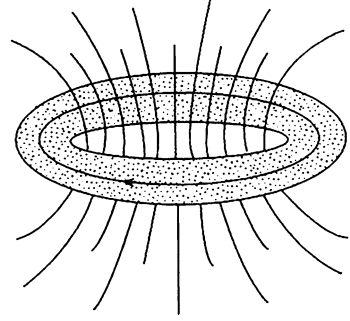
Equations (13.62) and (13.63) are referred to as GL differential equations. The quantities e^* and m^* in (13.61–13.63) are likely to be construed to have normal electronic values. However, assuming that the ‘particle’ is an electron pair, we consider $m^* = 2m$ and $e^* = 2e$ [7].

The first equation, (13.62) has a form similar to Schrödinger equation for a particle of charge $2e$ and mass $2m$ and an energy eigen value $-\alpha$. GL theory is a generalization of London’s theory to deal with the variation of n_s as n_s is equal to $|\psi(r)|^2$.

13.6.4.1 Flux Quantization

The total magnetic flux that passes through a superconducting material in the form of a ring assumes only quantized values.

Fig. 13.29 Magnetic flux lines about a ring-shaped superconductor



Let us consider a superconducting material in the shape of a ring as in Fig. 13.29. Since the magnetic field cannot penetrate through the material, the flux lines will be enclosed within the ring.

The current inside the ring takes a closed path. Hence, using (13.63) for current density, we have

$$\oint \mathbf{J} \cdot d\mathbf{l} = \oint \left(\hbar \nabla \phi - \frac{e^*}{c} \mathbf{A} \right) \cdot d\mathbf{l} = 0. \quad (13.64)$$

Using Stoke's theorem,

$$\oint \mathbf{A} \cdot d\mathbf{l} = \int \nabla \times \mathbf{A} \cdot d\mathbf{s} = \int \mathbf{B} \cdot d\mathbf{s} = \Phi. \quad (13.65)$$

where Φ is the flux enclosed in the ring. Since the order parameter $\psi(r)$ is single valued, its phase change around the close path must be an integral multiple of 2π . Thus

$$\oint \nabla \phi \cdot d\mathbf{l} = 2\pi n. \quad (13.66)$$

Combining (13.64), (13.65) and (13.66) we get

$$\Phi = \frac{n\hbar c}{2e} = n\Phi_0. \quad (13.67)$$

where $\Phi_0 = \hbar c/2e$. Φ_0 is known as a fluxoid or flux quantum. On substitution of numerical values, $\Phi_0 = 2.067 \times 10^{-7}$ gauss-cm².

13.6.4.2 Type II Superconductors

In Sect. 13.4.3 we have discussed some of the characteristics of type II superconductors. An important difference between type I and type II superconductors is in the mean free path of conduction electrons in the normal state. As mentioned earlier, the mean free path changes on alloying. Figure 13.24 shows the penetration depth λ and the coherence length ξ as a function of the mean free path l . If $\xi < \lambda$, the superconductivity will be of type II.

The ratio λ/ξ is termed Ginzburg–Landau parameter κ . Abrikosov [36] classified the superconductors in terms of the GL parameter. For pure metals which are of type I, $\kappa < 1/\sqrt{2}$ and for type II superconductors $\kappa > 1/\sqrt{2}$. The magnetization curves for some superconductors with the same H_c and different values of κ are shown in Fig. 13.30. The curve for $\kappa < 1/\sqrt{2}$ is of type I and those for $\kappa \sim 0.8$ and $\kappa = 2$ are of type II. The area under the curve is given by $H_c^2/8\pi$ for all the cases.

The magnetic phase diagram for a type II superconductor is shown in Fig. 13.31. Below H_{c1} all the flux is expelled as it is the Meissner state. Between H_{c1} and H_{c2} it is the vortex state. In the vortex state, the vortices act as tiny magnets and repel each other. The superconducting currents will be in the form of vortices within the bulk

Fig. 13.30 Magnetization curves for different values of the G–L parameter κ

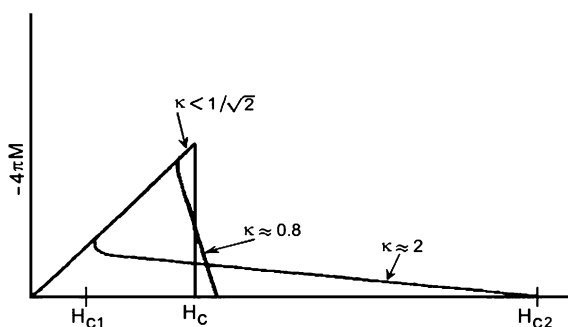


Fig. 13.31 Magnetic phase diagram for a type II superconductor

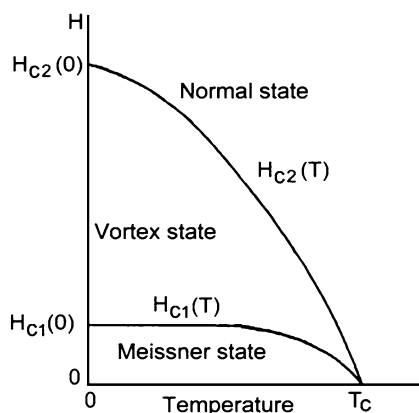


Fig. 13.32 Schematic diagram of a triangular vortex array

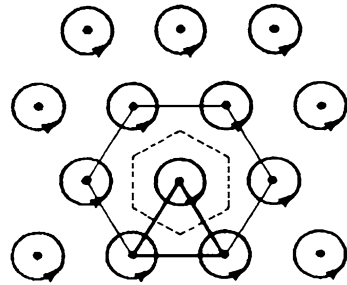
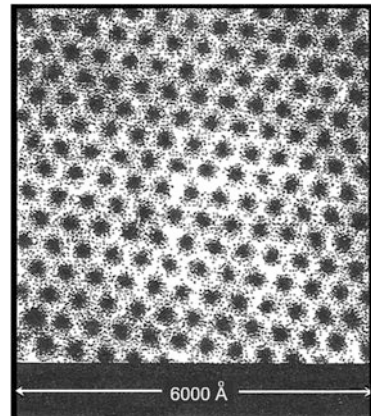


Fig. 13.33 Scanning electron micrograph of a vortex pattern for NbSe₂



of the specimen. In the vortex state, the flux penetrates in the form of quantized flux $\Phi_0 = hc/2e$. The vortex state is stable when the penetration of the applied field causes the surface energy to become negative. The regions of high density states, ‘the vortices’, form symmetric arrays. The schematic picture of such an array is shown in Fig. 13.32. A scanning electron micrograph of a typical type II superconducting material reproduced from Abrikosov [37] is shown in Fig. 13.33. The black star-like regions are images of vortices.

13.7 Normal and Josephson Tunneling

Tunneling of electrons through thin insulating barriers between metals and superconductors was investigated for the first time by Giaever [38]. Josephson [39] showed that tunneling of electron pairs is also possible with very thin insulator layers between two superconductors; his work led to the startling result that in suitable junctions, an applied voltage V results in an AC current with frequency $(2eV/\hbar)$.

In this section, we shall discuss both normal and Josephson tunneling followed by the principle and working of a SQUID.

13.7.1 Normal Tunneling

When two normal metals are separated by an insulator, the insulator acts as a barrier for the conduction of electrons. If the barrier is a thin oxide layer of the order of 10–20 Å, there is a finite probability of the electrons to pass through the barrier. Schematic diagrams of energy versus density of states are shown in Fig. 13.34. Figure 13.34a shows two normal metals *n*, *n* with an insulator *i* at $V = 0$. Both the metals have the same Fermi level E_F . In this situation the current is zero. If a potential difference V is applied across the metals, it results in the shift of energy levels as shown in Fig. 13.34b. Tunneling also takes place with more electrons traversing from one metal to the other. The plot between current (I_{nn}) and voltage (V) is shown in Fig. 13.34c. The variation obeys Ohm's law.

Let us now consider a combination of a normal metal *n* and a superconductor *s* with an insulating layer *i* between. This device *s*–*i*–*n* shown in Fig. 13.35a is fabricated by depositing first a thin film of superconductor on a smooth glass surface. The surface is coated by an aluminium layer which is oxidized to form the insulator. A second coating of a normal metal is then applied.

For depicting the energy levels, we shall adopt a semiconductor model for the superconductor i.e. we show the energy gap $E_g = 2\Delta$ as if it is the gap between two bands (Fig. 13.35a). Initially, the Fermi level E_F of the metal is at the middle of the

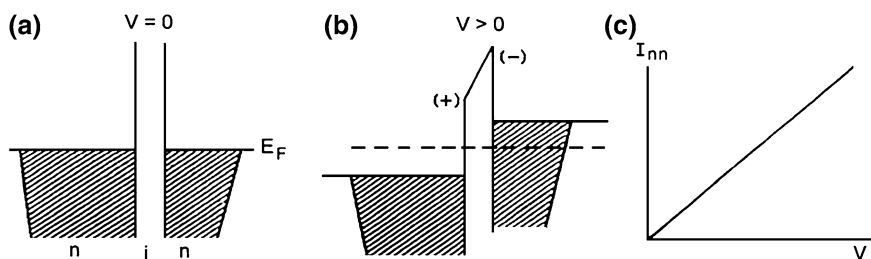


Fig. 13.34 Energy levels of electrons on the two sides of a metal-insulator- metal junction: **a** at $V = 0$, **b** at $V > 0$ and **c** plot of I_{nn} versus V

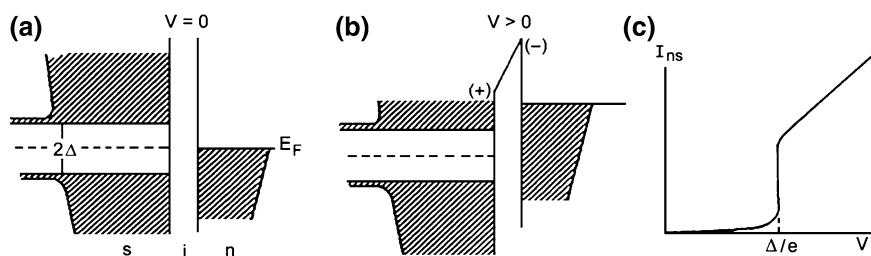


Fig. 13.35 Energy levels of electrons on the two sides of a superconductor- insulator- metal junction: **a** at $V = 0$, **b** at $V > 0$ and **c** plot of I_{ns} versus V

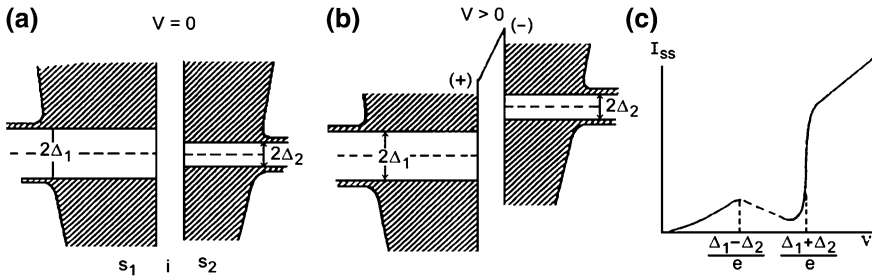


Fig. 13.36 Energy levels of electrons on the two sides of a superconductor-insulator-superconductor metal junction: **a** at $V = 0$, **b** at $V > 0$ and **c** plot of I_{ss} versus V

energy gap of the superconductor. The shaded regions indicate occupied states. When a voltage is applied, the Fermi level of the metal is shifted upwards (higher) with respect to the middle of the energy gap of s (Fig. 13.35b). However, since s is a superconductor, current cannot pass until the voltage V is equal to Δ/e or $E_g/2e$. The I_{ns} versus V plot for this device is shown in Fig. 13.35c. The rise of the current at V is sharp at $T = 0$; at higher temperatures, the onset of the current is less sharp and there is a small current even at $V < (\Delta/e)$; this is due to the thermally excited electrons. The important point is that this provides a simple method for the determination of the energy gap E_g .

We now consider tunneling between two superconductors having different energy gaps $2\Delta_1$ and $2\Delta_2$. The experiments by Giaever [38] were done with aluminium and lead with aluminium oxide barrier. Figure 13.36a shows the initial situation where s_1 , s_2 correspond to lead and aluminium respectively. They have their centres of energy gap at the same level. A potential difference is applied to raise the energy levels of Al with respect to those of Pb (Fig. 13.36b). This encourages tunneling from Al to Pb (i.e. s_2 to s_1), resulting in current I_{ss} . The I_{ss} - V curve is shown in Fig. 13.36c. As the top of the s_2 -gap passes the top of the s_1 -gap a peak is observed at a voltage $V_1 = (\Delta_1 - \Delta_2)/e$. Thereafter, the current decreases as the number of available states decreases. Again when the bottom of the energy gap of s_2 passes the top of s_1 -gap there is an increase in current corresponding to a voltage $V_2 = (\Delta_1 + \Delta_2)/e$. From the voltages V_1 and V_2 , the energy gaps of both the superconductors are obtained.

In the above cases, it is essentially a single electron that tunnels through the barrier. Hence the process is called ‘single particle tunneling’.

13.7.2 Josephson Tunneling

We have discussed single particle tunneling where the oxide barrier is 10–20 Å thick. Josephson [39] predicted that in addition to the ‘normal tunneling’ of single electrons, supercurrents consisting of pairs of electrons can flow across a barrier. Such an effect

was experimentally observed and is called ‘quantum tunneling’. Physically it means that the Cooper pairs might tunnel under suitable conditions with zero potential difference across very thin barriers; the bound electrons traverse through the junction from one superconductor to the other without dissociating. In effect, a significant overlap of the phases occurs to establish a weak link between the two superconductors. This results in a current flow and is referred to as DC Josephson effect.

In normal tunneling, the barrier is 10–20 Å thick; it is made of Al_2O_3 . Josephson effect requires thinner barriers. Al_2O_3 is not suitable for making such thin barriers. Instead oxides of Sn and Pb in the form of films are employed [40].

Josephson also predicted that if a DC voltage is applied across the junctions, it causes an alternating current flow. This is referred to as AC Josephson effect.

13.7.2.1 DC Josephson Effect

Let ψ_1 be the probability amplitude of electron pairs on one side of a junction and ψ_2 on the other. Initially we assume that the two superconductors are at the same potential.

The time dependent Schrödinger equations applied to ψ_1 and ψ_2 are

$$i\hbar \frac{\partial \psi_1}{\partial t} = \hbar k_0 \psi_2 \text{ and } i\hbar \frac{\partial \psi_2}{\partial t} = \hbar k_0 \psi_1 \quad (13.68)$$

where $\hbar k_0$ represents electron pair coupling interaction across the insulator. If the insulator is very thick $k_0 = 0$ and pair tunneling cannot take place.

Let us assume the solution to be of the form

$$\psi_1 = n_1^{1/2} e^{i\phi_1} \text{ and } \psi_2 = n_2^{1/2} e^{i\phi_2} \quad (13.69)$$

where n_1 and n_2 are the Cooper pair concentrations in the two superconductors and ϕ_1 and ϕ_2 are the phases. From (13.68) in (13.69) we get

$$\frac{\partial \psi_1}{\partial t} = \frac{1}{2} n_1^{-1/2} e^{i\phi_1} \frac{\partial n_1}{\partial t} + i\psi_1 \frac{\partial \phi_1}{\partial t} = -ik_0 \psi_2 \quad (13.70)$$

and

$$\frac{\partial \psi_2}{\partial t} = \frac{1}{2} n_2^{-1/2} e^{i\phi_2} \frac{\partial n_2}{\partial t} + i\psi_2 \frac{\partial \phi_2}{\partial t} = -ik_0 \psi_1 \quad (13.71)$$

Multiplying (13.70) by $n_1^{1/2} e^{-i\phi_1}$ and (13.71) by $n_2^{1/2} e^{-i\phi_2}$, we get

$$\frac{1}{2} \frac{\partial n_1}{\partial t} + in_1 \frac{\partial \phi_1}{\partial t} = -ik_0 n_1^{1/2} n_2^{1/2} e^{i\delta} \quad (13.72)$$

and

$$\frac{1}{2} \frac{\partial n_2}{\partial t} + in_2 \frac{\partial \phi_2}{\partial t} = -ik_0 n_1^{1/2} n_2^{1/2} e^{-i\delta} \quad (13.73)$$

where

$$(\phi_2 - \phi_1) = \delta. \quad (13.74)$$

Equating the real and imaginary parts in (13.72) and (13.73) gives

$$\frac{\partial n_1}{\partial t} = 2k_0(n_1 n_2)^{1/2} \sin \delta; \quad \frac{\partial n_2}{\partial t} = -2k_0(n_1 n_2)^{1/2} \sin \delta \quad (13.75)$$

and

$$\frac{\partial \phi_1}{\partial t} = -k_0 \left(\frac{n_2}{n_1}\right)^{1/2} \cos \delta; \quad \frac{\partial \phi_2}{\partial t} = -k_0 \left(\frac{n_1}{n_2}\right)^{1/2} \cos \delta. \quad (13.76)$$

Since $n_1 \approx n_2$, we have from (13.76)

$$\frac{\partial \phi_1}{\partial t} = \frac{\partial \phi_2}{\partial t}$$

or

$$\frac{\partial}{\partial t}(\phi_1 - \phi_2) = 0 \quad (13.77)$$

Further, from (13.75) we find that

$$\frac{\partial n_2}{\partial t} = -\frac{\partial n_1}{\partial t} \quad (13.78)$$

The current flow J from one superconductor to the other through the junction is proportional to $\partial n_2 / \partial t$ or $\partial n_1 / \partial t$. Equations (13.75) and (13.76) show that the current across the junction depends on the phase difference δ .

$$J = J_0 \sin \delta = J_0 \sin(\phi_2 - \phi_1). \quad (13.79)$$

The result shows that

- (i) J_0 is proportional to k_0 and represents the maximum zero voltage current that can be passed through the junction.
- (ii) with zero applied voltage a DC current will flow across the junction, with values between J_0 and $-J_0$ depending on the phase difference $\phi_2 - \phi_1$.

13.7.2.2 AC Josephson Effect

Let us now consider the case when a DC voltage V_0 is applied across the junction i.e. the insulator. An electron pair will be subjected to a potential energy difference $2eV_0$. This means that, on one side the pair is at a potential energy $-eV_0$ and on the other side the potential energy is $+eV_0$.

The equation of motion is given by

$$i\hbar \frac{\partial \psi_1}{\partial t} = \hbar k_0 \psi_2 - eV_0 \psi_1; \quad i\hbar \frac{\partial \psi_2}{\partial t} = \hbar k_0 \psi_1 - eV_0 \psi_2. \quad (13.80)$$

Proceeding in the same manner as in the case of DC Josephson effect, we obtain the following equations for $\partial n / \partial t$ and $\partial \phi / \partial t$.

$$\frac{\partial n_1}{\partial t} = 2k_0(n_1 n_2)^{1/2} \sin \delta \quad (13.81)$$

and

$$\frac{\partial \phi_1}{\partial t} = \frac{eV_0}{\hbar} - k_0(n_2/n_1)^{1/2} \cos \delta. \quad (13.82)$$

Similarly,

$$\frac{\partial n_2}{\partial t} = -2k_0(n_1 n_2)^{1/2} \sin \delta \quad (13.83)$$

and

$$\frac{\partial \phi_2}{\partial t} = -\frac{eV_0}{\hbar} - k_0(n_1/n_2)^{1/2} \cos \delta. \quad (13.84)$$

Assuming $n_1 \approx n_2$, we get from (13.82) and (13.84)

$$\frac{\partial}{\partial t}(\phi_2 - \phi_1) = \frac{\partial \delta}{\partial t} = -\frac{2eV_0}{\hbar}. \quad (13.85)$$

On integration (13.85) gives

$$\delta(t) = \delta(0) - \frac{2eV_0 t}{\hbar} \quad (13.86)$$

The superconducting current is given by

$$J = J_0 \sin[\delta(0) - 2eV_0 t / \hbar] \quad (13.87)$$

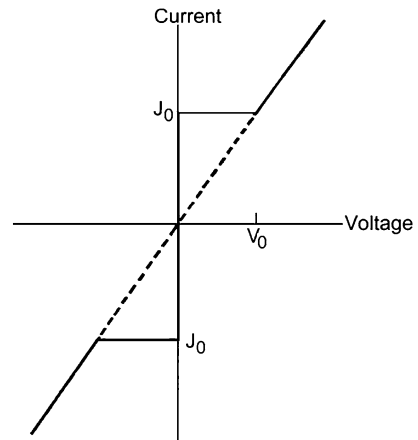
Equation (13.87) shows that the current oscillates with a frequency $\omega = 2eV_0/\hbar$. Thus if two superconducting elements are at different electric potentials, the transfer of a pair of electrons from one side to the other involves an energy change of eV_0 . This appears as a photon of energy $\hbar\omega = 2eV_0$.

The tunnel current arises from a transition between a state in which a pair is at one phase on one side and at a different phase on the other side. The relative phase of the two states oscillates at the beat frequency corresponding to the energy difference $\hbar\omega = 2eV_0$. This gives a current that oscillates at that frequency as in the case of atomic transitions.

A photon of energy $\hbar\omega = 2eV_0$ is emitted or absorbed whenever an electron pair crosses the barrier. Hence AC Josephson effect permits an accurate determination of $e/\hbar$ from the relationship between voltage and the frequency (13.87). It has been observed that a DC voltage of 1 mV across a junction produces a frequency of 483.6 GHz.

We shall summarize the behaviour of a Josephson junction with reference to the current-voltage diagram (Fig. 13.37). At zero voltage, a DC current of the order of mA passes through the junction with a value between $+J_0$ and $-J_0$ depending upon the phase difference ($\phi_2 - \phi_1$) according to (13.79). If now a small DC voltage (of the order of mV) is applied, it still results in a DC current between $-J_0$ and J_0 and the phase difference gets adjusted to suit (13.79). But no potential difference appears across the junction. This process which is called the DC Josephson effect continues as the voltage is increased up to a critical value V_0 ; the corresponding value of the DC current is now J_0 . At this voltage, suddenly a potential drop shows up across the junction. For DC voltages greater than V_0 , say V_1 , an AC current starts flowing through the junction with a frequency $\omega = 2eV_1/\hbar$ according to (13.87). This is the AC Josephson effect. It may be mentioned that the passage of the AC current is accompanied by emission of radiation of the same frequency $2eV_1/\hbar$; this frequency is ~ 100 GHz.

Fig. 13.37 Current-voltage characteristic for a DC Josephson junction according to (13.87)



13.7.3 Macroscopic Quantum Interference Effect

The quantum interference between two Josephson tunnel junctions results in a current flow even in the absence of voltage difference. The superconducting quantum interference device (SQUID) is used to study this effect. The principle of SQUID is discussed in this section.

We consider a closed circuit consisting of a ring of superconducting material (Fig. 13.38) with arms a and b . The current entering through i_1 passes through the arms a and b with Josephson junctions X. It is then collected at i_2 . Let the phase difference for the two paths be ϕ_a and ϕ_b . In the absence of a magnetic flux within the ring, the two phases are equal. When a magnetic flux Φ is allowed to pass through the space enclosed by the ring, there will be a phase difference between the two paths. This will allow a current flow J_a and J_b in the two arms.

The phase difference in the presence of Φ is given by $\phi_b - \phi_a = (2e/\hbar c)\Phi$ or,

$$\phi_b = \phi_0 + \frac{e}{\hbar c}\Phi \text{ and } \phi_a = \phi_0 - \frac{e}{\hbar c}\Phi. \quad (13.88)$$

where ϕ_0 is the initial phase. The total current due to the phase difference is given by

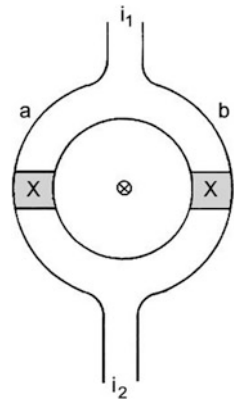
$$J_{\text{total}} = J_a + J_b = J_0 \left[\sin\left(\phi_0 + \frac{e}{\hbar c}\Phi\right) + \sin\left(\phi_0 - \frac{e}{\hbar c}\Phi\right) \right]. \quad (13.89)$$

Hence

$$J_{\text{total}} = 2(J_0 \sin \phi_0) \cos \frac{e\Phi}{\hbar c}. \quad (13.90)$$

Equation (13.90) shows that J_{total} varies with Φ and has maxima when $(e\Phi/\hbar c) = n\pi$ where n is an integer. The current flowing in the two junctions

Fig. 13.38 Principle of a DC SQUID



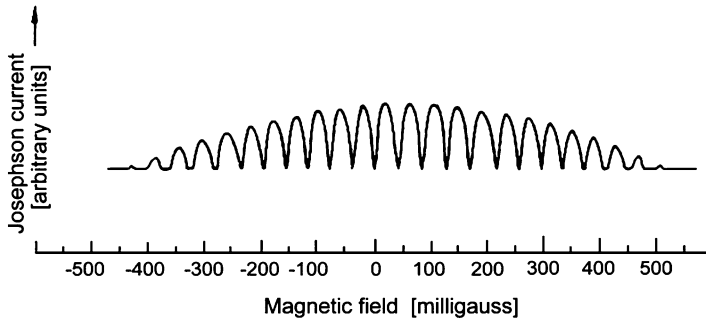
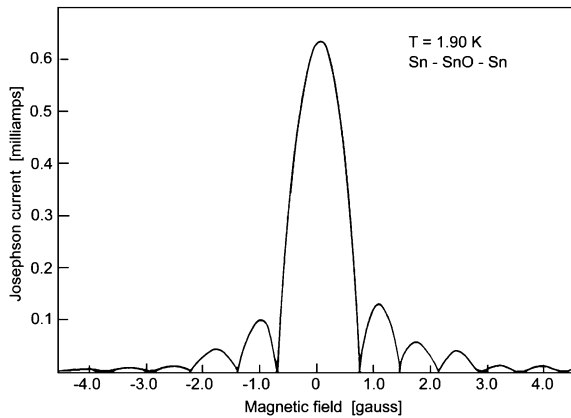


Fig. 13.39 Plot of Josephson current versus magnetic field for a two-junction interferometer showing interference effect

Fig. 13.40 Plot of Josephson current versus magnetic field for a single junction showing diffraction effect



oscillates with a period equal to a flux quantum. The oscillations are the result of interference between macroscopic wave functions at the two junctions. This is just like the interference between two coherent light beams. A long period variation is also observed which is due to the finite dimensions of the two junctions giving rise to diffraction effects.

The Josephson effects are manifestations of quantum interference phenomena on a macroscopic scale. The term quantum refers to the quantum state of the superconductors and the term interference to the properties due to difference in phase of a particular state. The experimental traces of Josephson current versus magnetic field recorded using Sn–SnO–Sn tunnel junctions are shown in Fig. 13.39 [41] and Fig. 13.40 [42]. Figure 13.39 shows the interference effects recorded with two junctions; the envelop arises because of finite size of individual junctions. Figure 13.40 shows diffraction effect recorded with a single junction.

Fig. 13.41 Comparison of currents due to Josephson tunneling, quasiparticle tunneling and normal tunneling types; the Shapiro steps are due to absorption of microwave field

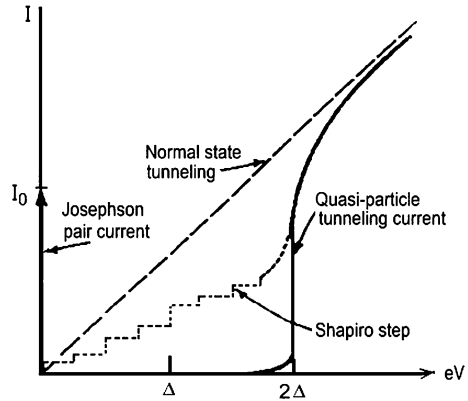


Figure 13.41 shows various tunneling currents and Shapiro steps, which are induced when a Josephson junction is exposed to microwave radiation, i.e. the radiation is absorbed at step interval of $\Delta V = h\nu/2e$ [43].

13.7.4 Electrical Characteristics of a SQUID

We have discussed the principle of DC SQUID in the earlier section. We now see the basic electrical characteristics. When the flux through the SQUID is changed the current-voltage characteristics change. The squid is a very sensitive detector of the change in flux. A SQUID is essentially a flux-to-voltage transducer that converts a change in magnetic flux to a change in voltage easily detectable with conventional electronics. The schematic and electrical characteristics of a DC SQUID is shown in Fig. 13.42. In Fig. 13.42a the DC SQUID is biased with a current I and flux Φ . Each of the two Josephson tunnel junctions X and X is shunted with the self capacitance C and resistance R . The inductance of the loop is L . The flux and current bias are selected such that the voltage is almost linear in applied flux. Figure 13.42b shows the current-voltage characteristics with flux $\Phi = n\Phi_0$ and $(n + \frac{1}{2})\Phi_0$. The plot in Fig. 13.42c shows the voltage across the SQUID as a function of the flux (Φ/Φ_0) at a fixed bias (the dashed line in Fig. 13.42b).

Thus when the flux through the SQUID is changed it is recorded as a change in the voltage. Hence the SQUID is called a sensitive magnetometer. It can record extremely feeble magnetic fields of the order of 10^{-15} T.

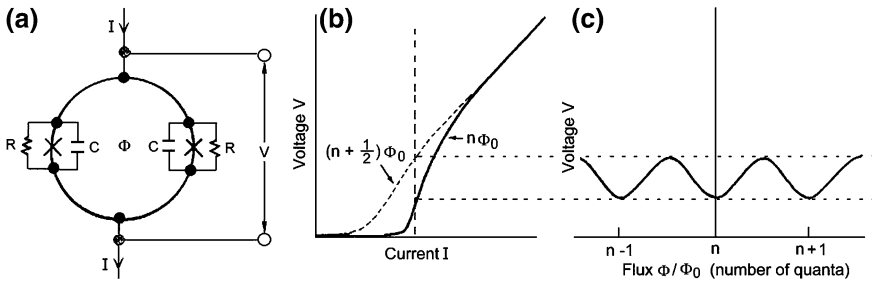


Fig. 13.42 Schematic and characteristics of a DC SQUID

13.8 High Temperature Superconductors

13.8.1 Structure and Transition Temperature

Till 1973, the highest temperature for superconducting phase transition T_c was ≈ 23 K for Nb_3Ge . The discovery of superconductivity in the family of cuprates by Bednorz and Muller [12] heralded the beginning of the era of high temperature superconductivity (HTS). These are La_2CuO_4 in which lanthanum is partially replaced by barium or strontium. Since then a large number of Cu oxides with $T_c \approx 30$ K were discovered. In early 1987, several investigators found that replacing lanthanum with yttrium raised the critical temperature to 92 K. This was the first superconductor of the type $\text{YBa}_2\text{Cu}_3\text{O}_7$ to show superconductivity with $T_c > 77$ K (liquid nitrogen temperature). It was a major step in the development of HTS technology as liquid nitrogen can be produced cheaply from air. The compound $\text{YBa}_2\text{Cu}_3\text{O}_7$ (YBCO) is referred to as 123 superconductor. $\text{YBa}_2\text{Cu}_3\text{O}_7$ has an orthorhombic structure. The structure has two Cu–O sheets in the ab plane and Cu–O chains along the b axis. The structure of $\text{YBa}_2\text{Cu}_3\text{O}_7$ and the variation of resistance with temperature is shown in Fig. 13.43. Most of the high T_c superconductors have tetragonal or nearly tetragonal (some have small distortion leading to orthorhombic) structure. The materials can be generally considered as quasi-tetragonal.

Around 1988, a transition temperature of $T_c = 125$ K was reached in the compound $\text{Tl}_2\text{Ba}_2\text{Ca}_2\text{Cu}_3\text{O}_{13}$. This was the beginning of high temperature superconductor technology era with possible commercial applications. Figure 13.44 shows year-wise record of T_c of some materials. The highest recorded T_c is 173 K for $\text{HgBa}_2\text{Ca}_2\text{Cu}_3\text{O}_8$ at 30 GPa pressure.

Several compounds with the general formula $\text{La}_{2-x}\text{M}_x\text{CuO}_4$ are HT superconductors. They have the tetragonal K_2NiF_4 structure at room temperature and becomes orthorhombic around 180 K (Fig. 13.45a). The lanthanum ion can be replaced by Pr, Nd or Gd ions; M represents either Sr or Ba. The temperature variation of resistivity of some of these compounds is shown in Fig. 13.45b.

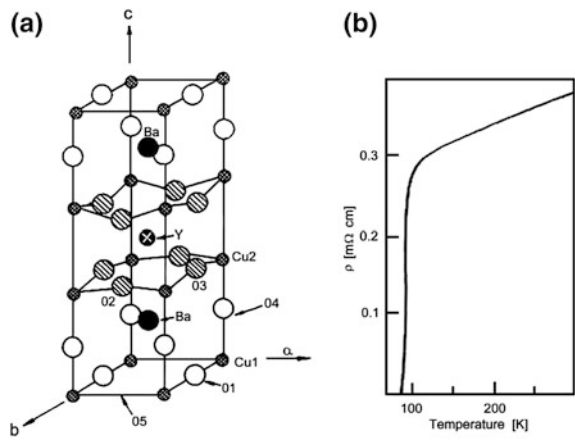


Fig. 13.43 YBa₂Cu₃O₇: **a** structure, **b** temperature variation of resistance

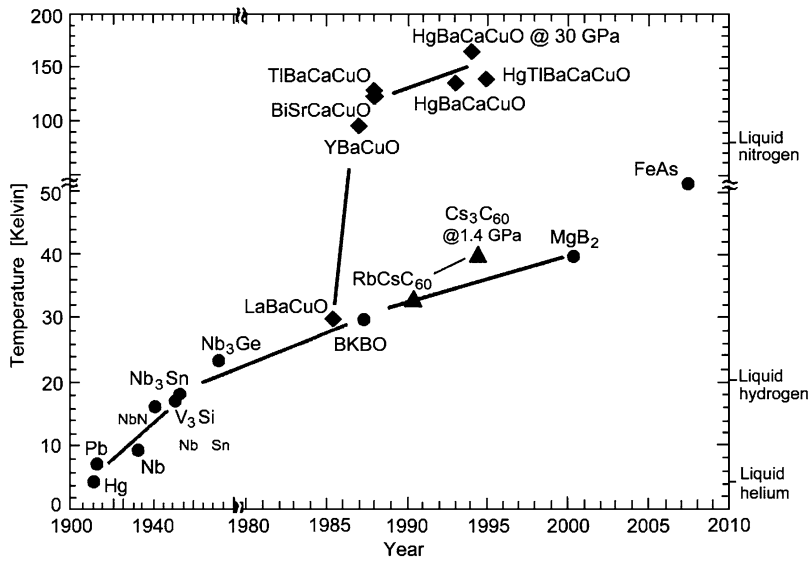


Fig. 13.44 Year-wise discovery of superconductivity

A list of some superconductors along with their T_c values and the number (n) of adjacent Cu–O planes in the unit cell are shown in Table 13.8. It is to be noted that larger the number of Cu–O planes the higher are the T_c values within a group. This is true up to a value of $n \geq 3$. It is also to be seen that for $n = 1, 2$ and 3 , values of T_c approximately lie in the range 20–85 K, 60–92 K and 105–125 K.

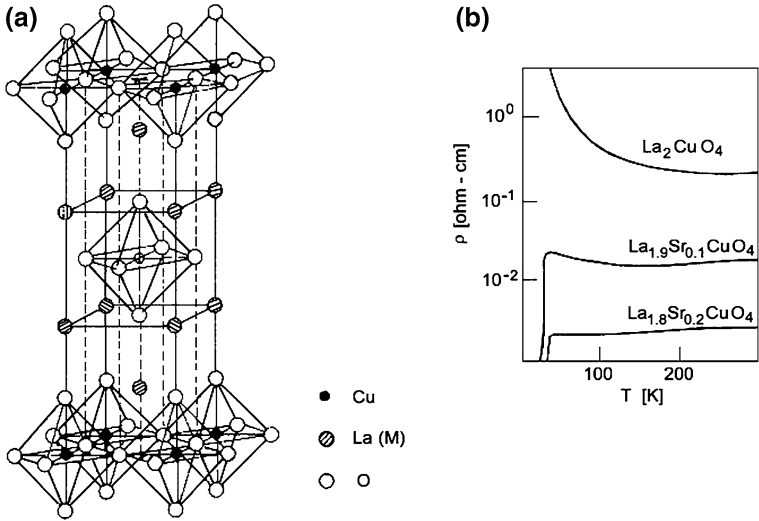


Fig. 13.45 $\text{La}_{2-x}\text{M}_x\text{CuO}_4$: **a** structure, **b** temperature variation of resistance

Table 13.8 Transition temperatures (T_c) and number n of Cu–O planes in some HT superconductors

Superconductor	T_c (K)	n
$\text{Tl}_2\text{Ba}_2\text{CuO}_6$	80	1
$\text{Tl}_2\text{Ba}_2\text{CaCuO}_8$	108	2
$\text{Tl}_2\text{Ba}_2\text{Ca}_2\text{Cu}_3\text{O}_{10}$	125	3
$\text{Bi}_2\text{Sr}_2\text{CuO}_6$	20	1
$\text{Bi}_2\text{Sr}_2\text{CaCuO}_8$	85	2
$\text{Bi}_2\text{Sr}_2\text{Ca}_2\text{Cu}_3\text{O}_{10}$	110	3
$\text{TlBa}_2\text{CuO}_5$	50	1
$\text{TlBa}_2\text{CaCu}_2\text{O}_7$	80	2
$\text{TlBa}_2\text{Ca}_2\text{Cu}_3\text{O}_9$	110	3
$\text{TlBa}_2\text{Ca}_3\text{Cu}_4\text{O}_{11}$	122	4

The Cu–O planes are responsible for many of the electrical properties of the HTS materials and also the structures. The anisotropic structures are the cause of anisotropic properties.

13.8.2 Properties

- (i) The HTS materials are all type II superconductors with two critical magnetic fields H_{c1} and H_{c2} .
- (ii) The transition temperature can be varied by changing the doping.
- (iii) The presence of two-dimensional Cu–O sheets in the ab plane and Cu–O chains along the perpendicular direction shows that the structures of HTS materials are highly anisotropic. This is reflected in many of the properties.

The important length scales are the coherence length and magnetic penetration depth. These are related to H_{c1} and H_{c2} by the relations

$$H_{c1} = \Phi_0 / 4\pi\lambda^2$$

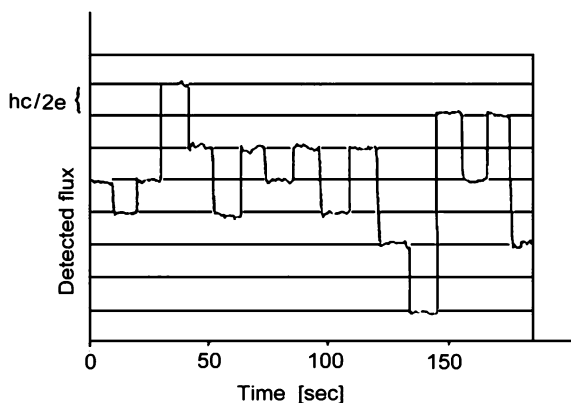
and

$$H_{c2} = \Phi_0 / 2\pi\xi^2. \quad (13.91)$$

For Y123 single crystal, values of the penetration depth λ in the ab plane and along the c axis at 0 K are $\sim 1,500$ and $\sim 7,000$ Å respectively. Values of the coherence length ξ in the ab plane and along the c axis at 0 K are 12–16 Å and 1.3–3 Å respectively. These are estimated from experimental results on magnetic fields using (13.91).

- (iv) Experiments have shown that the magnetic flux Φ is quantized in units of $\Phi_0 = hc/2e$ as in the case of conventional superconductors. This has been measured for Y123 superconductor. Magnetic flux through a ring of Y123 superconductor was obtained by using a ring of outer and inner diameters 10 and 5 mm respectively. The magnetic flux was measured as a function of time. The flux generated by electromagnetic noise below T_c was measured using a SQUID magnetometer. Figure 13.46 shows flux versus time. The flux

Fig. 13.46 Flux versus time in a Y123 superconducting ring



was calibrated with a long solenoid passing through the ring. The value of fluxoid quantum was found to be $\Phi_0 = (0.97 \pm 0.04) hc/2e$ [7].

- (v) Results on energy gap obtained from tunneling, infrared reflectivity and specific heat measurements indicate anisotropy in the energy gap. The results show large difference between the gaps in the c and ab directions. For Y123, IR reflectivity measurements have given values for the energy gap as $(2\Delta/k_B T_c) \approx 8$ in ab plane and ≈ 3.5 along the c direction.
- (vi) Experimental evidence for electron pairing has been established. The mechanism of electron pairing is not apparent.

13.9 Applications

There are numerous applications of superconductivity. Only some of them are mentioned here.

Power Transmission: For large scale applications, superconductors are advantageous over normal metals, because of low resistance i.e. low power loss. But the cost involved in production of low temperatures and in handling the superconducting materials make the conventional materials more advantageous. On the other hand the use of HTS materials eliminates iron cores in motors, generators and transformers resulting in reduction in size, weight and losses from iron cores. Also, HTS materials are of type II superconductors with high magnetic fields. These properties make them particularly useful.

Levitation: Superconductors display the property of levitation. This has led to the magnetically levitated high speed trains using liquid helium. These trains 'MAG-LEV' have been tried in Japan, Germany and USA.

SQUIDS: Early applications of superconducting materials in electronics were and continue to be instruments based primarily on superb sensitivity in magnetic fields. This has led to several small-scale applications where the currents involved are small. These are Josephson junction devices, infrared detectors and SQUID magnetometers.

Biomagnetism: An important area where superconductor devices are used is in the field of biomagnetism. Josephson junctions are fabricated with HTS materials for use in medical diagnostics such as checking the function of heart and brain. This technique is known as magnetic resonance imaging (MRI). By impinging a strong superconductor-derived magnetic field onto the human body, hydrogen atoms that exist in the body's water and fat molecules are forced to accept energy from the magnetic field. They then release this energy at a frequency that can be detected and displayed graphically by a computer. MRI was actually discovered in the mid-1940s but the first MRI on a human body was performed only in 1977. It took almost five hours to produce one image. The time taken nowadays is only a few seconds.

Satellite Communication: Besides being ultra-sensitive detectors of magnetic fields, superconductors also excel in detection of extremely faint electromagnetic

signals e.g. signals originating from outer space. SQUID detectors of high sensitivity are used in many radioastronomy observatories worldwide. HTS thin films have been fabricated into passive microwave devices such as filters and resonators for use in space technology.

The most significant change in satellite communication is the ‘all-digital receiver’ on a one centimeter superconducting niobium chip. The chip contains about 11,000 Josephson junctions laid out to form superconducting Rapid Signal Flux Quantum (RSFQ) circuits.

Computer Technology: In the field of computer technology, low temperature technology can take computing speed far beyond the theoretical limits of silicon while simultaneously effecting major reductions in size and power requirements. Another emerging area is in the field of communication. One of them is the superconducting router of high speed data communication up to 160 GHz. Since internet traffic is increasing exponentially, superconductor technology is expected to meet this requirement.

The development of RSFQ logic circuits has ultra-fast operating speed of several hundred GHz and very low power loss (nanowatts).

Other Developing Areas: Superconducting X-ray detectors and ultra fast superconducting light detectors.

13.10 Concluding Remarks

Superconductivity is one of the most important areas in physics. A variety of experimental observations have been made like the Meissner effect, the isotope effect and the Josephson effects. In order to explain the experimental observations, a number of theories were formulated like Londons’ theory, the BCS theory and the Ginzburg–Landau theory. These theories involved the introduction of new concepts like the energy gap, the coherence length, the critical field and the Cooper pairs. Over the years a number of applications of superconductivity have been discovered like superconducting magnets, maglev trains and SQUID devices.

The importance of superconductivity is indicated by the award of the Nobel Prize to a large number of investigators in this field. These are:

The 1913 Nobel Prize to Kammerlingh Onnes for his investigations of properties of matter at low temperatures.

The 1972 Nobel Prize to Bardeen, Cooper and Schrieffer for their theory of superconductivity.

The 1973 Nobel Prize to Giaever for the discovery of tunneling in superconductors and to Josephson for the discovery of phenomena known after him.

The 1987 Nobel Prize to Bednorz and Muller for the discovery of high temperature superconductors.

The 2003 Nobel Prize to Abrikosov and Ginzburg for contributions to the theory of superconductivity.

13.11 Problems

1. The critical temperature of mercury with isotopic mass 199.5 atomic mass units is 4.185 K. Calculate its T_c when the isotopic mass is changed to 203.4.
2. A superconducting metal has a critical temperature of 7.26 K at zero magnetic field and a critical field of 8×10^5 A/m at 0 K. Find the critical field at 4 K.
3. Calculate the ratio of penetration depths at 0 and 3.6 K for lead with critical temperature of 7.22 K.
4. Calculate the frequency of radiation emitted from a Josephson junction; the voltage across the junction is 5 μ V.
5. From thermodynamic considerations show that the entropy of a superconductor in the normal state is higher than the entropy in the superconducting state.
6. Describe the salient features of BCS theory and the BCS ground state.
7. Describe the working of a DC SQUID and mention some applications.

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